

**US Army Corps
of Engineers**
Louisville District

Solicitation For

Conroe Army Reserve Center Microgrid Conroe, TX

P2: 505014

Design-Bid-Build

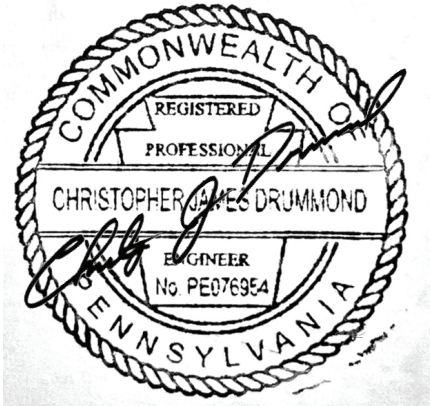
**Specifications Vol. 2 (Div. 26-33)
Certified Final Design**

31 March 2025

W912QR25RA023

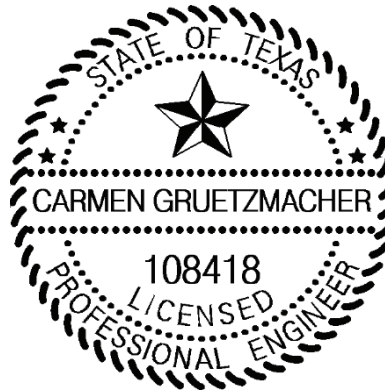
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Disposition: Maintain for 15yrs after construction**

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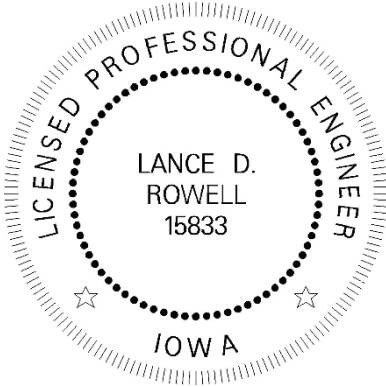
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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code
for Electricity Metering

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for
Hard-Drawn Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for
Concentric-Lay-Stranded Copper Conductors,
Hard, Medium-Hard, or Soft

ASTM D709 (2017) Standard Specification for
Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100 (2000; Archived) The Authoritative
Dictionary of IEEE Standards Terms

IEEE C2 (2023) National Electrical Safety Code

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing
Specifications for Electrical Power
Equipment and Systems

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1 (2015) Standard for Good Workmanship in
Electrical Construction

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C12.7 (2014) Requirements for Watthour Meter
Sockets

ANSI C80.1 (2020) American National Standard for
Electrical Rigid Steel Conduit (ERSC)

ANSI C80.3 (2020) American National Standard for

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Electrical Metallic Tubing (EMT)

ANSI C80.5	(2020) American National Standard for Electrical Rigid Aluminum Conduit
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA FU 1	(2012) Low Voltage Cartridge Fuses
NEMA ICS 1	(2022) Standard for Industrial Control and Systems: General Requirements
NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA ICS 4	(2015) Application Guideline for Terminal Blocks
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA KS 1	(2013) Enclosed and Miscellaneous Distribution Equipment Switches (600 V Maximum)
NEMA MG 1	(2021) Motors and Generators
NEMA MG 10	(2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors
NEMA MG 11	(1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors
NEMA RN 1	(2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
NEMA ST 20	(2014) Dry-Type Transformers for General Applications
NEMA TC 2	(2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit
NEMA TC 3	(2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing
NEMA WD 1	(1999; R 2020) Standard for General Color Requirements for Wiring Devices
NEMA WD 6	(2021) Wiring Devices Dimensions Specifications
NEMA Z535.4	(2011; R 2017) Product Safety Signs and Labels

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023) National Electrical Code
NFPA 70E	(2021) Standard for Electrical Safety in the Workplace

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-569	(2019e; Add 1 2022) Telecommunications Pathways and Spaces
TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 431	Energy Efficiency Program for Certain Commercial and Industrial Equipment
29 CFR 1910.147	The Control of Hazardous Energy (Lock Out/Tag Out)
29 CFR 1910.303	Electrical, General

UNDERWRITERS LABORATORIES (UL)

UL 1	(2005; Reprint Jan 2022) UL Standard for Safety Flexible Metal Conduit
UL 5	(2016; Reprint Jul 2022) UL Standard for Safety Surface Metal Raceways and Fittings
UL 5A	(2015; Reprint Aug 2020) Nonmetallic Surface Raceways and Fittings
UL 6	(2022) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 6A	(2008; Reprint Mar 2021) UL Standard for Safety Electrical Rigid Metal Conduit - Aluminum, Red Brass, and Stainless Steel
UL 20	(2018; Reprint Jan 2021) UL Standard for Safety General-Use Snap Switches
UL 44	(2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables
UL 50	(2015) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
UL 67	(2018; Reprint Jul 2020) UL Standard for Safety Panelboards
UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and

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Cables

UL 248-4	(2010; Reprint Apr 2019) Low-Voltage Fuses - Part 4: Class CC Fuses
UL 248-8	(2011; Reprint Aug 2020) Low-Voltage Fuses - Part 8: Class J Fuses
UL 248-10	(2011; Reprint Aug 2020) Low-Voltage Fuses - Part 10: Class L Fuses
UL 248-12	(2011; Reprint Aug 2020) Low Voltage Fuses - Part 12: Class R Fuses
UL 248-15	(2018) Low-Voltage Fuses - Part 15: Class T Fuses
UL 360	(2013; Reprint Aug 2021) UL Standard for Safety Liquid-Tight Flexible Metal Conduit
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 489	(2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 498	(2017) UL Standard for Safety Attachment Plugs and Receptacles
UL 506	(2017; Reprint Jan 2022) UL Standard for Safety Specialty Transformers
UL 508	(2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 674	(2022) UL Standard for Safety Electric Motors and Generators for Use in Hazardous (Classified) Locations
UL 797	(2007; Reprint Mar 2021) UL Standard for Safety Electrical Metallic Tubing -- Steel
UL 869A	(2006; Reprint Jun 2020) Reference Standard for Service Equipment
UL 870	(2016; Reprint Mar 2019) UL Standard for Safety Wireways, Auxiliary Gutters, and Associated Fittings
UL 943	(2016; Reprint Feb 2018) UL Standard for Safety Ground-Fault Circuit-Interrupters

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UL 984	(1996; Reprint Sep 2005) Hermetic Refrigerant Motor-Compressors
UL 1063	(2017; Reprint Jun 2022) UL Standard for Safety Machine-Tool Wires and Cables
UL 1203	(2013; Reprint Apr 2022) UL Standard for Safety Explosion-Proof and Dust-Ignition-Proof Electrical Equipment for Use in Hazardous (Classified) Locations
UL 1242	(2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel
UL 1449	(2021; Reprint Dec 2022) UL Standard for Safety Surge Protective Devices
UL 1660	(2019; Reprint Jan 2022) Liquid-Tight Flexible Nonmetallic Conduit
UL 1699	(2017; Reprint Feb 2022) UL Standard for Safety Arc-Fault Circuit-Interrupters
UL 4248-1	(2022) UL Standard for Safety Fuseholders - Part 1: General Requirements
UL 4248-12	(2018; Reprint Feb 2022) UL Standard for Safety Fuseholders - Part 12: Class R

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE 100.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Panelboards; G

Transformers; G

Wireways; G

SD-03 Product Data

Receptacles; G

Circuit Breakers; G

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Switches; G

Transformers; G

Enclosed Circuit Breakers; G

Motor Controllers; G

Metering; G

Meter Base Only; G

Secondary Bonding Busbar; G

Surge Protective Devices; G

SD-06 Test Reports

600-volt Wiring Test; G

Grounding System Test; G

Transformer Tests; G

Ground-fault Receptacle Test; G

Arc-fault Receptacle Test; G

SD-07 Certificates

Fuses; G

SD-09 Manufacturer's Field Reports

Transformer Factory Tests

SD-10 Operation and Maintenance Data

Electrical Systems, Data Package 5; G

Metering, Data Package 5; G

1.4 QUALITY ASSURANCE

1.4.1 Fuses

Submit coordination data as specified in paragraph, FUSES of this section.

1.4.2 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated. NECA NEIS 1 shall be considered the minimum standard for workmanship.

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1.4.3 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.4.3.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.4.3.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.5 MAINTENANCE

1.5.1 Electrical Systems

Submit operation and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein. Submit operation and maintenance manuals for electrical systems that provide basic data relating to the design, operation, and maintenance of the electrical distribution system for the building. Include the following:

- a. Single line diagram of the "as-built" building electrical system.
- b. Schematic diagram of electrical control system (other than HVAC, covered elsewhere).
- c. Manufacturers' operating and maintenance manuals on active electrical equipment.

1.6 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

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PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70 for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit

ANSI C80.1, UL 6.

2.2.1.2 Rigid Aluminum Conduit

ANSI C80.5, UL 6A.

2.2.2 Rigid Nonmetallic Conduit

PVC Type EPC-40 in accordance with NEMA TC 2, UL 651.

2.2.3 Intermediate Metal Conduit (IMC)

UL 1242, zinc-coated steel only.

2.2.4 Electrical, Zinc-Coated Steel Metallic Tubing (EMT)

UL 797, ANSI C80.3.

2.2.5 Plastic-Coated Rigid Steel and IMC Conduit

NEMA RN 1, Type 40(40 mils thick).

2.2.6 Flexible Metal Conduit

UL 1, limited to 6 feet.

2.2.6.1 Liquid-Tight Flexible Metal Conduit, Steel

UL 360, limited to 6 feet.

2.2.7 Fittings for Metal Conduit, EMT, and Flexible Metal Conduit

UL 514B. Ferrous fittings: cadmium- or zinc-coated in accordance with UL 514B.

2.2.7.1 Fittings for Rigid Metal Conduit and IMC

Threaded-type. Split couplings unacceptable.

2.2.7.2 Fittings for EMT

Die Cast compression type.

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2.2.8 Fittings for Rigid Nonmetallic Conduit

NEMA TC 3 for PVC, and UL 514B.

2.2.9 Liquid-Tight Flexible Nonmetallic Conduit

UL 1660.

2.3 SURFACE RACEWAY

2.3.1 Surface Metal Raceway

UL 5, two-piece painted steel, totally enclosed, snap-cover type. Provide multiple outlet-type raceway with grounding-type receptacle where indicated. Provide receptacles as specified herein, spaced a minimum of one every 18 inches.

2.3.2 Surface Nonmetallic Raceway

UL 5A, nonmetallic totally enclosed, snap-cover type.

2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES

UL 50; volume greater than 100 cubic inches, NEMA Type 1 enclosure; sheet steel, hot-dip, zinc-coated. Where exposed to wet, damp, or corrosive environments, NEMA Type 3R.

2.5 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of NFPA 70 and UL for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 12 months prior to date of delivery to site.

2.5.1 Conductors

Provide the following:

- a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.
- b. Conductors No. 8 AWG and larger diameter: stranded.
- c. Conductors No. 10 AWG and smaller diameter: solid.
- d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.
- e. All conductors: copper, unless type of conductor material is specifically indicated, or specified, or required by equipment manufacturer.

2.5.1.1 Equipment Manufacturer Requirements

When manufacturer's equipment requires copper conductors at the terminations or requires copper conductors to be provided between components of equipment, provide copper conductors or splices, splice boxes, and other work required to satisfy manufacturer's requirements.

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2.5.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

2.5.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

- a. Grounding conductors: Green.
- b. Neutral conductors: White.
- c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.5.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black
 - (2) Phase B - red
 - (3) Phase C - blue
- b. 480/277 volt, three-phase
 - (1) Phase A - brown
 - (2) Phase B - orange
 - (3) Phase C - yellow
- c. 120/240 volt, single phase: Black and red

2.5.3 Insulation

Unless specified or indicated otherwise or required by NFPA 70, provide power and lighting wires rated for 600-volts, Type THWN/THHN conforming to UL 83 or Type XHHW or RHW conforming to UL 44, except that grounding wire may be type TW conforming to UL 83; remote-control and signal circuits: Type TW or TF, conforming to UL 83. Where equipment or devices require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.5.4 Bonding Conductors

ASTM B1, solid bare copper wire for sizes No. 8 AWG and smaller diameter; ASTM B8, Class B, stranded bare copper wire for sizes No. 6 AWG and larger diameter.

2.6 DEVICE PLATES

Provide the following:

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- a. UL listed, one-piece device plates for outlets to suit the devices installed.
- b. For metal outlet boxes, plates on unfinished walls: zinc-coated sheet steel or cast metal having round or beveled edges.
- c. For nonmetallic boxes and fittings, other suitable plates may be provided.
- d. Plates on finished walls: satin finish stainless steel or brushed-finish aluminum, minimum 0.03 inch thick.
- e. Screws: machine-type with countersunk heads in color to match finish of plate.
- f. Sectional type device plates are not be permitted.
- g. Plates installed in wet locations: gasketed and UL listed for "wet locations."

2.7 SWITCHES

2.7.1 Toggle Switches

NEMA WD 1, UL 20, single pole, double pole, and three-way, totally enclosed with bodies of thermoplastic or thermoset plastic and mounting strap with grounding screw. Include the following:

- a. Handles: ivory thermoplastic.
- b. Wiring terminals: screw-type, side-wired.
- c. Contacts: silver-cadmium and contact arm - one-piece copper alloy.
- d. Switches: rated quiet-type ac only, 120/277 volts, with current rating and number of poles indicated.

2.7.2 Switch with Red Pilot Handle

NEMA WD 1. Provide the following:

- a. Pilot lights that are integrally constructed as a part of the switch's handle.
- b. Pilot light color: red and illuminate whenever the switch is closed or "on".
- c. Pilot lighted switch: rated 20 amps and 120 volts or 277 volts as indicated.
- d. The circuit's neutral conductor to each switch with a pilot light.

2.7.3 Breakers Used as Switches

For 120- and 277-Volt fluorescent fixtures, mark breakers "SWD" in accordance with UL 489.

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2.7.4 Disconnect Switches

NEMA KS 1. Provide heavy duty-type switches where indicated, where switches are rated higher than 240 volts, and for double-throw switches. Utilize Class R fuseholders and fuses for fused switches, unless indicated otherwise. Provide horsepower rated for switches serving as the motor-disconnect means. Provide switches in NEMA 1 3R, enclosure.

2.8 FUSES

NEMA FU 1. Provide complete set of fuses for each fusible switch panel and control center. Coordinate time-current characteristics curves of fuses serving motors or connected in series with circuit breakers or other circuit protective devices for proper operation. Submit coordination data for approval. Provide fuses with a voltage rating not less than circuit voltage.

2.8.1 Fuseholders

Provide in accordance with UL 4248-1.

2.8.2 Cartridge Fuses, Current Limiting Type (Class R)

UL 248-12, Class RK-1 RK-5 time-delay type. Provide only Class R associated fuseholders in accordance with UL 4248-12.

2.8.3 Cartridge Fuses, High-Interrupting Capacity, Current Limiting Type (Classes J, L, and CC)

UL 248-8, UL 248-10, UL 248-4, Class J for zero to 600 amperes, Class L for 601 to 6,000 amperes, and Class CC for zero to 30 amperes.

2.8.4 Cartridge Fuses, Current Limiting Type (Class T)

UL 248-15, Class T for zero to 1,200 amperes, 300 volts; and zero to 800 amperes, 600 volts.

2.9 RECEPTACLES

Provide the following:

- a. UL 498, general purpose specification grade. Residential grade receptacles are not acceptable.
- b. Ratings and configurations: as indicated.
- c. Bodies: ivory as per NEMA WD 1.
- d. Face and body: thermoplastic supported on a metal mounting strap.
- e. Dimensional requirements: per NEMA WD 6.
- f. Screw-type, side-wired wiring terminals or of the solderless pressure type having suitable conductor-release arrangement.
- g. Grounding pole connected to mounting strap.
- h. The receptacle: containing triple-wipe power contacts and double or triple-wipe ground contacts.

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2.9.1 Weatherproof Receptacles

Provide receptacles, UL listed for use in "wet locations" with integral GFCI protection. Include cast metal box with gasketed, hinged, lockable and weatherproof while-in-use, polycarbonate, UV resistant/stabilized cover plate.

2.9.2 Ground-Fault Circuit Interrupter Receptacles

UL 943, duplex type for mounting in standard outlet box. Provide device capable of detecting current leak when the current to ground is 6 milliamperes or higher, and tripping per requirements of UL 943 for Class A ground-fault circuit interrupter devices. Provide screw-type, side-wired wiring terminals or pre-wired (pigtail) leads.

2.10 PANELBOARDS

Provide panelboards in accordance with the following:

- a. UL 67 and UL 50.
- b. Panelboards for use as service disconnecting: additionally conform to UL 869A.
- c. Panelboards: circuit breaker-equipped.
- d. Designed such that individual breakers can be removed without disturbing adjacent units or without loosening or removing supplemental insulation supplied as means of obtaining clearances as required by UL.
- e. Where "space only" is indicated, make provisions for future installation of breaker sized as indicated.
- f. Directories: indicate load served by each circuit of panelboard.
- g. Directories: indicate source of service (e.g. upstream panel, switchboard, motor control center) to panelboard.
- h. Type directories and mount in holder behind transparent protective covering.
- i. Panelboard nameplates: provided in accordance with paragraph FIELD FABRICATED NAMEPLATES.

2.10.1 Enclosure

Provide panelboard enclosure in accordance with the following:

- a. UL 50.
- b. Cabinets mounted outdoors or flush-mounted: hot-dipped galvanized after fabrication.
- c. Cabinets: painted in accordance with paragraph PAINTING.
- d. Outdoor cabinets: NEMA 4x with conduit hubs welded to the cabinet a removable steel plate 1/4 inch thick in the bottom for field drilling

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for conduit connections.

- e. Front edges of cabinets: form-flanged or fitted with structural shapes welded or riveted to the sheet steel, for supporting the panelboard front.
- f. All cabinets: fabricated such that no part of any surface on the finished cabinet deviates from a true plane by more than 1/8 inch.
- g. Holes: provided in the back of indoor surface-mounted cabinets, with outside spacers and inside stiffeners, for mounting the cabinets with a 1/2 inch clear space between the back of the cabinet and the wall surface.
- h. Flush doors: mounted on hinges that expose only the hinge roll to view when the door is closed.
- i. Each door: fitted with a combined catch and lock latch.
- j. Keys: two provided with each lock, with all locks keyed alike.
- k. Finished-head cap screws: provided for mounting the panelboard fronts on the cabinets.

2.10.2 Panelboard Buses

Support bus bars on bases independent of circuit breakers. Design main buses and back pans so that breakers may be changed without machining, drilling, or tapping. Provide isolated neutral bus in each panel for connection of circuit neutral conductors. Provide separate ground bus identified as equipment grounding bus per UL 67 for connecting grounding conductors; bond to steel cabinet. In addition to equipment grounding bus, provide second "isolated" ground bus, where indicated.

2.10.2.1 Panelboard Neutrals for Non-Linear Loads

Provide in accordance with the following:.

- a. UL listed, with panelboard type specifically UL heat rise tested for use on non-linear loads.
- b. Panelboard: heat rise tested in accordance with UL 67, except with the neutral assembly installed and carrying 200 percent of the phase bus current during testing.
- c. Verification of the testing procedure: provided upon request.
- d. Two neutral assemblies paralleled together with cable is not acceptable.
- e. Nameplates for panelboard rated for use on non-linear loads: marked "SUITABLE FOR NON-LINEAR LOADS" and in accordance with paragraph FIELD FABRICATED NAMEPLATES.
- f. Provide a neutral label with instructions for wiring the neutral of panelboards rated for use on non-linear loads.

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2.10.3 Circuit Breakers

UL 489, thermal magnetic-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Where indicated on the drawings, provide circuit breakers with shunt trip devices. Series rated circuit breakers and plug-in circuit breakers are unacceptable.

2.10.3.1 Multipole Breakers

Provide common trip-type with single operating handle. Design breaker such that overload in one pole automatically causes all poles to open. Maintain phase sequence throughout each panel so that any three adjacent breaker poles are connected to Phases A, B, and C, respectively.

2.10.3.2 Circuit Breaker With Ground-Fault Circuit Interrupter

UL 943 and NFPA 70. Provide with auto-monitoring (self-test) and lockout features, "push-to-test" button, visible indication of tripped condition, and ability to detect and trip when current imbalance is 6 milliamperes or higher per requirements of UL 943 for Class A ground-fault circuit interrupter devices.

2.11 ENCLOSED CIRCUIT BREAKERS

UL 489. Individual molded case circuit breakers with voltage and continuous current ratings, number of poles, overload trip setting, and short circuit current interrupting rating as indicated. Enclosure type as indicated.

2.12 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)

Motor short-circuit protectors, also called motor circuit protectors (MCPs): UL 508 and UL 489, and provided as shown. Provide MSCPs that consist of an adjustable instantaneous trip circuit breaker used only in conjunction with a combination motor controller which provides coordinated motor branch-circuit overload and short-circuit protection. Rate MSCPs in accordance with the requirements of NFPA 70.

2.13 TRANSFORMERS

Provide transformers in accordance with the following:

- a. NEMA ST 20, general purpose, dry-type, self-cooled, ventilated.
- b. Provide transformers in NEMA 1 enclosure.
- c. Taps for transformers 15 kVA and larger: Two 2.5 percent taps Full Capacity Above Nominal (FCAN) and four 2.5 percent taps Full Capacity Below Nominal (FCBN).
- d. Transformer insulation system:
 - (1) 220 degrees C insulation system for transformers 15 kVA and greater, with temperature rise not exceeding 115 degrees C under full-rated load in maximum ambient of 40 degrees C.
 - (2) 180 degrees C insulation for transformers rated 10 kVA and less,

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with temperature rise not exceeding 80 degrees C under full-rated load in maximum ambient of 40 degrees C.

- e. Transformer of 150 degrees C temperature rise is not acceptable.
- f. Transformer of 115 degrees C temperature rise: capable of carrying continuously 115 percent of nameplate kVA without exceeding insulation rating.
- g. Transformer of 80 degrees C temperature rise: capable of carrying continuously 130 percent of nameplate kVA without exceeding insulation rating.
- h. Transformers: quiet type with maximum sound level at least 3 decibels less than NEMA standard level for transformer ratings indicated.

2.13.1 Specified Transformer Efficiency

Transformers, indicated and specified with: 480V primary, 80 degrees C or 115 degrees C temperature rise, kVA ratings of 37.5 to 100 for single phase or 30 to 500 for three phase, energy efficient type. The transformer is not acceptable if the calculated transformer efficiency is less than the efficiency indicated in 10 CFR 431, Subpart K.

2.14 MOTORS

Provide motors in accordance with the following:

- a. NEMA MG 1.
- b. Hermetic-type sealed motor compressors: Also comply with UL 984.
- c. Provide the size in terms of HP, or kVA, or full-load current, or a combination of these characteristics, and other characteristics, of each motor as indicated or specified.
- d. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters.
- e. Rate motors for operation on 208-volt, 3-phase circuits with a terminal voltage rating of 200 volts, and those for operation on 480-volt, 3-phase circuits with a terminal voltage rating of 460 volts.
- f. Use motors designed to operate at full capacity with voltage variation of plus or minus 10 percent of motor voltage rating.
- g. Unless otherwise indicated, use continuous duty type motors if rated 1 HP and above.
- h. Where fuse protection is specifically recommended by the equipment manufacturer, provide fused switches in lieu of non-fused switches indicated.
- i. Use Inverter-Rated motors designed to operate with adjustable speed drive (ASD).

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2.14.1 High Efficiency Single-Phase Motors

Single-phase fractional-horsepower alternating-current motors: high efficiency types are not acceptable. In exception, for special purpose motors and motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.14.2 Premium Efficiency Polyphase and Single-Phase Motors

Select polyphase and continuous-duty single phase motors based on high efficiency characteristics relative to typical characteristics and applications as listed in NEMA MG 10 and NEMA MG 11. In addition, continuous rated, polyphase squirrel-cage medium induction motors must meet the requirements for premium efficiency electric motors in accordance with NEMA MG 1, including the NEMA full load efficiency ratings. In exception, for motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.14.3 Motor Sizes

Provide size for duty to be performed, not exceeding the full-load nameplate current rating when driven equipment is operated at specified capacity under most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, make adjustments to wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

2.14.4 Wiring and Conduit

Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide power wiring and conduit for field-installed equipment using adjustable speed drive (ASD) manufacturer required wiring type and length, and motor control equipment forming part of motor control centers or switchgear assemblies, the conduit and wiring connecting such centers, assemblies, or other power sources to equipment as specified herein. Power wiring and conduit: conform to the requirements specified herein. Control wiring: provided under, and conform to, the requirements of the section specifying the associated equipment.

2.15 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

- a. UL 508, NEMA ICS 1, and NEMA ICS 2.
- b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.
- c. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.

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- d. Provide protection for motors from immediate restart by a time adjustable restart relay.
- e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.
- f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.
- g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.
- h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.
- i. Provide selector switch with the means for locking in any position.
- j. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.
- k. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual reset-type pushbutton on outside of motor controller case.
- l. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.

2.15.1 Control Wiring

Provide control wiring in accordance with the following:

- a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type SIS meeting UL 44, or Type MTW meeting UL 1063, and passing the VW-1 flame tests included in those standards.
- b. Hinge wire: Class K stranding.
- c. Current transformer secondary leads: not smaller than No. 10 AWG.
- d. Control wire minimum size: No. 14 AWG.
- e. Power wiring for 480-volt circuits and below: the same type as control wiring with No. 12 AWG minimum size.
- f. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

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2.15.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

- a. NEMA ICS 4.
- b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.
- c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.
- d. Terminals: not less than No. 10 in size with sufficient length and space for connecting at least two indented terminals for 10 AWG conductors to each terminal.
- e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.
- f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.
- g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity, and are constructed to assure positive contact between current-carrying parts.

2.15.2.1 Types of Terminal Blocks

- a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.
- b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide each connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

2.15.3 Control Circuits

Control circuits: maximum voltage of 120 volts derived from control transformer in same enclosure. Transformers: conform to UL 506, as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side. Provide one fused secondary lead with the other lead

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grounded. For designated systems, as indicated, provide backup power supply, including transformers connected to emergency power source. Provide for automatic switchover and alarm upon failure of primary control circuit.

2.15.4 Enclosures for Motor Controllers

NEMA ICS 6.

2.15.5 Multiple-Speed Motor Controllers and Reversible Motor Controllers

Across-the-line-type, electrically and mechanically interlocked. Multiple-speed controllers: include compelling relays and multiple-button, station-type with pilot lights for each speed.

2.15.6 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

2.15.7 Pilot and Indicating Lights

Provide LED cluster lamps.

2.16 LOCKOUT REQUIREMENTS

Provide circuit breakers, disconnecting means, and other devices that are electrical energy-isolating capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy in accordance with 29 CFR 1910.147, NFPA 70E and 29 CFR 1910.303. Comply with requirements of Division 23, "Mechanical" for mechanical isolation of machines and other equipment.

2.17 TELECOMMUNICATIONS SYSTEM

Provide system of telecommunications wire-supporting structures (pathway), including: outlet boxes, conduits with pull wires wireways, cable trays, and other accessories for telecommunications outlets and pathway in accordance with TIA-569 and as specified herein. Additional telecommunications requirements are specified in Section 27 10 00.00 48 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

2.18 GROUNDING AND BONDING EQUIPMENT

2.18.1 Ground Rods

UL 467. Ground rods: cone pointed copper-clad steel, with minimum diameter of 3/4 inch and minimum length 10 feet. Sectional type rods may be used for rods 20 feet or longer.

2.18.2 Ground Bus

Copper ground bus: provided in the electrical equipment rooms as indicated.

2.18.3 Secondary Bonding Busbar

Provide corrosion-resistant grounding busbar suitable for indoor

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installation in accordance with TIA-607. Busbars: plated for reduced contact resistance. If not plated, clean the busbar prior to fastening the conductors to the busbar and apply an anti-oxidant to the contact area to control corrosion and reduce contact resistance. Provide a Primary bonding busbar (PBB) in the telecommunications entrance facility. The Primary bonding busbar (PBB): sized in accordance with the immediate application requirements and with consideration of future growth. Provide Secondary bonding busbars with the following:

- a. Predrilled copper busbar provided with holes for use with standard sized lugs,
- b. Minimum dimensions of 0.25 in thick by 4 in wide for the PBB with length as indicated;
- c. Listed by a nationally recognized testing laboratory.

2.19 HAZARDOUS LOCATIONS

Electrical materials, equipment, and devices for installation in hazardous locations, as defined by NFPA 70: specifically approved by Underwriters' Laboratories, Inc., or Factory Mutual for particular "Class," "Division," and "Group" of hazardous locations involved. Boundaries and classifications of hazardous locations: as indicated. Equipment in hazardous locations: comply with UL 1203 for electrical equipment and industrial controls and UL 674 for motors.

2.20 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.21 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

- a. ASTM D709.
- b. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified or as indicated on the drawings.
- c. Each nameplate inscription: identify the function and, when applicable, the position.
- d. Nameplates: melamine plastic, 0.125 inch thick, white with black center core.
- e. Provide red laminated plastic label with white center core where indicated.
- f. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.
- g. Minimum size of nameplates: one by 2.5 inches.
- h. Lettering size and style: a minimum of 0.25 inch high normal block

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style.

2.22 WARNING SIGNS

Provide warning signs for flash protection in accordance with NFPA 70E and NEMA Z535.4 for switchboards, panelboards, industrial control panels, and motor control centers that are in other than dwelling occupancies and are likely to require examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.23 WIREWAYS

UL 870. Material: steel galvanized 16 gauge for heights and depths up to 6 by 6 inches, and 14 gauge for heights and depths up to 12 by 12 inches. Provide in length required for the application with screw-cover NEMA 1 enclosure per NEMA ICS 6.

2.24 METERING

ANSI C12.1. Provide a self-contained, socket-mounted, electronic programmable outdoor watthour meter. Meter: either programmed at the factory or programmed in the field. Turn field programming device over to the Contracting Officer at completion of project. Coordinate meter to system requirements. Coordinate meter, system components, and meter location to be compatible with the Activity's central advanced metering system.

- a. Design: Provide watthour meter designed for use on a single-phase, three-wire, 480/240 volt system. Include necessary KYZ pulse initiation hardware for Energy Monitoring and Control System (EMCS).
- b. Class: 200; Form: 2S, accuracy: plus or minus 1.0 percent; Finish: Class II.
- c. Cover: Polycarbonate and lockable to prevent tampering and unauthorized removal.
- d. Kilowatt-hour Register: five digit electronic programmable type.
- e. Demand Register:
 - (1) Provide solid state.
 - (2) Meter reading multiplier: Indicate multiplier on the meter face.
 - (3) Demand interval length: programmed for 15 minutes with rolling demand up to six subintervals per interval.
- f. Socket: ANSI C12.7. Provide NEMA Type 3R, box-mounted socket, ringless, having manual circuit-closing bypass and having jaws compatible with requirements of the meter. Provide manufacturers standard enclosure color unless otherwise indicated.

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2.25 METER BASE ONLY

ANSI C12.7. Provide NEMA Type 3R, box-mounted socket, ringless, having jaws compatible with requirements of a class: 200 and Form: 2S self contained watthour meter. Provide gray plastic closing cover and bypass links. Provide manufacturers standard enclosure color unless otherwise indicated.

2.26 SURGE PROTECTIVE DEVICES

Provide parallel type surge protective devices (SPD) which comply with UL 1449 at the service entrance, load centers, and panelboards. Provide surge protectors in a NEMA 1 enclosure per NEMA ICS 6. SPD must have the same short-circuit current rating as the protected equipment and must not be installed at a point of system where the available fault current is in excess of that rating. Use Type 1 or Type 2 SPD and connect on the load side of a dedicated circuit breaker. Submit performance and characteristic curves.

Provide the following modes of protection:

FOR SINGLE PHASE AND THREE PHASE WYE CONNECTED SYSTEMS-

Phase to phase (L-L)
Each phase to neutral (L-N)
Neutral to ground (N-G)
Phase to ground (L-G)

SPDs at the service entrance: provide with a minimum surge current rating of 80,000 amperes for L-L mode minimum and 40,000 amperes for other modes (L-N, L-G, and N-G) and downstream SPDs rated 40,000 amperes for L-L mode minimum and 20,000 amperes for other modes (L-N, L-G, and N-G).

Provide SPDs. Maximum L-N, L-G, and N-G Voltage Protection Rating:

700V for 120V, single phase system
700V for 120/240V, single phase system
700V for 208Y/120V, three phase system
1,200V for 480Y/277V, three phase system

Maximum L-L Voltage Protection Rating:

1,200V for 120V, single phase system
1,200V for 120/240V, single phase system
1,200V for 208Y/120V, three phase system
2,000V for 480Y/277V, three phase system

The minimum MCOV (Maximum Continuous Operating Voltage) rating for L-N and L-G modes of operation: 120 percent of nominal voltage for 240 volts and below; 115 percent of nominal voltage above 240 volts to 480 volts.

2.27 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

- a. NEMA 250 corrosion-resistance test and the additional requirements as specified herein.
- b. Interior and exterior steel surfaces of equipment enclosures:

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thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.

- c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.
- d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.
- e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.
- f. Equipment located indoors: ANSI Light Gray, and equipment located outdoors: ANSI Light Gray.
- g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.28 SOURCE QUALITY CONTROL

2.28.1 Transformer Factory Tests

Submittal: include routine NEMA ST 20 transformer test results on each transformer and also provide the results of NEMA "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

2.29 COORDINATED POWER SYSTEM PROTECTION

Prepare analyses as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of NFPA 70 and IEEE C2 and to requirements specified herein.

3.1.1 Underground Service

Underground service conductors and associated conduit: continuous from service entrance equipment to outdoor power system connection.

3.1.2 Hazardous Locations

Perform work in hazardous locations, as defined by NFPA 70, in strict accordance with NFPA 70 for particular "Class," "Division," and "Group" of hazardous locations involved. Provide conduit and cable seals where required by NFPA 70. Provide conduit with tapered threads.

3.1.3 Service Entrance Identification

Service entrance disconnect devices, switches, and enclosures: labeled and identified as such.

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3.1.3.1 Labels

Wherever work results in service entrance disconnect devices in more than one enclosure, as permitted by NFPA 70, label each enclosure, new and existing, as one of several enclosures containing service entrance disconnect devices. Label, at minimum: indicate number of service disconnect devices housed by enclosure and indicate total number of enclosures that contain service disconnect devices. Provide laminated plastic labels conforming to paragraph FIELD FABRICATED NAMEPLATES. Use lettering of at least 0.25 inch in height, and engrave on black-on-white matte finish. Service entrance disconnect devices in more than one enclosure: provided only as permitted by NFPA 70.

3.1.4 Wiring Methods

Provide insulated conductors installed in rigid steel conduit, IMC, rigid nonmetallic conduit, or EMT, except where specifically indicated or specified otherwise or required by NFPA 70 to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Minimum conduit size: 1/2 inch in diameter for low voltage lighting and power circuits. Vertical distribution in multiple story buildings: made with metal conduit in fire-rated shafts, with metal conduit extending through shafts for minimum distance of 6 inches.

3.1.4.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 200-pound force tensile strength. Leave minimum 36 inches of slack at each end of pull wire.

3.1.5 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Keep conduit minimum 6 inches away from parallel runs of flues and steam or hot water pipes. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project.

3.1.5.1 Restrictions Applicable to Aluminum Conduit

- a. Do not install underground or encase in concrete or masonry.
- b. Do not use brass or bronze fittings.
- c. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.5.2 Restrictions Applicable to EMT

- a. Do not install underground.
- b. Do not encase in concrete, mortar, grout, or other cementitious materials.
- c. Do not use in areas subject to physical damage including but not limited to equipment rooms where moving or replacing equipment could

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physically damage the EMT.

- d. Do not use in hazardous areas.
- e. Do not use outdoors.
- f. Do not use in fire pump rooms.
- g. Do not use when the enclosed conductors must be shielded from the effects of High-altitude Electromagnetic Pulse (HEMP).

3.1.5.3 Restrictions Applicable to Nonmetallic Conduit

a. PVC Schedule 40.

- (1) Do not use where subject to physical damage, including but not limited to, mechanical equipment rooms, electrical equipment rooms, fire pump rooms, and where restrictions are applying to both PVC Schedule 40 and PVC Schedule 80.
- (2) Do not use above grade, except where allowed in this section for rising through floor slab or indicated otherwise.

b. PVC Schedule 40 and Schedule 80.

- (1) Do not use where subject to physical damage, including but not limited to, hospitals, power plant, missile magazines, and other such areas.
- (2) Do not use in hazardous (classified) areas.
- (3) Do not use in penetrating fire-rated walls or partitions, or fire-rated floors.

-

3.1.5.4 Underground Conduit

Plastic-coated rigid steel; plastic-coated steel IMC; PVC, Type EPC-40; or fiberglass. Convert nonmetallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before rising through floor slab. Plastic coating: extend minimum 6 inches above floor.

3.1.6 Watthour Meters

ANSI C12.1.

3.1.7 Surge Protective Devices

Connect the surge protective devices in parallel to the power source, keeping the conductors as short and straight as practically possible. Maximum allowed lead length is 3 feet avoiding 90 degree bends. Do not locate surge protective devices inside a panelboard or switchboard enclosure.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

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3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria. Where field painting of enclosures for panelboards, load centers or the like is specified to match adjacent surfaces, to correct damage to the manufacturer's factory applied coatings, or to meet the indicated or specified safety criteria, provide manufacturer's recommended coatings and apply in accordance to manufacturer's instructions.

3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 10 working days notice prior to tests. Where applicable, test electrical equipment in accordance with NETA ATS.

3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring per NETA ATS to provide direct reading of resistance. All existing wiring to be reused must also be tested.

3.5.3 Transformer Tests

Perform the standard, not optional, tests in accordance with the Inspection and Test Procedures for transformers, dry type, air-cooled, 600 volt and below; as specified in NETA ATS. Measure primary and secondary voltages for proper tap settings. Tests need not be performed by a recognized independent testing firm or independent electrical consulting firm.

3.5.4 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in UL 943.

3.5.5 Arc-Fault Receptacle Test

Test arc-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in UL 1699.

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3.5.6 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

3.5.7 Watthour Meter

a. Visual and mechanical inspection

- (1) Examine for broken parts, shipping damage, and tightness of connections.
- (2) Verify that meter type, scales, and connections are in accordance with approved shop drawings.

b. Electrical tests

- (1) Determine accuracy of meter.
- (2) Calibrate watthour meters to one-half percent.
- (3) Verify that correct multiplier has been placed on face of meter, where applicable.

3.5.8 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code for Electricity Metering

ASTM INTERNATIONAL (ASTM)

ASTM A123/A123M (2017) Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products

ASTM A153/A153M (2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

ASTM A240/A240M (2022b) Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications

ASTM A653/A653M (2022) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process

ASTM A780/A780M (2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

ASTM D1535 (2014; R 2018) Standard Practice for Specifying Color by the Munsell System

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

IEEE 100 (2000; Archived) The Authoritative Dictionary of IEEE Standards Terms

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IEEE C2	(2023) National Electrical Safety Code
IEEE C37.13	(2015) Standard for Low-Voltage AC Power Circuit Breakers Used in Enclosures
IEEE C37.20.1A	(2020) Metal-Enclosed Low-Voltage (1000 Vac and below, 3200 Vdc and below) Power Circuit-Breaker Switchgear Amendment 1: Control and Secondary Circuits and Devices, and All Wiring
IEEE C37.20.7	(2017; Corr 2021) Guide for Testing Switchgear Rated Up to 52 kV for Internal Arcing Faults
IEEE C37.90.1	(2013) Standard for Surge Withstand Capability (SWC) Tests for Relays and Relay Systems Associated with Electric Power Apparatus
IEEE C57.12.28	(2014) Standard for Pad-Mounted Equipment - Enclosure Integrity
IEEE C57.13	(2016) Standard Requirements for Instrument Transformers

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA LI 1	(1998; R 2011) Industrial Laminating Thermosetting Products

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023) National Electrical Code
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UNDERWRITERS LABORATORIES (UL)

UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 1558	(2016; Reprint Nov 2019) UL Standard for Safety Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear

1.2 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

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1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE 100.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Switchgear Drawings; G, D0

SD-03 Product Data

Switchgear; G, D0

SD-06 Test Reports

Switchgear Design Tests; G, D0

Switchgear Production Tests; G, D0

Acceptance Checks and Tests; G, D0

SD-07 Certificates

Submit certification indicating conformance with the paragraph CYBERSECURITY EQUIPMENT CERTIFICATION.

Submit certification indicating conformance with the paragraph CYBERSECURITY INSTALLATION CERTIFICATION.

SD-10 Operation and Maintenance Data

Switchgear Operation and Maintenance, Data Package 5; G, D0

SD-11 Closeout Submittals

Assembled Operation and Maintenance Manuals; G, D0

Equipment Test Schedule; G, D0

Required Settings; G, D0

Service Entrance Available Fault Current Label; G, D0

1.5 QUALITY ASSURANCE

1.5.1 Product Data

Include manufacturer's information on each submittal for each component, device and accessory provided with the switchgear including:

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- a. Circuit breaker type, interrupting rating, and trip devices, including available settings.
- b. Manufacturer's instruction manuals and published time-current curves (in electronic format) of the main secondary breaker and largest secondary feeder device.

1.5.2 Switchgear Drawings

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that must be shown to ensure a coordinated installation. Identify circuit terminals on wiring diagrams and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Indicate on the drawings adequate clearance for operation, maintenance, and replacement of operating equipment devices. Include the nameplate data, size, and capacity on submittal. Also include applicable federal, military, industry, and technical society publication references on submittals. Include the following:

- a. One-line diagram including breakers, fuses, current transformers, and meters.
- b. Outline drawings including front elevation, section views, footprint, and overall dimensions.
- c. Bus configuration including dimensions and ampere ratings of bus bars.
- d. Markings and NEMA nameplate data, including fuse information (manufacturer's name, catalog number, and ratings).
- e. Circuit breaker type, interrupting rating, and trip devices, including available settings.
- f. Wiring diagrams and elementary diagrams with terminals identified, and indicating prewired interconnections between items of equipment and the interconnection between the items.
- g. Manufacturer's instruction manuals and published time-current curves (in electronic format) of the main secondary breaker and largest secondary feeder device. Use this information (designer of record) to provide breaker settings that ensures protection and coordination are achieved.

1.5.3 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "shall" or "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated.

1.5.4 Standard Products

Provide materials and equipment that are products of manufacturers

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regularly engaged in the production of such products which are of equal material, design and workmanship, and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.4.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site are not acceptable.

1.6 MAINTENANCE

1.6.1 Switchgear Operation and Maintenance Data

Submit Operation and Maintenance Manuals in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6.2 Assembled Operation and Maintenance Manuals

Assemble and securely bind manuals in durable, hard covered, water resistant binders. Assemble and index the manuals in the following order with a table of contents:

- a. Manufacturer's O&M information required by the paragraph SD-10, OPERATION AND MAINTENANCE DATA.
- b. Catalog data required by the paragraph SD-03, PRODUCT DATA.
- c. Drawings required by the paragraph SD-02, SHOP DRAWINGS.
- d. Prices for spare parts and supply list.
- e. Information on metering.
- f. Design test reports.
- g. Production test reports.

1.6.3 Spare Parts

Provide spare parts as specified below. Provide spare parts that are of the same material and workmanship, meet the same requirements, and are

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interchangeable with the corresponding original parts furnished.

- a. Quantity 2 - Fuses of each type and size.

1.7 WARRANTY

Provide equipment items that are supported by service organizations reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

Products and materials not considered to be switchgear and related accessories are specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION, Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and Section 25 08 20 MICROGRID CONTROL SYSTEM.

Low voltage switchgear must incorporate microgrid control system and functional requirements per Section 25 08 20 MICROGRID CONTROL SYSTEM.

2.2 SWITCHGEAR

IEEE C37.20.1A and UL 1558.

2.2.1 Ratings

Provide equipment with the following ratings:

- a. Voltage rating: 480Y/277 volts AC, 4-wire three-phase, 4-wire.
- b. Continuous current rating of the main bus: 3000 amperes.
- c. Short-circuit current rating: 100k rms symmetrical amperes.
- d. UL listed and labeled for its intended use and as service entrance equipment.

2.2.2 Construction

Provide the following:

- a. Switchgear: consisting of vertical sections bolted together to form a rigid assembly and aligned as indicated.
- b. All circuit breakers: front accessible with rear load connections.
- c. Compartmentalized switchgear: vertical insulating barriers between the front device section, the main bus section, and the cable compartment with full front to rear vertical insulating barriers between adjacent sections.
- d. Where indicated, "space for future" or "space" means to include all necessary components and hardware to be fully equipped for racking in a circuit breaker element.
- e. Insulating barriers: provided in accordance with NEMA LI 1, Type

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GPO-3, 0.25 inch minimum thickness.

- f. Moisture resistant coating: applied to all rough-cut edges of barriers.
- g. Switchgear: Arc-resistant Type 2B, tested in accordance with IEEE C37.20.7.

2.2.2.1 Enclosure

Provide the following:

- a. Enclosure: NEMA ICS 6 Type 1as indicated.
- b. Enclosure: bolted together with removable bolt-on side and hinged rear covers.
- c. Front doors: provided with padlockable vault handles with a three point catch.
- d. Bases, frames and channels of enclosure: corrosion resistant and fabricated of ASTM A240/A240M type 304 or 304L stainless steel or galvanized steel.
- e. Base: includes any part of enclosure that is within 3 inches of concrete pad.
- f. Galvanized steel: ASTM A123/A123M, ASTM A653/A653M G90 coating, and ASTM A153/A153M, as applicable. Galvanize after fabrication where practicable.
- g. Paint color: ASTM D1535 light gray No. 61 or No. 49 over rust inhibitor.
- h. Paint coating system: comply with IEEE C57.12.28 for galvanized steel.
- i. Infrared viewing windows: install to allow the use of an infrared camera or thermal imager direct line of site to inspect electrical connections without requiring the opening of panels and doors. These windows are intended to allow thermographers the ability to inspect the electrical equipment without directly exposing themselves to live electrical components and energized devices.

2.2.2.2 Bus Bars

Provide the following:

- a. Bus bars: copper with silver-plated contact surfaces.
 - (1) Phase bus bars: uninsulated.
 - (2) Neutral bus: rated 100 percent of the main bus continuous current rating.
- b. Make bus connections and joints with hardened steel bolts.
- c. Main-bus (through bus): rated at the full ampacity of the main throughout the switchgear.

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- d. Minimum one-quarter by 2 inch copper ground bus secured to each vertical section along the entire length of the switchgear.

2.2.2.3 Main Section

Provide the main section consisting of drawout air power circuit breaker.

2.2.2.4 Distribution Sections

Provide the distribution section consisting of individually mounted, drawout, air power circuit breakers as indicated.

2.2.2.5 Auxiliary Sections

Provide auxiliary sections consisting of indicated instruments, metering equipment, control equipment, and current transformer compartments as indicated.

2.2.2.6 Handles

Provide handles for individually mounted devices of the same design and method of external operation. Label handles prominently to indicate device ampere rating, color coded for device type. Identify ON-OFF indication by handle position and by prominent marking.

2.2.3 Protective Device

Provide switchgear circuit breakers with multifunction microprocessor relays and protective functions as indicated on single line diagram.

Provide the following:

- a. IEEE C37.13. 120 Vac operated drawout, unfused, low-voltage power circuit breaker with a short-circuit current rating as indicated at 480 volts.
- b. Breaker frame size: as indicated.
- c. Equip electrically operated breakers with motor-charged, stored-energy closing mechanism to permit rapid and safe closing of the breaker against fault currents within the short time rating of the breaker, independent of the operator's strength or effort in closing the handle.

2.2.4 Drawout Breakers

Equip drawout breakers with disconnecting contacts, wheels, and interlocks for drawout application. Provide main, auxiliary, and control disconnecting contacts with silver-plated, multifinger, positive pressure, self-aligning type. Provide drawout compartment shutters to protect operators from accidental contact with breaker stabs when the breaker is withdrawn from its cubicle. Provide each drawout breaker with four-position operation with each position clearly identified by an indicator on the circuit breaker front panel as follows.

- a. Connected Position: Primary and secondary contacts are fully engaged. Breaker must be tripped before racking into or out of position.
- b. Test Position: Primary contacts are disconnected but secondary

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contacts remain fully engaged. This position allows complete test and operation of the breaker without energizing the primary circuit.

- c. Disconnected Position: Primary and secondary contacts are disconnected.
- d. Withdrawn (Removed) Position: Places breaker completely out of compartment, ready for removal. Removal of the breaker actuates assembly that isolates the primary stabs.

Generator circuit breaker (IEEE device 52) ratings must be consistent with the generator rated voltage and frequency, with continuous, short circuit withstand, and interrupting current ratings. Generator circuit breakers are capable of synchronization and parallel operation.

2.2.5 Protective Device

Equip main and distribution breakers with a solid-state tripping system consisting of three current sensors and a microprocessor-based multifunction relay. Equip switchgear breakers with multifunction relay, including VT and CT as required to perform protective, control, and metering functions as specified and indicated on single line diagram. Include the following:

- a. Current sensors ampere rating: the same as the breaker frame rating.
- b. Trip unit ampere rating: as indicated.
- c. Ground fault protection: residual type sensing.
- d. Provide additional features:
 - (1) Breakers: include a digital display for phase and ground current.
 - (2) Breakers: include a digital display for watts, vars, VA, kWh, kvarh, and kVAh.
 - (3) Breakers: include a digital display for phase voltage, and percent THD voltage and current.
 - (4) Breakers: include provisions for communication via a network twisted pair cable for remote monitoring and control. Provide the following communications protocol: Modbus.
 - (5) For breakers that are rated for or can be adjusted to 1,200 amperes or higher, provide arc energy reduction capability with an energy-reducing maintenance switch with local status indicator.

2.2.6 Metering

2.2.6.1 Digital Meters

IEEE C37.90.1 for surge withstand. Provide true rms, plus/minus one percent accuracy, programmable, microprocessor-based meter enclosed in a sealed case with the following features.

- a. Display capability:
 - (1) Multi-Function Meter: Display a selected phase to neutral

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voltage, phase to phase voltage, percent phase to neutral voltage THD, percent phase to phase voltage THD; a selected phase current, neutral current, percent phase current THD, percent neutral current; selected total PF, kW, KVA, kVAR, FREQ, kVAh, kWh. Detected alarm conditions include over/under current, over/under voltage, over/under KVA, over/under frequency, over/under selected PF/kVAR, voltage phase reversal, voltage imbalance, reverse power, over percent THD. Include a Form C KYZ pulse output relay on the meter.

- (2) Power Meter: Display Watts, VARs, and selected KVA/PF. Detected alarm conditions include over/under KVA, over/under PF, over/under VARs, over/under reverse power.
 - (3) Volt Meter: Provide capability to be selectable between display of the three phases of phase to neutral voltages and simultaneous display of the three phases of the phase to phase voltages. Detected alarm conditions include over/under voltage, over/under voltage imbalance, over percent THD.
 - (4) Ammeter: Display phase A, B, and C currents. Detected alarm conditions include over/under current, over percent THD.
 - (5) Digital Watthour Meter: Provide a single selectable display for watts, total kilowatt hours (kWh) and watt demand (Wd). Include a Form C KYZ pulse output relay on the meter.
- b. Design meters to accept input from standard 5A secondary instrument transformers.
 - c. Provide programming via a front panel display and a communication interface accessible by a computer.
 - d. Provide password secured programming stored in non-volatile EEPROM memory.
 - e. Provide digital communications in a Modbus RTU protocol via a RS485 serial port and an independently addressable RS485 serial port.
 - f. Provide meter that calculates and stores average max/min demand values with time and date for all readings based on a user selectable sliding window averaging period.
 - g. Provide meter with programmable hi/low set limits with two Form C dry contact relays when exceeding alarm conditions.
 - h. Provide meter with a display of Total Harmonic Distortion (THD) measurement to a minimum of the thirty-first order.
 - i. Include historical trend logging capability with the ability to store up to 100,000 data points with intervals of 1 second to 180 minutes. Provide a unit that can store and time stamp up to 1000 programmable triggered conditions.
 - j. Provide event waveform recording triggered by the rms of 2 cycles of voltage or current exceeding programmable set points. Store waveforms for all 6 channels of voltage and current for a minimum of 10 cycles prior to the event and 50 cycles past the event.

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2.2.6.2 Electronic Watthour Meter

ANSI C12.1. Provide a switchgear style electronic programmable watthour meter, semi-flush mounted, as indicated. Meter can be either programmed at the factory or programmed in the field. Turn field programming device over to the Contracting Officer at completion of project. Coordinate meter to system requirements.

- a. Design: Provide meter designed for use on a 3-phase, 4-wire, 480Y/277 volt system with 3 current transformers. Include necessary KYZ pulse initiation hardware for Energy Monitoring and Control System (EMCS).
- b. Coordination: Provide meter coordinated with ratios of current transformers and transformer secondary voltage.
- c. Class: 20. Accuracy: plus or minus 1.0 percent. Finish: Class II.
- d. Kilowatt-hour Register: five digit electronic programmable type.
- e. Demand Register:
 - (1) Provide solid state.
 - (2) Display actual values and readings of the metered circuit. No multipliers must be required.
 - (3) Demand interval length: programmed for 15 minutes with rolling demand up to six subintervals per interval.
- f. Meter fusing: Provide a fuse block mounted in the metering compartment containing one fuse per phase to protect the voltage input to the watthour meter. Size fuses as recommended by the meter manufacturer.
- g. Provide meter with a communications port, RS485, with Modbus RTU serial or Ethernet, Modbus-TCP communications.

IEEE C57.13. Provide single ratio transformers, 60 hertz, 3000 to 5-ampere ratio, 1.5 rating factor, with a metering accuracy class of 0.3 through B-1.8.

2.2.7 Terminal Boards

Provide with engraved plastic terminal strips and screw type terminals for external wiring between components and for internal wiring between removable assemblies. Provide short-circuiting type terminal boards associated with current transformer. Terminate conductors for current transformers with ring-tongue lugs. Provide terminal board identification that is identical in similar units. Provide color coded external wiring that is color coded consistently for similar terminal boards.

2.2.8 Wire Marking

Mark control and metering conductors at each end. Provide factory installed, white, plastic tubing, heat stamped with black block type letters on factory-installed wiring. On field-installed wiring, provide white, preprinted, polyvinyl chloride (PVC) sleeves, heat stamped with black block type letters. Provide a single letter or number on each sleeve, elliptically shaped to securely grip the wire, and keyed in such a manner to ensure alignment with adjacent sleeves. Provide specific wire

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markings using the appropriate combination of individual sleeves. Indicate on each wire marker the device or equipment, including specific terminal number to which the remote end of the wire is attached.

2.3 MANUFACTURER'S NAMEPLATE

Provide a nameplate on each item of equipment bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is not acceptable. This nameplate and method of attachment may be the manufacturer's standard if it contains the required information.

2.4 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each switchgear, equipment enclosure, relay, switch, and device; as specified in this section or as indicated on the drawings. Identify on each nameplate inscription the function and, when applicable, the position. Provide nameplates of melamine plastic, 0.125 inch thick, white with black center core. Provide matte finish surface. Provide square corners. Accurately align lettering and engrave into the core. Provide nameplates with minimum size of one by 2.5 inches. Provide lettering that is a minimum of 0.25 inch high normal block style.

2.5 SOURCE QUALITY CONTROL

2.5.1 Equipment Test Schedule

The Government reserves the right to witness tests. Provide equipment test schedules for tests to be performed at the manufacturer's test facility. Submit required test schedule and location, and notify the Contracting Officer 30 calendar days before scheduled test date. Notify Contracting Officer 15 calendar days in advance of changes to scheduled date.

Provide the following as part of test equipment calibration:

- a. Provide a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- b. Accuracy: Traceable to the National Institute of Standards and Technology.
- c. Instrument calibration frequency schedule: less than or equal to 12 months for both test floor instruments and leased specialty equipment.
- d. Dated calibration labels: visible on all test equipment.
- e. Calibrating standard: higher accuracy than that of the instrument tested.
- f. Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
 - (1) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.

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- (2) Identify the third party/laboratory calibrated instrument to verify that calibrating standard is met.

2.5.2 Switchgear Design Tests

IEEE C37.20.1A and UL 1558 IEEE C37.20.1A, IEEE C37.20.7, and UL 1558.

2.5.2.1 Design Tests

Furnish documentation showing the results of design tests on a product of the same series and rating as that provided by this specification.

- a. Short-circuit current test.
- b. Enclosure tests.
- c. Dielectric test.

2.5.3 Switchgear Production Tests

IEEE C37.20.1A and UL 1558. Furnish reports which include results of production tests performed on the actual equipment for this project. These tests include:

- a. 60-hertz dielectric tests.
- b. Mechanical operation tests.
- c. Electrical operation and control wiring tests.
- d. Ground fault sensing equipment test.

2.6 COORDINATED POWER SYSTEM PROTECTION

Provide a power system study as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.

2.7 ARC FLASH WARNING LABEL

Provide warning label for switchgear. Locate this self-adhesive warning label on the outside of the enclosure warning of potential electrical arc flash hazards and appropriate PPE required. Provide label format as indicated.

2.8 SERVICE ENTRANCE AVAILABLE FAULT CURRENT LABEL

Provide label on exterior of switchgear used as service equipment listing the maximum available fault current at that location. Include on the label the date that the fault calculation was performed and the contact information for the organization that completed the calculation. Locate this self-adhesive warning label on the outside of the switchgear. Provide label format as indicated.

2.9 MIMIC BUS LABELING

Provide a mimic bus on the front of the equipment to diagrammatically show the internal bus structure of the lineup.

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PART 3 EXECUTION

3.1 INSTALLATION

Conform to IEEE C2, NFPA 70, and to the requirements specified herein. Provide new equipment and materials unless indicated or specified otherwise.

3.2 GROUNDING

NFPA 70 and IEEE C2, except that grounds and grounding systems with a resistance to solid earth ground not exceeding 25 ohms.

3.2.1 Grounding Electrodes

Provide driven ground rods as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Connect ground conductors to the upper end of the ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.2.2 Equipment Grounding

Provide bare copper cable not smaller than No. 4/0 AWG not less than 24 inches below grade connecting to the indicated ground rods. When work in addition to that indicated or specified is directed to obtain the specified ground resistance, the provision of the contract covering "Changes" applies.

3.2.3 Connections

Make joints in grounding conductors and loops by exothermic weld or compression connector. Install exothermic welds and compression connectors as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.2.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

Install and connect equipment furnished under this section as indicated on project drawings, the approved shop drawings, and as specified herein.

3.3.1 Switchgear

IEEE C37.20.1A.

3.3.2 Meters and Instrument Transformers

ANSI C12.1.

3.3.3 Field Applied Painting

Where field painting of enclosures is required to correct damage to the manufacturer's factory applied coatings, provide manufacturer's recommended coatings and apply in accordance with manufacturer's instructions.

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3.3.4 Galvanizing Repair

Repair damage to galvanized coatings using ASTM A780/A780M, zinc rich paint, for galvanizing damaged by handling, transporting, cutting, welding, or bolting. Do not heat surfaces that repair paint has been applied to.

3.3.5 Field Fabricated Nameplate Mounting

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.4 FOUNDATION FOR EQUIPMENT AND ASSEMBLIES

3.4.1 Interior Location

Mount switchgear on concrete slab as follows:

- a. Unless otherwise indicated, provide the slab with dimensions at least 4 inches thick.
- b. Install slab such that the top of the concrete slab is approximately 4 inches above the finished grade.
- c. Provide edges above grade 1/2 inch chamfer.
- d. Provide slab of adequate size to project at least 8 inches beyond the equipment.
- e. Provide conduit turnups and cable entrance space required by the equipment to be mounted.
- f. Seal voids around conduit openings in slab with water- and oil-resistant caulking or sealant.
- g. Cut off and bush conduits 3 inches above slab surface.
- h. Provide concrete work as specified in Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.5 FIELD QUALITY CONTROL

Submit Required Settings of breakers to the Contracting Officer after approval of switchgear and at least 30 days in advance of their requirement.

3.5.1 Performance of Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.5.1.1 Switchgear

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.

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- (2) Inspect physical, electrical, and mechanical condition.
- (3) Verify appropriate anchorage, required area clearances, and correct alignment.
- (4) Clean switchgear and verify shipping bracing, loose parts, and documentation shipped inside cubicles have been removed.
- (5) Inspect all doors, panels, and sections for paint, dents, scratches, fit, and missing hardware.
- (6) Verify that circuit breaker sizes and types correspond to approved shop drawings as well as to the circuit breaker's address for microprocessor-communication packages.
- (7) Verify that current transformer ratios correspond to approved shop drawings.
- (8) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (9) Confirm correct operation and sequencing of electrical and mechanical interlock systems.
- (10) Confirm correct application of manufacturer's recommended lubricants.
- (11) Inspect insulators for evidence of physical damage or contaminated surfaces.
- (12) Verify correct barrier and shutter installation and operation.
- (13) Exercise all active components.
- (14) Inspect all mechanical indicating devices for correct operation.
- (15) Verify that filters are in place and vents are clear.
- (16) Test operation, alignment, and penetration of instrument transformer withdrawal disconnects.
- (17) Inspect control power transformers.

b. Electrical Tests

- (1) Perform insulation-resistance tests on each bus section.
- (2) Perform dielectric withstand voltage tests.
- (3) Perform insulation-resistance test on control wiring; Do not perform this test on wiring connected to solid-state components.
- (4) Perform control wiring performance test.
- (5) Perform primary current injection tests on the entire current circuit in each section of assembly.

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- (6) Perform phasing check on double-ended switchgear to ensure correct bus phasing from each source.

3.5.1.2 Circuit Breakers - Low Voltage - Power

a. Visual and Mechanical Inspection

- (1) Compare nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Inspect anchorage, alignment, and grounding.
- (4) Verify that all maintenance devices are available for servicing and operating the breaker.
- (5) Inspect arc chutes.
- (6) Inspect moving and stationary contacts for condition, wear, and alignment.
- (7) Verify that primary and secondary contact wipe and other dimensions vital to satisfactory operation of the breaker are correct.
- (8) Perform all mechanical operator and contact alignment tests on both the breaker and its operating mechanism.
- (9) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (10) Verify cell fit and element alignment.
- (11) Verify racking mechanism.
- (12) Confirm correct application of manufacturer's recommended lubricants.

b. Electrical Tests

- (1) Perform contact-resistance tests on each breaker.
- (2) Perform insulation-resistance tests.
- (3) Adjust Breaker(s) for final settings in accordance with Government provided settings.
- (4) Determine long-time minimum pickup current by primary current injection.
- (5) Determine long-time delay by primary current injection.
- (6) Determine short-time pickup and delay by primary current injection.
- (7) Determine ground-fault pickup and delay by primary current

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injection.

- (8) Determine instantaneous pickup value by primary current injection.
- (9) Activate auxiliary protective devices, such as ground-fault or undervoltage relays, to ensure operation of shunt trip devices; Check the operation of electrically-operated breakers in their cubicle.
- (10) Verify correct operation of any auxiliary features such as trip and pickup indicators, zone interlocking, electrical close and trip operation, trip-free, and antipump function.
- (11) Verify operation of charging mechanism.

3.5.1.3 Current Transformers

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify correct connection.
- (4) Verify that adequate clearances exist between primary and secondary circuit.
- (5) Inspect all bolted electrical connections for high resistance using low-resistance ohmmeter, verifying tightness of accessible bolted electrical connections by calibrated torque-wrench method, or performing thermographic survey.
- (6) Verify that all required grounding and shorting connections provide good contact.

b. Electrical Tests

- (1) Perform resistance measurements through all bolted connections with low-resistance ohmmeter, if applicable.
- (2) Perform insulation-resistance tests.
- (3) Perform polarity tests.
- (4) Perform ratio-verification tests.

3.5.1.4 Metering and Instrumentation

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of electrical connections.

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b. Electrical Tests

- (1) Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.
- (2) Calibrate watthour meters according to manufacturer's published data.
- (3) Verify all instrument multipliers.
- (4) Electrically confirm that current transformer and voltage transformer secondary circuits are intact.

3.5.1.5 Grounding System

a. Visual and Mechanical Inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

- (1) IEEE 81. Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.
- (2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e., pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

3.5.2 Follow-Up Verification

Upon completion of acceptance checks, settings, and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. Trip circuit breakers by operation of each protective device. Test each item to perform its function not less than three times. As an exception to requirements stated elsewhere in the contract, provide the Contracting Officer 5 working days advance notice of the dates and times for checks, settings, and tests.

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C12.1 (2014; Errata 2016) Electric Meters - Code for Electricity Metering

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16 (2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures

ASTM INTERNATIONAL (ASTM)

ASTM D149 (2020) Dielectric Breakdown Voltage and Dielectric Strength of Solid Electrical Insulating Materials at Commercial Power Frequencies

ASTM D257 (2014) Standard Test Methods for D-C Resistance or Conductance of Insulating Materials

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

ASTM D882 (2012) Tensile Properties of Thin Plastic Sheeting

ASTM D903 (1998; R 2017) Standard Test Method for Peel or Stripping Strength of Adhesive Bonds

ASTM D1876 (2008; R 2015; E 2015) Standard Test Method for Peel Resistance of Adhesives (T-Peel Test)

ASTM D2244 (2016) Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates

ASTM D2765 (2016) Standard Test Methods for Determination of Gel Content and Swell Ratio of Crosslinked Ethylene Plastics

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ASTM D5870	(2016) Standard Practice for Calculating Property Retention Index of Plastics
ASTM D7567	(2009) Standard Test Method for Determining Gel Content in Crosslinked Ethylene Plastics Using Pressurized Liquid Extraction
ASTM E308	(2022) Standard Practice for Computing the Colors of Objects by Using the CIE System
ASTM E424	(1971; R 2015) Standard Test Methods for Solar Energy Transmittance and Reflectance (Terrestrial) of Sheet Materials
ASTM E772	(2015; R 2021) Standard Terminology of Solar Energy Conversion
ASTM E1171	(2015) Standard Test Methods for Photovoltaic Modules in Cyclic Temperature and Humidity Environments
ASTM F1249	(2020) Standard Test Method for Water Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor
ASTM G155	(2021) Standard Practice for Operating Xenon Arc Lamp Apparatus for Exposure of Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1547	(2018) Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces
IEEE C2	(2023) National Electrical Safety Code
IEEE Stds Dictionary	(2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL CODE COUNCIL (ICC)

ICC IBC	(2021) International Building Code
ICC IgCC	(2018) International Green Construction Code

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
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INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

ANSI IEC 60529	(2020) Degrees of Protection Provided by Enclosures
IEC 61853-1	(2011; ED 1.0) Photovoltaic (Pv) Module Performance Testing and Energy Rating - Part 1: Irradiance and Temperature Performance Measurements and Power Rating
IEC 62446	(2018) Photovoltaic (PV) Systems - Requirements for Testing, Documentation, and Maintenance - Part 1: Grid Connected Systems - Documentation, Commissioning Tests and Inspection

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001	(2015) Quality Management Systems- Requirements
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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA IEC 60529	(2004) Degrees of Protection Provided by Enclosures (IP Code)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1	(2021) Fire Code
NFPA 70	(2023) National Electrical Code
NFPA 70E	(2021) Standard for Electrical Safety in the Workplace
NFPA 780	(2023) Standard for the Installation of Lightning Protection Systems

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910	Occupational Safety and Health Standards
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UNDERWRITERS LABORATORIES (UL)

UL 969	(2017; Reprint Mar 2018) UL Standard for Safety Marking and Labeling Systems
UL 1703	(2002; Reprint Jun 2016) UL Standard for Safety Flat-Plate Photovoltaic Modules and Panels
UL 1741	(2010; Reprint Jan 2015) UL Standard for Safety Inverters, Converters, Controllers and Interconnection System Equipment for Use With Distributed Energy Resources

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UL 6703 (2014, ANSI Approved December 22, 2017)
Standard for Connectors for Use in
Photovoltaic Systems

UL Electrical Construction (2012) Electrical Construction Equipment
Directory

1.2 RELATED REQUIREMENTS

Sections 26 20 00 INTERIOR DISTRIBUTION SYSTEM apply to this section with additions and modifications specified herein.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in the IEEE Stds Dictionary.
- b. Unless otherwise specified or indicated, solar energy conversion terms used in these specifications, and on the drawings, are as defined in ASTM E772.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Commissioning Plan; G

Commissioning Schedule; G

SD-02 Shop Drawings

Schematic Diagrams; G

Interconnection Diagrams; G

Installation Drawings; G

Site Plan Drawings; G

Riser Diagram and General Notes; G

Installation and Assembly Details; G

Shop Drawings; G

Complete Solar PV System Diagrams; G

SD-03 Product Data

Combiner Boxes; G

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Disconnects; G

Inverters; G; S

String Inverter Efficiency; G; S

Roof Mounting Structure for Modules (Racking); G

Photovoltaic Module Backsheet; G

Photovoltaic Module Encapsulant; G

Photovoltaic Modules; G; S

Photovoltaic Wire; G

System Monitoring; G

System Wiring; G

SD-05 Design Data

System Operation; G

Calculations; G; S

System Performance Calculations; G; S

SD-06 Test Reports

NABCEP Acceptance Checks and Tests; G

NETA Acceptance Checks and Tests; G

Inverter Startup Tests; G

Functional Performance Testing; G

SD-07 Certificates

Installer; G

Materials; G

Warranty; G

Cybersecurity Equipment Certification; G

Commissioning Agent Qualification; G

SD-08 Manufacturer's Instructions

Manufacturer's Installation Instructions; G

SD-10 Operation and Maintenance Data

Electrical Systems, Data Package 5; G

Training Course; G

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SD-11 Closeout Submittals

Solar Posted Operating Instructions; G

Final Commissioning Report; G

Warranty; G

1.5 MAINTENANCE MATERIAL SUBMITTALS

Comply with requirements specified in Section 01 33 00 SUBMITTAL PROCEDURES.

1.6 QUALITY ASSURANCE

1.6.1 Regulatory Requirements

Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officers. Provide equipment, materials, installation, and workmanship in accordance with the mandatory and advisory provisions of NFPA 70 unless more stringent requirements are specified or indicated.

1.6.2 Installation Drawings

Submit a minimum of three hard copies of drawings for government approval prior to manufacturing and equipment construction or integration. Submit site plan drawings and riser diagram and general notes at a minimum of 24 by 36 inches. Submit installation and assembly details at a minimum of 24 by 36 inches. Submit at minimum scale of 1/2 inch per foot for overview and 2 inches per foot for detail.

In addition to requirements in Section 01 33 00 SUBMITTAL PROCEDURES, include the following:

- a. All details legible and all text no smaller than 0.1 inches in height on any drawing. As needed, provide enlargements to ensure clarity of intent.
- b. Submit shop drawings at a minimum of 11 by 17 inches in size using a minimum scale of 1/4 inch per foot, for the exception of drawings not required scale. Shop drawings must include wiring diagrams and installation details of photovoltaic (PV) system equipment indicating location as proposed in design drawings, layout and arrangement of PV modules, support and mounting mechanism, inverters, combiner boxes, AC and DC disconnects, equipment enclosures, conduits, monitors, meters, security systems, and all other accessories associated with the installation of the PV system. Wiring diagrams must identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each equipment item.
- c. Shop drawings may include legible copies of manufacturer's product literature, with selected items and specifications highlighted thereon.
- d. Modifications to original drawings made during installation must be immediately recorded for inclusion into the as-built drawings. When items have changed relative to the approved design, the designer must provide certification indicating that the changes will not negatively affect the system's operation or the structure supporting the system.

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1.6.3 System Operation

Provide a complete description of the function of each component including PV modules, DC wiring, combiner boxes, inverters, AC wiring, AC and DC disconnect switches, and monitoring system. Provide a discussion of the overall system operation.

1.6.4 Installer

Submit NABCEP (North American Board of Certified Energy Practitioners) PV Installation Professional certification, and a resume with references that details least four successful projects that, in aggregate, equal or exceed the size of the proposed project. Provide references for each of these referenced projects.

1.6.5 Standard Materials and Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Submit proof of compliance with requirements of UL, where material or equipment is specified to comply. The label of or listing in UL Electrical Construction Directory will be acceptable evidence. In lieu of the label or listing, a written certificate from an approved nationally recognized testing laboratory (NRTL) equipped to perform such services, stating that the items have been tested and conform to the requirements and testing methods of Underwriters Laboratories may be submitted.

1.6.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if the manufacturer has been regularly engaged in the design and production of solar photovoltaic products for a minimum of 5-years. Similar photovoltaic products must have been in satisfactory commercial or industrial use for 5-years prior to bid opening and must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 5-year period.

1.6.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 years prior to date of delivery to site must not be used, unless specified otherwise.

1.6.6 Cybersecurity Equipment Certification

Furnish a certification that control systems are designed and tested in accordance with DoD Instruction 8500.01, DoD Instruction 8510.01, and as required by individual Service Implementation Policy.

1.6.7 Operation and Maintenance Data

Submit Solar Photovoltaic Systems data package for the following items in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA.

- a. Troubleshooting guide.
- b. Warranty.

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- c. Operation instructions.
- d. Preventive maintenance and inspection data, including a schedule for system operators.

1.6.7.1 Electrical Systems

Submit operation and maintenance data in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. In addition to requirements of Data Package 5, include the following for the actual solar photovoltaic (PV) system provided:

- a. Service and maintenance information including preventive maintenance, assembly, and disassembly procedures.
- b. Complete operation, repair, and maintenance information, detailed to the smallest replaceable unit.
- c. Adjustment, trouble-shooting, configuration, tuning, and system calibration instructions.
- d. Programming information for the communications and monitoring interface.
- e. An instruction manual with pertinent items and information highlighted.
- f. A layout drawing showing locations as well as views of equipment; front, top, and side views.
- g. A one-line drawing showing all components and interfaces to the electrical system.
- h. Prices for spare parts and supply list including spare modules and inverters.
- i. Inverter efficiency report and field acceptance test reports.
- j. Actual nameplate diagram.
- k. Date of purchase.

1.6.7.2 Training Course

Provide training by a factory trained instructor to provide full instructions to designated Government personnel in the operation, maintenance and programming of the specified systems and equipment. Include safety training for first responders including fire department. The proposed Training Course Curriculum (including topics and dates of discussion) indicating that all of the items contained in the operating and maintenance instructions, as well as demonstrations of safety and routine maintenance operations, including testing procedures included in the maintenance instructions, are to be covered. The proposed Training Course must be video-recorded and provided with any PowerPoint slides as part of the final documentation for those that cannot attend. .

1.6.8 Bill of Materials

Submit a Bill of Materials listing each product being incorporated into the system. Bill of Materials includes a general description of the

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product, quantity, and exact manufacturer's model number. Where the manufacturer's model number does not fully identify the product, list options, accessories, or custom features by additional descriptions.

1.6.9 Qualified Testing Organization

Comply with requirements specified in Section 26 08 00 APPARATUS INSPECTION AND TESTING. Engage the services of a qualified testing organization, NABCEP-certified professional, or licensed electrician to provide inspection, testing, calibration, and adjustment of the solar photovoltaic electrical distribution system and equipment listed herein. Organization must be independent of the supplier, manufacturer, and installer of the equipment. The organization must be a first tier contractor.

Submit name and qualifications of organization. Organization must have been regularly engaged in the testing of electrical materials, devices, installations, and regularly engaged in solar PV systems for a minimum of five years.

Organization calibration program requirements:

- a. Provide a calibration program which assures that all applicable test instruments are maintained within rated accuracy.
- b. Accuracy: Traceable to the National Institute of Standards and Technology.
- c. Instrument calibration frequency schedule: Less than or equal to 12 months for both test floor instruments and leased specialty equipment.
- d. Dated calibration tables: Visible on all test equipment.
- e. Calibrating standard: Higher accuracy than that of the instrument tested.
- f. Keep up-to-date records that indicate dates and test results of instruments calibrated or tested. For instruments calibrated by the manufacturer on a routine basis, in lieu of third party calibration, include the following:
 - (1) Maintain up-to-date instrument calibration instructions and procedures for each test instrument.
 - (2) Identify the third party laboratory calibrated instrument to verify that calibrating standard is met.

1.6.10 System Performance Calculations

Submit system performance calculations to show that the components provided will produce the minimum required production of power in accordance with PERFORMANCE REQUIREMENTS paragraph.

1.7 DELIVERY, STORAGE, AND HANDLING

- a. Store solar PV modules in their original packaging according to the manufacturer's guidance, and do not remove from packaging until day of installation.

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- b. If a solar PV module is removed from its packaging, store it according to the manufacturer's guidance.
- c. Do not store solar PV modules on-site for more than 12 months.

1.8 WARRANTY

Provide a list of all applicable warranties for all equipment and components. Include warranty information, names, addresses, telephone numbers, and procedures for filing a claim and obtaining warranty services. The equipment items must be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.8.1 Solar Photovoltaic Modules

Furnish the solar photovoltaic module manufacturer's written warranty. The warranty must be a 25-year linear 80 percent (minimum) power warranty (at the end of the 25th year after purchase an actual minimum power output of 80 percent based on the nameplate rating must be achieved) and not less than 10-years for workmanship material and manufacturing defects from the date of manufacture.

The warranty must state that the malfunctioning solar photovoltaic module must be exchanged by the manufacturer and promptly shipped to the using Government facility. The replacement solar module must be identical to, or an improvement upon, the original design of the malfunctioning solar module. Provide an extra 10 percent of spare modules in the event of necessary replacement of malfunctioning installed module.

1.8.2 Inverters

Furnish the inverter manufacturer's warranty. Inverter to be free from defects in material and workmanship for a minimum of 20 years from the date of manufacture. Inverter device installation, transportation, and on-site storage must not exceed 12 months, thereby permitting 19 years of the 20 year warranty to be in service and energized.

The warranty must state that the malfunctioning inverter must be exchanged by the manufacturer and promptly shipped to the using Government facility, and arrive in no more than ten days. The replacement inverter must be identical to, or an improvement upon, the original design of the malfunctioning inverter. Provide an extra 2 spare inverters in the event of necessary replacement of malfunctioning installed inverter.

1.8.2.1 Inverter Software Updates Title

Provide, at no cost or charge, any inverter software upgrades that become available during the warranty period.

1.8.3 Combiner Boxes

Combiner boxes to be free from defects in material and workmanship for a period of 5 years.

1.8.4 Mounting System

Provide PV mounting system warranty of minimum 15 years.

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1.8.5 Warranty Exclusion

The warranty must cover all system malfunctions and failures except those resulting from misuse, abuse, neglect, fire, vandalism, acts of nature, or other causes beyond the control of the Contractor or manufacturer.

1.8.6 Cybersecurity During Warranty Period

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

1.9 CALCULATIONS

If construction deviates from design, provide relevant calculations to demonstrate that new design is satisfactory and approved by a licensed professional engineer.

1.10 HEALTH AND SAFETY RECOMMENDATIONS

Section 01 35 26.00 06 GOVERNMENT SAFETY REQUIREMENTS, applies to this section with additions and modifications specified herein.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

- a. The PV system described in this document is a facility-scale less than 1 megawatt with system voltage not exceeding 1,000 VDC, multiple PV systems with a single service, and is of the grid-connected type which provides a direct interconnection of PV system and grid power service supplying building PV system feeds AC power into the local services when solar energy is available and immediately disconnects from the grid upon loss of grid power to the service in accordance with IEEE 1547 and local utility regulations.
- b. PV system must comply with these specifications, all applicable construction document drawings, all applicable codes, and all local authorities having jurisdiction. System must comply with all policies and standards required by the electrical utility having jurisdiction and all applicable incentive program guidelines. PV system equipment includes, but is not limited to, PV modules and electrical insulating components such as encapsulants and backsheets, raceways, inverters, combiner boxes, disconnect switches, wire, conduit, junction boxes, mounting hardware, mounting structure for modules (racking), monitoring and communication equipment.

2.1.1 System Requirements

Conform electrical installations to IEEE C2, NFPA 70, and requirements specified herein.

- a. Solar photovoltaic system characteristics as indicated on the design drawings.
- b. All equipment must be listed and labeled in accordance with NFPA 70 and OSHA-listed nationally recognized testing laboratories (NRTL) and installed in accordance with the listing requirements and the

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manufacturer's instructions.

- c. Provide all accessories needed for a complete, secure, operational grid-tied PV system.
- d. Wiring and connections of inverters, PV source circuits, AC branch circuits, and all interconnections must be rated at a minimum for IP65 in accordance with NEMA IEC 60529.

2.1.1.1 System Wiring

System wiring must conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and must be in accordance with Section 690 of NFPA 70. Cabling exposed to sunlight must be UV resistant. All wiring must be copper conductor.

Provide conduits in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Use galvanized rigid steel conduit above grade and mount on UV resistant high-density polyethylene (HDPE) supports. Conduit below grade must be as required by Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

2.1.1.2 Site Design

Provide adequate space for personnel, vehicles and equipment throughout the PV array to facilitate installation, inspection and maintenance access to all modules.

2.1.2 Performance Requirements

System components provided must be selected to achieve a minimum calculated energy production as required by ICC IgCC, and minimum AC power output as indicated on the design drawings.

2.2 PHOTOVOLTAIC MODULES

- a. PV modules must be listed to UL 1703, and manufactured in an ISO 9001 certified facility.
- b. PV modules must be of monocrystalline technology and
 - (1) for carport.
- d. PV module efficiency must be greater than 15 percent for crystalline technology.
- e. PV modules must be of the same manufacturer and model number and consistent sub-components.
- f. Submit on cutsheets PV module performance data from the manufacturer that must include a flash test data in accordance with IEC 61853-1, and temperature coefficients at: STC, nominal operating cell temperature (NOCT), low irradiance conditions (LIC), high temperature conditions (HTC), and low temperature conditions (LTC).
- g. PV module bypass diodes must be inside the solar PV module's single conductor cable junction box.
- h. Photovoltaic wire, wiring methods, and utilization of locking-type connectors must comply with the requirements of NFPA 70 and UL 6703.

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Provide USE-2 or RHH or RHW-2 wire, and sunlight-resistant wire when exposed to sunlight.

2.2.1 Crystalline Photovoltaic Module Backsheet

- a. Backsheet component must consist of a tri-layer construction (minimum thickness of 9.8 mils) with outer layers of polyvinyl fluoride (PVF) and an inner layer of polyester for crystalline-silicon modules with a maximum system voltage of 1000 VDC.
- b. Alternate polymeric backsheets consisting of different chemical composition, thickness, or construction must fulfill the safety and performance specifications and acceptance criteria in Table 1. The required component properties in Table 1 must be verified by a test report provided by an OSHA-listed nationally recognized testing laboratory (NRTL) and a cutsheet submitted.

TABLE 1 - PV MODULE BACKSHEET COMPONENT SAFETY AND PERFORMANCE		
Items	Test Methods	Specification
Tensile Strength (MPa)	ASTM D882	≥ 100 (TD) ≥ 100 (MD)
Elongation at Break (percent)	ASTM D882	≥ 80 (TD) ≥ 100 (MD)
Dimensional Stability (percent, 150 degrees C, 0.5 h)	ASTM D882	≤ 1.0 (TD) ≤ 1.0 (MD)
Breakdown Voltage (kV)	ASTM D149	≥ 18
WVTR (g/m ² day, 37.8 degrees C, 100 percent RH)	ASTM F1249	≤ 2.5
Interlayer Peeling Strength (N/cm)	ASTM D1876	≥ 4
Peeling Strength with EVA (N/cm)	ASTM D903	≥ 40
Damp Heat (85 degrees C, 85 percent RH, 1000 hrs) -Color Change delta b -Elongation Retention (percent) -Appearance	ASTM E1171 ASTM E308/ASTM D2244 ASTM D882/ASTM D5870	≤ 2.5 ≥ 70 No cracking or delamination.

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TABLE 1 - PV MODULE BACKSHEET COMPONENT SAFETY AND PERFORMANCE		
UV Exposure Irradiance of 0.55 W/m ² at 340 nm (61 W/m ²) using a xenon lamp with a daylight filter (outer layer). Exposure is 4200 hours (260 kWh/m ² total UV (300-400 nm)) -Color Change delta b -Elongation Retention (percent) -Tensile Retention (percent) -Appearance	ASTM G155 ASTM E308/ASTM D2244 ASTM D882/ASTM D5870 ASTM D882/ASTM D5870	≤2.0 ≥70 ≥70 No cracking or delamination.

2.2.2 Crystalline Photovoltaic Module Encapsulant

- a. Encapsulant component must consist of ethyl vinyl acetate (EVA) with a total nominal (prelamination) thickness of 35 mils or greater in the completed module. The EVA must have a minimum of 28 percent VA content. Through statistical process control, the module manufacturer must ensure that the cured EVA has a minimum of 70 percent gel content per ASTM D7567 or ASTM D2765. The EVA must have a UV cutoff wavelength of 360 nm as measured according to ASTM E424. The EVA must have a minimum volume resistivity of 1X10¹⁵ ohm-cm per ASTM D257.
- b. Thermoplastic encapsulants consisting of different chemical composition, thickness, or construction must fulfill the safety and performance specifications and acceptance criteria described in Table 2. The required component properties described in Table 2 must be verified by a test report provided by an OSHA-listed nationally recognized testing laboratory (NRTL) and a cutsheet submitted.

TABLE 2 - PV MODULE ENCAPSULANT COMPONENT PROPERTIES		
Items	Test Methods	Specification
Appearance	Visual Examination	No bubble, crack, or delamination.
Gel Content (percent)	ASTM D7567/ASTM D2765	≥70
UV Cutoff Wavelength (nm)	ASTM E424	≥360
Volume Resistivity (ohm-cm)	ASTM D257	≥1X10 ¹⁵

2.2.3 Electrical Characteristics

Provide high-power type PV module(s), with typical peak power of not less than 305 watts, plus or minus 3 percent power tolerance, under Standard Test Conditions (STC). The AC output must not be less than 80 percent of the DC kWp rating. The individual current harmonics and TRD shall not exceed the limits specified in IEEE 1547.

The operating voltage corresponding to the power output mentioned above must be at least 32.73 volts. The open circuit voltage of the PV modules under STC should be at least 40.79 volts. Operate PV module at an ambient

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temperature range of minus 40 degrees F to plus 185 degrees F with 100 percent relative humidity.

2.2.4 Terminal Box

Include a terminal box on the module having a provision for opening for replacing the cable, if required.

2.2.5 Nameplate

Include the following on the module nameplate so as to be clearly visible:

- a. Name of the Manufacturer or distinctive logo;
- b. Model or Type Number;
- c. Serial Number;
- d. Year of make;
- e. Peak wattage rating;
- f. Peak voltage; and
- g. Peak current.

2.3 INVERTERS

- a. Array-to-inverter kW ratio must not exceed manufacturer recommendations. Inverter must be IEEE 1547 compliant, listed to UL 1741, comply with the latest applicable ANSI and FCC standards and addenda, and inspected before commissioning, testing, and operation of the system. Submit documentation validating system performance requirements.
- b. Inverter must be approved by FCC Part 15, Class A as an unintentional radiator.
- c. All same-sized inverters supplied must be of the same manufacturer and model number.
- d. Provide inverter utilizing a support structure mount system.
- e. Provide inverter utilizing a NEMA 4X enclosure in accordance with NEMA 250. Provide enclosure made of stainless steel.
- f. Provide inverter with anti-islanding protection to prevent back-feeding inverter generated power to the grid in the event of a utility outage. Anti-islanding protection must be listed to UL 1741 and IEEE 1547.
- g. Overcurrent protection, ground fault protection, arc fault circuit interrupter (AFCI), and rapid shutdown must comply with the requirements of NFPA 70.
- h. Provide inverter with self-diagnostics routines, and remote and local display of operating status and remote monitoring capabilities. Provide inverter compatible with monitoring system and metering system. If capability for remote monitoring and control does not

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exist, then it must be added.

- i. Provide inverter with integrated monitoring system. Data monitoring equipment must be able to sustain an overload across its output terminals up the 150 percent load, while supplying any load within its rating and without reducing its output voltage. Fuse power semiconductors in the inverter with fast acting fuses to prevent cascading failures. Provide each fuse with a blown fuse excluding String and Micro inverters and alarm indicating diodes on the control panel.
- j. Rate inverters output as 60 AC kW at unity (1), 3 phase, 480 volts, 422-528 maximum power point tracking (MPPT) voltage range. The peak inverters power conversion efficiency must be 96 percent or greater.
- k. Match inverter DC input to the design of the PV module array outputs and account for the following:
 - (1) The inverter low voltage is 50 percent of the maximum system voltage, to account for 25 year degradation.
 - (2) Voltage decrease due to high temperatures at project site. Operate inverter at an ambient temperature range of minus 22 degrees F to 140 degrees F with 95 percent humidity (non-condensing).
- l. Inverter must include AC and DC disconnecting means. DC and AC disconnecting means must be listed with ratings suitable for the intended use and purpose. System disconnecting means must meet the requirements of NFPA 70.

2.3.1 String Inverters

- a. Submit String Inverter Efficiency of having a weighted average inverter power conversion efficiency of 96 percent or greater.
- b. Allow the use of DC optimizers provided that a design which coordinates the DC optimizers and the inverter(s) is approved by the Contracting Officer.
- c. Inverter must feature maximum power point tracking (MPPT).

2.4 COMBINER BOXES

- a. All combiner boxes must be listed to UL 1741, and inspected before commissioning, testing, and operation of the system.
- b. Provide combiner boxes support structure mount, NEMA 4/4X outdoor stainless steel enclosures in accordance with NEMA 250.
- c. Supply combiner boxes designed for use with the inverter provided, and coordinated to the specific PV source circuit design.
- d. Include in the combiner boxes fuses and a bus to combine the outputs of the strings. Each combiner box must be UL 1741 listed and operate at an ambient temperature range of minus 13 degrees F to plus 135 degrees F. Provide combiner box capable of at least 12 inputs and an input fuse rating of 20 amps.

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- e. Provide combiner box output terminals for paralleling two conductors for the PV positive and negative, as well as the equipment ground conductors. Run set of wires from the combiner box to the inverter. Provide overcurrent protection and output disconnecting means listed for intended use and purpose that comply with the requirements of NFPA 70.

2.5 ROOF MOUNTING STRUCTURE FOR MODULES (RACKING)

- a. Racking and PV array, including modules, hardware, and attachments, must withstand gravity loads, seismic loads, snow loads, ice loads and wind loads as required by ASCE 7-16 and ICC IBC. Coordinate with structural engineer to insure roof will withstand the racking and PV array loads.
- b. Racking must be suitable for Seismic Design Category B as defined by ASCE 7-16 and ICC IBC.
- c. Submit gravity, seismic, wind, ice and snow load design calculations for the array mounting system and its attachment to the structure showing compliance with gravity, seismic, wind, ice and snow requirements while supporting the PV modules.
- d. Provide the mechanical hardware for mounting the PV arrays and all other hardware required for assembling the PV modules, and the attachments to the building structure.
- e. Use array mounting hardware compatible with the site considerations and environment. Select mechanical hardware for corrosion resistance and durability. Use a stainless steel, galvanized steel, or aluminum support structure. Do not use wood or plastic components for support.
- f. Use cathodic protection compatible with the site considerations and environment. Utilize galvanized anchor encased in concrete driven into ground.

2.5.1 Mounting System Base Supports

Fabricate with fastening points integral to the mounting structure. Mounting system supports must be permanently affixed stanchions that are anchored to the carport structure.

2.5.2 Flashing Boot

Fabricate for precision fit over base support. Coordinate height with base supports.

2.5.3 Base Cap

Fabricate to overlap base support and flashing boot a minimum of 2 inches.

2.5.4 Base Cap Gasket

EPDM with self-adhesive closed cell foam or other gasketing material compatible with the roofing material.

2.5.5 Framing

Provide with wall thickness as determined by structural calculations.

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2.5.6 Hardware

Bolts, nuts, washers, and screws must be 18-8 stainless steel.

2.6 PV SYSTEM MONITORING

- a. Provide a PV system monitoring panel mounted as indicated.
- b. The following quantities must be viewable from a touch screen display mounted at location as indicated:
 - (1) DC Input Voltage from PV array
 - (2) DC Input Power from PV system
 - (3) DC Input Current from PV system
 - (4) AC Phase Current from inverter (average)
 - (5) AC Voltage from inverter (average)
 - (6) AC Real Power from inverter
 - (7) Daily, Weekly, Monthly, Yearly, and Cumulative Energy Production
 - (8) Fault Status Report
 - (9) DC Ground Current Report
 - (10) AC Neutral Current from inverter
 - (11) AC Reactive Power from inverter
 - (12) AC Apparent Power from inverter
 - (13) AC Power Factor
 - (14) AC Phase Current from inverter (A, B, C)
 - (15) AC Voltage from inverter (A, B, C)
 - (16) AC Voltage and Current Balance.
- c. Provide additional data acquisition sensors to measure irradiance wind speed ambient temperature PV module temperature. Any additional data acquisition sensors require a conduit separate from the current conductor conduit.

2.7 PV SYSTEM METERING

- a. Provide a revenue-grade non-revenue-grade Interval Data Recording (IDR) meter complete with industry standard telemetry for communications with Ethernet, cellular, or other common output capabilities. Conform to CSI requirements and electrical utility requirements.
- b. Connect to a monitoring/data collection recording solar production through time increments applicable to installation and utility standards, with a minimum of 15-minute intervals and 30-day memory.

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- c. UL listed and conform to ANSI C12.1.
- d. Measure kWh, demand, instantaneous power, volts, amps, and watts.
- e. Provide UL listed communication and annunciator panel.

2.8 POSTED OPERATING INSTRUCTIONS

Provide for each system and principal item of equipment as specified in the technical sections for use by the operation and maintenance personnel. The operating instructions include the following:

- a. Wiring diagrams, schematic diagrams, interconnection diagrams, control diagrams, and control sequence for each principal system and item of equipment.
- b. Array layout showing the locations of all DC and AC disconnects.
- c. Start up, proper adjustment, operating, and shutdown procedures.
- d. Safety precautions.
- e. The procedure in the event of equipment failure.
- f. Other items of instruction as recommended by the manufacturer of each system or item of equipment.

Print operating instructions and frame under glass or in approved laminated plastic. Post instructions where directed. For operating instructions exposed to the weather, provide weather-resistant materials or weatherproof enclosures. Operating instructions do not fade when exposed to sunlight and secure to prevent easy removal or peeling.

2.9 MANUFACTURER'S NAMEPLATE

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable. For PV modules, a label on the back of the module is acceptable.

2.10 FIELD FABRICATED NAMEPLATES

ASTM D709. Provide laminated plastic nameplates for each equipment enclosure, relay, switch, and device; as specified. Each nameplate inscription identifies the function and, when applicable, the position. Nameplates are of melamine plastic, 0.125 inch thick, white with black center core. Surface is of matte finish. Square corners. Accurately align lettering and engrave into the core. Minimum size of nameplates is 1 inch by 2.5 inches. Lettering is a minimum of 0.25 inch high normal block style.

2.11 PV EQUIPMENT MARKING AND WARNING LABELS

Provide PV equipment listed or be evaluated for the application and have a field label applied in compliance with NFPA 70.

Provide warning signs for the enclosures of electrical equipment having a

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nominal rating exceeding 600 volts.

- a. Provide PV equipment with UL 969 weather-resistant marking and warning labels in compliance with NFPA 1 and NFPA 70.
- b. When such equipment is guarded by a fence, mount signs on the fence. Provide metal signs having nominal dimensions of 14 inches by 10 inches with the legend "DANGER HIGH VOLTAGE KEEP OUT" printed in three lines of nominal 3 inches high white letters on a red and black field.
- c. Provide warning signs for arc flash protection in accordance with NFPA 70E as indicated for all electrical equipment and components that are requiring examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electrical arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible and readable from each accessible side to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.12 CABLE TAGS IN MANHOLES, HANDHOLES, AND VAULTS

Provide tags for each power cable or wire located in manholes, handholes, and vaults. The tags must be polyethylene or sheet lead. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identify the circuit. The next to last position denotes the phase of the circuit and must include the Greek "phi" symbol. The last position denotes the cable size.

2.13 GROUNDING AND BONDING

- a. Provide properly sized equipment grounding conductors. Equipment grounding conductors must be insulated stranded copper, except that sizes 10 AWG and smaller must be solid copper. Insulation color must be continuous green for all equipment grounding conductors, except that wire sizes 4 AWG and larger shall be identified per NFPA 70.
- b. Provide grounding lugs for aluminum PV solar module frames of either stainless steel or tin-coated copper.
- c. Bonding conductors must be bare stranded copper, except that sizes 10 AWG and smaller must be bare solid copper. Bonding conductors must be stranded for final connection to motors, transformers, and vibrating equipment.
- d. Provide bonding fittings on concentric/eccentric knockouts with metal conduits for circuits over 250 volts in accordance with NFPA 70.
- e. Provide bonding fittings for ferrous metal conduits enclosing grounding electrode conductors in accordance with NFPA 70.

2.14 PV LIGHTNING PROTECTION SYSTEM

Provide PV Lightning Protection for electrical and mechanical systems in accordance with Section 26 41 00 LIGHTNING PROTECTION SYSTEM and NFPA 780.

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PART 3 EXECUTION

3.1 MANUFACTURER'S INSTALLATION INSTRUCTIONS AND INSTALLATION DRAWINGS

- a. Complete all electrical work in accordance with NFPA 70.
- b. Provide all permanent and temporary shoring, anchoring, and bracing required by the nature of this work in order to make all parts absolutely stable and rigid, even when such shoring, anchoring, and bracing are not explicitly called for.
- c. Install the solar PV system in accordance with this section, installation drawings, and the printed installation instructions of the manufacturer.
- d. Follow the manufacturer's installation recommendations to ensure no electricity is being fed to the grid and that all available disconnects are in the open position and fuses are not installed during wiring operations. Utilize on-site measurements in conjunction with engineering designs to accurately cut wires and layout before making permanent connections. Locate wires out of the way of windows, doors, openings, and other hazards. Ensure wires are free of snags and sharp edges that have the potential to compromise the wire insulation. If the system is roof-mounted, it must have direct current ground fault protection in accordance with NFPA 70. Ensure breakers in combiner box are in the off position (or fuses removed) during combiner box wiring. Ensure wires and conduit are not installed as a trip hazard.
- e. Attach solar PV modules to the mounting structure according to the manufacturer's instructions and approved plans.
- f. Install instrumentation according to the manufacturer's instructions, with control panels located as indicated.

3.1.1 Wiring Methods

Furnish and install conductors required to connect incoming and outgoing circuits. Install conductors with conduits, boxes, and terminal cabinets in a totally enclosed installation. Install wiring in accordance with NFPA 70 and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

3.1.2 Electrical Connections

- a. Use twist on wire connectors listed for the environment (i.e. wet, damp, direct burial) and installed per manufacturer's instructions.
- b. Use listed power distribution blocks.
- c. Use terminals containing more than one conductor listed for multiple conductors.
- d. Use connectors and terminals used for fine strand conductors that are listed for use with such conductors.
- e. Utilize appropriate tools for connector type as recommended by the manufacturer.
- f. Tighten and secure module connectors.

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- g. Provide corrosion protection in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and by adding a stainless steel isolating washer between components of incompatible metals on the racking structure.
- h. Rate all enclosures for electrical connections and interconnections for NEMA 6 in accordance with NEMA 250 or IP67 in accordance with ANSI IEC 60529.

3.1.3 Disconnects

Provide disconnecting means in accordance with NFPA 70 requirements.

- a. Install disconnects for all current carrying conductors of the PV source.
- b. Install disconnects for the PV equipment. For inverters and other equipment that are energized from more than one source, group and identify the disconnecting means. Equipment disconnecting means or its remote operating device or the enclosure providing access to the disconnecting means must be capable of being locked in the open position when not within sight or not within 10 ft of the equipment.
- c. Install disconnects and overcurrent protection for all ungrounded conductors in ungrounded (transformerless) PV power systems.
- d. Install disconnecting means with a rapid shutdown function using the rapid shutdown initiation devices as specified in NFPA 70. Each device's "off" position must indicate that the rapid shutdown function has been initiated for all PV systems connected to that rapid shutdown initiation device.
- e. Disconnecting means equipment that performs the rapid shutdown function, other than initiating devices, must be listed for providing rapid shutdown protection.
- f. Buildings with rapid shutdown disconnecting means must have a permanent label as specified in NFPA 70.

3.1.4 Overcurrent and Overvoltage Protection

- a. Install the PV interconnect overcurrent protective device as indicated in accordance with NFPA 70. Overcurrent devices used in PV system dc circuits must be listed for use in PV systems.
- b. Install overvoltage surge protective device (SPD) as indicated and in accordance with NFPA 780 and NFPA 70. PV surge protective devices must be listed for use in PV system and marked "DC" or "PV SPD." If the system inverter is more than 100 ft from the closest combiner or recombiner box, provide additional PV SPDs at the PV output circuit adjacent to the inverter.

3.2 GROUNDING

3.2.1 PV System Grounding

NFPA 70 and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms. Ground according to racking

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manufacturer's recommendations.

Install grounding lugs in locations on the solar PV module as designated by the module manufacturer, using stainless steel machine screws of the thread size provided in the pre-tapped holes, along with a stainless steel star washer placed between the grounding lug and the solar module frame.

3.2.2 Grounding Electrodes

Provide driven ground rods as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Connect ground conductors to the upper end of ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.3 INSTALLATION OF EQUIPMENT AND ASSEMBLIES

3.3.1 Carport Mounted Systems

- a. Install and support PV system on carport structures in accordance with PV system manufacturing requirements.

3.4 WARNING SIGN MOUNTING

- a. Display calculated maximum and minimum voltages and their respective amperages on engraved warning labels.
- b. Display information on the arc flash warning labels in accordance with NFPA 70E.
- c. Provide the number of signs required to be clearly visible and readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.5 CABLE TAG INSTALLATION

Install cable tags in each manhole, handhole, and vault as specified, including each splice. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes, handholes, and vaults.

3.6 FIELD QUALITY CONTROL

Perform in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.6.1 Performance of NABCEP Acceptance Checks and Tests

Perform all inspections using a NABCEP-certified professional and in accordance with NABCEP inspection procedures, and in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests.

3.6.1.1 PV Modules

- a. Visual and Mechanical Inspection

(1) Solar PV module manufacturer, model, and number of modules must

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match the approved plans.

- (2) Solar PV modules must be in good conditions (including but not limited to no broken glass or cells, no discoloration, frames not damaged).

b. Electrical Tests

- (1) Verify output of PV modules according to manufacturer's recommendations and NABCEP practices.

3.6.1.2 Inverters

a. Visual and Mechanical Inspection

- (1) Inverter manufacturer, model, and number of inverters must match the approved plans.
- (2) Inverters must be in good condition.

b. Electrical Tests

- (1) Verify output of inverters according to manufacturer's recommendations and NABCEP practices.

3.6.2 Performance of NETA Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.6.2.1 Grounding System

a. Visual and Mechanical Inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod, perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. Use an instrument equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test.
- (2) Submit the measured ground resistance of each ground rod and grounding system, indicating the location of the rod and grounding system. Include the test method and test setup (i.e. pin location) used to determine ground resistance and soil conditions at the time the measurements were made.

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3.6.3 Functional Acceptance Tests

- a. Provide final and complete commissioning of the solar PV system in accordance with IEEE 1547.
- b. Verify that all electrical components are installed and connected according to the requirements of the PV electrical drawings, specifications, and manufacturer's written instructions.
- c. Before starting or operating the system, check continuity of all conductors and grounding conductors to verify that there are no faults and that all equipment has been properly installed according to the manufacturer's recommendations. Check factory instructions to see that installations have been made accordingly. Check equipment for any damage that may have occurred during shipment, after delivery, or during installation. Replace damaged equipment.
- d. Before starting or operating the system, obtain a final inspection approval and final inspection from the Contracting Officer. Be present on site for both of these inspections.
- e. Make final adjustments to all inverters and monitoring equipment so that they will be placed in an acceptable operating condition. Adjustable parameters must be set so that the PV system will produce the maximum possible amount of energy on an annual basis.

3.7 COMMISSIONING

Conduct Commissioning, after the system is installed and is ready for operation, in accordance with Section 01 46 00.00 07 COMMISSIONING - MICROGRID, item (6) renewable energy generation, to verify that the completed and installed system meets the requirements of IEEE 1547. Tailor for non-building systems.

3.7.1 Commissioning Agent Qualification

Individual qualified in testing protective equipment (e.g., professional engineer, factory-certified technician, licensed electrician with experience in testing protective equipment) must perform or directly supervise commissioning tests.

3.7.2 Commissioning Plan and Schedule

Develop and implement a commissioning plan and commissioning schedule in accordance with Section 01 46 00.00 07 COMMISSIONING - MICROGRID.

3.7.3 Start-up Pre-functional Checklists

Carry out a checklist of startup requirements and conduct a series of safety tests to ensure proper installation, safe operation, and performance conforming to specification.

3.7.4 Functional Performance Testing

Prepare test procedures and conduct functional performance testing of the installed system. Include the following test requirements:

- a. All inverter startup tests as specified by the inverter manufacturer in the inverter operation manual;

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- b. Actual power;
- c. Loss of grid;
- d. Grid resume;
- e. Data monitoring check out;
- f. V_{oc} measurement of every source circuit and log it;
- g. Verify tightness of all wiring terminations;
- h. Verify proper markings and labeling of all wire terminations and enclosures;
- i. Verify startup/shut down procedures;
- j. Verify system 5 minutes delay upon restart;
- k. Verify PV array quick connectors are fully mated and wires are neatly secured;
- l. Verify no debris on the modules, no damaged or broken modules;
- m. Verification and inspections (see IEEE 1547.1 7.2)
- n. Field-conducted type and production tests (see IEEE 1547 7.3)
- o. Unintentional islanding functionality test (see IEEE 1547.1)
- p. Cease-to-energize functionality test (see IEEE 1547.1)
- q. Unintentional islanding functionality test (see IEEE 1547.1)
- r. Cease-to-energize functionality test (see IEEE 1547.1 7.5)
- s. Revised settings (see IEEE 1547.1 7.6)

3.7.5 Functional Performance Testing Results

Coordinate, observe and record the results of the functional performance testing. Coordinate retesting as necessary until satisfactory performance is verified. Verify the intended operation of individual components and system interactions under various conditions and modes of operation.

Document items of non-compliance in materials, installation or operation. Immediately address observed non-conformance and deficiencies in terms of notification to responsible parties, and provide recommended actions to correct deficiencies.

3.7.6 Final Commissioning Report

Prepare and submit final commissioning report. Summarize all tasks, findings, conclusions, and recommendations of the commissioning process in accordance with IEC 62446. Include the results of all tests and a listing of the final settings.

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3.8 FINAL ACCEPTANCE

The acceptance of the solar PV system occurs only after all deficiencies identified by the functional acceptance tests and commissioning report are corrected, and the system operates successfully during a 30 day initial testing period.

The Contracting Officer must sign appropriate certificates, if equipment and systems are operating satisfactorily in accordance with the specifications, stating the system's operation has been tested and accepted at the end of the final start-up and testing.

3.9 CLOSEOUT ACTIVITIES

3.9.1 Demonstration

Upon completion of the work and at a time approved by the Contracting Officer, provide instructions by a qualified instructor to the Government personnel in the proper adjustment, system operation, and maintenance of the specified systems and equipment, including pertinent safety requirements as required. Government personnel must receive training comparable to the equipment manufacturer's factory training. Instructor must provide a separate training course for the monitoring system.

3.9.2 Instructor's Qualification Resume

Instructor(s) must be employee(s) of installer, manufacturer, or certified solar photovoltaic system training program. Instructors must be thoroughly familiar with all parts of the installation and trained in operating theory as well as practical operation and maintenance work. Submit the name(s) and qualification resume(s) of instructor(s) to the Contracting Officer for approval.

3.9.3 Training

Furnish training service by a factory-trained representative. Document that each qualified employee has received the required training in accordance with 29 CFR 1910. Maintain all training documentation in a central location for the entire employee's employment duration. Minimum documentation data includes employee's name, training name, and date(s) of training.

The training period must consist of a total of 2 hours of normal working time and begin after the system is functionally completed but prior to final acceptance tests. Submit the training course curriculum for approval, along with the proposed training date, at least 14 days prior to the date of proposed conduction of the training course. Instruction must be given during the first regular work week after the equipment or system has been accepted and turned over to the Government for regular operation. Provide video recording and any PowerPoint slides as part of the final documentation for those that cannot attend. Extend safety training to fire department representatives. Coordinate with Contracting Officer for Fire Department first responder training.

-- End of Section --

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1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2022) Forged Fittings, Socket-Welding and Threaded
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B31.1	(2022) Power Piping
ASME B31.3	(2022) Process Piping
ASME BPVC SEC IX	(2017; Errata 2018) BPVC Section IX-Welding, Brazing and Fusing Qualifications
ASME BPVC SEC VIII D1	(2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A106/A106M	(2019a) Standard Specification for Seamless Carbon Steel Pipe for High-Temperature Service
ASTM A126	(2004; R 2019) Standard Specification for Gray Iron Castings for Valves, Flanges, and Pipe Fittings

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ASTM A181/A181M	(2014; R 2020) Standard Specification for Carbon Steel Forgings, for General-Purpose Piping
ASTM A193/A193M	(2023) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2022a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A234/A234M	(2023) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service

ELECTRICAL GENERATING SYSTEMS ASSOCIATION (EGSA)

EGSA 101P	(1995) Performance Standard for Engine Driven Generator Sets
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INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1	(2000; R 2011) General Principles for Temperature Limits in the Rating of Electric Equipment and for the Evaluation of Electrical Insulation
IEEE 43	(2013) Recommended Practice for Testing Insulation Resistance of Rotating Machinery
IEEE 81	(2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE 115	(2019) Guide for Test Procedures for Synchronous Machines: Part I Acceptance and Performance Testing; Part II Test Procedures and Parameter Determination for Dynamic Analysis
IEEE 120	(1989; R 2007) Master Test Guide for Electrical Measurements in Power Circuits
IEEE 519	(2022) Standard for Harmonic Control in Electrical Power Systems
IEEE C2	(2023) National Electrical Safety Code
IEEE C50.12	(2005; R 2010) Standard for Salient Pole 50 HZ and 60 Hz Synchronous Generators and Generation/Motors for Hydraulic Turbine

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Applications Rated 5 MVA and above

- IEEE C57.13 (2016) Standard Requirements for Instrument Transformers
- IEEE C57.13.1 (2006; R 2012) Guide for Field Testing of Relaying Current Transformers

INTERNATIONAL CODE COUNCIL (ICC)

- ICC IBC (2021) International Building Code

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

- NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

- ISO 3046 (2002, 2006, 2009, 2001) Reciprocating Internal Combustion Engines - Performance--Part 1, 3, 4, 5, 6
- ISO 8528 (1993; R 2018) Reciprocating Internal Combustion Engine Driven Alternating Current Generator Sets--Part 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

- MSS SP-58 (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation
- MSS SP-70 (2011) Gray Iron Gate Valves, Flanged and Threaded Ends
- MSS SP-71 (2018) Gray Iron Swing Check Valves, Flanged and Threaded Ends
- MSS SP-80 (2019) Bronze Gate, Globe, Angle and Check Valves
- MSS SP-85 (2011) Gray Iron Globe & Angle Valves Flanged and Threaded Ends

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- NEMA ICS 2 (2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
- NEMA ICS 6 (1993; R 2016) Industrial Control and Systems: Enclosures
- NEMA MG 1 (2021) Motors and Generators

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NEMA PB 1	(2011) Panelboards
NEMA PB 2	(2011) Deadfront Distribution Switchboards
NEMA/ANSI C12.11	(2006; R 2019) Instrument Transformers for Revenue Metering, 10 kV BIL through 350 kV BIL (0.6 kV NSV through 69 kV NSV)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30	(2021; TIA 20-1; TIA 20-2; TIA 21-3) Flammable and Combustible Liquids Code
NFPA 37	(2021) Standard for the Installation and Use of Stationary Combustion Engines and Gas Turbines
NFPA 54	(2021) National Fuel Gas Code
NFPA 58	(2020; TIA 20-1; TIA 20-2; TIA 20-3) Liquefied Petroleum Gas Code
NFPA 70	(2023) National Electrical Code
NFPA 110	(2022) Standard for Emergency and Standby Power Systems

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE ARP892	(1965; R 1994) DC Starter-Generator, Engine
SAE J537	(2016) Storage Batteries

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-461	(2015; Rev G) Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
UFC 1-200-01	(2022) DoD Building Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 60	Standards of Performance for New Stationary Sources
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UNDERWRITERS LABORATORIES (UL)

UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 489	(2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 891	(2019) UL Standard for Safety Switchboards
UL 1236	(2015; Reprint Feb 2021) UL Standard for Safety Battery Chargers for Charging

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Engine-Starter Batteries

UL 1437

(2006) Electrical Analog Instruments -
Panel Board Types

1.2 RELATED MATERIALS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, and Section 26 08 00 APPARATUS INSPECTION AND TESTING apply to this section, except as modified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Engine-Generator Set and Auxiliary Equipment; G, D
Auxiliary Systems; G, DA
Detailed Drawings; G, DA
Acceptance; G, D0

SD-03 Product Data

Harmonic Requirements; G, D0
Engine-Generator Set Efficiencies; G, D0
Emissions; G, DA
filters; G, D0
special tools; G, D0
Remote Alarm Annunciator; G, D0
Engine-Generator Parameter Schedule
Heat Exchanger
Generator
Manufacturer's Catalog
Site Welding
Spare Parts
Onsite Training
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Posted Data and Instructions; G, D0

Instructions; G, D0

Experience

Field Engineer

General Installation

Exciter

SD-05 Design Data

Performance Criteria

Sound Limitations; G, DA

Power Factor

Heat Exchanger

Time-Delay on Alarms

Cooling System

Vibration Isolation

Battery Charger

Capacity Calculations for Engine-Generator Set; G, D0

Brake Mean Effective Pressure (BMEP) Calculations; G, D0

Torsional Vibration Stress Analysis Computations; G, D0

Capacity Calculations for Batteries; G, D0

Turbocharger Load Calculations; G, D0

Governor Performance Model

Excitation System Model

Generator Performance Model

Generator Dynamic Model

Generator Capability Curve

Generator Impedance Model

SD-06 Test Reports

Performance Tests

Factory Inspection and Tests

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Factory Tests

Onsite Inspection and Tests; G, D0

Acceptance Checks and Tests; G, D0

Functional Acceptance Tests; G, D0

Maintenance Procedures; G, D0

Operation and Maintenance Manuals; G, DA

Inspections; G, D0

Functional Acceptance Test Procedure; G, D0

SD-07 Certificates

Cooling System

Vibration Isolation

Prototype Test

Reliability and Durability

Start-Up Engineer; G, D0

Instructor's Qualification Resume; G, D0

Engine Emission Limits; G, DA

Sound Limitations

Site Visit

Current Balance

Materials and Equipment

Factory Inspection and Tests

SD-09 Manufacturer's Field Reports

Engine Tests; G, DA

Generator Tests; G, D0

Assembled Engine-Generator Set Tests; G, D0

SD-10 Operation and Maintenance Data

Preliminary Assembled Operation and Maintenance Manuals; G, D0

Submit in accordance with Section 01 78 23 OPERATION AND
MAINTENANCE DATA and the paragraph ASSEMBLED OPERATION AND
MAINTENANCE MANUALS.

SD-11 Closeout Submittals

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Posted Data and Instructions; G, D0

Training Plan; G, DA

1.4 QUALITY ASSURANCE

1.4.1 Conformance to Codes and Standards

Where equipment is specified to conform to requirements of any code or standard such as UL, NEMA, etc., the design, fabrication and installation must also conform to the code.

1.4.2 Site Welding

Weld structural members in accordance with Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS. For all other welding, qualify procedures and welders in accordance with ASME BPVC SEC IX.

- a. Welding procedures qualified by others, and welders and welding operators qualified by a previously qualified employer may be accepted as permitted by ASME B31.1.
- b. Submit a copy of qualifying procedures and a list of names and identification symbols of qualified welders and welding operators.
- c. Submit a letter listing the welder qualifying procedures for each welder, complete with supporting data such as test procedures used, what was tested to, and a list of the names of all welders and their identification symbols.
- d. Perform welder qualification tests for each welder whose qualifications are not in compliance with the referenced standards. Notify the Contracting Officer 24 hours in advance of qualification tests which must be performed at the work site, if practical.
- e. The welder or welding operator must apply the personally assigned symbol near each weld made as a permanent record.

1.4.3 Vibration Limitation

Limit the maximum engine-generator set vibration in the horizontal, vertical, and axial directions to 6 mils (peak-peak RMS), with an overall velocity limit of 0.95 inches/second RMS, at rated speed for all loads through 110 percent of rated speed. The engine-generator set must be provided with vibration isolation in accordance with the manufacturer's standard recommendation. Where the vibration isolation system does not secure the base to the structure floor or unit foundation, provide seismic restraints in accordance with the seismic parameters specified.

1.4.4 Torsional Analysis

Submit torsional analysis including prototype testing or calculations which certify and demonstrate that no damaging or dangerous torsional vibrations will occur when the prime mover is connected to the generator, at synchronous speeds, plus/minus 10 percent.

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1.4.5 Performance Data

Submit vibration isolation system performance data for the range of frequencies generated by the engine-generator set during operation from no load to full load and the maximum vibration transmitted to the floor. Also submit a description of seismic qualification of the engine-generator mounting, base, and vibration isolation.

1.4.6 Experience

Each component manufacturer must have a minimum of 3 years' experience in the manufacture, assembly and sale of components used with stationary engine-generator sets for commercial and industrial use. The engine-generator set manufacturer/assembler must have a minimum of 3 years' experience in the manufacture, assembly and sale of stationary engine-generator sets for commercial and industrial use. Submit a statement showing and verifying these requirements.

1.4.7 Field Engineer

The engine-generator set manufacturer or assembler must furnish a qualified field engineer to supervise the complete installation of the engine-generator set, assist in the performance of the onsite tests, and instruct personnel as to the operational and maintenance features of the equipment. The field engineer must have attended the engine generator manufacturer's training courses on installation and operation and maintenance of engine generator sets. Submit a letter listing the qualifications, schools, formal training, and experience of the field engineer.

1.4.8 Detailed Drawings

Submit detailed drawings showing the following:

- a. Base-mounted equipment, complete with base and attachments, including anchor bolt template and recommended clearances for maintenance and operation.
- b. Starting system.
- c. Fuel system.
- d. Cooling system.
- e. Exhaust system.
- f. Electric wiring of relays, breakers, programmable controllers, and switches including single line and wiring diagrams.
- g. Lubrication system, including piping, pumps, strainers, filters, heat exchangers for lube oil and turbocharger cooling, electric heater, controls and wiring.
- h. Location, type, and description of vibration isolation devices for all applications.
- i. The safety system, including wiring schematics.
- j. One-line schematic and wiring diagrams of the generator, exciter,

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regulator, governor, and instrumentation.

- k. Panel layouts.
- l. Mounting and support for each panel and major piece of electrical equipment.
- m. Engine-generator set rigging points and lifting instructions.

1.4.9 Auxiliary Systems Engine-Generator Set and Auxiliary Equipment Drawing Requirements

Submit drawings pertaining to the engine-generator set and auxiliary equipment, including but not limited to the following:

- a. Certified outline, general arrangement (setting plan), and anchor bolt details. Show total weight and center of gravity of assembled equipment on the steel sub-base.
- b. Detailed elementary, schematic wiring, and interconnection diagrams of the engine starting system, jacket coolant heating system, engine protective devices, engine alarm devices, engine speed governor system, generator and excitation system, and other integral devices.
- c. Detailed elementary, schematic wiring; and interconnection diagrams of the fuel system, starting battery system, engine-generator control panel, generator circuit breaker.
- d. Dimensional drawings or catalog cuts of exhaust silencers, radiator, fuel day tanks, fuel oil cooler, valves and pumps, intake filters, vibration isolators, and other auxiliary equipment not integral with the engine-generator set.

1.4.10 Auxiliary Systems Drawing Requirements

Submit drawings showing floor plan arrangement of exhaust, air intake, and jacket coolant water systems including arrangement of piping and pipe sizes.

1.4.11 Vibration Isolation System Certification

Submit certification from the manufacturer that the vibration isolation system will reduce the vibration to the limits specified in the paragraph VIBRATION ISOLATION.

1.5 DELIVERY, STORAGE, AND HANDLING

Properly protect materials and equipment, in accordance with the manufacturers recommended storage procedures, before, during, and after installation. Protect stored items from the weather and contamination. During installation, cap piping and similar openings to keep out dirt and other foreign matter.

Deliver equipment on pallets or blocking wrapped in heavy-duty plastic, sealed to protect parts and assemblies from moisture and dirt. Protect and prepare batteries for shipment as recommended by the battery manufacturer. Store auxiliary equipment at the site in covered enclosures, protected from atmospheric moisture, dirt, and ground water.

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1.6 EXTRA MATERIALS

Provide two sets of special tools and two sets of filters required for maintenance. Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts. One handset must be provided for each electronic governor when required to indicate and/or change governor response settings. Furnish 4 liters one gallon of identical paint used on engine-generator set in manufacturer's sealed container with each engine-generator set.

Wrenches and tools specifically designed and required to work on the new equipment, which are not commercially available as standard mechanic's tools, must be furnished to the Contracting Officer.

Provide proposed operating instructions for the engine-generator set and auxiliary equipment laminated between matte-surface thermoplastic sheets and suitable for placement adjacent to corresponding equipment. After approval, install operating instructions where directed.

1.7 MAINTENANCE SERVICES

Submit the operation and maintenance manuals and have them approved prior to commencing onsite tests.

1.7.1 Operation Manual

Provide three copies of the operation manual in 8-1/2 by 11 inch three-ring binders. Sections must be separated by heavy plastic dividers with tabs which identify the material in the section. Fold drawings with the title block visible, and placed in 8-1/2 by 11 inch plastic pockets with reinforced holes. The manual must include:

- a. Step-by-step procedures for system startup, operation, and shutdown;
- b. Drawings, diagrams, and single-line schematics to illustrate and define the electrical, mechanical, and hydraulic systems with their controls, alarms, and safety systems;
- c. Procedures for interface and interaction with related systems to include fire alarm systems.

1.7.2 Maintenance Manual

Provide three copies of the maintenance manual containing the information described below in 8-1/2 x 11 inch three-ring binders. Separate each section by a heavy plastic divider with tabs. Fold drawings with the title block visible, and placed in plastic pockets with reinforced holes. The manual must include:

- a. Procedures for each routine maintenance item. Procedures for troubleshooting.
- b. The manufacturer's recommended maintenance schedule.
- c. A component list which includes the manufacturer's name, address, type or style, model or serial number, rating, and catalog number for the major components.
- d. A list of spare parts for each piece of equipment and a complete list

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of materials and supplies needed for operation.

1.7.3 Assembled Operation and Maintenance Manuals

The contents of the assembled operation and maintenance manuals must include the manufacturer's O&M information required by the paragraph SD-10, OPERATION AND MAINTENANCE DATA and the manufacturer's O&M information.

- a. Manuals must be in separate books or volumes, assembled and bound securely in durable, hard covered, water resistant binder, and indexed by major assembly and components in sequential order.
- b. A table of contents (index) must be made part of the assembled O&M. The manual must be assembled in the order noted in table of contents.
- c. The cover sheet or binder on each volume of the manuals must be identified and marked with the words, "Operation and Maintenance Manual."

1.8 SITE CONDITIONS

Protect the components of the engine-generator set, including cooling system components, pumps, fans, and similar auxiliaries when not operating and provide components capable of the specified outputs in the following environment:

- a. Site Location: Conroe, Texas
- b. Site Elevation: 205 feet above mean sea level.
- c. Ambient Temperatures:
 - (1) Maximum 100 degrees F dry bulb, 84.4 degrees F wet bulb.
 - (2) Minimum 30 degrees F dry bulb.
- d. Design Wind Velocity: 122 mph.
- e. Prevailing Wind Direction: West
- f. Wind Exposure: C.
- g. Seismic: Seismic Design Category B, Site Class D as defined by ICC IBC.
- h. Structural Design Risk Category: II.
- i. Environmental Severity Classification: C3, as defined by UFC 1-200-01.

PART 2 PRODUCTS

2.1 SYSTEM REQUIREMENTS

- a. Provide and install each engine-generator set complete and totally functional, with all necessary ancillary equipment to include: air filtration; starting system; generator controls, protection, and isolation; instrumentation; lubrication; fuel system; cooling system; and engine exhaust system. Each engine-generator set must satisfy the requirements specified in the Engine-Generator Parameter Schedule.

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Submit certification that the engine-generator set and cooling system function properly in the ambient temperatures specified.

- b. Provide each engine-generator set consisting of one engine, one generator, and one exciter mounted, assembled, and aligned on one base; and all other necessary ancillary equipment which may be mounted separately. Assemble sets having a capacity of 750 kW or smaller and attach to the base prior to shipping. Sets over 750 kW capacity may be shipped in sections. Provide set components that are environmentally suitable for the locations shown and that are the manufacturer's standard product offered in catalogs for commercial or industrial use. Provide a generator strip heater for moisture control when the generator is not operating. Identify any nonstandard products or components and the reason for their use.

2.1.1.1 Engine-Generator Parameter Schedule

Engine-Generator Set and Auxiliary Equipment Capacity Calculations for
Engine-Generator Set

ENGINE-GENERATOR PARAMETER SCHEDULE	
Identification	Make/Model
Electrical Characteristics	
Power Rating	Prime Gross bhp rating / Net brake power rating 340 kW at 0.8 power factor
Governor Type	Type Make / Model Isochronous
Overload Capacity (Prime applications only)	110 percent of Service Load for 1 hour in 12 consecutive hours
Service Load	340 kVA (maximum) 340 kVA (continuous)
Motor Starting kVA (Max.)	340 kVA
Power Factor	0.8 lagging
Engine-Generator Applications	parallel with other generators on an infinite bus
Voltage Regulation (No Load to Full Load)(Stand-alone applications)	plus or minus 2 percent (maximum)

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ENGINE-GENERATOR PARAMETER SCHEDULE	
Voltage Bandwidth (steady state)	plus or minus 2 percent
Frequency	60 Hz
Voltage	480 volts
Phases	3 Phase, Delta
Minimum Generator Sub-transient Reactance	0.12 percent Sub-transient
Max Step Load Increase	100 percent of Service Load at 0.8 PF
Transient Recovery Time with Step Load Increase (Voltage)	15 seconds
Transient Recovery	15 seconds
Time with Step Load Increase (Frequency)	
Maximum Voltage Deviation with Step Load Increase	10 percent of rated voltage
Maximum Frequency Deviation with Step Load Increase	2 percent of rated frequency
Max Step Load Decrease (without shutdown)	100 percent of Service Load at .8 PF
Frequency Bandwidth (steady state)	plus or minus 0.4 percent
Frequency Regulation (droop) (No Load to Full Load)	3 percent (maximum)
Frequency Bandwidth (steady state)	plus or minus 0.4 percent

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ENGINE-GENERATOR PARAMETER SCHEDULE	
Reactances	Synchronous reactance, Xd Transient reactance, X'd Sub-transient reactance, X''d Negative sequence reactance, X2 Zero sequence reactance, Xo
Capacity Calculations for Batteries	
Calculation must verify that the engine starting battery capacity exceeds dc power requirements.	
Mechanical Characteristics	
Engine Description	Strokes/cycle Number of cylinders Bore and Stroke, inches
Engine Speed	1800 rpm
Heat Exchanger Type	fin-tube (radiator)
Engine Cooling Type	water/ethylene glycol
Intercooler Type	Air-to-Air / Jacket Water
Induction Method	Naturally Aspirated / Turbocharged
Turbocharger	Make / Model
Max Time to Start and be Ready to Assume Load	10 seconds
Max Summer Outdoor Temp (Ambient)	100 degrees F
Min Winter Outdoor Temp (Ambient)	30 degrees F
Installation Elevation	232 above sea level
Engine-Generator Set Efficiencies	
Fuel Consumption	At 0.8 power factor, Gallons / hour for: 1 / 2 load 3 / 4 load Full Load
Generator Efficiency	At 0.8 power factor,(per cent) for: 1 / 2 load 3 / 4 load Full Load

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ENGINE-GENERATOR PARAMETER SCHEDULE	
Radiator Capacity	Coolant Type gpm coolant cfm air through radiator Btu per hr of heat exchange based on optimum coolant temperature to and from engine
Engine-Generator Set Emissions Data	
Exhaust Temperature	Degrees F at full load
Weight of Exhaust Gas	lb per hr at full load
Weight of Intake Air	lb per hr at full load
Total Heat Rejected	Btu per hr, at full load to: Jacket Coolant System Fuel Oil Cooling System
Emissions	lb per hr, at full load Total Suspended Particulate Particulate Matter with an average aerodynamic diameter of 10 microns Sulfur Dioxides Nitrogen Oxides (as NO ₂) Carbon Monoxide Volatile Organic Compounds
Visible Emissions	Percent opacity at full load
Brake Mean Effective Pressure (BMEP) Calculations	
Calculation must verify that the engine meets the specified maximum BMEP, as follows: $\text{BMEP psi} = \frac{(120,000 \times \text{bkw}) \times (792,000 \times \text{bhp})}{(\text{rpm} \times \text{cu. in.})}$ Where: $\text{bkw bhp} = \text{bkw}' + \text{bkw}'' \text{ bhp}' + \text{bhp}''$ $\text{bkw}'' \text{ bhp}''$ is the Brake kw horsepower required by engine driven fan for cooling radiator or motor driven fan for cooling radiator. $\text{bkw}' \text{ bhp}' = \text{kw}/\text{GEN.EFF. kw}/(\text{GEN.EFF. times } 0.746)$ GEN.EFF. = Generator efficiency cu. in. = Total engine piston displacement in cubic inches rpm = Engine revolutions per minute kw = Minimum power rating	
Torsional Vibration Stress Analysis Computations	

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ENGINE-GENERATOR PARAMETER SCHEDULE
Torsional vibrational stresses in the crankshaft and generator shaft of assembled engine and driven generator must not exceed 5000 psi when engine is driving generator at rated speed while assembled unit is loaded to rated engine-generator set power. Computations must be based on a mathematical model of the assembled generator set provided or based on calculations using measured values from tests on a unit identical to the one provided. Calculations based on models of, or measured data from, the unassembled engine and generator will not be acceptable. Calculations must include: a. A description of the system relating information pertinent to analysis such as operating speed range and identification plate data. b. A mass elastic assembly drawing, showing the arrangement of the units in the generator set and dimensions of shafting, including minimum diameters (or section moduli) of shafting in the system. c. A labeled line diagram of the mass elastic system indicating values of masses, stiffness, equivalent lengths, and equivalent diameters including basic assumptions and definition of terms. d. Sample computations showing procedures used to obtain resulting stress values. e. Computations indicating assembled engine-generator speed of 1800 rpm with assembly loaded to rated generator power and the resulting computed critical torsional stress values in the assembled engine crankshaft and generator shaft.
Turbocharger Load Calculations
NOTE: When the engine-generator set installation includes field installed exhaust system (i.e., the engine-generator set is installed internal to a building in lieu of in a self-contained outdoor enclosure), include the following paragraph. When the proposed exhaust system layout is different from that shown on the contract drawings, submit calculations showing that the external loads from the exhaust system such as weight and thermal expansion do not exceed the engine manufacturer's maximum allowed forces and moments on the turbocharger.

2.1.2 Rated Output Capacity

Provide each engine-generator-set with power equal to the sum of service load plus the machine's efficiency loss and associated ancillary equipment loads. Rated output capacity must also consider engine and/or generator oversizing required to meet requirements in paragraph Engine-Generator Parameter Schedule.

The engine must meet the specified maximum BMEP requirements at rated speed as calculated in accordance with the calculations in the

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engine-generator parameter schedule. The engine capacity must be based on the following:

- a. Engine cooled by a radiator fan mechanically driven by the engine or remote with a motor driven fan.
- b. Engine cooled by coolant mixture of water and ethylene glycol, 50 percent by volume of each.

Maximum BMEP, psi	
	Turbocharged
Four-cycle engines	150 - 225
Engine speed, rpm:	1800

2.1.2.1 Engine Emission Limits

Engine must be certified by the manufacturer to meet applicable EPA emission standards found in 40 CFR 60 Subpart IIII. In addition, engine must meet any applicable state or local emission requirements (ex: California SCAQMD).

2.1.2.2 Performance Class

The voltage and frequency behavior of the generator set must be in accordance with ISO 8528 operating limit values for performance Class G3.

2.1.3 Power Ratings

Power ratings must be in accordance with EGSA 101P.

2.1.4 Transient Response

The engine-generator set governor and voltage regulator must cause the engine-generator set to respond to the maximum step load changes such that output voltage and frequency recover to and stabilize within the operational bandwidth within the transient recovery time. The engine-generator set must respond to maximum step load changes such that the maximum voltage and frequency deviations from bandwidth are not exceeded.

2.1.5 Reliability and Durability

Each prime engine-generator set must have both an engine and a generator capable of delivering the specified power on a standby basis with an anticipated mean time between overhauls of no less than 5,000 hours operating with a load factor of 70 percent. Cite two like engines and two like generators that have performed satisfactorily in a stationary power plant, independent and separate from the physical location of the manufacturer's and assembler's facilities, for standby without any failure to start, including all periodic exercise. Provide like engines and generators that have had no failures resulting in downtime for repairs in excess of 72 hours during two consecutive years of service. Provide engines that are the same model, speed, bore, stroke, number and configuration of cylinders, and rated output capacity. Provide generators that are the same model, speed, pitch, cooling, exciter, voltage regulator and rated output capacity.

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Submit a reliability and durability certification letter from the manufacturer and assembler to prove that existing facilities are and have been successfully utilizing the same components proposed to meet this specification, in similar service. Certification may be based on components, i.e. engines used with different models of generators and generators used with different engines, and does not exclude annual technological improvements made by a manufacturer in the basic standard-model component on which experience was obtained, provided parts interchangeability has not been substantially affected and the current standard model meets the performance requirements specified. Provide a list with the name of the installations, completion dates, and name and telephone number of a point of contact.

2.1.6 Parallel Operation

Configure each engine-generator set specified for parallel operation for automatic parallel operation. Each set must be capable of parallel operation with a commercial power source on an infinite bus.

2.1.7 Load Sharing

Configure each engine-generator set specified for parallel operation to automatically load share with other sets by proportional loading. Proportional loading must load each set to within 5 percent of its fair share. A set's fair share is its nameplate-rated capacity times the total load, divided by the sum of all nameplate-rated capacities of on-line sets. Incorporate both the real and reactive components of the load.

2.1.8 Engine-Generator Set Enclosure

Provide engine-generator set enclosures that are corrosion resistant and fully weather resistant. The enclosure must contain all set components and provide ventilation to permit operation at Service Load under secured conditions. Provide access doors to controls and equipment requiring periodic maintenance or adjustment. Provide removable panels for access to components requiring periodic replacement. The enclosure must be capable of being removed without disassembly of the engine-generator set or removal of components other than the exhaust system. The enclosure must reduce the noise of the generator set to within the limits specified in the paragraph SOUND LIMITATIONS.

2.1.9 Vibration Isolation

Provide an engine-generator set with a vibration isolation system in accordance with the manufacturer's standard recommendation. Submit vibration isolation system performance data for the range of frequencies generated by the engine-generator set during operation from no load to full load and the maximum vibration transmitted to the floor plus description of seismic qualification of the engine-generator mounting, base, and vibration isolation. Submit torsional analysis including prototype testing or and calculations which certify and demonstrate that no damaging or dangerous torsional vibrations will occur when the prime mover is connected to the generator, at synchronous speeds, plus 10 percent. Design and qualify vibration isolation systems as an integral part of the base and mounting system in accordance with the seismic parameters specified. Where the vibration isolation system does not secure the base to the structure floor or unit foundation, provide seismic restraints in accordance with the seismic parameters specified.

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2.1.10 Harmonic Requirements

Non-linear loads to be served by each engine-generator set are as indicated. The maximum linear load demand (kVA at PF) when non-linear loads will also be in use is as indicated.

2.1.11 Starting Time Requirements

Upon receipt of a signal to start, each engine generator set will start, reach rated frequency and voltage and be ready to assume load within the time specified. For standby sets used in emergency power applications, each engine generator set will start, reach rated frequency and voltage, and power will be supplied to the load terminals of the automatic transfer switch within the starting time specified.

2.2 NAMEPLATES

Provide the manufacturer's name, type or style, model or serial number and rating on a plate secured to the equipment for each major component of this specification. Provide plates and tags sized so that inscription is readily legible to operating or maintenance personnel and securely mounted to or attached in proximity of their identified controls or equipment. Lettering must be normal block lettering, a minimum of 0.25 inch high. As a minimum, provide nameplates for:

Engines	Relays
Generators	Transformers (CT & PT)
Regulators	Day tanks
Pumps and pump motors	Governors
Generator Breaker	Air Starting System
Economizers	Heat exchangers (other than base mounted)

Where the following equipment is not provided as a standard component by the engine generator set manufacturer, the nameplate information may be provided in the maintenance manual in lieu of nameplates.

Battery charger	Heaters
Switchboards	Exhaust mufflers
Switchgear	Silencers
Battery	Exciters

2.2.1 Materials

Construct ID plates and tags of 16 gage minimum thickness bronze or stainless steel sheet metal engraved or stamped with inscription. Construct plates and tags not exposed to the weather or high operational temperature of the engine of laminated plastic, 0.125 inch thick, matte

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white finish with black center core, with lettering accurately aligned and engraved into the core.

2.2.2 Control Devices and Operation Indicators

Provide ID plates or tags for control devices and operation indicators, including valves, off-on switches, visual alarm annunciators, gages and thermometers, that are required for operation and maintenance of provided mechanical systems. Plates or tags must be minimum of 0.5 inch high and 2 inches long and must indicate component system and component function.

2.2.3 Equipment

Provide ID plates of a minimum size of high and 5 inches long on provided equipment indicating the following information:

- a. Manufacturer's name, address, type and model number, serial number, and certificate of compliance with applicable EPA mission standards;
- b. Contract number and accepted date;
- c. Capacity or size;
- d. System in which installed; and
- e. System which it controls.

2.3 SAFETY DEVICES

Exposed moving parts, parts that produce high operating temperatures, parts which may be electrically energized, and parts that may be a hazard to operating personnel must be insulated, fully enclosed, guarded, or fitted with other types of safety devices. Install safety devices such that proper operation of the equipment is not impaired.

2.4 MATERIALS AND EQUIPMENT

Submit certification stating that where materials or equipment are specified to comply with requirements of UL, written proof of such compliance has been obtained. The label or listing of the specified agency, or a written certificate from an approved, nationally recognized testing organization equipped to perform such services, stating that the items have been tested and conform to the requirements and testing methods of the specified agency are acceptable as proof.

2.4.1 Auxiliary Power Supply Circuit Breakers, Low Voltage Distribution System

UL 489.

Manufacturer shall pre-install all equipment, enclosures, panelboard, internal wiring, wire markers, terminal blocks, raceways, auxiliary equipment, and accessories as required for generation set. Manufacturer must provide a 3-phase, 208Y/120V, auxiliary power panel installed in enclosure to serve auxiliary equipment. Provide a single point service will be provided from an auxiliary panel located in the microgrid switchgear house to the generator set auxiliary power panel.

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2.4.2 Filter Elements

Provide the manufacturer's standard fuel-oil, lubricating-oil, and combustion-air filter elements.

2.4.3 Instrument Transformers

NEMA/ANSI C12.11.

2.4.4 Revenue Metering

IEEE C57.13.

2.4.5 Pipe (Fuel/Lube-Oil, Compressed Air, Coolant, and Exhaust)

ASTM A53/A53M, or ASTM A106/A106M steel pipe. Pipe smaller than 2 inches must be Schedule 80. Pipe 2 inches and larger must be Schedule 40.

2.4.5.1 Flanges and Flanged Fittings

ASTM A181/A181M, Class 60, or ASME B16.5, Grade 1, Class 150.

2.4.5.2 Pipe Welding Fittings

ASTM A234/A234M, Grade WPB or WPC, Class 150 or ASME B16.11, 3000 lb.

2.4.5.3 Threaded Fittings

ASME B16.3, Class 150.

2.4.5.4 Valves

MSS SP-80, Class 150.

2.4.5.5 Gaskets

Manufacturer's standard.

2.4.6 Pipe Hangers

MSS SP-58.

2.4.7 Electrical Enclosures

NEMA ICS 6.

2.4.7.1 Switchboards

NEMA PB 2.

2.4.7.2 Panelboards

NEMA PB 1.

2.4.8 Electric Motors

Provide electric motors that conform to the requirements of NEMA MG 1. Motors must have sealed ball bearings and a maximum speed of 1800 rpm. Motors used indoors must have drip-proof frames; enclose those that are

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used outside. Alternating current motors larger than 1/2 Hp must be of the squirrel-cage induction type for operation on 208 volts or higher, 60 Hz, and three-phase power. Alternating current motors 1/2 Hp or smaller, must be suitable for operation on 120 volts, 60 Hz, and single-phase power.

2.4.9 Motor Controllers

Provide motor controllers and starters that conform to the requirements of NFPA 70 and NEMA ICS 2.

2.5 ENGINE

Each engine must operate on Natural Gas, must be designed for stationary applications and must be complete with ancillaries. The engine must be a standard production model shown in the manufacturer's catalog describing and depicting each engine-generator set and all ancillary equipment in sufficient detail to demonstrate complete specification compliance. The engine must be naturally aspirated, supercharged, or turbocharged. The engine must be 2- or 4-stroke-cycle and compression-ignition type. The engine must be vertical in-line, V- or opposed-piston type, with a solid cast block or individually cast cylinders. The engine must have a minimum of two cylinders. Opposed-piston type engines must have more than four cylinders. Each block must have a coolant drain port. Equip each engine with an over-speed sensor.

2.5.1 Sub-base Mounting

Mount each engine-generator set on a structural steel sub-base sized to support the engine, generator, and necessary accessories, auxiliaries and control equipment to produce a complete self-contained unit as standard with the manufacturer. Design the structural sub-base to properly support the equipment and maintain proper alignment of the engine-generator set in the specified seismic zone. In addition, provide sub-base with both lifting rings and jacking pads properly located to facilitate shipping and installation of the unit. Factory align engine and generator on the sub-base and securely bolt into place in accordance with the manufacturer's standard practice. Crankshaft must have rigid coupling for connection to the generator.

2.5.2 Assembly

Completely shop assemble each engine-generator set on its structural steel sub-base. Paint entire unit with manufacturer's standard paints and colors. After factory tests and before shipping, thoroughly clean and retouch painting as necessary to provide complete protection.

2.5.3 Turbocharger

If required by the manufacturer to meet the engine-generator set rating, provide turbine type driven by exhaust gas from engine cylinders, and direct connected to the blower supplying air to the engine intake manifold.

2.5.4 Intercooler

Provide manufacturer's standard intercooler for engine size specified.

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2.5.5 Crankcase Protection

2.5.6 Miscellaneous Engine Accessories

Provide the following engine accessories where the manufacturer's standard design permits:

- a. Piping on engine to inlet and outlet connections, including nonstandard companion flanges.
- b. Structural steel sub-base and vibration isolators, foundation bolts, nuts, and pipe sleeves.
- c. Level jack screws or shims, as required.
- d. Rails, chocks, and shims for installation of sub-base on the foundation.
- e. Removable guard, around fan. Support guard, on engine sub-base, to suit manufacturer's standard.

2.5.7 Intercooler

Provide manufacturer's standard intercooler for engine size specified.

2.6 FUEL SYSTEM

Provide fuel system conforming to the requirements of NFPA 54 and NFPA 37 and containing the following elements.

2.6.1 Relief/Bypass Valve

Provide a relief/bypass valve to regulate pressure in the fuel supply line, return excess fuel to a return line and prevent the build-up of excessive pressure in the fuel system.

2.7 LUBRICATION

Provide engine with a separate lube-oil system conforming to NFPA 30 and NFPA 37. Pressurize each system by engine-driven pumps. Regulate system pressure as recommended by the engine manufacturer. Provide a pressure relief valve on the crankcase for closed systems. Vent the crankcase in accordance with the manufacturer's recommendation. Do not vent the crankcase to the engine exhaust system. Crankcase breathers, if provided on engines installed in buildings or enclosures, must be piped to vent to the outside. The system must be readily accessible for service such as draining, refilling, etc. Each system must permit addition of oil and have oil-level indication with the set operating. The system must utilize an oil cooler as recommended by the engine manufacturer.

2.7.1 Lube-Oil Filter

Provide one full-flow filter for each pump. The filter must be readily accessible and capable of being changed without disconnecting the piping or disturbing other components. Mark inlet and outlet connections.

2.7.2 Lube-Oil Sensors

Equip each engine with lube-oil pressure sensors located downstream of the

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filters and provide signals for required indication and alarms. Submit two complete sets of filters, required for maintenance, supplied in a suitable storage box. Provide these filters in addition to filters replaced after testing.

2.7.3 Precirculation Pump

Provide a motor-driven precirculation pump powered by the station battery, complete with motor starter, if recommended by the engine manufacturer.

2.8 COOLING SYSTEM

Provide each engine with its own cooling system to operate automatically while its engine is running. The cooling system coolant must use a combination of water and ethylene-glycol sufficient for freeze protection at the minimum winter outdoor temperature specified. The maximum temperature rise of the coolant across each engine must not exceed that recommended below. Submit a letter which certifies that the engine-generator set and cooling system function properly in the ambient temperature specified, stating the following values:

- a. The maximum allowable inlet temperature of the coolant fluid.
- b. The minimum allowable inlet temperature of the coolant fluid through the engine.
- c. The maximum allowable temperature rise in the coolant fluid through the engine.
- d. The minimum allowable inlet fuel temperature.

2.8.1 Heat Exchanger

Provide heat exchanger with the size and capacity to limit the maximum allowable temperature rise in the coolant across the engine to that recommended and submitted for the maximum summer outdoor design temperature and site elevation. Submit manufacturer's data to quantify heat rejected to the space with the engine generator set at rated capacity. Provide heat exchangers that are corrosion resistant, suitable for service in ambient conditions of application.

2.8.1.1 Fin-Tube-Type Heat Exchanger (Radiator)

Heat exchanger may be factory coated with corrosion resistant film, provided that corrective measures are taken to restore the heat rejection capability of the radiator to the initial design requirement via oversizing, or other compensating methods. Provide internal surfaces that are compatible with liquid fluid coolant used. Materials and coolant are subject to approval by the Contracting Officer. Provide heat exchangers that are pressure type incorporating a pressure valve, vacuum valve and a cap. Design caps for pressure relief prior to removal. Provide heat exchanger and cooling system that is capable of withstanding a minimum pressure of 7 psi and protect with a strong grille or screen guard. Provide heat exchanger with at least two tapped holes; equip one tapped hole with a drain cock, and plug the rest.

Provide for each engine-generator set, as standard with the manufacturer.

- a. Design Conditions: Each radiator unit must have ample capacity to

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remove not less than the total Btu per hour of heat rejected by its respective engine at 100 percent full-rated load to the jacket water, fuel oil, and lubricating oil system, and intercooler. Radiator capacity must be rated at optimum temperature of coolant leaving the engine and intercooler as recommended by the engine manufacturer with an ambient dry bulb air temperature outside the enclosure of 115 degrees F maximum, and 20 degrees F minimum at the site elevation specified in the paragraph SITE CONDITIONS, and with the coolant mixture specified in the paragraph ENGINE CAPACITY. Pressure drop through the radiator must not exceed 6 psi when circulating the maximum required coolant flow. Radiator air velocity must be a maximum of 1500 feet per minute.

- b. Engine Mounted Radiator Construction: Radiator fan must direct airflow from the engine outward through the radiator. Fan must be V-belt driven directly from the engine crankshaft. Radiator fan must have sufficient capacity to meet design conditions. Fan static capacity must be adjusted to suit the ductwork furnished. Cooling section must have a tube and fin-type core consisting of copper or copper base alloy tubes with nonferrous fins. Select engine-driven fans for quiet vibration-free operation. Make provision for coolant expansion either by self-contained expansion tanks or separately mounted expansion tanks, as standard with the manufacturer. Provide suitable guards for each fan and drive.
- c. Remote Radiator Construction: Provide radiators as described above, except radiators must be remotely piped and provided with electric motor driven fan. Drive must be multiple V-belt or reduction gears. Expansion tanks must be separately mounted. Air flow must be vertical or horizontal as indicated. Interlock fan with engine operation such that fan must operate when engine operates when recommended by engine manufacturer. Provide controls and control devices complete which must cycle fan on and off based upon coolant temperature. Provide motors and controllers in accordance with the paragraphs ELECTRIC MOTORS and MOTOR CONTROLLERS, respectively. Motors, controllers, contactors, and disconnects must conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.
- d. Coolant solution must be a mixture of clean water and ethylene glycol, 50 percent by volume each. Provide an anti-freeze solution tester suitable for the mixture.

Field installed jacket coolant water piping must conform to the following:

- a. Piping: Provide seamless steel pipe, Schedule 40, conforming to ASTM A53/A53M, Grade A.
- b. Fittings and Flanges: Fittings, 1 1/2 inches or smaller, must be malleable iron conforming to ASME B16.3 for Class 300 threaded type. Fittings, 2 inches and larger, must be steel butt welding conforming to ASME B16.9. Utilize either ASME B16.1 or Class A of ASTM A126 for Class 125 cast-iron flanged fittings. Flanges must be Class 150 slip-on forged steel welding flanges in accordance with ASME B16.5, with material in accordance with ASTM A181/A181M, Grade I. Provide flat face flanges for connecting to Class 125 standard cast-iron valves, fittings, and equipment connections.
- c. Valves

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- (1) Gate Valves: For valves, 1 1/2 inches and smaller, provide double disk, rising stem, inside screw, union bonnet type, Class 125 bronze material conforming to MSS SP-80. For valves, 2 inches and larger, provide double-disk, parallel seat type, hydraulic-rated, Class 125, outside screw and yoke type with flanged ends and bronze trim conforming to MSS SP-70. Provide stem packing of material compatible with the system coolant.
- (2) Globe Valves: For valves, 1 1/2 inches and smaller, provide rising stem, inside screw, union bonnet type, Class 125 bronze valves conforming to MSS SP-80. For valves, 2 inches and larger, provide Class 125 cast iron, flanged ends, bronze trim globe valves conforming to MSS SP-85. Valves must have renewable composition or cast iron discs compatible with the system coolant.
- (3) Check Valves: MSS SP-71 or MSS SP-80, swing check type.

d. Hangers and Supports: MSS SP-58.

2.8.2 Expansion Tank

The cooling system must include an air expansion tank which will accommodate the expanded water of the system generated within the normal operating temperature range, limiting the pressure increase at all components in the system to the maximum allowable pressure at those components. The tank must be suitable for operating temperature of 250 degrees F and a working pressure of 125 psi. Provide welded steel tank, tested and stamped in accordance with ASME BPVC SEC VIII D1 for the stated working pressure. Do not use a bladder type tank. Support the tank by steel legs or bases for vertical or steel saddles for horizontal installation.

2.8.3 Thermostatic Control Valve

Provide a modulating type, thermostatic control valve in the coolant system to maintain the coolant temperature range submitted in paragraph SUBMITTALS.

2.8.4 Temperature Sensors

Equip each engine with coolant temperature sensors. Provide temperature sensors with signals for pre-high and high indication and alarms.

2.9 SOUND LIMITATIONS

Submit sound power level data for the packaged unit operating at 100 percent load in a free field environment. The data should demonstrate compliance with the sound limitation requirements of this specification. Submit certification from the manufacturer stating that the sound emissions meet the specification. Do not exceed the following sound pressure levels in any of the indicated frequencies when measured in a free field at a radial distance of 22.9 feet 7 meters at 45 degrees apart in all directions when operating at 100 percent load.

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Frequency Band (Hz)	Maximum Acceptable Sound Level (Decibels)
31	52
63	66
125	70
250	67
500	69
1,000	64
2,000	58
4,000	54
8,000	48

2.10 AIR INTAKE EQUIPMENT

Locate filters and silencers in locations that are convenient for servicing. Provide high-frequency filter type silencers and locate in the air intake system as recommended by the engine manufacturer. Provide silencer to reduce the noise level at the air intake so that the indicated pressure levels specified in paragraph SOUND LIMITATIONS will not be exceeded. A combined filter-silencer unit meeting requirements for the separate filter and silencer items may be provided. Provide rubber expansion elements in air-intake lines.

Provide intake filter assemblies for each engine of the oil bath or dry type, as standard with the manufacturer. Filters must be capable of removing a minimum of 92 percent of dirt and abrasive 3 microns and larger from intake air. Size filters to suit engine requirements at 100 percent of rated full load. Design unit for field access for maintenance purposes.

2.11 EXHAUST SYSTEM

Provide a separate and complete system for each engine. Support piping to minimize vibration. Where a V-type engine is provided, use a V-type connector, with necessary flexible sections and hardware, to connect the engine exhaust outlets.

2.11.1 Flexible Sections and Expansion Joints

Provide a flexible section at each engine and an expansion joint at each muffler. Provide flexible sections and expansion joints that have flanged connections. Provide flexible sections made of convoluted seamless tube without joints or packing. Provide bellows type expansion joints. Provide stainless steel expansion and flexible elements suitable for engine exhaust gas at the maximum exhaust temperature that is specified by the engine manufacturer. Provide expansion and flexible elements that are capable of absorbing vibration from the engine and compensation for thermal expansion and contraction.

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2.11.2 Exhaust Muffler

Provide a chamber type exhaust muffler. Provide welded steel muffler designed for outside vertical mounting. Provide eyebolts, lugs, flanges, or other items as necessary for support in the location and position indicated. Do not exceed the engine manufacturer's recommended pressure drop. Outside mufflers must be zinc coated or painted with high temperature 400 degrees F resisting paint. The muffler and exhaust piping together must reduce the noise level to less than the maximum acceptable level listed for sound limitations in paragraph SOUND LIMITATIONS. Provide muffler with a drain valve, nipple, and cap at the low-point of the muffler.

A critical class silencer must be provided for each engine which will reduce the exhaust sound spectrum by the following listed values at a 75 foot radius from the outlet, with generator set loaded to rated capacity and clear weather. Inlet and outlet connections must be flanged.

Octave Band Center Frequency (Hertz)								
Minimum Silencer Attenuation Decibels	63	125	250	500	1000	2000	4000	8000
Critical Class	15	32	37	36	30	36	37	37

2.11.3 Exhaust Piping

Slope horizontal sections of exhaust piping downward away from the engine to a drip leg for collection of condensate with drain valve and cap. Changes in direction must be long radius. Insulate exhaust piping, mufflers and silencers installed inside any building in accordance with paragraph THERMAL INSULATION and covered to protect personnel. Provide vertical exhaust piping with a hinged, gravity-operated, self-closing, rain cover.

Field installed exhaust piping must conform to the following:

- a. Exhaust Piping: Provide flanges for connections to engines, exhaust mufflers, and flexible connections. Provide steel pipe conforming to ASTM A53/A53M for each engine complete with necessary fittings, flanges, gaskets, bolts, and nuts. Exhaust piping must be Schedule 40 pipe for 12 inches and smaller, standard weight for sizes 14 inches through 24 inches, and 0.25 inch wall thickness for sizes larger than 24 inches. Flanges must be Class 150 slip-on forged steel welding flanges in accordance with ASME B16.5, with material in accordance with ASTM A181/A181M, Grade I. Fittings must be butt welding conforming to ASTM A234/A234M, with wall thickness same as adjoining piping. Fittings must be of same material and wall thickness as pipe. Built-up miter welded fittings may be used. Miter angles of each individual section must not exceed 22.5 degrees total and not more than 11.25 degrees relative to the axis of the pipe at any one cut. Gaskets for exhaust piping must be of high temperature asbestos-free material suitable for the service and must be ASME B16.21, composition ring, 0.0625 inch thick. Bolting material for exhaust flanges must be alloy-steel bolt-studs conforming to ASTM A193/A193M, Grade B7 bolts and alloy-steel nuts conforming to ASTM A194/A194M, Grade 7. Bolts must be of sufficient length to obtain full bearing on

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the nuts and must project not more than two full threads beyond the nut. Provide stainless steel counterbalance type rain caps at termination of each exhaust pipe.

- b. Expansion (Flexible) Joints: Provide sections of multiple corrugated stainless steel expansion joints with liners in the engine exhaust piping for each engine to absorb expansion strains and vibration transmitted to the piping. Flexible joints must be suitable for operation at 200 degrees F above normal exhaust gas temperature at 100 percent load, 10,000 cycles, minimum. Joints must be flanged and located between engine exhaust manifold and exhaust piping, must be the same size as exhaust piping size, and must be designed and constructed for engine exhaust service.
- c. Hangers and Supports: MSS SP-58.
- d. Piping Sleeves: Provide where piping passes through masonry or concrete walls, floors, roofs, and partitions. Sleeves must be placed during construction. Unless indicated otherwise, pipe sleeves must comply with following requirements: sleeves in outside walls below and above grade, in floor, or in roof slabs, must be standard weight zinc coated steel pipe. Sleeves in partitions must be zinc coated sheet steel having a nominal weight of not less than 0.90 pound per square foot. Space between piping insulation and the sleeve must not be less than 0.25 inch. Sleeves must be held securely in proper position and location during construction. Sleeves must be sufficient length to pass through entire thickness of walls, partitions, or slabs. Sleeves in floor slabs must extend 2 inches above the finished floor. Space between the pipe and the sleeve must be firmly packed with insulation and caulked at both ends of the sleeve with plastic waterproof cement.
- e. Piping Insulation: Provide exhaust piping insulation in accordance with Section 23 07 00 THERMAL INSULATION FOR MECHANICAL SYSTEMS.

2.12 PYROMETER

Provide a pyrometer, multi-point selector switch, and individual thermocouples and thermocouple with calibrated leads to show the temperature of the combined exhaust. For a supercharged engine, provide additional points, thermocouples and leads to show the temperature in the turbocharger exhaust gas outlet and combustion air discharge passages. Graduated scale length less than 6 inches is not acceptable. Provide double pole selector switch with an "off" position, one set of points for each thermocouple, and suitable indicating dial. Calibrate the pyrometer, thermocouples, leads and compensating devices to show true exhaust temperature within plus or minus 1 percent above the highest temperature encountered at 110 percent load conditions.

2.13 EMISSIONS

The finished installation must comply with Federal, state, and local regulations and restrictions regarding the limits of emissions. Submit certification from the engine manufacturer stating that the engine exhaust emissions meet the federal, state, and local regulations and restrictions specified. At a minimum this certification must include emission factors for criteria pollutants including nitrogen oxides, carbon monoxide, particulate matter, sulfur dioxide, non-methane hydrocarbon, and for hazardous air pollutants (HAPs).

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2.14 STARTING SYSTEM

Provide starting system for engine generator sets used in non-emergency applications as follows.

2.14.1 Controls

Provide an engine control switch with functions including: run/start (manual), off/reset, and, automatic mode. Provide start-stop logic for adjustable cycle cranking and cool-down operation. Arrange the logic for manual starting and fully automatic starting in accordance with paragraph AUTOMATIC ENGINE-GENERATOR-SET SYSTEM OPERATION. Provide electrical starting systems with an adjustable cranking limit device to limit cranking periods from 1 second up to the maximum duration.

Controls must also provide the following:

Generator control features must include:

- a. Local/remote monitoring and control of gas and diesel gensets
- b. Local/remote genset diagnostics and operating information
- c. Local/remote annunciation of genset conditions
- d. Automatic transfer/automatic mains failure capability
- e. Automatic synchronization and load sharing of at least 16 gensets
- f. Multi-generator paralleling to utility and paralleling functionality with multiple circuit breaker control
- g. Supervisory control panel for centralized system monitoring and control
- h. Remote communications capabilities

Control Parameters

- a. Run / Auto / Stop control
- b. Operation modes
- c. Power output control setpoints
- d. Speed and voltage adjustment
- e. Local and remote emergency stop
- f. Remote start/stop
- g. Cycle crank

Monitoring Parameters

- a. Voltage (L-L, L-N)
- b. Current (Phase)

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- c. Average Volt, Amp, Frequency
- d. kW, kVAr, kVA (Average, Phase, Percent)
- e. Power Factor (Average, Phase)
- f. kW-hr, kVAr-hr (total)
- g. Excitation voltage and current (with CDVR)
- h. Desired Voltage, Excitation Command, Operating Mode (with IVR)
- i. Generator stator and bearing temp (with optional module)
- j. kW load histogram

Generator Protection

- a. Generator phase sequence
- b. Over/Under voltage (27/59)
- c. Over/Under frequency (81 O/U)
- d. Reverse Power (kW) (32)
- e. Reverse Reactive Power (kVAr) (32RV)
- f. Overcurrent (50/51)
- g. Thermal Damage Curve

Engine Monitoring

- a. Coolant temperature
- b. Oil pressure
- c. Engine speed (RPM)
- d. Battery voltage
- e. Run hours
- f. Crank attempt and successful start counter
- g. Enhanced engine monitoring (with electronic engines)

Engine Protection

- a. Control switch not in auto (alarm)
- b. High coolant temp (alarm and shutdown)
- c. Low coolant temp (alarm)
- d. Low coolant level (alarm)
- e. High engine oil temp (alarm and shutdown)

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- f. Low, high, and weak battery voltage
- g. Overspeed
- h. Overcrank
- i. Low Oil Pressure

2.14.2 Capacity

Provide starting system with sufficient capacity, at the maximum outdoor summer temperature specified to crank the engine without damage or overheating. The system must provide a minimum of three cranking periods with 15 second intervals between cranks. Each cranking period must have a maximum duration of 15 seconds. Starting must be accomplished using an adequately sized dc starter system with a positive shift solenoid to engage the starter motor and to crank the engine continuously for 60 seconds without overheating.

2.14.3 Electrical Starting

Manufacturers recommended dc system, utilizing a negative circuit ground. Starting motors must be in accordance with SAE ARP892.

2.14.3.1 Battery

Provide a starting battery system including the battery, battery rack, intercell connectors, spacers, automatic battery charger with overcurrent protection, metering and relaying. Provide battery in accordance with SAE J537. Size critical system components (rack, protection, etc.) to withstand the seismic acceleration forces specified. Provide lead-acid battery with sufficient capacity, at the minimum outdoor and maximum outdoor temperature specified, to provide the specified cranking periods. Valve-regulated lead-acid batteries are not acceptable.

Provide maintenance free, sealed, lead-acid, SAE Type D engine starting batteries. Batteries must have sufficient capacity to provide 60 seconds of continuous cranking of the engine in an ambient temperature of 37 degrees F.

2.14.3.2 Battery Charger

Provide a current-limiting battery charger, conforming to UL 1236, that automatically recharges the batteries. Submit battery charger sizing calculations. The charger must be capable of an equalize charging rate which is manually adjustable in a continuous range and a floating charge rate for maintaining the batteries at fully charged condition. Provide an ammeter to indicate charging rate. Provide a voltmeter to indicate charging voltage. Provide a timer for the equalize charging-rate setting. A battery is considered to be fully depleted when the output voltage falls to a value which will not operate the engine generator set and its components.

Provide 120 volt ac, enclosed, automatic equalizing, dual-rate, solid-state, constant voltage type battery charger with automatic ac line compensation. DC output must be voltage regulated and current limited. Charger must have two ranges, float and equalize, and must provide continuous taper charging. The charger must have a continuous output

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rating of not less than 10 amperes and must be sized to recharge the engine starting batteries in a minimum of 8 hours while providing the control power needs of the engine-generator set. Enclosure must be NEMA ICS 6, Type 4X. The following accessories must be included:

- a. DC ammeter
- b. DC voltmeter
- c. Equalize light
- d. AC on light
- e. Low voltage light
- f. High voltage light
- g. Equalize test button/switch
- h. AC circuit breaker
- i. Low dc voltage alarm relay
- j. High dc voltage alarm relay
- k. Current failure relay
- l. AC power failure relay

2.14.4 Storage Batteries

Provide storage batteries of suitable rating and capacity to supply and maintain power for the remote alarm annunciator for a period of 90 minutes minimum without the voltage applied falling below 87.5 percent of normal. Provide a 120 volt ac automatic battery charger.

2.14.5 Starting Aids

Provide one or more of other following methods to assist engine starting.

2.14.5.1 Jacket-Coolant Heaters

Mount a thermostatically controlled electric heater in the engine coolant jacketing to automatically maintain the coolant within plus or minus 3 degrees F of the control temperature. The heater must operate independently of engine operation so that starting times are minimized. Power for the heaters must be 120 volts ac. Include necessary equipment, piping, controls, wiring, and accessories.

2.14.5.1.1 Prime Rated Sets

The control temperature must be the higher of the manufacturer's recommended temperature or the minimum coolant inlet temperature of the engine recommended in paragraph SUBMITTALS.

2.14.5.1.2 Standby Rated Sets

The control temperature must be the temperature recommended by the engine manufacturer to meet the starting time specified at the minimum winter

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outdoor temperature.

2.14.5.2 Lubricating-Oil Heaters

Mount a thermostatically controlled electric heater in the engine lubricating-oil system to automatically maintain the oil temperature within plus or minus 3 degrees F of the control temperature. The heater must operate independently of engine operation so that starting times are minimized. Power for the heaters must be 120 volts ac.

2.15 GOVERNOR

Provide a forward acting type engine speed governor system. Steady-state frequency band and frequency regulation (droop) must be in accordance with the operating limit values of the performance class specified in the paragraph PERFORMANCE CLASS. Provide governor performance model for generator set.

Provide engine with a governor which maintains the frequency within a bandwidth of the rated frequency, over a steady-state load range of zero to 100 percent of rated output capacity. Configure the governor for safe manual adjustment of the speed/frequency during operation of the engine-generator set, without special tools, from 90 to 110 percent of the rated speed/frequency, over a steady state load range of 0 to 100 percent or rated capacity. Submit two complete sets of special tools required for maintenance (except for electronic governor handset). Special tools are those that only the manufacturer provides, for special purposes, or to reach otherwise inaccessible parts. Provide a suitable tool box for tools. Provide one handset for each electronic governor when required to indicate and/or change governor response settings. Maintain the midpoint of the frequency bandwidth at the same value for steady-state loads over the range of zero to 100 percent of rated output capacity for isochronous governors.

2.16 GENERATOR

Provide synchronous type, one or two bearing, generator conforming to the performance criteria in NEMA MG 1, equipped with winding terminal housings in accordance with NEMA MG 1, equipped with an amortisseur winding, and directly connected to the engine. Submit calculations of the engine and generator output power capability, including efficiency and parasitic load data. Provide Class F insulation.

- a. Select NEMA MG 1, Part 16, standby duty, and temperature rise of 130 degrees C for engine-generator sets which are expected to operate for less than 300 hours per year. Select NEMA MG 1, Part 22, continuous duty, and temperature rise of 105 degrees C for engine-generator sets expected to operate 300 hours or greater per year or rated 300 kW and above.
- b. Select 2/3 pitch design option for engine-generator sets rated 300 kW and above.
- c. Select 10-12 lead re-connectable for engine-generator sets rated 300 kW to 800 kW.
- d. For applications requiring high SCR loading or in harsh environments laden with salts and chemicals, select vacuum pressure impregnation (VPI) insulated coils. When engine-generator sets are rated 800 kW

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and larger, also select form wound coils.

- e. Provide salient-pole type, ac, brushless-excited, revolving field, air-cooled, self-ventilated, drip-proof guarded, coupled type, synchronous generator conforming to NEMA MG 1, Part 22, and IEEE C50.12. Generator must be rated for continuous duty at 100 percent of the power rating of the engine-generator set as specified in paragraph ENGINE-GENERATOR SET RATINGS AND PERFORMANCE. Temperature rise of each of the various parts of the generator must not exceed 130 degrees C as measured by resistance, based on a maximum ambient temperature of 40 degrees C. Winding insulation must be Class H.
- f. Stator: Stator windings must be 2/3 pitch design, 10-12 lead re-connectable.
- g. Rotor: The rotor must have connected amortisseur windings.
- h. Generator Space Heater: Provide 120 volt ac heaters. Heater capacity must be as recommended by the generator manufacturer to aid in keeping the generator insulation dry.
- i. Grounding: Provide non-corrosive steel grounding pads located at two opposite mounting legs.
- j. Filters: Provide manufacturer's standard generator cooling air filter assembly.
- k. Design generator to protect against mechanical, electrical and thermal damage due to vibration, 25 percent overspeeds, or voltages and temperatures at a rated output capacity of 110 percent for prime applications and 100 percent for standby applications.
- l. Provide generator ancillary equipment meeting the short circuit requirements of NEMA MG 1. Select drip-proof guarded option for generators without weatherproof enclosures.
- m. Submit manufacturer's standard data for each generator (prototype data at the specified rating or above is acceptable), listing the following information:
 - (1) Direct-Axis sub-transient reactance (per unit).
 - (2) The generator kW rating and short circuit current capacity (both symmetric and asymmetric).

2.16.1 Current Balance

At 100 percent rated output capacity, and load impedance equal for each of the 3 phases, the permissible current difference between any 2 phases must not exceed 2 percent of the largest current on either of the 2 phases. Submit certification stating that the flywheel has been statically and dynamically balanced and is capable of being rotated at 125 percent of rated speed without vibration or damage.

2.16.2 Voltage Balance

At any balanced load between 75 and 100 percent of rated output capacity, the difference in line-to-neutral voltage among the 3 phases must not exceed 1 percent of the average line-to-neutral voltage. For a single

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phase load condition, consisting of 25 percent load at unity power factor placed between any phase and neutral with no load on the other 2 phases, the maximum simultaneous difference in line-to-neutral voltage between the phases must not exceed 3 percent of rated line to neutral voltage. The single-phase load requirement must be valid utilizing normal exciter and regulator control. The interpretation of the 25 percent load for single phase load conditions means 25 percent of rated current at rated phase voltage and unity power factor.

2.16.3 Waveform

The deviation factor of the line-to-line voltage at zero load and at balanced rated output capacity must not exceed 10 percent. The RMS of all harmonics must be less than 5.0 percent and that of any one harmonic less than 3.0 percent of the fundamental at rated output capacity. Design and configure engine-generator to meet the total harmonic distortion limits of IEEE 519.

2.17 EXCITER

Provide brushless generator exciter. Provide semiconductor rectifiers that have a minimum safety factor of 300 percent for peak inverse voltage and forward current ratings for all operating conditions, including 110 percent generator output at 104 degrees F ambient. The exciter and regulator in combination must maintain generator-output voltage within the limits specified.

Provide a brushless excitation system consisting of an exciter and rotating rectifier assembly, and permanent magnet generator integral with the generator and a voltage regulator. Insulation class for parts integral with the generator must be as specified in paragraph GENERATOR. System must provide a minimum short circuit of 300 percent rated engine-generator set current for at least 10 seconds. Steady state voltage regulation must be in accordance with the operating limit values of the performance class specified in the paragraph PERFORMANCE CLASS. Provide excitation system model for generator set.

- a. Exciter and Rotating Rectifier Assembly: Rectifiers must be provided with surge voltage protection.
- b. Permanent Magnet Generator: Provide a voltage spike suppression device for permanent magnet generator (PMG) excitation systems.
- c. Voltage Regulator: Voltage regulator must be solid state or digital, automatic, three-phase sensing, volts per hertz type regulator. Regulator must receive its input power from a PMG. Voltage variation for any 40 degree C change over the operating temperature range must be less than plus or minus 1.0 percent. Operating temperature must be minus 40 degree C to plus 70 degree C. Voltage adjust range must be plus to minus 5.0 percent of nominal. Inherent regulator features must include over excitation shutdown.

2.17.1 Electromagnetic Interference (EMI) Suppression

Provide as an integral part of the generator and excitation system, EMI suppression complying with MIL-STD-461.

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2.18 VOLTAGE REGULATOR

Provide a solid-state voltage regulator, separate from the exciter, for each generator. Maintain the voltage within a bandwidth of the rated voltage, over a steady-state load range of zero to 100 percent of rated output capacity. Configure regulator for safe manual adjustment of the engine-generator voltage output without special tools, during operation, from 90 to 110 percent of the rated voltage over the steady state load range of 0 to 100 percent of rated output capacity. Regulation drift exceeding plus or minus 0.5 percent for an ambient temperature change of 68 degrees F is not acceptable. Reactive droop compensation or reactive differential compensation must load share the reactive load proportionally between sets during parallel operation. Provide voltage regulator with a maximum droop of 2 percent of rated voltage over a load range from 0 to 100 percent of rated output capacity and automatically maintain the generator output voltage within the specified operational bandwidth.

2.19 GENERATOR ISOLATION AND PROTECTION

Provide necessary devices for electrical protection and isolation of each engine-generator set and its ancillary equipment. The generator circuit breaker (IEEE Device 52) ratings must be consistent with the generator rated voltage and frequency, with continuous, short circuit withstand, and interrupting current ratings to match the generator capacity. Provide electrically operated generator circuit breaker for synchronization and parallel operation. Mount a set of surge capacitors at the generator terminals. Provide monitoring and control devices as specified in paragraph GENERATOR PANEL.

The generator circuit breaker must comply with UL 489 requirements. The circuit breaker continuous current rating must be adequate for the power rating of the engine-generator set and the circuit breaker must be rated to withstand the short circuit current provided by the generator set. Provide circuit breaker in a NEMA ICS 6, Type 4X enclosure mounted on the engine-generator set.

2.19.1 Switchboards

Provide free-standing, metal-enclosed, general purpose, 3-phase, 4-wire, 600 volt rated, with neutral bus and continuous ground bus, switchboards conforming to NEMA PB 2 and UL 891. Neutral bus and ground bus capacity must be full capacity. Provide panelboards conforming to NEMA PB 1. Provide enclosure designs, construction, materials and coatings suitable for the application and environment. Bus continuous current rating must be as indicated. Current withstand (short circuit rating) must be as indicated. Provide copper buses.

2.19.2 Devices

Provide switches, circuit breakers, switchgear, fuses, relays, and other protective devices as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.

Furnish with respective pieces of equipment. Motors, controllers, contactors, and disconnects must conform to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide electrical connections under Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide controllers and contactors with maximum of 120-volt control circuits, and auxiliary contacts for use with controls furnished. When motors and equipment furnished are larger than

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size indicated, the cost of providing additional electrical service and related work must be included under this section.

2.20 SAFETY SYSTEM

Provide and install devices, wiring, remote panels, and local panels, etc., as a complete system to automatically activate the appropriate signals and initiate the appropriate actions. Provide a safety system with a self-test method to verify its operability. Provide alarm signals that have manual acknowledgment and reset devices. The alarm signal systems must reactivate for new signals after acknowledgment is given to any signal. Configure the systems so that loss of any monitoring device will be dealt with as an alarm on that system element.

2.20.1 Audible Signal

Provide audible alarm signal sound at a frequency of 70 Hz at a volume of 75 dB at 10 feet. The sound must be continuously activated upon alarm and silenced upon acknowledgment. Locate signal devices as shown.

2.20.2 Visual Signal

The visual alarm signal must be a panel light. The light must be normally off, activated to be blinking upon alarm. The light must change to continuously lit upon acknowledgement. If automatic shutdown occurs, the display must maintain activated status to indicate the cause of failure and must not be reset until cause of alarm has been cleared and/or restored to normal condition. Shutdown alarms must be red; all other alarms must be amber.

2.20.3 Alarms and Action Logic

2.20.3.1 Shutdown

Accomplish simultaneous activation of the audible signal, activation of the visual signal, stopping the engine, and opening the generator main circuit breakers.

2.20.3.2 Problem

Accomplish activation of the visual signal.

2.20.4 Safety Indications and Shutdowns

Provide a local alarm panel with the following shutdown and alarm functions in accordance with NFPA 110 level 2 mounted either on or adjacent to the engine generator set.

A remote alarm panel is required for audible alarms, e.g., in the control room.

Indicator Function (at battery voltage)	NFPA 110 Level 2 CV S RA
Overcrank	X X 0
Low water temperature	X NA 0

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Indicator Function (at battery voltage)	NFPA 110 Level 2 CV S RA
High engine temperature pre-alarm	0 NA NA
High engine temperature	X X 0
Low lube oil pressure pre-alarm	NA NA NA
Low lube oil pressure	X X 0
Overspeed	X X 0
Low fuel main tank	0 NA 0
Low coolant level	X 0 X
EPS supplying load	0 NA NA
Control switch not in automatic position	X NA X
High battery voltage	0 NA NA
Low cranking voltage	0 NA NA
Low voltage in battery	0 NA NA
Battery charger ac failure	0 NA NA
Lamp test	X NA NA
Contacts for local and remote common alarm	X NA X
Audible alarm silencing switch	NA NA 0
Low starting air pressure	0 NA NA
Low starting hydraulic pressure	0 NA NA
Air shutdown damper when used	X X 0
Remote emergency stop	NA X NA

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Indicator Function (at battery voltage)	NFPA 110 Level 2 CV S RA
Symbology: CV: Control panel-mounted visual. S: Shutdown of EPS indication. RA: Remote audible. Symbology: CV: Control panel-mounted visual. S: Shutdown of EPS indication. RA: Remote audible. X: Required. O: Optional. NA: Not applicable.	

2.20.5 Time-Delay on Alarms

For startup of the engine-generator set, install time-delay devices bypassing the low lubricating oil pressure alarm during cranking, and the coolant-fluid outlet temperature alarm. Submit the magnitude of monitored values which define alarm or action set points, and the tolerance (plus and/or minus) at which the devices activate the alarm or action for items contained within the alarm panels. The lube-oil time-delay device must return its alarm to normal status after the engine starts. The coolant time-delay device must return its alarm to normal status 5 minutes after the engine starts.

2.21 SYNCHRONIZING PANEL

Provide panel as specified in paragraph PANELS and provide controls, gauges, meters, and displays to include:

- a. Frequency meters, dial type, with a range of 90 to 110 percent of rated frequency. Do not use vibrating-reed type meters. One must monitor generator output frequency ("Generator Frequency Meter") and the other must monitor the frequency of the parallel source ("Bus Frequency Meter").
- b. Voltmeters, ac, dial type, 3-phase, with 4-position selector switch for the generator output ("Generator Volt Meter") and for the parallel power source ("Bus volt meter").
- c. Automatic synchronizer.
- d. Manual synchronizing controls.
- e. Indicating lights for supplementary indication of synchronization.
- f. Synchroscope.
- g. Wattmeter, indicating.

2.22 PANELS

Each panel must be of the type and kind necessary to provide specified

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functions. Mount panels as shown. Mount instruments flush or semiflush. Provide convenient access to the back of panels to facilitate maintenance. Calibrate instruments using recognized industry calibration standards. Provide a panel identification plate identifying the panel function. Provide a plate identifying the device and its function for each instrument and device on the panel. Provide switch plates identifying the switch-position function.

2.22.1 Enclosures

Design enclosures for the application and environment, conforming to NEMA ICS 6. Locking mechanisms must be keyed alike.

Provide for each engine-generator set and fabricate from zinc coated or phosphatized and shop primed 16 gage minimum sheet steel in accordance with the manufacturer's standard design. Provide a complete, weatherproof enclosure for the engine, generator, and auxiliary systems and equipment. Support exhaust piping and silencer so that the turbocharger is not subjected to exhaust system weight or lateral forces generated in connecting piping that exceed the engine manufacturer's maximum allowed forces and moments. The housing must have sufficient louvered openings to allow entrance of outside air for engine and generator cooling at full load. Design louvered openings to exclude driving rain and snow. Provide properly arranged and sized, hinged panels in the enclosure to allow convenient access to the engine, generator, and control equipment for maintenance and operational procedures. Provide hinged panels with spring type latches which must hold the panels closed securely and will not allow them to vibrate. Brace the housing internally to prevent excessive vibration when the set is in operation

2.22.2 Analog

Provide analog electrical indicating instruments in accordance with UL 1437 with semi-flush mounting. Switchboard, switchgear, and control-room panel-mounted instruments must have 250 degree scales with an accuracy of not less than 99 percent. Unit-mounted instruments must be the manufacturer's standard with an accuracy of not less than 98 percent. The instrument's operating temperature range must be minus 4 to plus 158 degrees F. Distorted generator output voltage waveform of a crest factor less than 5 must not affect metering accuracy for phase voltages, hertz and amps.

2.22.3 Electronic

Electronic indicating instruments must be true RMS indicating instruments, 100 percent solid state, state-of-the-art, microprocessor controlled to provide specified functions. Provide control, logic, and function devices that are compatible as a system, sealed, dust and water tight, and that utilize modular components with metal housings and digital instrumentation. Provide an interface module to decode serial link data from the electronic panel and translate alarm, fault and status conditions to set of relay contacts. Instrument accuracy less than 98 percent for unit mounted devices and 99 percent for control room, panel mounted devices, throughout a temperature range of minus 4 to 158 degrees F is not acceptable. Provide LED or back lit LCD data display. Additionally, the display must provide indication of cycle programming and diagnostic codes for troubleshooting. Numeral height must be 0.5 inch.

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2.22.4 Parameter Display

Provide indication or readouts of the tachometer, lubricating-oil pressure, ac voltmeter, ac ammeter, frequency meter, and safety system parameters. Specify a momentary switch for other panels.

2.23 SURGE PROTECTION

Electrical and electronic components must be protected from, or designed to withstand the effects of surges from switching and lightning.

2.24 AUTOMATIC ENGINE-GENERATOR-SET SYSTEM OPERATION

Provide fully automatic operation for the following operations: engine-generator set starting and load transfer upon loss of preferred source; retransfer upon restoration of the preferred source; sequential starting; paralleling, and load-sharing for multiple engine-generator sets; and stopping of each engine-generator set after cool-down. Devices must automatically reset after termination of their function.

2.24.1 Monitoring and Transfer

Provide devices to monitor voltage and frequency for the preferred power source and each engine-generator set, and control transfer from the preferred source and retransfer upon restoration of the preferred source. Describe functions, actuation, and time delays as described in Section 25 08 20 MICROGRID CONTROL SYSTEM.

2.24.2 Automatic Paralleling and Loading of Engine-Generator Sets

Provide an automatic loading system to load and unload engine-generator sets in the sequence indicated. Monitor the system load and cause additional engine-generator sets to start, synchronize, and be connected in parallel with the system bus with increasing load. Actuation of the additional engine-generator set start logic must occur when the load exceeds a percentage set-point of the operating set's rating for a period of approximately 10 seconds. Provide an adjustable set-point range from 50 to 100 percent.

2.25 MANUAL ENGINE-GENERATOR-SET SYSTEM OPERATION

Provide complete facilities for manual starting and testing of each set without load, loading and unloading of each set, and synchronization of each set with an energized bus.

2.26 BASE

Provide a steel base. Design the base to rigidly support the engine-generator set, ensure permanent alignment of rotating parts, be arranged to provide easy access to allow changing of lube-oil, and ensure that alignment is maintained during shipping and normal operation. The base must permit skidding in any direction during installation and must withstand and mitigate the affects of synchronous vibration of the engine and generator. Provide base with suitable holes for anchor bolts and jacking screws for leveling.

2.27 THERMAL INSULATION

Provide thermal insulation as specified in Section 23 07 00 THERMAL

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INSULATION FOR MECHANICAL SYSTEMS.

2.28 PAINTING AND FINISHING

Clean, prime and paint the engine-generator set in accordance with the manufacturer's standard color and practice.

2.29 FACTORY INSPECTION AND TESTS

Submit six complete reproducible copies of the factory inspection result on the checklist format specified below. Perform the factory tests on each engine-generator set. The component manufacturer's production line test is acceptable as noted. Run each engine-generator set for at least 1 hour at rated output capacity prior to inspections. Complete inspections and make all necessary repairs prior to testing. Use engine generator controls and protective devices that are provided by the generator set manufacturer as part of the standard package for factory tests. When controls and switchgear are not provided as part of the generator set manufacturer's standard package, the actual controls and protective devices provided for the project are not required to be used during the factory test. The Contracting Officer may provide one or more representatives to witness inspections and tests. Provide generator performance model, generator dynamic model, generator capability curve, and the generator impedance model.

2.29.1 Factory Inspection

Perform inspections prior to beginning and after completion of testing of the assembled engine-generator set. Look for leaks, looseness, defects in components, proper assembly, etc. and note any item found to be in need of correction as a necessary repair. Use the following checklist for the inspection:

INSPECTION ITEM	GOOD	BAD	NOTES
Drive belts			
Governor and adjustments			
Engine timing mark			
Starting motor			
Starting aids			
Coolant type and concentration			
Radiator drains			
Block coolant drains			
Coolant fill level			
All coolant line connections			

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INSPECTION ITEM	GOOD	BAD	NOTES
All coolant hoses			
Combustion air filter			
Combustion air silencer			
Lube oil type			
Lube oil sump drain			
Lube-oil filter			
Lube-oil-level indicator			
Lube-oil-fill level			
All lube-oil line connections			
All lube-oil lines			
All fuel-line connections			
All fuel lines			
Fuel filter			
Coupling and shaft alignment			
Voltage regulators			
Battery-charger connections			
All wiring connections			
Instrumentation			
Hazards to personnel			
Base			
Nameplates			
Paint			
Switchboard			
Switchgear			

2.29.2 Factory Tests

Submit a letter giving notice of the proposed dates of factory inspections and tests at least 14 days prior to beginning tests, including:

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- a. A detailed description of the manufacturer's procedures for factory tests at least 14 days prior to beginning tests.
- b. Six copies of the Factory Test data described below in 8-1/2 by 11 inch binders having a minimum of 3 rings from which material may readily be removed and replaced, including a separate section for each test. Separate sections by heavy plastic dividers with tabs. Provide full size (8-1/2 by 11 inch minimum) data plots showing grid lines, with full resolution.
 - (1) A detailed description of the procedures for factory tests.
 - (2) A list of equipment used, with calibration certifications.
 - (3) A copy of measurements taken, with required plots and graphs.
 - (4) The date of testing.
 - (5) A list of the parameters verified.
 - (6) The condition specified for the parameter.
 - (7) The test results, signed and dated.
 - (8) A description of adjustments made.

On engine-generator set tests where the engine and generator are required to be connected and operated together, the load power factor must be the power factor specified in the engine generator set parameter schedule . For engine-generator set with dual-fuel operating capability, perform the following tests using the primary fuel type. Perform electrical measurements in accordance with IEEE 120. Temperature limits in the rating of electrical equipment and for the evaluation of electrical insulation must be in accordance with IEEE 1. In the following tests where measurements are to be recorded after stabilization of an engine-generator set parameter (voltage, frequency, current, temperature, etc.), stabilization is considered to have occurred when measurements are maintained within the specified bandwidths or tolerances, for a minimum of four consecutive readings. Tests specifically for the generator may be performed utilizing any prime mover.

- a. Insulation Resistance for Stator and Exciter Test, IEEE 115 and IEEE 43, to the performance criteria in NEMA MG 1, Part 22. Generator manufacturer's production line test is acceptable.
- b. High Potential Test, in accordance with IEEE 115 and NEMA MG 1, test voltage in accordance with NEMA MG 1. Generator manufacturer's production line test is acceptable.
- c. Winding Resistance Test, Stator and Exciter, in accordance with IEEE 115. Generator manufacturer's production line test is acceptable.
- d. Phase Balance Voltage Test, to the performance criteria specified in paragraph GENERATOR. This test can be performed with any prime mover. Generator manufacturer's production line test results are acceptable.
 - (1) Start and operate the generator at no load.

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- (2) Adjust a regulated phase voltage (line-to-neutral) to rated voltage.
 - (3) Read and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
 - (4) Apply 75 percent rated load and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
 - (5) Apply rated load and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
 - (6) Calculate average line-neutral voltage and percent deviation of individual line-neutral voltages from average for each load condition.
- e. Current Balance on Stator Winding Test, by measuring the current on each phase of the winding with the generator operating at 100 percent of Rated Output Capacity, with the load impedance equal for each of the three phases: to the performance criteria specified in paragraph GENERATOR.
- f. Voltage Waveform Deviation and Distortion Test in accordance with IEEE 115 to the performance criteria specified in paragraph GENERATOR. Use high-speed recording instruments capable of recording voltage waveform deviation and all distortion, including harmonic distortion. Include appropriate scales to provide a means to measure and interpret results.
- g. Voltage and Frequency Droop Test. Verify that the output voltage and frequency are within the specified parameters as follows:
- (1) With the generator operating at no load, adjust voltage and frequency to rated voltage and frequency. Record the generator output frequency and line-line and line-neutral voltages.
 - (2) Increase load to Rated Output Capacity. Record the generator output frequency and line-line and line-neutral voltages.
 - (3) Calculate the percent droop for voltage and frequency with the following equations:

$$\text{Voltage droop percent} = \frac{(\text{No-Load Volts}) - (\text{Rated Capacity Volts})}{(\text{Service-Load Volts})} \times 100$$

$$\text{Frequency droop percent} = \frac{(\text{No-Load Hertz}) - (\text{Rated Capacity Hertz})}{(\text{Service-Load Hertz})} \times 100$$

- (4) Repeat steps 1 through 3 two additional times without making any adjustments.
- h. Frequency and Voltage Stability and Transient Response. Verify that the engine-generator set responds to addition and dropping of blocks of load in accordance with the transient response requirements. Document maximum voltage and frequency variation from bandwidth and

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verify that voltage and frequency return to and stabilize within the specified bandwidth, within the specified response time period. Document results in tabular form and with high resolution, high speed strip chart recorders or comparable digital recorders, as approved by the Contracting Officer. Include the following tabular data:

- (1) Ambient temperature (at 15 minute intervals).
- (2) Generator output current (before and after load changes).
- (3) Generator output voltage (before and after load changes).
- (4) Frequency (before and after load changes).
- (5) Generator output power (before and after load changes).
- (6) Graphic representations must include the actual instrument trace of voltage and frequency showing: charts marked at start of test; observed steady-state band; mean of observed band; momentary overshoot and undershoot (generator terminal voltage and frequency) and recovery time for each load change together with the voltage and frequency maximum and minimum trace excursions for each steady state load condition prior to and immediately following each load change. Generator terminal voltage and frequency transient recovery time for each step load increase and decrease.
 - (a) Perform and record engine manufacturer's recommended pre-starting checks and inspections.
 - (b) Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period and no load. Verify stabilization of voltage and frequency within specified bandwidths.
 - (c) With the unit at no load, apply the Maximum Step Load Increase.
 - (d) Apply load in steps equal to the Maximum Step Load Increase until the addition of one more step increase will exceed the Service Load.
 - (e) Decrease load to the unit such that addition of the Maximum Step Load Increase will load the unit to 100 percent of Service Load.
 - (f) Apply the Maximum Step Load Increase.
 - (g) Decrease load to zero percent in steps equal to the Maximum Step Load Decrease.
 - (h) Repeat steps (c) through (g).
- i. Test Voltage Unbalance with Unbalanced Load (Line-to-Neutral) to the performance criteria specified in paragraph GENERATOR. Prototype test data is acceptable in lieu of the actual test. Submit manufacturer's standard certification that prototype tests were performed for the generator model proposed. This test may be performed using any prime mover.

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- (1) Start and operate the generator set at rated voltage, no load, rated frequency, and under control of the voltage regulator. Read and record the generator frequency, line-to-neutral voltages, and the line-to-line voltages.
- (2) Apply the specified load between terminals L_1-L_2 , L_2-L_0 , and L_3-L_0 in turn. Record all instrument readings at each line-neutral condition.
- (3) Express the greatest difference between any two of the line-to-line voltages and any two of the line-to-neutral voltages as a percent of rated voltage.
- (4) Compare the largest differences expressed in percent with the maximum allowable difference specified.

PART 3 EXECUTION

3.1 EXAMINATION

After becoming familiar with all details of the job, perform a Site Visit to verify the information shown on the drawings, before performing any work. Submit a letter stating the date the site was visited and listing discrepancies found. Notify the Contracting Officer in writing of any discrepancies.

3.2 GENERAL INSTALLATION

Provide clear space for operation and maintenance in accordance with NFPA 70 and IEEE C2. Submit a copy of the manufacturer's installation procedures and a detailed description of the manufacturer's recommended break-in procedure. Install pipe, duct, conduit, and ancillary equipment to facilitate easy removal and replacement of major components and parts of the engine-generator set.

3.3 PIPING INSTALLATION

Weld piping. Provide flanged valve connections. Provide flanged connections at equipment. Provide threaded connections to the engine if the manufacturers standard connection is threaded. Except where otherwise specified, use welded flanged fittings to allow for complete dismantling and removal of each piping system from the facility without disconnecting or removing any portion of any other system's equipment or piping. Make connections to equipment with vibration isolation-type flexible connectors. Support and align piping and tubing to prevent stressing of flexible hoses and connectors. Flash pipes extending through the roof. Install piping clear of windows, doors and openings, to permit thermal expansion and contraction without damage to joints or hangers, and install a 1/2 inch drain valve with cap at each low point.

The installation of gas engines must conform to the requirements of NFPA 37 and its references therein, including NFPA 54, NFPA 58, and ASME B31.3.

3.3.1 Support

Provide hangers, inserts, and supports to accommodate any insulation and conforming to MSS SP-58. Space supports no more than 7 feet on center for pipes 2 inches in diameter or less, no more than 12 feet on center for

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pipes larger than 2 inches but smaller than 4 inches in diameter, and not more than 17 feet on center for pipes larger than 4 inches in diameter. Provide supports at pipe bends or change of direction.

3.3.1.1 Ceiling and Roof

Support exhaust piping with appropriately sized Type 41 single pipe roll and threaded rods; support all other piping with appropriately sized Type 1 clevis and threaded rods.

3.3.1.2 Wall

Make wall supports for pipe by suspending the pipe from appropriately sized Type 33 brackets with the appropriate ceiling and roof pipe supports.

3.3.2 Flanged Joints

Provide flanges that are Class 125 type, drilled, and of the proper size and configuration to match the equipment and engine connections. Provide gasketed flanged joints that are square and tight.

3.3.3 Cleaning

After fabrication and before assembly, piping interiors must be manually wiped clean of debris.

3.3.4 Pipe Sleeves

Fit pipes passing through construction such as ceilings, floors, or walls with sleeves. Extend each sleeve through and fasten in its respective structure and cut flush with each surface. Build the structure tightly to the sleeve. The inside diameter of each sleeve must be minimum 1/2 inch, and where pipes pass through combustible materials 1 inch larger than the outside diameter of the passing pipe or pipe insulation/covering.

3.4 ELECTRICAL INSTALLATION

Perform electrical installation in compliance with NFPA 70, IEEE C2, and Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. For vibration isolation, provide flexible fittings for conduit, cable trays, and raceways attached to engine-generator sets; provide flexible stranded conductor for metallic conductor cables installed on the engine generator set and from the engine generator set to equipment not mounted on the engine generator set; and provide crimp-type terminals or lugs for terminations of conductors on the engine generator set.

3.5 FIELD PAINTING

Perform field painting as specified in Section 09 90 00 PAINTS AND COATINGS.

3.6 ONSITE INSPECTION AND TESTS

Perform and report on factory tests and inspections prior to shipment. Provide certified copies of manufacturer's test data and results. Test procedures must conform to ASME, IEEE, and ANSI standards, and to ISO requirements on testing, as appropriate and applicable. The manufacturer performing the tests must provide equipment, labor, and consumables necessary for tests and measuring and indicating devices must be certified

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to be within calibration. Tests must indicate satisfactory operation and attainment of specified performance. If satisfactory, equipment tested will be given a tentative approval. Equipment must not be shipped before approval of the factory test reports for the following tests.

Submit a letter giving notice of the proposed dates of onsite inspections and tests at least 14 days prior to beginning tests.

- a. Submit a detailed description of the Contractor's procedures for onsite tests including the test plan and a listing of equipment necessary to perform the tests at least 14 days prior to beginning tests.
- b. Submit six copies of the onsite test data described below in 8-1/2 by 11 inch binders having a minimum of 3 rings from which material may readily be removed and replaced, including a separate section for each test. Separate sections by heavy plastic dividers with tabs. Provide full size (8-1/2 by 11 inch minimum) data plots showing grid lines, with full resolution.
 - (1) A detailed description of the procedures for onsite tests.
 - (2) A list of equipment used, with calibration certifications.
 - (3) A copy of measurements taken, with required plots and graphs.
 - (4) The date of testing.
 - (5) A list of the parameters verified.
 - (6) The condition specified for the parameter.
 - (7) The test results, signed and dated.
 - (8) A description of adjustments made.

3.6.1 Test Conditions

3.6.1.1 Data

Make and record measurements of all parameters necessary to verify that each set meets specified parameters. If the results of any test step are not satisfactory, make adjustments, replacements, or repairs and repeat the step until satisfactory results are obtained. Unless otherwise indicated, record data in 15 minute intervals during engine-generator set operation and include: readings of all engine-generator set meters and gauges for electrical and power parameters; oil pressure; ambient temperature; and engine temperatures available from meters and gauges supplied as permanent equipment on the engine-generator set. Perform electrical measurements in accordance with IEEE 120. Definitions of terms are in accordance with IEEE 100. Provide temperature limits in the rating of electrical equipment and for the evaluation of electrical insulations in accordance with IEEE 1.

3.6.1.2 Power Factor

Submit the generator capability curve showing generator kVA output capability (kW vs. kvar) for both leading and lagging power factors ranging from 0 to 1.0. For all engine-generator set operating tests the

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load power factor must be the power factor specified in the engine-generator set parameter schedule power factor.

3.6.1.3 Contractor Supplied Items

Provide equipment and supplies required for inspections and tests including fuel, test instruments, and loadbanks at the specified power factors.

3.6.1.4 Instruments

Verify readings of panel gauges, meters, displays, and instruments provided as permanent equipment during test runs, using test instruments of greater precision and accuracy. Test instrument accuracy must be within the following: current plus or minus 1.5 percent, voltage plus or minus 1.5 percent, real power plus or minus 1.5 percent, reactive power plus or minus 1.5 percent, power factor plus or minus 3 percent, frequency plus or minus 0.5 percent. Calibrate test instruments by a recognized standards laboratory within 30 days prior to testing.

3.6.1.5 Sequence

Provide the sequence of testing as specified in the approved testing plan unless variance is authorized by the Contracting Officer. Perform field testing in the presence of the Contracting Officer. Schedule and sequence tests in order to optimize run-time periods; however, follow the general order of testing: Construction Tests; Inspections; Pre-operational Tests; Safety Run Tests; Performance Tests; and Final Inspection.

3.6.2 Construction Tests

Perform individual component and equipment functional tests for fuel piping, coolant piping, and lubricating-oil piping, electrical circuit continuity, insulation resistance, circuit protective devices, and equipment not provided by the engine-generator set manufacturer prior to connection to the engine-generator set.

3.6.2.1 Piping Test

- a. Flush lube-oil and fuel-oil piping with the same type of fluid intended to flow through the piping, until the outflowing fluid has no obvious sediment or emulsion.
- b. Test fuel piping which is external to the engine-generator set in accordance with NFPA 30. Pressure all remaining piping which is external to the engine-generator set with air pressure at 150 percent of the maximum anticipated working pressure, but not less than 150 psi, for a period of 2 hours to prove the piping has no leaks. If piping is to be insulated, perform the test before the insulation is applied.

3.6.2.2 Electrical Equipment Tests

- a. Perform low-voltage cable insulation integrity tests for cables connecting the generator breaker to the panelboard, main disconnect switch, and distribution bus. Test low-voltage cable, complete with splices, for insulation resistance after the cables are installed, in their final configuration, ready for connection to the equipment, and prior to energization. Apply a test voltage of 500 volts dc for one minute between each conductor and ground and between all possible

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combinations conductors in the same trench, duct, or cable, with all other conductors in the same trench, duct, or conduit. Provide the minimum value of insulation as follows:

- (1) $R \text{ in meg-ohms} = (\text{rated voltage in kV plus } 1) \times 304.8 / (\text{length of cable in meters})$
- (2) $R \text{ in meg-ohms} = (\text{rated voltage in kV plus } 1) \times 1000 / (\text{length of cable in feet})$
- (3) Each cable failing this test must be repaired or replaced. The repair cable must be retested until failures have been eliminated.

b. Ground-Resistance Tests. Measure the resistance of each grounding electrode and the ground ring using the fall-of-potential method defined in IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. Take measurements in normally dry weather, not less than 48 hours after rainfall. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance. The combined resistance of separate systems may be used to meet the requirements resistance, but the specified number of electrodes must still be provided as follows:

- (1) Single rod electrode - 25 ohms.
- (2) Multiple rod electrodes - 25 ohms.

c. Examine and test circuit breakers and switchgear in accordance with the manufacturer's published instructions for functional testing.

3.6.3 Inspections

Perform the following inspections jointly by the Contracting Officer and the Contractor, after complete installation of each engine-generator set and its associated equipment, and prior to startup of the engine-generator set. Submit a letter certifying that all facilities are complete and functional; that each system is fully functional; and that each item of equipment is complete, free from damage, adjusted, and ready for beneficial use. Perform checks applicable to the installation. Document and submit the results of those which are physical inspections (I) in accordance with paragraph SUBMITTALS. Present manufacturer's data for the inspections designated (D) at the time of inspection. Verify that equipment type, features, accessibility, installation and condition are in accordance with the contract specification. Provide manufacturer's statements to certify provision of features which cannot be verified visually.

Drive belts	I
Governor type and features	I
Engine timing mark	I
Starting motor	I

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Starting aids	I
Coolant type and concentration	D
Radiator drains	I
Block coolant drains	I
Coolant fill level	I
Coolant line connections	I
Coolant hoses	I
Combustion air filter	I
Intake air silencer	I
Lube oil type	D
Lube oil sump drain	I
Lube-oil filter	I
Lube-oil level indicator	I
Lube-oil fill level	I
Lube-oil line connections	I
Lube-oil lines	I
Fuel type	D
Fuel level	I
Fuel-line connections	I
Fuel filter	I
Access for maintenance	I
Voltage regulator	I
Battery-charger connections	I
Wiring and terminations	I
Instrumentation	I
Hazards to personnel	I
Base	I
Nameplates	I

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Paint	I
Exhaust muffler	I
Switchboard	I
Switchgear	I
Access provided to controls	I
Enclosure is weather resistant	I
Engine and generator mounting bolts (application)	I

3.6.4 Engine Tests

Perform customary commercial factory tests in accordance with ISO 3046 on each engine and associated engine protective device, including, but not limited to the following:

- a. Perform dynamometer test at rated power. Record horsepower at rated speed and nominal characteristics such as lubricating oil pressure, jacket water temperature, and ambient temperature.
- b. Test and record the values that the low oil pressure alarm and protective shutdown devices actuate prior to assembly on the engine.
- c. Test and record values that the high jacket water temperature alarm and protective shutdown devices actuate prior to assembly on the engine.

3.6.5 Generator Tests

Tests must be performed on the complete factory assembled generator prior to shipment. Conduct tests in accordance with IEEE 115, and NEMA MG 1.

3.6.5.1 Routine Tests

Perform the following routine tests on the generators and their exciters:

- a. Resistance of armature and field windings.
- b. Mechanical balance.
- c. Phases sequence.
- d. Open circuit saturation curve and phase (voltage) balance test.
- e. Insulation resistance of armature and field windings.
- f. High potential test

3.6.5.2 Design Tests

Submit the following design tests made on prototype machines that are physically and electrically identical to the generators specified.

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- a. Temperature rise test
- b. Short circuit saturation curve and current balance test

3.6.6 Assembled Engine-Generator Set Tests

Perform the following tests on the assembled engine-generator set.

3.6.6.1 Initial Stabilization Readings

Operate the engine-generator set and allow the set to stabilize at rated kW at rated power factor, rated voltage, and rated frequency. During this period record instrument readings for output power (kW), terminal voltage, line current, power factor, frequency (rpm) generator (exciter) field voltage and current, lubricating oil pressure, jacket water temperature, and ambient temperature at minimum intervals of 15 minutes. Adjust the load, voltage, and frequency to maintain rated load at rated voltage and frequency. Adjustments to load, voltage, or frequency controls must be recorded on the data sheet at the time of adjustment. Stabilization must be considered to have occurred when four consecutive voltage and current recorded readings of the generator (or exciter) field either remain unchanged or have only minor variations about an equilibrium condition with no evident continued increase or decrease in value after the last adjustment to the load, voltage, or frequency has been made.

3.6.6.2 Regulator Range Test

Remove load and record instrument readings (after transients have subsided). Adjust voltage to the maximum attainable value or to a value just prior to actuation of the overvoltage protection device. Apply rated load and adjust voltage to the minimum attainable value or a value just prior to activation of the under-voltage protection device. The data sheets must indicate the voltage regulation as a percent of rated voltage and the maximum and minimum voltages attainable. Voltage regulation must be defined as follows:

$$\text{Percent Regulation} = \frac{((\text{No-Load Voltage}) - (\text{Rated-Load Voltage})) \times 100}{(\text{Rated-Load Voltage})}$$

3.6.6.3 Frequency Range Test

Adjust the engine-generator set frequency for the maximum attainable frequency at rated load. Record instrument readings. Adjust the engine-generator set frequency for the specified minimum attainable frequency at rated load. Record instrument readings. Reduce the load to zero and adjust the engine-generator set frequency for the maximum attainable frequency. Record instrument readings. Adjust the engine-generator set frequency for the minimum attainable frequency. Record instrument readings. The data sheet must show the maximum and minimum frequencies attained at rated load, and at no load.

3.6.6.4 Transient Response Test

Drop the load to no load and re-apply rated load three times to ensure that the no load and rated load voltage and frequency values are repeatable and that the frequency and voltage regulation is within the limits specified. Record generator terminal voltage and frequency using a high speed strip chart recorder. The data sheet must show the following

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results:

a. Frequency

- (1) Stability bandwidth or deviation in percent of rated frequency.
- (2) Recovery time.
- (3) Overshoot and undershoot.

b. Voltage

- (1) Stability bandwidth or deviation in percent of rated voltage.
- (2) Recovery time.
- (3) Overshoot and undershoot.

3.6.7 Pre-operational Tests

3.6.7.1 Protective Relays

Visually and mechanically inspect, adjust, test, and calibrate protective relays in accordance with the manufacturer's published instructions. Include pick-up, timing, contact action, restraint, and other aspects necessary to ensure proper calibration and operation. Implement relay settings in accordance with the installation coordination study. Manually or electrically operate relay contacts to verify that the proper breakers and alarms initiate. Field test relaying current transformers in accordance with IEEE C57.13.1.

3.6.7.2 Insulation Test

Test generator and exciter circuits insulation resistance in accordance with IEEE 43. Take stator readings including generator leads to switchgear at the circuit breaker. Record results of insulation resistance tests. Readings must be within limits specified by the manufacturer. Verify mechanical operation, insulation resistance, protective relay calibration and operation, and wiring continuity of switchgear assembly. Do not damage generator components during test.

3.6.7.3 Engine-Generator Connection Coupling Test

When the generator provided is a two-bearing machine, inspect and check the engine-generator connection coupling by dial indicator to prove that no misalignment has occurred. Use the dial indicator to measure variation in radial positioning and axial clearance between the coupling halves. Take readings at four points, spaced 90 degrees apart. Align solid couplings and pin-type flexible couplings within a total indicator reading of 0.0005 to 0.001 inch for both parallel and angular misalignment. For gear-type or grid-type couplings, 0.002 inch will be acceptable.

3.6.8 Safety Run Test

For the following tests, repeat the associated safety tests if any parts are changed, or adjustments made to the generator set, its controls, or auxiliaries.

- a. Perform and record engine manufacturer's recommended prestarting

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checks and inspections.

- b. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.
- c. Activate the manual emergency stop switch and verify that the engine stops.
- d. Remove the high and pre-high lubricating oil temperature sensing elements from the engine and temporarily install a temperature gauge in their normal locations on the engine (required for safety, not for recorded data). Where necessary provide temporary wiring harness to connect the sensing elements to their permanent electrical leads.
- e. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period. Operate the engine-generator set at no load until the output voltage and frequency stabilize. Monitor the temporarily installed temperature gauges. If either temperature reading exceeds the value required for an alarm condition, activate the manual emergency stop switch.
- f. Immerse the elements in a vessel containing controlled-temperature hot oil and record the temperature at which the pre-high alarm activates and the temperature at which the engine shuts down. Remove the temporary temperature gauges and reinstall the temperature sensors on the engine.
- g. Remove the high and pre-high coolant temperature sensing elements from the engine and temporarily install a temperature gauge in their normal locations on the engine (required for safety, not for recorded data). Where necessary provide temporary wiring harness to connect the sensing elements to their permanent electrical leads.
- h. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period. Operate the engine generator-set at no load until the output voltage and frequency stabilize.
- i. Immerse the elements in a vessel containing controlled-temperature hot oil and record the temperature at which the pre-high alarm activates and the temperature at which the engine shuts down. Remove the temporary temperature gauges and reinstall the temperature sensors on the engine.
- j. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.
- k. Operate the engine generator-set for at least 2 hours at 75 percent of Service Load.
- l. Verify proper operation and set-points of gauges and instruments.
- m. Verify proper operation of ancillary equipment.
- n. Manually adjust the governor to increase engine speed past the over-speed limit. Record the RPM at which the engine shuts down.

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- o. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 75 percent of Service Load.
- p. Manually adjust the governor to increase engine speed to within 2 percent of the over-speed trip speed previously determined and operate at that point for 5 minutes. Manually adjust the governor to the rated frequency.
- q. Shut down the engine. Remove the time-delay low lube oil pressure alarm bypass and try to start the engine.
- r. Attach a manifold to the engine oil system (at the oil pressure sensor port) that contains a shutoff valve in series with a connection for the engine's oil pressure sensor followed by an oil pressure gauge ending with a bleed valve. Move the engine's oil pressure sensor from the engine to the manifold. Open the manifold shutoff valve and close the bleed valve.
- s. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 75 percent of Service Load.
- t. Close the manifold shutoff valve. Slowly allow the pressure in the manifold to bleed off through the bleed valve while watching the pressure gauge. Record the pressure at which the engine shuts down. Catch oil spillage from the bleed valve in a container. Add the oil from the container back to the engine, remove the manifold, and reinstall the engine's oil pressure sensor on the engine.
- u. Start the engine, record the starting time, make and record engine manufacturer's after-starting checks and inspections and operate the engine generator-set for at least 15 minutes at 100 percent of Service Load. Record the maximum sound level in each frequency band at a distance of 22 feet from the end of the exhaust and air intake piping directly along the path of intake and discharge for horizontal piping; or at a radius of 22 feet from the engine at 45 degrees apart in all directions for vertical piping. If a sound limiting enclosure is provided, modify or replace the enclosure, the muffler, and intake silencer must be modified or replaced as required to meet the sound requirements contained within this specification.

3.6.9 Performance Tests

In the following tests, where measurements are to be recorded after stabilization of an engine-generator set parameter (voltage, frequency, current, temperature, etc.), stabilization is considered to have occurred when measurements are maintained within the specified bandwidths or tolerances, for a minimum of four consecutive readings. For the following tests, repeat the associated tests if any parts are changed, or adjustments made to the generator set, its controls, or auxiliaries.

3.6.9.1 Continuous Engine Load Run Test

Test the engine-generator set and ancillary systems at service load to demonstrate durability; verify that heat of extended operation does not

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adversely affect or cause failure in any part of the system; and check all parts of the system. If the engine load run test is interrupted for any reason, repeat the entire test. After each change in load in the following test, measure the vibration at the end bearings (front and back of engine, outboard end of generator) in the horizontal, vertical, and axial directions. Verify that the vibration is within the allowable range. Take data taken at 15 minute intervals and include the following:

Electrical: Output amperes, voltage, real and reactive power, power factor, frequency.

Pressure: Lube-oil.

Temperature: Coolant, Lube-oil, Exhaust, Ambient.

- a. Perform and record engine manufacturer's recommended prestarting checks and inspections. Include as a minimum checking of coolant fluid, fuel, and lube-oil levels.
- b. Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warmup period.
- c. Operate the engine generator-set for 2 hours at 75 percent of Service Load.
- d. Increase load to 100 percent of Service Load and operate the engine generator-set for 4 hours.
- e. For prime rated units, increase load to 110 percent of Service Load and operate the engine generator-set for 2 hours.
- f. Decrease load to 100 percent of Service Load and operate the engine generator-set for 2 hours or until all temperatures have stabilized.
- g. Remove load from the engine-generator set.

3.6.9.2 Voltage and Frequency Droop Test

For the following steps, verify that the output voltage and frequency return to and stabilize within the specified bandwidth values following each load change. Record the generator output frequency and line-line and line-neutral voltages following each load change.

- a. With the generator operating at no load, adjust voltage and frequency to rated voltage and frequency.
- b. Increase load to 100 percent of Rated Output Capacity. Record the generator output frequency and line-line and line-neutral voltages.
- c. Calculate the percent droop for voltage and frequency with the following equations.

$$\text{Voltage droop percent} = \frac{\text{No-load volts} - \text{rated output capacity volts}}{\text{Rated output capacity volts}} \times 100$$

$$\text{Frequency droop percent} = \frac{\text{No load hertz} - \text{rated output capacity hertz}}{\text{rated output capacity hertz}} \times 100$$

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Rated output capacity volts

- d. Repeat steps a. through c. two additional times without making any adjustments.

3.6.9.3 Voltage Regulator Range Test

- a. While operating at no load, verify that the voltage regulator adjusts from 90 to 110 percent of rated voltage.
- b. Increase load to 100 percent of Rated Output Capacity. Verify that the voltage regulator adjusts from 90 to 110 percent of rated voltage.

3.6.9.4 Governor Adjustment Range Test

- a. While operating at no load, verify that the governor adjusts from 90 to 110 percent of rated frequency.
- b. Increase load to 100 percent of Rated Output Capacity. Verify that the governor adjusts from 90 to 110 percent of rated frequency.

3.6.9.5 Frequency and Voltage Stability and Transient Response

Verify that the engine-generator set responds to addition and dropping of blocks of load in accordance with the transient response requirements. Document maximum voltage and frequency variation from bandwidth and verify that voltage and frequency return to and stabilize within the specified bandwidth, within the specified response time period. Document results in tabular form and with high resolution, high speed strip chart recorders or comparable digital recorders, as approved by the Contracting Officer. Include the following tabular data:

- (1) Ambient temperature (at 15 minute intervals).
- (2) Generator output current (before and after load changes).
- (3) Generator output voltage (before and after load changes).
- (4) Frequency (before and after load changes).
- (5) Generator output power (before and after load changes).
- (6) Include the actual instrument trace of voltage and frequency in graphic representations showing:

Charts marked at start of test; observed steady-state band; mean of observed band; momentary overshoot and undershoot (generator terminal voltage and frequency) and recovery time for each load change together with the voltage and frequency maximum and minimum trace excursions for each steady state load condition prior to and immediately following each load change. Generator terminal voltage and frequency transient recovery time for each step load increase and decrease.

- a. Perform and record engine manufacturer's recommended prestarting checks and inspections.
- b. Start the engine, make and record engine manufacturer's after-starting checks and inspections during a reasonable warm-up period and no load. Verify stabilization of voltage and frequency within specified

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bandwidths.

- c. With the unit at no load, apply the Maximum Step Load Increase.
- d. Apply load in steps equal to the Maximum Step Load Increase until the addition of one more step increase will exceed the Service Load.
- e. Decrease load to the unit such that addition of the Maximum Step Load Increase will load the unit to 100 percent of Service Load.
- f. Apply the Maximum Step Load Increase.
- g. Decrease load to zero percent in steps equal to the Maximum Step Load Decrease.
- h. Repeat steps c. through g.

3.6.10 Parallel Operation Test

Test the capability of each engine-generator set to parallel and share load with other generator sets, individually and in all combinations. This test must be performed with the voltage regulator and governor adjustment settings used for the Frequency and Voltage Stability and Transient Response test. If settings are changed during the performance of this test, a voltage and frequency stability and transient response test must be performed for each engine generator set using the setting utilized in this test. During operations record load-sharing characteristics of each set in parallel operation. Include the following data:

- (1) Ambient temperature (at 15 minute intervals).
- (2) Generator output current (before and after load changes).
- (3) Generator output voltage (before and after load changes).
- (4) Power division and exchange between generator sets.
- (5) Real power (watts) and reactive power (vars) on each set.

3.6.10.1 Combinations

Connect each set, while operating at no load, parallel with one other set in the system, operating at service load, until all possible combinations have been achieved. Verify stabilization of voltage and frequency within specified bandwidths and proportional sharing of real and reactive loads. Document stabilization of voltage and frequency within specified bandwidth, the active power division, active power exchange, reactive power division, and voltage and frequency stability and transient response in the following steps for each combination.

- a. Divide the load proportionally between the sets and operate in parallel for 15 minutes.
- b. Increase the load, in steps equal to the Maximum Step Increase, until each set is loaded to its service load.
- c. Decrease the load, in steps equal to the Maximum Step Decrease, until each set is loaded to approximately 25 percent of its service load.

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- d. Increase the load, in steps equal to the Maximum Step Increase, until each set is loaded to approximately 50 percent of its service load. Verify stabilization of voltage and frequency within specified bandwidths and proportional sharing of real and reactive load.
- e. Reduce the sum of the loads on both sets to the output rating of the smaller set.
- f. Transfer a load equal to the output rating of the smaller of the 2 sets to and from each set. Verify stabilization of voltage and frequency within specified bandwidths and proportional sharing of real and reactive load.
- g. Document the active power division, active power exchange, reactive power division, and voltage and frequency stability and transient response.

3.6.10.2 Multiple Combinations

Connect each set, while operating at no load, parallel with all multiple combinations of all other set in the system, while operating at service load, until all multiple combinations of parallel operations have been achieved.

3.6.11 Parallel Operation Test (Commercial Source)

Connect each set parallel with the commercial power source. Operate in parallel for 15 minutes. Verify stabilization of voltage and frequency within specified bandwidths. Record the output voltage, frequency, and loading to demonstrate ability to synchronize with the commercial power source.

3.6.12 Automatic Operation Tests

Test the automatic operating system to demonstrate automatic starting, loading and unloading, the response to loss of operating engine-generator sets, and paralleling of each engine-generator set. Utilize load banks at the indicated power factor for this test, and the loading sequence is the indicated sequence. Record load-sharing characteristics during all operations. Perform this test for a minimum of two successive, successful tests. Include the following data:

- (1) Ambient temperature (at 15 minute intervals).
 - (2) Generator output current (before and after load changes).
 - (3) Generator output voltage (before and after load changes).
 - (4) Generator output frequency (before and after load changes).
 - (5) Power division and exchange between generator sets.
 - (6) Real and reactive power on each set.
- a. Initiate loss of the preferred power source and verify the specified sequence of operation.
 - b. Verify resetting of automatic starting and transfer logic.

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3.6.13 Automatic Operation Tests for Stand-Alone Operation

Test the automatic loading system to demonstrate automatic starting, and loading and unloading of each engine-generator set. Utilize the actual loads to be served for this test, and the loading sequence is the indicated sequence. Perform this test for a minimum of two successive, successful tests. Include the following data:

- (1) Ambient temperature (at 15 minute intervals).
 - (2) Generator output current (before and after load changes).
 - (3) Generator output voltage (before and after load changes).
 - (4) Generator output frequency (before and after load changes).
- a. Initiate loss of the primary power source and verify automatic sequence of operation.
 - b. Restore the primary power source and verify sequence of operation.
 - c. Verify resetting of controls to normal.

3.7 GROUNDING

NFPA 70 and IEEE C2, except that grounding systems must have a resistance to solid earth ground not exceeding 5 ohms.

3.7.1 Grounding Electrodes

Provide driven ground rods as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Connect ground conductors to the upper end of ground rods by exothermic weld or compression connector. Provide compression connectors at equipment end of ground conductors.

3.7.2 Engine-Generator Set Grounding

Provide separate copper grounding conductors and connect them to the ground system as indicated. When work in addition to that indicated or specified is required to obtain the specified ground resistance, the provision of the contract covering "Changes" must apply.

3.7.3 Connections

Make joints in grounding conductors by exothermic weld or compression connector. Exothermic welds and compression connectors must be installed as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION paragraph regarding GROUNDING.

3.7.4 Grounding and Bonding Equipment

UL 467, except as indicated or specified otherwise.

3.8 START-UP ENGINEER

Provide the services of a qualified factory trained start-up engineer, regularly employed by the engine-generator set manufacturer. The start-up services must include conducting preliminary operations and functional

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acceptance tests. The start-up engineer must be present at the engine generator set installation-site, full-time, while preliminary operations and functional acceptance tests are being conducted.

3.9 PREREQUISITES FOR FUNCTIONAL ACCEPTANCE TESTING

Completion of the following requirements is mandatory prior to scheduling functional acceptance tests for the engine-generator set and auxiliary equipment.

3.9.1 Piping Tests

Complete as specified in Section 33 11 23 NATURAL GAS AND LIQUID PETROLEUM PIPING.

3.9.2 Performance of Acceptance Checks and Tests

The acceptance checks and tests must be accomplished by the testing organization as described in Section 26 08 00 APPARATUS INSPECTION AND TESTING.

3.9.3 Generator Sets

Complete as specified in the paragraph ACCEPTANCE CHECKS AND TESTS.

3.9.4 Preliminary Operations

The start-up engineer must conduct manufacturer recommended start-up procedures and tests to verify that the engine-generator set and auxiliary equipment are ready for functional acceptance tests. Give the Contracting Officer 15 days' advance notice that preliminary operations will be conducted. After preliminary operation has been successfully conducted, the start-up engineer will notify the Contracting Officer in writing stating the engine-generator set and auxiliary equipment are ready for functional acceptance tests.

3.9.5 Preliminary Assembled Operation and Maintenance Manuals

Preliminary assembled operation and maintenance manuals must have been submitted to and approved by the Contracting Officer. Manuals must be prepared as specified in the paragraph ASSEMBLED OPERATION AND MAINTENANCE MANUALS.

3.9.6 Functional Acceptance Test Procedure

Test procedure must be prepared by the start-up engineer specifically for the engine-generator set and auxiliary equipment. The test agenda must cover the requirements specified in the paragraph FUNCTIONAL ACCEPTANCE TESTS. The test procedure must indicate in detail how tests are to be conducted. A statement of the tests that are to be performed without indicating how the tests are to be performed is not acceptable. Indicate what work is planned on each workday and identify the calendar dates of the planned workdays. Specify what additional technical support personnel is needed such as factory representatives for major equipment. Specify on which testing workday each technical support personnel is needed. Data recording forms to be used to document test results are to be submitted with the proposed test procedure. A list of test equipment and instruments must also be included in the test procedure.

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3.9.7 Test Equipment

Test equipment and instruments must be on hand prior to scheduling field tests or, subject to Contracting Officer approval, evidence must be provided to show that arrangements have been made to have the necessary equipment and instruments on-site prior to field testing.

3.10 FIELD QUALITY CONTROL

Give Contracting Officer 30 days' notice of dates and times scheduled for tests which require the presence of the Contracting Officer. The Contracting Officer will coordinate with the using activity and schedule a time that will eliminate or minimize interruptions and interference with the activity operations. The Contractor must be responsible for costs associated with conducting tests outside of normal working hours and with incorporating special arrangements and procedures, including temporary power conditions. The Contractor must provide labor, equipment, fuel, test load, and consumables required for the specified tests. The test load must be a cataloged product. Calibration of measuring devices and indicating devices must be certified. Refer to Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, for requirements for a cataloged product. Perform the following field tests.

3.10.1 Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.10.1.1 Circuit Breakers - Low Voltage Insulated Case/Molded Case

a. Visual and Mechanical Inspection

- (1) Compare nameplate data with specifications and approved shop drawings.
- (2) Inspect circuit breaker for correct mounting.
- (3) Operate circuit breaker to ensure smooth operation.
- (4) Inspect case for cracks or other defects.
- (5) Verify tightness of accessible bolted connections and cable connections by calibrated torque-wrench method. Thermo-graphic survey is not required.
- (6) Inspect mechanism contacts and arc chutes in unsealed units.

b. Electrical Tests

- (1) Perform contact-resistance tests.
- (2) Perform insulation-resistance tests.
- (3) Adjust breaker(s) for final settings in accordance with engine-generator set manufacturer's requirements.

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3.10.1.2 Current Transformers

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify correct connection.
- (4) Verify that adequate clearances exist between primary and secondary circuit.
- (5) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermo-graphic survey is not required.
- (6) Verify that all required grounding and shorting connections provide good contact.

b. Electrical Tests

- (1) Perform insulation-resistance tests.
- (2) Perform polarity tests.
- (3) Perform ratio-verification tests.

3.10.1.3 Metering and Instrumentation

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of electrical connections.

b. Electrical Tests

- (1) Determine accuracy of meters at 25, 50, 75, and 100 percent of full scale.
- (2) Calibrate watt-hour meters according to manufacturer's published data.
- (3) Verify all instrument multipliers.
- (4) Electrically confirm that current transformer secondary circuits are intact.

3.10.1.4 Battery Systems

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.

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- (2) Inspect physical and mechanical condition.
- (3) Verify tightness of accessible bolted electrical connections by calibrated torque-wrench method. Thermo-graphic survey is not required.
- (4) Measure electrolyte specific gravity and temperature and visually check fill level.
- (5) Verify adequacy of battery support racks, mounting, anchorage, and clearances.

b. Electrical Tests

- (1) Set charger float and equalizing voltage levels.
- (2) Verify all charger functions and alarms.
- (3) Measure each cell voltage and total battery voltage with charger energized and in float mode of operation.
- (4) Perform a capacity load test.

3.10.1.5 Engine-Generator Set

a. Visual and Mechanical Inspection

- (1) Compare equipment nameplate data with specifications and approved shop drawings.
- (2) Inspect physical and mechanical condition.
- (3) Inspect for correct anchorage and grounding.

b. Electrical and Mechanical Tests

- (1) Perform an insulation-resistance test on generator winding with respect to ground. Calculate polarization index.
- (2) Perform phase rotation test to determine compatibility with load requirements.

3.10.1.6 Grounding System

a. Visual and Mechanical Inspection

- (1) Inspect ground system for compliance with contract plans and specifications.

b. Electrical Tests

- (1) Perform ground-impedance measurements utilizing the fall-of-potential method defined in IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. Take measurements in normally dry weather, not less than 48 hours after rainfall. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

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3.10.2 Functional Acceptance Tests

The tests must be performed by the start-up engineer. Upon successful test completion, the start-up engineer must provide the Contracting Officer with a written test report within 15 calendar days showing the tests performed and the results of each test. The report must include the completed approved test data forms and certification from the start-up engineer that the test results fall within the manufacturer's recommended limits and meet the specified requirements performance. The report must be dated and signed by the start-up engineer, and submitted for approval by the Contracting Officer. The Contracting Officer will witness final acceptance tests. Testing must include, but not be limited to, the following:

- a. Verify proper functioning of each engine protective shutdown device and pre-shutdown alarm device. Testing of the devices must be accomplished by simulating device actuation and observing proper alarm and engine shutdown operation.
- b. Verify proper functioning of the engine over-speed trip device. Testing of the over-speed trip device must be accomplished by raising the speed of the engine-generator set until an over-speed trip is experienced.
- c. Verify proper functioning of the crank cycle/terminate relay. Testing of the relay must be accomplished by engaging the starter motor with the engine being prevented from running. Observe the complete crank/rest cycle as described in the paragraph STARTING SYSTEM.
- d. Verify proper functioning of the following automatic and manual operations. Testing must include, but not be limited to, the following:
 - (1) Loss of Utility: Initiate a normal power failure with connected test load of rated kW at 1.0 power factor. Record time delay on start, cranking time until engine starts and runs, time to come up to operating speed, voltage and frequency overshoot, and time to achieve steady state conditions with all switches transferred to emergency position.
 - (2) Return of Utility: Return normal power and record time delay on retransfer for each automatic transfer switch, and time delay on engine cool-down and shutdown.
 - (3) Manual starting.
 - (4) Emergency stop.
 - (5) Perform generation tests in all modes of microgrid operation.
- e. Operate the engine-generator set at rated current (amperes) until the jacket water temperature stabilizes. Stabilization will be considered to have occurred when three consecutive temperature readings remain unchanged. Continue to operate the generator set for an additional 2 hours. Record instrument readings for terminal voltage, line current, frequency (Hz), engine speed rpm, lubricating oil pressure, jacket water temperature, and ambient temperature at 5 minute intervals for first 15 minutes and at 15 minute intervals thereafter.

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3.11 DEMONSTRATION

Upon completion of the work and at a time approved by the Contracting Officer, the Contractor must provide instructions by a qualified instructor to the Government personnel in the proper operation and maintenance of the equipment. Government personnel must receive training comparable to the equipment manufacturer's factory training. The duration of instruction must be for not less than one 8 hour working day for instruction of operating personnel and not less than one 8 hour working day for instruction of maintenance personnel.

3.11.1 Instructor's Qualification Resume

Instructors must be regular employees of the engine-generator set manufacturer. The instruction personnel provided to satisfy the requirements above must be factory certified by the related equipment manufacturer to provide instruction services. Submit the name and qualification resume of instructor to the Contracting Officer for approval.

3.11.2 Training Plan

Submit training plan 30 calendar days prior to training sessions. Training plan must include scheduling, content, outline, and training material (handouts). Content must include, but not be limited to, the following:

3.11.2.1 Operating Personnel Training

This instruction includes operating the engine-generator set, auxiliary equipment including automatic transfer switches in all modes, and the use of all functions and features specified.

3.11.2.2 Maintenance Personnel Training

Training must include mechanical, hydraulic, electrical, and electronic instructions for the engine-generator set and auxiliary equipment including automatic transfer switches.

a. Mechanical Training: Must include at least the following:

- (1) A review of mechanical diagrams and drawings.
- (2) Component location and functions.
- (3) Troubleshooting procedures and techniques.
- (4) Repair procedures.
- (5) Assembly/disassembly procedures.
- (6) Adjustments (how, when, and where).
- (7) Preventive maintenance procedures.
- (8) Review of flow diagram.
- (9) Valve locations and function.

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- (10) Valve and hydraulic equipment adjustment and maintenance procedures.
- (11) Hydraulic system maintenance and servicing.
- (12) Lubrication points, type, and recommended procedures and frequency.
- b. Electrical and Electronic Maintenance Training: Must include at least the following:
 - (1) A review of electrical and electronic systems including wiring diagrams and drawings.
 - (2) Troubleshooting procedures for the machine and control systems.
 - (3) Electrical and electronic equipment servicing and care.
 - (4) Use of diagnostics to locate the causes of malfunction.
 - (5) Procedures for adjustments (locating components, adjustments to be made, values to be measured, and equipment required for making adjustments).
 - (6) Maintenance and troubleshooting procedures for microprocessor or minicomputer where applicable.
 - (7) Circuit board repair procedures where applicable (with schematics provided).
 - (8) Use of diagnostic tapes.
 - (9) Recommended maintenance servicing and repair for motors, switches, relays, solenoids, and other auxiliary equipment and devices.

3.12 ONSITE TRAINING

Conduct a training course for the operating staff as designated by the Contracting Officer. The training period must consist of a total 8 hours of normal working time and must start after the system is functionally completed but prior to final acceptance.

- a. Submit a letter giving the date proposed for conducting the onsite training course, the agenda of instruction, a description of the digital video recording to be provided. The course instructions must cover pertinent points involved in operating, starting, stopping, servicing the equipment, as well as major elements of the operation and maintenance manuals. Additionally, the course instructions must demonstrate routine maintenance procedures as described in the operation and maintenance manuals.
- b. Submit a digital video recording of the manufacturers operating and maintenance training course.
- c. One full size reproducible Mylar ach drawing must accompany the booklets. Mylars must be rolled and placed in a heavy cardboard tube with threaded caps on each end. The manual must include step-by-step procedures for system startup, operation, and shutdown; drawings, diagrams, and single-line schematics to illustrate and define the

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electrical, mechanical, and hydraulic systems together with their controls, alarms, and safety systems; the manufacturer's name, model number, and a description of equipment in the system. The instructions must include procedures for interface and interaction with related systems to include fire alarm/suppression systems, load shedding systems, power supplies, and microgrid controllers. Each booklet must include a CD containing an ASCII file of the procedures.

- d. Provide approved operation and maintenance manuals for the training course. Post approved instructions prior to the beginning date of the training course. Coordinate the training course schedule with the using service's work schedule, and submit for approval 14 days prior to beginning date of proposed beginning date of training.

3.13 INSTALLATION

Installation must conform to the applicable requirements of IEEE C2, NFPA 30, NFPA 37, and NFPA 70.

3.14 FINAL TESTING AND INSPECTION

- a. Start the engine, record the starting time, make and record all engine manufacturer's after-starting checks and inspections during a reasonable warm-up period.
- b. Increase the load in steps no greater than the Maximum Step Load Increase to 100 percent of Service Load, and operate the engine-generator set for at least 30 minutes. Measure the vibration at the end bearings (front and back of engine, outboard end of generator) in the horizontal, vertical, and axial directions. Verify that the vibration is within the same range as previous measurements and is within the required range.
- c. Remove load and shut down the engine-generator set after the recommended cool down period.
- d. Remove the lube oil filter and have the oil and filter examined by the engine manufacturer for excessive metal, abrasive foreign particles, etc. Verify any corrective action for effectiveness by running the engine for 8 hours at Service Load, then re-examine the oil and filter.
- e. Remove the fuel filter and examine the filter for trash, abrasive foreign particles, etc.
- f. Visually inspect and check engine and generator mounting bolts for tightness and visible damage.
- g. Replace air, oil, and fuel filters with new filters.

3.15 MANUFACTURER'S FIELD SERVICE

The engine generator-set manufacturer must furnish a qualified representative to supervise the installation of the engine generator-set, assist in the performance of the onsite tests, and instruct personnel as to the operational and maintenance features of the equipment.

3.16 POSTED DATA AND INSTRUCTIONS

Post Data and Instructions prior to field acceptance testing of the engine

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generator set. Provide two sets of instructions/data, typed and framed under weatherproof laminated plastic, and post side-by-side where directed. Include a one-line diagram, wiring and control diagrams and a complete layout of the system in the first set. Include the condensed operating instructions describing manufacturer's pre-start checklist and precautions; startup procedures for test-mode, manual-start mode, and automatic-start mode (as applicable); running checks, procedures, and precautions; and shutdown procedures, checks, and precautions in the second set. Submit posted data including wiring and control diagrams showing the key mechanical and electrical control elements, and a complete layout of the entire system.

- a. Include procedures for interrelated equipment (such as heat recovery systems, co-generation, load-shedding, and automatic transfer switches).
- b. Include a one-line diagram, wiring and control diagrams and a complete layout of the system in the first set. Include the condensed operating instructions describing manufacturer's pre-start checklist and precautions; startup procedures for test-mode, manual-start mode, and automatic-start mode (as applicable); running checks, procedures, and precautions; and shutdown procedures, checks, and precautions in the second set. Include procedures for interrelated equipment (such as heat recovery systems, co-generation, load-shedding, and automatic transfer switches).
- c. Submit instructions including: the manufacturers pre-start checklist and precautions; startup procedures for test-mode, manual-start mode, and automatic-start mode (as applicable); running checks, procedures, and precautions; and shutdown procedures, checks, and precautions. Include procedures for interrelated equipment (such as heat recovery systems, co-generation, load-shedding, and automatic transfer switches). Provide weatherproof instructions, laminated in plastic, and post where directed.

3.17 ACCEPTANCE

Submit drawings which accurately depict the as-built configuration of the installation, upon acceptance of the engine-generator set installation. Revise layout drawings to reflect the as-built conditions and submit them with the as-built drawings. Final acceptance of the engine-generator set will not be given until the Contractor has successfully completed all tests and all defects in installation material or operation have been corrected.

-- End of Section --

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SECTION 26 41 00

LIGHTNING PROTECTION SYSTEM 11/13

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

NFPA 780 (2023) Standard for the Installation of Lightning Protection Systems

UNDERWRITERS LABORATORIES (UL)

UL 96 (2016) UL Standard for Safety Lightning Protection Components

UL 467 (2022) UL Standard for Safety Grounding and Bonding Equipment

UL Electrical Construction (2012) Electrical Construction Equipment Directory

1.2 RELATED REQUIREMENTS

1.2.1 Verification of Dimensions

Confirm all details of work, verify all dimensions in field, and advise Contracting Officer of any discrepancy before performing work. Obtain prior approval of Contracting Officer before making any departures from the design.

1.2.2 System Requirements

Provide a system furnished under this specification consisting of the latest UL Listed products of a manufacturer regularly engaged in production of lightning protection system components. Comply with NFPA 70, NFPA 780, and UL 96.

1.2.3 Lightning Protection System Installers Documentation

Provide documentation showing that the installer is certified with a commercial third-party inspection company whose sole work is lightning

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protection, or is a UL Listed Lightning Protection Installer. In either case, the documentation must show that they have completed and passed the requirements for certification or listing, and have a minimum of 2 years documented experience installing lightning protection systems for DoD projects of similar scope and complexity.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Overall lightning protection system; G

Each major component; G

SD-06 Test Reports

Lightning Protection and Grounding System Test Plan; G

Lightning Protection and Grounding System Test; G

SD-07 Certificates

Lightning Protection System Installers Documentation; G

Component UL Listed and Labeled; G

Lightning protection system inspection certificate; G

Roof manufacturer's warranty; G

1.4 QUALITY ASSURANCE

In each standard referred to herein, consider the advisory provisions to be mandatory, as though the word "shall" or "must" has been substituted for "should" wherever it appears. Interpret references in these standards to "authority having jurisdiction," or words of similar meaning, to mean Contracting Officer.

1.4.1 Installation Drawings

1.4.1.1 Overall System Drawing

Submit installation shop drawing for the overall lightning protection system. Include on the drawings the physical layout of the equipment (plan view and elevations), mounting details, relationship to other parts of the work, and wiring diagrams.

1.4.1.2 Major Components

Submit detail drawings for each major component including manufacturer's descriptive and technical literature, catalog cuts, and installation instructions.

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1.4.2 Component UL Listed and Labeled

Submit proof of compliance that components are UL Listed and Labeled. Listing alone in UL Electrical Construction, which is the UL Electrical Construction Directory, is not acceptable evidence. In lieu of Listed and Labeled, submit written certificate from an approved, nationally recognized testing organization equipped to perform such services, stating that items have been tested and conform to requirements and testing methods of Underwriters Laboratories.

1.4.3 Lightning Protection and Grounding System Test Plan

Provide a lightning protection and grounding system test plan. Detail both the visual inspection and electrical testing of the system and components in the test plan. Identify (number) the system test points/locations along with a listing or description of the item to be tested and the type of test to be conducted. As a minimum, include a sketch of the facility and surrounding lightning protection system as part of the specific test plan for each structure. Include the requirements specified in paragraph, "Testing of Integral Lightning Protection System" in the test plan.

1.4.4 Lightning Protection System Inspection Certificate

Provide certification from a commercial third-party inspection company whose sole work is lightning protection, stating that the lightning protection system complies with NFPA 780. Third party inspection company cannot be the system installer or the system designer. Alternatively, provide a UL Lightning Protection Inspection Master Label Certificate for each facility indicating compliance to NFPA 780.

Inspection must cover every connection, air terminal, conductor, fastener, accessible grounding point and other components of the lightning protection system to ensure 100% system compliance. This includes witnessing the tests for the resistance measurements for ground rods with test wells, and for continuity measurements for bonds. It also includes verification of proper surge protective devices for power, data and telecommunication systems. Random sampling or partial inspection of a facility is not acceptable.

1.5 SITE CONDITIONS

Confirm all details of work, verify all dimensions in field, and advise Contracting Officer of any discrepancy before performing work. Obtain prior approval of Contracting Officer before changing the design.

PART 2 PRODUCTS

2.1 MATERIALS

Do not use a combination of materials that forms an electrolytic couple of such nature that corrosion is accelerated in the presence of moisture unless moisture is permanently excluded from the junction of such metals. Where unusual conditions exist which would cause corrosion of conductors, provide conductors with protective coatings, such as tin or lead, or oversize conductors. Where a mechanical hazard is involved, increase conductor size to compensate for the hazard or protect conductors. When metallic conduit or tubing is provided, electrically bond conductor to

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conduit or tubing at the upper and lower ends by clamp type connectors or welds (including exothermic). All lightning protection components, such as bonding plates, air terminals, air terminal supports and braces, chimney bands, clips, connector fittings, and fasteners are to comply with the requirements of UL 96 classes as applicable.

2.1.1 Main and Bonding Conductors

NFPA 780 and UL 96 Class I, Class II, or Class II modified materials as applicable.

2.1.2 Copper Only

Provide copper conductors, except where aluminum conductors are required for connection to aluminum equipment.

2.2 COMPONENTS

2.2.1 Air Terminals

Provide solid air terminals with a blunt tip. Tubular air terminals are not permitted. Support air terminals more than 24 inches in length by suitable brace, supported at not less than one-half the height of the terminal.

2.2.2 Ground Rods

Provide ground rods made of solid copper conforming to conform to UL 467. Provide ground rods that are not less than 3/4 inch in diameter and 10 feet in length. Do not mix ground rods of copper-clad steel or solid copper on the job.

2.2.3 Connections and Terminations

Provide connectors for splicing conductors that conform to UL 96, class as applicable. Conductor connections can be made by clamps or welds (including exothermic). Provide style and size connectors required for the installation.

2.2.4 Connector Fittings

Provide connector fittings for "end-to-end", "Tee", or "Y" splices that conform to NFPA 780 and UL 96.

PART 3 EXECUTION

3.1 INTEGRAL SYSTEM

Provide a lightning protection system that meets the requirements of NFPA 780, including tie-ins to existing lightning protection systems. Lightning protection system consists of air terminals, roof conductors, down conductors, ground connections, grounding electrodes and ground ring electrode conductor. Expose conductors on the structures except where conductors are required to be in protective sleeves. Bond secondary conductors with grounded metallic parts within the building. Make interconnections within side-flash distances at or below the level of the grounded metallic parts.

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3.1.1 Roof-Mounted Components

Coordinate with the roofing manufacturer and provide certification that the roof manufacturer's warranty is not violated by the installation methods for air terminals and roof conductors.

3.1.1.1 Air Terminals

Use a standing seam base for installation of air terminals on a standing seam metal roof that does not produce any roof penetrations.

3.1.2 Down Conductors

Protect exposed down conductors from physical damage as required by NFPA 780. Use Schedule 80 PVC to protect down conductors. Paint the Schedule 80 PVC to match the surrounding surface with paint that is approved for use on PVC.

3.1.3 Ground Connections

Attach each down conductor and ground ring electrode to ground rods by welding (including exothermic), brazing, or compression. All connections to ground rods below ground level must be by exothermic weld connection or with a high compression connection using a hydraulic or electric compression tool to provide the correct circumferential pressure. Accessible connections above ground level and in test wells can be accomplished by mechanical clamping.

3.1.4 Grounding Electrodes

Extend driven ground rods vertically into the existing undisturbed earth for a distance of not less 10 feet. Set ground rods not less than 3 feet nor more than 8 feet, from the structure foundation, and at least beyond the drip line for the facility. After the completed installation, measure the total resistance to ground using the fall-of-potential method described in IEEE 81. Maximum allowed resistance of a driven ground rod is 25 ohms, under normally dry conditions. Contact the Contracting Officer for direction on how to proceed when two of any three ground rods, driven not less than 10 feet into the ground, a minimum of 10 feet apart, and equally spaced around the perimeter, give a combined value exceeding 50 ohms immediately after having driven. For ground ring electrode, provide continuous No. 1/0 bare stranded copper cable. Lay ground ring electrode around the perimeter of the structure in a trench not less than 3 feet nor more than 8 feet from the nearest point of the structure foundation, and at least beyond the drip line for the facility. Install ground ring electrode to a minimum depth of 30 inches. Install a ground ring electrode in earth undisturbed by excavation, not earth fill, and do not locate beneath roof overhang, or wholly under paved areas or roadways where rainfall cannot penetrate to keep soil moist in the vicinity of the cable.

3.2 APPLICATIONS

3.2.1 Nonmetallic Exterior Walls with Metallic Roof

Bond metal roof sections together which are insulated from each other so that they are electrically continuous, having a surface contact of at least 3 square inches.

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3.3 INTERFACE WITH OTHER STRUCTURES

3.3.1 Exterior Overhead Systems

Bond to the nearest down conductor as close to grade as possible. This includes overhead pipes, conduits, cable trays, or any other metallic objects on the exterior of the building that enter a building. In addition, bond pipes, conduits, and cable trays to any metallic objects (such as steel structural support of air handling units or cooling towers) that are within 6 feet.

3.4 RESTORATION

Where sod has been removed, place sod as soon as possible after completing the backfilling. Restore, to original condition, the areas disturbed by trenching, storing of dirt, cable laying, and other work. Overfill to accommodate for settling. Include necessary topsoil, fertilizing, liming, seeding, sodding, sprigging or mulching in any restoration. Maintain disturbed surfaces and replacements until final acceptance.

3.5 FIELD QUALITY CONTROL

3.5.1 Lightning Protection and Grounding System Test

Test the lightning protection and grounding system to ensure continuity is not in excess of 1 ohm and that resistance to ground is not in excess of 25 ohms. Provide documentation for the measured values at each test point. Test the ground rod for resistance to ground before making connections to the rod. Tie the grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Include in the written report: locations of test points, measured values for continuity and ground resistances, and soil conditions at the time that measurements were made. Submit results of each test to the Contracting Officer.

-- End of Section --

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SECTION 26 51 00

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A580/A580M	(2018) Standard Specification for Stainless Steel Wire
ASTM A641/A641M	(2019) Standard Specification for Zinc-Coated (Galvanized) Carbon Steel Wire
ASTM A653/A653M	(2022) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2021a) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B164	(2003; R 2014) Standard Specification for Nickel-Copper Alloy Rod, Bar, and Wire
ASTM B633	(2023) Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel

EUROPEAN UNION (EU)

Directive 2011/65/EU	(2011) Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment
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ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79	(2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products
ANSI/IES LM-80	(2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules
ANSI/IES LS-1	(2020) Lighting Science: Nomenclature and Definitions for Illuminating Engineering

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ANSI/IES TM-15	(2020) Technical Memorandum: Luminaire Classification System for Outdoor Luminaires
ANSI/IES TM-21	(2021) Technical Memorandum: Projecting Long-Term Luminous, Photon, and Radiant Flux Maintenance of LED Light Sources
ANSI/IES TM-30	(2020) Technical Memorandum: IES Method for Evaluating Light Source Color Rendition
IES Lighting Library	IES Lighting Library

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE C2	(2023) National Electrical Safety Code

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C78.54	(2019) Specification Sheet for Tubular Fluorescent Replacement and Retrofit LED Lamps
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ANSLG C78.377	(2017) Electric Lamps— Specifications for the Chromaticity of Solid State Lighting Products
NEMA C82.77-10	(2020) Harmonic Emission Limits - Related Power Quality Requirements
NEMA SSL 1	(2016) Electronic Drivers for LED Devices, Arrays, or Systems
NEMA SSL 3	(2011) High-Power White LED Binning for General Illumination

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2023) National Electrical Code
NFPA 101	(2021; TIA 21-1) Life Safety Code

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

47 CFR 15	Radio Frequency Devices
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UNDERWRITERS LABORATORIES (UL)

UL 844	(2012; Reprint Oct 2021) UL Standard for Safety Luminaires for Use in Hazardous (Classified) Locations
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UL 924	(2016; Reprint Dec 2022) UL Standard for Safety Emergency Lighting and Power Equipment
UL 1598	(2021; Reprint Jun 2021) Luminaires
UL 1993	(2017) Self-Ballasted Lamps and Lamp Adapters
UL 8750	(2015; Reprint Sep 2021) UL Standard for Safety Light Emitting Diode (LED) Equipment for Use in Lighting Products

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Luminaires and accessories that are mounted in exterior environments and not attached to the exterior of the building are specified in Section 26 56 00 EXTERIOR LIGHTING. Cybersecurity requirements are specified in Sections 25 05 11.01 CYBERSECURITY FOR MICROGRID FACILITY-RELATED CONTROL SYSTEMS, 25 05 11.02 CYBERSECURITY FOR EEDRS FACILITY-RELATED CONTROL SYSTEMS, 25 05 11.03 CYBERSECURITY FOR FLS FACILITY-RELATED CONTROL SYSTEMS, and 25 05 11.04 CYBERSECURITY FOR HVAC DDC FACILITY-RELATED CONTROL SYSTEMS.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings, must be as defined in IEEE 100 and ANSI/IES LS-1.
- b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined in ANSI/IES LM-80.
- c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.
- d. Total harmonic distortion (THD) is the root mean square (RMS) of all the harmonic components divided by the total fundamental current.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

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Luminaire Drawings; G

Occupancy/Vacancy Sensor Coverage Layout; G; S

Lighting Control System One-Line Diagram; G

SD-03 Product Data

Luminaires; G

Light Sources; G

LED Drivers; G

Luminaire Warranty; G

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Switches; G

Digital Switch Timers; G

Photosensors; G

Power Hook Luminaire Hangers; G

Exit Signs; G

Emergency Drivers; G

Linear LED Lamps; G

SD-05 Design Data

Luminaire Design Data; G

Photometric Plan; G

SD-06 Test Reports

ANSI/IES LM-79 Test Report; G

ANSI/IES LM-80 Test Report; G

ANSI/IES TM-21 Test Report; G

ANSI/IES TM-30 Test Report; G

Occupancy/Vacancy Sensor Verification Test; G

Photosensor Verification Test; G

SD-07 Certificates

LED Driver and Dimming Switch Compatibility Certificate; G

SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G

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Lighting Control System, Data Package 5; G

Maintenance Staff Training Plan; G

End-User Training Plan; G

1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the IES Lighting Library as applicable, for the lighting system specified.

1.5.1 Luminaire Drawings

Include dimensions, accessories installation details, and construction details. Photometric data, including CRI, CCT, LED driver type, zonal lumen data, and candlepower distribution data must accompany shop drawings.

1.5.2 Luminaire Design Data

- a. Provide safety certification and file number for the luminaire family that must be listed, labeled, or identified in accordance with the NFPA 70. Applicable testing bodies are determined by the US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).
- b. Provide long term lumen maintenance projections for each LED luminaire in accordance with ANSI/IES TM-21. Data used for projections must be obtained from testing in accordance with ANSI/IES LM-80.

1.5.3 ANSI/IES LM-79 Test Report

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data in IES format as outlined under "14.0 Test Report" in ANSI/IES LM-79.

1.5.4 ANSI/IES LM-80 Test Report

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in ANSI/IES LM-80.

1.5.5 ANSI/IES TM-21 Test Report

Submit test report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in ANSI/IES TM-21.

1.5.6 ANSI/IES TM-30 Test Report

Submit color vector graphic in accordance with ANSI/IES TM-30 on

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manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Include spectral distribution of test LED light source.

1.5.7 LED Driver and Dimming Switch Compatibility Certificate

Submit certification from the luminaire, driver, or dimmer switch manufacturer that ensures compatibility and operability between devices without flickering and to specified dimming levels.

1.5.8 Photometric Plan

1.5.8.1 Computer-generated Photometric Plans

Computer-generated photometric plans for each space are required to verify proposed luminaires and locations meet the required performance criteria of the design using the applicable light loss factor (LLF).

Target illumination levels are provided for each Interior Application. Depending on the application and the recommendations provided by the IES, values are given as one of the following:

- a. Minimum: No values anywhere on the calculation grid may be less than this value, within a 10 percent margin of error.
- b. Minimum Average: An average, taken over the entire task area for the application, may not be less than this value, within a 10 percent margin of error.
- c. Maximum: No values anywhere on the calculation grid may be greater than this value, within a 10 percent margin of error.
- d. Maximum Average: An average, taken over the entire task area for the application, may not be greater than this value, within a 10 percent margin of error.
- e. Uniformity: Unless otherwise noted, uniformity is calculated as a ratio of the average calculated illuminance over the minimum calculated illuminance of the calculation grid.

1.5.8.2 Schematic Photometric Plan Calculations

Schematic photometric plan calculations must include:

- a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every one foot across the task area.
- b. Average maintained illuminance level.
- c. Minimum and maximum maintained illuminance levels.
- d. Lighting power density (Watts per square foot).
- e. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

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1.5.8.3 Final Photometric Plan Calculations

Final photometric plan calculations must include:

- a. Horizontal illuminance measurements at workplane or other designated height above finished floor, taken at a maximum of every one foot across the task area.
- b. Where applicable, vertical illuminance measurements at designated surface, taken at a maximum of every one foot across task area.
- c. Minimum and maximum maintained illuminance levels.
- d. Average maintained illuminance level.
- e. Average to minimum and maximum to minimum ratios for horizontal illuminance.
- f. Lighting power density (Watts per square foot).
- g. LLF. Recommended LLF is 0.81 for LED luminaires but LLF varies based on environment and application.

1.5.9 Occupancy/Vacancy Sensor Coverage Layout

Provide floor plans showing coverage layouts of all devices using manufacturer's product information.

1.5.10 Test Laboratories

Test laboratories for the ANSI/IES LM-79 and ANSI/IES LM-80 test reports must be one of the following:

- a. National Voluntary Laboratory Accreditation Program (NVLAP) accredited for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program for both LM-79 and LM-80 testing.
- b. One of the qualified labs listed on the Department of Energy - LED Lighting Facts Approved Testing Laboratories List for LM-79 testing.
- c. One of the EPA-Recognized Laboratories listed for LM-80 testing.

1.5.11 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of NFPA 70, unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

1.5.12 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design, and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have

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been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.12.1 Alternative Qualifications

Products having less than a six-month field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.12.2 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

- a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return authorization number is required, return shipping information, and closest return location to the luminaire location.
 - (1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.
 - (2) Material warranty must include:
 - (a) All LED drivers and integral control equipment.
 - (b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.
- b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.
- c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and assembly.

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1.6.2 Lighting Controls Warranty

Provide and transfer to the government the original lighting controls manufacturers standard commercial warranty for each different lighting controls manufacturer used in the project. Warranty coverage must begin from date of final system commissioning or three months from date of delivery, whichever is the earliest. Warranty service must be performed by a factory-trained engineer or technician.

- a. Unless otherwise noted, provide a written five year minimum warranty on the complete system for all systems with factory commissioning. Provide warranty that covers 100 percent of the cost of any replacement parts and services required over the five years which are directly attributable to the product failure. Failures include, but are not limited to, the following:
 - (1) Software: Failure of input/output to execute switching or dimming commands.
 - (2) Damage of electronic components due to transient voltage surges.
 - (3) Failure of control devices, including but not limited to occupancy sensors, photosensors, and manual wall station control devices.
- b. Provide a written five year minimum warranty on all input devices against defect in workmanship or materials provided by device manufacturer.
- c. Provide a written five year minimum warranty on all control components attached to luminaires against defect in workmanship or materials.

1.7 OPERATION AND MAINTENANCE MANUALS

1.7.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting system for the building. Include the following:

- a. Manufacturers' operating and maintenance manuals.
- b. Luminaire shop drawings for modified and custom luminaires.
- c. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

1.7.2 Lighting Control System

Provide operation and maintenance manuals for the lighting control system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting control system for the building. Include the following:

- a. Lighting control system layout and wiring plan.
- b. Lighting control system one-line diagram.
- c. Product data for all devices, including installation and programming

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instructions.

- d. Occupancy/vacancy sensor coverage layout.
- e. Training materials, such as videos or in-depth manuals, that cover basic operation of the lighting control system and instructions on modifying the lighting control system. Training materials must include calibration, adjustment, troubleshooting, maintenance, repair, and replacement.
- f. Sequence of operation descriptions for each typical room type, including final programming, schedules, and calibration settings.
- g. "As-built" lighting control panel schedules.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and NL plates or details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the driver and light source provided.

2.2.1 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

- a. Complete system with LED drivers and light sources.
- b. Housings constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.
- c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

Luminaire Style	Minimum Luminaire Efficacy
Recessed 1 by 4, 2 by 4, and 2 by 2	100 LPW
Recessed Downlight (fixed, adjustable, wallwash)	80 LPW
Linear, Accent (undercabinet, cove)	45 LPW
Linear, Ambient (indirect wall mount, linear pendent)	100 LPW

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Luminaire Style	Minimum Luminaire Efficacy
High Bay, Low Bay, and Industrial Locations	100 LPW
Food Service and Hazardous Locations	60 LPW
Other (track, residential diffusers)	50 LPW
Exterior Wall Sconce	50 LPW
Steplight	30 LPW
Parking Garage Luminaire	100 LPW

- e. UL listed for dry or damp location typical of interior installations. Any luminaire mounted on the exterior of the building must be UL listed for wet location typical of exterior installations.
- f. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.
- g. Lenses constructed of heat tempered borosilicate glass, UV-resistant acrylic, or silicone. Provide polycarbonate vandal-resistant lenses as indicated. Sandblasting, etching and polishing must be performed as indicated in the luminaire description.
- h. ANSI/IES TM-15. Provide exterior building-mounted luminaires that do not exceed the BUG ratings as listed in the luminaire schedule. If BUG ratings are not listed in the luminaire schedule, provide luminaires that meet the following minimum values for each application and mounting conditions:

Lighting Application	Mounting Conditions	BUG Rating
Exterior Wall Sconce	Above 4 feet AFF	B1-U0-G2
Exterior Wall Sconce	Below or at 4 feet AFF	B4-U0-G4
Steplight	Above 4 feet AFF	B1-U1-G2
Steplight	Below or at 4 feet AFF	B4-U1-G4
Parking Garage Luminaire	Ceiling mounted	B4-U4-G3

2.2.2 Luminaires for Hazardous Locations

In addition to requirements stated herein, provide LED luminaires for hazardous locations which conform to UL 844 or which have Factory Mutual certification for the class and division indicated.

2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, delivered lumen output, and wattage as indicated in the luminaire schedule on project plans.

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2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

- a. NEMA ANSLG C78.377. Emit white light and have a nominal CCT of 4000 Kelvin.
- b. Minimum Color Rendering Index (CRI) of 80 Fidelity index greater than or equal to 80, gamut index between 97 and 110, determined in accordance with ANSI/IES TM-30.
- c. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- d. Light source color consistency by utilizing a binning tolerance within a 3-step McAdam ellipse.

2.3.1.1 Linear LED Lamps

Provide linear LED Lamps that are compatible with existing instant-start or programmed-start ballasts, and meet the following additional requirements:

- a. UL 1993 UL Type A linear LED lamp.
- b. Power Factor greater than or equal to 0.90 at full input power and across specified dimming range.
- c. Maximum Total Harmonic Distortion (THD) less than or equal to 20 percent at full input power and across specified dimming range.
- d. Lumen per watt efficacy no less than 120.
- e. Minimum beam angle of 180 degrees.
- f. Lamp datasheet complies with ANSI C78.54. Manufacturer must provide list of all ballasts that are compatible for use with lamp.

2.4 LED DRIVERS

NEMA SSL 1, UL 8750. Provide LED drivers that are electronic, UL Class 1 or Class 2, constant-current type and that comply with the following requirements:

- a. The combined driver and LED light source system does not exceed the minimum luminaire efficacy values as listed in the luminaire schedule provided.
- b. Operates at a voltage of 120-277 volts at 50/60 hertz, with input voltage fluctuations of plus/minus 10 percent.
- c. Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.
- d. Maximum Total Harmonic Distortion (THD) less than 20 percent at full input power and across specified dimming range.
- e. Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.

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- f. Withstands Category A surges of 2 kV without impairment of performance. Provide surge protection that is integral to the driver.
- g. Integral thermal protection that reduces the output power to protect the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.
- h. 47 CFR 15. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- i. Class A sound rating.
- j. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.

2.5 LIGHTING CONTROLS

Provide network certification for all networked lighting control systems and devices in accordance with the requirements of Sections 25 05 11.01 CYBERSECURITY FOR MICROGRID FACILITY-RELATED CONTROL SYSTEMS, 25 05 11.02 CYBERSECURITY FOR EEDRS FACILITY-RELATED CONTROL SYSTEMS, 25 05 11.03 CYBERSECURITY FOR FLS FACILITY-RELATED CONTROL SYSTEMS, and 25 05 11.04 CYBERSECURITY FOR HVAC DDC FACILITY-RELATED CONTROL SYSTEMS. Provide lighting control systems that do not switch off battery-operated or emergency backup luminaires or exit signs in path of egress. Provide system with override of lighting control devices controlling luminaires in path of egress with activation of fire alarm system.

2.5.1 System

2.5.1.1 Localized Control Systems

Provide room or area-wide lighting control system capable of manual control, time-based control, and receiving input from photosensors and occupancy/vacancy sensors.

2.5.2 Devices

2.5.2.1 Switches

Provide line-voltage toggle switches as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. When used for non-digital loads, devices must be rated at 20 Amps inductive load, and be compatible with the lighting control systems.

2.5.2.2 Digital Switch Timers

Provide line-voltage toggle switches that allow manual control to ON and automatically switches lighting load to OFF. Device operates with the use of paddle, button, or toggle, and operates at 120-277 volts. Device allows for programming of auto off timer from 10 minutes to 12 hours.

2.5.2.3 Photosensors

Provide photosensors that meet the following requirements:

- a. Detect changes in ambient lighting level and enable dimming as

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required by sequence of operation by operating in a closed-loop system.

- b. Contain a detection cone, where the base of the cone may be circular or an elongated shape, and where the smallest angle between the edge and the axis of the cone is between 20 and 50 degrees. The cone axis may be tilted to the vertical when installed to give the sensor preferred directionality.
- c. Sensors are wall-mounted with sensitivity, filtering, range and viewing angle to meet requirements of sequence of operation, scope of work and construction documents.
- d. Time delay that is adjustable from 1 to 30 seconds ON delay, and 1 to 30 minutes OFF delay to prevent cycling, with deadband adjustment of 25 percent to 100 percent above lower setpoint.
- e. Output dimming signal is linear to light level with less than 1 percent variation. Cadmium sulfide photo-resistors are not acceptable.
- f. Sensor is not combined in the same housing or location with occupancy or vacancy sensors if the proper location for one function compromises the successful operation of the other function, or in any way reduces the system's ability to meet the design intent.

2.6 EXIT AND EMERGENCY LIGHTING EQUIPMENT

2.6.1 Exit Signs

UL 924, NFPA 101. Provide wattage as indicated in the luminaire schedule on project plans. Provide LED Exit Signs that meet the following criteria:

- a. Housing constructed of UV-stable, thermo-plastic finish.
- b. UL listed for damp location.
- c. Configured for universal mounting.
- d. 6 inch high, 3/4 inch stroke red lettering on face of sign with chevrons on either side of lettering to indicate direction.
- e. Single or double face as indicated in project plans and luminaire schedule.

2.6.1.1 Exit Signs with Battery Backup

Equip with automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 1-1/2 hours. LEDs must have a minimum rated life of 10 years. Provide self-diagnostic circuitry integral to emergency LED driver. In lieu of battery, can use a nonradioactive photoluminescent plate.

2.6.2 Emergency Lighting Unit (ELU)

UL 924, NFPA 101. Provide emergency lighting units (ELUs) completely assembled with wiring and mounting devices, ready for installation at the locations indicated. Provide in UV-stable, thermo-plastic housing with UL

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damp label as indicated. Emergency lighting units must be rated for 12 volts, except units having no remote-mounted light sources and having no more than two unit-mounted light sources may be rated six volts. Equip units with brown-out sensitive circuit to activate battery when input voltage falls to 75 percent of normal. Equip with two LED light sources, automatic power failure device, test switch, and pilot light, and fully automatic high/low trickle charger in a self-contained power pack. Battery must be sealed, maintenance free nickel-cadmium type, and must operate unattended for a period of not less than five years. Emergency run time must be a minimum of 90 minutes. LEDs must have a minimum rated life of 10 years. Provide self-diagnostic circuitry integral to emergency LED driver.

2.6.3 LED Emergency Drivers

UL 924, NFPA 101. Provide LED emergency driver with automatic power failure detection, test switch and LED indicator (or combination switch/indicator) located on luminaire exterior, and fully-automatic solid-state charger, battery and inverter integral to a self-contained housing. Provide self-diagnostic function integral to emergency driver. Integral nickel-cadmium battery is required to supply a minimum of 90 minutes of emergency power at 10 watts, 10-50 VDC, constant output. Driver must be RoHS compliant, rated for installation in plenum-rated spaces and damp locations, and be warranted for a minimum of five years.

2.6.4 Self-Diagnostic Circuitry for LED Drivers

UL 924, NFPA 101. Provide emergency lighting unit with fully-automatic, integral self-testing/diagnostic electronic circuitry. Circuitry must provide for a one minute diagnostic test every 28 days, and a 30 minute diagnostic test every six months, minimum. Any malfunction of the unit must be indicated by LED(s) visible from the exterior of the luminaire. A manual test switch must also be provided to perform a diagnostic test at any given time.

2.7 LUMINAIRE MOUNTING ACCESSORIES

2.7.1 Suspended luminaires

- a. Provide hangers capable of supporting twice the combined weight of luminaires supported by hangers.
- b. Hangers must allow luminaires to swing within an angle of 45 degrees. Brace pendants 4 feet or longer to limit swinging.
- c. Single-unit suspended luminaires must have cable hangers. Multiple-unit or continuous row luminaires with a separate power supply cord must have a tubing or stem for wiring at one point and a tubing or rod suspension provided for each unit length of chassis, including one at each end.
- d. Provide all linear pendent and surface mounted luminaires with two supports per four-foot section or three per eight-foot section unless otherwise recommended by manufacturer.
- e. Provide rods in minimum 0.18 inch diameter.

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2.7.2 Recess and Surface Mounted Luminaires

Provide access to light source and LED driver from bottom of luminaire. Provide trim and lenses for the exposed surface of flush-mounted luminaires as indicated on project drawings and specifications. Luminaires recessed in ceilings which have a fire resistive rating of one hour or more must be enclosed in a box which has a fire resistive rating equal to that of the ceiling. For surface mounted luminaires with brackets, provide flanged metal stem attached to outlet box, with threaded end suitable for supporting the luminaire rigidly in design position. Flanged part of luminaire stud must be of broad base type, secured to outlet box at not fewer than three points.

2.7.3 Luminaire Support Hardware

2.7.3.1 Wire

ASTM A641/A641M. Galvanized, soft tempered steel, minimum 0.11 inches in diameter, or galvanized, braided steel, minimum 0.08 inches in diameter.

2.7.3.2 Wire for Humid Spaces

ASTM A580/A580M. Composition 302 or 304, annealed stainless steel, minimum 0.11 inches in diameter.

ASTM B164. UNS N04400, annealed nickel-copper alloy, minimum 0.11 inches in diameter.

2.7.3.3 Threaded Rods

Threaded steel rods, 3/16 inch diameter, zinc or cadmium coated.

2.7.3.4 Straps

Galvanized steel, one by 3/16 inch, conforming to ASTM A653/A653M, with a light commercial zinc coating or ASTM A1008/A1008M with an electrodeposited zinc coating conforming to ASTM B633, Type RS.

2.7.4 Power Hook Luminaire Hangers

UL 1598. Provide an assembly consisting of through-wired power hook housing, interlocking plug and receptacle, power cord, and luminaire support loop. Power hook housing must be cast aluminum having two 3/4 inch threaded hubs. Support hook must have safety screw. Luminaire support loop must be cast aluminum with provisions for accepting 3/4 inch threaded stems. Power cord must include 16 inches of 3 conductor No. 16 Type SO cord. Assembly must be rated 120 volts or 277 volts, 15 amperes.

2.8 EQUIPMENT IDENTIFICATION

2.8.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

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2.8.2 Labels

UL 1598. All luminaires must be clearly marked for operation of specific light sources and LED drivers. The labels must be easy to read when standing next to the equipment, and durable to match the life of the equipment to which they are attached. Note the following light source characteristics in the format "Use Only _____":

- a. Correlated Color Temperature (CCT) and Color Rendering Index (CRI) for all luminaires.
- b. Driver and dimming protocol.

All markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

2.9 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum, meets requirements of corrosion-resistance testing.

PART 3 EXECUTION

3.1 INSTALLATION

IEEE C2, NFPA 70.

3.1.1 Light Sources

When light sources are not provided as an integral part of the luminaire, deliver light sources of the type, wattage, lumen output, color temperature (CCT), color rendering index (CRI), and voltage rating indicated to the project site and install just prior to project completion, if not already installed in the luminaires from the factory.

3.1.2 Luminaires

Set luminaires plumb, square, and level with ceiling and walls, in alignment with adjacent luminaires and secure in accordance with manufacturers' directions and approved drawings. Provide accessories as required for ceiling construction type indicated on Finish Schedule. Luminaire catalog numbers do not necessarily denote specific mounting accessories for type of ceiling in which a luminaire may be installed. Provide wires, straps, or rods for luminaire support in this section. Install luminaires with vent holes free of air blocking obstacles.

3.1.2.1 Suspended Luminaires

Measure mounting heights from the bottom of the luminaire for ceiling-mounted luminaires and to center of luminaire for wall-mounted luminaires. Obtain architect approval of the exact mounting height on the job before commencing installation and, where applicable, after coordinating with the type, style, and pattern of the ceiling being installed. Support suspended luminaires from structural framework of ceiling or from inserts cast into slab.

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- a. Provide suspended luminaires with 45 degree swivel hangers so that they hang plumb and level.
- b. Locate so that there are no obstructions within the 45 degree range in all directions.
- c. The stem, canopy and luminaire must be capable of 45 degree swing.
- d. Rigid pendent stem, aircraft cable, rods, or chains 4 feet or longer excluding luminaire must be braced to prevent swaying using three cables at 120 degree separation.
- e. Suspended luminaires in continuous rows must have internal wireway systems for end to end wiring and must be properly aligned to provide a straight and continuous row without bends, gaps, light leaks or filler pieces.
- f. Utilize aligning splines on extruded aluminum luminaires to assure minimal hairline joints.
- g. Support steel luminaires to prevent "oil-canning" effects.
- h. Match supporting pendants with supported luminaire. Aircraft cable must be stainless steel.
- i. Match finish of canopies to match the ceiling, and provide low profile canopies unless otherwise shown.
- j. Maximum distance between suspension points must be 10 feet or as recommended by the manufacturer, whichever is less.

3.1.2.2 Recessed and Semi-Recessed Luminaires

- a. Support recessed and semi-recessed luminaires independently from the building structure by a minimum of two wires, straps or rods per luminaire and located near opposite corners of the luminaire. Secure horizontal movement with clips provided by manufacturer. Ceiling grid clips are not allowed as an alternative to independently supported luminaires.
- b. Support round luminaires or luminaires smaller in size than the ceiling grid independently from the building structure by a minimum of four wires, straps or rods per luminaire, spaced approximately equidistant around.
- c. Do not support luminaires by acoustical tile ceiling panels.
- d. Where luminaires of sizes less than the ceiling grid are indicated to be centered in the acoustical panel, support each independently and provide at least two 3/4 inch metal channels spanning, and secured to, the ceiling tees for centering and aligning the luminaire.
- e. Adjust aperture rings on all applicable ceiling recessed luminaires to accommodate various ceiling material thickness. Coordinate cut-out size in ceiling to ensure aperture covers cut-out entirely. Install aperture rings such that the bottom of the ring is flush with finished ceiling or not more than 1/16 inch above. Do not install luminaires such that the aperture ring extends below the finished ceiling surface.

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3.1.3 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

3.1.4 Exit Signs

NFPA 101. Wire exit signs and emergency lighting units ahead of the local switch, to the normal lighting circuit located in the same room or area.

3.1.5 Lighting Controls

Refer to Sections 25 05 11.01 CYBERSECURITY FOR MICROGRID FACILITY-RELATED CONTROL SYSTEMS, 25 05 11.02 CYBERSECURITY FOR EEDRS FACILITY-RELATED CONTROL SYSTEMS, 25 05 11.03 CYBERSECURITY FOR FLS FACILITY-RELATED CONTROL SYSTEMS, and 25 05 11.04 CYBERSECURITY FOR HVAC DDC FACILITY-RELATED CONTROL SYSTEMS for additional lighting control installation requirements.

3.1.5.1 Photosensors

Locate and aim sensor as indicated and in accordance with the manufacturer's recommendations. Adjust sensor set-point in accordance with the manufacturer's recommendations and for the indicated light level of the area of coverage, measured at the work plane.

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

3.2.1.1 Lighting Control Verification Tests

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

- a. Verify occupancy/vacancy sensors operate as described in sequence of operations. Provide testing of sensor coverage, sensitivity, and time-out settings in all spaces where sensors are placed. This is to be completed only after all furnishings have been installed. Submit occupancy/vacancy sensor verification test.
- b. Verify photosensors operate as described in sequence of operations. Provide testing of sensor coverage, aiming, and calibration in all spaces where sensors are placed. This is to be completed only after all furnishings have been installed. Submit photosensor verification test.

3.2.1.2 Emergency Lighting Test

Interrupt power supply to demonstrate proper operation of emergency lighting. If adjustments are made to the lighting system, re-test system to show compliance with standards.

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3.3 CLOSEOUT ACTIVITIES

3.3.1 Commissioning

NFPA 101. Commission all components of the lighting system and lighting control system in accordance with Section 01 46 00.00 07 COMMISSIONING - MICROGRID.

3.3.2 Training

3.3.2.1 Maintenance Staff Training

Submit a Maintenance Staff Training Plan at least 30 calendar days prior to training session that describes training procedures for Owner's personnel in the operation and maintenance of lighting and lighting control system. Provide on-site training which demonstrates full system functionality, assigning schedules, calibration adjustments for light levels and sensor sensitivity, integration procedures for connecting to third-party devices, and manual override including information on appropriate use. Provide protocols for troubleshooting, maintenance, repair, and replacement, and literature on available system updates and process for implementing updates.

3.3.2.2 End-User Training

Submit an End-User Training Plan at least 30 calendar days prior to training session that describes training procedures for end-users on the lighting control system. Provide users with a list of control devices located within user-occupied spaces, such as photosensors and occupancy and vacancy sensors, including information on the proper operation and schedule for each device. Provide demonstration for each type of interface. Provide users with the building schedule as currently commissioned, including conditional programming based on astronomic time clock functionality. Provide users with the correct contact information for maintenance personnel who will be available to address any lighting control issues.

Provide laminated instructions to the user at each scene wallstation. Provide only instructions relevant to the functionality of the specific scene wallstation. Provide a description of each labeled scene control button. If the room utilizes occupancy/vacancy sensors or photosensors, include a description of this functionality on the instruction sheet.

-- End of Section --

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SECTION 26 56 00

EXTERIOR LIGHTING 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B117	(2019) Standard Practice for Operating Salt Spray (Fog) Apparatus
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EUROPEAN UNION (EU)

Directive 2011/65/EU	(2011) Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment
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ILLUMINATING ENGINEERING SOCIETY (IES)

ANSI/IES LM-79	(2019) Approved Method: Electrical and Photometric Measurements of Solid State Lighting Products
ANSI/IES LM-80	(2020) Approved Method: Measuring Luminous Flux and Color Maintenance of LED Packages, Arrays and Modules
ANSI/IES LS-1	(2020) Lighting Science: Nomenclature and Definitions for Illuminating Engineering
ANSI/IES RP-8	(2018; Addenda 1 2020; Errata 1-2 2021) Recommended Practice for Design and Maintenance of Roadway and Parking Facility Lighting
ANSI/IES TM-15	(2020) Technical Memorandum: Luminaire Classification System for Outdoor Luminaires
ANSI/IES TM-21	(2021) Technical Memorandum: Projecting Long-Term Luminous, Photon, and Radiant Flux Maintenance of LED Light Sources
IES Lighting Library	IES Lighting Library

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
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IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C136.3 (2020) Roadway and Area Lighting Equipment
- Luminaire Attachments

NEMA 250 (2020) Enclosures for Electrical Equipment
(1000 Volts Maximum)

NEMA ANSLG C78.377 (2017) Electric Lamps— Specifications for
the Chromaticity of Solid State Lighting
Products

NEMA C82.77-10 (2020) Harmonic Emission Limits - Related
Power Quality Requirements

NEMA C136.10 (2023) American National Standard for
Roadway and Area Lighting
Equipment-Locking-Type Photocontrol
Devices and Mating Receptacles--Physical
and Electrical Interchangeability and
Testing

NEMA C136.31 (2018) Roadway and Area Lighting Equipment
- Luminaire Vibration

NEMA ICS 2 (2000; R 2020) Industrial Control and
Systems Controllers, Contactors, and
Overload Relays Rated 600 V

NEMA ICS 6 (1993; R 2016) Industrial Control and
Systems: Enclosures

NEMA IEC 60529 (2004) Degrees of Protection Provided by
Enclosures (IP Code)

NEMA SSL 1 (2016) Electronic Drivers for LED Devices,
Arrays, or Systems

NEMA SSL 3 (2011) High-Power White LED Binning for
General Illumination

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 773 (2016; Reprint Jul 2020) UL Standard for
Safety Plug-In, Locking Type Photocontrols
for Use with Area Lighting

UL 773A (2016; Reprint Jun 2020) UL Standard for
Safety Nonindustrial Photoelectric
Switches for Lighting Control

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UL 1310	(2018; Reprint Jun 2022) UL Standard for Safety Class 2 Power Units
UL 1598	(2021; Reprint Jun 2021) Luminaires
UL 8750	(2015; Reprint Sep 2021) UL Standard for Safety Light Emitting Diode (LED) Equipment for Use in Lighting Products

1.2 RELATED REQUIREMENTS

Materials not considered to be luminaires, luminaire accessories, or lighting equipment are specified in Section(s) 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Luminaires and accessories installed in interior of buildings or attached to the exterior of a building are specified in Section 26 51 00 INTERIOR LIGHTING.

1.3 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications and on the drawings must be as defined in IEEE 100 and ANSI/IES LS-1.
- b. For LED luminaire light sources, "Useful Life" is the operating hours before reaching 70 percent of the initial rated lumen output (L70) with no catastrophic failures under normal operating conditions. This is also known as 70 percent "Rated Lumen Maintenance Life" as defined in ANSI/IES LM-80.
- c. For LED luminaires, "Luminaire Efficacy" (LE) is the appropriate measure of energy efficiency, measured in lumens/watt. This is gathered from LM-79 data for the luminaire, in which absolute photometry is used to measure the lumen output of the luminaire as one entity, not the source separately and then the source and housing together.
- d. Total Harmonic Distortion (THD) is the Root Mean Square (RMS) of all the harmonic components divided by the total fundamental current.
- e. The "Groundline Section" of wood poles is that portion of the pole between one foot above, and 2 feet below the groundline.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Luminaire Drawings; G-D0

Control System One-Line Diagram; G-D0

Sequence of Operation for Exterior Lighting Control System; G-D0

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SD-03 Product Data

Luminaires; G-D0
Light Sources; G-D0
LED Drivers; G-D0
Luminaire Warranty; G-D0
Lighting Controls Warranty; G-D0
Photosensors; G-D0
Time Clock; G-D0
Lighting Contactor; G-D0

SD-05 Design Data

Luminaire Design Data; G-D0

SD-06 Test Reports

ANSI/IES LM-79 Test Report; G-D0
ANSI/IES LM-80 Test Report; G-D0
ANSI/IES TM-21 Test Report; G-D0

SD-10 Operation and Maintenance Data

Lighting System, Data Package 5; G-D0
Exterior Lighting Control System, Data Package 5; G-D0
Maintenance Staff Training Plan; G-D0
End-User Training Plan; G-D0

1.5 QUALITY ASSURANCE

Data, drawings, and reports must employ the terminology, classifications and methods prescribed by the IES Lighting Library as applicable, for the lighting system specified.

1.5.1 Drawing Requirements

1.5.1.1 Luminaire Drawings

Include dimensions, accessories, and installation and construction details. Photometric data, including CRI, CCT, TM-15-11 BUG rating, LED driver type, aiming diagram, zonal lumen data, and candlepower distribution data per LM-79 must accompany shop drawings.

1.5.2 Luminaire Design Data

- a. Provide distribution data according to IES classification type as defined in IES Lighting Library and ANSI/IES RP-8.

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- b. B.U.G. rating for the installed position as defined by ANSI/IES TM-15 and shielding as defined by ANSI/IES RP-8.
- c. Provide safety certification and file number for the luminaire family. Include listing, labeling and identification in accordance with NFPA 70 (NEC). Applicable testing bodies are determined by the US Occupational Safety Health Administration (OSHA) as Nationally Recognized Testing Laboratories (NRTL) and include: CSA (Canadian Standards Association), ETL (Edison Testing Laboratory), and UL (Underwriters Laboratories).
- d. Provide long term lumen maintenance projections for each LED luminaire in accordance with ANSI/IES TM-21. Data used for projections must be obtained from testing in accordance with ANSI/IES LM-80.
- e. Provide wind loading calculations for luminaires mounted on poles. Weight and effective projected area (EPA) of luminaires and mounting brackets must not exceed maximum rating of pole as installed in particular wind zone area.

1.5.3 ANSI/IES LM-79 Test Report

Submit test report on manufacturer's standard production model of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "14.0 Test Report" in ANSI/IES LM-79.

1.5.4 ANSI/IES LM-80 Test Report

Submit report on manufacturer's standard production LED light source (package, array, or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data as outlined under "8.0 Test Report" in ANSI/IES LM-80.

1.5.5 ANSI/IES TM-21 Test Report

Submit test report on manufacturer's standard production LED light source (package, array or module) of specified luminaire. Testing must be performed at the same operating drive current as specified luminaire. Include all applicable and required data, as well as required interpolation information as outlined under "7.0 Report" in ANSI/IES TM-21.

1.5.6 Test Laboratories

Test laboratories for the ANSI/IES LM-79 and ANSI/IES LM-80 test reports must be one of the following:

- a. National Voluntary Laboratory Accreditation Program (NVLAP) accredited for solid-state lighting testing as part of the Energy-Efficient Lighting Products laboratory accreditation program.
- b. One of the qualified labs listed on the Department of Energy - Energy Efficiency & Renewable Energy, Solid-State Lighting web site.
- c. One of the EPA-Recognized Laboratories listed at for LM-80 testing.

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1.5.7 Regulatory Requirements

Equipment, materials, installation, and workmanship must be in accordance with the mandatory provisions of NFPA 70 unless more stringent requirements are specified or indicated. Provide luminaires and assembled components that are approved by and bear the label of UL for the applicable location and conditions unless otherwise specified.

1.5.8 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for six months prior to bid opening. The six-month period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the six-month period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.5.8.1 Alternative Qualifications

Products having less than a six-month field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.5.8.2 Material and Equipment Manufacturing Date

Do not use products manufactured more than six months prior to date of delivery to site, unless specified otherwise.

1.6 WARRANTY

Support all equipment items by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

1.6.1 Luminaire Warranty

Provide and transfer to the government the original LED luminaire manufacturers standard commercial warranty for each different luminaire manufacturer used in the project.

- a. Provide a written five year minimum replacement warranty for material, luminaire finish, and workmanship. Provide written warranty document that contains all warranty processing information needed, including customer service point of contact, whether or not a return authorization number is required, return shipping information, and closest return location to the luminaire location.

(1) Finish warranty must include failure and substantial deterioration such as blistering, cracking, peeling, chalking, or fading.

(2) Material warranty must include:

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(a) All LED drivers and integral control equipment.

(b) Replacement when more than 15 percent of LED sources in any lightbar or subassembly(s) are defective, non-starting, or operating below 70 percent of specified lumen output.

- b. Warranty period must begin in accordance with the manufacturer's standard warranty starting date.
- c. Provide replacements that are promptly shipped, without charge, to the using Government facility point of contact and that are identical to or an improvement upon the original equipment. All replacements must include testing of new components and installation.

1.6.2 Lighting Controls Warranty

Provide and transfer to the government the original lighting controls manufacturers standard commercial warranty for each different lighting controls manufacturer used in the project. Warranty coverage must begin from date of final system commissioning or three months from date of delivery, whichever is the earliest. Warranty service must be performed by a factory-trained engineer or technician.

- a. Unless otherwise noted, provide a written five year minimum warranty on the complete system for all systems with factory commissioning. Provide warranty that covers 100 percent of the cost of any replacement parts and services required over the five years which are directly attributable to the product failure. Failures include, but are not limited to, the following:
 - (1) Software: Failure of input/output to execute switching or dimming commands.
 - (2) Damage of electronic components due to transient voltage surges.
 - (3) Failure of control devices, including but not limited to photosensors and motion sensors.
- b. Provide a written five year minimum warranty on all input devices against defect in workmanship or materials provided by device manufacturer.
- c. Provide a written five year minimum warranty on all control components attached to luminaires against defect in workmanship or materials.

1.7 OPERATION AND MAINTENANCE MANUALS

1.7.1 Lighting System

Provide operation and maintenance manuals for the lighting system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the lighting system. Include the following:

- a. Manufacturers' operating and maintenance manuals.
- b. Luminaire shop drawings for modified and custom luminaires.

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- c. Luminaire Manufacturers' standard commercial warranty information as specified in paragraph LUMINAIRE WARRANTY.

1.7.2 Exterior Lighting Control System

Provide operation and maintenance manuals for the exterior lighting control system in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA that provide basic data relating to the design, operation, and maintenance of the exterior lighting control system. Include the following:

- a. Control System One-Line Diagram
- b. Product data for all devices, including installation and programming instructions.
- c. Training materials, such as videos or in-depth manuals, that cover basic operation of the lighting control system and instructions on modifying the control system. Training materials must include calibration, adjustment, troubleshooting, maintenance, repair, and replacement.

PART 2 PRODUCTS

2.1 PRODUCT COORDINATION

2.2 LUMINAIRES

UL 1598, NEMA C82.77-10. Provide luminaires as indicated in the luminaire schedule and XL plates or details on project plans, complete with light source, wattage, and lumen output indicated. All luminaires of the same type must be provided by the same manufacturer. Luminaires must be specifically designed for use with the LED driver and light source provided.

2.2.1 Luminaires

UL 8750, ANSI/IES LM-79, ANSI/IES LM-80. For all luminaires, provide:

- a. Complete system with LED drivers and light sources.
- b. Housing constructed of non-corrosive materials. All new aluminum housings must be anodized or powder-coated. All new steel housings must be treated to be corrosion resistant.
- c. ANSI/IES TM-21, ANSI/IES LM-80. Minimum L70 lumen maintenance value of 50,000 hours unless otherwise indicated in the luminaire schedule. Luminaire drive current value must be identical to that provided by test data for luminaire in question.
- d. Minimum efficacy as specified in the luminaire schedule. Theoretical models of initial lamp lumens per watt are not acceptable. If efficacy values are not listed in the luminaire schedule, provide luminaires that meet the following minimum values:

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Luminaire Style	Minimum Luminaire Efficacy
Area and Roadway (pole mounted, arm mounted)	119 LPW
Pedestrian Post-Top (pole mounted, arm mounted)	97 LPW
Bollard	45 LPW
Accent (adjustable landscape, sign lighting)	35 LPW
Linear Accent (facade, wallwash)	80 LPW
Exterior Wall Sconce	50 LPW
Steplight	30 LPW
Parking Garage Luminaire	113 LPW

- e. Product rated for operation within an ambient temperature range of minus 22 degrees F to 122 degrees F.
- f. UL listed for wet locations. Optical compartment for LED luminaires must be sealed and rated a minimum of IP65 per NEMA IEC 60529.
- g. IES Lighting Library. Light distribution and NEMA field angle classifications as indicated in luminaire schedule on project plans.
- h. Housing finish that is baked-on enamel, anodized, or baked-on powder coat paint. Finish must be capable of surviving ASTM B117 salt fog environment testing for 2500 hours minimum without blistering or peeling.
- i. LED driver and light source package, array, or module are accessible for service or replacement without removal or destruction of luminaire.
- j. ANSI/IES TM-15. Does not exceed the BUG ratings as listed in the luminaire schedule. If BUG ratings are not listed in the luminaire schedule, provide luminaires that meet the following minimum values for each application and mounting conditions:

Lighting Application	Mounting Conditions	BUG Rating
Area and Roadway	All	B3-U0-G3
Pedestrian Post-Top	All	B2-U1-G1
Exterior Wall Sconce	Above 4 feet AFF	B1-U0-G2
Exterior Wall Sconce	Below or at 4 feet AFF	B4-U0-G4
Steplight	Above 4 feet AFF	B1-U1-G2
Steplight	Below or at 4 feet AFF	B4-U1-G4

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Lighting Application	Mounting Conditions	BUG Rating
Parking Garage Luminaire	Ceiling mounted	B4-U4-G3

- k. Fully assembled and electrically tested prior to shipment from factory.
- l. Finish color is as indicated in the luminaire schedule or detail on the project plans.
- m. Lenses constructed of clear tempered glass or UV-resistant acrylic.
- n. All factory electrical connections are made using crimp, locking, or latching style connectors. Twist-style wire nuts are not acceptable.
- o. NEMA C136.31. Comply with 3G vibration testing.
- p. Luminaire arm bolts constructed from 304 stainless steel or zinc-plated steel.
- q. Incorporate modular electrical connections, and construct luminaires to allow replacement of all or any part of the optics, heat sinks, LED drivers, surge suppressors and other electrical components using only a simple tool, such as a manual or cordless electric screwdriver.
- r. ANSI C136.3. For all roadway and area luminaires, provide products with an integral tilt adjustment of plus or minus 5 degrees to allow the unit to be leveled.

2.3 LIGHT SOURCES

NEMA ANSLG C78.377, NEMA SSL 3. Provide type, lumen rating, and wattage as indicated in luminaire schedule on project plans.

2.3.1 LED Light Sources

Provide LED light sources that meet the following requirements:

- a. NEMA ANSLG C78.377. Emit white light and have a nominal Correlated Color Temperature (CCT) of 4000 Kelvin.
- b. Minimum Color Rendering Index (CRI) of 80.
- c. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- d. Light source color consistency by utilizing a binning tolerance within a 4-step McAdam ellipse.

2.4 LED DRIVERS

NEMA SSL 1, UL 1310. Provide LED Drivers that are electronic, UL Class 1 or Class 2, constant-current type and meet the following requirements:

- a. The combined LED driver and LED light source system is greater than or equal to the minimum luminaire efficacy values as listed in the luminaire schedule provided.
- b. Operate at a voltage of 120-277 volts at 50/60 hertz, with input

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voltage fluctuations of plus or minus 10 percent.

- c. Power Factor (PF) greater than or equal to 0.90 at full input power and across specified dimming range.
- d. Maximum Total Harmonic Distortion (THD) less than or equal to 20 percent at full input power and across specified dimming range.
- e. Operates for at least 50,000 hours at maximum case temperature and 90 percent non-condensing relative humidity.
- f. Meets the "Elevated" (10kV/10kA) requirements per IEEE C62.41.2 -2002. Manufacturer must indicate whether failure of the electrical immunity system can possibly result in disconnect of power to luminaire. Provide surge protection that is integral to the LED driver.
- g. Contains integral thermal protection that reduces the output power to protect the driver and light source from damage if the case temperature approaches or exceeds the driver's maximum operating temperature.
- h. Complies with the requirements of the Federal Communications Commission (FCC) rules and regulations, Title 47 CFR part 15, Non-Consumer (Class A) for EMI/RFI (conducted and radiated).
- i. Class A sound rating for all drivers mounted under a covered structure, such as a canopy, or where otherwise appropriate.
- j. Directive 2011/65/EU. Restriction of Hazardous Substances (RoHS) compliant.
- k. UL listed for wet locations typical of exterior installations.
- l. Rated to operate between ambient temperatures of minus 22 degrees F and 122 degrees F.

2.5 LIGHTING CONTROLS

Provide a control system interface within each luminaire that is compatible with the energy management or control system used by the utility department in charge of the project area for control of site lighting.

2.5.1 System

Provide exterior lighting control system that operates the exterior lighting system as described in the exterior lighting control strategies in the project plans. Submit Sequence of Operation for Exterior Lighting Control System describing the operation of the proposed exterior lighting control system and devices. Sequence of Operation must provide the strategies identified in the exterior lighting control strategies.

2.5.2 Devices

2.5.2.1 Time Clock

NEMA ICS 6. House time clock in a surface-mounted, lockable NEMA 3R enclosure constructed of painted steel or plastic polymer. Provide

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electronic type time clock that meets the following criteria:

- a. Astronomic programming function, providing a total of 56 on/off set points.
- b. 24 hour type digital clock display format.
- c. Power outage back-up for switch utilizing lithium battery which provides coverage for a minimum of three years.
- d. Capable of controlling a minimum of channels or loads.
- e. Contacts are rated for 30 amps at 120-277 VAC resistive load in a SPST normally open (NO) configuration.
- f. Contains function that allows automatic control to be skipped on certain selected days of the week manual bypass or remote override control daylight savings time automatic adjustability for photosensor input.

2.5.2.2 Photosensors

UL 773, UL 773A. Provide Photosensors that meet the following requirements:

- a. Hermetically sealed, silicon diode light sensor.
- b. Turns ON at 1 to 3 footcandles and turns OFF at 3 to 15 footcandles.
- c. Designed to fail to the ON position.
- d. Housing is constructed of polycarbonate, rated to operate within a temperature range of minus 40 to 158 degrees F.
- e. Time delay that prevents accidental switching from transient light sources.
- f. Directional lens in front of the cell to prevent fixed light sources from creating a turnoff condition.
- g. Designed for 20-year service to match life expectancy of long-life LED fixtures and exceed 15,000 operations at full load. Provide photosensors with zero-cross technology to withstand severe in-rush current and extend relay life.
- h. NEMA C136.10. Twist-lock receptacle type. Provide with solid brass prongs and voltage markings and color coding on exterior of housing.
- i. Provide photosensors with metal oxide varistor (MOV) type surge protection.

2.5.2.3 Lighting Contactor

NEMA ICS 2. Provide a electrically-held lighting contactor housed in a NEMA 1 enclosure conforming to NEMA ICS 6. Contactor must have 4 poles, configured as normally open (NO). Contacts must be rated 600 volts, 30 amperes for a resistive load. Coil operating voltage must be 120 volts. Contactor must have silver cadmium oxide double-break contacts and must require no arcing contacts. Provide contactor with hand-off-automatic

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selector switch. Provide contactor as specified above along with circuit breaker in integral NEMA 1 enclosure with flange-mounted handle to satisfy requirement for a "combination lighting contactor" when specified.

2.6 EQUIPMENT IDENTIFICATION

2.6.1 Manufacturer's Nameplate

Each item of equipment must have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.6.2 Labels

UL 1598. Luminaires must be clearly marked for operation of specific light sources and drivers according to proper light source type. Note the following luminaire characteristics in the format "Use Only _____":

- a. Correlated color temperature (CCT) and color rendering index (CRI) for all luminaires.
- b. Driver and dimming protocol.

Markings related to light source type must be clear and located to be readily visible to service personnel, but unseen from normal viewing angles when light sources are in place. LED drivers must have clear markings indicating dimming type and indicate proper terminals for the various outputs.

2.7 FACTORY APPLIED FINISH

NEMA 250. Provide all luminaires and lighting equipment with factory-applied painting system that as a minimum meets requirements of corrosion-resistance testing.

PART 3 EXECUTION

3.1 INSTALLATION

3.1.1 Luminaires

Install all luminaires in accordance with the luminaire manufacturer's written instructions. Install all luminaires at locations and heights as indicated on the project plans. Level all luminaires in accordance to manufacturer's written instructions.

3.1.2 LED Drivers

Provide LED drivers integral to luminaire as constructed by the manufacturer.

3.1.3 Lighting Controls

3.1.3.1 Photosensors

Aim photosensor according to manufacturer's recommendations.

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3.1.3.2 Motion Sensors

Locate sensors in accordance with the manufacturer's recommendation. Locate sensors to achieve coverage of areas as indicated on project plans. Coverage patterns must be derated as recommended by manufacturer based on mounting height of sensor and any obstructions such as trees. Do not use gross rated coverage in manufacturer's product literature.

3.1.4 Grounding

Ground noncurrent-carrying parts of equipment including metal poles, luminaires, mounting arms, brackets, and metallic enclosures as specified in Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Where copper grounding conductor is connected to a metal other than copper, provide specially treated or lined connectors suitable for this purpose.

3.2 FIELD QUALITY CONTROL

3.2.1 Tests

Upon completion of installation, verify that equipment is properly installed, connected, and adjusted. Perform initial operational test, consisting of the entire system energized for 72 consecutive hours without any failures of any kind occurring in the system. All circuits must test clear of faults, grounds, and open circuits.

3.2.1.1 Lighting Control Verification Test

Verify lighting control system and devices operate according to approved sequence of operations. Verification tests are to be completed after commissioning.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Training

Provide on-site training to the Owner's personnel in the operation and maintenance of lighting and lighting control system. Provide training that includes calibration, adjustment, troubleshooting, maintenance, repair, and replacement.

3.3.1.1 Maintenance Staff Training

Submit a Maintenance Staff Training Plan at least 30 calendar days prior to training session that describes training procedures for Owner's personnel in the operation and maintenance of lighting and lighting control system. Provide on-site training which demonstrate full system functionality, assigning schedules, calibration adjustments for light levels and sensor sensitivity, integration procedures for connecting to third-party devices, and manual override including information on appropriate use. Provide protocols for troubleshooting, maintenance, repair, and replacement, and literature on available system updates and process for implementing updates.

3.3.1.2 End-User Training

Submit a End-User Training Plan at least 30 calendar days prior to training session that describes training procedures for end-users on the lighting control system. Provide demonstration for each type of user

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interface. Provide users with the curfew schedule as currently commissioned, including conditional programming based on astronomic time clock functionality. Provide users with the correct contact information for maintenance personnel who will be available to address any lighting control issues.

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BUILDING TELECOMMUNICATIONS CABLING SYSTEM

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BUILDING TELECOMMUNICATIONS CABLING SYSTEM 10/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ARMY RESERVE NETWORK ENTERPRISE CENTER (ARNEC)

ARNEC	Army Reserve Network Enterprise Center Infrastructure Technical Criteria
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ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

ECIA EIA/ECA 310-E	(2005) Cabinets, Racks, Panels, and Associated Equipment
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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(ERTA 1-2 2017; TIA 17-1; TIA 17-2; TIA 17-3; TIA 17-4; TIA 17-5; TIA 17-6; TIA 17-7; TIA 17-8; TIA 17-9; TIA 17-10; TIA 17-11; TIA 17-12; TIA 17-13; TIA 17-14; TIA 17-15; TIA 17-16; TIA 17-17) National Electrical Code
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TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-21	(1988a; R 2012) FOTP-21 - Mating Durability of Fiber Optic Interconnecting Devices
TIA-455-46A	(1990) FOTP-46 Spectral Attenuation Measurement for Long-Length, Graded-Index Optical Fibers
TIA-455-78-B	(2002) FOTP-78 Optical Fibres - Part 1-40: Measurement Methods and Test Procedures - Attenuation
TIA-526-7	OFSTP-7 Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant
TIA-526-14	OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant
TIA-568.0	Generic Telecommunications Cabling for Customer Premises
TIA-568.1	(Commercial Building Telecommunications Cabling Standard

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TIA-568.2	Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	Optical Fiber Cabling Components Standard
TIA-568.4	Broadband Coaxial Cabling and Components Standard
TIA-569	Commercial Building Standard for Telecommunications Pathways and Spaces
TIA-606	Administration Standard for the Telecommunications Infrastructure
TIA-607	Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
TIA/EIA-455	(1998b) Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components
TIA/EIA-604-3	(2004b; R 2014) Fiber Optic Connector Intermateability Standard (FOCIS), Type SC and SC-APC, FOCIS-3

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-580-01	Telecommunications Interior Infrastructure Planning and Design
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UNDERWRITERS LABORATORIES (UL)

UL 50	(2015) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
UL 467	(2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment
UL 1286	(2008; Reprint Jan 2018) UL Standard for Safety Office Furnishings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications; G-D0

Telecommunications Space; G-D0

SD-03 Product Data

Submit manufacturer's name, trade name, place of manufacture, and

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catalog model or number. Include performance and characteristic curves; applicable federal, military, industry, and technical society publication references. Should manufacturer's data require supplemental information for clarification, submit the supplemental information as specified in paragraph "Regulatory Requirements." All product submissions must have the exact product that is intended to be used specifically highlighted or called out.

Patch Panels; G-D0

Telecommunications Outlet/Connector Assemblies; G-D0

Equipment Support Frame; G-D0

Cable Support; G-D0

Connector Blocks; G-D0

Grounding and Bonding Products; G-D0

T5 Drawings; G-D0

SD-06 Test Reports

Telecommunications Cabling Testing; G-D0

SD-07 Certificates

Telecommunications Qualifications; G-D0

Key Personnel Qualifications; G-D0

Manufacturer Qualifications; G-D0

Test Plan; G-D0

Certificate; G-D0

1.3 GENERAL DESIGN REQUIREMENTS

Telecommunication products, product quality, and product execution in accordance with ARNEC and UFC 3-580-01. If a difference arises between these specification or the drawings, or both, then adherence to these criteria documents will take precedence for such products.

1.4 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification are as defined in the ARNEC, TIA-568.0, TIA-568.1 TIA-568.2 TIA-568.3 TIA-568.4 TIA-569, TIA-606, TIA-607 and as stated.

1.4.1 Entrance Facility (EF) (Telecommunications)

An environmentally controlled centralized space that serves as an entrance to the building for both private and public network service cables (including wireless) including the entrance point at the building wall.

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1.4.2 Telecommunications Equipment Room (TER) (Telecommunications Equipment Room)

An environmentally controlled centralized space for telecommunications equipment that serves the occupants of a building. The space will also house all backbone cable terminations that serve all other telecommunications spaces on an Army Reserve site. Equipment housed therein is considered distinct from a telecommunications room because of the nature of its complexity. The room is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.4.3 Telecommunications Room (TR)

An environmentally controlled space for housing telecommunications equipment, cable, terminations, and cross-connects. The room is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.4.4 Main Telecommunications Room (MTR)

An environmentally controlled space for housing telecommunications equipment, cable, terminations, and cross-connects. This room will be utilized when a new building or buildings with more than one TR within the new building(s) is being constructed on an existing site where the site already has an existing TER in an existing building. This room will serve as space where all backbone cable from other TRs within the building will terminate. This will also be the room where outside plant conduit will terminate. The room is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.4.5 Telecommunications Enclosure(TE)

An environmentally controlled space for housing telecommunications equipment, cable, terminations, and cross-connects. This is a non-dedicated telecommunications space. This enclosure will only be utilized when there is no available opportunity to build out a dedicated telecommunications room. A lockable floor or wall mounted telecommunications cabinet will be required in the room and will house all of the telecommunications terminations and network equipment. The enclosure is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.4.6 Open Cable

Cabling that is not run in a raceway as defined by NFPA 70, ARNEC. This refers to cabling that is "open" to the space in which the cable has been installed and is therefore exposed to the environmental conditions associated with that space.

1.4.7 Open Office

A floor space division provided by furniture, moveable partitions, or other means instead of by building walls.

1.4.8 Pathway

A physical infrastructure used for the placement and routing of telecommunications cable.

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1.5 SYSTEM DESCRIPTION

The backbone cabling and pathway system includes intra-building and inter-building interconnecting cabling, pathway, and terminal hardware. The intra-building and inter-building backbone provides connectivity from all TRs on an Army Reserve site to the TER as required. Wire the backbone system in a star topology with the TER for the site at the center or hub of the star. Provide telecommunications pathway systems referenced as stated. This section is required to specify optical fiber termination patch panels and associated accessories (cabinet to mount patch panel in and termination hardware) for microgrid control system outside plant fiber optic cable.

1.6 QUALITY ASSURANCE

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items to be shown to ensure a coordinated installation. Wiring diagrams to identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings to indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Include the nameplate data, size, and capacity; and all applicable federal, military, industry, and technical society publication references.

1.6.1 Telecommunications

Provide drawings in accordance with ARNEC and TIA-606. Show the identifier for each termination and cable; depict final telecommunications installed wiring system infrastructure in accordance with TIA-606. Provide details required to prove that the distribution system will properly support connectivity from the EF and TER telecommunications, TRs, TEs, to the telecommunications work area outlets. Submit the following as a minimum:

- a. T1 - Layout of complete building per floor - Building Area/Serving Zone Boundaries, Backbone Systems, and Horizontal Pathways. Layout of complete building per floor. The drawing indicates location of building areas, serving zones, vertical backbone diagrams, telecommunications rooms, access points, pathways, grounding system, and other systems that need to be viewed from the complete building perspective.
- b. T2 - Serving Zones/Building Area Drawings - Drop Locations and Cable Identification (ID'S). Shows a building area or serving zone. These drawings show drop locations, telecommunications rooms, access points and detail call outs for common equipment rooms and other congested areas.
- c. T4 - Typical Detail Drawings - Faceplate Labeling. Detailed drawings of symbols and typical such as faceplate labeling, faceplate types, faceplate population installation procedures, detail racking, and raceways.
- d. Where this design conflicts with the Contract Drawings, request clarification and direction from the Contracting Officer prior to proceeding.

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1.6.2 Telecommunications Space

Provide T3 drawings in accordance with ARNEC and TIA-606 that includes telecommunications rooms plan views, pathway layout (cable tray, racks, ladder-racks, etc.), mechanical/electrical layout, and cabinet, rack, backboard and wall elevations. Show layout of applicable equipment including incoming cable stub or connector blocks, building protector assembly, outgoing cable connector blocks, patch panels and equipment spaces, and cabinet/racks. Include a complete list of equipment and material, equipment rack details, proposed layout and anchorage of equipment and appurtenances, and equipment relationship to other parts of the work including clearance for maintenance and operation. Drawings may also be an enlargement of a congested area of T1 or T2 drawings. Where this design conflicts with the Contract Drawings, request clarification and direction from the Contracting Officer prior to proceeding.

1.6.3 Telecommunications Qualifications

Perform work by the approved telecommunications contractor and key personnel. A minimum of 30 days prior to installation, submit qualifications for the telecommunications system contractor, the telecommunications system installer, the supervisor (if different from the installer) and key personnel.

1.6.3.1 Telecommunications Contractor

Provide a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor to demonstrate experience in providing successful telecommunications systems within the past 3 years of similar scope and size. Submit documentation for a minimum of 3 and a maximum of 5 successful telecommunication system installations for the telecommunications contractor.

1.6.3.2 Key Personnel Qualifications

Provide professional personnel engaged in the business of the application, installation and testing of the specified telecommunications systems and equipment. There may be one key person or more proposed depending upon how many of the key roles each has successfully provided. Each of the key personnel to demonstrate experience in providing successful telecommunications systems within the past 3 years.

Supervisors and installers assigned to the installation of this system or any of its components to be in accordance with qualified in accordance with ARNEC. Submit documentation of current BICSI certification for each of the key personnel.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work by the date of Award of Contract. Only the key personnel approved will work on the telecommunications system. Any substitutions for the telecommunications contractor's key personnel requires approval from the Contracting Officer.

1.6.3.3 Manufacturer Qualifications

Cabling, equipment, and hardware manufacturers to have a minimum of 3

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years experience in the manufacturing, assembly, and factory testing of components in accordance with TIA-568.0, TIA-568.1, TIA-568.2, TIA-568.3, TIA-568.4.

1.6.4 Test Plan

Provide a complete and detailed test plan for the telecommunications cabling system including a complete list of test equipment for the components and accessories for each cable type specified, 60 calendar days prior to the proposed test date. Include procedures for certification, validation, and testing. Test plan in accordance with ARNEC.

1.6.5 Regulatory Requirements

Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship in accordance with NFPA 70, unless more stringent requirements are specified or indicated.

1.6.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products to have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period to include applications of equipment and materials under similar circumstances and of similar size. The product to have been on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items to be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless otherwise specified.

1.6.6.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.6.6.2 Material and Equipment Manufacturing Date

Products manufactured more than 1 year prior to date of delivery to site are prohibited for use.

1.7 DELIVERY AND STORAGE

Provide protection from weather, moisture, extreme heat and cold, dirt, dust, and other contaminants for telecommunications cabling and equipment placed in storage.

1.8 ENVIRONMENTAL REQUIREMENTS

Connecting hardware to be rated for operation under ambient conditions of 32 to 140 degrees F and in the range of 0 to 95 percent relative humidity, noncondensing.

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1.9 WARRANTY

The equipment items to be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the Contract. Submit warranty information in accordance with Section 00 80 00.00 06 SPECIAL PROVISIONS.

1.10 MAINTENANCE

1.10.1 Record Documentation

Submit T5 drawings including documentation on cables and termination hardware in accordance with TIA-606. T5 drawings to include schedules to show information for cut-overs and cable plant management, patch panel layouts and cover plate assignments, cross-connect information and connecting terminal layout as a minimum. Provide the following T5 drawing documentation as a minimum:

- a. Cables - Provide a record of installed cable in accordance with TIA-606. The cable records to include only the required data fields in accordance with TIA-606. Include manufacture date of cable with submittal.
- b. Termination Hardware - A record of installed patch panels, cross-connect points, distribution frames, terminating block arrangements and type, and outlets in accordance with TIA-606. Documentation to include the required data fields as a minimum in accordance with TIA-606.

PART 2 PRODUCTS

2.1 COMPONENTS

Cabling and interconnecting hardware and components for telecommunications systems to be UL listed or third party independent testing laboratory certified, and in accordance with NFPA 70 and requirements as specified. Where equipment or materials are specified to conform to industry and technical society reference standards of the organizations, submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate to state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard. Provide a complete system of telecommunications cabling and pathway components using star topology. Provide support structures and pathways, complete with outlets, cables, connecting hardware and telecommunications cabinets/racks.

2.2 TELECOMMUNICATIONS PATHWAY

Provide telecommunications pathways in accordance with ARNEC, UFC 3-580-01, and TIA-569. Provide system furniture pathways in accordance with UL 1286.

2.3 TELECOMMUNICATIONS SPACES

Provide connecting hardware and termination equipment in the

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telecommunications entrance facility and telecommunication equipment room(s) to facilitate installation as shown for terminating and cross-connecting permanent cabling. Provide telecommunications interconnecting hardware color coding in accordance with TIA-606.

2.3.1 Backboards

Provide void-free, interior grade A-C plywood 3/4 inch thick. Backboards to be fire rated by manufacturing process. The wood facing out to be fire stamped and clearly visible. Paint each sheet of plywood with fire-retardant paint. Paint color of plywood to match the paint color of the wall the plywood is being installed onto. Provide backboards on a minimum of two adjacent walls in the telecommunication spaces.

2.3.2 Equipment Support Frame

Provide in accordance with ARNEC, ECIA EIA/ECA 310-E, and UL 50.

- a. Racks, floor mounted modular type, 16 gauge steel or 11 gauge aluminum construction, minimum, treated to resist corrosion. Provide rack with vertical and horizontal cable management channels, top and bottom cable troughs, grounding lug. Rack to be compatible with 19 inch panel mounting.
- b. Cabinets, wall or floor mounted modular type, 16 gauge steel or 11 gauge aluminum construction, minimum, treated to resist corrosion. Cabinet to have lockable hinged cover, ground lug, and top and bottom cable access.

2.3.3 Connector Blocks

Provide insulation displacement connector (IDC) Type 110 for Category 3 copper backbone systems. Provide blocks for the number of horizontal and backbone cables terminated on the block plus 10 percent spare.

2.3.4 Cable Support

Provide cable support specifically manufactured for the purpose of routing cables, wires and patch cords horizontally and vertically on 19 inch equipment racks and telecommunications backboards. Cable guides or rings or bracket type devices backboard for horizontal cable management and individually mounted for vertical cable management. Mount ladder style cable tray for vertical cable management on telecommunication backboards. Mount cable guides with screws, and or nuts and lockwashers.

2.3.5 Patch Panels

Provide ports for the number of horizontal and backbone cables terminated on the panel plus 10 percent spare. Provide fiber optic patch panel for maintenance and cross-connecting of optical fiber cables. Panel to be constructed of 16 gauge steel or 11 gauge aluminum minimum and be rack or wall mounted in accordance with design and ARNEC requirements and in accordance with ECIA EIA/ECA 310-E 19 inch equipment rack. Each panel to provide a minimum of 12 duplex single-mode adapters as SC in accordance with TIA/EIA-604-3 with zirconia ceramic alignment sleeves. Provide dust cover for unused adapters. The rear of each panel to have a cable management tray a minimum of 8 inches deep with removable cover, incoming cable strain-relief and routing guides. Panels to have each adapter factory numbered and be equipped with laminated plastic nameplates above

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each adapter.

2.4 TELECOMMUNICATIONS OUTLET/CONNECTOR ASSEMBLIES

2.4.1 Optical Fiber Adapters (Couplers)

Provide optical fiber adapters suitable for duplex SC in accordance with TIA/EIA-604-3 with zirconia ceramic alignment sleeves. Provide dust cover for adapters. Optical fiber adapters in accordance with TIA-455-21 for 500 mating cycles.

2.4.2 Optical Fiber Connectors

Provide in accordance with ARNEC, UFC 3-580-01, TIA-455-21, TIA-568.3. Optical fiber connectors to be duplex SC in accordance with TIA/EIA-604-3 with zirconia ceramic alignment sleeves compatible with 8/125 single-mode fiber. The connectors to provide a maximum attenuation of 0.3 dB at 1300nm with less than a 0.2 dB change after 500 mating cycles. All FOC splices shall be fusion type to factory produced pigtail connectors. The use of angled polish connectors (APC) is not allowed.

2.5 GROUNDING AND BONDING PRODUCTS

Provide in accordance with UL 467, TIA-607, NFPA 70, and ARNEC. Identify components in accordance with TIA-606.

2.6 FACTORY REEL TESTS

Provide documentation of the testing and verification actions taken by manufacturer in accordance with TIA-568.2 and TIA-526-7 for single mode optical fiber cables.

PART 3 EXECUTION

3.1 INSTALLATION

Install telecommunications cabling and pathway systems, including the horizontal and backbone cable, pathway systems, telecommunications outlet or connector assemblies, and associated hardware in accordance with TIA-568.0, TIA-568.1, TIA-568.2, TIA-568.3, TIA-568.4, TIA-569, NFPA 70, ARNEC, UFC 3-580-01, and UL standards as applicable.

Install cables required in workstation (cubicle) panel bases and conference room tables as indicated. Installation in workstation panel bases to include termination with appropriate outlets and outlet face plates.

Installation to be scheduled and coordinated with the furnishings contractor, through the Contracting Officer, to take place at the time workstation panels are erected and before workstation components are installed which could prohibit access to the panel bases. Quantity and location of telecommunications outlets in the workstation bases are as shown. There will be 2 meetings required prior to installation. The first meeting will be a coordination and clarification meeting with the Contracting Officer and a US Army Corps of Engineers (USACE) project engineer prior to the Bid Opening Date. The second meeting will be held during the furniture installation kick-off meeting to include the representatives of the Contractor, the furniture installer, and the Contracting Officer.

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3.1.1 Cabling

Install UTP and optical fiber telecommunications cabling system in accordance with TIA-568.0, TIA-568.1, TIA-568.2, TIA-568.3, ARNEC, and UFC 3-580-01. Screw terminals are prohibited for use except where specifically indicated. Use an approved insulation displacement connection (IDC) tool kit for copper cable terminations. Do not exceed manufacturers' cable pull tensions for copper and optical fiber cables. Provide a device to monitor cable pull tensions. Do not exceed 25 pounds pull tension for four pair copper cables. Do not chafe or damage outer jacket materials. Use only lubricants approved by cable manufacturer. Do not over cinch cables, or crush cables with staples. For UTP cable, bend radii not to be less than 4 times the cable diameter. Terminate cables; no cable to contain unterminated elements. Cables not to be spliced. Label cabling in accordance with paragraph "LABELING."

3.1.2 Pathway Installations

Provide in accordance with TIA-569, NFPA 70, ARNEC, and UFC 3-580-01. Provide building pathway as specified.

3.1.3 Telecommunications Space Termination

Install termination hardware required for UTP and optical fiber system. Use an insulation displacement tool for terminating copper cable to insulation displacement connectors.

3.1.3.1 Patch Panels

Mount in equipment racks with sufficient ports to accommodate the installed cable plant plus 10 percent spares.

- a. Fiber Optic Patch Panel. Fiber optic cable loop to be 3 feet in length. Secure the outer jacket of each cable entering a patch panel to the panel to prevent movement of the fibers within the panel, using clamps or brackets specifically manufactured for that purpose.

3.1.3.2 Equipment Support Frames

Install in accordance with ARNEC, UFC 3-580-01, and TIA-569.

- a. Racks, and cabinet floor mounted modular type. Permanently anchor rack to the floor.
- b. Cabinets, wall-mounted modular type. Mount cabinet at a height of the highest panel not to exceed 78 inches above floor.

3.1.4 Electrical Penetrations

Seal openings around electrical penetrations through wall, partitions, floors, or ceilings.

3.1.5 Grounding and Bonding

Provide in accordance with ARNEC, UFC 3-580-01, TIA-607, and NFPA 70. All Telecommunications bonding to originate at the PBB and be distributed to the SBBs.

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3.1.5.1 Telecommunications Bonding Backbone (TBB)

Telecommunication Bonding Backbone (TBB) is required between the primary bonding bus-bar (PBB) and all secondary bonding bus-bars (SBBs). A TBB is not required for installation with only a single PBB.

Provide a copper conductor TBB in accordance with TIA-607, and NFPA 70. Provide insulated TBB, UL listed with the fire ratings of the pathway.

3.1.5.2 Telecommunications Bonding Conductor

Provide a copper Telecommunications Bonding Conductor (TBC) between the primary bonding bus-bar (PBB) and the electrical service ground in accordance with TIA-607. Size the Telecommunications Bonding Conductor (TBC) so that it is no smaller than the Telecommunications Bonding Backbone (TBB) to a maximum size of 3/0 AWG.

3.2 LABELING

3.2.1 Labels

Provide labeling in accordance with ARNEC and TIA-606. Handwritten labeling is prohibited.

3.2.2 Cable

Label on both ends with identifiers in accordance with ARNEC and TIA-606.

3.2.3 Telecommunications and Signal System Pathway

Install in accordance with ARNEC and TIA-569.

- a. Horizontal Pathway: Install telecommunications pathways from the work area to the telecommunications room and cabling length requirements in accordance with ARNEC, TIA-568.0 AND TIA-568.1. Size of conduits and cable trays in accordance with ARNEC, TIA-569, and as indicated.
- b. Backbone Pathway: Install telecommunication pathways from the telecommunications entrance facility to telecommunications rooms, and, telecommunications equipment rooms (backbone cabling) in accordance with ARNEC and TIA-569. Size of conduits, and cable trays for telecommunications risers in accordance with ARNEC, TIA-569, and as indicated.
- c. Place pull boxes in conduit runs where continuous conduit length exceeds 100 feet or where more than two 90-degree bends occurs. Place pull boxes in straight runs and not to be used in lieu of a bend.

3.2.4 Termination Hardware

Label workstation outlets and patch panel connections with identifiers in accordance with ARNEC and TIA-606.

3.3 TESTING

3.3.1 Telecommunications Cabling Testing

Perform telecommunications cabling inspection, verification, and performance tests in accordance with ARNEC, TIA-568.0, TIA-568.2, TIA-568.3,

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and TIA-568.4. Perform optical fiber field inspection tests via attenuation measurements on factory reels and provide results along with manufacturer certification for factory reel tests. Remove failed cable reels from project site upon attenuation test failure.

3.3.1.1 Inspection

Visually inspect optical fiber jacket materials for UL or third party certification markings. Inspect cabling terminations in telecommunications rooms and at workstations to confirm color code for T568A or T568B pin assignments, and inspect cabling connections to confirm compliance with TIA-568.0, TIA-568.2, and TIA-568.3.

3.3.1.2 Performance/Acceptance Tests

Perform acceptance testing of all installed cabling in accordance with ARNEC, UFC 3-580-01, TIA-568.0, TIA-568.2, TIA-568.3, and TIA-568.4 and as specified. Provide personnel, equipment, instrumentation, and supplies necessary to perform required testing. Notification of any planned testing to be given to the Contracting Officer at least 14 days prior to any test unless specified otherwise. Testing not to proceed until after the Contractor has received written Contracting Officer's approval of the test plans as specified. Test plans to define the tests required to ensure that the system meets technical, operational, and performance specifications. The test plans to define milestones for the tests, equipment, personnel, facilities, and supplies required. The test plans to identify the capabilities and functions to be tested. Provide test reports in booklet form showing all field tests performed, upon completion and testing of the installed system. Measurements to be tabulated on a pair by pair or strand by strand basis.

3.3.1.2.1 Backbone Copper Cable Tests

Perform the following acceptance tests in accordance with TIA-568.0 and TIA-568.0:

- a. Wire map (pin to pin continuity);
- b. Continuity to remote end;
- c. Crossed pairs;
- d. Reversed pairs;
- e. Split pairs;
- f. Shorts between two or more conductors

3.3.1.2.2 Fiber Optic Cable Cable Tests

Test fiber optic cable in accordance with ARNEC, TIA-568.0, TIA-568.3, TIA-526-7, TIA/EIA-455, and as specified. Perform 2 optical tests on all optical fibers: Optical Time Domain Reflectometry (OTDR) Test and Attenuation Test. In addition, perform a width test on all multimode optical fibers. These tests to be performed on the completed end-to-end spans which include the near-end pre-connectorized single fiber cable assembly, outside plant as specified, and the far-end pre-connectorized single fiber cable.

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- a. OTDR Test: Used to determine the adequacy of the cable installations by showing any irregularities, such as discontinuities, micro-bendings or improper splices for the cable span under test. A reference length of fiber, 150 feet minimum, used as the delay line to be placed before the new end connector and after the far end patch panel connectors for inspection of connector signature. Conduct OTDR test and provide calculation or interpretation of results in accordance with TIA-526-7 for single-mode fiber and TIA-526-14 for multimode fiber. Splice losses not to exceed 0.3 db.
- b. Attenuation Test: End-to-end attenuation measurements to be made on all fibers, in both directions, using an 850 and 1300 (multimode fiber) or 1310 and 1550 (single-mode) nanometer light source at one end and the optical power meter on the other end to verify that the cable system attenuation requirements are met in accordance with TIA-455-46A for multimode and TIA-526-7 for single-mode fiber optic cables. The measurement method in accordance with TIA-455-78-B. Attenuation losses not to exceed 0.5 db/km at 1310 nm and 1550 nm for single-mode fiber. Attenuation losses not to exceed 5.0 db/km at 850 nm and 1.5 db/km at 1300 nm for multimode fiber.

3.3.1.3 Earth to Ground Tests

Perform Earth to Ground tests in accordance with ARNEC.

-- End of Section --

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08/20

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INTERIOR FIRE ALARM SYSTEM, ADDRESSABLE 08/20

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

FM GLOBAL (FM)

FM APP GUIDE (updated on-line) Approval Guide
<http://www.approvalguide.com/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE C62.41.1 (2002; R 2008) Guide on the Surges
Environment in Low-Voltage (1000 V and
Less) AC Power Circuits

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 4 (2018) Standard for Integrated Fire
Protection and Life Safety System Testing

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 72 (2019; TIA 19-1; ERTA 1 2019) National
Fire Alarm and Signaling Code

NFPA 170 (2021) Standard for Fire Safety and
Emergency Symbols

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-601-02 (2010) Operations and Maintenance:
Inspection, Testing, and Maintenance of
Fire Protection Systems

UFC 3-600-01 (2016; with Change 6, 2021) Fire
Protection Engineering for Facilities

UFC 4-010-06 (2016; with Change 1, 2017) Cybersecurity
of Facility-Related Control Systems

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UNDERWRITERS LABORATORIES (UL)

UL 268	(2016; Reprint Oct 2019) UL Standard for Safety Smoke Detectors for Fire Alarm Systems
UL 464	(2016; Reprint Sep 2017) UL Standard for Safety Audible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 497B	(2004; Reprint Dec 2012) Protectors for Data Communication Circuits
UL 864	(2014; Reprint May 2020) UL Standard for Safety Control Units and Accessories for Fire Alarm Systems
UL 1283	(2017) UL Standard for Safety Electromagnetic Interference Filters
UL 1449	(2021) UL Standard for Safety Surge Protective Devices
UL 1638	(2016; Reprint Sep 2017) UL Standard for Safety Visible Signaling Devices for Fire Alarm and Signaling Systems, Including Accessories
UL 1971	(2002; Reprint Oct 2008) Signaling Devices for the Hearing Impaired
UL Fire Prot Dir	(2012) Fire Protection Equipment Directory

1.2 RELATED SECTIONS

Section 25 05 11.01 through 25 05 11.03 Cybersecurity for Facility-Related Control Systems, applies to this section, with the additions and modifications specified herein.

1.3 SUMMARY

1.3.1 Scope

- a. This work includes designing and providing a new, complete, fire alarm system as described herein and on the contract drawings for monitoring of the switchgear building. Include system wiring, raceways, pull boxes, terminal cabinets, outlet and mounting boxes, control equipment, initiating devices, notification appliances, programming Reserve Center FACU to accept signals from RFACU and transmit signals to supervising station, and other accessories and miscellaneous items required for a complete operational system even though each item is not specifically mentioned or described. Provide system complete and ready for operation. Existing interior fire alarm system within the Reserve Center Building was manufactured by Notifier. Design and installation must comply with UFGS 25 05 11.01 through 25 05 11.03, UFC 4-010-06, UFC 3-600-01, NFPA 72, and AFGM 2019-320-02.
- b. Provide equipment, materials, installation, workmanship, inspection,

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and testing in strict accordance with NFPA 72, except as modified herein. The system layout on the drawings show the intent of coverage and suggested locations. Final quantity, system layout, and coordination are the responsibility of the Contractor.

- c. Each remote fire alarm control unit must be powered from a wiring riser specifically for that use or from a local emergency power panel located on the same floor as the remote fire alarm control unit. Where remote fire control units are provided, equipment for notification appliances may be located in the remote fire alarm control units.
- d. Where an emergency generator provides standby power supply for life safety system circuits, the generator must be monitored by the Remote Fire Alarm Control Unit (FACU) and transmit emergency generator signals in accordance with NFPA 72.
- e. The fire alarm system must be independent of the building security, building management, and energy/utility monitoring systems other than for control functions.

1.3.2 Qualified Fire Protection Engineer (QFPE)

Services of the QFPE must include:

- a. Reviewing SD-02, SD-03, and SD-05 submittal packages for completeness and compliance with the provisions of this specification. Construction (shop) drawings and calculations must be prepared by, or prepared under the immediate supervision of, the QFPE. The QFPE must affix their professional engineering stamp with signature to the shop drawings, calculations, and material data sheets, indicating approval prior to submitting the shop drawings to the DFPE.
- b. Providing a letter documenting that the SD-02, SD-03, and SD-05 submittal package has been reviewed and noting any outstanding comments.
- c. Performing in-progress construction surveillance prior to installation of ceilings (rough-in inspection).
- d. Witnessing pre-Government and final Government functional performance testing and performing a final installation review.
- e. Signing applicable certificates under SD-07.

1.4 DEFINITIONS

Wherever mentioned in this specification or on the drawings, the equipment, devices, and functions must be defined as follows:

1.4.1 Interface Device

An addressable device that interconnects hard wired systems or devices to an analog/addressable system.

1.4.2 Fire Alarm Control Unit (FACU)

A master control unit having the features of a fire alarm control unit (FACU).

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1.4.3 Remote Fire Alarm Control Unit (RFACU)

A control unit, physically remote from the fire alarm control unit, that receives inputs from automatic and manual fire alarm devices; may supply power to detection devices and interface devices; may provide transfer of power to the notification appliances; may provide transfer of condition to relays or devices connected to the control unit; and reports to and receives signals from the fire alarm and mass notification control unit.

1.4.4 Terminal Cabinet

A steel cabinet with locking, hinge-mounted door where terminal strips are securely mounted inside the cabinet.

1.4.5 Control Module and Relay Module

Terms utilized to describe emergency control function interface devices as defined by NFPA 72.

1.4.6 Designated Fire Protection Engineer (DFPE)

The DoD fire protection engineer that oversees that Area of Responsibility for that project. This is sometimes referred to as the "cognizant" fire protection engineer. Interpret reference to "authority having jurisdiction" and/or AHJ in referenced standards to mean the Designated Fire Protection Engineer (DFPE). The DFPE may be responsible for review of the contractor submittals having a "G" designation, and for witnessing final inspection and testing.

1.4.7 Qualified Fire Protection Engineer (QFPE)

A QFPE is an individual who is a licensed professional engineer (P.E.), who has passed the fire protection engineering written examination administered by the National Council of Examiners for Engineering and Surveying (NCEES) and has relevant fire protection engineering experience.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval.

Shop drawings (SD-02), product data (SD-03) and calculations (SD-05) must be prepared by the fire alarm designer and combined and submitted as one complete package. The QFPE must review the SD-02/SD-03/SD-05 submittal package for completeness and compliance with the Contract provisions prior to submission to the Government. The QFPE must provide a Letter of Confirmation that they have reviewed the submittal package for compliance with the contract provisions. This letter must include their registered professional engineer stamp and signature. Partial submittals and submittals not reviewed by the QFPE will be returned by the Government disapproved without review.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

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Qualified Fire Protection Engineer (QFPE); G

Fire alarm system designer; G

Supervisor; G

Technician; G

Installer; G

Test Technician; G

Fire Alarm System Site-Specific Software Acknowledgement; G

SD-02 Shop Drawings

Nameplates; G

Instructions; G

Wiring Diagrams; G

Explosion Control System Sequence and Integration; G

System Layout; G

Notification Appliances; G

Initiating devices; G

Battery Power; G

Voltage Drop Calculations; G

SD-03 Product Data

Manual Stations; GSmoke Detectors; G

Flame Detectors; G

Addressable Interface Devices; G

Notification Appliances; G

Batteries; G

Battery Chargers; G

Surge Protective Devices; G

Alarm Wiring; G

Back Boxes and Conduit; G

Terminal Cabinets; G

Document Storage Cabinet; G

Remote Fire Alarm Control Units; G

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SD-05 Design Data

SD-06 Test Reports

Test Procedures; G

SD-07 Certificates

Verification of Compliant Installation; G

Request for Government Final Test; G

SD-10 Operation and Maintenance Data

Operation and Maintenance (O&M) Instructions; G

Instruction of Government Employees; G

SD-11 Closeout Submittals

As-Built Drawings

Spare Parts

1.6 SYSTEM OPERATION

Fire alarm system components requiring power, except for the RFACU(s) power supply, must operate on 24 volts DC unless noted otherwise in this section.

The interior fire alarm system must be a complete, supervised, noncoded, analog/addressable fire alarm system conforming to NFPA 72, and UL 864. Systems meeting UL 2017 only are not acceptable. The system must be activated into the alarm mode by actuation of an alarm initiating device. The system must remain in the alarm mode until the initiating device is reset and the control unit is reset and restored to normal.

1.6.1 Alarm Initiating Devices and Notification Appliances (Visual, Audible)

- a. Connect alarm initiating devices to initiating device circuits (IDC) Class "B", or to signaling line circuits (SLC) Class "B" and installed in accordance with NFPA 72.
- b. Connect notification appliances to notification appliance circuits (NAC) Class "B".

1.6.2 Functions and Operating Features

The system must provide the following functions and operating features:

- a. Power, annunciation, supervision, and control for the system. Addressable systems must be microcomputer (microprocessor or microcontroller) based with a minimum word size of eight bits with sufficient memory to perform as specified.
- b. Visual alarm notification appliances must be synchronized as required by NFPA 72.

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- c. Electrical supervision of the primary power (AC) supply, presence of the battery, battery voltage, and placement of system modules within the control unit.
- d. An audible and visual trouble signal to activate upon a single break or open condition, or ground fault. The trouble signal must also operate upon loss of primary power (AC) supply, absence of a battery supply, low battery voltage, or removal of alarm or supervisory control unit modules. After the system returns to normal operating conditions, the trouble signal must again sound until the trouble is acknowledged. A smoke detector in the process of being verified for the actual presence of smoke must not initiate a trouble condition.
- e. A trouble signal silence feature that must silence the audible trouble signal, without affecting the visual indicator.
- f. Program capability via switches in a locked portion of the RFACU to bypass the automatic notification appliance circuits, fire reporting system features. Operation of this programmed action must indicate on the RFACU display as a supervisory or trouble condition.
- g. Alarm functions must override trouble or supervisory functions. Supervisory functions must override trouble functions.
- h. The system must be capable of being programmed from the control unit keyboard. Programmed information must be stored in non-volatile memory.
- i. The system must be capable of operating, supervising, and/or monitoring non-addressable alarm and supervisory devices.
- j. There must be no limit, other than maximum system capacity, as to the number of addressable devices that may be in alarm simultaneously.
- k. Where the fire alarm system is responsible for initiating an action in another emergency control device or system the addressable fire alarm relay must be located in the vicinity of the emergency control device.
- l. An alarm signal must automatically initiate the following functions:
 - (1) Transmission of an alarm signal to the Reserve Center FACU. The Reserve Center FACU shall transmit a signal to a remote supervising station.
 - (2) Visual indication of the device operated on the RFACU .
 - (3) Actuation of alarm notification appliances.
 - (4) Recording of the event electronically in the history log of the RFACU and Reserve Center FACU.
- m. A supervisory signal must automatically initiate the following functions:
 - (1) Visual indication of the device operated on the RFACU.
 - (2) Transmission of a supervisory signal to the Reserve Center FACU. The Reserve Center FACU shall transmit a signal to a remote

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supervising station.

- (3) Recording of the event electronically in the history log of the RFACU and Reserve Center FACU.

n. A trouble condition must automatically initiate the following functions:

- (1) Visual indication of the device operated on the RFACU.
- (2) Transmission of a trouble signal to the Reserve Center FACU. The Reserve Center FACU shall transmit a signal to a remote supervising station.
- (3) Recording of the event electronically in the history log of the RFACU.

1.7 TECHNICAL DATA AND SITE-SPECIFIC SOFTWARE

Technical data and site-specific software (meaning technical data that relates to computer software) that are specifically identified in this project, and may be required in other specifications, must be delivered, strictly in accordance with the CONTRACT CLAUSES. The fire alarm system manufacturer must submit written confirmation of this contract provision as "Fire Alarm System Site-Specific Software Acknowledgement". Identify data delivered by reference to the specification paragraph against which it is furnished. Data to be submitted must include complete system, equipment, and software descriptions. Descriptions must show how the equipment will operate as a system to meet the performance requirements of this contract. The site-specific software data package must also include the following:

- a. Items identified in NFPA 72, titled "Site-Specific Software".
- b. Identification of programmable portions of the system equipment and capabilities.
- c. Description of system revision and expansion capabilities and methods of implementation detailing both equipment and software requirements.
- d. Provision of operational software data on all modes of programmable portions for fire alarm .
- e. Description of Reserve Center Fire Alarm Control Unit equipment operation.
- f. Description of auxiliary and remote equipment operations.
- g. Library of application software.
- h. Operation and maintenance manuals.

1.8 EXISTING EQUIPMENT

- a. Equipment and devices must be compatible and operable with the existing fire alarm mass notification system and must not impair reliability or operational functions of existing supervising station fire alarm system.

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- b. Equipment and devices must be compatible and operable with the existing fire alarm system within the Army Reserve Center. Equipment must not impair reliability or operational functions of the existing system. The existing fire alarm control unit is in Electrical 194A of the Army Reserve Center. The contractor shall verify the type and manufacturer of the existing fire alarm system.

1.9 QUALITY ASSURANCE

1.9.1 Submittal Documents

1.9.1.1 Preconstruction Submittals

Within 36 days of contract award but not less than 14 days prior to commencing any work on site, the Contractor must submit the following for review and approval. SD-02, SD-03 and SD-05 submittals received prior to the review and approval of the qualifications of the fire alarm subcontractor and QFPE must be returned disapproved without review. All resultant delays must be the sole responsibility of the Contractor.

1.9.1.2 Shop Drawings

Shop drawings must not be smaller than the Contract Drawings. Drawings must comply with the requirements of NFPA 72 and NFPA 170. Minimum scale for floor plans must be 1/8"=1'.

1.9.1.3 Nameplates

Nameplate illustrations and data to obtain approval by the Contracting Officer before installation.

1.9.1.4 Wiring Diagrams

Four copies of point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the Reserve Center FACU and remote FACU, initiating circuits, switches, relays and terminals, including pathway diagrams between the control unit and shared communications equipment within the protected premises. Point-to-point wiring diagrams must be job specific and must not indicate connections or circuits not being utilized. Provide complete riser diagrams indicating the wiring sequence of all devices and their connections to the control equipment. Include a color-code schedule for the wiring.

1.9.1.5 System Layout

Four copies of plan view drawing showing device locations, terminal cabinet locations, junction boxes, other related equipment, conduit routing, conduit sizes, wire counts, conduit fill calculations, wire color-coding, circuit identification in each conduit, and circuit layouts for all floors. Indicate candle rating of each visual notification appliance. Clearly identify the locations of isolation modules. Indicate the addresses of all devices, modules, relays, and similar. Indicate if the environment for the RFACU is within its environmental listing (e.g. temperature/humidity).

Provide a complete description of the system operation in matrix format

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similar to the "Typical Input/Output Matrix" included in the Annex of NFPA 72.

1.9.1.6 Notification Appliances

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for notification appliances. Annotate data for each circuit on the drawings.

1.9.1.7 Initiating Devices

Calculations and supporting data on each circuit to indicate that there is at least 25 percent spare capacity for initiating devices. Annotate data for each circuit on the drawings.

1.9.1.8 Battery Power

Calculations and supporting data as required in paragraph Battery Power Calculations for alarm, alert, and supervisory power requirements. Calculations including ampere-hour requirements for each system component and each control unit component, and the battery recharging period, must be included on the drawings.

1.9.1.9 Voltage Drop Calculations

Voltage drop calculations for each notification circuit indicating that sufficient voltage is available for proper operation of the system and all components, at a minimum rated voltage of the system operating on batteries. Include the calculations on the system layout drawings.

1.9.1.10 Product Data

Four copies of annotated descriptive data to show the specific model, type, and size of each item. Catalog cuts must also indicate the NRTL listing. The data must be highlighted to show model, size, and options that are intended for consideration. Data must be adequate to demonstrate compliance with all contract requirements. Product data for all equipment must be combined into a single submittal.

Provide an equipment list identifying the type, quantity, make, and model number of each piece of equipment to be provided under this submittal. The equipment list must include the type, quantity, make and model of spare equipment. Types and quantities of equipment submitted must coincide with the types and quantities of equipment used in the battery calculations and those shown on the shop drawings.

1.9.1.11 Operation and Maintenance (O&M) Instructions

Six copies of the Operation and Maintenance Instructions. The O&M Instructions must be prepared in a single volume or in multiple volumes, with each volume indexed, and may be submitted as a Technical Data Package. Manuals must be approved prior to training. The Interior Fire Alarm System Operation and Maintenance Instructions must include the following:

- a. "Manufacturer Data Package five" as specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA.
- b. Operating manual outlining step-by-step procedures required for system

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startup, operation, and shutdown. The manual must include the manufacturer's name, model number, service manual, parts list, and preliminary equipment list complete with description of equipment and their basic operating features.

- c. Maintenance manual listing routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. The manuals must include conduit layout, equipment layout and simplified wiring, and control diagrams of the system as installed.
- d. Complete procedures for system revision and expansion, detailing both equipment and software requirements.
- e. Software submitted for this project on CD/DVD media utilized.
- f. Printouts of configuration settings for all devices.
- g. Routine maintenance checklist. The routine maintenance checklist must be arranged in a columnar format. The first column must list all installed devices, the second column must state the maintenance activity or state no maintenance required, the third column must state the frequency of the maintenance activity, and the fourth column provided for additional comments or reference. All data (devices, testing frequencies, and similar) must comply with UFC 3-601-02.
- h. A final Equipment List must be submitted with the Operating and Maintenance (O&M) manual.

1.9.1.12 As-Built Drawings

The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of the as-built (marked-up) drawings must be provided at the time of, or prior to the final Government test.

1.9.2 Qualifications

1.9.2.1 Fire Alarm System Designer

The fire alarm system designer must be certified as a Level III or IV (minimum) Technician by National Institute for Certification in Engineering Technologies (NICET) in the Fire Alarm Systems subfield of Fire Protection Engineering Technology or meet the qualifications for a QFPE.

1.9.2.2 Supervisor

A NICET Level III or IV fire alarm technician must supervise the installation of the fire alarm system. The fire alarm technicians supervising the installation of equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.3 Technician

Fire alarm technicians with a minimum of four years of experience must be utilized to install and terminate fire alarm devices, cabinets and control units. The fire alarm technicians installing the equipment must be

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factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.4 Installer

Fire alarm installer with a minimum of two years of experience utilized to assist in the installation of fire alarm devices, cabinets and control units, or a NICET Level II technician to assist in the installation of fire alarm devices, cabinets and control units. A licensed electrician must be allowed to install wire, cable, conduit and backboxes for the fire alarm system. The fire alarm installer must be factory trained in the installation, adjustment, testing, and operation of the equipment specified herein and on the drawings.

1.9.2.5 Test Technician

Fire alarm technicians with a minimum of eight years of experience and NICET Level III or IV utilized in testing and certification of the installation of the fire alarm devices, cabinets and control units. The fire alarm technicians testing the equipment must be factory trained in the installation, adjustment, testing, and operation of the equipment installed as part of this project.

1.9.2.6 Manufacturer

Components must be of current design and must be in regular and recurrent production at the time of installation. Provide design, materials, and devices for a protected premises fire alarm system, complete, conforming to NFPA 72, except as specified herein.

1.9.3 Regulatory Requirements

Equipment and material must be listed or approved. Listed or approved, as used in this Section, means listed, labeled or approved by a Nationally Recognized Testing Laboratory (NRTL) such as UL Fire Prot Dir or FM APP GUIDE. The omission of these terms under the description of any item of equipment described must not be construed as waiving this requirement. All listings or approvals by testing laboratories must be from an existing ANSI or UL published standard. The recommended practices stated in the manufacturer's literature or documentation must be considered as mandatory requirements.

1.10 DELIVERY, STORAGE, AND HANDLING

Protect equipment delivered and placed in storage from the weather, humidity, and temperature variation, dirt and dust, and other contaminants.

1.11 MAINTENANCE

1.11.1 Spare Parts

Furnish the following spare parts in the manufacturers original unopened containers:

- a. Five complete sets of system keys.
- b. Two of each type of fuse required by the system.
- c. One manual stations.

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- d. One of each type of detector installed.
- e. One of each type of detector base and head installed.
- f. One of each type of audible and visual alarm device installed.
- g. One of each type of addressable monitor module installed.
- h. One of each type of addressable control module installed.
- i. One 120 VAC surge protective device.

1.11.2 Special Tools

Software, connecting cables and proprietary equipment, necessary for the maintenance, testing, and reprogramming of the equipment must be furnished to the Contracting Officer, prior to the instruction of Government employees.

PART 2 PRODUCTS

2.1 GENERAL PRODUCT REQUIREMENT

All fire alarm equipment must be listed for use under the applicable reference standards.

2.2 MATERIALS AND EQUIPMENT

2.2.1 Standard Products

Provide materials, equipment, and devices that have been tested by a nationally recognized testing laboratory and listed for fire protection service when so required by NFPA 72 or this specification. Select material from one manufacturer, where possible, and not a combination of manufacturers, for any particular classification of materials. Material and equipment must be the standard products of a manufacturer regularly engaged in the manufacture of the products for at least 2 years prior to bid opening.

2.2.2 Nameplates

Major components of equipment must have the manufacturer's name, address, type or style, model or serial number, catalog number, date of installation, installing Contractor's name and address, and the contract number provided on a new name plate permanently affixed to the item or equipment. Major components include, but are not limited to, the following:

- a. RFACU

Nameplates must be etched metal or plastic, permanently attached by screws to control units or adjacent walls.

2.2.3 Keys

Keys and locks for equipment, control units and devices must be identical. Master all keys and locks to a single key as required by the Installation Fire Department. Keys must be CAT 60.

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2.2.4 Instructions

Provide a typeset printed or typewritten instruction card mounted behind a Lexan plastic or glass cover in a stainless steel or aluminum frame. Install the instructions on the interior of the RFACU. The card must show those steps to be taken by an operator when a signal is received as well as the functional operation of the system under all conditions, normal, alarm, supervisory, and trouble. The instructions and their mounting location must be approved by the Contracting Officer before being posted.

2.3 REMOTE FIRE ALARM CONTROL UNIT

Provide a complete Remote fire alarm control units (RFACU) fully enclosed in a lockable steel cabinet as specified herein. Operations required for testing or for normal care, maintenance, and use of the system must be performed from the front of the enclosure. If more than a single unit is required at a location to form a complete control unit, the unit cabinets must match exactly. The system must be capable of defining any module as an alarm module and report alarm trouble, loss of polling, or as a supervisory module, and reporting supervisory short, supervisory open or loss of polling such as waterflow switches, valve supervisory switches, fire pump monitoring, independent smoke detection systems, relays for output function actuation. The RFACU must communicate directly with the Reserve Center FACU located in Electrical Room 194A of the Reserve Center Building.

- a. Each control unit must provide power, supervision, control, and logic for the entire system, utilizing solid state, modular components, internally mounted and arranged for easy access. Each control unit must be suitable for operation on a 120 volt, 60 hertz, normal building power supply. Provide each control unit with supervisory functions for power failure, internal component placement, and operation.
- b. Visual indication of alarm, supervisory, or trouble initiation on the RFACU must be by liquid crystal display or similar means with a minimum of 80 characters.

2.3.1 Cabinet

Install control unit components in cabinets large enough to accommodate all components and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and surface or semi-recessed mounting provisions. The enclosure must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must say "Fire Alarm control unit" and must not be less than 1-inch high. Provide prominent rigid plastic or metal identification plates for lamps, circuits, meters, fuses, and switches.

2.3.2 Silencing Switches

2.3.2.1 Alarm Silencing Switch

Provide an alarm silencing switch at the RFACU that must silence the audible and visual notification appliances. Subsequent activation of initiating devices must cause the notification appliances to re-activate.

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2.3.2.2 Supervisory/Trouble Silencing Switch

Provide supervisory and trouble silencing switch(es) that must silence the audible trouble and supervisory signal(s), but not extinguish the visual indicator. This switch must be overridden upon activation of a subsequent supervisory or trouble condition. Audible trouble indication must resound automatically every 24 hours after the silencing feature has been operated if the supervisory or trouble condition still exists.

2.3.3 Non-Interfering

Power and supervise each circuit such that a signal from one device does not prevent the receipt of signals from any other device. Initiating devices must be manually reset by switch from the RFACU after the initiating device or devices have been restored to normal.

2.3.4 Memory

Provide each control unit with non-volatile memory and logic for all functions. The use of long life batteries, capacitors, or other age-dependent devices must not be considered as equal to non-volatile processors, PROMS, or EPROMS.

2.3.5 Field Programmability

Provide control units and control units that are fully field programmable for both input and output of control, initiation, notification, supervisory, and trouble functions. The system program configuration must be menu driven. System changes must be password protected. Any proprietary equipment and proprietary software needed by qualified technicians to implement future changes to the fire alarm system must be provided as part of this contract.

2.3.6 Input/Output Modifications

The RFACU must contain features that allow the bypassing of input devices from the system or the modification of system outputs. These control features must consist of a control unit mounted keypad. Any bypass or modification to the system must indicate a trouble condition on the RFACU.

2.3.7 Resetting

Provide the necessary controls to prevent the resetting of any alarm, supervisory, or trouble signal while the alarm, supervisory or trouble condition on the system still exists.

2.3.8 Walk Test

The RFACU must have a walk test feature. When using this feature, operation of initiating devices must result in limited system outputs, so that the notification appliances operate for only a few seconds and the event is indicated in the history log, but no other outputs occur.

2.3.9 History Logging

The control unit must have the ability to store a minimum of 400 events in a log. These events must be stored in a battery-protected memory and must remain in the memory until the memory is downloaded or cleared manually. Resetting of the control unit must not clear the memory.

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2.3.10 Manual Access

An operator at the control unit, having a proper access level, must have the capability to manually access the following information for each initiating device.

- a. Primary status.
- b. Device type.
- c. Present average value.
- d. Present sensitivity selected.
- e. Detector range (normal, dirty).

2.4 MANUAL STATIONS

Provide metal or plastic, semi-flush or flush mounted, double-action, addressable manual stations, that are not subject to operation by jarring or vibration. Stations must be equipped with screw terminals for each conductor. Stations that require the replacement of any portion of the device after activation are not permitted. Stations must be finished in red with molded raised lettering operating instructions of contrasting color. The use of a key must be required to reset the station.

2.5 SMOKE DETECTORS

2.5.1 Spot Type Detectors

Provide addressable photoelectric smoke detectors as follows:

- a. Provide analog/addressable photoelectric smoke detectors utilizing the photoelectric light scattering principle for operation in accordance with UL 268. Smoke detectors must be listed for use with the RFACU.
- b. Provide self-restoring type detectors that do not require any readjustment after actuation at the RFACU to restore them to normal operation. The detector must have a visual indicator to show actuation.
- c. Vibration must have no effect on the detector's operation. Protect the detection chamber with a fine mesh metallic screen that prevents the entrance of insects or airborne materials. The screen must not inhibit the movement of smoke particles into the chamber.
- d. Provide twist lock bases with screw terminals for each conductor. The detectors must maintain contact with their bases without the use of springs.
- e. The detector address must identify the particular unit, its location within the system, and its sensitivity setting. Detectors must be of the low voltage type rated for use on a 24 VDC system.

2.6 FLAME DETECTORS

Detectors must be sensitive to the micron range best suited for their intended use. Detectors must operate over electrically supervised wiring

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circuits and the loss of power to the detector must result in a trouble signal. A self-test feature must be provided for each detector to be individually tested.

2.6.1 Infrared (IR) Single Frequency Flame Detector

The detector must be sensitive in the range as required by NFPA 72 and manufacturer recommendations specific to Li-Ion battery applications.

2.6.2 Infrared (IR) Multi Frequency Flame Detector

The IR detector must consist of three or more IR sensors, each selected for a different IR frequency. The primary sensor must have a sensitivity as required by NFPA 72 and manufacturer recommendations specific to Li-Ion battery applications. Secondary sensors are tuned to different IR wavelengths to null out the effect of black body radiation to the primary sensor.

2.6.3 Ultraviolet (UV) Flame Detectors

UV flame detector must be of the narrow band response type which operates on radiated ultraviolet energy and must have a sensitivity as required by NFPA 72 and manufacturer recommendations specific to Li-Ion battery applications. The cone of vision must be 80 degrees or greater. Each detector must be completely insensitive to light sources in the visible frequency range.

2.6.4 Combination UV/IR Flame Detector

The UV/IR detector must provide discrimination against false alarms by requiring both UV and IR flame detection before an alarm is sent. The UV sensor must be sensitive in the range of 0.185 to 0.265 micrometers only. The IR sensor must have a sensitivity as required by NFPA 72 and manufacturer recommendations specific to Li-Ion battery applications. Detectors must be completely insensitive to light sources in the visible frequency range.

2.7 ADDRESSABLE INTERFACE DEVICES

The initiating device being monitored must be configured as a Class "B" initiating device circuits. The module must be listed as compatible with the control unit. The module must provide address setting means compatible with the control unit's SLC supervision and store an internal identifying code. Monitor module must contain an integral LED that flashes each time the monitor module is polled and is visible through the device cover plate. Pull stations with a monitor module in a common backbox are not required to have an LED. Modules must be listed for the environmental conditions in which they will be installed.

2.8 NOTIFICATION APPLIANCES

2.8.1 Audible Notification Appliances

Audible appliances must conform to the applicable requirements of UL 464. Appliances must be connected into notification appliance circuits. Surface mounted audible appliances must be painted red or white. Recessed audible appliances must be installed with a grill that is painted red or white.

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2.8.1.1 Horns

Horns must be semi-flush mounted. Horns must produce a sound rating of at least 85 dBA at 10 feet. Horns used in exterior locations must be specifically listed or approved for outdoor use and be provided with metal housing and protective grilles. Horns must be weatherproof where indicated by the contract drawings.

2.8.2 Visual Notification Appliances

Visual notification appliances must conform to the applicable requirements of UL 1638, UL 1971 and conform to the Architectural Barriers Act (ABA). Visual Notification Appliances must have clear high intensity optic lens, xenon flash tubes, or light emitting diode (LED) and be marked "FIRE" in letters of contrasting color. The light pattern must be disbursed so that it is visible above and below the strobe and from a 90 degree angle on both sides of the strobe. Strobe flash rate must be 1 flash per second and a minimum of 15 candela based on the UL 1971 test. Strobe must be surface or semi-flush mounted.

2.9 ELECTRIC POWER

2.9.1 Primary Power

Power must be 120 VAC 60 Hz service for the FACU from the AC service to the building in accordance with NFPA 72.

2.10 SECONDARY POWER SUPPLY

Provide for system operation in the event of primary power source failure. Transfer from normal to auxiliary (secondary) power or restoration from auxiliary to normal power must be automatic and must not cause transmission of a false alarm.

2.10.1 Batteries

Battery size must be the greater of the following capacity. This capacity applies to every control unit associated with this system.

- a. SWITCHGEAR ENCLOSURE FIRE ALARM SYSTEM: Sufficient capacity to operate the fire alarm system under supervisory and trouble conditions, including audible trouble signal devices for 48 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes.

2.10.1.1 Capacity

Battery size must be the following capacity. This capacity applies to every control unit associated with this system.

- a. Sufficient capacity to operate the fire alarm system under supervisory and trouble conditions, including audible trouble signal devices for 24 hours and audible and visual signal devices under alarm conditions for an additional 15 minutes.

2.10.1.2 Battery Power Calculations

- a. Verify that battery capacity exceeds supervisory and alarm power requirements for the criteria noted in the paragraph "Capacity" above.

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- (1) Substantiate the battery calculations for alarm and supervisory power requirements. Include ampere-hour requirements for each system component and each control unit component, and compliance with UL 864.
 - (2) Provide complete battery calculations for both the alarm and supervisory power requirements. Submit ampere-hour requirements for each system component with the calculations.
 - (3) Provide voltage drop calculations to indicate that sufficient voltage is available for proper operation of the system and all components. Calculations must be performed using the minimum rated voltage of each component.
- b. For battery calculations assume a starting voltage of 24 VDC for starting the calculations to size the batteries. Calculate the required Amp-Hours for the specified standby time, and then calculate the required Amp-Hours for the specified alarm time. Using 20.4 VDC as starting voltage, perform a voltage drop calculation for circuits containing device and/or appliances remote from the power sources.

2.10.2 Battery Chargers

Provide a solid state, fully automatic, variable charging rate battery charger. The charger must be capable of providing 125 percent of the connected system load and must maintain the batteries at full charge. In the event the batteries are fully discharged (20.4 Volts dc), the charger must recharge the batteries back to 95 percent of full charge within 48 hours after a single discharge cycle as described in paragraph CAPACITY above. Provide pilot light to indicate when batteries are manually placed on a high rate of charge as part of the unit assembly if a high rate switch is provided.

2.11 SURGE PROTECTIVE DEVICES

Surge protective devices must be provided to suppress all voltage transients which might damage fire alarm control unit components. Systems having circuits located outdoors, communications equipment must be protected against surges induced on any signaling line circuit. Cables and conductors, that serve as communications links, must have surge protection circuits installed at each end. The surge protective device must wire in series to the power supply of the protected equipment with screw terminations. Line voltage surge arrester must be installed directly adjacent to the power panel where the RFACU breaker is located.

- a. Surge protective devices for nominal 120 VAC must be UL 1449 listed with a maximum 500 volt suppression level and have a maximum response time of 5 nanoseconds. The surge protective device must also meet IEEE C62.41.1 and IEEE C62.41.2 category B tests for surge capacity. The surge protective device must feature multi-stage construction and be provided with a long-life indicator lamp (either light emitting diode or neon) which extinguishes upon failure of protected components. Any unit fusing must be externally accessible.
- b. All surge protective devices (SPD) must be the standard product of a single manufacturer and be equal or better than the following:
 - (1) For 120 VAC nominal line voltage: UL 1449 and UL 1283 listed,

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series connected 120 VAC, 20A rated, surge protective device in a NEMA 4x enclosure. Minimum 50,000 amp surge current rating with EMI/RFI filtering and a dry contact circuit for remote monitoring of surge protection status.

- (2) For 24-volt nominal line voltage: UL 497B listed, series connected low voltage, 24-volt, 5A rated, loop circuit protector, base and replaceable module.

2.12 WIRING

Provide wiring materials under this section as specified in Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM with the additions and modifications specified herein.

2.12.1 Alarm Wiring

IDC and SLC wiring must be solid copper cable in accordance with the manufacturers requirements. At a minimum, copper signaling line circuits and initiating device circuit field wiring must be No. 18 AWG and 16 AWG size conductors, respectively. Visual notification appliance circuit conductors, that contain audible alarm appliances, must be copper No. 14 AWG size conductors at a minimum. Wire size must be sufficient to prevent voltage drop problems. Circuits operating at 24 VDC must not operate at less than the listed voltages for the detectors and/or appliances. Power wiring, operating at 120 VAC minimum, must be a minimum No. 12 AWG solid copper having similar insulation. Acceptable power-limited cables are FPL, FPLR or FPLP as appropriate with red colored covering. Nonpower-limited cables must comply with NFPA 70.

2.13 REMOTE FIRE ALARM SIGNALING AND COMMUNICATION

The Remote Fire Alarm Control panel is not required to communicate with the supervising station. The RFACU shall be capable of communicating directly with the Reserve Center FACU located in Electrical Room 194A.

2.13.1 Signals to Be Transmitted to the Remote Fire Alarm Control Unit:

The following signals must be sent to the Remote Fire Alarm Control Unit:

- a. Manual pull stations
- b. Smoke detectors
- Emergency Generators supervisory and trouble
- d. Generator Fault
- e. Generator Run
- f. Generator Switch to Nonautomatic
- g. Fire Extinguishing System

PART 3 EXECUTION

3.1 VERIFYING ACTUAL FIELD CONDITIONS

Before commencing work, examine all adjoining work on which the contractor's work is in any way dependent for perfect workmanship according to the intent of this specification section, and report to the Contracting Officer's Representative any condition which prevents performance of first class work. No "waiver of responsibility" for incomplete, inadequate or defective adjoining work will be considered unless notice has been filed before submittal of a proposal.

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3.2 INSTALLATION

3.2.1 Remote Fire Alarm Control Unit (RFACU)

Locate the RFACU where indicated on the drawings. Recess, Semi-recess, or Surface mount the enclosure with the top of the cabinet 6 feet above the finished floor or center the cabinet at 5 feet, whichever is lower. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the RFACU. Locate the document storage cabinet adjacent to the RFACU unless the Contracting Officer directs otherwise.

3.2.2 Battery Cabinets

When batteries will not fit in the RFACU, locate battery cabinets below or adjacent to the FACU. Battery cabinets must be installed at an accessible location when standing at floor level. Battery cabinets must not be installed lower than 12 inches above finished floor, measured to the bottom of the cabinet, nor higher than 36 inches above the floor, measured to the top of the cabinet. Installing batteries above drop ceilings or in inaccessible locations is prohibited. Battery cabinets must be large enough to accommodate batteries and also to allow ample gutter space for interconnection of control units as well as field wiring. The cabinet must be provided in a sturdy steel housing, complete with back box, hinged steel door with cylinder lock, and surface mounting provisions. The cabinet must be identified by an engraved phenolic resin nameplate. Lettering on the nameplate must indicate the control unit(s) the batteries power and must not be less than 1-inch high.

3.2.3 Manual Stations

Locate manual stations as required by NFPA 72 and as indicated on the drawings. Mount stations so they are located no farther than 5 feet from the exit door they serve, measured horizontally. Manual stations must be mounted at a minimum of 42 inches measured to the operating handle.

3.2.4 Notification Appliances

- a. Locate notification appliance devices as required by NFPA 72 and where indicated. Where more than two visual notification appliances are located in the same room or corridor or field of view, provide synchronized operation. Devices must use screw terminals for all field wiring.
- b. Audible and visual notification appliances mounted on the exterior of the building, within unconditioned spaces, or in the vicinity of showers must be listed weatherproof appliances installed on weatherproof backboxes.
- c. Speakers must not be located in close proximity to the FACU or LOC so as to cause feedback when the microphone is in use.

3.2.5 Smoke Detectors

Locate detectors as required by NFPA 72 and their listing and as indicated on the drawings on a 4-inch mounting box. Smoke detectors are permitted to be on the wall no lower than 12 inches from the ceiling with no minimum distance from the ceiling. Install smoke detectors no closer than 3 feet from air handling supply diffusers. Detectors installed in acoustical

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ceiling tiles must be centered in the tiles plus or minus 2 inches.

3.3 SYSTEM FIELD WIRING

3.3.1 Wiring within Cabinets, Enclosures, and Boxes

Provide wiring installed in a neat and workmanlike manner and installed parallel with or at right angles to the sides and back of any box, enclosure, or cabinet. Conductors that are terminated, spliced, or otherwise interrupted in any enclosure, cabinet, mounting, or junction box must be connected to screw-type terminal blocks. Mark each terminal in accordance with the wiring diagrams of the system. The use of wire nuts or similar devices is prohibited. Wiring to conform with NFPA 70.

Indicate the following in the wiring diagrams:

- a. Point-to-point wiring diagrams showing the points of connection and terminals used for electrical field connections in the system, including interconnections between the equipment or systems that are supervised or controlled by the system. Diagrams must show connections from field devices to the FACU and remote fire alarm control units, initiating circuits, switches, relays and terminals.
- b. Complete riser diagrams indicating the wiring sequence of devices and their connections to the control equipment. Include a color code schedule for the wiring. Include floor plans showing the locations of devices and equipment.

3.3.2 Terminal Cabinets

Provide a terminal cabinet at point where circuits leave the Switchgear Building to connect to the Reserve Center FACU, and where indicated on the drawings. Terminal size must be appropriate for the size of the wiring to be connected. Conductor terminations must be labeled and a drawing containing conductors, their labels, their circuits, and their interconnection must be permanently mounted in the terminal cabinet. Minimum size is 8 inches by 8 inches. Only screw-type terminals are permitted. Provide an identification label, that displays "FIRE ALARM TERMINAL CABINET" with 2-inch lettering, on the front of the terminal cabinet.

3.3.3 Alarm Wiring

- a. Voltages must not be mixed in any junction box, housing or device, except those containing power supplies and control relays.
- b. Utilize shielded wiring where recommended by the manufacturer. For shielded wiring, ground the shield at only one point, in or adjacent to the RFACU.
- c. Pigtail or T-tap connections to signal line circuits, initiating device circuits, supervisory alarm circuits, and notification appliance circuits are prohibited.
- d. Color coding is required for circuits and must be maintained throughout the circuit. Conductors used for the same functions must be similarly color coded. Conform wiring to NFPA 70.
- e. Pull all conductors splice free. The use of wire nuts, crimped

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connectors, or twisting of conductors is prohibited. Where splices are unavoidable, the location of the junction box or pull box where they occur must be identified on the as-built drawings. The number and location of splices must be subject to approval by the Designated Fire Protection Engineer (DFPE).

3.3.4 Back Boxes and Conduit

In addition to the requirements of Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM, provide all wiring in rigid metal conduit or intermediate metal conduit unless specifically indicated otherwise. Minimum conduit size must be 3/4-inch in diameter. Do not use electrical non-metallic tubing (ENT) or flexible non-metallic tubing and associated fittings.

- a. Galvanized rigid steel (GRS) conduit must be utilized where exposed to weather, where subject to physical damage, and where exposed on exterior of buildings. Intermediate metal conduit (IMC) may be used in lieu of GRS as allowed by NFPA 70.
- b. Electrical metallic tubing (EMT) is permitted above suspended ceilings or exposed where not subject to physical damage. Do not use EMT underground, encased in concrete, mortar, or grout, in hazardous locations, where exposed to physical damage, outdoors or in fire pump rooms. Use die-cast compression connectors.
- c. For rigid metallic conduit (RMC), only threaded type fitting are permitted for wet or damp locations.
- d. Flexible metal conduit is permitted for initiating device circuits 6 feet in length or less. Flexible metal conduit is prohibited for notification appliance circuits and signaling line circuits. Use liquid tight flexible metal conduit in damp and wet locations.
- e. Schedule 40 (minimum) polyvinyl chloride (PVC) is permitted where conduit is routed underground or underground below floor slabs. Convert non-metallic conduit, other than PVC Schedule 40 or 80, to plastic-coated rigid, or IMC, steel conduit before turning up through floor slab.
- f. Exterior wall penetrations must be weathertight. Conduit must be sealed to prevent the infiltration of moisture.

3.3.5 Conductor Terminations

Labeling of conductors at terminal blocks in terminal cabinets, FACU, and remote FACU must be provided at each conductor connection. Each conductor or cable must have a shrink-wrap label to provide a unique and specific designation. Each terminal cabinet, FACU, and remote FACU must contain a laminated drawing that indicates each conductor, its label, circuit, and terminal. The laminated drawing must be neat, using 12 point lettering minimum size, and mounted within each cabinet, control unit, or unit so that it does not interfere with the wiring or terminals. Maintain existing color code scheme where connecting to existing equipment.

3.4 PAINTING

- a. In unfinished areas (including areas above drop ceilings), paint all exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceway, junction boxes and covers red. In

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lieu of painting conduit, the contractor may utilize red conduit with a factory applied finish.

- b. In finished areas, paint exposed electrical conduit (serving fire alarm equipment), fire alarm conduit, surface metal raceways, junction boxes, and electrical boxes to match adjacent finishes. The inside cover of the junction box must be identified as "Fire Alarm" and the conduit must have painted red bands 3/4-inch wide at 10-foot centers and at each side of a floor, wall, or ceiling penetration.
- c. Painting must comply with Section 09 90 00 PAINTS AND COATINGS.

3.5 FIELD QUALITY CONTROL

3.5.1 Test Procedures

Submit detailed test procedures, prepared and signed by the NICET Level III or IV Fire Alarm Technician, and the representative of the installing company, and reviewed by the QFPE 30 days prior to performing system tests. Detailed test procedures must list all components of the installed system such as initiating devices and circuits, notification appliances and circuits, signaling line devices and circuits, control devices/equipment, batteries, transmitting and receiving equipment, power sources/supply, annunciators, special hazard equipment, emergency communication equipment, interface equipment, and surge protective devices. Test procedures must include sequence of testing, time estimate for each test, and sample test data forms. The test data forms must be in a check-off format (pass/fail with space to add applicable test data; similar to the forms in NFPA 72 and NFPA 4.) The test procedures and accompanying test data forms must be used for the pre-Government testing and the Government testing. The test data forms must record the test results and must:

- a. Identify the NFPA Class of all Initiating Device Circuits (IDC), and Notification Appliance Circuits (NAC), and Signaling Line Circuits (SLC).
- b. Identify each test required by NFPA 72 Test Methods and required test herein to be performed on each component, and describe how these tests must be performed.
- c. Identify each component and circuit as to type, location within the facility, and unique identity within the installed system. Provide necessary floor plan sheets showing each component location, test location, and alphanumeric identity.
- d. Identify all test equipment and personnel required to perform each test (including equipment necessary for smoke detector testing. The use of magnets is not permitted.
- e. Provide space to identify the date and time of each test. Provide space to identify the names and signatures of the individuals conducting and witnessing each test.

3.5.2 Pre-Government Testing

3.5.2.1 Verification of Compliant Installation

Conduct inspections and tests to ensure that devices and circuits are

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functioning properly. Tests must meet the requirements of paragraph entitled "Minimum System Tests" as required by NFPA 72. The contractor and an authorized representative from each supplier of equipment must be in attendance at the pre-Government testing to make necessary adjustments. After inspection and testing is complete, provide a signed Verification of Compliant Installation letter by the QFPE that the installation is complete, compliant with the specification and fully operable. The letter must include the names and titles of the witnesses to the pre-Government tests. Provide all completion documentation as required by NFPA 72 including all referenced annex sections and the test reports noted below.

- a. NFPA 72 Record of Completion.
- b. NFPA 72 Record of Inspection and Testing.
- c. Fire Alarm and Emergency Communication System Inspection and Testing Form.
- d. Audibility test results with marked-up test floor plans.
- e. Documentation that all tests identified in the paragraph "Minimum System Tests" are complete.

3.5.2.2 Request for Government Final Test

When the verification of compliant installation has been completed, submit a formal request for Government final test to the Designated Fire Protection Engineer (DFPE). Government final testing will not be scheduled until the DFPE has received copies of the request for Government final testing and Verification of Compliant Installation letter with all required reports. Government final testing will not be performed until after the connections to the installation-wide fire reporting system has been completed and tested to confirm communications are fully functional. Submit request for test at least 15 calendar days prior to the requested test date.

3.5.3 Correction of Deficiencies

If equipment was found to be defective or non-compliant with contract requirements, perform corrective actions and repeat the tests. Tests must be conducted and repeated if necessary until the system has been demonstrated to comply with all contract requirements.

3.5.4 Government Final Tests

The tests must be performed in accordance with the approved test procedures in the presence of the DFPE. Furnish instruments and personnel required for the tests. The following must be provided at the job site for Government Final Testing:

- a. The manufacturer's technical representative.
- b. The contractor's Qualified Fire Protection Engineer (QFPE).
- c. Marked-up red line drawings of the system as actually installed.
- d. Loop resistance test results.

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- e. Complete program printout including input/output addresses.
- f. Copy of pre-Government Test Certificate, test procedures and completed test data forms.
- g. Audibility test results with marked-up floor plans.

Government Final Tests will be witnessed by the Designated Fire Protection Engineer and Qualified Fire Protection Engineer (QFPE). At this time, any and all required tests noted in the paragraph "Minimum System Tests" must be repeated at their discretion.

3.6 MINIMUM SYSTEM TESTS

3.6.1 System Tests

Test the system in accordance with the procedures outlined in NFPA 72. The required tests are as follows:

- a. Loop Resistance Tests: Measure and record the resistance of each circuit with each pair of conductors in the circuit short-circuited at the farthest point from the circuit origin. The tests must be witnessed by the Contracting Officer and test results recorded for use at the final Government test.
- b. Verify the absence of unwanted voltages between circuit conductors and ground. The tests must be accomplished at the pre-Government test with results available at the final system test.
- c. Verify that the control unit is in the normal condition as detailed in the manufacturer's O&M manual.
- d. Test each initiating device and notification appliance and circuit for proper operation and response at the control unit. Smoke detectors must be tested in accordance with manufacturer's recommended calibrated test method. Use of magnets is prohibited. Testing of duct smoke detectors must comply with the requirements of NFPA 72 except disconnect at least 20 percent of devices. If there is a failure at these devices, then supervision must be tested at each device.
- e. Test the system for specified functions in accordance with the contract drawings and specifications and the manufacturer's O&M manual.
- f. Test both primary power and secondary power. Verify, by test, the secondary power system is capable of operating the system for the time period and in the manner specified.
- g. Determine that the system is operable under trouble conditions as specified.
- h. Visually inspect wiring.
- i. Test the battery charger and batteries.
- j. Verify that software control and data files have been entered or programmed into the RFACU. Hard copy records of the software must be provided to the Contracting Officer.

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- k. Verify that red-line drawings are accurate.
- l. Measure the current in circuits to ensure there is the calculated spare capacity for the circuits.
- m. Measure voltage readings for circuits to ensure that voltage drop is not excessive.
- n. Disconnect the verification feature for smoke detectors during tests to minimize the amount of smoke needed to activate the sensor. Testing of smoke detectors must be conducted using real smoke or the use of canned smoke which is permitted.
- o. Measure the voltage drop at the most remote appliance (based on wire length) on each notification appliance circuit.
- p. Verify the documentation cabinet is installed and contains all as-built shop drawings, product data sheets, design calculations, site-specific software data package, and all documentation required by paragraph titled "Test Reports".

3.6.2 Audibility Tests

Sound pressure levels from audible notification appliances must be a minimum of 15 dBA over ambient with a maximum of 110 dBA in any occupiable area. The provisions for audible notification (audibility and intelligibility) must be met with doors, fire shutters, movable partitions, and similar devices closed.

3.7 SYSTEM ACCEPTANCE

Following acceptance of the system, as-built drawings and O&M manuals must be delivered to the Contracting Officer for review and acceptance. The drawings must show the system as installed, including deviations from both the project drawings and the approved shop drawings. These drawings must be submitted within two weeks after the final Government test of the system. At least one set of as-built (marked-up) drawings must be provided at the time of, or prior to the Final Government Test.

- a. The drawings must be prepared electronically and sized no less than the contract drawings. Furnish one set of CDs or DVDs containing software back-up and CAD based drawings in latest version of AutoCAD, DXF and portable document formats of as-built drawings and schematics.
- b. Include complete wiring diagrams showing connections between devices and equipment, both factory and field wired.
- c. Include a riser diagram and drawings showing the as-built location of devices and equipment.
- d. Provide Operation and Maintenance (O&M) Instructions.

3.8 INSTRUCTION OF GOVERNMENT EMPLOYEES

3.8.1 Instructor

Provide the services of an instructor, who has received specific training from the manufacturer for the training of other persons regarding the operation, inspection, testing, and maintenance of the system provided.

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The instructor must train the Government employees designated by the Contracting Officer, in the care, adjustment, maintenance, and operation of the fire alarm system. The instructor must be thoroughly familiar with all parts of this installation. The instructor must be trained in operating theory as well as in practical O&M work. Submit the instructors information and qualifications including the training history.

3.8.2 Required Instruction Time

Provide 8 hours of instruction after final acceptance of the system. The instruction must be given during regular working hours on such dates and times selected by the Contracting Officer. The instruction may be divided into two or more periods at the discretion of the Contracting Officer. The training must allow for rescheduling for unforeseen maintenance and/or fire department responses.

3.8.3 Technical Training Manual

Provide, in manual format, lesson plans, operating instructions, maintenance procedures, and training data for the training courses. The operations training must familiarize designated government personnel with proper operation of the installed system. The maintenance training course must provide the designated government personnel adequate knowledge required to diagnose, repair, maintain, and expand functions inherent to the system.

3.9 EXTRA MATERIALS

3.9.1 Repair Service/Replacement Parts

Repair services and replacement parts for the system must be available for a period of 10 years after the date of final acceptance of this work by the Contracting Officer. During the warranty period, the service technician must be on-site within 24 hours after notification. All repairs must be completed within 24 hours of arrival on-site.

During the warranty period, the installing fire alarm contractor is responsible for conducting all required testing and maintenance in accordance with the requirements and recommended practices of NFPA 72 and the system manufacturer. Installing fire alarm contractor is NOT responsible for any damage resulting from abuse, misuse, or neglect of equipment by the end user.

3.9.2 Spare Parts

Spare parts furnished must be directly interchangeable with the corresponding components of the installed systems. Spare parts must be suitably packaged and identified by nameplate, tagging, or stamping. Spare parts must be delivered to the Contracting Officer at the time of the Government testing and must be accompanied by an inventory list.

3.9.3 Document Storage Cabinet

Upon completion of the project, but prior to project close-out, place in the document storage cabinet copies of the following record documentation:

- a. As-built shop drawings
- b. Product data sheets

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- c. Design calculations
- d. Site-specific software data package
- e. All documentation required by SD-06.

-- End of Section --

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SECTION 31 00 00

EARTHWORK 08/23

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 180 (2017) Standard Method of Test for
Moisture-Density Relations of Soils Using
a 4.54-kg (10-lb) Rammer and a 457-mm
(18-in.) Drop

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C600 (2017) Installation of Ductile-Iron Mains
and Their Appurtenances

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M (2020; Errata 1 2021) Structural Welding
Code - Steel

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2018) Standard Specification for Concrete
Aggregates

ASTM C117 (2017) Standard Test Method for Materials
Finer than 75-um (No. 200) Sieve in
Mineral Aggregates by Washing

ASTM C136/C136M (2019) Standard Test Method for Sieve
Analysis of Fine and Coarse Aggregates

ASTM D698 (2012; E 2014; E 2015) Laboratory
Compaction Characteristics of Soil Using
Standard Effort (12,400 ft-lbf/cu. ft.
(600 kN-m/cu. m.))

ASTM D1140 (2017) Standard Test Methods for
Determining the Amount of Material Finer
than 75-µm (No. 200) Sieve in Soils by
Washing

ASTM D1557 (2012; E 2015) Standard Test Methods for
Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000
ft-lbf/ft³) (2700 kN-m/m³)

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ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D4253	(2016; E 2019) Standard Test Methods for Maximum Index Density and Unit Weight of Soils Using a Vibratory Table
ASTM D4254	(2016) Standard Test Methods for Minimum Index Density and Unit Weight of Soils and Calculation of Relative Density
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4829	(2021) Standard Test Method for Expansion Index of Soils
ASTM D5268	(2019) Topsoil Used for Landscaping Purposes

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual
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U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA 600/4-79/020	(1983) Methods for Chemical Analysis of Water and Wastes
EPA SW-846.3-3	(1999, Third Edition, Update III-A) Test Methods for Evaluating Solid Waste: Physical/Chemical Methods

1.2 DEFINITIONS

1.2.1 Structural Fill

Soil material placed to support buildings, walls, pads, and other similar facilities.

1.2.2 Embankment Fill

Soil material placed to construct embankment.

1.2.3 Porous Fill

Free-draining material placed for subsurface drainage, as a capillary break, or another specific purpose.

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1.2.4 Topsoil

Surface layer of primarily organic soil capable of supporting vegetation growth.

1.2.5 Utility Bedding Material

Fill placed to directly support pipes, conduits, cables, and appurtenant structures. Bedding may also be used to provide a cushion between utilities and bedrock, obstacles, obstructions and other unyielding materials.

1.2.6 Flowable Fill

Fill placed in a plastic or liquid form that flows to near its final placement location with limited assistance and subsequently cures or solidifies to provide a stable or impermeable barrier.

1.2.7 Satisfactory Materials

Satisfactory materials for fill, backfill, and/or any in-situ soils to remain in place comprise any materials classified by ASTM D2487 and as directed by the Geotechnical Engineer. Maximum particle size to be no greater than 3 inches in any dimension.

1.2.8 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials are unsatisfactory. Unsatisfactory materials also include man-made fills; trash; refuse; backfills from previous construction; roots and other organic matter or frozen material. Notify the Contracting Officer when encountering any contaminated materials.

1.2.9 Cohesionless Materials

Cohesionless materials include materials classified in ASTM D2487 as GW, GP, SW, and SP. Materials classified as GM and SM will be identified as cohesionless only when the fines are nonplastic. Perform testing, required for classifying materials, in accordance with ASTM D4318, ASTM C117, ASTM C136/C136M and ASTM D1140.

1.2.10 Cohesive Materials

Cohesive materials include materials classified as GC, SC, ML, CL, MH, and CH. Materials classified as GM and SM will be identified as cohesive only when the fines are plastic. Perform testing, required for classifying materials, in accordance with ASTM D4318, ASTM C117, ASTM C136/C136M and ASTM D1140.

1.2.11 Hard/Unyielding Materials

Hard/Unyielding materials comprise weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" with stones greater than 8 inches in any dimension or as defined by the pipe manufacturer, whichever is smaller. These materials usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

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1.2.12 Unstable Material

Unstable materials are too weak to adequately support the utility pipe, conduit, equipment, or appurtenant structure. Satisfactory material may become unstable due to ineffective drainage, dewatering, becoming frozen, excessive loading.

1.2.13 Expansive Soils

Expansive soils are defined as soils that have an expansion index greater than 20 when tested in accordance with ASTM D4829.

1.2.14 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling and blasting, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 1/2 cubic yard in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling and blasting that is performed merely to increase production.

1.2.15 Capillary Water Barrier

A layer of clean, poorly graded crushed rock, stone, or natural sand or gravel having a high porosity which is placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below a slab.

1.2.16 Degree of Compaction (Proctor)

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum density obtained by the test procedure presented in ASTM D1557 or ASTM D698 abbreviated as a percent of laboratory maximum density. Since ASTM D1557 applies only to soils that have 30 percent or less by weight of their particles retained on the 3/4 inch sieve, express the degree of compaction for material having more than 30 percent by weight of their particles retained on the 3/4 inch sieve as a percentage of the maximum density in accordance with AASHTO T 180-21 paragraph 1.5, Note 1.

1.2.17 Degree of Compaction (Relative Density)

Degree of compaction required for soils with less than 5 percent passing the No. 200 sieve, is expressed as a relative percentage of the maximum index density/dry unit weight and minimum index density/dry unit weight, obtained by the test procedures in accordance with ASTM D4253 and ASTM D4254, respectively, abbreviated as a percent of laboratory relative density.

1.2.18 Overhaul

Overhaul is the authorized transportation of satisfactory excavation or borrow materials in excess of the free-haul limit as determined by the COR. Overhaul is the product of the quantity of materials hauled beyond the free-haul limit, and the distance such materials are hauled beyond the free-haul limit, expressed in station yards.

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1.2.19 Borrow

Soil brought to the project site from an external location for the purposes of project construction.

1.2.20 Subgrade

Earth materials directly below foundations and directly below granular base materials in building slab and pavement areas including shoulders.

1.3 SUBSURFACE DATA

Subsurface soil boring logs are shown in project plans. These data represent available subsurface information; however, variations may exist between boring locations.

1.4 CRITERIA FOR BIDDING

Base bids on the following criteria:

- a. Surface elevations are as indicated.
- b. Pipes or other artificial obstructions, except those indicated, will not be encountered.
- c. Ground water elevations indicated by the boring log were those existing at the time subsurface investigations were made and do not necessarily represent ground water elevation at the time of construction.
- d. Material character is indicated by the boring logs.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Excavation and Trenching Plan; G-D0

Borrow Plan; G-D0

Dewatering Work Plan; G-D0

Jacking, Boring, and Tunneling Plan; G-D0

Preconstruction Meeting; G-D0

SD-03 Product Data

Geotextiles

SD-06 Test Reports

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Borrow Site Testing; G-D0

Pipe Inspection Report; G-D0

Geotechnical Evaluation Report; G-D0

1.6 QUALITY CONTROL

1.6.1 Geotechnical Engineer

Provide a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer is responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer is responsible for preparing and updating the Excavation and Trenching Plan and Dewatering Work Plan as construction progresses to reflect changing conditions and submit an updated plan if necessary. Submit a monthly Geotechnical Evaluation report, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Contractor is responsible for arranging meetings with the Geotechnical Engineer and Contracting Officer throughout the contract duration.

1.6.2 Qualified Technician

Provide a Qualified Technician to inspect, monitor, sample, and performing field testing. The technician qualifications need to be one of the following: a current National Institute for Certification in Engineering Technologies (NICET) Level II minimum certification in Construction Materials Testing Soils; a Geologist-in-Training with minimum one-year experience; an Engineer-in-Training with minimum one-year experience; a Registered Geologist; or a Professional Engineer.

1.6.3 Lab Validation

Perform testing by a Corps validated commercial testing laboratory or Contractor established testing laboratory meeting the requirements of Section 01 45 04.10 06 (or similar number) entitled CONTRACTOR QUALITY CONTROL and approved by the Contracting Officer. Submit testing laboratory validation for the testing to be performed. Do not permit work requiring testing until testing facilities have been inspected, Corps validated and approved by the Contracting Officer.

1.6.4 Preconstruction Meeting

Conduct a preconstruction meeting at the jobsite at least five business days prior to the start of earthwork operations on the project. The preconstruction meeting is to be arranged by the Contractor and is to follow the written agenda submitted prior to the meeting. The purpose of this meeting is to review the requirements of this specification and the associated plans. The following individuals must be in attendance at this meeting: Contractor's Project Manager and Project Superintendent, earthwork subcontractor's Project Manager and Site Foreman, Contractor's Geotechnical Engineer and Testing Agency, Government Geotechnical Engineer and Civil Engineer, and Government Construction Manager and Engineering Technician.

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The minutes of this meeting are to be recorded by the Contractor and published via email within 48 hours to all attendees. The minutes must be re-published within 48 hours via email pending any subsequent comments from the attendees.

PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Structural Fill

Materials classified in accordance with ASTM D2487. Select material type appropriate for the intended purpose.

2.1.2 Embankment Fill

Materials classified in accordance with ASTM D2487. Select material type appropriate for the intended purpose.

2.1.3 Porous Fill

Materials containing less than 5 percent passing the No. 200 sieve. Provide the gradation as appropriate for the intended purpose.

2.1.4 Topsoil

Material suitable for topsoil obtained from offsite areas is defined as: Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than one inch diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. Amend topsoil pH range to obtain a pH of 5.5 to 7. Topsoil material will be in accordance with ASTM D5268.

2.1.5 Capillary Water Barrier

Conform to ASTM C33/C33M for fine aggregate grading with a maximum of 3 percent by weight passing ASTM D1140, No. 200 sieve.

2.1.6 Utility Bedding Material

Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with AWWA C600 and ASTM D2321. Install bedding for plastic piping to spring line of pipe. Provide geotextile fabric below bedding layer where indicated. Utility bedding material may include the following:

2.1.6.1 Class I

Angular, 0.25 to 1.5 inch, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.

2.1.6.2 Class II

Coarse sands and gravels with maximum particle size of 1.5 inch, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types GW, GP, SW, and SP are included in this class as specified in ASTM D2487.

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2.1.6.3 Sand

Clean, coarse-grained sand classified by ASTM D2487.

2.1.6.4 Gravel and Crushed Stone

Clean, coarsely graded natural gravel, crushed stone or a combination thereof having a classification of GW GP in accordance with ASTM D2487. Do not exceed maximum particle size of 3 inches.

2.2 BURIED WARNING AND IDENTIFICATION MARKERS

Provide polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 3 inches minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Provide permanent color and printing, unaffected by moisture or soil.

Warning Tape Color Codes	
Red	Electric
Yellow	Gas, Oil; Dangerous Materials
Orange	Telephone and Other Communications
Blue	Water Systems
Green	Sewer Systems
White	Steam Systems
Gray	Compressed Air

2.2.1 Warning Tape for Metallic Piping

Provide acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.003 inch and a minimum strength of 1500 psi lengthwise, and 1250 psi crosswise, with a maximum 350 percent elongation.

2.2.2 Detectable Warning Tape for Non-Metallic Piping

Provide polyethylene plastic tape conforming to the width, color, and printing requirements specified above, with a minimum thickness of 0.004 inch, and a minimum strength of 1500 psi lengthwise and 1250 psi crosswise. Manufacture tape with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 3 feet deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

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2.2.3 Detection Wire for Non-Metallic Piping

Insulate a single strand, solid copper detection wire with a minimum of 12 AWG.

2.3 MATERIAL FOR RIP-RAP

Provide bedding material, grout filter, fabric and rock conforming to TXDOT for construction as indicated.

2.3.1 Bedding Material

Provide bedding material consisting of sand, gravel, or crushed rock, well graded, with a maximum particle size of 2 inches. Compose material of tough, durable particles. Allow fines passing the No. 200 standard sieve with a plasticity index less than 6.

2.3.2 Grout

Provide durable grout composed of cement, water, an air-entraining admixture, and sand mixed in proportions of one-part Portland cement to two parts of sand, sufficient water to produce a workable mixture, and an amount of admixture which will entrain sufficient air, as determined by the Contracting Officer. Mix grout in a concrete mixer. Allow a sufficient mixing time to produce a mixture having a consistency permitting gravity flow into the interstices of the riprap with limited spading and brooming.

2.3.3 Rock

Provide rock fragments which ensure permanence in the structure and the environment in which it is to be used. Use rock fragments free from cracks, seams, and other defects that would increase the risk of deterioration from natural causes. Provide fragments sized such that no individual fragment exceeds a weight of 150 pounds and that no more than 10 percent of the mixture, by weight, consists of fragments weighing 2 pounds or less each. Provide rock with a minimum specific gravity of 2.50 . Do not permit the inclusion of more than trace 1 percent quantities of dirt, sand, clay and rock fines.

2.4 BORROW

Provide borrow materials from sources located within and/or outside of Government property meeting the requirements of paragraph STRUCTURAL FILL, EMBANKMENT FILL, and TOPSOIL.

2.5 GEOTEXTILE

Provide a pervious sheet of polyester, nylon, glass or polypropylene ultraviolet resistant filaments woven, spun bonded, fused, or otherwise manufactured into a non-raveling fabric with uniform thickness and strength. Fabric must have manufacturer certified minimum average roll properties that conform with TXDOT requirements. Submit a sample and material product data for all Geotextiles utilized.

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PART 3 EXECUTION

3.1 PROTECTION

Perform all work specified in accordance with applicable requirements of the Corps of Engineers publication EM 385-1-1 Safety and Health Requirements Manual. Provide a Geotechnical Engineer to monitor construction activities and to prepare necessary work plans and reports; see paragraph QUALITY CONTROL.

Use equipment of type and size appropriate for the site conditions (soil character and moisture content). Maintenance of exposed subgrades and fills is the responsibility of the Contractor. The Contractor is required to prevent damage by ineffective drainage, dewatering, and heavy loads and equipment by implementing precautionary measures. Repair or replace any defects or damage.

3.1.1 Underground Utilities

Location of the existing utilities indicated is approximate. Physically verify the location and elevation of the existing utilities indicated prior to starting construction. The Contractor is responsible for protecting utilities from damage during construction.

3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

Provide for the collection and disposal of surface and subsurface water encountered during construction. Construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity and provide temporary ditches, swales, and other drainage features and equipment as required to keep soils from becoming unstable, prevent erosion, or undermining of foundations. Remove unstable material from working platforms for equipment operation and soil support for subsequent construction features and provide new material as specified herein. It is the responsibility of the Contractor to assess the site conditions to employ necessary measures to permit construction to proceed.

3.1.2.2 Dewatering

Control groundwater flowing toward or into excavations to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches are not allowed within 3 feet of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Perform control measures by the time the excavation reaches the water level in order to maintain the integrity of the in-situ material. While the excavation is open, maintain the water level continuously, at least 3 feet below the working level. Submit a Dewatering Work Plan outlining procedures for accomplishing dewatering work. Operate dewatering system continuously until construction work below existing water levels is complete. Measure

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and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system.

3.1.3 Shoring and Sheet piling

Submit an Excavation and Trenching Plan to stabilize features, prevent undermining or unintended horizontal and vertical movement of adjacent structures, and prevent slippage or movement in banks or slopes adjacent to the excavation. Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheet piling of excavations. Drawings to include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations are to include data and references used.

3.1.4 Protection of Graded Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.2 BORROW

Select borrow material to meet the requirements and conditions of the fill or embankment for which it is to be used. Obtain borrow material from COR approved private sources. Submit a Borrow Plan that includes materials to be excavated, stockpile locations, proposed slopes, drainage, and closure.

Unless otherwise provided in the contract, the Contractor is responsible for obtaining the right to procure material, pay royalties and other charges involved, and bear the expense of developing the sources, including rights-of-way for hauling from the owners. Unless specifically provided, do not obtain borrow within the limits of the project site without prior written approval.

3.2.1 Government Furnished Borrow Area(s)

3.2.1.1 Stripping and Stockpiling Operations in Borrow Area

Strip in accordance with paragraph STRIPPING. Strip at least 5 feet beyond the limits of the borrow excavation and any stockpiles of fill and embankment materials.

Stockpile materials within the borrow area work limits such that the stockpiles does not interfere with borrow operations. Stockpile borrow material awaiting transport in approved segregated piles. Maintain a minimum of 30 feet between all stockpile toes and the top of the borrow cut.

3.2.1.2 Drainage of Borrow Excavations

Provide adequate drainage of borrow area. Ensure that borrow operations result in minimum detrimental effects on natural environmental conditions.

3.2.1.3 Borrow Area Closure

Complete borrow areas final grading, so that slopes are not steeper than 3 vertical on 1 horizontal, except as otherwise indicated. Avoid abrupt changes in grade. Distribute stripped material and stockpiles of unstable materials over the disturbed borrow area, as directed. Final grade the

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borrow area to drain.

3.2.2 Contractor Furnished Borrow Area(s)

Obtain approved borrow materials from approved offsite sources. If a borrow source is selected that is not a commercial entity from which soil material is directly purchased, submit a Borrow Plan that includes the borrow source location, geotechnical test results showing the fill material meets the Contract requirements, environmental test results in accordance with paragraph ENVIRONMENTAL REQUIREMENTS FOR OFF-SITE SOIL, and any Federal, State, and local permits required for excavation and reclamation of the borrow area.

3.2.3 Environmental Requirements for Off-Site Soil

Test offsite soils brought in for use as backfill for Total Petroleum Hydrocarbons (TPH), Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and full Toxicity Characteristic Leaching Procedure (TCLP) including ignitability, corrosivity and reactivity. Backfill may not contain concentrations above appropriate State and EPA criteria, and for hazardous waste characteristics 10 parts per million (ppm) of total petroleum hydrocarbons (TPH) and 10 ppm of the sum of Benzene, Toluene, Ethyl Benzene, and Xylene (BTEX) and pass the TCPL test. Determine TPH concentrations by using EPA 600/4-79/020 Method 418.1. Determine BTEX concentrations by using EPA SW-846.3-3 Method 5030/8020/8260B. Perform TCLP in accordance with EPA SW-846.3-3 Method 1311. Perform hazardous waste characteristic tests for ignitability, corrosivity, and reactivity in accordance with accepted standard methods. Perform PCB testing in accordance with accepted standard methods for sampling and analysis of bulk solid samples. Provide borrow site testing for petroleum hydrocarbons and BTEX from a grab sample of material from the area most likely to be contaminated at the borrow site (as indicated by visual or olfactory evidence), with at least one test from each borrow site. Provide borrow site testing for hazardous waste characteristics (TPH, BTEX and TCLP) from a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring borrow material to project site until Borrow Plan containing environmental test results has been received and approved by the Contracting Officer.

3.3 SURFACE PREPARATION

3.3.1 Clearing and Grubbing

Clear and grub as specified in Section 31 11 00 CLEARING AND GRUBBING.

3.3.2 Stripping

Strip site where indicated on the plans. Strip existing surface materials to a depth of 4 inches below the existing ground surface in areas designated as Clear and Grub on the plans. Strip existing surficial soils to a depth of 4 inches in all other areas. Strip in all areas within the planned limits of disturbance. All stripped materials not suitable for reuse as topsoil will be wasted in specified disposal area. Screen all stripped soils to remove roots and organic materials prior disposal.

Strip suitable soil from the site where excavation or grading is indicated and stockpile separately from other excavated material. Protect topsoil and keep in segregated piles until needed.

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3.3.3 Proof Rolling

Perform proof rolling on exposed subgrade that is unfrozen and free of surface water (wet conditions resulting from rainfall). Notify the Contracting Officer a minimum of three days prior to proof rolling. Perform proof rolling in the presence of the Contracting Officer.

After stripping, excavating, and rough grading to the planned elevation, proof roll the existing subgrade of all building, pavement and embankment locations with six passes of a 15 ton, pneumatic-tired or smooth drum roller. Operate the roller truck in a systematic manner to ensure the number of passes over all areas, and at speeds between 2.5 to 3.5 miles per hour. Subgrade materials that exhibit excessive deflection and/or rutting during proof rolling need to be scarified, aerated, and re-compacted to specified density at 2 percent of optimum moisture content prior to being considered for remedial action by the Contracting Officer. When proof rolling under buildings, the building subgrade is considered to extend 5 feet beyond the building lines, and make one-half of the passes with the roller in a direction perpendicular to the other passes.

3.3.4 Stockpiling Operations

Place and grade stockpiles of satisfactory as specified. Keep stockpiles in a neat and well drained condition, giving due consideration to drainage at all times. Clear, grub, and seal by rubber-tired equipment, the ground surface at stockpile locations; separately stockpile excavated satisfactory and unsatisfactory materials. Protect stockpiles of satisfactory materials from contamination which may destroy the quality and fitness of the stockpiled material. Do not create stockpiles that could obstruct the flow of any stream, endanger a partly finished structure, impair the efficiency or appearance of any structure, or be detrimental to the completed work in any way. If the Contractor fails to protect the stockpiles, and any material becomes unsatisfactory, remove and replace such material with satisfactory material from approved sources.

3.4 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Excavate soil disturbed or weakened by Contractor's operations, and soils softened or made unstable for subsequent construction due to exposure to weather. Use material removed from excavations meeting the specified requirements in the construction of fills, embankments, subgrades, shoulders, bedding (as backfill), and for similar purposes to minimize surplus material and to minimize additional material brought on site. Do not excavate below indicated depths except to remove unstable material as determined by the Government and confirmed by the Contracting Officer. Remove and replace excavations below the grades shown with appropriate materials as directed by the Contracting Officer.

If at any time during excavation, including excavation from borrow areas, the Contractor encounters material that may be classified as rock or as hard/unyielding material, uncover such material, and notify the Contracting Officer. Do not proceed with the excavation of this material until the Contracting Officer has classified the materials as common excavation or rock excavation. Failure on the part of the Contractor to uncover such material, notify the Contracting Officer, and allow sufficient time for classification and delineation of the undisturbed surface of such material will cause the forfeiture of the Contractor's right of claim to any classification or volume of material to be paid for

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other than that allowed by the Contracting Officer for the areas of work in which such deposits occur.

3.4.1 Ditches, Gutters, and Channel Changes

Finish excavation of ditches, gutters, and channel changes by cutting accurately to the cross sections, grades, and elevations shown. Do not excavate below grades shown. Backfill excessive excavation as directed by the Contracting Officer, with satisfactory, compacted, material or with suitable stone or cobble to grades shown. Dispose excavated material as shown or as directed. Do not allow material to be deposited within 4 feet from edge of a ditch. Maintain excavations free from detrimental quantities of leaves, brush, sticks, trash, and other debris until final acceptance of the work.

3.4.2 Trench Excavation Requirements

Excavate the trench as recommended by the manufacturer of the pipe to be installed. Slope trench walls below the top of the pipe, or make vertical, and of such width as recommended by the manufacturer. Provide vertical trench walls where no manufacturer installation instructions are available. Do not exceed the trench width of 24 inches below the top pipe plus pipe outside diameter (O.D.) for pipes of less than 24 inches inside diameter, and do not exceed 36 inches plus pipe outside diameter for pipe sizes larger than 24 inches inside diameter. Where recommended trench widths are exceeded, provide redesign, stronger pipe, or special installation procedures. The Contractor is responsible for the cost of redesign, stronger pipe, or special installation procedures without any additional cost to the Government.

3.4.2.1 Bottom Preparation

Grade the bottoms of trenches accurately to provide uniform bearing and support for the bottom quadrant of each section of the pipe. Excavate bell holes to the necessary size at each joint or coupling to eliminate point bearing. Remove stones of 2 inch or greater in any dimension, or as recommended by the pipe manufacturer, whichever is smaller, to avoid point bearing.

3.4.2.2 Removal of Unyielding Material

Where overdepth is not indicated and unyielding material is encountered in the bottom of the trench, notify the Contracting Officer. Following approval, remove such material 4 inch below the required grade and replaced with suitable materials as provided in paragraph FILLING AND COMPACTION.

3.4.2.3 Removal of Unstable Material

Where unstable material is encountered in the bottom of the trench, remove such material to the depth directed and replace it to the proper grade with suitable material as provided in paragraph FILLING AND COMPACTION. When removal of unstable material is required due to the Contractor's fault or neglect in performing the work, the Contractor is responsible for excavating the resulting material and replacing it without additional cost to the Government.

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3.4.2.4 Excavation for Appurtenances

Provide excavation for manholes, catch-basins, inlets, or similar structures sufficient to leave at least 12 inches clear between the outer structure surfaces and the face of the excavation or support members.

3.4.2.5 Gas Distribution

Excavate trenches to a depth that will provide a minimum 18 inches of cover in rock excavation and a minimum 24 inches of cover in other excavation.

3.4.2.6 Water Lines

Excavate trenches to a depth that provides a minimum cover of 3 feet from the existing ground surface, or from the indicated finished grade, whichever is lower, to the top of the pipe.

3.4.3 Jacking, Boring, and Tunneling

Unless otherwise indicated, provide excavation by open cut except those sections requested and if, in the opinion of the Contracting Officer, can be safely and properly installed and backfill. Provide a Jacking, Boring, and Tunneling Plan, signed and sealed by a licensed Professional Engineer that includes a work site layout, methods and procedures, equipment, alignment control, monitoring, and contingency for responding to unplanned movements.

3.4.3.1 Pipeline Casing

Provide natural gas pipeline casing per specifications by the Mechanical Engineer and as shown on the Mechanical Drawings.

3.4.3.2 Bore Holes

Mechanically bore holes and case through the soil with a cutting head on a continuous auger mounted inside the casing pipe. Weld lengths of pipe together in accordance with AWS D1.1/D1.1M. Do not use water or other fluids in connection with the boring operation.

3.4.3.3 Cleaning

Clean inside of the pipeline casing of dirt, weld splatters, and other foreign matter which would interfere with insertion of the piped utilities by attaching a pipe cleaning plug to the boring rig and passing it through the pipe.

3.4.3.4 End Seals

After installation of piped utilities in pipeline casing, provide watertight end seals at each end of pipeline casing between pipeline casing and piping utilities. Provide watertight end seals as indicated.

3.4.4 Underground Utilities

Perform work adjacent to utilities in accordance with procedures outlined by utility owner. Excavation made with power-driven equipment is not permitted within 2 feet of known utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or

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other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

3.4.5 Rock Excavation

Excavate rock encountered in the cut section to a depth of 6 inches below finished grade and replace with satisfactory material.

3.5 SUBGRADE PREPARATION

3.5.1 General Requirements

Shape subgrade to line, grade, and cross section as indicated. Remove unsatisfactory and unstable material in surfaces to receive fill or in excavated areas, as determined by proof rolling, and replaced with satisfactory materials. Do not place material on surfaces that are muddy, frozen, contain frost, or otherwise containing unstable material. Scarify the surface to a depth of 4 inches prior to placing fill. Step or bench sloped surfaces steeper than 1 vertical to 4 horizontal prior to scarifying. Place 4 inches of loose fill and blend with scarified material. When subgrade is part fill and part excavation or natural ground, scarify to a depth of 8 inches.

3.5.2 Subgrade for Structures, Spread Footings, and Concrete Slabs

Do not excavate below depth shown for structures, spread footings, and concrete slabs. If over excavation occurs, notify the Contracting Officer and remove, replace, and compact as directed. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.3 Subgrade for Pavements

Compact top 12 inches of subgrade for pavements to at least 95 percent of ASTM D698. After final rolling, the surface of the subgrade for buildings and pavements must not show deviations greater than 0.05 foot when tested with a 12-foot straightedge applied both parallel and at right angles to the centerline of the area.

3.5.4 Subgrade Filter Fabric

Place filter fabric as indicated directly on prepared subgrade free of vegetation, stumps, rocks larger than 2 inch diameter and other debris which may puncture or otherwise damage the fabric. Repair damaged fabric by placing an additional layer of fabric to cover the damaged area a minimum of 3 feet overlap in all directions. Overlap fabric at joints a minimum of 3 feet. Obtain approval of filter fabric installation before placing fill or backfill. Place fill or backfill on fabric in the direction of overlaps and compact as specified herein. Follow manufacturer's recommended installation procedures.

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3.6 FILLING AND COMPACTION

Prepare ground surface on which backfill is to be placed and provide compaction requirements for backfill materials in conformance with the applicable portions of paragraphs for SUBGRADE PREPARATION. Do not place material on surfaces that are muddy, frozen, or contain frost. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Moisten material as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used. Fill and backfill to contours, elevations, and dimensions indicated. Compact and test each lift before placing overlaying lift.

3.6.1 Trench Backfill

Backfill trenches to the grade shown. Do not backfill the trench until all specified tests are performed.

3.6.1.1 Replacement of Unyielding Material

Replace unyielding material removed from the bottom of the trench with satisfactory material or initial backfill material.

3.6.1.2 Replacement of Unstable Material

Replace unstable material removed from the bottom of the trench or excavation with satisfactory material placed in layers not exceeding 6 inches loose thickness.

3.6.1.3 Bedding and Initial Backfill

Place initial backfill material and compact it with approved tampers to a height of at least one foot above the utility pipe or conduit. Bring up the backfill evenly on both sides of the pipe for the full length of the pipe. Take care to ensure thorough compaction of the fill under the haunches of the pipe. Except where shown or when specified otherwise in the individual piping section, provide bedding for buried piping in accordance with PART 2 paragraph UTILITY BEDDING MATERIAL. Compact backfill to top of pipe to 95 percent of ASTM D698 maximum density. Provide plastic piping with bedding to spring line of pipe.

3.6.1.4 Final Backfill

Do not begin backfill until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Bring backfill to indicated finish grade. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 2 feet above sewer lines and one foot above other utility lines need to be free from stones larger than one inch in any dimension. Heavy equipment for spreading and compacting backfill are not to be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; compact remaining area in layers not more than 4 inches in compacted thickness with power-driven hand tampers suitable for the material being compacted. Place backfill carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Do not place backfill against foundation walls prior to 7 days after completion of the walls. As far as practicable, bring backfill up evenly on each side of the wall and sloped to drain away from the wall.

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3.6.1.4.1 Turfed or Seeded Areas and Miscellaneous Areas

Deposit backfill in layers of a maximum of 12 inches loose thickness, and compact it to 85 percent maximum density for cohesive soils and 90 percent maximum density for cohesionless soils. Do not permit compaction by water flooding or jetting. Apply this requirement to all other areas not specifically designated above.

3.6.1.5 Electrical Distribution System

Provide a minimum cover of 24 inches from the finished grade to direct burial cable and conduit or duct line, unless otherwise indicated.

3.6.1.6 Displacement of Features

After other required tests have been performed and the trench backfill compacted to the finished grade surface, inspect the pipe to determine whether unexpected or damaging displacement has occurred. Conduct walk-through inspection of pipe sizes larger than 48 inches. Inspect pipes smaller than 48 inches using remote methods using closed circuit television, sonar, or hybrid that can provide a 360-degree inspection of the pipe. Prepare and submit a pipe inspection report consisting of digital video or photos. If, in the judgment of the Contracting Officer, the interior of the pipe shows poor alignment or any other defects that would cause improper functioning of the system, replace or repair the defects as directed at no additional cost to the Government.

3.6.1.7 Buried Tape And Detection Wire

3.6.1.7.1 Buried Warning and Identification Tape

Provide buried utility lines with utility identification tape. Bury tape 12 inches below finished grade; under pavements and slabs, bury tape 6 inches below top of subgrade.

3.6.1.7.2 Buried Detection Wire

Bury detection wire directly above non-metallic piping at a distance not to exceed 12 inches above the top of pipe. Extend the wire continuously and unbroken, from manhole to manhole. Terminate the ends of the wire inside the manholes at each end of the pipe, with a minimum of 3 feet of wire, coiled, remaining accessible in each manhole. Furnish insulated wire over its entire length. Install wires at manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, terminate the wire in the valve pit at the pump station end of the pipe.

3.6.2 Structural Fill Placement

Place fill and backfill beneath and adjacent to structures in successive horizontal layers of loose material not more than 8 inches in depth, or in loose layers not more than 4 inches in depth when using hand-operated compaction equipment. Do not place over wet or frozen materials. Compact to at least 90 percent of laboratory maximum density for cohesive materials or 95 percent of laboratory maximum density for cohesionless materials, except as otherwise specified. Perform compaction in such a manner as to prevent wedging action or eccentric loading upon or other damage to the structure. Moisture condition fill and backfill material to

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within range of plus 2 or minus 2 percent of optimum moisture content at the time of compaction.

3.6.3 Backfill for Appurtenances

After the manhole, catchbasin, inlet, or similar structure has been constructed and the concrete has been allowed to cure for 7 days, place backfill in such a manner that the structure is not be damaged by the shock of falling earth. Deposit the backfill material, compact it as specified for final backfill, and bring up the backfill evenly on all sides of the structure to prevent eccentric loading and excessive stress.

3.6.4 Porous Fill Placement

Provide under floor and area-way slabs on a compacted subgrade. Place in a single lift and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.6.5 Flowable Fill

Place fill in a manner to completely fill voids in the location indicated. Do not place when atmospheric temperatures are expected to be below 33 degrees F at any time during the 3 day period following placement.

3.6.6 Compaction

3.6.6.1 General Site

Compact underneath areas designated for vegetation and areas outside the 5 foot line of the paved area or structure to 85 percent of ASTM D698.

3.6.6.2 Adjacent Areas

Compact areas within 5 feet of structures to 95 percent of ASTM D698.

3.7 EMBANKMENTS

3.7.1 Earth Embankments

Construct earth embankments from satisfactory materials free of organic or frozen material and rocks with any dimension greater than 3 inches. Place the material in successive horizontal layers of loose material not more than 8 inches in depth. Spread each layer uniformly on a soil surface that has been moistened or aerated as necessary and scarified or otherwise broken up so that the fill will bond with the surface on which it is placed. After spreading, plow, disk, or otherwise break up each layer; moisten or aerate as necessary; thoroughly mix; and compact to at least 90 percent laboratory maximum density for cohesive materials or 95 percent laboratory maximum density for cohesionless materials. Backfill and fill material are to be within the range of minus 2 to plus 2 percent of optimum moisture.

Compaction requirements for the upper portion of earth embankments forming subgrade for pavements are identical with those requirements specified in paragraph SUBGRADE PREPARATION. Finish compaction by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, vibratory compactors, or other approved equipment.

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3.8 RIP-RAP CONSTRUCTION

Construct rip-rap in accordance with TXDOT in the areas indicated. Trim and dress indicated areas to conform to cross sections, lines and grades shown within a tolerance of 0.1 foot.

3.8.1 Bedding Placement

Spread filter fabric and bedding material uniformly to a thickness of at least 3 inches on prepared subgrade as indicated.

3.8.2 Stone Placement

Place rock for rip-rap on prepared bedding material to produce a well graded mass with the minimum practicable percentage of voids in conformance with lines and grades indicated. Distribute larger rock fragments, with dimensions extending the full depth of the rip-rap throughout the entire mass and eliminate "pockets" of small rock fragments. Rearrange individual pieces by mechanical equipment or by hand as necessary to obtain the distribution of fragment sizes specified above.

3.9 FINISHING/FINISH OPERATIONS

During construction, keep embankments and excavations shaped and drained. Maintain ditches and drains along subgrade to drain effectively at all times. Do not disturb the finished subgrade by traffic or other operation. Protect and maintain the finished subgrade in a satisfactory condition until ballast, subbase, base, or pavement is placed. Do not permit the storage or stockpiling of materials on finished subgrade. Do not lay subbase, base course, ballast, or pavement until the subgrade has been checked and approved, and in no case place subbase, base, surfacing, pavement, or ballast on a muddy, spongy, frozen or otherwise unstable subgrade.

Finish the surface of excavations, embankments, and subgrades to a smooth and compact surface in accordance with the lines, grades, and cross sections or elevations shown. Provide the degree of finish for graded areas within 0.1 foot of the grades and elevations indicated except as indicated for subgrades specified in paragraph SUBGRADE PREPARATION. Finish gutters and ditches in a manner that will result in effective drainage. Finish the surface of areas to be turfed to a smoothness suitable for the application of turfing materials. Repair graded, topsoiled, or backfilled areas prior to acceptance of the work, and re-established grades to the required elevations and slopes.

3.9.1 Capillary Water Barrier

Place a capillary water barrier under concrete floor and area-way slabs grade directly on the subgrade and compact with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.9.2 Grading Around Structures

Construct areas within 5 feet outside of each building and structure line true-to-grade, shape to drain, and maintain free of trash and debris until final inspection has been completed and the work has been accepted.

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3.9.3 Grading

Finish grades as indicated within one-tenth of one foot. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

3.9.4 Topsoil and Seed

Provide as specified in Section 32 92 19 SEEDING.

3.10 DISPOSITION OF SURPLUS MATERIAL

Remove from Government property all surplus or other soil material not required or not suitable for filling or backfilling, along with brush, refuse, stumps, roots, and timber. Properly disposed of in accordance with all applicable laws and regulations.

-- End of Section --

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SECTION 31 10 00

SITE CLEARING 02/21

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Clearing (Timber and Structure)

1.1.1.1 Payment

Payment will be made for costs associated with furnishing plant, labor, materials and equipments, and performing all operations necessary for clearing (timber and structures) as specified.

1.1.1.2 Unit of Measure

Unit of measure: lump sum.

1.2 DEFINITIONS

1.2.1 Trees

The line of demarcation between brush and trees, for the purpose of distinguishing clearing requirements, is that trees, as used, will be considered as that woody growth not falling within the limits of brush as defined below.

1.2.2 Brush

Brush is that growth which is less than 2 inches in diameter measured 6 inches from the ground on the uphill side and is less than 6 feet in height measured from the ground on the uphill side.

1.2.3 Structures

The term "structures" includes buildings or portions thereof, walls, silos, storm or root cellars, cisterns, wells, windmills, pit silos, water towers, etc. Remove or fill structures to the ground surface.

1.3 PROJECT/SITE CONDITIONS

1.3.1 Aesthetics and Pollution Control

1.3.1.1 Ground Areas

All ground areas in the zone of normal pool level fluctuations which are disturbed by clearing operations and which would become subject to erosion will be protected or restored.

1.3.1.2 Construction Roads

All construction roads proposed for use by the Contractor for removing salvaged timber or for access to the work area must be approved, as to location and alignment, prior to construction. Where such roads are

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determined to be of no value to project operation or will not serve recreational access needs after project construction, the areas occupied by these roads will be restored as nearly as possible to pre-construction conditions by reasonable grading and seeding of a native cover crop or seed mix as approved by TXDEQ along with the planting of seedling trees if in a tree cover area.

1.3.2 Existing Conditions

1.3.2.1 Boundaries

The areas to be cleared are shown on the plan drawings.

1.3.2.2 Fences

No fences shall be cleared as part of the proposed project. Temporary removal of fencing is permitted as approved by the COR.

1.3.2.3 Structures

Where filling of structures is required, fill to within 18 inches of the ground surface and made with noncombustible materials such as masonry rubble, and other debris. When all available debris has been used in filling, all remaining unfilled portions, together with the above 18 inches, completely fill to the ground surface with earth, borrowed as directed by the Contracting Officer. The top surfaces of fills must be neat in appearance and smooth enough not to constitute a hazard to persons or livestock.

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

3.1 CLEARING REQUIREMENTS

Clear brush and trees as shown on the plan drawings.

3.1.1 Equipment

A tree crushing machine may be used at the option of the Contractor in all clearing operations.

3.2 DISPOSAL OF MATERIAL

3.2.1 General

Completely remove material cleared from the areas by transporting from the Government property approved by the Contracting Officer. All timber from which saw logs, posts, ties or cordwood can be produced will become the property of the Contractor and in the interest of conservation it is required that the Contractor make a reasonable effort to dispose of material for these purposes. The Contractor may cut timber into convenient lengths at the site but approval must be secured prior to the operation of saw mills within the Government lands. Do not throw or leave cleared material in the creeks or river. Remove all felled timber. However, it is intended that all existing down timber will remain in place except dispose solid, floatable material that is larger than 4 inches in

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diameter (regardless of length) and/or over 8 feet in length (regardless of diameter) in the manner prescribed for cleared material. Clean up floatable debris. The cutting of branches and debris remaining after clean-up, to reduce their length in order to avoid removal, will not be permitted.

3.2.2 Removal from Site

Except as otherwise provided, the Contractor will be permitted to remove felled and trimmed timber from the site of the work. The Contractor will be allowed to stockpile salvaged timber near contour line at the laydown storage area as shown on the plan drawings. The Government will assume no responsibility for the protection and safekeeping of such material. All stockpiled timber must be removed from Government lands before final acceptance of the work will be made.

3.3 DEBRIS

Dispose burn combustible debris in accordance with paragraph DISPOSAL OF MATERIAL. Debris includes trash of all kinds.

3.4 MARKETABLE MATERIALS

Any of the cleared materials which the Contractor considers marketable becomes its property and remove from the reservoir area.

-- End of Section --

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SECTION 31 11 00

CLEARING AND GRUBBING 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. DEPARTMENT OF DEFENSE (DOD)

DODI 4150.07 (2019) DOD Pest Management Program

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Herbicide Application Plan

SD-03 Product Data

Tree Wound Paint

Herbicides; G-D0

SD-07 Certificates

Qualifications; G-D0

SD-11 Closeout Submittals

Pest Management Report

1.3 QUALITY CONTROL

1.3.1 Regulatory Requirements

Comply with DODI 4150.07 for requirements on Contractor's licensing, certification, and record keeping. Maintain daily records using the Pest Management Maintenance Record, DD Form 1532-1, or a computer generated equivalent. These forms may be obtained from the main web site:
<https://www.acq.osd.mil/eie/afpmb/docs/standardlists/dd1532-1.xlsm>.

1.3.2 Qualifications

For the application of herbicides, use the services of an applicator who

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is commercially certified in the state where the work is to be performed as required by DODI 4150.07. Submit a copy of the pesticide applicator certificates.

1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to the site, and handle in a manner which will maintain the materials in their original manufactured or fabricated condition until ready for use.

1.4.1 Storage

Storage of herbicides on the installation will not be permitted unless it is written into the contract.

1.4.2 Handling

Handle herbicides in accordance with the manufacturer's label and Safety Data Sheet (SDS), preventing contamination by dirt, water, and organic material. Protect herbicides from weather elements as recommended by the manufacturer's label and SDS. Spill kits must be maintained on herbicide control vehicles. Mixing of herbicides on the installation will not be permitted unless it is written into the contract.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Tree Wound Paint

Use bituminous based paint from standard manufacture specially formulated for tree wounds.

2.1.2 Herbicide

Provide herbicides currently registered by the EPA or approved for such use by the appropriate agency of the host county and approved by the Contracting Officer. Select a herbicide that is suitable for the climatic conditions at the project site. Submit manufacturer's label and SDS for herbicides proposed for use.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Herbicide Application Plan

Prior to commencing application of herbicide, submit a herbicide application plan with proposed sequence of treatment work including dates and times of application. Include the herbicide trade name, EPA registration number, chemical composition, formulation, application rate of active ingredients, method of application, area or volume treated, and amount applied. Include a copy of the pesticide applicator certificates.

3.1.2 Protection

3.1.2.1 Roads and Walks

Keep roads and walks free of dirt and debris at all times.

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3.1.2.2 Trees, Shrubs, and Existing Facilities

Protect trees and vegetation to be left standing from damage incident to clearing, grubbing, and construction operations by the erection of barriers or by such other means as the circumstances require.

3.1.2.3 Utility Lines

Protect existing utility lines that are indicated to remain from damage. Notify the Contracting Officer immediately of damage to or an encounter with an unknown existing utility line. The Contractor is responsible for the repair of damage to existing utility lines that are indicated or made known to the Contractor prior to start of clearing and grubbing operations. When utility lines which are to be removed are encountered within the area of operations, notify the Contracting Officer in ample time to minimize interruption of the service. Refer to Section 01 57 19.00 06 TEMPORARY ENVIRONMENTAL CONTROLS AND PERMITS for additional utility protection.

3.2 APPLICATION

3.2.1 Herbicide Application

Adhere to safety precautions as recommended by the manufacturer concerning handling and application of the herbicide.

3.2.1.1 Clean Up, Disposal, And Protection

Once application has been completed, proceed with clean up and protection of the site without delay. Clean the site of all material associated with the treatment measures, according to label instructions, and as indicated. Remove and dispose of excess and waste material off Government property.

3.2.1.1.1 Disposal of Herbicide

Dispose of residual herbicides and containers off Government property, and in accordance with the approved disposal plan, label instructions and EPA requirements.

3.3 CLEARING

Clearing consists of the felling, trimming, and cutting of trees into sections and the satisfactory disposal of the trees and other vegetation designated for removal, including downed timber, snags, brush, and rubbish occurring within the areas to be cleared. Cut off flush with or below the original ground surface trees, stumps, roots, brush, and other vegetation in areas to be cleared, except such trees and vegetation as may be indicated or directed to be left standing. Trim dead branches 1-1/2 inches or more in diameter on trees designated to be left standing within the cleared areas and trim all branches to the heights indicated or directed. Neatly cut close to the bole of the tree or main branches, limbs and branches to be trimmed. Paint, with an approved tree-wound paint, cuts more than 1-1/2 inches in diameter. Apply herbicide in accordance with the manufacturer's label to the top surface of stumps designated not to be removed.

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3.3.1 Tree Removal

Where indicated or directed, remove trees and stumps that are designated as trees from areas outside those areas designated for clearing and grubbing. This work includes the felling of such trees and the removal of their stumps and roots as specified in paragraph GRUBBING. Dispose of trees as specified in paragraph DISPOSAL OF MATERIALS.

3.3.2 Pruning

Trim trees designated to be left standing within the cleared areas of dead branches 1-1/2 inches or more in diameter; and trim branches to heights and in a manner as indicated. Neatly cut limbs and branches to be trimmed close to the bole of the tree or main branches. Paint cuts more than 1-1/4 inches in diameter with an approved tree wound paint.

3.3.3 Grubbing

Grubbing consists of the removal and disposal of stumps, roots larger than 3 inches in diameter, and matted roots from the designated grubbing areas. Remove material to be grubbed, together with logs and other organic or metallic debris not suitable for foundation purposes, to a depth of not less than 18 inches below the original surface level of the ground in areas indicated to be grubbed and in areas indicated as construction areas under this contract, such as areas for buildings, and areas to be paved. Fill depressions made by grubbing with suitable material and compact to make the surface conform with the original adjacent surface of the ground.

3.4 DISPOSAL OF MATERIALS

Dispose of excess materials in accordance with the approved solid waste management permit and include those materials in the solid waste management report.

All wood or wood like materials, except for salable timber, remaining from clearing, pruning or grubbing such as limbs, tree tops, roots, stumps, logs, rotten wood, and other similiar materials is the property of the Contractor and dispose of as specified. All non-saleable timber and wood or wood like materials remaining from timber harvesting such as limbs, tree tops, roots, stumps, logs, rotten wood, and other similiar materials is the property of the Contractor and dispose of as specified.

3.5 CLOSEOUT ACTIVITIES

3.5.1 Herbicides

Upon completion of this work, submit the Pest Management Report DD Form 1532, or an equivalent computer product, to the Integrated Pest Management Coordinator. This form identifies the type of operation, brand name and manufacturer of herbicide, formulation, concentration or rate of application used.

-- End of Section --

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 38-22 (2022) Standard Guideline for
Investigating and Documenting Existing
Utilities

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C600 (2017) Installation of Ductile-Iron Mains
and Their Appurtenances

ASTM INTERNATIONAL (ASTM)

ASTM C33/C33M (2018) Standard Specification for Concrete
Aggregates

ASTM C136/C136M (2019) Standard Test Method for Sieve
Analysis of Fine and Coarse Aggregates

ASTM D422 (1963; R 2007; E 2014; E 2014)
Particle-Size Analysis of Soils

ASTM D698 (2012; E 2014; E 2015) Laboratory
Compaction Characteristics of Soil Using
Standard Effort (12,400 ft-lbf/cu. ft.
(600 kN-m/cu. m.))

ASTM D1140 (2017) Standard Test Methods for
Determining the Amount of Material Finer
than 75- μ m (No. 200) Sieve in Soils by
Washing

ASTM D1556/D1556M (2015; E 2016) Standard Test Method for
Density and Unit Weight of Soil in Place
by Sand-Cone Method

ASTM D1557 (2012; E 2015) Standard Test Methods for
Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000
ft-lbf/ft³) (2700 kN-m/m³)

ASTM D1883 (2016) Standard Test Method for California
Bearing Ratio (CBR) of
Laboratory-Compacted Soils

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ASTM D2216	(2019) Standard Test Methods for Laboratory Determination of Water (Moisture) Content of Soil and Rock by Mass
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D3786/D3786M	(2018) Standard Test Method for Bursting Strength of Textile Fabrics-Diaphragm Bursting Strength Tester Method
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4355/D4355M	(2014) Deterioration of Geotextiles from Exposure to Light, Moisture and Heat in a Xenon-Arc Type Apparatus
ASTM D4491/D4491M	(2017) Standard Test Methods for Water Permeability of Geotextiles by Permittivity
ASTM D4533/D4533M	(2015) Standard Test Method for Trapezoid Tearing Strength of Geotextiles
ASTM D4632/D4632M	(2015a) Grab Breaking Load and Elongation of Geotextiles
ASTM D4759	(2011; R 2018) Standard Practice for Determining the Specification Conformance of Geosynthetics
ASTM D4829	(2021) Standard Test Method for Expansion Index of Soils
ASTM D4833/D4833M	(2007; R 2020) Standard Test Method for Index Puncture Resistance of Geomembranes and Related Products
ASTM D5084	(2016a) Standard Test Methods for Measurement of Hydraulic Conductivity of Saturated Porous Materials Using a Flexible Wall Permeameter
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health
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Requirements Manual

U.S. ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA SW-846.3-3

(1999, Third Edition, Update III-A) Test
Methods for Evaluating Solid Waste:
Physical/Chemical Methods

1.2 DEFINITIONS

1.2.1 Capillary Water Barrier

A layer of clean, poorly graded crushed rock, stone, or natural sand or gravel having a high porosity which is placed beneath a building slab with or without a vapor barrier to cut off the capillary flow of pore water to the area immediately below a slab.

1.2.2 Degree of Compaction

Degree of compaction is expressed as a percentage of the maximum density obtained by the test procedure presented in ASTM D698 and ASTM D1557, for general soil types, abbreviated as percent laboratory maximum density.

1.2.3 Hard Materials

Weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" but which usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.2.4 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling and blasting, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 1/2 cubic yard in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling and blasting that is performed merely to increase production.

1.2.5 Pile Supported Structure

As used herein, a structure where pile supported.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Shoring and Sheet piling Plan

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Dewatering work plan

Submit 15 days prior to starting work.

SD-06 Test Reports

Borrow Site Testing; G-D0

Fill and backfill test

Select material test

Porous fill test for capillary water barrier

Density tests

Moisture Content Tests

Copies of all laboratory and field test reports within 24 hours of the completion of the test.

1.4 DELIVERY, STORAGE, AND HANDLING

Perform in a manner to prevent contamination or segregation of materials.

1.5 CRITERIA FOR BIDDING

Base bids on the following criteria:

- a. Surface elevations are as indicated.
- b. Pipes or other artificial obstructions, except those indicated, will not be encountered.
- c. Ground water elevations indicated by the boring log were those existing at the time subsurface investigations were made and do not necessarily represent ground water elevation at the time of construction.
- d. Material character is indicated by the boring logs.
- e. Borrow material, Suitable backfill, and bedding material in the quantities required is not available at the project site
- f. Blasting will not be permitted. Remove material in an approved manner.

1.6 REQUIREMENTS FOR OFF SITE SOIL

Soils brought in from off site for use as backfill shall be tested for petroleum hydrocarbons, BTEX, PCBs and HW characteristics (including toxicity, ignitability, corrosivity, and reactivity). Backfill shall not contain concentrations of these analytes above the appropriate State and/or EPA criteria, and shall pass the tests for HW characteristics. Determine petroleum hydrocarbon concentrations by using appropriate State protocols. Determine BTEX concentrations by using EPA SW-846.3-3 Method 5035/8260B. Perform complete TCLP in accordance with EPA SW-846.3-3 Method 1311. Perform HW characteristic tests for ignitability, corrosivity, and reactivity in accordance with accepted standard methods.

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Perform PCB testing in accordance with accepted standard methods for sampling and analysis of bulk solid samples. Provide borrow site testing for petroleum hydrocarbons and BTEX from a grab sample of material from the area most likely to be contaminated at the borrow site (as indicated by visual or olfactory evidence), with at least one test from each borrow site. For each borrow site, provide borrow site testing for HW characteristics from a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring material onsite until tests results have been received and approved by the Contracting Officer.

1.7 QUALITY ASSURANCE

1.7.1 Shoring and Sheet Piling Plan

Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheet piling of excavations. Drawings shall include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations shall include data and references used.

The Contractor is required to hire a Professional Geotechnical Engineer to provide inspection of excavations and soil/groundwater conditions throughout construction. The Geotechnical Engineer shall be responsible for performing pre-construction and periodic site visits throughout construction to assess site conditions. The Geotechnical Engineer shall update the excavation, sheet piling and dewatering plans as construction progresses to reflect changing conditions and shall submit an updated plan if necessary. A written report shall be submitted, at least monthly, informing the Contractor and Contracting Officer of the status of the plan and an accounting of the Contractor's adherence to the plan addressing any present or potential problems. The Geotechnical Engineer shall be available to meet with the Contracting Officer at any time throughout the contract duration.

1.7.2 Dewatering Work Plan

Submit procedures for accomplishing dewatering work.

1.7.3 Utilities

Movement of construction machinery and equipment over pipes and utilities during construction shall be at the Contractor's risk. Perform work adjacent to non-Government utilities as indicated in accordance with procedures outlined by utility company. Excavation made with power-driven equipment is not permitted within two feet of known Government-owned utility or subsurface construction. For work immediately adjacent to or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

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PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Satisfactory Materials

Any materials classified by ASTM D2487 free of debris, roots, wood, scrap material, vegetation, refuse, soft unsound particles, expansive soils, and deleterious, or objectionable materials. Unless specified otherwise, the maximum particle diameter shall be one-half the lift thickness at the intended location.

2.1.2 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials. Unsatisfactory materials also include man-made fills, trash, refuse, or backfills from previous construction. Unsatisfactory material also includes material classified as satisfactory which contains root and other organic matter, frozen material, expansive soils, and stones larger than 3/4 inches. The Contracting Officer shall be notified of any contaminated materials.

2.1.3 Cohesionless and Cohesive Materials

Cohesionless materials include materials classified in ASTM D2487 as GW, GP, SW, and SP. Cohesive materials include materials classified as GC, SC, ML, CL, MH, and CH. Materials classified as GM, GP-GM, GW-GM, SW-SM, SP-SM, and SM shall be identified as cohesionless only when the fines are nonplastic (plasticity index equals zero). Materials classified as GM and SM will be identified as cohesive only when the fines have a plasticity index greater than zero.

2.1.4 Expansive Soils

Soils that have a plasticity index equal to or greater than 15 when tested in accordance with ASTM D4318, more than 10 percent soil particles pass a No. 200 sieve when tested in accordance with ASTM D422, more than 10 percent of soil particles are less than 5 micrometers in size when tested in accordance with ASTM D422, and expansion index is greater than 20 when tested in accordance with ASTM D4829.

2.1.5 Common Fill

Approved, unclassified soil material with the characteristics required to compact to the soil density specified for the intended location.

2.1.6 Backfill and Fill Material

ASTM D2487, classification GW, GP, GM, SW, SP, SM, with a maximum ASTM D4318 liquid limit of 35, maximum ASTM D4318 plasticity index of 12, and a maximum of 25 percent by weight passing ASTM D1140, No. 200 sieve. Material must not include expansive soils.

2.1.7 Select Material

Provide materials classified as silty or sandy clay by ASTM D2487 where indicated. The liquid limit of such material shall not exceed 35 percent when tested in accordance with ASTM D4318. The plasticity index shall not be greater than 18 percent when tested in accordance with ASTM D4318, and

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not more than 35 percent by weight shall be finer than No. 200 sieve when tested in accordance with ASTM D1140. Coefficient of permeability shall be a minimum of 0.002 feet per minute when tested in accordance with ASTM D5084. Select material must not include expansive soils.

Bearing Ratio: At 0.1 inch penetration, the bearing ratio shall be as determined by the Geotechnical Engineer at 95 percent ASTM D1557 maximum density as determined in accordance with ASTM D1883 for a laboratory soaking period of not less than 4 days.

2.1.8 Topsoil

Natural, friable soil representative of productive, well-drained soils in the area, free of subsoil, stumps, rocks larger than one inch diameter, brush, weeds, toxic substances, and other material detrimental to plant growth. Amend topsoil pH range to obtain a pH of 5.5 to 7.

2.2 POROUS FILL FOR CAPILLARY WATER BARRIER

ASTM C33/C33M fine aggregate grading with a maximum of 3 percent by weight passing ASTM D1140, No. 200 sieve, or 1-1/2 inches and no more than 2 percent by weight passing the No. 4 size sieve or coarse aggregate Size 57, 67, or 77 and conforming to the general soil material requirements specified in paragraph entitled "Satisfactory Materials."

2.3 UTILITY BEDDING MATERIAL

Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with AWWA C600, Type 4, except as specified herein. Backfill to top of pipe shall be compacted to 95 percent of ASTM D698 maximum density. Plastic piping shall have bedding to spring line of pipe. Provide ASTM D2321 materials as follows:

- a. Class I: Angular, 0.25 to 1.5 inches, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.
- b. Class II: Coarse sands and gravels with maximum particle size of 1.5 inches, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types GW, GP, SW, and SP are included in this class as specified in ASTM D2487.

2.3.1 Sand

Clean, coarse-grained sand classified as or SW or SP by ASTM D2487 for bedding and backfill as indicated.

2.3.2 Gravel

Clean, coarsely graded natural gravel, crushed stone or a combination thereof or having a classification of GW or GP in accordance with ASTM D2487 for bedding and backfill as indicated. Maximum particle size shall not exceed 3 inches.

2.4 BORROW

Obtain borrow materials required in excess of those furnished from excavations from sources outside of Government property.

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2.5 FILTER FABRIC

Provide a pervious sheet of polyester, nylon, glass or polypropylene , ultraviolet resistant filaments woven, spun bonded, fused, or otherwise manufactured into a nonraveling fabric with uniform thickness and strength. Fabric shall have the following manufacturer certified minimum average roll properties as determined by ASTM D4759:

	<u>Class A</u>	<u>Class B</u>
a. Grab tensile strength (ASTM D4632/D4632M) machine and transversed direction	min. 180	80 lbs.
b. Grab elongation (ASTM D4632/D4632M) machine and transverse direction	min. 15	15 percent
c. Puncture resistance (ASTM D4833/D4833M)	min. 80	25 lbs.
d. Mullen burst strength (ASTM D3786/D3786M)	min. 290	130 psi
e. Trapezoidal Tear (ASTM D4533/D4533M)	min. 50	25 lbs.
f. Permeability (ASTM D4491/D4491M)	k fabric greater than k Soil	
g. Ultraviolet Degradation (ASTM D4355/D4355M)	70 percent Strength retained at 150 hours	

2.6 MATERIAL FOR RIP-RAP

Bedding material, grout, filter fabric, and grout conforming to TXDOT standards.

2.7 BURIED WARNING AND IDENTIFICATION TAPE

Polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 3 inch minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Color and printing shall be permanent, unaffected by moisture or soil.

Warning Tape Color Codes	
Red:	Electric

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Warning Tape Color Codes	
Yellow:	Gas, Oil; Dangerous Materials
Orange:	Telephone and Other Communications
Blue:	Potable Water Systems
Green:	Sewer Systems
White:	Steam Systems
Gray:	Compressed Air
Purple:	Non Potable, Reclaimed Water, Irrigation and Slurry lines

2.7.1 Warning Tape for Metallic Piping

Acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above. Minimum thickness of tape shall be 0.003 inch. Tape shall have a minimum strength of 1500 psi lengthwise, and 1250 psi crosswise, with a maximum 350 percent elongation.

2.7.2 Detectable Warning Tape for Non-Metallic Piping

Polyethylene plastic tape conforming to the width, color, and printing requirements specified above. Minimum thickness of the tape shall be 0.004 inch. Tape shall have a minimum strength of 1500 psi lengthwise and 1250 psi crosswise. Tape shall be manufactured with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 3 feet deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.8 DETECTION WIRE FOR NON-METALLIC PIPING

Detection wire shall be insulated single strand, solid copper with a minimum of 12 AWG.

PART 3 EXECUTION

3.1 PROTECTION

3.1.1 Shoring and Sheeting

Provide shoring bracing, cribbing, trench boxes, underpinning, and sheeting where indicated. In addition to Section 25 A and B of EM 385-1-1, include provisions in the shoring and sheeting plan that will accomplish the following:

- a. Prevent undermining of pavements, foundations and slabs.
- b. Prevent slippage or movement in banks or slopes adjacent to the excavation.

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3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

So that construction operations progress successfully, completely drain construction site during periods of construction to keep soil materials sufficiently dry. The Contractor shall establish/construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity and/or provide temporary ditches, swales, and other drainage features and equipment as required to maintain dry soils, prevent erosion and undermining of foundations. When unsuitable working platforms for equipment operation and unsuitable soil support for subsequent construction features develop, remove unsuitable material and provide new soil material as specified herein. It is the responsibility of the Contractor to assess the soil and ground water conditions presented by the plans and specifications and to employ necessary measures to permit construction to proceed. Excavated slopes and backfill surfaces shall be protected to prevent erosion and sloughing. Excavation shall be performed so that the site, the area immediately surrounding the site, and the area affecting operations at the site shall be continually and effectively drained.

3.1.2.2 Dewatering

Groundwater flowing toward or into excavations shall be controlled to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches will not be permitted within 3 feet of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Control measures shall be taken by the time the excavation reaches the water level in order to maintain the integrity of the in situ material

Operate dewatering system continuously until construction work below existing water levels is complete. Submit performance records weekly.

3.1.3 Underground Utilities

Location of the existing utilities indicated is approximate. The Contractor shall physically verify the location and elevation of the existing utilities indicated prior to starting construction. Perform SUE QL-A in accordance with ASCE 38-22 to locate underground utilities within project limits. The Contractor shall contact the Public Works Department for assistance in locating existing utilities.

3.1.4 Machinery and Equipment

Movement of construction machinery and equipment over pipes during construction shall be at the Contractor's risk. Repair, or remove and provide new pipe for existing or newly installed pipe that has been displaced or damaged.

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3.2 SURFACE PREPARATION

3.2.1 Clearing and Grubbing

Unless indicated otherwise, remove trees, stumps, logs, shrubs, brush and vegetation and other items that would interfere with construction operations within the clearing limits. Remove stumps entirely. Grub out matted roots and roots over 2 inches in diameter to at least 18 inches below existing surface.

3.2.2 Stripping

Strip suitable soil from the site where excavation or grading is indicated and stockpile separately from other excavated material. Material unsuitable for use as topsoil shall be stockpiled and used for backfilling. Locate topsoil so that the material can be used readily for the finished grading. Where sufficient existing topsoil conforming to the material requirements is not available on site, provide borrow materials suitable for use as topsoil. Protect topsoil and keep in segregated piles until needed.

3.2.3 Unsuitable Material

Remove vegetation, debris, decayed vegetable matter, sod, mulch, and rubbish underneath paved areas or concrete slabs.

3.3 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Reuse excavated materials that meet the specified requirements for the material type required at the intended location. Keep excavations free from water. Excavate soil disturbed or weakened by Contractor's operations, soils softened or made unsuitable for subsequent construction due to exposure to weather. Excavations below indicated depths will not be permitted except to remove unsatisfactory material. Unsatisfactory material encountered below the grades shown shall be removed as directed by the COR. Unless specified otherwise, refill excavations cut below indicated depth with backfill and fill material, satisfactory material, select material, and porous fill and compact to 95 percent of ASTM D698 and ASTM D1557 maximum density. Satisfactory material removed below the depths indicated, without specific direction of the Contracting Officer, shall be replaced with satisfactory materials to the indicated excavation grade; except as specified for spread footings. Determination of elevations and measurements of approved overdepth excavation of unsatisfactory material below grades indicated shall be done under the direction of the Contracting Officer.

3.3.1 Structures With Spread Footings

Ensure that footing subgrades have been inspected and approved by the Contractor's Geotechnical Engineer and Contracting Officer prior to concrete placement. Fill over excavations with concrete during foundation placement.

3.3.2 Pipe Trenches

Excavate to the dimension indicated. Grade bottom of trenches to provide uniform support for each section of pipe after pipe bedding placement. Tamp if necessary to provide a firm pipe bed. Recesses shall be excavated

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to accommodate bells and joints so that pipe will be uniformly supported for the entire length. Rock, where encountered, shall be excavated to a depth of at least 6 inches below the bottom of the pipe.

3.3.3 Hard Material and Rock Excavation

Remove hard material and rock to elevations indicated in a manner that will leave foundation material in an unshattered and solid condition. Roughen level surfaces and cut sloped surfaces into benches for bond with concrete. Protect shale from conditions causing decomposition along joints or cleavage planes and other types of erosion. Removal of hard material and rock beyond lines and grades indicated will not be grounds for a claim for additional payment unless previously authorized by the Contracting Officer. Excavation of the material claimed as rock shall not be performed until the material has been cross sectioned by the Contractor and approved by the Contracting Officer. Common excavation shall consist of all excavation not classified as rock excavation.

3.3.4 Excavated Materials

Satisfactory excavated material required for fill or backfill shall be placed in the proper section of the permanent work required or shall be separately stockpiled if it cannot be readily placed. Satisfactory material in excess of that required for the permanent work and all unsatisfactory material shall be disposed of as specified in Paragraph "DISPOSITION OF SURPLUS MATERIAL."

3.3.5 Final Grade of Surfaces to Support Concrete

Excavation to final grade shall not be made until just before concrete is to be placed. Only excavation methods that will leave the foundation rock in a solid and unshattered condition shall be used. Approximately level surfaces shall be roughened, and sloped surfaces shall be cut as indicated into rough steps or benches to provide a satisfactory bond. Shales shall be protected from slaking and all surfaces shall be protected from erosion resulting from ponding or flow of water.

3.4 SUBGRADE PREPARATION

Unsatisfactory material in surfaces to receive fill or in excavated areas shall be removed and replaced with satisfactory materials as directed by the Contracting Officer. The surface shall be scarified to a depth of 6 inches before the fill is started. Sloped surfaces steeper than 1 vertical to 4 horizontal shall be plowed, stepped, benched, or broken up so that the fill material will bond with the existing material. When subgrades are less than the specified density, the ground surface shall be broken up to a minimum depth of 6 inches, pulverized, and compacted to the specified density. When the subgrade is part fill and part excavation or natural ground, the excavated or natural ground portion shall be scarified to a depth of 12 inches and compacted as specified for the adjacent fill. Material shall not be placed on surfaces that are muddy, frozen, or contain frost. Compaction shall be accomplished by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Material shall be moistened or aerated as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used. Minimum subgrade density shall be as specified herein.

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3.4.1 Proof Rolling

Proof rolling shall be done on an exposed subgrade free of surface water (wet conditions resulting from rainfall) which would promote degradation of an otherwise acceptable subgrade. After stripping, proof roll the existing subgrade with six passes of a dump truck loaded with 212 cubic feet of soil. Operate the truck in a systematic manner to ensure the number of passes over all areas, and at speeds between 2 1/2 to 3 1/2 miles per hour. When proof rolling under buildings, the building subgrade shall be considered to extend 5 feet beyond the building lines, and one-half of the passes made with the roller shall be in a direction perpendicular to the other passes. Notify the Contracting Officer a minimum of 3 days prior to proof rolling. Proof rolling shall be performed in the presence of the Contracting Officer. Rutting or pumping of material shall be undercut as directed by the Contracting Officer.

3.5 SUBGRADE FILTER FABRIC

Place synthetic fiber filter fabric as indicated directly on prepared subgrade free of vegetation, stumps, rocks larger than 2 inches diameter and other debris which may puncture or otherwise damage the fabric. Repair damaged fabric by placing an additional layer of fabric to cover the damaged area a minimum of 3 feet overlap in all directions. Overlap fabric at joints a minimum of 3 feet. Obtain approval of filter fabric installation before placing fill or backfill. Place fill or backfill on fabric in the direction of overlaps and compact as specified herein. Follow manufacturer's recommended installation procedures.

3.6 FILLING AND BACKFILLING

Fill and backfill to contours, elevations, and dimensions indicated. Compact each lift before placing overlaying lift.

3.6.1 Common Fill Placement

Provide for general site. Use satisfactory materials. Place in 6 inch lifts. Compact areas not accessible to rollers or compactors with mechanical hand tampers. Aerate material excessively moistened by rain to a satisfactory moisture content. Finish to a smooth surface by blading, rolling with a smooth roller, or both.

3.6.2 Backfill and Fill Material Placement

Provide for paved areas and under concrete slabs, except where select material is provided. Place in 6 inch lifts. Do not place over wet or frozen areas. Place backfill material adjacent to structures as the structural elements are completed and accepted. Backfill against concrete only when approved. Place and compact material to avoid loading upon or against the structure.

3.6.3 Select Material Placement

Provide under porous fill of structures not pile supported. Place in 6 inch lifts. Do not place over wet or frozen areas. Backfill adjacent to structures shall be placed as structural elements are completed and accepted. Backfill against concrete only when approved. Place and compact material to avoid loading upon or against structure.

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3.6.4 Backfill and Fill Material Placement Over Pipes and at Walls

Backfilling shall not begin until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Backfill shall be brought to indicated finish grade. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 2 feet above sewer lines and 1 foot above other utility lines shall be free from stones larger than 1 inch in any dimension. Heavy equipment for spreading and compacting backfill shall not be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; the area remaining shall be compacted in layers not more than 4 inches in compacted thickness with power-driven hand tampers suitable for the material being compacted. Backfill shall be placed carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Backfill shall not be placed against foundation walls prior to 7 days after completion of the walls. As far as practicable, backfill shall be brought up evenly on each side of the wall and sloped to drain away from the wall.

3.6.5 Porous Fill Placement

Provide under floor and area-way slabs on a compacted subgrade. Place in 4 inch lifts with a minimum of two passes of a hand-operated plate-type vibratory compactor.

3.6.6 Trench Backfilling

Backfill as rapidly as construction, testing, and acceptance of work permits. Place and compact backfill under structures and paved areas in 6 inch lifts to top of trench and in 6 inch lifts to one foot over pipe outside structures and paved areas.

3.7 BORROW

Where satisfactory materials are not available in sufficient quantity from required excavations, approved borrow materials shall be obtained as specified herein.

3.8 BURIED WARNING AND IDENTIFICATION TAPE

Provide buried utility lines with utility identification tape. Bury tape 12 inches below finished grade; under pavements and slabs, bury tape 6 inches below top of subgrade.

3.9 BURIED DETECTION WIRE

Bury detection wire directly above non-metallic piping at a distance not to exceed 12 inches above the top of pipe. The wire shall extend continuously and unbroken, from manhole to manhole. The ends of the wire shall terminate inside the manholes at each end of the pipe, with a minimum of 3 feet of wire, coiled, remaining accessible in each manhole. The wire shall remain insulated over its entire length. The wire shall enter manholes between the top of the corbel and the frame, and extend up through the chimney seal between the frame and the chimney seal. For force mains, the wire shall terminate in the valve pit at the pump station end of the pipe.

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3.10 COMPACTION

Determine in-place density of existing subgrade; if required density exists, no compaction of existing subgrade will be required.

3.10.1 General Site

Compact underneath areas designated for vegetation and areas outside the 5 foot line of the paved area or structure to 85 percent of ASTM D698.

3.10.2 Structures, Spread Footings, and Concrete Slabs

Compact top 12 inches of subgrades to 95 percent of ASTM D698. Compact fill and backfill material, and select material to 98 percent of ASTM D698.

3.10.3 Adjacent Area

Compact areas within 5 feet of structures to 95 percent of ASTM D698.

3.10.4 Paved Areas

Compact top 12 inches of subgrades to 95 percent of ASTM D698 and ASTM D1557. Compact fill and backfill materials to 95 percent of ASTM D698 and ASTM D1557.

3.11 RIP-RAP CONSTRUCTION

Construct riprap per TXDOT standards.

3.11.1 Preparation

Trim and dress indicated areas to conform to cross sections, lines and grades shown within a tolerance of 0.1 foot.

3.11.2 Bedding Placement

Spread filter fabric bedding material uniformly to a thickness of at least 3 inches on prepared subgrade as indicated.

3.11.3 Stone Placement

Place rock for rip-rap on prepared bedding material to produce a well graded mass with the minimum practicable percentage of voids in conformance with lines and grades indicated. Distribute larger rock fragments, with dimensions extending the full depth of the rip-rap throughout the entire mass and eliminate "pockets" of small rock fragments. Rearrange individual pieces by mechanical equipment or by hand as necessary to obtain the distribution of fragment sizes specified above.

3.12 FINISH OPERATIONS

3.12.1 Grading

Finish grades as indicated within one-tenth of one foot. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

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3.12.2 Topsoil and Seed

Provide as specified in Section 32 92 19 SEEDING.

3.12.3 Protection of Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.13 DISPOSITION OF SURPLUS MATERIAL

Remove from Government property surplus or other soil material not required or suitable for filling or backfilling, and brush, refuse, stumps, roots, and timber.

3.14 FIELD QUALITY CONTROL

3.14.1 Sampling

Take the number and size of samples required to perform the following tests.

3.14.2 Testing

Perform one of each of the following tests for each material used. Provide additional tests for each source change.

3.14.2.1 Fill and Backfill Material Testing

Test fill and backfill material in accordance with ASTM C136/C136M for conformance to ASTM D2487 gradation limits; ASTM D1140 for material finer than the No. 200 sieve; ASTM D4318 for liquid limit and for plastic limit; ASTM D698 or ASTM D1557 for moisture density relations, as applicable.

3.14.2.2 Select Material Testing

Test select material in accordance with ASTM C136/C136M for conformance to ASTM D2487 gradation limits; ASTM D1140 for material finer than the No. 200 sieve; ASTM D698 or ASTM D1557 for moisture density relations, as applicable.

3.14.2.3 Porous Fill Testing

Test porous fill in accordance with ASTM C136/C136M for conformance to gradation specified in ASTM C33/C33M.

3.14.2.4 Density Tests

Test density in accordance with ASTM D1556/D1556M, or ASTM D6938. When ASTM D6938 density tests are used, verify density test results by performing an ASTM D1556/D1556M density test at a location already ASTM D6938 tested as specified herein. Perform an ASTM D1556/D1556M density test at the start of the job, and for every 10 ASTM D6938 density tests thereafter. Test each lift at randomly selected locations every 2000 square feet of existing grade in fills for structures and concrete slabs, and every 2500 square feet for other fill areas and every 2000 square feet of subgrade in cut. Include density test results in daily report.

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Bedding and backfill in trenches: One test per 50 linear feet in each lift.

3.14.2.5 Moisture Content Tests

In the stockpile, excavation or borrow areas, a minimum of two tests per day per type of material or source of materials being placed is required during stable weather conditions. During unstable weather, tests shall be made as dictated by local conditions and approved moisture content shall be tested in accordance with ASTM D2216. Include moisture content test results in daily report.

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DRILLED CONCRETE PIERS AND SHAFTS 11/19

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117	(2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary
ACI 301	(2016) Specifications for Structural Concrete
ACI 304R	(2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete
ACI 305R	(2020) Guide to Hot Weather Concreting
ACI 306.1	(1990; R 2002) Standard Specification for Cold Weather Concreting
ACI 318	(2014; Errata 1-2 2014; Errata 3-5 2015; Errata 6 2016; Errata 7-9 2017) Building Code Requirements for Structural Concrete (ACI 318-14) and Commentary (ACI 318R-14)
ACI 336.1	(2001) Specification for the Construction of Drilled Piers
ACI SP-66	(2004) ACI Detailing Manual

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303	(2016) Code of Standard Practice for Steel Buildings and Bridges
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AMERICAN SOCIETY OF CIVIL ENGINEERS (ASCE)

ASCE 7-16	(2017; Errata 2018; Supp 1 2018) Minimum Design Loads and Associated Criteria for Buildings and Other Structures
ASCE 38-22	(2022); Standard Guideline for Investigating and Documenting Existing Utilities

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ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2022) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
ASTM A1064/A1064M	(2022) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
ASTM C31/C31M	(2022) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C39/C39M	(2021) Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C94/C94M	(2022a) Standard Specification for Ready-Mixed Concrete
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C150/C150M	(2022) Standard Specification for Portland Cement
ASTM C172	(2010) Standard Practice for Sampling Freshly Mixed Concrete

CONCRETE REINFORCING STEEL INSTITUTE (CRSI)

CRSI 10MSP	(2018) Manual of Standard Practice
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U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1	(2014) Safety -- Safety and Health Requirements Manual
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U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

FHWA NHI-10-016	(2010) Drilled Shafts: Construction Procedures and LRFD Design Methods
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U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926.651	Specific Excavation Requirements
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1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

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Installation Plan

SD-02 Shop Drawings

Drilled Shaft Diameters

Top and Bottom of Shaft Elevations

Steel Reinforcement

Anchor Bolt Locations

Accessories

SD-05 Design Data

Drilled Shaft Foundation Design Analysis

Mix Design Data

SD-06 Test Reports

Soils Report

Ground Water Conditions

Slump

Concrete

Compressive Strength

SD-07 Certificates

Bill of Lading for Ready-Mix Concrete Deliveries

Steel Reinforcement

Excavation and Drilling Equipment

Qualifications of Excavator

Qualifications of Engineer

1.3 QUALITY CONTROL

1.3.1 General

Install drilled shaft foundations in accordance with applicable requirements as described by ACI 336.1, and FHWA NHI-10-016.

1.3.2 Sequencing and Scheduling

Submit a detailed installation plan describing the schedule for drilling and/or excavation, installation of steel reinforcement and concrete placement with anticipated site conditions so that each excavated shaft is poured the same day that the drilling is performed.

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1.3.3 Inspection Criteria

Design inspection activities to minimize delays while insuring the intent of the Industry Standard Specifications.

1.3.4 Qualification of Excavation Contractor

An experienced excavator with five (5) years experience and licensed in the State of Texas, specialized in excavating and installing work similar in material, design, and extent to that indicated for this Project. Submit certificates substantiating the Qualifications of Excavator.

1.3.5 Qualification of Professional Engineer

Provide engineering services by an authorized engineer currently licensed in the State of Texas; having a minimum of four (4) years experience as an engineer knowledgeable in drilled shaft foundation design analysis, protocols and procedures for the ACI 336.1, FHWA NHI-10-016, ASCE 7-16, and the International Building Code. Submit certificates substantiating the Qualifications of Engineer.

1.3.6 Pre-Construction Conference

After submittals are received and approved but before drilled shaft excavation and foundation work, including associated work, is performed, the Contracting Officer will hold a pre-construction conference to review the following:

- a. The drawings, specifications and the geotechnical report.
- b. Finalize construction schedule and verify availability of materials, Excavator's personnel, equipment, and facilities needed to make progress and avoid delays.
- c. Methods and procedures related to drilled shaft foundation installation, including engineer's written instructions.
- d. Support conditions for compliance with requirements, including alignment between foundation system and erection of structural members.
- e. Governing regulations and requirements for, certificates, insurance, tests and inspections if applicable.
- f. Temporary protection requirements for foundation assembly during and after installation.

1.4 PROJECT CONDITIONS

1.4.1 Existing Conditions

Locate existing underground utilities before excavating drilled shaft foundations. Perform SUE QL-A in accordance with ASCE 38-22 to locate underground utilities within project limits. If existing utilities are to remain in place, provide protection during drilled shaft operations.

1.4.2 Interruption of Existing Utilities

Do not interrupt any utility to occupied facilities unless directed in writing by the Contracting Officer.

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1.4.3 Weather Limitations

Proceed with installation preparation only when existing and forecasted weather conditions permit work to proceed without water entering into the area of excavation.

PART 2 PRODUCTS

2.1 DESIGN REQUIREMENTS

Submit design data for the following:

- a. Drilled shaft foundation design analysis to include, but not limited to the following:
 - (1) Applicable Building code criteria for the excavation's geographic area
 - (2) Prepare and submit following design documents if PV carport column base loads exceed drilled pier design loads indicated on drawings:
 - (a) Carport column base load schedule.
 - (b) Drilled pier computations and drawings showing required embedment and reinforcement.
 - (c) Documents must be signed and sealed by licensed engineer retained by Contractor. Contractor is solely responsible for additional design and construction costs if carport column base loads exceed drilled pier design loads indicated on drawings.
 - (3) Casing description
- b. Mix design data in accordance with paragraph READY-MIX CONCRETE accompanied by the Bill of Lading for Ready Mix Concrete deliveries.

2.1.1 Assembly

Installation drawings are to include, but not limited to, the following items indicating a completely dimensioned layout and location of drilled shafts and concrete placement for foundation system. Submit detailed shop drawings for the following:

- a. Drilled shaft diameters
- b. Top and bottom of shaft elevations
- c. Steel reinforcement
- d. Anchor bolt locations
- e. Accessories

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2.2 EQUIPMENT

2.2.1 Drilling and Excavation Equipment

Provide drilling and excavation equipment having adequate capacity, including but not limited to, power, torque and down thrust to excavate a hole of diameter and depth indicated. Also provide excavation and over-reaming tools of adequate design, size and strength to perform the work indicated.

Provide special drilling equipment including, but not limited to, rock core barrels, rock tools, air tools and other equipment as necessary to construct the shaft excavation to the size and depth indicated when materials encountered can not be drilled using earth augers and/or over-reaming tools.

Submit certificates substantiating appropriate selection of excavation and drilling equipment.

2.3 MATERIALS

2.3.1 Steel Reinforcement

2.3.1.1 Deformed Steel Bars

Steel bars conforming to ASTM A615/A615M, Grade 60 ksi and ACI 318.

2.3.1.2 Plain Steel Wire

Steel wire conforming to ASTM A1064/A1064M.

2.3.2 Ready-Mix Concrete

Ready-Mix concrete and mix design conforming to ACI 117, ACI 301, and ACI 304R, minimum compressive strength 5,000 psi at 28 days. Slump results between 5 to 6 inches, according to ASTM C143/C143M.

Portland cements conforming to ASTM C150/C150M, Type II. Provide one brand and type of cement for formed concrete having exposed-to-view finished surfaces.

Potable water conforming to ASTM C94/C94M.

Measure, batch, mix and deliver concrete according to ASTM C94/C94M and furnish batch ticket information.

PART 3 EXECUTION

3.1 PREPARATION

Protect existing structures, utilities, sidewalks, pavements, and other facilities from damage caused by settlement, lateral movement, vibration, and other hazards created by drilled shaft foundation operations.

Provide safety and Fall Protection as required by EM 385-1-1 and 29 CFR 1926.651.

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3.2 INSTALLATION

3.2.1 Construction Criteria

Provide equipment for checking the dimensions and alignment of each shaft excavation. Determine dimensions and alignment jointly with the contractor and engineer. Measure final shaft depths with appropriate weighted tape measure or other approved method after cleaning.

Provide and install monolithically cast-in-place concrete drilled shaft foundation to the sizes indicated.

Provide and install straight cylindrical shaft foundation of the type indicated.

Tolerances:

- a. Maximum variation of the center of any shaft foundation from the required location: 3 inches, measured at the ground surface.
- b. Bottom Diameter: Minus zero, plus 6 inches, measured in any direction.
- c. Maximum variation from plumb: 1:40.
- d. Maximum bottom level: Plus or minus 2 inches.
- e. Anchor bolt tolerances must comply with AISC 303-10 requirements.

3.2.2 Excavation

Accomplish excavation of shaft foundations by standard excavation methods including, but not limited to, conventional augers fitted with soil and/or rock teeth, or under-reaming tools attached to drilling equipment of adequate size, power, torque and down thrust necessary for the work.

Perform excavation through whatever materials that are encountered to the dimensions, depths and applicable ACI 336.1 tolerances.

Protect excavated walls with temporary watertight steel casings of sufficient length to prevent water intrusion, cave-ins, displacement of surrounding earth, and injury to personnel and damage to construction operations.

Excavate shafts for drilled foundations to indicated elevations. Remove loose debris, materials and/or muck to make bottom surfaces level within ACI 336.1 tolerances.

Remove water from excavated shaft prior to concrete placement.

3.2.3 Steel Reinforcement

Comply with recommendations in the CRSI "Manual of Standard Practice" CRSI 10MSP for fabricating, placing and supporting reinforcement. Shop fabricate steel reinforcement in accordance with ACI SP-66.

When practicable, deliver the reinforcement cage assembly to the jobsite as a complete unit ready for installation. Should it be necessary to make any additional connections and/or splices, provide as indicated on the approved shop drawings, at-grade level prior to lowering the complete

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assembly into the hole.

Clean reinforcement of loose rust, mill scale, earth and other foreign materials. Do not tack weld crossing reinforcing bars. Set wire ties with ends directed into concrete, not toward exposed concrete surfaces.

Lower reinforcement steel into the hole in such a manner as to prevent damage to the walls of the excavation. Place, tie and/or clip cage symmetrically about the axis of the shaft. Use centering devices securely attached to the cage to clear the shaft walls and maintain the cage in place throughout the concrete placement operations.

Cooperate with other trades in setting of anchor bolts, inserts, and other embedded items. Where conflicts occur between reinforcing and embedded items, notify the Contracting Officer in order to reconcile conflicts before concrete placement. Position and support anchors and embedded items with appropriate accessories.

Use templates to set anchor bolts, leveling plates and other accessories required for structure erection. Provide blocking and/or holding devices to maintain required anchoring positions during final concrete placement.

3.2.4 Concrete Placement

Keep all equipment, including but not limited to, mixers, pumps, hoses, tools and screeds clean and free of set concrete throughout the placement operation.

Convey concrete from the mixer to place of deposit by best industry methods that prevents segregation and loss of material. Size and design the equipment for conveying concrete to ensure uniform, continuous placement of concrete.

Place concrete in accordance with ACI 318.

Place concrete in a continuous operation and without segregation into dry excavations whenever possible after inspection and written approval by the Contracting Officer. Use all practicable means to obtain a dry excavation before and during concrete placement.

Protect freshly placed concrete from premature drying and excessive cold or hot temperatures. When hot weather conditions exist that would impair quality and strength of placed concrete, comply with ACI 305R. Comply with ACI 306.1 for cold-weather protection.

A minimum of 50 percent of the base for each shaft is to be less than 1/2 inch of sediment at the time of concrete placement. Maximum depth of sediment or debris at any place on the base of the shaft is not to exceed 1-1/2 inches. Shaft cleanliness is to be determined by the engineer by visual inspection.

3.3 FIELD QUALITY CONTROL

3.3.1 Test Reports

As a minimum, submit the following test reports and data.

a. Soils Report

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- b. Ground Water conditions
- c. Slump
- d. Concrete
- e. Compressive Strength

Sample and test concrete for quality control during placement.

Sample freshly placed concrete for testing in accordance with ASTM C172.

Make concrete test specimens for compressive strength at 7 and 28 days for each design mix conforming to ASTM C31/C31M. Compression test concrete in accordance with ASTM C39/C39M.

Test Slump at plant for each design mix in accordance with ASTM C143/C143M.

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AGGREGATE BASE COURSE FOR FLEXIBLE PAVING 05/22

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Measurement

1.1.1.1 Area

Measure the quantity of 8 inch thick ABC completed and accepted, in square yards.

1.1.1.2 Volume

Measure the quantity of ABC completed and accepted, in cubic yards. Determine the volume of material in-place and accepted by the average job thickness obtained in accordance with paragraph LAYER THICKNESS and the dimensions shown on the drawings.

1.1.2 Payment

1.1.2.1 Base Course Material

Quantities of ABC, determined as specified above, will be paid for at the respective contract unit prices, which will constitute full compensation for the construction and completion of the ABC.

1.1.2.2 Stabilization

Cohesionless subgrade or subbase courses to be stabilized, as specified in paragraph PREPARATION OF UNDERLYING COURSE OR SUBGRADE, will be paid for as a special item on a tonnage basis including extra manipulation as required.

1.1.3 Waybills and Delivery Tickets

Submit copies of waybills and delivery tickets during progress of the work. Before the final payment is allowed, file certified waybills and certified delivery tickets for all aggregates actually used.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C29/C29M	(2017a) Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate
ASTM C88	(2018) Standard Test Method for Soundness

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	of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C117	(2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C127	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate
ASTM C128	(2022) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
ASTM C131/C131M	(2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM C1252	(2017) Standard Test Methods for Uncompacted Void Content of Fine Aggregate (as Influenced by Particle Shape, Surface Texture, and Grading)
ASTM D75/D75M	(2019) Standard Practice for Sampling Aggregates
ASTM D1556/D1556M	(2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
ASTM D1557	(2012; E 2015) Standard Test Methods for Laboratory Compaction Characteristics of Soil Using Modified Effort (56,000 ft-lbf/ft ³) (2700 kN-m/m ³)
ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D3665	(2012; R 2017) Standard Practice for Random Sampling of Construction Materials
ASTM D4318	(2017; E 2018) Standard Test Methods for Liquid Limit, Plastic Limit, and Plasticity Index of Soils
ASTM D4718/D4718M	(2015) Standard Practice for Correction of Unit Weight and Water Content for Soils Containing Oversize Particles
ASTM D4791	(2019) Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate

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ASTM D5821	(2013; R 2017) Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate
ASTM D6938	(2017a) Standard Test Method for In-Place Density and Water Content of Soil and Soil-Aggregate by Nuclear Methods (Shallow Depth)
ASTM D7928	(2017) Standard Test Method for Particle-Size Distribution (Gradation) of Fine-Grained Soils Using the Sedimentation (Hydrometer) Analysis
ASTM E11	(2022) Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

1.3 DEFINITIONS

For the purposes of this specification, the following definitions apply.

1.3.1 Aggregate Base Course

Aggregate base course (ABC) is well graded, durable aggregate uniformly moistened and mechanically stabilized by compaction.

1.3.2 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in ASTM D1557 abbreviated as a percent of laboratory maximum dry density. Since ASTM D1557 applies only to soils that have 30 percent or less by weight of their particles retained on the 3/4 inch sieve, express the degree of compaction for material having more than 30 percent by weight of their particles retained on the 3/4 inch sieve as a percentage of the laboratory maximum dry density in accordance with ASTM D1557 Method C and corrected with ASTM D4718/D4718M.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Plant, Equipment, and Tools; G-D0

Waybills and Delivery Tickets

SD-06 Test Reports

Initial Tests; G-D0

In-Place Tests; G-D0

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1.5 QUALITY ASSURANCE

Perform sampling and testing using a laboratory approved in accordance with Section 01 45 04.10 06 CONTRACTOR QUALITY CONTROL. Do not start work requiring testing until the testing laboratory has been inspected and approved. All contractor quality control testing laboratories performing acceptance testing require USACE validation by the Material Testing Center (MTC) for both parent laboratory and on-site laboratory. Validation on all laboratories is required to remain current throughout the duration of the paving project. Contact the MTC manager listed at <https://mtc.erdcdren.mil/requestvalidation.aspx#> for costs and scheduling.

Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. Furnish copies of test results within 24 hours of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with ASTM D75/D75M.

1.5.2 Tests

1.5.2.1 Gradation Analysis

Perform gradation analysis in conformance with ASTM C117 and ASTM C136/C136M using sieves conforming to ASTM E11. Perform particle-size analysis of the soils in conformance with ASTM D7928.

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with ASTM D4318.

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with ASTM D1556/D1556M, or ASTM D6938. For the method presented in ASTM D1556/D1556M use the base plate as shown in the drawing. For the method presented in ASTM D6938 check the calibration curves and adjust them, if necessary, using only the sand cone method as described in Annex A2 of ASTM D6938. Use ASTM D6938 to determine the moisture content of the soil. Check the calibration curves furnished with the moisture gauges along with density calibration checks as described in ASTM D6938. Make the calibration checks of both the density and moisture gauges using the prepared containers of material method, as described in Annex A2 of ASTM D6938, on each different type of material being tested at the beginning of a job and at intervals as directed. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.5.2.5 Wear Test

Perform wear tests on ABC course material in conformance with ASTM C131/C131M.

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1.5.2.6 Flat and Elongated Pieces

Determine flat and elongated pieces on ABC course material in conformance with ASTM D4791, Method A.

1.5.2.7 Soundness

Perform soundness tests on GCA in accordance with ASTM C88.

1.5.2.8 Fractured Faces

Perform fractured faces test on ABC coarse aggregate in conformance with ASTM D5821. Determine uncompacted void content of the GCA fine aggregate in accordance with ASTM C1252, Method A.

1.5.2.9 Weight of Slag

Determine weight per cubic foot of slag in accordance with ASTM C29/C29M on the ABC course material.

1.6 ENVIRONMENTAL REQUIREMENTS

Perform construction when the atmospheric temperature is above 35 degrees F. When the temperature falls below 35 degrees F, protect all completed areas by approved methods against detrimental effects of freezing. Correct completed areas damaged by freezing, rainfall, or other weather conditions to meet specified requirements.

1.7 ACCEPTANCE

1.7.1 Tolerances

Acceptance of ABC is based on compliance with the tolerances presented in Table 1. Remove any materials found to be non-compliant and replace with compliant material or rework, as directed, to meet the requirements of this specification

TABLE 1	
Measurement	Tolerance
Grade	Plus 1/4 inch, Minus 1/2 inch
Smoothness	Plus/Minus 3/8 inch
Individual Test Total Thickness	Plus/Minus
Average Job Thickness	Plus/Minus
Compaction	Minimum 100 percent

1.7.2 Test Section

A test section is not required.

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PART 2 PRODUCTS

2.1 AGGREGATES

Provide ABC consisting of clean, sound, durable particles of crushed stone, crushed gravel, angular sand, or other approved material. Provide ABC that is free of lumps of clay, organic matter, and other objectionable materials or coatings. The portion retained on the No. 4 sieve is known as coarse aggregate; that portion passing the No. 4 sieve is known as fine aggregate. When the coarse and fine aggregate is supplied from more than one source, provide aggregate from each source that meets the specified requirements.

2.1.1 Coarse Aggregate

Provide coarse aggregates with angular particles of uniform density. Separately stockpile coarse aggregate supplied from more than one source.

- a. Crushed Gravel: Provide crushed gravel that has been manufactured by crushing gravels and that meets all the requirements specified below.
- b. Crushed Stone: Provide crushed stone consisting of freshly mined quarry rock, meeting all the requirements specified below.

2.1.1.1 Aggregate Base Course

Limit the percentage of loss of ABC coarse aggregate to a maximum of 50 percent when tested in accordance with ASTM C131/C131M. Provide aggregate that contains a maximum of 30 percent flat and elongated particles when tested in accordance with ASTM D4791, Method A. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, provide crushed aggregates containing a minimum of 50 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with ASTM D5821. When two fractures are contiguous, the angle between planes of the fractures is required to be a minimum of 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from gravel particles 50 percent of which, by weight, are retained on the maximum size sieve listed in TABLE 2.

2.1.1.2 Graded-Crushed Aggregate Base Course

Limit the percentage of loss of GCA coarse aggregate to a maximum of 40 percent when tested in accordance with ASTM C131/C131M. Provide GCA coarse aggregate that does not exhibit a loss greater than 18 percent weighted average, at five cycles, when tested for soundness in magnesium sulfate, or 12 percent weighted average, at five cycles, when tested in sodium sulfate in accordance with ASTM C88. Provide aggregate that contains a maximum of 20 percent flat and elongated particles for the fraction retained on the 1/2 inch sieve nor 20 percent for the fraction passing the 1/2 inch sieve when tested in accordance with ASTM D4791, Method A. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, provide crushed aggregate containing a minimum of 90 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with ASTM D5821. When two fractures are contiguous, the angle between planes of the fractures is required to be a minimum of 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from

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gravel particles 90 percent of which by weight are retained on the maximum size sieve listed in TABLE 2.

2.1.2 Fine Aggregate

Provide fine aggregates consisting of angular particles of uniform density.

2.1.2.1 Aggregate Base Course

Provide ABC fine aggregate that consists of screenings, angular sand, crushed recycled concrete fines, or other finely divided mineral matter processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Apply the specified gradation requirements to the completed base course. Provide aggregates that are continuously well graded within the limits specified in TABLE 2. Use sieves that conform to ASTM E11.

TABLE 2. GRADATION OF AGGREGATES			
Percentage By Weight Passing Square-Mesh Sieve			
Sieve Designation	No. 1	No. 2	No.3
2 inch	100	---	---
1-1/2 inch	70-100	100	---
1 inch	45-80	60-100	100
1/2 inch	30-60	30-65	40-70
No. 4	20-50	20-50	20-50
No. 10	15-40	15-40	15-40
No. 40	5-25	5-25	5-25
No. 200	0-8	0-8	0-8

NOTE 1: Limit particles having diameters less than 0.02 mm to a maximum of 3 percent by weight of the total sample tested as determined in accordance with ASTM D7928.

NOTE 2: The values are based on aggregates of uniform specific gravity. If materials from different sources are used for the coarse and fine aggregates, test the materials in accordance with ASTM C127 and ASTM C128 to determine their specific gravities. Correct the percentages passing the various sieves as directed if the specific gravities vary by more than 10 percent.

NOTE 3: Gradations containing more than 30 percent retained on the ¾ inch sieve can produce inconsistent compacted density values when tested in

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accordance with paragraph DEGREE OF COMPACTION.

2.2 LIQUID LIMIT AND PLASTICITY INDEX

Apply liquid limit and plasticity index requirements to the completed course and to any component that is blended to meet the required gradation. Limit the portion of any component or of the completed course passing the No. 40 sieve to be either nonplastic or have a maximum liquid limit of 25 and a maximum plasticity index of 5.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following initial tests on the proposed material prior to commencing construction to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed. Submit certified copies of test results for approval a minimum of 30 days before material is required for the work.

- a. Gradation Analysis including 0.02 mm material.
- b. Liquid limit and plasticity index.
- c. Moisture-density relationship.
- d. Wear.
- e. Flat and Elongated Pieces.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

2.4 EQUIPMENT, TOOLS, AND MACHINES

All plant, equipment, and tools used in the performance of the work are subject to approval by the Government before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

When the ABC is constructed in more than one lift, clean the previously constructed lift of loose and foreign matter by sweeping with power sweepers or power brooms. Use hand brooms in areas where power cleaning is not practicable. Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 OPERATION OF AGGREGATE SOURCES

Clear, strip, and excavate as required. Condition aggregate sources on

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Government property to readily drain and leave in a satisfactory condition upon completion of the work.

3.3 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated to prevent segregation. Stockpile materials obtained from different sources separately.

3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the base course(s). Do not construct base course(s) on underlying course or subgrade that is frozen. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in ASTM D2487, stabilize the surface prior to placement of the base course(s). Stabilize by mixing ABC into the underlying course and compacting by approved methods. Proof roll in accordance with paragraph PROOF ROLLING. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a compliant condition until the base course is placed.

3.5 GRADE CONTROL

Provide a finished and completed base course conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.6 MIXING AND PLACING MATERIALS

3.6.1 Mixing

Mix the coarse and fine aggregates in a stationary plant. Make adjustments in mixing procedures or in equipment to obtain true grades, to minimize segregation or degradation, to obtain the required water content, and to produce a satisfactory base course meeting all requirements of this specification.

3.6.2 Placing

Place the mixed material on the prepared subgrade or subbase in lifts of uniform thickness with an approved spreader. Place the lifts so that when compacted they are true to the grades or levels required with the least possible surface disturbance. Where the base course is placed in more than one lift, clean the previously constructed lift of loose and foreign matter by sweeping with power sweepers, power brooms, or hand brooms. Make adjustments in placing procedures or equipment to obtain true grades, to minimize segregation and degradation, to adjust the water content, and to produce an acceptable base course.

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3.7 LAYER THICKNESS

Compact the completed base course to the thickness indicated. Limit individual compacted lifts to a maximum thickness of 6 inches and a minimum thickness of 3 inches. Compact the base course(s) to a total thickness that is within the tolerances of paragraph ACCEPTANCE of the thickness indicated. Where the measured thickness is more than 1/2 inch deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompact as directed. Where the measured thickness is more than 1/2 inch thicker than indicated, the course will be considered as conforming to the specified thickness requirements. However, the requirements for wearing course thickness and plan grade are still applicable. The average job thickness will be the average of all thickness measurements taken for the job and within the tolerances of paragraph ACCEPTANCE of the thickness indicated.

3.8 COMPACTION

Compact each lift of the base course, as specified, with approved compaction equipment. For cohesive soils, maintain water content during the compaction procedure to within plus or minus 2 percent of the optimum water content determined from laboratory tests as specified and for cohesionless soils, maintain the water content to facilitate compaction without bulking. Begin rolling at the outside edge of the surface and proceed to the center, overlapping on successive trips at least one-half the width of the roller. Slightly vary the length of alternate trips of the roller. Adjust speed of the roller as needed so that displacement of the aggregate does not occur. Compact mixture with hand-operated power tampers in all places not accessible to the rollers. Continue compaction until each lift is compacted through the full depth to meet the compaction requirements of Table 1. Make such adjustments in compacting or finishing procedures to obtain true grades, to minimize segregation and degradation, to reduce or increase water content, and to produce a compliant base course. Remove any materials found to be non-compliant and replace with compliant material or rework, as directed, to meet the requirements of this specification.

3.9 PROOF ROLLING

In addition to the compaction specified, proof roll areas designated on the drawings by application of a heavy pneumatic-tired roller having four or more tires abreast, each tire loaded to a minimum of 30,000 pounds and inflated to a minimum of 125 psi. A coverage is defined as the application of one tire print over the designated area. In the areas designated, apply proof rolling to the top of the underlying material on which the base course is laid and to the top of each lift of base course. Maintain water content of the underlying material and each lift of the base course as specified in Paragraph COMPACTION from start of compaction to completion of proof rolling of that lift. Remove any base course materials or any underlying materials that produce permanent deformation exceeding 3/8 inch by proof rolling and replace with satisfactory materials. Then recompact and proof roll to meet these specifications.

3.10 EDGES OF BASE COURSE

Place approved material along the outer edges of the base course in sufficient quantity to compact to the thickness of the course being constructed. When the course is being constructed in two or more lifts,

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simultaneously roll and compact at least a 2 foot width of this shoulder material with the rolling and compacting of each lift of the base course.

3.11 FINISHING

Finish the surface of the top lift of base course after final compaction and proof rolling by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin lifts of material to the top lift of base course to meet grade. If the elevation of the top lift of base course exceeds the tolerances of paragraph ACCEPTANCE, scarify the top lift to a depth of at least 3 inches and blend new material in and compact and proof roll to bring to grade. Make adjustments to rolling and finishing procedures to minimize segregation and degradation, obtain grades, maintain moisture content, and produce an acceptable base course. If the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the non-compliant portion and rework and recompact it or replace as directed.

3.12 SMOOTHNESS TEST

Construct the top lift so that the surface shows no deviations exceeding the tolerances of paragraph ACCEPTANCE when tested with a 12 foot straightedge. Test the entire area in both a longitudinal and a transverse direction on parallel lines. Perform the transverse lines at a maximum spacing of 15 feet or less apart, as directed. Perform the longitudinal lines at the centerline of each placement lane, regardless of whether multiple lanes are allowed to be paved at the same time, and at the 1/8th point in from each side of the lane. Hold the straightedge in contact with the surface and moved ahead one-half the length of the straightedge for each successive measurement. Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and measuring the maximum gap between the straightedge and the pavement surface. Determine measurements along the entire length of the straight edge. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.13 FIELD QUALITY CONTROL

3.13.1 In-Place Tests

Perform each of the following in-place tests on samples taken from the placed and compacted ABC. Determine sample locations using random sampling in accordance with ASTM D3665. Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square yards, or portion thereof, of completed area. Gradations containing more than 30 percent retained on the $\frac{3}{4}$ inch sieve can produce inconsistent compacted density values when tested in accordance with paragraph DEGREE OF COMPACTION.
- b. Perform gradation analysis including 0.02 mm size material on every lift of material placed and at a frequency of one sieve analysis for every 500 square yards, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.

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- d. Measure the thickness of the base course at intervals providing at least one measurement for each 500 square yards of base course or part thereof. Measure the thickness using test holes, at least 3 inch in diameter through the base course.

3.13.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.14 TRAFFIC

Completed portions of the base course can be opened to limited traffic, provided there is no marring or distorting of the surface by the traffic. Do not allow heavy equipment on the completed base course except when necessary for construction. When it is necessary for heavy equipment to travel on the completed base course, protect the area against marring or damage to the completed work. Repair damage to meet these specifications.

3.15 MAINTENANCE

Maintain the base course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact. Retest any base course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area of base course that is damaged as necessary to comply with this specification.

3.16 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed outside the limits of Government-controlled land. No additional payments will be made for materials that have to be replaced.

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BITUMINOUS TACK AND PRIME COATS 05/17

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Measurement

The bituminous material paid for will be the measured quantities of residual bituminous material used in the accepted work, provided that the measured quantities are not 10 percent over the specified quantities. Any amount of bituminous material more than 10 percent over the specified quantity will be deducted from the measured quantities. Express measured quantities in gallons at 60 degrees F. Correct volumes measured at temperatures other than 60 degrees F in accordance with ASTM D1250 using a coefficient of expansion of 0.00025 per degree F for asphalt emulsion.

1.1.2 Payment

The quantities of bituminous material, determined as specified above, will be paid for at the respective contract unit prices. Payment will constitute full compensation for all operations necessary to complete the work as specified herein.

1.1.3 Waybills and Delivery Tickets

Submit waybills and delivery tickets, during progress of the work. Before the final statement is allowed, file with the Contracting Officer certified waybills and certified delivery tickets for all bituminous materials used in the construction of the pavement covered by the contract. These submittals are required for Unit Pricing bid only. Do not remove bituminous material from storage until the initial outage and temperature measurements have been taken. The delivery or storage units will not be released until the final outage has been taken.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO T 102 (2009; R 2013) Standard Method of Test for
Spot Test of Asphaltic Materials

ASTM INTERNATIONAL (ASTM)

ASTM D140/D140M (2016) Standard Practice for Sampling
Asphalt Materials

ASTM D946/D946M (2020) Standard Specification for
Penetration-Graded Asphalt Cement for Use

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ASTM D977	(2019a; E 2019) Standard Specification for Emulsified Asphalt
ASTM D1250	(2019; E 2020) Standard Guide for Use of the Joint API and ASTM Adjunct for Temperature and Pressure Volume Correction Factors for Generalized Crude Oils, Refined Products, and Lubricating Oils: API MPMS Chapter 11.1
ASTM D2026/D2026M	(2015) Cutback Asphalt (Slow-Curing Type)
ASTM D2028/D2028M	(2015) Cutback Asphalt (Rapid-Curing Type)
ASTM D2995	(1999; R 2009) Determining Application Rate of Bituminous Distributors
ASTM D6373	(2016) Standard Specification for Performance Graded Asphalt Binder

U.S. GREEN BUILDING COUNCIL (USGBC)

LEED BD+C	(2009; R 2010) Leadership in Energy and Environmental Design(tm) Building Design and Construction (LEED-NC)
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1.3 SUBMITTALS

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SD-03 Product Data

Waybills and Delivery Tickets

SD-06 Test Reports

Sampling and Testing

1.4 QUALITY ASSURANCE

Certificates of compliance for asphalt materials delivered will be obtained and checked to ensure that specification requirements are met. Quantities of applied material will be determined. Payment will be for amount of residual asphalt applied. Tack coat materials will not be diluted. Prime coat materials when emulsions are used can be diluted on site with potable water up to 1 part emulsion to 1 part water.

1.5 DELIVERY, STORAGE, AND HANDLING

Inspect the materials delivered to the site for contamination and damage. Unload and store the materials with a minimum of handling.

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1.6 EQUIPMENT, TOOLS AND MACHINES

1.6.1 General Requirements

Equipment, tools and machines used in the work are subject to approval. Maintain in a satisfactory working condition at all times. Calibrate equipment such as asphalt distributors, scales, batching equipment, spreaders and similar equipment within 12 months of their use. If the calibration expires during project, recalibrate the equipment before work can continue.

1.6.2 Bituminous Distributor

Provide a self propelled distributor with pneumatic tires of such size and number to prevent rutting, shoving or otherwise damaging the surface being sprayed. Calibrate the distributor in accordance with ASTM D2995. Design and equip the distributor to spray the bituminous material in a uniform coverage at the specified temperature, at readily determined and controlled total liquid rates from 0.03 to 1.0 gallons per square yard, with a pressure range of 25 to 75 psi and with an allowable variation from the specified rate of not more than plus or minus 5 percent, and at variable widths. Include with the distributor equipment a separate power unit for the bitumen pump, full-circulation spray bars, tachometer, pressure gauges, volume-measuring devices, adequate heaters for heating of materials to the proper application temperature, a thermometer for reading the temperature of tank contents, and a hand hose attachment suitable for applying bituminous material manually to areas inaccessible to the distributor. The distributor will be capable of circulating and agitating the bituminous material during the heating process.

1.6.3 Heating Equipment for Storage Tanks

Use steam, electric, or hot oil heaters for heating the bituminous material. Provide steam heaters consisting of steam coils and equipment for producing steam, so designed that the steam cannot come in contact with the bituminous material. Fix an armored thermometer to the tank with a temperature range from 40 to 400 degrees F so that the temperature of the bituminous material may be determined at all times.

1.6.4 Power Brooms and Power Blowers

Use power brooms and power blowers suitable for cleaning the surfaces to which the bituminous coat is to be applied.

1.7 ENVIRONMENTAL REQUIREMENTS

Apply bituminous coat only when the surface to receive the bituminous coat is dry. A limited amount of moisture (approximately 0.03 gallon/square yard) can be sprayed on the surface of unbound material when prime coat is used to improve coverage and penetration of asphalt material. Apply bituminous coat only when the atmospheric temperature in the shade is 50 degrees F or above and when the temperature has not been below 35 degrees F for the 12 hours prior to application, unless otherwise directed.

PART 2 PRODUCTS

2.1 PRIME COAT

Provide asphalt conforming to one of the following grades:

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2.1.1 Cutback Asphalt

Provide cutback asphalt conforming to ASTM D2026/D2026M, Grade SC-70.

2.1.2 Emulsified Asphalt

Provide emulsified asphalt conforming to ASTM D977, Type SS-1. Asphalt emulsion can be diluted up to 1 part water to 1 part emulsion for prime coat use. Do not dilute asphalt emulsion for tack coat use.

2.2 TACK COAT

2.2.1 Asphalt Cement

Provide asphalt cement conforming to ASTM D946/D946M or ASTM D6373 Grade.

2.2.2 Cutback Asphalt

Provide cutback asphalt conforming to ASTM D2028/D2028M, Grade RC-70.

2.2.3 Emulsified Asphalt

Provide emulsified asphalt conforming to ASTM D977. For prime coats the emulsified asphalt can be diluted with up to 1 part emulsion to 1 part water. No dilution is allowed for tack coat applications. The base asphalt used to manufacture the emulsion is required to show a negative spot when tested in accordance with AASHTO T 102 using standard naphtha.

2.2.4 Local/Regional Materials

Submit documentation indicating distance between manufacturing facility and the project site. Indicate distance of raw material origin from the project site. Indicate relative dollar value of local/regional materials to total dollar value of products included in project in accordance with LEED BD+C.

PART 3 EXECUTION

3.1 PREPARATION OF SURFACE

Immediately before applying the bituminous coat, remove all loose material, dirt, clay, or other objectionable material from the surface to be treated by means of a power broom or blower supplemented with hand brooms. Apply treatment only when the surface is dry and clean.

3.2 APPLICATION RATE

The exact quantities within the range specified, which may be varied to suit field conditions, will be determined by the Contracting Officer.

3.2.1 Tack Coat

Apply bituminous material for the tack coat in quantities of not less than 0.03 gallons nor more than 0.10 gallons per square yard of residual asphalt onto the pavement surface as approved by the Contracting Officer. Do not dilute asphalt emulsion when used as a tack coat.

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3.2.2 Prime Coat

Apply bituminous material for the prime coat in quantities of not less than 0.05 gallons nor more than 0.12 gallons per square yard of residual asphalt for asphalt emulsion up to a 1 to 1 dilution rate or for residual asphalt for cutback asphalt.

3.3 APPLICATION TEMPERATURE

3.3.1 Viscosity Relationship

Apply asphalt at a temperature that will provide a viscosity between 10 and 60 seconds, Saybolt Furol, or between 20 and 120 centistokes, kinematic. Furnish the temperature viscosity relation to the Contracting Officer.

3.3.2 Temperature Ranges

The viscosity requirements determine the application temperature to be used. The following is a normal range of application temperatures:

Cutback Asphalts	
MC-30	85-190 degrees F
SC-70, MC-70, RC-70	120-225 degrees F
SC-250, MC-250, RC-250	165-270 degrees F
Asphalt Emulsion	
All Grades	70-160 degrees F
Asphalt Cement	
All Grades	275-350 degrees F

Some of these temperatures for rapid cure cutbacks are above the flash point of the material and care should be taken in their heating.

3.4 APPLICATION

3.4.1 General

Following preparation and subsequent inspection of the surface, apply the bituminous prime or tack coat with the bituminous distributor at the specified rate with uniform distribution over the surface to be treated. Properly treat all areas and spots, not capable of being sprayed with the distributor, with the hand spray. Until the succeeding layer of pavement is placed, maintain the surface by protecting the surface against damage and by repairing deficient areas at no additional cost to the Government. If required, spread clean dry sand to effectively blot up any excess bituminous material. No smoking, fires, or flames other than those from the heaters that are a part of the equipment are permitted within 25 feet of heating, distributing, and transferring operations of cutback materials. Prevent all traffic, except for paving equipment used in constructing the surfacing, from using the underlying material, whether

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primed or not, until the surfacing is completed. The bituminous coat requirements are described herein.

3.4.2 Prime Coat

The prime coat is required if it will be at least 7 days before the asphalt mixture is constructed on the underlying (base course, etc.) compacted material. The type of liquid asphalt and application rate will be as specified herein. Protect the underlying layer from any damage (water, traffic, etc.) until the surfacing is placed. If the Contractor places the surfacing within seven days, the choice of protection measures or actions to be taken is at the Contractor's option. Repair (recompact or replace) damage to the underlying material caused by lack of, or inadequate, protection by approved methods at no additional cost to the Government. If the Contractor opts to use the prime coat, apply as soon as possible after consolidation of the underlying material. Apply the bituminous material uniformly over the surface to be treated at a pressure range of 25 to 75 psi; the rate will be as specified above in paragraph APPLICATION RATE. To obtain uniform application of the prime coat on the surface treated at the junction of previous and subsequent applications, spread building paper on the surface for a sufficient distance back from the ends of each application to start and stop the prime coat on the paper and to ensure that all sprayers will operate at full force on the surface to be treated. Immediately after application remove and destroy the building paper.

3.4.3 Tack Coat

Apply tack coat at the locations shown on the drawings. A tack coat should be applied to every bound surface (asphalt or concrete pavement) that is being overlaid with asphalt mixture and at transverse and longitudinal joints. Apply the tack coat when the surface to be treated is clean and dry. Immediately following the preparation of the surface for treatment, apply the bituminous material by means of the bituminous distributor, within the limits of temperature specified herein and at a rate as specified above in paragraph APPLICATION RATE. Apply the bituminous material so that uniform distribution is obtained over the entire surface to be treated. Treat lightly coated areas and spots missed by the distributor by spraying with a hand wand or using other approved method. Following the application of bituminous material, allow the surface to cure without being disturbed for period of time necessary to permit setting of the tack coat. Apply the bituminous tack coat only as far in advance of the placing of the overlying layer as required for that day's operation. Maintain and protect the treated surface from damage until the succeeding course of pavement is placed.

3.5 CURING PERIOD

Following application of the bituminous material and prior to application of the succeeding layer of asphalt mixture allow the bituminous coat to cure and water or volatiles to evaporate prior to overlaying. Maintain the tacked surface in good condition until the succeeding layer of pavement is placed, by protecting the surface against damage and by repairing and recoating deficient areas. Allow the prime coat to cure without being disturbed for a period of at least 48 hours or longer, as may be necessary to attain penetration into the treated course. Furnish and spread enough sand to effectively blot up excess bituminous material.

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3.6 FIELD QUALITY CONTROL

Obtain certificates of compliance for all asphalt material delivered to the project. Obtain samples of the bituminous material under the supervision of the Contracting Officer. The sample may be retained and tested by the Government at no cost to the Contractor.

3.7 SAMPLING AND TESTING

Furnish certified copies of the manufacturer's test reports indicating temperature viscosity relationship for cutback asphalt or asphalt cement, compliance with applicable specified requirements, not less than 5 days before the material is required in the work.

3.7.1 Sampling

Unless otherwise specified, sample bituminous material in accordance with ASTM D140/D140M.

3.7.2 Calibration Test

Furnish all equipment, materials, and labor necessary to calibrate the bituminous distributor. Calibrate using the approved job material and prior to applying the bituminous coat material to the prepared surface. Calibrate the bituminous distributor in accordance with ASTM D2995.

3.7.3 Trial Applications

Before applying the spray application of tack or prime coat, apply three lengths of at least 100 feet for the full width of the distributor bar to evaluate the amount of bituminous material that can be satisfactorily applied.

3.7.3.1 Tack Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous tack coat materials in the amount of 0.05 gallons per square yard. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.3.2 Prime Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous materials in the amount of 0.15 gallon per square yard. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.4 Sampling and Testing During Construction

Perform quality control sampling and testing as required in paragraph FIELD QUALITY CONTROL.

3.8 TRAFFIC CONTROLS

Keep traffic off surfaces freshly treated with bituminous material. Provide sufficient warning signs and barricades so that traffic will not travel over freshly treated surfaces.

-- End of Section --

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ROAD-MIX ASPHALT PAVING 11/20

PART 1 GENERAL

1.1 PERCENT PAYMENT

1.1.1 Method of Measurement

The amount paid for will be the number of tons of hot-mix warm-mix asphalt pavement mixture used in the accepted work. Weigh the hot-mix warm-mix asphalt pavement mixture after mixing. No separate payment will be made for weight of asphalt cement material incorporated herein.

1.1.2 Basis of Payment

The measured quantity of hot-mixed warm-mixed asphalt pavement will be paid for and included in the lump sum contract price.

1.2 PAYMENT

1.2.1 Method of Measurement

The amount paid for will be the number of tons of hot-mix warm-mix asphalt pavement mixture used in the accepted work. Weigh the hot-mix warm-mix asphalt pavement mixture after mixing. No separate payment will be made for weight of asphalt cement material incorporated herein.

1.2.2 Basis of Payment

The measured quantity of hot-mixed warm-mixed asphalt pavement will be paid for and included in the lump sum contract price.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 156	(2013; R 2017) Standard Specification for Requirements for Mixing Plants for Hot-Mixed, Hot-Laid Bituminous Paving Mixtures
AASHTO T 304	(2011; R 2015) Standard Method of Test for Uncompacted Void Content of Fine Aggregate
AASHTO T 329	(2015) Standard Test Method for Moisture Content of Hot Mix Asphalt (HMA) by Oven Method

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ASPHALT INSTITUTE (AI)

AI MS-2 (2015) Asphalt Mix Design Methods

ASTM INTERNATIONAL (ASTM)

ASTM C29/C29M	(2017a) Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate
ASTM C88	(2018) Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C117	(2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C127	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate
ASTM C128	(2022) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
ASTM C131/C131M	(2020) Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM C142/C142M	(2017) Standard Test Method for Clay Lumps and Friable Particles in Aggregates
ASTM C566	(2013) Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying
ASTM D75/D75M	(2019) Standard Practice for Sampling Aggregates
ASTM D242/D242M	(2009; R 2014) Mineral Filler for Bituminous Paving Mixtures
ASTM D979/D979M	(2015) Sampling Bituminous Paving Mixtures
ASTM D2041/D2041M	(2011) Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures
ASTM D2172/D2172M	(2017; E 2018) Standard Test Methods for Quantitative Extraction of Asphalt Binder from Asphalt Mixtures
ASTM D2419	(2014) Sand Equivalent Value of Soils and Fine Aggregate

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ASTM D2726/D2726M	(2019) Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures
ASTM D3203/D3203M	(2017) Standard Test Method for Percent Air Voids in Compacted Asphalt Mixtures
ASTM D3665	(2012; R 2017) Standard Practice for Random Sampling of Construction Materials
ASTM D3666	(2016) Standard Specification for Minimum Requirements for Agencies Testing and Inspecting Road and Paving Materials
ASTM D4791	(2019) Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate
ASTM D4867/D4867M	(2009; R 2014) Effect of Moisture on Asphalt Concrete Paving Mixtures
ASTM D5361/D5361M	(2016) Standard Practice for Sampling Compacted Asphalt Mixtures for Laboratory Testing
ASTM D5444	(2015) Mechanical Size Analysis of Extracted Aggregate
ASTM D5821	(2013; R 2017) Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate
ASTM D6084/D6084M	(2018) Standard Test Method for Elastic Recovery of Asphalt Materials by Ductilometer
ASTM D6307	(2019) Standard Test Method for Asphalt Content of Asphalt Mixture by Ignition Method
ASTM D6925	(2014) Standard Test Method for Preparation and Determination of the Relative Density of Hot Mix Asphalt (HMA) Specimens by Means of the Superpave Gyratory Compactor
ASTM E1274	(2018) Standard Test Method for Measuring Pavement Roughness Using a Profilograph

1.4 SUBMITTALS

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SD-03 Product Data

Mix Design; G-D0

Contractor Quality Control; G-D0

SD-04 Samples

Aggregates

Asphalt Cement Binder

SD-06 Test Reports

Aggregates; G-D0

QC Monitoring

SD-07 Certificates

Asphalt Cement Binder; G-D0

Laboratory Accreditation and Validation

1.5 ACCEPTANCE

1.5.1 Acceptability of Work

Acquire the services of an independent commercial laboratory to perform acceptance testing. Acceptance of the plant produced mix and in-place requirements will be on a lot to lot basis. The materials and the pavement itself will be accepted on the basis of production testing. The Government may make check tests from split samples to validate the results of the production testing. Testing performed by the Government does not reduce the required testing of the independent commercial laboratory. Split samples will be taken for Government testing to reduce the variability between the independent commercial laboratory and the Government's test results. When the difference between the independent commercial laboratory and the Government's test results for split samples exceed the acceptable range of two results for multilaboratory precision for the appropriate test method (i.e. ASTM) then at least one of the laboratories is determined to be in error. An evaluation of procedures and equipment in both laboratories will be made to determine the cause(s) for the differences. Develop steps to correct procedures and equipment to bring multilaboratory precision to within acceptable limits.

1.5.2 Acceptance Requirements

Provide all sampling and testing required for acceptance and payment adjustment. Where appropriate, adjustments in percent payment acceptance for individual lots of asphalt pavement will be made based on laboratory air voids, in-place density, smoothness, and grade in accordance with the following paragraphs. Surface smoothness and grade determinations will be made on the lot as a whole. Exceptions or adjustments to this will be made in situations where the mix within one lot is placed as part of both the intermediate and surface courses, thus smoothness and grade measurements for the entire lot cannot be made.

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1.5.3 Pavement Lots

A standard lot for all requirements is equal to one day's production or 2,000 tons, whichever is smaller. Divide each lot into four equal sublots in order to evaluate laboratory air voids and in-place density. When operational conditions cause a lot to be terminated before the specified four sublots have been completed, use the following procedure to adjust the lot size and number of tests for the lot. Where three sublots have been completed, they constitute a lot. Where one or two sublots have been completed, incorporate them into the next lot and the total number of sublots (i.e. 5 or 6 sublots) is used for acceptance criteria. Include partial lots at the end of asphalt production into the previous lot. Complete and report all theoretical maximum density, laboratory air voids, and in-place density testing within 24 hours after construction of each lot.

1.5.4 Sublot Sampling

Take one mixture sample for each sublot in accordance with ASTM D979/D979M from a random truck or another location for determining theoretical maximum density, laboratory air voids, any additional testing the Government desires, and Contractor Quality Control. All samples will be selected randomly, using commonly recognized methods of assuring randomness conforming to ASTM D3665 and employing tables of random numbers or computer programs.

1.5.5 Additional Sampling and Testing

The Government reserves the right to direct additional samples and tests for any area which appears to deviate from the specification requirements. The cost of any additional testing will be paid for by the Government. Testing in these areas will be treated as a separate lot. Payment Acceptance will be made for the quantity of asphalt pavement represented by these tests in accordance with the provisions of this section.

1.5.6 Theoretical Maximum Density (TMD)

Measure theoretical maximum density one time for each sublot in accordance with ASTM D2041/D2041M for purposes of calculating laboratory air voids and determining in-place density. The average TMD for each lot will be determined as the average TMD of the random sublot samples. When the TMD on both sides of a longitudinal joint is different, the average of these two TMD values will be used as the TMD needed to calculate the percent joint density.

1.5.7 Laboratory Air Voids

Prepare one set of laboratory compacted specimens for each sublot in accordance with ASTM D6925 using the Superpave gyratory compactor. Provide three test specimens prepared from the same sample for each set of laboratory compacted specimens. Compact the specimens within 2 hours of the time the mixture was loaded into trucks at the asphalt plant. Do not reheat samples prior to compaction. Provide insulated containers as necessary to maintain the sample temperature. Measure the bulk density of laboratory compacted specimens in accordance with ASTM D2726/D2726M. Determine laboratory air voids from one set (three laboratory compacted specimens) for each sublot sample in accordance with ASTM D3203/D3203M.

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1.5.7.1 Tolerance

Provide laboratory air voids with a mean absolute deviation of 1.00 percent or less from the JMF for each lot. Remove and replace lots that do not meet the laboratory air voids requirement at least 4 inches into the cold (existing) lane adjacent to the longitudinal joint, at no additional cost to the Government. The mean absolute deviation of the laboratory air void contents from the JMF air void content will be evaluated as shown in the example below.

1.5.7.2 Calculating Laboratory Air Voids

Laboratory air void calculations for each lot will use the average theoretical maximum density values obtained for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The mean absolute deviation of the laboratory air void contents (one from each subplot) from the JMF air void content will be evaluated as in the following example:

Assume that the laboratory air voids are determined from 4 sublots where one set of laboratory compacted specimens is from a single subplot. The laboratory air voids for the 4 sublots are determined to be 3.5, 3.0, 4.0, and 3.7. Assume that the target air voids from the JMF is 4.0. The mean absolute deviation is then:

$$\text{Mean Absolute Deviation} = (|3.5 - 4.0| + |3.0 - 4.0| + |4.0 - 4.0| + |3.7 - 4.0|)/4$$

$$\text{Mean Absolute Deviation} = (0.5 + 1.0 + 0.0 + 0.3)/4 = (1.8)/4 = 0.45$$

The mean absolute deviation for laboratory air voids is determined to be 0.45. It can be seen that 0.45 is less than 1.00 percent. The lot is acceptable for laboratory air voids.

1.5.8 In-place Density

Obtain one random 4 inch or 6 inch diameter core from the mat and joint of each subplot in accordance with ASTM D5361/D5361M for determining in-place density. Cut samples neatly with a diamond core drill bit. Obtain random cores that are the full thickness of the layer being placed. Select core locations randomly using the procedures contained in ASTM D3665. Locate cores for mat density no closer than 12 inches from a transverse or longitudinal joint including the pavement edge. Center all cores for joint density on the joint. Discard samples that are clearly defective as a result of sampling and take an additional random core. When the random core is less than 1 inch thick, it will not be included in the analysis. In this case, obtain another random core sample. Clean and tack coat dry core holes before filling with asphalt mixture. Fill all core holes with asphalt mixture and compact using a standard Marshall hammer to the density specified. Provide all tools, labor, and materials for cutting samples, cleaning, and filling the cored pavement. Measure in-place density in accordance with ASTM D2726/D2726M using each core obtained from the mat and joint.

1.5.8.1 Tolerance

Provide a minimum in-place mat density of 93.0 percent and a minimum in-place joint density of 90.0 percent for each lot. The average in-place mat and joint densities are expressed as a percentage of the average

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theoretical maximum density (TMD) for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). Remove and replace lots that do not meet the in-place mat density requirement at least 4 inches into the cold (existing) lane adjacent to the longitudinal joint, at no additional cost to the Government. Remove and replace the longitudinal joint when the lot does not meet the in-place joint density, at no additional cost to the Government. Use a 10 feet wide paving lane that is centered over the joint.

1.5.9 Surface Smoothness

Use a straightedge and profilograph for measuring surface smoothness. Use the profilograph method for all longitudinal testing, except for paving lanes less than 0.25 miles in length. Use the straightedge method for transverse testing, for longitudinal testing where the length of each pavement lane is less than 0.25 miles, and at the ends of the paving limits for the project. Smoothness requirements do not apply over crowns or grade breaks. Maintain detailed notes of the testing results and provide a copy to the Government immediately after each day's testing.

1.5.9.1 Smoothness Requirements

1.5.9.1.1 Straightedge Testing

Provide finished surfaces of the pavements with no abrupt change of 1/4 inch or more when checked with an approved 12 foot straightedge. Remove and replace surface lift lots when the surface smoothness exceeds 3/8 inch, at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet surface smoothness requirements at individual locations.

1.5.9.1.2 Profilograph Testing

Provide finished surfaces with a Profile Index not greater than 9 inches per mile when tested with an approved California-type profilograph. Remove and replace the lot when the Profile Index exceeds the tolerance by 4.0 inches per mile or more, at no additional cost to the Government. Correct any small individual area with surface deviation which exceeds the tolerance given above by more than 5.0 inches per mile or more by diamond grinding to meet the specification requirements above or remove and replace at no additional cost to the Government.

1.5.9.2 Testing Method

After the final rolling, but not later than 24 hours after placement, test the surface of the pavement in each entire lot in a manner to reveal surface irregularities exceeding the tolerances specified above. If any pavement areas are diamond ground, retest these areas immediately after diamond grinding. The maximum area allowed to be corrected by diamond grinding is 10 percent of the total area of the lot. Test the entire area of the pavement with a profilograph. Check a number of random locations along with any observed suspicious locations primarily at transverse and longitudinal joints with the straightedge.

1.5.9.2.1 Straightedge Testing

Use the straightedge to measure abrupt changes in surface smoothness. Hold the straightedge in contact with the pavement surface and measure the maximum distance between the straightedge and the pavement surface.

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Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length, and measuring the maximum gap between the straightedge and the pavement surface in the area between these two high points.

1.5.9.2.2 Profilograph Testing

Perform profilograph testing using an approved California profilograph and procedures described in ASTM E1274. Provide equipment that utilizes electronic recording and automatic computerized reduction of data to indicate "must-grind" bumps and the Profile Index for the pavement. Use a "blanking band" that is 0.2 inch wide and the "bump template" spanning 1 inch with an offset of 0.4 inch. Provide profilograph operated by an approved, factory-trained operator on the alignments specified above. Provide a copy of the reduced tapes to the Government at the end of each day's testing.

1.5.9.2.3 Bumps ("Must Grind" Areas)

Reduce any bumps ("must grind" areas) shown on the profilograph trace which exceed 0.4 inch in height by diamond grinding until they do not exceed 0.3 inch when retested. Taper diamond grinding in all directions to provide smooth transitions to areas not requiring diamond grinding. The following will not be permitted: (1) skin patching for correcting low areas, (2) planing or milling for correcting high areas.

1.5.10 Plan Grade

Provide a final wearing surface of pavement conforming to the elevations and cross sections shown and not vary more than 0.05 foot from the plan grade established and approved at site of work. Within 5 working days after completion of a particular lot incorporating the final wearing course, test the final wearing surface of the pavement for conformance with specified plan grade requirements. Match finished surfaces at juncture with other pavements with finished surfaces of abutting pavements. Deviation from the plan elevation will not be permitted in areas of pavements where closer conformance with planned elevation is required for the proper functioning of drainage and other appurtenant structures involved. For roads, the grade will be determined by running lines of levels along the centerline at intervals of 25 feet or less longitudinally to determine the elevation of the completed pavement surface. Measure transverse grades at appropriate intervals. For parking lots, the grade will be determined by running lines of levels at intervals of 25 feet or less longitudinally and transversely to determine the elevation of the completed pavement surface. Diamond grinding can be used to remove high spots to meet grade requirements. Skin patching for correcting low areas or planing or milling for correcting high areas will not be permitted. Maintain detailed notes of the results of the testing and provide a copy to the Government immediately after each day's testing.

Remove and replace surface lift lots when individual locations exceed 0.05 foot tolerance, at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet plan grade requirements at individual locations.

1.5.11 Laboratory Accreditation and Validation

Provide laboratories used to develop the Job Mix Formula (JMF), perform acceptance testing, and Contractor Quality Control testing that meet the

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requirements of ASTM D3666. Provide laboratories with a masonry saw having a diamond blade for trimming pavement cores and samples. Perform all required test methods by an accredited laboratory. Schedule and provide payment for laboratory inspections. Additional payment or a time extension due to failure to acquire the required laboratory accreditation is not allowed. The Government will inspect the laboratory equipment and test procedures prior to the start of hot-mix warm-mix operations for conformance with ASTM D3666. In addition, all testing laboratories performing acceptance testing require USACE validation by the Material Testing Center (MTC) for both parent laboratory and plant testing laboratory. Validation on all laboratories is required to remain current throughout the duration of the paving project. Contact the MTC manager listed at <https://mtc.erdc.dren.mil> for costs and scheduling. Submit a certificate of compliance signed by the manager of the laboratory stating that it meets these requirements to the Government prior to the start of construction. At a minimum, include the following certifications:

- a. Qualifications of personnel; laboratory manager, supervising technician, and testing technicians.
- b. A listing of equipment to be used in developing the job mix.
- c. A copy of the laboratory's quality control system.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not place the asphalt mixture upon a wet surface or when the surface temperature of the underlying course is less than specified in Table 3 Table 1. The temperature requirements may be waived by the Government, if requested; however, meet all other requirements including compaction.

Table 3. Table 1. Surface Temperature Limitations of Underlying Course	
Mat Thickness, inches	Degrees F
3 or greater	40
Less than 3	45

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Perform the work consisting of pavement courses composed of mineral aggregate and asphalt material heated and mixed in a central mixing plant and placed on a prepared course. Provide asphalt pavement designed and constructed in accordance with this section conforming to the lines, grades, thicknesses, and typical cross sections shown on the drawings. Construct each course to the depth, section, or elevation required by the drawings and rolled, finished, and approved before the placement of the next course.

2.1.1 Asphalt Mixing Plant

Provide plants used for the preparation of asphalt mixture conforming to the requirements of AASHTO M 156 with the following changes:

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2.1.1.1 Truck Scales

Weigh the asphalt mixture on approved scales, or on certified public scales at no additional expense to the Government. Inspect and seal scales at least annually by an approved calibration laboratory.

2.1.1.2 Inspection of Plant

Provide access to the Government at all times, to all areas of the plant for checking adequacy of equipment; inspecting operation of the plant; verifying weights, proportions, and material properties; checking the temperatures maintained in the preparation of the mixtures and for taking samples. Provide assistance as requested, for the Government to procure any desired samples.

2.1.1.3 Storage bins

The asphalt mixture can be stored in non-insulated storage bins for a period of time not exceeding 3 hours. The asphalt mixture can be stored in insulated storage bins for a period of time not exceeding 8 hours. Provide the mix drawn from bins that meets the same requirements as mix loaded directly into trucks.

2.1.2 Hauling Equipment

Provide trucks used for hauling asphalt mixture that have tight, clean, and smooth metal beds. To prevent the mixture from adhering to them, lightly coat the truck beds with a minimum amount of paraffin oil, lime solution, or other approved material. Do not use petroleum based products as a release agent. Provide each truck with a suitable cover to protect the mixture from adverse weather, contamination, and loss of material during hauling. When necessary due to long haul distance and cold weather, provide insulated truck beds with covers (tarps) that are securely fastened.

2.1.3 Asphalt Pavers

Provide mechanical spreading and finishing equipment consisting of a self-powered paver, capable of spreading and finishing the mixture to the specified line, grade, and cross section. Provide paver screed capable of laying a uniform mixture to meet the specified thickness, smoothness, and grade without physical or temperature segregation, the full width of the material being placed. Provide a paver with a vibrating screed to be used during all placement.

2.1.3.1 Receiving Hopper

Provide paver with a receiving hopper of sufficient capacity to permit a uniform spreading operation and a distribution system to place the mixture uniformly in front of the screed without segregation. Provide a screed that effectively produces a finished surface of the required evenness and texture without tearing, shoving, or gouging the mixture.

2.1.3.2 Automatic Grade Controls

Provide a paver equipped with a control system capable of maintaining the specified screed elevation. One of three methods can be used to control grade: stringline, laser, or computerized elevations along with GPS. For multiple layers it is acceptable to control the grade in the underlying

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layer and control the grade of the surface layer by applying a constant thickness over the underlying layer which has been placed to the desired grade. Slope control can also be used to control the grade of the surface for roads, but is not acceptable for wide pavements such as parking lots. Provide transverse slope controller capable of maintaining the screed at the desired slope within plus or minus 0.1 percent. A ski-type device of not less than 30 ft can be used to provide improved smoothness. Use a shoe on one side of the paver to match an existing paved surface to provide a smooth joint.

2.1.4 Rollers

Provide rollers in good condition and operate at slow speeds to avoid displacement of the asphalt mixture. Provide sufficient number, type, and weight of rollers to compact the mixture to the required density while it is still in a workable condition. Do not use equipment which causes excessive crushing of the aggregate.

2.2 AGGREGATES

Notify the Government at least 7 days before sampling aggregates. Obtain samples in accordance with ASTM D75/D75M that are representative of the materials to be used for the project. Provide aggregates consisting of crushed stone, crushed gravel, crushed slag, screenings, natural sand, and mineral filler as required. The portion of material retained on the No. 4 sieve is coarse aggregate. The portion of material passing the No. 4 sieve and retained on the No. 200 sieve is fine aggregate. The portion passing the No. 200 sieve is defined as mineral filler. Submit sufficient materials to produce 200 pounds of blended mixture for mix design verification. Submit all aggregate test results and samples to the Government at least 14 days prior to start of construction. Perform job aggregate testing no earlier than 6 months before contract award.

2.2.1 Coarse Aggregate

Provide coarse aggregate consisting of sound, tough, durable particles, free from films of material that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances. Provide coarse aggregate particles meeting the following requirements and TXDOT requirements:

- a. The percentage of loss not greater than 40 percent after 500 revolutions when tested in accordance with ASTM C131/C131M.
- b. The sodium sulfate soundness loss not exceeding 12 percent, or the magnesium sulfate soundness loss not exceeding 18 percent after five cycles when tested in accordance with ASTM C88.
- c. At least 75 percent by weight of coarse aggregate containing two or more fractured faces when tested in accordance with ASTM D5821 with fractured faces produced by crushing.
- d. The particle shape essentially cubical and the aggregate containing not more than 10 percent, by weight, of flat and elongated particles (5:1 ratio of length to thickness) when tested in accordance with ASTM D4791, Method B.
- e. Slag consisting of air-cooled, blast furnace slag with a compacted weight of not less than 75 lb/cu ft when tested in accordance with

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ASTM C29/C29M.

- f. Clay lumps and friable particles not exceeding 0.3 percent, by weight, when tested in accordance with ASTM C142/C142M.

2.2.2 Fine Aggregate

Provide fine aggregate consisting of clean, sound, tough, durable particles. Provide aggregate particles that are free from coatings of clay, silt, or any objectionable material, contain no clay balls, and meet the following requirements and TXDOT requirements:

- a. Quantity of natural sand (noncrushed material) added to the aggregate blend not exceeding 15 percent by weight of total aggregate.
- b. Individual fine aggregate sources with a sand equivalent value greater than 45 when tested in accordance with ASTM D2419.
- c. Fine aggregate portion of the blended aggregate with an uncompacted void content greater than 45.0 percent when tested in accordance with AASHTO T 304 Method A.
- d. Clay lumps and friable particles not exceeding 0.3 percent, by weight, when tested in accordance with ASTM C142/C142M.

2.2.3 Mineral Filler

Provide mineral filler consisting of a nonplastic material meeting the requirements of ASTM D242/D242M.

2.2.4 Aggregate Gradation

Provide a combined aggregate gradation that conforms to gradations specified in Table 4 Table 2, when tested in accordance with ASTM C136/C136M and ASTM C117, and does not vary from the low limit on one sieve to the high limit on the adjacent sieve or vice versa, but grades uniformly from coarse to fine. Provide a JMF within the specification limits; however, the gradation can exceed the limits when the allowable deviation from the JMF shown in Tables 6 and 7 Tables 4 and 5 are applied.

Table 4. Table 2. Aggregate Gradations			
Sieve Size, inch	Gradation 1 Percent Passing by Mass	Gradation 2 Percent Passing by Mass	Gradation 3 Percent Passing by Mass
1	100	---	---
3/4	90-100	100	---
1/2	68-88	90-100	100
3/8	60-82	69-89	90-100
No. 4	45-67	53-73	58-78
No. 8	32-54	38-60	40-60
No. 16	22-44	26-48	28-48
No. 30	15-35	18-38	18-38

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Table 4. Table 2. Aggregate Gradations			
Sieve Size, inch	Gradation 1 Percent Passing by Mass	Gradation 2 Percent Passing by Mass	Gradation 3 Percent Passing by Mass
No. 50	9-25	11-27	11-27
No. 100	6-18	6-18	6-18
No. 200	3-6	3-6	3-6

2.3 ASPHALT CEMENT BINDER

Provide an asphalt binder with a minimum elastic recovery of 75 percent in accordance with ASTM D6084/D6084M and TXDOT specifications. Provide test data indicating grade certification by the supplier at the time of delivery of each load to the mix plant. When warm-mix asphalt technology involves additives, grade the asphalt binder with the asphalt binder additive included. Submit copies of these certifications to the Government. The supplier is defined as the last source of any modification to the binder. The Government may sample and test the binder at the mix plant at any time before or during mix production.

2.4 WARM-MIX ASPHALT TECHNOLOGIES/PRODUCTS

Provide warm-mix asphalt technologies/products that have a record of good performance and are included on the local state DOT's qualified products list, if the DOT maintains a qualified products list. These qualified products lists can be found at each state DOT's website.

2.5 MIX DESIGN

Develop the mix design. Perform Job Mix formula (JMF) and aggregates testing no earlier than 6 months before contract award. Provide asphalt mixture composed of well-graded aggregate, mineral filler if required, and asphalt material that meets TXDOT requirements and specifications. Provide aggregate fractions sized, handled in separate size groups, and combined in such proportions that the resulting mixture meets the grading requirements of Table 4 Table 2, the JMF, and TXDOT specifications for mix design. Do not produce asphalt pavement for payment acceptance until a JMF has been approved. Use laboratory compaction temperatures for Polymer Modified Asphalts as recommended by the asphalt binder manufacturer. Determine the Tensile Strength Ratio (TSR) of the composite mixture in accordance with ASTM D4867/D4867M. Compact the TSR specimens to an air void content of 7 percent plus or minus 1 percent. If the Tensile Strength Ratio (TSR) of the composite mixture is less than 75, reject the aggregates or treat the asphalt mixture with an anti-stripping agent. Add a sufficient amount of anti-stripping agent to produce a TSR of not less than 75. If an antistrip agent is required, provide it at no additional cost to the Government. Provide sufficient materials to produce 200 pound of blended mixture to the Government for verification of mix design at least 14 days prior to construction of test section.

2.5.1 JMF Requirements

Submit the proposed JMF in writing, for approval and in accordance with

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TXDOT Division III Surface Courses or Pavements specifications, at least 14 days prior to the start of the test section including, as a minimum:

- a. Percent passing each sieve size.
- b. Percent of asphalt cement.
- c. Percent of each aggregate and mineral filler to be used.
- d. Asphalt performance grade or penetration grade.
- e. Laboratory mixing temperature.
- f. Laboratory compaction temperature.
- g. Temperature-viscosity relationship of the asphalt cement
- h. Plot of the combined gradation on the 0.45 power gradation chart, stating the nominal maximum size.
- i. Specific gravity and absorption of each aggregate.
- j. Percent natural sand.
- k. Percent particles with two or more fractured faces (in coarse aggregate).
- l. Fine aggregate angularity.
- m. Percent flat or elongated particles in coarse aggregate.
- n. Tensile Strength Ratio and wet/dry specimen test results.
- o. Antistrip agent (if required).
- p. List of all modifiers.
- q. Percentage and properties (asphalt content, aggregate gradation, and aggregate properties) of RAP in accordance with paragraph RECYCLED ASPHALT PAVEMENT, if RAP is used.

2.5.2 Adjustments to JMF

The JMF for each mixture is in effect until a new formula is approved in writing by the Government. Should a change in sources of any materials be made, perform a new mix design and a new JMF approved before the new material is used. Make minor adjustments within the specification limits to the JMF to optimize mix volumetric properties. Adjustments to the original JMF are limited to plus or minus 4 percent on the No. 4 and coarser sieves; plus or minus 3 percent on the No. 8 to No. 50 sieves; and plus or minus 1 percent on the No. 100 sieve and No. 200 sieve. Asphalt content adjustments are limited to plus or minus 0.40 from the original JMF. If adjustments are needed that exceed these limits, develop a new mix design.

2.6 RECYCLED HOT MIX ASPHALT

Recycled asphalt mixture is not allowed for the project.

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PART 3 EXECUTION

3.1 CONTRACTOR QUALITY CONTROL

3.1.1 General Quality Control Requirements

Submit the Quality Control Plan. Do not produce hot-mix warm-mix asphalt for payment acceptance until the quality control plan has been approved. In the quality control plan, address all elements which affect the quality of the pavement including, but not limited to:

- a. Mix Design
- b. Aggregate Grading
- c. Quality of Materials
- d. Stockpile Management and procedures to prevent contamination
- e. Proportioning
- f. Mixing and Transportation
- g. Mixture Volumetrics
- h. Moisture Content of Mixtures
- i. Placing and Compaction
- j. Joints
- k. Surface Smoothness
- l. Truck bed release agent

3.1.2 Testing Laboratory

Provide a fully equipped asphalt laboratory located at the plant or job site that is equipped with heating and air conditioning units to maintain a temperature of 75 plus or minus 5 degrees F. Provide laboratory facilities that are kept clean and all equipment maintained in proper working condition. Provide the Government with unrestricted access to inspect the laboratory facility, to witness quality control activities, and to perform any check testing desired. The Government will advise in writing of any noted deficiencies concerning the laboratory facility, equipment, supplies, or testing personnel and procedures. When the deficiencies are serious enough to adversely affect test results, immediately suspend the incorporation of the materials into the work. Incorporation of the materials into the work will not be permitted to resume until the deficiencies are corrected.

3.1.3 Quality Control Testing

Perform all quality control tests applicable to these specifications and as set forth in the Quality Control Program. Use the independent commercial laboratory for acceptance testing in paragraph ACCEPTANCE. Use in-house capabilities or the independent commercial laboratory for quality control testing. Required elements of the testing program include, but are not limited to tests for the control of asphalt content, aggregate

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gradation, aggregate moisture, moisture in the asphalt mixture, temperatures, VMA, and in-place density. Develop a Quality Control Testing Plan as part of the Quality Control Program.

3.1.3.1 Asphalt Content

Determine asphalt content a minimum of twice per lot (a lot is defined in paragraph PAVEMENT LOTS) using the ignition method in accordance with ASTM D6307. Use the extraction method in accordance with ASTM D2172/D2172M if the correction factor for the ignition method in ASTM D6307 is greater than 1.0. The asphalt content for the lot will be determined by averaging the test results.

3.1.3.2 Aggregate Properties

Determine aggregate gradations a minimum of twice per lot from mechanical analysis of extracted aggregate in accordance with ASTM D5444, ASTM C136/C136M, and ASTM C117. Determine the specific gravity of each aggregate size grouping for each 20,000 tons in accordance with ASTM C127 or ASTM C128. Determine fractured faces for gravel sources for each 20,000 tons in accordance with ASTM D5821. Determine the uncompacted void content of natural sand, manufactured sand, and blended aggregate for each 20,000 tons in accordance with AASHTO T 304 Method A.

3.1.3.3 Moisture Content of Aggregate

Determine the moisture content of aggregate used for production a minimum of once per lot in accordance with ASTM C566.

3.1.3.4 Moisture Content of Asphalt Mixture

Determine the moisture content of the asphalt mixture at least once per lot in accordance with AASHTO T 329.

3.1.3.5 Temperatures

Check temperatures at least four times per lot, at necessary locations to determine the temperature at the dryer, the asphalt cement binder in the storage tank, the asphalt mixture at the plant, and the asphalt mixture at the job site.

3.1.3.6 VMA

Obtain mixture samples at least four times per lot. Calculate the VMA of each specimen in accordance with AI MS-2 based on ASTM C127 and ASTM C128 bulk specific gravity for the aggregate.

3.1.3.7 In-Place Density

Conduct any necessary testing to ensure the specified density is achieved. A nuclear gauge or other non-destructive testing device can be used to monitor pavement density.

3.1.3.8 Additional Testing

Perform any additional testing deemed necessary to control the process.

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3.1.3.9 QC Monitoring

Submit all QC test results to the Government on a daily basis as the tests are performed. The Government reserves the right to monitor any of the Contractor's quality control testing and to perform duplicate testing as a check to the Contractor's quality control testing.

3.1.4 Sampling

When directed by the Government, sample and test any material which appears to not meet specification requirements unless such material is voluntarily removed and replaced or deficiencies corrected. Perform all sampling in accordance with standard procedures specified.

3.2 PREPARATION OF ASPHALT BINDER MATERIAL

Heat the asphalt cement material while avoiding local overheating. Provide a continuous supply of the asphalt material to the mixer at a uniform temperature. Maintain the temperature of the asphalt delivered to the mixer to provide a suitable viscosity for adequate coating of the aggregate particles. For hot-mix, do not heat unmodified asphalt to a temperature exceeding 325 degrees F when added to the aggregate. Do not heat modified asphalt to a temperature exceeding 350 degrees F when added to the aggregate. For warm-mix, do not heat asphalt binder to a temperature exceeding 270 degrees F when added to the aggregate.

3.3 PREPARATION OF MINERAL AGGREGATE

Heat and dry the aggregate prior to mixing. Provide a rate of heating and a maximum temperature that does not damage the aggregates. Do not heat the aggregate to a temperature exceeding 350 degrees F when the asphalt binder is added. Maintain the temperature no lower than is required to obtain complete coating and uniform distribution on the aggregate particles and to provide a mixture of satisfactory workability.

3.4 PREPARATION OF ASPHALT MIXTURE

Weigh or meter the aggregates and the asphalt cement and introduce into the mixer the amount specified by the JMF. Mix the combined materials until the aggregate obtains a uniform coating of asphalt binder and is thoroughly distributed throughout the mixture. The moisture content of all asphalt mixture upon discharge from the plant is not to exceed 0.5 percent by total weight of mixture as measured by AASHTO T 329.

3.5 PREPARATION OF THE UNDERLYING SURFACE

Immediately before placing the asphalt mixture, clean the underlying course of dust and debris. Apply a prime coat or tack coat in accordance with Section 32 12 13 BITUMINOUS TACK AND PRIME COATS.

3.6 TRANSPORTING AND PLACING

3.6.1 Transporting

Transport asphalt mixture from the mixing plant to the site in clean, tight vehicles. Schedule deliveries so that placing and compacting of mixture is uniform with minimum stopping and starting of the paver. Provide adequate artificial lighting for night placements. Hauling over freshly placed material will not be permitted until the material has been

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compacted as specified, and allowed to cool to 140 degrees F.

3.6.2 Placing

Place the mix in lifts of adequate thickness and compact at a temperature suitable for obtaining density, surface smoothness, and other specified requirements. Upon arrival, place the mixture to the full width by an asphalt paver; strike off in a uniform layer of such depth that, when the work is completed, the required thickness is obtained and the surface conforms to the grade and contour indicated. Do not broadcast waste mixture onto the mat or recycle into the paver hopper. Collect waste mixture and dispose off site. Regulate the speed of the paver to eliminate pulling and tearing of the asphalt mat. Begin placement of the mixture along the centerline of a crowned section or on the high side of areas with a one-way slope. Place the mixture in consecutive adjacent strips having a minimum width of 10 feet. Offset the longitudinal joint in one course from the longitudinal joint in the course immediately below by at least 1 foot; however, locate the joint in the surface course at the centerline of the pavement. Offset transverse joints in one course by at least 10 feet from transverse joints in the previous course. Offset transverse joints in adjacent lanes a minimum of 10 feet. On isolated areas where irregularities or unavoidable obstacles make the use of mechanical spreading and finishing equipment impractical, the mixture can be spread and luted by hand tools.

3.7 COMPACTION OF MIXTURE

3.7.1 General

- a. After placing, thoroughly and uniformly compact the mixture by rolling. Compact the surface as soon as possible without causing displacement, cracking, or shoving. Determine the sequence of rolling operations and the type of rollers used with the exception that application of more than three passes with a vibratory roller in the vibrating mode is prohibited. Maintain the speed of the roller, at all times, sufficiently slow to avoid displacement of the asphalt mixture and to be effective in compaction. Correct at once any displacement occurring as a result of reversing the direction of the roller, or from any other cause.
- b. Furnish sufficient rollers to handle the output of the plant. Continue rolling until the surface is of uniform texture, true to grade and cross section, and the required field density is obtained. To prevent adhesion of the mixture to the roller, keep the wheels properly moistened, but excessive water is not permitted. In areas not accessible to the roller, thoroughly compact the mixture with hand tampers or small compactors. Remove the full depth of any mixture that becomes loose and broken, mixed with dirt, contains check-cracking, or is in any way defective. Replace with fresh asphalt mixture and immediately compact to conform to the surrounding area. Perform this work at no expense to the Government. Skin patching is not allowed.

3.7.2 Segregation

The Government can sample and test any material that looks deficient. When the in-place material appears to be segregated, the Government has the option to sample the material and have it tested and compared to the in-place density requirements in Table 2 paragraph ACCEPTANCE. If the

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material fails to meet these specification requirements, remove and replace the extent of the segregated material the full depth of the layer of asphalt mixture at no additional cost to the Government. When segregation occurs in the mat, take appropriate action to correct the process so that additional segregation does not occur.

3.8 JOINTS

Construct joints to ensure a continuous bond between the courses and to obtain the required density. Provide all joints with the same texture as other sections of the course and meet the requirements for smoothness and grade.

3.8.1 Transverse Joints

Do not pass the roller over the unprotected end of the freshly laid mixture, except when necessary to form a transverse joint. When necessary to form a transverse joint, construct by means of placing a bulkhead or by tapering the course. Utilize a dry saw cut on the transverse joint full depth and width on a straight line to expose a vertical face prior to placing the adjacent lane. Remove the cutback material from the project. In both methods, provide a light tack coat of asphalt material to all contact surfaces before placing any fresh mixture against the joint.

3.8.2 Longitudinal Joints

Provide a joint that meets density and smoothness requirements for joints and has uniform texture. Cut back longitudinal joints which are irregular, damaged, uncompacted, cold (less than 175 degrees F at the time of placing adjacent lanes), or otherwise defective, a maximum of 3 inches from the top of the course with a cutting wheel to expose a clean, sound, near vertical surface for the full depth of the course. Remove all cutback material from the project. Provide a light tack coat of asphalt material to all contact surfaces prior to placing any fresh mixture against the joint.

-- End of Section --

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SECTION 32 16 19

CONCRETE CURBS, GUTTERS AND SIDEWALKS 05/18

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Measurement

1.1.1.1 Sidewalks

The quantities of sidewalks to be paid for will be the number of square yards of each depth of sidewalk constructed as indicated.

1.1.1.2 Curbs and Gutters

The quantities of curbs and gutters to be paid for will be the number of linear feet of each cross section constructed as indicated, measured along the face of the curb at the gutter line.

1.1.2 Payment

1.1.2.1 Sidewalks

Payment of the quantities of sidewalks measured as specified will be at the Contract unit price per square yard of the thickness specified.

1.1.2.2 Curbs and Gutters

Payment of the quantities of curbs and gutters measured as specified will be at the Contract unit price per linear foot of each cross section.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 182	(2005; R 2017) Standard Specification for Burlap Cloth Made from Jute or Kenaf and Cotton Mats
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ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2022) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
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ASTM A1064/A1064M	(2022) Standard Specification for Carbon-Steel Wire and Welded Wire Reinforcement, Plain and Deformed, for Concrete
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ASTM C31/C31M	(2022) Standard Practice for Making and Curing Concrete Test Specimens in the Field
ASTM C94/C94M	(2022a) Standard Specification for Ready-Mixed Concrete
ASTM C143/C143M	(2020) Standard Test Method for Slump of Hydraulic-Cement Concrete
ASTM C171	(2020) Standard Specification for Sheet Materials for Curing Concrete
ASTM C172/C172M	(2017) Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	(2016) Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method
ASTM C231/C231M	(2022) Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C309	(2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete
ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction
ASTM D5893/D5893M	(2016) Standard Specification for Cold Applied, Single Component, Chemically Curing Silicone Joint Sealant for Portland Cement Concrete Pavements

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1	(2017) Standard And Commentary Accessible and Usable Buildings and Facilities
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1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL

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PROCEDURES:

SD-03 Product Data

Concrete

SD-06 Test Reports

Field Quality Control

1.4 EQUIPMENT, TOOLS, AND MACHINES

1.4.1 General Requirements

Plant, equipment, machines, and tools used in the work will be subject to approval and must be maintained in a satisfactory working condition at all times. Use equipment capable of producing the required product, meeting grade controls, thickness control and smoothness requirements as specified. Discontinue using equipment that produces unsatisfactory results. Allow the Contracting Officer access at all times to the plant and equipment to ensure proper operation and compliance with specifications.

1.4.2 Slip Form Equipment

Slip form paver or curb forming machines, will be approved based on trial use on the job and must be self-propelled, automatically controlled, crawler mounted, and capable of spreading, consolidating, and shaping the plastic concrete to the desired cross section in one pass.

1.5 ENVIRONMENTAL REQUIREMENTS

1.5.1 Placing During Cold Weather

Do not place concrete when the air temperature reaches 40 degrees F and is falling, or is already below that point. Placement may begin when the air temperature reaches 35 degrees F and is rising, or is already above 40 degrees F. Make provisions to protect the concrete from freezing during the specified curing period. If necessary to place concrete when the temperature of the air, aggregates, or water is below 35 degrees F, placement and protection must be approved in writing. Approval will be contingent upon full conformance with the following provisions. Prepare and protect the underlying material so that it is entirely free of frost when the concrete is deposited. Heat mixing water and aggregates as necessary to result in the temperature of the in-place concrete being between 50 and 85 degrees F. Methods and equipment for heating must be approved. Use only aggregates that are free of ice, snow, and frozen lumps before entering the mixer. Provide covering or other means as needed to maintain the concrete at a temperature of at least 50 degrees F for not less than 72 hours after placing, and at a temperature above freezing for the remainder of the curing period.

1.5.2 Placing During Warm Weather

The temperature of the concrete as placed must not exceed 85 degrees F except where an approved retarder is used. Cool the mixing water and aggregates as necessary to maintain a satisfactory placing temperature. The placing temperature must not exceed 95 degrees F at any time.

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PART 2 PRODUCTS

2.1 CONCRETE

Provide concrete conforming to the applicable requirements of ASTM C94/C94M except as otherwise specified. Concrete must have a minimum compressive strength of 3500 psi at 28 days. Size of aggregate must not exceed 1-1/2 inches. Submit copies of certified delivery tickets for all concrete used in the construction.

2.1.1 Air Content

Use concrete mixtures that have an air content by volume of concrete of 5 to 7 percent, based on measurements made immediately after discharge from the mixer.

2.1.2 Slump

Use concrete with a slump of 3 inches plus or minus 1 inch for hand placed concrete or 1 inch plus or minus 1/2 inch for slipformed concrete as determined in accordance with ASTM C143/C143M.

2.1.3 Reinforcement Steel

Use reinforcement bars conforming to ASTM A615/A615M. Use wire mesh reinforcement conforming to ASTM A1064/A1064M.

2.2 CONCRETE CURING MATERIALS

2.2.1 Impervious Sheet Materials

Use impervious sheet materials conforming to ASTM C171, type optional, except that polyethylene film, if used, must be white opaque.

2.2.2 Burlap

Use burlap conforming to AASHTO M 182.

2.2.3 White Pigmented Membrane-Forming Curing Compound

Use white pigmented membrane-forming curing compound conforming to ASTM C309, Type 2.

2.3 CONCRETE PROTECTION MATERIALS

Use concrete protection materials consisting of a linseed oil mixture of equal parts, by volume, of linseed oil and either mineral spirits, naphtha, or turpentine. At the option of the Contractor, commercially prepared linseed oil mixtures, formulated specifically for application to concrete to provide protection against the action of deicing chemicals may be used, except that emulsified mixtures are not acceptable.

2.4 JOINT FILLER STRIPS

2.4.1 Contraction Joint Filler for Curb and Gutter

Use hard-pressed fiberboard contraction joint filler for curb and gutter.

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2.4.2 Expansion Joint Filler, Premolded

Unless otherwise indicated, use 1/2 inch thick premolded expansion joint filler conforming to ASTM D1751 or ASTM D1752.

2.5 JOINT SEALANTS

Use cold-applied joint sealant conforming to ASTM C920 or ASTM D5893/D5893M.

2.6 FORM WORK

Design and construct form work to ensure that the finished concrete will conform accurately to the indicated dimensions, lines, and elevations, and within the tolerances specified. Use wood or steel forms that are straight and of sufficient strength to resist springing during depositing and consolidating concrete.

2.6.1 Wood Forms

Use forms that are surfaced plank, 2 inches nominal thickness, straight and free from warp, twist, loose knots, splits or other defects. Use forms with a nominal length of 10 feet. Radius bends may be formed with 3/4 inch boards, laminated to the required thickness.

2.6.2 Steel Forms

Use channel-formed sections with a flat top surface and welded braces at each end and at not less than two intermediate points. Use forms with interlocking and self-aligning ends. Provide flexible forms for radius forming, corner forms, form spreaders, and fillers as needed. Use forms with a nominal length of 10 feet and that have a minimum of 3 welded stake pockets per form. Use stake pins consisting of solid steel rods with chamfered heads and pointed tips designed for use with steel forms.

2.6.3 Sidewalk Forms

Use sidewalk forms that are of a height equal to the full depth of the finished sidewalk.

2.6.4 Curb and Gutter Forms

Use curb and gutter outside forms that have a height equal to the full depth of the curb or gutter. Use rigid forms for curb returns, except that benders or thin plank forms may be used for curb or curb returns with a radius of 10 feet or more, where grade changes occur in the return, or where the central angle is such that a rigid form with a central angle of 90 degrees cannot be used. Back forms for curb returns may be made of 1-1/2 inch benders, for the full height of the curb, cleated together. In lieu of inside forms for curbs, a curb "mule" may be used for forming and finishing this surface, provided the results are approved.

2.7 Detectable Warning System

Detectable Warning Systems shown on the Contract plans are to meet requirements of ICC A117.1 - Section 705.

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PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Construct subgrade to the specified grade and cross section prior to concrete placement.

3.1.1 Sidewalk Subgrade

Place and compact the subgrade in accordance with Section 31 00 00 EARTHWORK. Test the subgrade for grade and cross section with a template extending the full width of the sidewalk and supported between side forms.

3.1.2 Curb and Gutter Subgrade

Place and compact the subgrade in accordance with Section 31 00 00 EARTHWORK and Section 32 11 23 AGGREGATE BASE COURSE FOR FLEXIBLE PAVING. Test the subgrade for grade and cross section by means of a template extending the full width of the curb and gutter. Use subgrade materials equal in bearing quality to the subgrade under the adjacent pavement.

3.1.3 Maintenance of Subgrade

Maintain subgrade in a smooth, compacted condition in conformity with the required section and established grade until the concrete is placed. The subgrade must be in a moist condition when concrete is placed. Prepare and protect subgrade so that it is free from frost when the concrete is deposited.

3.2 FORM SETTING

Set forms to the indicated alignment, grade and dimensions. Hold forms rigidly in place by a minimum of 3 stakes per form placed at intervals not to exceed 4 feet. Use additional stakes and braces at corners, deep sections, and radius bends, as required. Use clamps, spreaders, and braces where required to ensure rigidity in the forms. Remove forms in a manner that will not injure the concrete. Do not use bars or heavy tools against the concrete when removing the forms. Promptly and satisfactorily repair concrete found to be defective after form removal. Clean forms and coat with form oil each time before concrete is placed. Wood forms may, instead, be thoroughly wetted with water before concrete is placed, except that with probable freezing temperatures, oiling is mandatory.

3.2.1 Sidewalks

Set forms for sidewalks with the upper edge true to line and grade with an allowable tolerance of 1/8 inch in any 10 foot long section. After forms are set, grade and alignment must be checked with a 10 foot straightedge. Sidewalks must have a transverse slope of 1/4 inch per foot. Unless otherwise indicated, construct sidewalks that are located adjacent to curbs with the low side adjacent to the curb. Do not remove side forms less than 12 hours after finishing has been completed.

3.2.2 Curbs and Gutters

Remove forms used along the front of the curb not less than 2 hours nor more than 6 hours after the concrete has been placed. Do not remove forms used along the back of curb until the face and top of the curb have been finished, as specified for concrete finishing. Do not remove gutter forms

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while the concrete is sufficiently plastic to slump in any direction.

3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING

3.3.1 Formed Sidewalks

Place concrete in the forms in one layer. When consolidated and finished, the sidewalks must be of the thickness indicated. Use a strike-off guided by side forms after concrete has been placed in the forms to bring the surface to proper section to be compacted. Consolidate concrete by tamping and spading or with an approved vibrator. Finish the surface to grade with a strike off.

3.3.2 Concrete Finishing

After straightedging, when most of the water sheen has disappeared, and just before the concrete hardens, finish the surface with a wood or magnesium float or darby to a smooth and uniformly fine granular or sandy texture free of waves, irregularities, or tool marks. Produce a scored surface by brooming with a fiber-bristle brush in a direction transverse to that of the traffic, followed by edging.

3.3.3 Edge and Joint Finishing

Finish all slab edges, including those at formed joints, with an edger having a radius of 1/8 inch. Edge transverse joints before brooming. Eliminate the flat surface left by the surface face of the edger with brooming. Clean and solidly fill corners and edges which have crumbled and areas which lack sufficient mortar for proper finishing with a properly proportioned mortar mixture and then finish.

3.3.4 Surface and Thickness Tolerances

Finished surfaces must not vary more than 5/16 inch from the testing edge of a 10-foot straightedge. Permissible deficiency in section thickness will be up to 1/4 inch.

3.4 CURB AND GUTTER CONCRETE PLACEMENT AND FINISHING

3.4.1 Formed Curb and Gutter

Place concrete to the required section in a single lift. Consolidate concrete using approved mechanical vibrators. Curve shaped gutters must be finished with a standard curb "mule".

3.4.2 Curb and Gutter Finishing

Approved slipformed curb and gutter machines may be used in lieu of hand placement.

3.4.3 Concrete Finishing

Float and finish exposed surfaces with a smooth wood float until true to grade and section and uniform in texture. Brush floated surfaces with a fine-hair brush using longitudinal strokes. Round the edges of the gutter and top of the curb with an edging tool to a radius of 1/2 inch. Immediately after removing the front curb form, rub the face of the curb with a wood or concrete rubbing block and water until blemishes, form marks, and tool marks have been removed. Brush the front curb surface,

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while still wet, in the same manner as the gutter and curb top. Finish the top surface of gutter to grade with a wood float.

3.4.4 Joint Finishing

Finish curb edges at formed joints as indicated.

3.4.5 Surface and Thickness Tolerances

Finished surfaces must not vary more than 1/4 inch from the testing edge of a 10-foot straightedge. Permissible deficiency in section thickness will be up to 1/4 inch.

3.5 SIDEWALK JOINTS

Construct sidewalk joints to divide the surface into rectangular areas. Space transverse contraction joints at a distance equal to the sidewalk width or 5 feet on centers, whichever is less, and continuous across the slab. Construct longitudinal contraction joints along the centerline of all sidewalks 10 feet or more in width. Construct transverse expansion joints at sidewalk returns and opposite expansion joints in adjoining curbs. Where the sidewalk is not in contact with the curb, install transverse expansion joints as indicated. Form expansion joints around structures and features which project through or into the sidewalk pavement, using joint filler of the type, thickness, and width indicated. Expansion joints are not required between sidewalks and curb that abut the sidewalk longitudinally.

3.5.1 Sidewalk Contraction Joints

Form contraction joints in the fresh concrete by cutting a groove in the top portion of the slab to a depth of at least one-fourth of the sidewalk slab thickness. Unless otherwise approved or indicated, either use a jointer to cut the groove or saw a groove in the hardened concrete with a power-driven saw. Construct sawed joints by sawing a groove in the concrete with a 1/8 inch blade. Provide an ample supply of saw blades on the jobsite before concrete placement is started. Provide at least one standby sawing unit in good working order at the jobsite at all times during the sawing operations.

3.5.2 Sidewalk Expansion Joints

Form expansion joints using 1/2 inch joint filler strips. Joint filler in expansion joints surrounding structures and features within the sidewalk may consist of preformed filler material conforming to ASTM D1752 or building paper. Hold joint filler in place with steel pins or other devices to prevent warping of the filler during floating and finishing. Immediately after finishing operations are completed, round joint edges using an edging tool having a radius of 1/8 inch. Remove any concrete over the joint filler. At the end of the curing period, clean the top of expansion joints and fill with cold-applied joint sealant. Use joint sealant that is gray or stone in color. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Apply joint sealing material only when the concrete at the joint is surface dry and atmospheric and concrete temperatures are above 50 degrees F. Immediately remove any excess material on exposed surfaces of the concrete and clean the concrete surfaces.

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3.5.3 Reinforcement Steel Placement

Accurately and securely fasten reinforcement steel in place with suitable supports and ties before the concrete is placed.

3.6 CURB AND GUTTER JOINTS

Construct curb and gutter joints at right angles to the line of curb and gutter.

3.6.1 Contraction Joints

Construct contraction joints directly opposite contraction joints in abutting portland cement concrete pavements and spaced so that monolithic sections between curb returns will not be less than 5 feet nor greater than 15 feet in length.

- a. Construct contraction joints (except for slip forming) by means of 1/8 inch thick separators and of a section conforming to the cross section of the curb and gutter. Remove separators as soon as practicable after concrete has set sufficiently to preserve the width and shape of the joint and prior to finishing.
- b. When slip forming is used, cut the contraction joints in the top portion of the gutter/curb hardened concrete in a continuous cut across the curb and gutter, using a power-driven saw. Cut the contraction joint to a depth of at least one-fourth of the gutter/curb depth using a 1/8 inch saw blade.

3.6.2 Expansion Joints

Form expansion joints by means of preformed expansion joint filler material cut and shaped to the cross section of curb and gutter. Construct expansion joints in curb and gutter directly opposite expansion joints of abutting portland cement concrete pavement using the same type and thickness of joints as joints in the pavement. Where curb and gutter do not abut portland cement concrete pavement, provide expansion joints at least 1/2 inch in width at intervals not less than 30 feet nor greater than 120 feet. Seal expansion joints immediately following curing of the concrete or as soon thereafter as weather conditions permit. Seal expansion joints and the top 1 inch depth of curb and gutter contraction-joints with joint sealant. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Concrete at the joint must be surface dry and atmospheric and concrete temperatures must be above 50 degrees F at the time of application of joint sealing material. Immediately remove excess material on exposed surfaces of the concrete and clean concrete surfaces.

3.7 CURING AND PROTECTION

3.7.1 General Requirements

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days from the beginning of the curing operation. Protect unhardened concrete from rain and flowing water. All equipment needed for adequate curing and protection of the concrete must be on hand and ready for use before actual concrete placement begins. Protect concrete as necessary to prevent cracking of the pavement due to temperature changes

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during the curing period.

3.7.1.1 Mat Method

Cover the entire exposed surface with two or more layers of burlap. Overlap mats at least 6 inches. Thoroughly wet the mat with water prior to placing on concrete surface and keep the mat continuously in a saturated condition and in intimate contact with concrete for not less than 7 days.

3.7.1.2 Impervious Sheeting Method

Wet the entire exposed surface with a fine spray of water and then cover with impervious sheeting material. Lay sheets directly on the concrete surface with the light-colored side up and overlapped 12 inches when a continuous sheet is not used. Use sheeting that is not less than 18-inches wider than the concrete surface to be cured. Secure sheeting using heavy wood planks or a bank of moist earth placed along edges and laps in the sheets. Satisfactorily repair or replace sheets that are torn or otherwise damaged during curing. Sheeting must remain on the concrete surface to be cured for not less than 7 days.

3.7.1.3 Membrane Curing Method

Apply a uniform coating of white-pigmented membrane-curing compound to the entire exposed surface of the concrete as soon after finishing as the free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after the removal of forms. Do not allow concrete surface to dry before application of the membrane. If drying has occurred, moisten the surface of the concrete with a fine spray of water and apply the curing compound as soon as the free water disappears. Apply curing compound in two coats by hand-operated pressure sprayers at a coverage of approximately 200 square feet/gallon for the total of both coats. Apply the second coat in a direction approximately at right angles to the direction of application of the first coat. The compound must form a uniform, continuous, coherent film that will not check, crack, or peel and must be free from pinholes or other imperfections. If pinholes, abrasion, or other discontinuities exist, apply an additional coat to the affected areas within 30 minutes. Respray concrete surfaces that are subjected to heavy rainfall within 3 hours after the curing compound has been applied by the method and at the coverage specified above. Respray areas where the curing compound is damaged by subsequent construction operations within the curing period. Take precautions necessary to ensure that the concrete is properly cured at sawed joints, and that no curing compound enters the joints. Tightly seal the top of the joint opening and the joint groove at exposed edges before the concrete in the region of the joint is resprayed with curing compound. Use a method used for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing period. Provide approved standby facilities for curing concrete pavement at a location accessible to the jobsite for use in the event of mechanical failure of the spraying equipment or other conditions that might prevent correct application of the membrane-curing compound at the proper time. Adequately protect concrete surfaces to which membrane-curing compounds have been applied during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from other possible damage to the continuity of the membrane.

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3.7.2 Backfilling

After curing, remove debris and backfill, grade, and compact the area adjoining the concrete to conform to the surrounding area in accordance with lines and grades indicated.

3.7.3 Protection

Protect completed concrete from damage until accepted. Repair damaged concrete and clean concrete discolored during construction. Remove and reconstruct concrete that is damaged for the entire length between regularly scheduled joints. Refinishing the damaged portion will not be acceptable. Dispose of removed material as directed.

3.7.4 Protective Coating

Apply a protective coating of linseed oil mixture to the exposed-to-view concrete surface after the curing period, if concrete will be exposed to de-icing chemicals within 6 weeks after placement. Moist cure concrete to receive a protective coating.

3.7.4.1 Application

Complete curing and backfilling operation prior to applying two coats of protective coating. Concrete must be surface dry and clean before each application. Spray apply at a rate of not more than 50 square yards/gallon for first application and not more than 70 square yards/gallon for second application, except that the number of applications and coverage for each application for commercially prepared mixture must be in accordance with the manufacturer's instructions. Protect coated surfaces from vehicular and pedestrian traffic until dry.

3.7.4.2 Precautions

Do not heat protective coating by direct application of flame or electrical heaters and protect the coating from exposure to open flame, sparks, and fire adjacent to open containers or applicators. Do not apply material at ambient or material temperatures lower than 50 degrees F.

3.8 FIELD QUALITY CONTROL

Submit copies of all test reports within 24 hours of completion of the test.

3.8.1 General Requirements

Perform the inspection and tests described and meet the specified requirements for inspection details and frequency of testing. Based upon the results of these inspections and tests, take the action and submit reports as required below, and additional tests to ensure that the requirements of these specifications are met.

3.8.2 Concrete Testing

3.8.2.1 Strength Testing

Take concrete samples in accordance with ASTM C172/C172M not less than once a day nor less than once for every 250 cubic yards of concrete placed. Mold cylinders in accordance with ASTM C31/C31M for strength

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testing by an approved laboratory. Each strength test result must be the average of 2 test cylinders from the same concrete sample tested at 28 days, unless otherwise specified or approved. Concrete specified on the basis of compressive strength will be considered satisfactory if the averages of all sets of three consecutive strength test results equal or exceed the specified strength, and no individual strength test result falls below the specified strength by more than 500 psi.

3.8.2.2 Air Content

Determine air content in accordance with ASTM C173/C173M or ASTM C231/C231M. Use ASTM C231/C231M with concretes and mortars made with relatively dense natural aggregates. Make two tests for air content on randomly selected batches of each class of concrete placed during each shift. Make additional tests when excessive variation in concrete workability is reported by the placing foreman or the Government inspector. Notify the placing foreman if results are out of tolerance. The placing foreman must take appropriate action to have the air content corrected at the plant. Additional tests for air content will be performed on each truckload of material until such time as the air content is within the tolerance specified.

3.8.2.3 Slump Test

Perform two slump tests on randomly selected batches of each class of concrete for every 250 cubic yards, or fraction thereof, of concrete placed during each shift. Perform additional tests when excessive variation in the workability of the concrete is noted or when excessive crumbling or slumping is noted along the edges of slip-formed concrete.

3.8.3 Thickness Evaluation

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section or by measuring the depth of opening of the extrusion template of the curb forming machine. If a slip form paver is used for sidewalk placement, construct the subgrade true to grade prior to concrete placement. The thickness will be determined by measuring each edge of the completed slab.

3.8.4 Surface Evaluation

Provide finished surfaces for each category of the completed work that are uniform in color and free of blemishes and form or tool marks.

3.9 SURFACE DEFICIENCIES AND CORRECTIONS

3.9.1 Thickness Deficiency

When measurements indicate that the completed concrete section is deficient in thickness by more than 1/4 inch the deficient section will be removed, between regularly scheduled joints, and replaced.

3.9.2 High Areas

In areas not meeting surface smoothness and plan grade requirements, reduce high areas either by rubbing the freshly finished concrete with carborundum brick and water when the concrete is less than 36 hours old or by grinding the hardened concrete with an approved surface grinding machine after the concrete is 36 hours old or more. The area corrected by

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grinding the surface of the hardened concrete must not exceed 5 percent of the area of any integral slab, and the depth of grinding must not exceed 1/4 inch. Remove and replace pavement areas requiring grade or surface smoothness corrections in excess of the limits specified.

3.9.3 Appearance

Exposed surfaces of the finished work will be inspected by the Contracting Officer and deficiencies in appearance will be identified. Remove and replace areas which exhibit excessive cracking, discoloration, form marks, or tool marks or which are otherwise inconsistent with the overall appearances of the work.

3.10 DETECTABLE WARNING SYSTEM

Install Detectable Warning Systems required by Contract plans in accordance with ICC A117.1, Section 705, and by manufacturers' installation instructions.

-- End of Section --

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CHAIN LINK FENCES AND GATES 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A392	(2011; R 2022a) Standard Specification for Zinc-Coated Steel Chain-Link Fence Fabric
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM A824	(2001; R 2022) Standard Specification for Metallic-Coated Steel Marcellled Tension Wire for Use With Chain Link Fence
ASTM C94/C94M	(2022a) Standard Specification for Ready-Mixed Concrete
ASTM F567	(2014a; R 2019) Standard Practice for Installation of Chain Link Fence
ASTM F626	(2014; R 2019) Standard Specification for Fence Fittings
ASTM F883	(2013; R 2022) Standard Performance Specification for Padlocks

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

FS RR-F-191/1	(Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Fabric)
FS RR-F-191/2	(Rev E) Fencing, Wire and Post, Metal (Chain-Link Fence Gates)
FS RR-F-191/3	(2023; Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Posts, Top Rails and Braces)
FS RR-F-191/4	(Rev F) Fencing, Wire and Post, Metal (Chain-Link Fence Accessories)

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for

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information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Assembly; G-D0

Location of Gate, Corner, End, and Pull Posts; G-D0

Gate Assembly; G-D0

Gate Hardware and Accessories; G-D0

Hardware Assembly; G-D0

Accessories; G-D0

Erection/Installation Drawings; G-D0

SD-03 Product Data

Fence Assembly; G-D0

Gate Assembly; G-D0

Hardware Assembly; G-D0

Accessories; G-D0

Gate Hardware and Accessories; G-D0

SD-07 Certificates

Certificates of Compliance

1.3 QUALITY CONTROL

1.3.1 Certificates of Compliance

Submit certificates of compliance in accordance with the applicable reference standards and descriptions of this section for the following:

- a. Zinc coating
- b. PVC coating
- c. Aluminum alloy coating
- d. Fabric
- e. Stretcher bars
- f. Gate hardware and accessories
- g. Concrete

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1.4 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials off the ground to provide protection against oxidation caused by ground contact.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Submit reports of listing chain-link fencing and accessories regarding weight in ounces for zinc coating.

Submit manufacturer's catalog data for complete fence assembly, gate assembly, hardware assembly and accessories.

2.2 FENCES

2.2.1 Fabric

FS RR-F-191/1; Type I, zinc-coated steel, 9 gauge conforming to ASTM A392, with 1.20 ounces per square foot of coated surface. Provide selvage. Height of fabric, as indicated.

Provide fabric consisting of wires woven into a 1 inch diamond mesh. Provide one-piece fabric widths for fence heights up to 12 feet.

2.2.1.1 Top and Bottom Selvages

Provide knuckled selvages at top and bottom for fabric with 2 inch mesh and up to 60 inches high, and if over 60 inches high, provide twisted and barbed top selvage and knuckled bottom selvage.

Knuckle top and bottom selvages for 1-3/4 inch and 1 inch mesh fabric.

2.2.2 Posts

2.2.2.1 Metal

2.2.2.1.1 Line Posts

Provide line posts complying with FS RR-F-191/3. Provide Class 1, steel pipe, Grade A Regular Strength.

2.2.2.1.2 End, Corner and Pull Posts

Provide end, corner, and pull posts in minimum sizes listed in FS RR-F-191/3 for each class and grade. Provide Class 1, steel pipe, Grade A Regular Strength.

2.2.2.2 Post Tops

Provide steel or aluminum tops and designed as a weathertight closure cap. Post top to have finish and coating matching rails and posts. Steel type to be pressed steel galvanized after fabrication having a minimum zinc coating of 1.20 ounces per square foot. Provide one cap for each post, unless equal protection is provided by a combination post-cap and wire supporting arm. Provide caps with an opening to permit through passage of the top rail.

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2.2.3 Braces and Rails

Class 1, steel pipe, Grade A Regular Strength in minimum sizes listed in FS RR-F-191/3 for each class and grade.

2.2.3.1 Top Rail

Provide top rail conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade unless members are to be oversized. Provide expansion couplings 6 inches long at each joint in top rails.

2.2.3.2 Post-Brace Assembly

Provide bracing conforming to minimum sizes specified in FS RR-F-191/3 for each class and grade, and 3/8 inch adjustable truss rods and turnbuckles.

2.2.4 Wire Ties

Provide 9-gauge wire for tying fabric to line posts, spaced 12 inches on center. For tying fabric to rails and braces, space wire ties 24 inches on center. For tying fabric to tension wire, space 0.105-inch hog rings 24 inches on center. Manufacturer's standard procedure will be accepted if of equal strength and durability.

FS RR-F-191/4. Provide wire ties constructed of the same material and coating as the fencing fabric.

2.2.5 Sleeves

Provide sleeves for setting into concrete construction of the same material as post sections, sized 1 inch greater than the diameter or dimension of the post. Weld flat plates to each sleeve base to provide anchorage and prevent intrusion of concrete.

2.2.6 Stretcher Bars

Provide bars that have one-piece lengths equal to the full height of the fabric with a minimum cross section of 3/16 by 3/4 inch, in accordance with ASTM F626.

2.2.7 Stretcher Bar Bands

Provide bar bands for securing stretcher bars to posts that are steel, wrought iron, or malleable iron spaced not over 15 inches on center. Bands may also be used in conjunction with special fittings for securing rails to posts. Provide bands with projecting edges chamfered or eased.

2.2.8 Tension Wire

Provide metallic coated steel marcelled tension wire, (No. 7-gauge) complying with ASTM A824.

2.2.9 Miscellaneous Hardware

Provide miscellaneous hot-dip galvanized hardware as required.

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2.3 GATES

FS RR-F-191/2; Type II, double swing. Shape and size of gate frame, as indicated. Provide gate frames and braces of minimum sizes listed in FS RR-F-191/3 for each Class and Grade, except that steel pipe frames are a minimum of 1.90 inches o.d., 0.120 inches minimum wall thickness and aluminum pipe frames and intermediate braces are 1.869 inches o.d. minimum, 0.940 lb/ft of length. Provide intermediate members as necessary for gate leaves more than 8 feet wide, to provide rigid construction, free from sag or twist.

2.3.1 Gate Posts

Provide gate posts for supporting each gate leaf in minimum sizes listed in FS RR-F-191/3 for each material class and grade. Gate post material class, grade and finish to match other fence posts.

2.3.2 Gate Fabric

Gate fabric, is as specified for fencing fabric. Attach gate fabric to gate frame in accordance with manufacturer's standards, except that welding is not permitted.

2.3.3 Gate Frame

Provide gate frame assembly that is welded or assembled with special malleable or pressed-steel fittings and rivets to provide rigid connections. Install fabric with stretcher bars at vertical edges; stretcher bars may also be used at top and bottom edges. Attach stretcher bars and fabric to gate frames on all sides at intervals not exceeding 15 inches. Attach hardware with rivets or by other means which provides equal security against breakage or removal.

2.3.4 Gate Bracing

Provide diagonal cross-bracing, consisting of 3/8 inch diameter adjustable-length truss rods on welded gate frames, where necessary to obtain frame rigidity without sag or twist. Provide nonwelded gate frames with diagonal bracing.

2.3.5 Padlocks

Provide padlocks conforming to ASTM F883, with chain.

2.3.6 Gate Hardware and Accessories

Provide gate hardware and accessories that conforms to ASTM A392 and ASTM F626, and as specified. Coating for steel latches, stops, hinges, keepers, and accessories, is galvanized.

- a. Provide pressed steel hinges to suit gate size, non-lift-off type, offset to permit 180-degree opening. Provide hinge with stainless steel pin.
- b. Provide latch that permits accessibility and operation from either side of the gate regardless of the latching arrangement, and with a padlock eye provided as an integral part of the latch. Provide fork type gate latches.

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- c. Provide stops and holders of malleable iron for vehicular gates. Provide stops that automatically engage the gate and hold it in the open position until manually released.
- d. Provide double gates with a cane bolt and ground-set keeper, with latch or locking device and padlock eye designed as an integral part.

2.4 MATERIALS

2.4.1 Zinc Coating

Provide zinc-coated ferrous metal components and accessories that are factory coated after fabrication, except as otherwise specified.

For galvanizing field repairs, provide material that is cold-applied zinc-rich coating conforming to ASTM A780/A780M.

2.4.2 Concrete

Provide concrete conforming to ASTM C94/C94M, and obtaining a minimum 28-day compressive strength of 3,000 psi.

2.4.3 Grout

Provide grout of proportions one part portland cement to three parts clean, well-graded sand and a minimum amount of water to produce a workable mix.

PART 3 EXECUTION

Submit manufacturer's erection/installation drawings and instructions that detail proper assembly and materials in the design for fence, gate, hardware and accessories.

Provide complete installation conforming to ASTM F567.

3.1 PREPARATION

Ensure final grading and established elevations are complete prior to commencing fence installation.

3.1.1 Clearing and Grading

Clear fence line of trees, brush, and other obstacles to install fencing within the Limits of Disturbance permit boundary. Establish a graded, compacted fence line prior to fencing installation.

3.2 INSTALLATION

3.2.1 Security

Install new chain link fencing, remove existing fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer and cognizant Security Officer.

3.2.2 Fence Installation

Install fence on prepared surfaces to line and grade indicated. Install

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fence in accordance with fence manufacturer's written installation instructions except as modified herein.

3.2.2.1 Post Spacing

Provide line posts spaced equidistantly apart, not exceeding 10 feet on center. Provide gate posts spaced as necessary for size of gate openings. Do not exceed 500 feet on straight runs between braced posts. Provide corner or pull posts, with bracing in both directions, for changes in direction of 15 degrees or more, or for abrupt changes in grade. Submit drawings showing location of gate, corner, end, and pull posts.

3.2.2.2 Top and Bottom Tension Wire

Install top and bottom tension wires before installing chain-link fabric, and pull wires taut. Place top and bottom tension wires within 8 inches of respective fabric line.

3.2.3 Excavation

Provide excavations for post footings which are in virgin or compacted soil, of minimum sizes as indicated. Space footings for line posts 10 feet on center maximum and at closer intervals when indicated, with bottoms of the holes approximately 3 inches below the bottoms of the posts. Set bottom of each post not less than 36 inches below finished grade when in firm, undisturbed soil. Set posts deeper, as required, in soft and problem soils and for heavy, lateral loads. Remove excavated soil from Government property.

When solid rock is encountered near the surface, drill into the rock at least 12 inches for line posts and at least 18 inches for end, pull, corner, and gate posts. Drill holes at least 1 inch greater in diameter than the largest dimension of the placed post. If solid rock is below the soil overburden, drill to the full depth required except that penetration into rock need not exceed the minimum depths specified above.

3.2.4 Setting Posts

Remove loose and foreign materials from holes and moisten the soil prior to placing concrete. Provide tops of footings that are trowel finished and sloped or domed to shed water away from posts. Set hold-open devices, sleeves, and other accessories in concrete.

Keep exposed concrete moist for at least 7 calendar days after placement or cured with a membrane curing material, as approved. Grout all posts set into sleeved holes in concrete with an approved grouting material. Maintain vertical alignment of posts in concrete construction until concrete has set.

3.2.4.1 Earth and Bedrock

Provide concrete bases of dimensions indicated on the manufactures installation drawings, except in bedrock. Compact concrete to eliminate voids, and finish to a dome shape. In bedrock, set posts with a minimum of 1 inch of grout around each post. Work grout into hole to eliminate voids, and finish to a dome shape.

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3.2.4.2 Bracing

Brace gate, corner, end, and pull posts to nearest post with a horizontal brace used as a compression member, placed at least 12 inches below top of fence, and a diagonal tension rod.

3.2.4.3 Tolerances

Provide posts that are straight and plumb within a vertical tolerance of 1/4 inch after the fabric has been stretched. Provide fencing and gates that are true to line with no more than 1/2 inch deviation from the established centerline between line posts. Repair defects as directed.

3.2.5 Concrete Strength

Provide concrete that has attained at least 75 percent of its minimum 28-day compressive strength, but in no case sooner than 7 calendar days after placement, before rails, tension wire, or fabric are installed. Do not stretch fabric and wires or hang gates until the concrete has attained its full design strength.

3.2.6 Top Rails

Provide top rails that run continuously through post caps or extension arms, bending to radius for curved runs. Provide expansion couplings as recommended by the fencing manufacturer.

3.2.7 Center Rails

Provide single piece center rails between posts set flush with posts on the fabric side, using special offset fittings where necessary.

3.2.8 Brace Assembly

Provide bracing assemblies at end and gate posts and at both sides of corner and pull posts, with the horizontal brace located at midheight of the fabric.

Install brace assemblies so posts are plumb when the diagonal rod is under proper tension. Provide two complete brace assemblies at corner and pull posts where required for stiffness and as indicated.

3.2.9 Tension Wire Installation

Install tension wire by weaving them through the fabric and tying them to each post with not less than 7-gauge galvanized wire or by securing the wire to the fabric with 10-gauge ties or clips spaced 24 inches on center.

3.2.10 Fabric Installation

Provide fabric in single lengths between stretch bars with bottom barbs placed approximately 1-1/2 inches above the ground line. Pull fabric taut and tied to posts, rails, and tension wire with wire ties and bands.

Install fabric on the security side of fence, unless otherwise directed. Ensure fabric remains under tension after the pulling force is released.

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3.2.11 Stretcher Bar Installation

Thread stretcher bars through or clamped to fabric 4 inches on center and secured to posts with metal bands spaced 15 inches on center.

3.2.12 Gate Installation

Install gates plumb, level, and secure, with full opening without interference. Install ground set items in concrete for anchorage as recommended by the fence manufacturer. Adjust hardware for smooth operation and lubricated where necessary.

3.2.13 Tie Wires

Provide tie wires that are U-shaped to the pipe diameters to which attached. Twist ends of tie wires not less than two full turns and bent so as not to present a hazard.

3.2.14 Fasteners

Install nuts for tension bands and hardware on the side of the fence opposite the fabric side. Peen ends of bolts to prevent removal of nuts.

3.2.15 Zinc-Coating Repair

Clean and repair galvanized surfaces damaged by welding or abrasion, and cut ends of fabric, or other cut sections with specified galvanizing repair material applied in strict conformance with the manufacturer's printed instructions.

3.2.16 Accessories Installation

3.2.16.1 Post Caps

Install post caps as recommended by the manufacturer.

3.2.16.2 Padlocks

Provide padlocks for gate openings and provide chains that are securely attached to gate or gate posts. Provide padlocks keyed alike, and provide two keys for each padlock.

3.2.17 Grounding

Ground fencing as indicated on drawings and specified.

Ground all fences crossed by overhead power lines in excess of 600 volts, and all electrical equipment attached to the fence. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 50 feet of the fence, and where the fence alignment changes more than 15 degrees. Grounding locations can not exceed 650 feet. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by power lines of 600 volts or more at or near the point of crossing and at distances not exceeding 150 feet on each side of crossing. Provide ground conductor consisting of No. 6 AWG solid copper wire. Provide copper-clad steel rod grounding electrodes 3/4 inch by 10 foot long. Drive electrodes into the earth so that the top of the electrode is at least 6 inches below the grade. Where driving is impracticable, bury electrodes a minimum of 12 inches deep and radially

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from the fence, with top of the electrode not less than 2 feet or more than 8 feet from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Total resistance of the fence to ground cannot exceed 25 ohms.

3.3 CLOSEOUT ACTIVITIES

Remove waste fencing materials and other debris from the work site.

-- End of Section --

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SEEDING

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C602	(2020) Agricultural Liming Materials
ASTM D4427	(2018) Standard Classification of Peat Samples by Laboratory Testing
ASTM D4972	(2018) Standard Test Methods for pH of Soils

U.S. DEPARTMENT OF AGRICULTURE (USDA)

AMS Seed Act	(1940; R 1988; R 1998) Federal Seed Act
DOA SSIR 42	(1996) Soil Survey Investigation Report No. 42, Soil Survey Laboratory Methods Manual, Version 3.0

1.2 DEFINITIONS

1.2.1 Stand of Turf

95 percent ground cover of the established species.

1.3 RELATED REQUIREMENTS

Section 31 00 00 EARTHWORK applies to this section for pesticide use and plant establishment requirements, with additions and modifications herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Wood Cellulose Fiber Mulch

Fertilizer

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Include physical characteristics, and recommendations.

SD-06 Test Reports

Topsoil Composition Tests (reports and recommendations).

SD-07 Certificates

State Certification and Approval for Seed

SD-08 Manufacturer's Instructions

Erosion Control Materials

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

1.5.1.1 Seed Protection

Protect from drying out and from contamination during delivery, on-site storage, and handling.

1.5.1.2 Fertilizer and Lime Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to state and federal laws. Instead of containers, fertilizer and lime may be furnished in bulk with certificate indicating the above information.

1.5.2 Storage

1.5.2.1 Seed, Fertilizer, and Lime Storage

Store in cool, dry locations away from contaminants.

1.5.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

1.5.2.3 Handling

Do not drop or dump materials from vehicles.

1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS

1.6.1 Restrictions

Do not plant when the ground is frozen, snow covered, muddy, or when air temperature exceeds 90 degrees Fahrenheit.

1.7 TIME LIMITATIONS

1.7.1 Seed

Apply seed within twenty four hours after seed bed preparation.

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PART 2 PRODUCTS

2.1 SEED

2.1.1 Classification

Provide State-certified seed of the latest season's crop delivered in original sealed packages, bearing producer's guaranteed analysis for percentages of mixtures, purity, germination, weedseed content, and inert material. Label in conformance with AMS Seed Act and applicable state seed laws. Wet, moldy, or otherwise damaged seed will be rejected. Field mixes will be acceptable when field mix is performed on site in the presence of the Contracting Officer.

2.1.2 Planting Dates - TXDOT District 12 (Houston)

<u>Planting Season</u>	<u>Planting Dates</u>
Permanent Seeding - All Seasons	January 15-May 15
Temporary - Warm Season	May 1- August 31
Temporary - Cool Season	September 1 - November 30

2.1.3 Seed Purity

Seed purity shall be per TXDOT District 12 Houston, Table 2 Permanent Urban Seed Mix, Table 3 Temporary Cool Season Seeding, and Table 4 Temporary Warm Season Seeding as stated in TXDOT Construction Specification Item 164 Seeding for Erosion Control.

2.1.4 Seed Mixture by Weight

Seed mixes shall be per TXDOT District 12 Houston, Table 2 Permanent Urban Seed Mix, Table 3 Temporary Cool Season Seeding, and Table 4 Temporary Warm Season Seeding as stated in TXDOT Construction Specification Item 164 Seeding for Erosion Control.

2.2 TOPSOIL

2.2.1 On-Site Topsoil

Surface soil stripped and stockpiled on site and modified as necessary to meet the requirements specified for topsoil in paragraph COMPOSITION. When available topsoil must be existing surface soil stripped and stockpiled on-site in accordance with Section 31 00 00 EARTHWORK.

2.2.2 Off-Site Topsoil

Conform to requirements specified in paragraph COMPOSITION. Additional topsoil must be furnished by the Contractor.

2.2.3 Composition

Containing from 5 to 10 percent organic matter as determined by the topsoil composition tests of the Organic Carbon, 6A, Chemical Analysis Method described in DOA SSIR 42. Maximum particle size, 3/4 inch, with

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maximum 3 percent retained on 1/4 inch screen. The pH must be tested in accordance with ASTM D4972. Topsoil must be free of sticks, stones, roots, and other debris and objectionable materials. Other components must conform to the following limits:

Silt	25-50 percent
Clay	10-30 percent
Sand	20-35 percent
pH	5.5 to 7.0
Soluble Salts	600 ppm maximum

2.3 SOIL CONDITIONERS

Add conditioners to topsoil as required to bring into compliance with "composition" standard for topsoil as specified herein.

2.3.1 Lime

Commercial grade hydrate or burnt limestone containing a calcium carbonate equivalent (C.C.E.) as specified in ASTM C602.

2.3.2 Aluminum Sulfate

Commercial grade.

2.3.3 Sulfur

100 percent elemental

2.3.4 Iron

100 percent elemental

2.3.5 Peat

Natural product of peat moss derived from a freshwater site and conforming to ASTM D4427. Shred and granulate peat to pass a 1/2 inch mesh screen and condition in storage pile for minimum 6 months after excavation.

2.3.6 Sand

Clean and free of materials harmful to plants.

2.3.7 Perlite

Horticultural grade.

2.3.8 Composted Derivatives

Ground bark, nitrolized sawdust, humus or other green wood waste material free of stones, sticks, and soil stabilized with nitrogen and having the following properties:

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2.3.8.1 Particle Size

Minimum percent by weight passing:

No. 4 mesh screen	95
No. 8 mesh screen	80

2.3.8.2 Nitrogen Content

Minimum percent based on dry weight:

Fir Sawdust	0.7
Fir or Pine Bark	1.0

2.3.9 Gypsum

Coarsely ground gypsum comprised of calcium sulfate dihydrate 80 percent, calcium 18 percent, sulfur 14 percent; minimum 96 percent passing through 20 mesh screen, 100 percent passing thru 16 mesh screen.

2.3.10 Calcined Clay

Calcined clay must be granular particles produced from montmorillonite clay calcined to a minimum temperature of 1200 degrees F. Gradation: A minimum 90 percent must pass a No. 8 sieve; a minimum 99 percent must be retained on a No. 60 sieve; and material passing a No. 100 sieve must not exceed 2 percent. Bulk density: A maximum 40 pounds per cubic foot.

2.4 FERTILIZER

2.4.1 Granular Fertilizer

Fertilizer shall meet the specifications and requirements listed in TXDOT Construction Specification Item 166 Fertilizer.

2.5 MULCH

Mulch must be free from noxious weeds, mold, and other deleterious materials.

2.5.1 Straw

Stalks from oats, wheat, rye, barley, or rice. Furnish in air-dry condition and of proper consistency for placing with commercial mulch blowing equipment. Straw must contain no fertile seed.

2.5.2 Hay

Air-dry condition and of proper consistency for placing with commercial mulch blowing equipment. Hay must be sterile, containing no fertile seed.

2.5.3 Wood Cellulose Fiber Mulch

Use recovered materials of either paper-based (100 percent post-consumer content) or wood-based (100 percent total recovered content) hydraulic mulch. Processed to contain no growth or germination-inhibiting factors and dyed an appropriate color to facilitate visual metering of materials application. Composition on air-dry weight basis: 9 to 15 percent moisture, pH range from 5.5 to 8.2. Use with hydraulic application of

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grass seed and fertilizer.

2.6 WATER

Source of water must be approved by Contracting Officer and of suitable quality for irrigation, containing no elements toxic to plant life.

2.7 EROSION CONTROL MATERIALS

Erosion control material must conform to the following:

2.7.1 Erosion Control Blanket

100 percent agricultural straw or 70 percent agricultural straw/30 percent coconut fiber matrix stitched with a degradable nettings, designed to degrade within 12 months.

2.7.2 Erosion Control Fabric

Fabric must be knitted construction of polypropylene yarn with uniform mesh openings 3/4 to 1 inch square with strips of biodegradable paper. Filler paper strips must have a minimum life of 6 months.

2.7.3 Erosion Control Net

Net must be heavy, twisted jute mesh, weighing approximately 1.22 pounds per linear yard and 4 feet wide with mesh openings of approximately one inch square.

2.7.4 Erosion Control Material Anchors

Erosion control anchors must be as recommended by the manufacturer.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 EXTENT OF WORK

Provide soil preparation prior to planting (including soil conditioners as required), fertilizing, seeding, and surface topdressing of all newly graded finished earth surfaces, unless indicated otherwise, and at all areas inside or outside the limits of construction that are disturbed by the Contractor's operations.

3.1.1.1 Topsoil

Provide 4 inches of off-site topsoil to meet indicated finish grade. After areas have been brought to indicated finish grade, incorporate fertilizer into soil a minimum depth of 4 inches by disking, harrowing, tilling or other method approved by the Contracting Officer. Remove debris and stones larger than 3/4 inch in any dimension remaining on the surface after finish grading. Correct irregularities in finish surfaces to eliminate depressions. Protect finished topsoil areas from damage by vehicular or pedestrian traffic.

3.1.1.2 Fertilizer Application Rates

Fertilizer rates shall meet the specifications and requirements listed in

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TXDOT Construction Specification Item 166 Fertilizer.

3.2 SEEDING

3.2.1 Seed Application Seasons and Conditions

Immediately before seeding, restore soil to proper grade. Do not seed when ground is muddy frozen, snow covered, or in an unsatisfactory condition for seeding. If special conditions exist that may warrant a variance in the above seeding dates or conditions, submit a written request to the Contracting Officer stating the special conditions and proposed variance. Apply seed within twenty four hours after seedbed preparation. Sow seed by approved sowing equipment. Sow one-half the seed in one direction, and sow remainder at right angles to the first sowing.

3.2.2 Seed Application Method

Seeding method must be broadcasted and drop seeding.

3.2.2.1 Broadcast and Drop Seeding

Seed must be uniformly broadcast at the rates listed in TXDOT District 12 Houston, Table 2 Permanent Urban Seed Mix, Table 3 Temporary Cool Season Seeding, and Table 4 Temporary Warm Season Seeding as stated in TXDOT Construction Specification Item 164 Seeding for Erosion Control. Use broadcast or drop seeders. Sow one-half the seed in one direction, and sow remainder at right angles to the first sowing. Cover seed uniformly to a maximum depth of 1/4 inch in clay soils and 1/2 inch in sandy soils by means of spike-tooth harrow, cultipacker, raking or other approved devices.

3.2.3 Mulching

3.2.3.1 Hay or Straw Mulch

Hay or straw mulch must be spread uniformly at the rate of 2 tons per acre. Mulch must be spread by hand, blower-type mulch spreader, or other approved method. Mulching must be started on the windward side of relatively flat areas or on the upper part of steep slopes, and continued uniformly until the area is covered. The mulch must not be bunched or clumped. Sunlight must not be completely excluded from penetrating to the ground surface. All areas installed with seed must be mulched on the same day as the seeding. Mulch must be anchored immediately following spreading.

3.2.3.2 Mechanical Anchor

Mechanical anchor must be a V-type-wheel land packer; a scalloped-disk land packer designed to force mulch into the soil surface; or other suitable equipment.

3.2.3.3 Asphalt Adhesive Tackifier

Asphalt adhesive tackifier must be sprayed at a rate between 10 to 13 gallons per 1000 square feet. Sunlight must not be completely excluded from penetrating to the ground surface.

3.2.3.4 Non-Asphaltic Tackifier

Hydrophilic colloid must be applied at the rate recommended by the

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manufacturer, using hydraulic equipment suitable for thoroughly mixing with water. A uniform mixture must be applied over the area.

3.2.4 Rolling

Immediately after seeding, firm entire area except for slopes in excess of 3 to 1 with a roller not exceeding 90 pounds for each foot of roller width.

3.2.5 Erosion Control Material

Install in accordance with manufacturer's instructions, where indicated or as directed by the Contracting Officer.

3.2.6 Watering

Start watering areas seeded as required by temperature and wind conditions. Apply water at a rate sufficient to insure thorough wetting of soil to a depth of 2 inches without run off. During the germination process, seed is to be kept actively growing and not allowed to dry out.

3.3 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use.

3.4 RESTORATION

Restore to original condition existing turf areas which have been damaged during turf installation operations at the Contractor's expense. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

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PART 1 GENERAL

1.1 UNIT PRICES

Measurement and payment will be based on completed work performed in accordance with the drawings, specifications, and the Contract payment schedules.

1.1.1 Measurement

The length of water lines will be determined by measuring along the centerlines of the various sizes of pipe provided. Pipe will be measured from center of fitting to center of fitting, and from center of water main to end of service connection. No deduction will be made for the space occupied by valves or fittings.

1.1.2 Payment

Payment will be made for water lines at the Contract unit price per linear foot for the various types and sizes of water lines, and will be full compensation for all pipes, joints, specials, and fittings, complete and in place. Payment for fire hydrants, valves, and valve boxes will be made at the respective Contract unit price each for such items complete and in place. Payment will include providing all testing, plant, labor, and material and incidentals necessary to complete the work, as specified and as shown.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1	(2020) Gray Iron Pipe Flanges and Flanged Fittings Classes 25, 125, and 250
ASME B16.26	(2018) Standard for Cast Copper Alloy Fittings for Flared Copper Tubes

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA B300	(2018) Hypochlorites
AWWA B301	(2018) Liquid Chlorine
AWWA C104/A21.4	(2022) Cement-Mortar Lining for Ductile-Iron Pipe and Fittings for Water
AWWA C105/A21.5	(2018) Polyethylene Encasement for Ductile-Iron Pipe Systems

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AWWA C110/A21.10	(2021) Ductile-Iron and Gray-Iron Fittings
AWWA C111/A21.11	(2017) Rubber-Gasket Joints for Ductile-Iron Pressure Pipe and Fittings
AWWA C115/A21.15	(2020) Flanged Ductile-Iron Pipe With Ductile-Iron or Gray-Iron Threaded Flanges
AWWA C151/A21.51	(2017) Ductile-Iron Pipe, Centrifugally Cast
AWWA C153/A21.53	(2019) Ductile-Iron Compact Fittings for Water Service
AWWA C500	(2019) Metal-Seated Gate Valves for Water Supply Service
AWWA C509	(2015) Resilient-Seated Gate Valves for Water Supply Service
AWWA C515	(2020) Reduced-Wall, Resilient-Seated Gate Valves for Water Supply Service
AWWA C600	(2017) Installation of Ductile-Iron Mains and Their Appurtenances
AWWA C605	(2021) Underground Installation of Polyvinyl Chloride (PVC) and Molecularly Oriented Polyvinyl Chloride (PVC0) Pressure Pipe and Fittings
AWWA C606	(2015) Grooved and Shouldered Joints
AWWA C651	(2014) Standard for Disinfecting Water Mains
AWWA C655	(2009) Field Dechlorination
AWWA C800	(2021) Underground Service Line Valves and Fittings
AWWA C900	(2016) Polyvinyl Chloride (PVC) Pressure Pipe, and Fabricated Fittings, 4 In. Through 60 In. (100 mm Through 1,500 mm)
AWWA M23	(2020) Manual: PVC Pipe - Design and Installation - Third Edition
AWWA M41	(2009; 3rd Ed) Ductile-Iron Pipe and Fittings

ASTM INTERNATIONAL (ASTM)

ASTM A536	(1984; R 2019; E 2019) Standard Specification for Ductile Iron Castings
ASTM B61	(2015; R 2021) Standard Specification for Steam or Valve Bronze Castings

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ASTM B62	(2017) Standard Specification for Composition Bronze or Ounce Metal Castings
ASTM C94/C94M	(2022a) Standard Specification for Ready-Mixed Concrete
ASTM D1784	(2020) Standard Specification for Rigid Poly(Vinyl Chloride) (PVC) Compounds and Chlorinated Poly(Vinyl Chloride) (CPVC) Compounds
ASTM D1785	(2015; E 2018) Standard Specification for Poly(Vinyl Chloride) (PVC), Plastic Pipe, Schedules 40, 80, and 120
ASTM D2241	(2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Pressure-Rated Pipe (SDR Series)
ASTM D2466	(2017) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 40
ASTM D2467	(2015) Standard Specification for Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80
ASTM D2774	(2021) Underground Installation of Thermoplastic Pressure Piping
ASTM D2855	(2015) Standard Practice for Making Solvent-Cemented Joints with Poly(Vinyl Chloride) (PVC) Pipe and Fittings
ASTM D3139	(2019) Joints for Plastic Pressure Pipes Using Flexible Elastomeric Seals
ASTM F402	(2005; R 2012) Safe Handling of Solvent Cements, Primers, and Cleaners Used for Joining Thermoplastic Pipe and Fittings
ASTM F477	(2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe
ASTM F1674	(2011) Standard Test Method for Joint Restraint Products for Use with PVC Pipe
ASTM F3190	(2021) Standard Practice for Heat Fusion Equipment (HFE) Operator Qualification on Polyethylene (PE) and Polyimide (PA) Pipe and Fittings

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 24	(2022) Standard for the Installation of Private Fire Service Mains and Their Appurtenances
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NSF INTERNATIONAL (NSF)

NSF/ANSI 14	(2022) Plastics Piping System Components and Related Materials
NSF/ANSI 61	(2022) Drinking Water System Components - Health Effects

UNDERWRITERS LABORATORIES (UL)

UL 262	(2004; Reprint Oct 2011) Gate Valves for Fire-Protection Service
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1.3 DEFINITIONS

1.3.1 Water Mains

Water mains include water piping having diameters 4 through 14 inch, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.3.2 Water Service Lines

Water service lines include water piping from a water main to a building service at a point approximately 5 feet from building or the point indicated on the drawings, specific materials, methods of joining and any appurtenances deemed necessary for a satisfactory system.

1.3.3 Additional Definitions

For additional definitions refer to the definitions in the applicable referenced standard.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Connections; G-D0

SD-03 Product Data

Pipe, Fittings, Joints and Couplings; G-Pipe Restraint; G-D0

Corporation Stops; G, D0

Precast Concrete Thrust Blocks; G, D0

Disinfection Procedures; G, D0

SD-06 Test Reports

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Leakage Test

Hydrostatic Test

SD-07 Certificates

Pipe, Fittings, Joints and Couplings
Lining

Lining for Fittings

Fusion Technician Qualifications; G-D0

SD-08 Manufacturer's Instructions

Ductile-Iron Piping

PVC Piping

PVC Piping For Service Lines

1.5 QUALITY CONTROL

1.5.1 Regulatory Requirements

Comply with NFPA 24 for materials, installation, and testing of fire main piping and components.

1.5.2 Qualifications

1.5.2.1 Fusion Technician Qualifications

Submit a certificate from the manufacturer of the fusible pipe that shows the fusion technician is fully qualified to install fusible pipe of the types and sizes being used. Qualification must be current as of the actual date of fusion performance on the project.

1.5.2.1.1 Fusion Technician Qualification on Polyethylene (PE) Pipe and Fittings

Provide certification for PE Pipe heat fusion in accordance with ASTM F3190.

1.6 DELIVERY, STORAGE, AND HANDLING

1.6.1 Delivery and Storage

Inspect materials delivered to site for required pipe markings and damage. Unload and store with minimum handling and in accordance with manufacturer's instructions to prevent cuts, scratches and other damage. Store materials on site in enclosures or under protective covering. Store plastic piping, jointing materials and rubber gaskets under cover out of direct sunlight. Do not store materials directly on the ground. Keep inside of pipes, fittings, valves, and other accessories free of dirt and debris or other contaminants.

1.6.2 Handling

Handle pipe, fittings, valves, and other accessories in accordance with applicable AWWA standard, manufacturer's instructions and in a manner to

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ensure delivery to the trench in sound undamaged condition. Avoid injury to coatings and linings on pipe and fittings; make repairs if coatings or linings are damaged. Do not place other material, hooks, or pipe inside a pipe or fitting after the coating has been applied. Inspect the pipe for defects before installation. Carry, do not drag pipe to the trench. Use of pinch bars and tongs for aligning or turning pipe will be permitted only on the bare ends of the pipe. Clean the interior of pipe and accessories of foreign matter before being lowered into the trench and keep them clean during laying operations by plugging. Replace defective material without additional expense to the Government. Store rubber gaskets, not immediately installed, under cover or out of direct sunlight.

Handle ductile iron pipe, fittings, and accessories in accordance with AWWA C600 and AWWA M41. Handle PVC pipe, fittings, and accessories in accordance with AWWA C605.

PART 2 PRODUCTS

2.1 MATERIALS

All materials are intended for potable water use unless otherwise indicated. Comply with NSF/ANSI 61 or NSF/ANSI 14 for all potable water pipe, fittings and other applicable materials. Provide pipe, fittings and other applicable materials bearing NSF/ANSI 61 or NSF/ANSI 14 markings for potable water service.

Provide all materials in accordance with AWWA C800 and as indicated herein. Provide valves and fittings with pressure ratings equivalent to the pressure ratings of the pipe.

2.1.1 Pipe, Fittings, Joints And Couplings

Submit manufacturer's standard drawings or catalog cuts, except submit both drawings and cuts for push-on and rubber-gasketed bell-and-spigot joints. Include information concerning gaskets with submittal for joints and couplings.

2.1.1.1 Ductile-Iron Piping

2.1.1.1.1 Pipe and Fittings

Pipe, except flanged pipe, AWWA C151/A21.51, . Flanged pipe, AWWA C115/A21.15. Fittings, AWWA C110/A21.10 or AWWA C153/A21.53; fittings with push-on joint ends are to meet the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design. Provide fittings with pressure ratings equivalent to that of the pipe. Provide compatible pipe ends and fittings for the specified joints. Provide cement-mortar lining, AWWA C104/A21.4, twice the standard thickness on pipe and fittings.

2.1.1.1.2 Joints and Jointing Material

Provide push-on joints or mechanical joints for pipe and fittings unless otherwise indicated. Provide mechanical joints where indicated. Provide flanged joints where indicated. Provide mechanically coupled type joints using a sleeve-type mechanical coupling where indicated. Provide grooved or shouldered type joints where indicated. Provide insulating joints where indicated. Sleeve-type mechanical couplings in lieu of push-on joints are acceptable, subject to the limitations specified in the

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paragraph SLEEVE-TYPE MECHANICAL COUPLINGS. Utilize grooved or shouldered type joints in lieu of flanged joint or push-on joint, except where joint is buried.

- a. Push-On Joints: Shape of pipe ends and fitting ends, gaskets, and lubricant for joint assembly as recommended in AWWA C111/A21.11.
- b. Mechanical Joints: Dimensional and material requirements for pipe ends, glands, bolts and nuts, and gaskets as recommended in AWWA C111/A21.11.
- c. Flanged Joints: Bolts, nuts, and gaskets for flanged connections as recommended in Appendix A of AWWA C115/A21.15. Provide AWWA C115/A21.15 ductile iron flanges and conform to ASME B16.1, Class 125. Provide ASTM A536 epoxy coated steel set screw flanges. Gasket and lubricants for set screw flanges, in accordance with mechanical-joint gaskets specified in AWWA C111/A21.11.
- d. Insulating Joints: Designed to prevent metal-to-metal contact at the joint between adjacent sections of piping. Provide flanged type joint with insulating gasket, insulating bolt sleeves, and insulating washers. Provide full face dielectric type gaskets, as recommended in the Appendix to AWWA C115/A21.15. Bolts and nuts, as recommended in the Appendix to AWWA C115/A21.15.
- e. Sleeve-Type Mechanical Coupled Joints: As specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS.
- f. Grooved and Shouldered Type Joints: Grooved and shouldered pipe ends and couplings, AWWA C606. Joint dimension as specified in AWWA C606 for rigid joints, joint dimensions as specified in AWWA C606 for flexible joints.

2.1.1.1.2 Plastic Piping

2.1.1.1.2.1 PVC Piping

2.1.1.1.2.1.1 PVC Piping

AWWA C900 plain end or gasket bell end pipe meeting or exceeding ASTM D1784 cell class 12454, with a minimum Pressure Class 150 (DR27.5) with ductile iron outside diameter (DIOD).

2.1.1.1.2.1.2 Fittings for PVC Pipe

Ductile iron fittings, AWWA C110/A21.10 or compact fittings in accordance with AWWA C153/A21.53, with cement-mortar lining for fittings, AWWA C104/A21.4, standard thickness. Fittings with push-on joint ends are to conform to the same requirements as fittings with mechanical-joint ends, except for the factory modified bell design compatible for use with PVC pipe as specified.

Fittings from material that meets or exceeds ASTM D1784 cell class 12454 and is the same material as the pipe with elastomeric gaskets, in conformance with AWWA C605 and AWWA C900.

2.1.1.1.2.1.3 Joints and Jointing Material for PVC

- a. Push-on joints: Use jointing material in accordance with ASTM D3139

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and AWWA C111/A21.11 between pipes, pipes and metal fittings, valves, and other accessories or compression-type joints/mechanical joints. Provide each joint connection with an elastomeric gasket compatible for the bell or coupling used. Gaskets for push-on joints for pipe, ASTM F477. Gaskets for push-on joints and compression-type joints/mechanical joints for joint connections between pipe and metal fittings, valves, and other accessories, AWWA C111/A21.11, respectively, for push-on joints and mechanical joints.

- b. Mechanical Joint: Use mechanically coupled joints having a sleeve-type mechanical coupling, as specified in the paragraph SLEEVE-TYPE MECHANICAL COUPLINGS, as an optional jointing method for plain-end PVC pipe, subject to the limitations specified for mechanically coupled joints using a sleeve-type mechanical coupling as specified for compression-type joints in ASTM D3139. Provide jointing material in accordance with AWWA C111/A21.11 between pipe and sleeve-type mechanical couplings.

2.1.1.2.2 PVC Piping for Service Lines

2.1.1.2.2.1 Pipe and Fittings

Provide ASTM D1784 cell class 12454 pipe and fittings of the same PVC material.

- a. ASTM D1785, Schedule 40 with ASTM D2466 Schedule 40 or ASTM D2467 Schedule 80 fittings.
- b. ASTM D2241 pipe and fittings with SDR as necessary to provide 150 psi minimum pressure rating with ASTM D2466 Schedule 40 or ASTM D2467 Schedule 80 fittings.

2.1.1.2.2.2 Joints and Connections

Fittings may be joined by the solvent-cement method or threading.

2.1.1.2.2.3 Solvent Joining

Provide solvent joints in accordance with ASTM D2855.

2.1.2 Disinfection

Chlorinating materials are to conform to: Chlorine, Liquid: AWWA B301; Hypochlorite, Calcium and Sodium: AWWA B300.

2.2 ACCESSORIES

2.2.1 Pipe Restraint

2.2.1.1 Thrust Blocks

Use ASTM C94/C94M concrete having a minimum compressive strength of 2,500 psi at 28 days or use concrete of a mix not leaner than one part cement, two and one half parts sand, and five parts gravel, having the same minimum compressive strength.

2.2.1.2 Precast Thrust Blocks

Provide precast concrete thrust blocks.

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2.2.1.3 Joint Restraint

Provide restrained joints in accordance with NFPA 24, Chapter 10 and in accordance with ASTM F1674.

Provide mechanical joint restraint devices with gripper wedges incorporated into a follower gland and specifically designed for the pipe material and meeting the requirements of AWWA C110/A21.10 or metal harness fabricated by the pipe manufacturer.

2.2.2 Tracer Wire for Nonmetallic Piping

Provide a continuous bare copper or aluminum wire not less than 0.10 inch in diameter in sufficient length over each separate run of nonmetallic pipe.

2.2.3 Water Service Line Appurtenances

2.2.3.1 Corporation Stops

Ground key type; lead-free bronze, ASTM B61 or ASTM B62; compatible with the working pressure of the system and solder-joint, or flared tube compression type joint. Threaded ends for inlet and outlet of corporation stops, AWWA C800; coupling nut for connection to flared copper tubing, ASME B16.26.

2.2.3.2 Curb or Service Stops

Ground key, round way, inverted key type; made of lead-free bronze, ASTM B61 or ASTM B62; and compatible with the working pressure of the system. Provide compatible ends for connection to the service piping. Cast an arrow into body of the curb or service stop indicating direction of flow.

2.2.3.3 Curb Boxes

Provide a curb box for each curb or service stop manufactured from cast iron, size capable of containing the stop where it is used. Provide a round head. Cast the word "WATER" on the lid. Factory coat the box with a heavy coat of bituminous paint.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 Connections to Existing System

Perform all connections to the existing water system in the presence of the Contracting Officer.

3.1.2 Operation of Existing Valves

Do not operate valves within or directly connected to the existing water system unless expressly directed to do so by the Contracting Officer.

3.1.3 Earthwork

Perform earthwork operations in accordance with Section 31 00 00 EARTHWORK.

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3.2 INSTALLATION

Install all materials in accordance with the applicable reference standard, manufacturers instructions and as indicated herein.

3.2.1 Piping

3.2.1.1 General Requirements

Install pipe, fittings, joints and couplings in accordance with the applicable referenced standard, the manufacturer's instructions and as specified herein.

3.2.1.1.1 Termination of Water Lines

Terminate the work covered by this section at a point approximately 5 feet from the building, unless otherwise indicated.

Do not lay water lines in the same trench with gas lines, fuel lines, electric wiring, or any other utility. Where nonferrous metallic pipe (i.e., copper tubing) crosses any ferrous piping, provide a minimum vertical separation of 12 inches between pipes.

3.2.1.1.2 Pipe Laying and Jointing

Remove fins and burrs from pipe and fittings. Before placing in position, clean pipe, fittings, valves, and accessories, and maintain in a clean condition. Provide proper facilities for lowering sections of pipe into trenches. Under no circumstances is it permissible to drop or dump pipe, fittings, valves, or other water line material into trenches. Cut pipe cleanly, squarely, and accurately to the length established at the site and work into place without springing or forcing. Replace a pipe or fitting that does not allow sufficient space for installation of jointing material. Blocking or wedging between bells and spigots is not permitted. Lay bell-and-spigot pipe with the bell end pointing in the direction of laying. Grade the pipeline in straight lines; avoid the formation of dips and low points. Support pipe at the design elevation and grade. Secure firm, uniform support. Wood support blocking is not permitted. Lay pipe so that the full length of each section of pipe and each fitting rests solidly on the pipe bedding; excavate recesses to accommodate bells, joints, and couplings. Provide anchors and supports for fastening work into place. Make provision for expansion and contraction of pipelines. Keep trenches free of water until joints have been assembled. At the end of each work day, close open ends of pipe temporarily with wood blocks or bulkheads. Do not lay pipe when conditions of trench or weather prevent installation. Provide a minimum of 3 feet depth of cover over top of pipe.

3.2.1.1.3 Tracer Wire

Install a continuous length of tracer wire for the full length of each run of nonmetallic pipe. Attach wire to top of pipe in such manner that it will not be displaced during construction operations.

3.2.1.1.4 Connections to Existing Water Lines

Make connections to existing water lines after coordination with the facility and with a minimum interruption of service on the existing line.

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Make connections to existing lines under pressure in accordance with the recommended procedures of the manufacturer of the pipe being tapped and as indicated.

3.2.1.1.5 Sewer Manholes

No water piping is to pass through or come in contact with any part of a sewer manhole.

3.2.1.1.6 Water Piping Parallel With Sewer Piping

Where the location of the water line is not clearly defined by dimensions on the drawings, do not lay water line closer than 10 feet, horizontally, from any sewer line.

3.2.1.1.7 Water Piping Crossing Sewer Piping

Provide at least 18 inches above the top (crown) of the sewer piping and the bottom (invert) of the water piping whenever possible. Measure the distance edge-to-edge. Where water lines cross under gravity sewer lines, construct sewer line of AWWA compliant ductile iron water piping with rubber-gasketed joints and no joint located within 10 feet, horizontally, of the crossing. Lay water lines which cross sewer force mains and inverted siphons at least 2 feet above these sewer lines; when joints in the sewer line are closer than 3 feet horizontally from the water line relay the sewer line to ensure no joint closer than 3 feet.

3.2.1.1.8 Penetrations

Provide ductile-iron or Schedule 40 steel wall sleeves for pipe passing through walls of valve pits and structures. Fill annular space between walls and sleeves with rich cement mortar. Fill annular space between pipe and sleeves with mastic.

3.2.1.1.9 Flanged Pipe

Only install flanged pipe aboveground or with the flanges in valve pits.

3.2.1.2 Ductile-Iron Piping

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of AWWA C600 for pipe installation, joint assembly, valve-and-fitting installation, and thrust restraint.

- a. Jointing: Make push-on joints with the gaskets and lubricant specified for this type joint; assemble in accordance with the applicable requirements of AWWA C600 and AWWA M41 for joint assembly. Make mechanical joints with the gaskets, glands, bolts, and nuts specified for this type joint; assemble in accordance with the applicable requirements of AWWA C600 and AWWA M41 for joint assembly and the recommendations of Appendix A to AWWA C111/A21.11. Make flanged joints with the gaskets, bolts, and nuts specified for this type joint. Make flanged joints up tight; avoid undue strain on flanges, fittings, valves, and other equipment and accessories. Align bolt holes for each flanged joint. Use full size bolts for the bolt holes; use of undersized bolts will not be permitted. Do not allow adjoining flange faces to be out of parallel to such degree that the flanged joint cannot be made watertight without overstraining the

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flange. When flanged pipe or fitting has dimensions that do not allow the making of a flanged joint as specified, replace it. Use set screw flanges to make flanged joints where conditions prevent the use of full-length flanged pipe and assemble in accordance with the recommendations of the set screw flange manufacturer. During installation of set screw gasket provide for confinement and compression of gasket when joint to adjoining flange is made. Assemble joints made with sleeve-type mechanical couplings in accordance with the recommendations of the coupling manufacturer. Make grooved and shouldered type joints with the couplings previously specified for this type joint connecting pipe with the grooved or shouldered ends specified for this type joint; assemble in accordance with the recommendations of the coupling manufacturer. Groove pipe in the field only with groove cutting equipment designed especially for the purpose and produced by a manufacturer of grooved joint couplings; secure approval for field-cut grooves before assembling the joint. Make insulating joints with the gaskets, sleeves, washers, bolts, and nuts previously specified for this type joint. Assemble insulating joints as specified for flanged joints, except that bolts with insulating sleeves are to be full size for the bolt holes. Ensure that there is no metal-to-metal contact between dissimilar metals after the joint has been assembled.

- b. Allowable Deflection: Follow AWWA C600 and AWWA M41 for the maximum allowable deflection. If the alignment requires deflection in excess of the above limitations, provide special bends or a sufficient number of shorter lengths of pipe to achieve angular deflections within the limit set forth.
- c. Exterior Protection: Completely encase buried ductile iron pipelines using Method A or B, with polyethylene film, in accordance with AWWA C105/A21.5.

3.2.1.3 PVC Water Main Pipe

Unless otherwise specified, install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the requirements of AWWA C605 for laying of pipe, joining PVC pipe to fittings and accessories, valves, and fittings; and with the recommendations for pipe joint assembly and appurtenance installation in AWWA M23, Chapter 7, "Installation."

- a. Jointing: Make push-on joints with the elastomeric gaskets specified for this type joint, using either elastomeric-gasket bell-end pipe or elastomeric-gasket couplings. For pipe-to-pipe push-on joint connections, use only pipe with push-on joint ends having factory-made bevel; for push-on joint connections to metal fittings, valves, and other accessories, cut spigot end of pipe off square and re-bevel pipe end to a bevel approximately the same as that on ductile-iron pipe used for the same type of joint. Use a lubricant recommended by the pipe manufacturer for push-on joints. Assemble push-on joints for pipe-to-pipe joint connections in accordance with the requirements of AWWA C605 for laying the pipe and the recommendations in AWWA M23, Chapter 7, "Installation," for pipe joint assembly. Assemble push-on joints for connection to fittings, valves, and other accessories in accordance with the requirements of AWWA C605 for joining PVC pipe to fittings and accessories and with the requirements of AWWA C600 for joint assembly. Make compression-type joints/mechanical joints with the gaskets, glands, bolts, nuts, and internal stiffeners previously specified for this type joint; assemble in accordance with the

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requirements of AWWA C605 for joining PVC pipe to fittings and accessories, with the requirements of AWWA C600 for joint assembly, and with the recommendations of Appendix A to AWWA C111/A21.11. Cut off spigot end of pipe for compression-type joint/mechanical-joint connections and do not re-bevel. Assemble joints made with sleeve-type mechanical couplings in accordance with the recommendations of the coupling manufacturer using internal stiffeners as previously specified for compression-type joints.

- b. Joint Offset: Construct joint offset in accordance AWWA C605. Do not exceed the minimum longitudinal bending as indicated by AWWA C605.
- c. Fittings: Install in accordance with AWWA C605.

3.2.1.4 Plastic Service Piping

Install pipe and fittings in accordance with the paragraph GENERAL REQUIREMENTS and with the applicable requirements of ASTM D2774 and ASTM D2855, unless otherwise specified. Handle solvent cements used to join plastic piping in accordance with ASTM F402.

3.2.1.4.1 Jointing

Make solvent-cemented joints for PVC piping using the solvent cement previously specified for this material; assemble joints in accordance with ASTM D2855. Make plastic pipe joints to other pipe materials in accordance with the recommendations of the plastic pipe manufacturer.

3.2.1.4.2 Plastic Pipe Connections to Appurtenances

Connect plastic service lines to corporation stops and gate valves in accordance with the recommendations of the plastic pipe manufacturer.

3.2.1.5 Fire Protection Service Lines for Sprinkler Supplies

Connect water service lines used to supply building sprinkler systems for fire protection to the water main in accordance with NFPA 24.

3.2.2 Disinfection

Prior to disinfection, provide disinfection procedures, proposed neutralization and disposal methods of waste water from disinfection as part of the disinfection submittal. Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with AWWA C651. Disinfect new water piping using the AWWA C651 .

3.2.3 Flushing

Perform bacteriological tests prior to flushing. Flush solution from the systems with domestic water until maximum residual chlorine content is within the range of 0.2 to 0.5 parts per million, the residual chlorine content of the distribution system, or acceptable for domestic use. Use AWWA C655 neutralizing chemicals.

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3.2.4 Pipe Restraint

3.2.4.1 Concrete Thrust Blocks

Install concrete thrust blocks where indicated.

3.2.4.2 Restrained Joints

Install restrained joints in accordance with the manufacturer's instructions and NFPA 24 where indicated. For metal harness use tie rods and clamps as shown in NFPA 24. Provide structural welded, skip welded, clamp type harness, bell bolt harness, snap ring harness for pipe anchorage. Provide metal harness fabricated by the pipe manufacturer and furnished with the pipe.

3.2.5 Valves

3.2.5.1 Gate Valves

Install gate valves, AWWA C500 and UL 262, in accordance with the requirements of AWWA C600 for valve-and-fitting installation and with the recommendations of the Appendix ("Installation, Operation, and Maintenance of Gate Valves") to AWWA C500. Install gate valves, AWWA C509 or AWWA C515, in accordance with the requirements of AWWA C600 for valve-and-fitting installation and with the recommendations of the Appendix ("Installation, Operation, and Maintenance of Gate Valves") to AWWA C509 or AWWA C515. Install gate valves on PVC water mains in accordance with the recommendations for appurtenance installation in AWWA M23, Chapter 7, "Installation." Make and assemble joints to gate valves as specified for making and assembling the same type joints between pipe and fittings.

3.2.5.2 Check Valves

Install check valves in accordance with the applicable requirements of AWWA C600 for valve-and-fitting installation. Make and assemble joints to check valves as specified for making and assembling the same type joints between pipe and fittings.

3.2.5.3 Air Release, Air/Vacuum, and Combination Air Valves

Install pressure vacuum assemblies of type, size, and capacity indicated. Include valves and test cocks. Install according to the requirements of plumbing and health department and authorities having jurisdiction. Do not install pressure vacuum breaker assemblies in vault or other space subject to flooding.

3.2.6 Blowoff Valve Assemblies

Install blowoff valve assemblies as indicated on the drawings or in accordance with the manufacturer's recommendations. Install discharge fitting on the end of riser pipe to direct the flow of water so as to minimize damage to surrounding areas.

3.3 FIELD QUALITY CONTROL

3.3.1 Tests

Notify the Contracting Officer a minimum of five days in advance of hydrostatic testing. Coordinate the proposed method for disposal of waste

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water from hydrostatic testing. Perform field tests, and provide labor, equipment, and incidentals required for testing. Provide documentation that all items of work have been constructed in accordance with the Contract documents.

3.3.1.1 Hydrostatic Test

Test the water system in accordance with the applicable AWWA standard specified below. Where water mains provide fire service, test in accordance with the special testing requirements given in the paragraph SPECIAL TESTING REQUIREMENTS FOR FIRE SERVICE. Test ductile-iron water mains in accordance with the requirements of AWWA C600 for hydrostatic testing. The amount of leakage on ductile-iron pipelines with mechanical-joints or push-on joints is not to exceed the amounts given in AWWA C600; no leakage will be allowed at joints made by any other methods. Test PVC plastic water systems made with PVC pipe in accordance with the requirements of AWWA C605 for pressure and leakage tests. The amount of leakage on pipelines made of PVC water main pipe is not to exceed the amounts given in AWWA C605, except that at joints made with sleeve-type mechanical couplings, no leakage will be allowed. Test steel water mains in accordance with applicable requirements of AWWA C600 for hydrostatic testing. The amount of leakage on steel pipelines with rubber-gasketed bell-and-spigot joints is not to exceed 20 gallons per 24 hours per inch of pipe diameter per mile of pipeline; no leakage will be allowed at joints made by any other method. To stop leakage, repair welded joints only by welding. Test water service lines in accordance with requirements of AWWA C600 for hydrostatic testing. No leakage will be allowed at plastic service pipe joints, flanged joints, and screwed joints. Do not backfill utility trench or begin testing on any section of a pipeline where concrete thrust blocks have been provided until at least 7 days after placing of the concrete.

3.3.1.2 Leakage Test

For leakage test, use a hydrostatic pressure not less than the maximum working pressure of the system. Leakage test may be performed at the same time and at the same test pressure as the pressure test.

3.3.1.3 Special Testing Requirements for Fire Service

Test water mains and water service lines providing fire service or water and fire service in accordance with NFPA 24. The additional water added to the system must not exceed the limits given in NFPA 24

3.3.1.4 Tracer Wire Continuity Test

Test tracer wire for continuity after service connections have been completed and prior to final pavement or restoration. Verify that tracer wire is locatable with electronic utility locating equipment. Repair breaks or separations and re-test for continuity.

3.4 SYSTEM STARTUP

Water mains and appurtenances must be completely installed, disinfected, flushed, and satisfactory bacteriological sample results received prior to permanent connections being made to the active distribution system. Obtain approval by the Contracting Officer prior to the new water piping being placed into service.

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3.5 CLEANUP

Upon completion of the installation of water lines and appurtenances,
remove all debris and surplus materials resulting from the work.

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NATURAL GAS AND LIQUID PETROLEUM PIPING 11/09, CHG 1: 08/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN GAS ASSOCIATION (AGA)

AGA ANSI B109.1	(2000) Diaphragm Type Gas Displacement Meters (Under 500 cubic ft./hour Capacity)
AGA ANSI B109.3	(2019) Rotary-Type Gas Displacement Meters
AGA ANSI B109.4	(2016) Self-Operated Diaphragm-Type Natural Gas Service Regulators for Nominal Pipe Size 1¼ inches (32 mm) and Smaller with Outlet Pressures of 2 psig (13.8 kPa) and Less
AGA XR0603	(2006; 8th Ed) AGA Plastic Pipe Manual for Gas Service

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	(2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B1.20.1	(2013; R 2018) Pipe Threads, General Purpose (Inch)
ASME B16.3	(2021) Malleable Iron Threaded Fittings, Classes 150 and 300
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Butt welding Fittings
ASME B16.33	(2012; R 2017) Manually Operated Metallic Gas Valves for Use in Gas Piping Systems Up to 125 psi, (Sizes NPS 1/2 - NPS 2)
ASME B16.38	(2012; R 2017) Large Metallic Valves for Gas Distribution Manually Operated, NPS 2 1/2 (DN 65) to NPS 12 (DN 300), 125 psig 8.6 bar) Maximum
ASME B16.39	(2020) Standard for Malleable Iron Threaded Pipe Unions; Classes 150, 250,

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and 300

ASME B18.2.1 (2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)

ASME B18.2.2 (2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASME B31.8 (2022; Supplement 2022) Gas Transmission and Distribution Piping Systems

ASME BPVC SEC VIII D1 (2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M (2022) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A193/A193M (2023) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications

ASTM A194/A194M (2022a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both

ASTM D2513 (2018a) Standard Specification for Polyethylene (PE) Gas Pressure Pipe, Tubing, and Fittings

ASTM D2683 (2020) Standard Specification for Socket-Type Polyethylene Fittings for Outside Diameter-Controlled Polyethylene Pipe and Tubing

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY (MSS)

MSS SP-58 (2018) Pipe Hangers and Supports - Materials, Design and Manufacture, Selection, Application, and Installation

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 54 (2021) National Fuel Gas Code

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-101 (2014; Rev C) Color Code for Pipelines and for Compressed Gas Cylinders

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

49 CFR 192 Transportation of Natural and Other Gas by

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Pipeline: Minimum Federal Safety Standards

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Pressure Regulator; G-AE

Valves; G-AE

Warning and Identification Tape

Risers; G-AE

Transition Fittings; G-AE

Gas meter; G-AE

SD-07 Certificates

Welder's Qualifications

PE Welder's Qualifications

Welder's Identification Symbols

SD-08 Manufacturer's Instructions

PE Pipe and Fittings

Submit manufacturer's installation instructions and manufacturer's visual joint appearance chart.

1.3 QUALITY ASSURANCE

1.3.1 Welder's Qualifications

Comply with ASME B31.8. The steel welder shall have a copy of a certified ASME B31.8 qualification test report. The PE welder shall have a certificate from a PE pipe manufacturer's sponsored training course. Contractor shall also conduct a qualification test. Submit each welder's identification symbols, assigned number, or letter, used to identify work of the welder. Affix symbols immediately upon completion of welds. Welders making defective welds after passing a qualification test shall be given a requalification test and, upon failing to pass this test, shall not be permitted to work this contract.

1.3.2 PE Welder's Qualifications

Prior to installation, Contractor shall have supervising and installing personnel trained by a PE pipe manufacturer's sponsored course of not less than one week duration, or present proof satisfactory to the Contracting

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Officer that personnel are currently working in the installation of PE gas distribution lines.

1.3.3 Safety Standards

49 CFR 192.

1.4 DELIVERY, STORAGE, AND HANDLING

Handle, transport, and store plastic pipe and fittings carefully. Plug or cap pipe ends during transportation or storage to minimize dirt and moisture entry. Do not subject to abrasion or concentrated external loads. Discard PE pipe sections and fittings that have been damaged.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

Conform to NFPA 54 and with requirements specified herein. Supply piping to appliances or equipment shall be at least as large as the inlets thereof.

2.2 PIPE AND FITTINGS

2.2.1 Aboveground and Within Buildings and Vaults

- a. Pipe: Black steel in accordance with ASTM A53/A53M, Schedule 80, threaded ends for sizes 2 inches and smaller; otherwise, plain end beveled for butt welding.
- b. Threaded Fittings: ASME B16.3, black malleable iron.
- c. Butt-Welding Fittings: ASME B16.9, with backing rings of compatible material.
- d. Unions: ASME B16.39, black malleable iron.
- e. Flanges and Flanged Fittings: ASME B16.5 steel flanges or convoluted steel flanges conforming to ASME BPVC SEC VIII D1. Flange faces shall have integral grooves of rectangular cross sections which afford containment for self-energizing gasket material.

2.2.2 Underground Polyethylene (PE)

PE pipe and fittings are as follows:

- a. Pipe: ASTM D2513, 100 psig working pressure, Standard Dimension Ratio (SDR), the ratio of pipe diameter to wall thickness, 11.5 maximum.
- b. Socket Fittings: ASTM D2683.
- c. Butt-Fusion Fittings: ASTM D2513, molded.

2.2.3 Risers

Use anodeless risers with flanged connection to above grade piping.

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2.2.4 Transition Fittings

- a. Steel to Plastic (PE): As specified for "riser" except designed for steel-to-plastic with tapping tee or sleeve. Coat or wrap exposed steel pipe with heavy plastic coating.

2.3 VALVES, ABOVEGROUND

Provide lockable valves where indicated.

2.3.1 Shutoff Valves, Sizes Larger Than 2 Inches

Steel body ball valve with flanged ends in accordance with ASME B16.38. Provide PTFE seats.

2.3.2 Shutoff Valves, Sizes 2 Inches and Smaller

Steel body ball valve in accordance with ASME B16.33, full port pattern, reinforced PTFE seals, threaded ends, and PTFE seat.

2.3.3 Line Appliance Pressure Regulator and Shutoff Valve

Provide regulators conforming to ANSI Z21.80/CSA 6.22 for line pressure regulators. Provide shutoff valves conforming to ANSI Z21.15/CSA 9.1 for manually controlled gas shutoff valves.

2.3.4 Service Regulators

- a. Provide ferrous bodied pressure regulators for individual service lines, capable of reducing distribution line pressure to pressures required for users. Provide service regulators conforming to AGA ANSI B109.4 CGA-6.18-M95 with full capacity internal relief and overpressure shutoff. Set pressure relief at a lower pressure than would cause unsafe operation of any connected user.
- c. Provide regulator(s) having a single port with orifice diameter no greater than that recommended by the manufacturer for the maximum gas flow rate at the regulator inlet pressure. Provide regulator valve vent of resilient materials designed to withstand flow conditions when pressed against the valve port, capable of regulating downstream pressure within limits of accuracy and limiting the buildup of pressure under no-flow conditions to 50 percent or less of the discharge pressure maintained under flow conditions. Provide a self-contained service regulator, and pipe not exceeding 2 inch size.

2.4 GAS METER

AGA ANSI B109.1 AGA ANSI B109.3 pipe mounted, enamel coated steel case. Provided with a strainer immediately upstream. Provide rotary-type displacement meter conforming to AGA ANSI B109.3 as required by local gas utility supplier. Provide combined odometer-type register totalizer index, UV-resistant index cover, water escape hole in housing, and means for sealing against tampering. Provide temperature-compensated type meters sized for the required volumetric flow rate and suitable for accurately measuring and handling gas at pressures, temperatures, and flow rates indicated. Provide meters with over-pressure protection as specified in 49 CFR 192 and ASME B31.8. Provide meters that are tamper-proof with fungus protection. Provide meters with a pulse switch initiator capable of operating up to speeds of 500 maximum pulses per

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minute with no false pulses and requiring no field adjustments. Provide not less than one pulse per 100 cubic feet of gas. Minimum service life shall be 30,000,000 cycles.

2.4.1 Utility Monitoring and Control System (UMCS) / Energy Monitoring and Control (EMCS) or Automatic Meter Reading Interfaces

Provide gas meters capable of interfacing the output signal, equivalent to volumetric flow rate, with the existing UMCS / EMCS for data gathering in units of cubic meters cubic feet. Provide meters that do not require power to function and deliver data. Output signal shall be either a voltage or amperage signal that can be converted to volumetric flow by using an appropriate scaling factor.

2.5 BURIED UTILITY WARNING AND IDENTIFICATION TAPE

Provide detectable aluminum-foil plastic-backed tape or detectable magnetic plastic tape manufactured specifically for warning and identification of buried piping. Tape shall be detectable by an electronic detection instrument. Provide tape in rolls, 3 inch minimum width, color-coded yellow for natural gas, with warning and identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning and identification shall be "CAUTION BURIED GAS PIPING BELOW" or similar wording. Use permanent code and letter coloring unaffected by moisture and other substances contained in trench backfill material.

2.6 HANGERS AND SUPPORTS

MSS SP-58.

2.7 WELDING FILLER METAL

ASME B31.8.

2.8 PIPE-THREAD TAPE

Antiseize and sealant tape of polytetrafluoroethylene (PTFE).

2.9 BOLTING (BOLTS AND NUTS)

Stainless steel bolting; ASTM A193/A193M, Grade B8M or B8MA, Type 316, for bolts; and ASTM A194/A194M, Grade 8M, Type 316, for nuts. Dimensions of bolts, studs, and nuts shall conform with ASME B18.2.1 and ASME B18.2.2 with coarse threads conforming to ASME B1.1, with Class 2A fit for bolts and studs and Class 2B fit for nuts. Bolts or bolt-studs shall extend through the nuts and may have reduced shanks of a diameter not less than the diameter at root of threads. Bolts shall have American Standard regular square or heavy hexagon heads; nuts shall be American Standard heavy semifinished hexagonal.

2.10 GASKETS

Fluorinated elastomer, compatible with flange faces.

2.11 IDENTIFICATION FOR ABOVEGROUND PIPING

MIL-STD-101 for legends and type and size of characters. For pipes 3/4 inch od and larger, provide printed legends to identify contents of pipes

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and arrows to show direction of flow. Color code label backgrounds to signify levels of hazard. Make labels of plastic sheet with pressure-sensitive adhesive suitable for the intended application. For pipes smaller than 3/4 inch od, provide brass identification tags 1 1/2 inches in diameter with legends in depressed black-filled characters.

PART 3 EXECUTION

3.1 INSTALLATION

Install gas piping, appliances, and equipment in accordance with NFPA 54.

3.1.1 Excavating and Backfilling

Perform excavating and backfilling of pipe trenches as specified in Section 31 00 00 EARTHWORK. Place pipe directly in trench bottom and cover with minimum 3 inches of sand to top of pipe. If trench bottom is rocky, place pipe on a 3 inch bed of sand and cover as above. Provide remaining backfilling. Coordinate provision of utility warning and identification tape with backfill operation. Bury utility warning and identification tape with printed side up at a depth of 12 inches below the top surface of earth or the top surface of the subgrade under pavements.

3.1.2 Piping

Cut pipe to actual dimensions and assemble to prevent residual stress. Within buildings, run piping parallel to structure lines and conceal in finished spaces. Terminate each vertical supply pipe to burner or appliance with tee, nipple and cap to form a sediment trap. To supply multiple items of gas-burning equipment, provide manifold with inlet connections at both ends.

3.1.2.1 Cleanliness

Clean inside of pipe and fittings before installation. Blow lines clear using 80 to 100 psig clean dry compressed air. Rap steel lines sharply along entire pipe length before blowing clear. Cap or plug pipe ends to maintain cleanliness throughout installation.

3.1.2.2 Aboveground Steel Piping

Determine and establish measurements for piping at the job site and accurately cut pipe lengths accordingly. For 2 inch diameter and smaller, use threaded or socket-welded joints. For 2 1/2 inch diameter and larger, use flanged or butt-welded joints.

- a. Threaded Joints: Where possible use pipe with factory-cut threads, otherwise cut pipe ends square, remove fins and burrs, and cut taper pipe threads in accordance with ASME B1.20.1. Provide threads smooth, clean, and full-cut. Apply anti-seize paste or tape to male threads portion. Work piping into place without springing or forcing. Backing off to permit alignment of threaded joints will not be permitted. Engage threads so that not more than three threads remain exposed. Use unions for connections to meters for which a means of disconnection is not otherwise provided.
- b. Welded Joints: Weld by the shielded metal-arc process, using covered electrodes and in accordance with procedures established and qualified in accordance with ASME B31.8.

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- c. Flanged Joints: Use flanged joints for connecting welded joint pipe and fittings to valves to provide for disconnection. Install joints so that flange faces bear uniformly on gaskets. Engage bolts so that there is complete threading through the nuts and tighten so that bolts are uniformly stressed and equally torqued.
- d. Pipe Size Changes: Use reducing fittings for changes in pipe size. Size changes made with bushings will not be accepted.
- e. Painting: Paint new ferrous metal piping, including supports, in accordance with Section 09 90 00 PAINTS AND COATINGS. Do not apply paint until piping tests have been completed.
- f. Identification of Piping: Identify piping aboveground in accordance with MIL-STD-101, using adhesive-backed or snap-on plastic labels and arrows. In lieu of labels, identification tags may be used. Apply labels or tags to finished paint at intervals of not more than 50 feet. Provide two copies of the piping identification code framed under glass and install where directed.

3.1.2.3 Buried Plastic Lines

Provide totally PE piping. Prior to installation, obtain printed instructions and technical assistance in proper installation techniques from pipe manufacturer.

- a. Jointing Procedures: Use jointing procedures conforming to AGA XR0603 and 49 CFR 192 that have been qualified by test in accordance with 49 CFR 192.283 and proven to make satisfactory joints. Personnel make joints in plastic pipe shall be qualified in accordance with 49 CFR 192.285, under the submitted and approved procedure by making a satisfactory specimen joint that passes the required inspection and test. Joints in plastic pipe shall be inspected by a person qualified by 49 CFR 192.287 under the applicable procedure. Certificates that qualify the applicable procedures, joining personnel, and inspectors shall be submitted and approved and shall be on file with the Contracting Officer prior to making these joints.
- b. PE Piping: Prior to installation, Contractor shall have supervising and installing personnel, certified in accordance with paragraph WELDER'S QUALIFICATIONS. Provide fusion-welded joints except where transitions have been specified. Use electrically heated tools, thermostatically controlled and equipped with temperature indication. Where connection must be made to existing plastic pipe, contractor shall be responsible for determination of compatibility of materials and procedural changes in fusion process necessary to attain maximum integrity of bond.
- c. Laying PE Pipe: Bury pipe 24 inches below finish grade or deeper unless noted otherwise. Lay in accordance with manufacturer's printed instructions.

3.1.2.4 Connections to Existing Pipeline

When making connections to live gas mains, use pressure tight installation equipment operated by workmen trained and experienced in making hot taps. For connections to existing underground pipeline or service branch, use transition fittings for dissimilar materials.

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3.1.2.5 Wrapping

Where connection to existing steel line is made underground, tape wrap new steel transition fittings and exposed existing pipe having damaged coating. Clean pipe to bare metal. Initially stretch first layer of tape to conform to the surface while spirally half-lapping. Apply a second layer, half-lapped and spiraled as the first layer, but with spirals perpendicular to first wrapping. Use 10 mil minimum thick polyethylene tape. In lieu of tape wrap, heat shrinkable 10 mil minimum thick polyethylene sleeve may be used.

3.1.3 Valves

Install valves approximately at locations indicated. Orient stems vertically, with operators on top, or horizontally.

3.1.4 Gas Service Installation

Contractor shall submit and use only tested and approved work procedures. Contractor shall use only welders and jointers who have been recently qualified by training and test for joining and installing the gas pipe material used on this job. The finished product shall be inspected by a person qualified to inspect joints made by the particular procedures used to make joints.

Contractor shall contract with Centerpoint Energy to install new piping from Airport Road to new meter station. Centerpoint Energy will procure and install the piping upstream of metering station and the gas regulator/meter.

3.2 FIELD QUALITY CONTROL

3.2.1 Metal Welding Inspection

Inspect for compliance with NFPA 54 ASME B31.8 and 49 CFR 192. Replace, repair, and then re-inspect defective welds.

3.2.2 PE Fusion Welding Inspection

Visually inspect butt joints by comparing with, manufacturer's visual joint appearance chart. Inspect fusion joints for proper fused connection. Replace defective joints by cutting out defective joints or replacing fittings. Inspect 100 percent of all joints and reinspect all corrections. Arrange with the pipe manufacturer's representative in the presence of the Contracting Officer to make first time inspection.

3.2.3 Pressure Tests

Use test pressure of 1 1/2 times maximum working pressure, but in no case less than 50 psig. Do not test until every joint has set and cooled at least 8 hours at temperatures above 50 degrees F. Conduct testing before backfilling; however, place sufficient backfill material between fittings to hold pipe in place during tests. Test system gas tight in accordance with NFPA 54 or ASME B31.8. Use clean dry air or inert gas, such as nitrogen or carbon dioxide, for testing. Systems which may be contaminated by gas shall first be purged as specified. Make tests on entire system or on sections that can be isolated by valves. After pressurization, isolate entire piping system from sources of air during

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test period. Maintain test pressure for at least 8 hours between times of first and last reading of pressure and temperature. Take first reading at least one hour after test pressure has been applied. Do not take test readings during rapid weather changes. Provide temperature same as actual trench conditions. There shall be no reduction in the applied test pressure other than that due to a change in ambient temperature. Allow for ambient temperature change in accordance with the relationship $PF + 14.7 = (P1 + 14.7) (T2 + 460) / T1 + 460$, in which "T" and "PF" represent Fahrenheit temperature and gage pressure, respectively, subscripts "1" and "2" denote initial and final readings, and "PF" is the calculated final pressure. If "PF" exceeds the measured final pressure (final gage reading) by 1/2 psi or more, isolate sections of the piping system, retest each section individually, and apply a solution of warm soapy water to joints of each section for which a reduction in pressure occurs after allowing for ambient temperature change. Repair leaking joints and repeat test until no reduction in pressure occurs. In performing tests, use a test gage calibrated in one psi increments and readable to 1/2 psi.

3.2.4 System Purging

After completing pressure tests, and before testing a gas contaminated line, purge line with nitrogen at junction with main line to remove all air and gas. Clear completed line by attaching a test pilot fixture at capped stub-in line at building location and let gas flow until test pilot ignites. Procedures shall conform to NFPA 54 and ASME B31.8.

-CAUTION-
Failure to purge may result in explosion within line when air-to-gas is at correct mixture.

-- End of Section --

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STORMWATER UTILITIES 11/21

PART 1 GENERAL

1.1 UNIT PRICES

1.1.1 Pipe Culverts and Storm Drains

The length of pipe installed will be measured along the centerlines of the pipe from end to end of pipe without deductions for diameter of manholes. Pipe will be paid for at the contract unit price for the number of linear feet of culverts or storm drains placed in the accepted work.

1.1.2 Storm Drainage Structures

The quantity of manholes and inlets will be measured as the total number of manholes and inlets of the various types of construction, complete with frames and gratings or covers and, where indicated, with fixed side-rail ladders, constructed to the depth of as shown on the plan drawings. The depth of manholes and inlets will be measured from the top of grating or cover to invert of outlet pipe. Manholes and inlets constructed to depths greater than the depth specified above will be paid for as units at the contract unit price for manholes and inlets.

1.1.3 Sheet piling and Bracing

Payment will be made for that sheet piling and bracing ordered to be left in place, based on the number of square feet of sheet piling and bracing remaining below the surface of the ground.

1.1.4 Rock Excavation

Payment will be made for the number of cubic yards of material acceptably excavated, as specified and defined as rock excavation in Section 31 00 00 EARTHWORK, measured in the original position, and computed by allowing actual width of rock excavation with the following limitations: maximum rock excavation width, 30 inches for pipe of 12 inch or less nominal diameter; maximum rock excavation width, 16 inches greater than outside diameter of pipe of more than 12 inch nominal diameter. Measurement will include authorized overdepth excavation. Payment will also include all necessary drilling and blasting, and all incidentals necessary for satisfactory excavation and disposal of authorized rock excavation. No separate payment will be made for backfill material required to replace rock excavation; include this cost in the unit price bid per cubic yard for rock excavation. In rock excavation for manholes and other appurtenances, 1 foot will be allowed outside the wall lines of the structures.

1.1.5 Backfill Replacing Unstable Material

Payment will be made for the number of cubic yards of select granular material required to replace unstable material for foundations under pipes or drainage structures, which will constitute full compensation for this backfill material, including removal and disposal of unstable material and

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all excavating, hauling, placing, compacting, and all incidentals
necessary to complete the construction of the foundation satisfactorily.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th
Edition) Standard Specifications for
Highway Bridges

ASTM INTERNATIONAL (ASTM)

ASTM A48/A48M (2003; R 2021) Standard Specification for
Gray Iron Castings

ASTM A123/A123M (2017) Standard Specification for Zinc
(Hot-Dip Galvanized) Coatings on Iron and
Steel Products

ASTM A536 (1984; R 2019; E 2019) Standard
Specification for Ductile Iron Castings

ASTM A716 (2018; R 2022) Standard Specification for
Ductile Iron Culvert Pipe

ASTM A807/A807M (2019) Standard Practice for Installing
Corrugated Steel Structural Plate Pipe for
Sewers and Other Applications

ASTM A929/A929M (2018) Standard Specification for Steel
Sheet, Metallic-Coated by the Hot-Dip
Process for Corrugated Steel Pipe

ASTM B26/B26M (2018; E 2018) Standard Specification for
Aluminum-Alloy Sand Castings

ASTM C76 (2022a) Standard Specification for
Reinforced Concrete Culvert, Storm Drain,
and Sewer Pipe

ASTM C76M (2022a) Standard Specification for
Reinforced Concrete Culvert, Storm Drain,
and Sewer Pipe (Metric)

ASTM C231/C231M (2022) Standard Test Method for Air
Content of Freshly Mixed Concrete by the
Pressure Method

ASTM C270 (2019a; E 2019) Standard Specification for
Mortar for Unit Masonry

ASTM C425 (2021) Standard Specification for
Compression Joints for Vitrified Clay Pipe

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and Fittings

ASTM C443	(2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets
ASTM C443M	(2021) Standard Specification for Joints for Concrete Pipe and Manholes, Using Rubber Gaskets (Metric)
ASTM C478/C478M	(2022) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections
ASTM C828	(2023) Standard Test Method for Low-Pressure Air Test of Vitrified Clay Pipe Lines
ASTM C923/C923M	(2020) Standard Specification for Resilient Connectors Between Reinforced Concrete Manhole Structures, Pipes and Laterals
ASTM C990	(2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants
ASTM C990M	(2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric)
ASTM C1103	(2022) Standard Practice for Joint Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines
ASTM C1103M	(2019) Standard Practice for Joint Acceptance Testing of Installed Precast Concrete Pipe Sewer Lines (Metric)
ASTM D1751	(2018) Standard Specification for Preformed Expansion Joint Filler for Concrete Paving and Structural Construction (Nonextruding and Resilient Bituminous Types)
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction
ASTM D2321	(2020) Standard Practice for Underground Installation of Thermoplastic Pipe for Sewers and Other Gravity-Flow Applications
ASTM D2564	(2020) Standard Specification for Solvent Cements for Poly(Vinyl Chloride) (PVC) Plastic Piping Systems

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ASTM D3034	(2016) Standard Specification for Type PSM Poly(Vinyl Chloride) (PVC) Sewer Pipe and Fittings
ASTM D3212	(2020) Standard Specification for Joints for Drain and Sewer Plastic Pipes Using Flexible Elastomeric Seals
ASTM F477	(2014; R 2021) Standard Specification for Elastomeric Seals (Gaskets) for Joining Plastic Pipe
ASTM F679	(2016) Standard Specification for Poly(Vinyl Chloride) (PVC) Large-Diameter Plastic Gravity Sewer Pipe and Fittings
ASTM F794	(2021) Standard Specification for Poly(Vinyl Chloride) (PVC) Profile Gravity Sewer Pipe and Fittings Based on Controlled Inside Diameter
ASTM F949	(2020) Standard Specification for Poly(Vinyl Chloride) (PVC) Corrugated Sewer Pipe with a Smooth Interior and Fittings
ASTM F1417	(2011a; E 2020) Standard Practice for Installation Acceptance of Plastic Non-pressure Sewer Lines Using Low-Pressure Air

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Leakage Test; G-D0

SD-07 Certificates

Hydrostatic Test on Watertight Joints

SD-08 Manufacturer's Instructions

Placing Pipe

1.4 DELIVERY, STORAGE, AND HANDLING

1.4.1 Delivery and Storage

Inspect materials delivered to site for damage and unload and store materials with minimal handling. Do not store materials directly on the

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ground. Keep the inside of pipes and fittings free of dirt and debris. Before, during, and after installation, protect plastic pipe and fittings from any environment that would result in damage or deterioration to the material. Keep a copy of the manufacturer's instructions available at the construction site at all times and follow these instructions unless directed otherwise by the Contracting Officer. Store solvents, solvent compounds, lubricants, elastomeric gaskets, and any similar materials required to install plastic pipe in accordance with the manufacturer's recommendations and discard if the storage period exceeds the recommended shelf life. Discard solvents in use when the recommended pot life is exceeded.

1.4.2 Handling

Handle materials in a manner that ensures delivery to the trench in sound, undamaged condition. Carry pipe to the trench.

PART 2 PRODUCTS

2.1 PIPE FOR CULVERTS AND STORM DRAINS

Pipe sizes for culverts and storm drains are indicated on the drawings.

2.1.1 Concrete Pipe

2.1.1.1 Reinforced Culvert and Storm Drain Pipe

Manufactured in accordance with and conforming to ASTM C76M ASTM C76, Class as indicated.

2.1.2 Ductile Iron Culvert Pipe

Provide ductile iron culvert pipe conforming to ASTM A716.

2.1.3 Poly Vinyl Chloride (PVC) Pipe

2.1.3.1 Type PSM PVC Pipe

ASTM D3034, maximum SDR 35.

2.1.3.2 Profile PVC Pipe

ASTM F794, Series 46.

2.1.3.3 Smooth Wall PVC Pipe

ASTM F679.

2.1.3.4 Corrugated PVC Pipe

ASTM F949.

2.2 PIPE JOINTS

Provide joints that have been tested for and meet the requirements of paragraph HYDROSTATIC TEST ON WATERTIGHT JOINTS.

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2.2.1 Concrete Pipe

2.2.1.1 Rubber Gasket Joints

Provide rubber gasket joints of a design and physical requirements conforming to ASTM C443.

2.2.1.2 Preformed Flexible Sealant Joints

Provide joints made with preformed flexible joint sealant conforming to ASTM C990.

2.2.2 Ductile Iron Pipe

Provide push-on type joints with rubber gaskets.

2.2.3 PVC Plastic Pipe

Provide solvent cement or elastomeric gasket type joints in accordance with the specification for the pipe and as recommended by the pipe manufacturer. Use solvent cement conforming to ASTM D2564. Provide gaskets for elastomeric joints conforming to ASTM F477.

2.3 MISCELLANEOUS MATERIALS

2.3.1 Concrete

Unless otherwise specified, reinforced concrete pipe shall conform to TXDOT Specifications Item 464 Reinforced Concrete Pipe. Provide air content by volume of concrete mixture, based on measurements made immediately after discharge from the mixer, of 5 to 7 percent when maximum size of coarse aggregate exceeds 1-1/2 inches. Determine air content in accordance with ASTM C231/C231M. Provide a minimum concrete covering over steel reinforcing of not less than 1 inch thick for covers and not less than 1-1/2 inches thick for walls and flooring. For concrete deposited directly against the ground, provide a covering thickness of at least 3 inches between steel and ground. Provide expansion-joint filler material conforming to ASTM D1751, or ASTM D1752, or provide be resin-impregnated fiberboard conforming to the physical requirements of ASTM D1752.

2.3.2 Mortar

Mortar is not allowed for pipe joints. Provide mortar for pipe connections to drainage structures conforming to ASTM C270, Type M, except that the maximum placement time will be 1 hour. Provide a sufficient quantity of water in the mixture to produce a stiff workable mortar but in no case may the quantity exceed 5 gallons of water per sack of cement. Use water that is clean and free of harmful acids, alkalis, and organic impurities. Use the mortar within 30 minutes after the ingredients are mixed with water.

2.3.3 Precast Reinforced Concrete Manholes

Provide precast reinforced concrete manholes conforming to ASTM C478/C478M . Provide joints between precast concrete risers and tops that are full-bedded in cement mortar and smoothed to a uniform surface on both interior and exterior of the structure.

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2.3.4 Frame and Cover or Gratings

Provide frame and cover or gratings made of cast gray iron, ASTM A48/A48M, Class 35B; cast ductile iron, ASTM A536, Grade 65-45-12; or cast aluminum, ASTM B26/B26M, Alloy 356.0-T6. Provide curb inlet grates conforming to the weight, shape, size, and waterway openings indicated on the plans. Stamp or cast the word "Storm Sewer" into covers so that it is plainly visible.

2.3.5 Steel Ladder

Provide a steel ladder where the depth of the storm drainage structure exceeds 12 feet. Provide ladders not less than 16 inches in width, with 3/4 inch diameter rungs spaced 12 inches apart. Provide two stringers that are a minimum 3/8 inch thick and 2-1/2 inches wide. Galvanize ladders and inserts after fabrication in conformance with ASTM A123/A123M.

2.3.6 Resilient Connectors

Provide flexible, watertight connectors conforming to ASTM C923/C923M for connecting pipe to manholes and inlets.

2.3.7 Flared End Sections

2.3.7.1 Metal Flared End Sections

Provide sections of a standard design fabricated from zinc or aluminum (Type 2) coated steel sheets meeting requirements of ASTM A929/A929M.

2.3.7.2 Concrete Flared End Sections

Provide sections of a standard design fabricated with reinforced concrete.

2.4 TESTS, INSPECTIONS, AND VERIFICATIONS

2.4.1 Hydrostatic Test on Watertight Joints

Perform a hydrostatic test on the watertight joint types as proposed. This test will be conducted at the plant or by an independent laboratory. Only one sample joint of each type needs testing; however, if the sample joint fails because of faulty design or workmanship, an additional sample joint may be tested.

2.4.1.1 Concrete, Clay, PVC, PE, SRPE and PP Pipe

Provide joints in reinforced and nonreinforced concrete pipe meeting the performance requirements in ASTM C990M ASTM C990 or ASTM C443M ASTM C443. Provide joints in clay pipe meeting the test requirements in ASTM C425. Provide joints in PVC, PE, SRPE, and PP plastic pipe meeting the test requirements in ASTM D3212.

PART 3 EXECUTION

3.1 EXCAVATION FOR PIPE CULVERTS, BOX CULVERTS, STORM DRAINS, AND DRAINAGE STRUCTURES

Excavate trenches, excavate for appurtenances and backfill for culverts and storm drains, in accordance with the applicable portions of Section 31 00 00 EARTHWORK and the requirements specified below.

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3.1.1 Trenching

Excavate trenches to the width indicated on the drawings or as specified herein. Trench width should permit satisfactory jointing and thorough tamping of the bedding material under and around the pipe. Place sheeting and bracing, where required, within the trench width as specified, without any overexcavation.

3.1.2 Removal of Rock

Replace rock in either ledge or boulder formation with suitable materials to provide a compacted earth cushion. Provide a compacted earth cushion between unremoved rock and the pipe with a thickness of at least 8 inches or 1/2 inch for each foot of fill over the top of the pipe, whichever is greater, but not more than three-fourths the nominal diameter of the pipe. Maintain the cushion under the bell as well as under the straight portion of the pipe where bell-and-spigot pipe is used. Provide a compacted earth cushion between unremoved rock and the box culvert of at least 8 inches in thickness for concrete box culverts. Excavate rock as specified and defined in Section 31 00 00 EARTHWORK.

3.1.3 Removal of Unstable Material

Where wet or otherwise unstable soil incapable of properly supporting the pipe or box culvert, as determined by the Contracting Officer, is unexpectedly encountered in the bottom of a trench, remove such material to the depth required and replace with select granular material to the proper grade. Compact select granular material as specified in paragraph FINAL BACKFILL. When removal of unstable material is due to the fault or neglect of the Contractor while performing shoring and sheeting, water removal, or other specified requirements, perform such removal and replacement at no additional cost to the Government.

3.2 BEDDING AND INITIAL BACKFILL

Provide a firm bedding foundation of uniform density throughout the entire length of the pipe or box culvert.

3.2.1 Concrete Pipe

Use select granular material conforming to Section 31 00 00 EARTHWORK for haunch and bedding material. Compact haunch and outer bedding to at least 90 percent laboratory maximum density and place in layers not exceeding 6 inch loose thickness for compaction by hand-operated compactors and 200 mm 8 inches for other than hand-operated machines. Loosely place middle bedding and do not compact. After the pipe has been properly bedded, place haunch material, at a moisture content that will facilitate compaction, evenly along both sides of the pipe and thoroughly compact each layer with mechanical tampers or rammers to the springline of the pipe. Thoroughly compact the haunch material under the haunches of the pipe. For bell and spigot pipe, form a depression in bedding material for bells so entire barrel of pipe is uniformly supported. Minimize the length, depth, and width of bell depressions to that required for properly making the particular type of joint.

3.2.1.1 Trenches

After the pipe has been properly bedded and haunch material placed to the

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midpoint (springline) of the pipe, backfill and compact the remainder of the trench by spreading and rolling or compacting by mechanical rammers or tampers in layers not exceeding 6 inches. Test for density as necessary to ensure conformance to the compaction requirements specified below. Leave untreated sheeting in place beneath structures or pavements.

3.2.1.2 Fill Sections

For pipe placed in fill sections, uniformly spread fill material longitudinally on both sides of the pipe in layers not exceeding 6 inches in compacted depth, and compact by rolling parallel with pipe or by mechanical tamping or ramming. Prior to commencing normal filling operations, the crown width of the fill at a height of 12 inches above the top of the pipe must extend a distance of not less than twice the outside pipe diameter on each side of the pipe or 12 feet, whichever is less. After the backfill has reached at least 12 inches above the top of the pipe, place and thoroughly compact the remainder of the fill in layers not exceeding 8 inches.

3.2.2 Ductile Iron Pipe

Provide bedding for ductile iron pipe as shown on the drawings.

3.2.3 Plastic Pipe

Provide bedding for PVC, PE, SRPE and PP pipe meeting the requirements of ASTM D2321. Use Class IB or II material for PVC, PE, SRPE pipe bedding, haunching, and initial backfill. Use Class I, II, or III material for PP pipe bedding, haunching and initial backfill.

3.3 PLACING PIPE

Submit printed copies of the pipe or box culvert manufacturer's recommended pipe or box culvert installation procedures prior to installation. Thoroughly examine each section of pipe or box culvert before being laid; do not use defective or damaged pipe. Protect plastic pipe, excluding SRPE pipe, from exposure to direct sunlight prior to laying, if necessary to maintain adequate pipe stiffness and meet installation deflection requirements. Lay pipelines to the grades and alignment indicated. Provide proper facilities for lowering sections of pipe into trenches. Do not lay pipe in water or when trench conditions or weather are unsuitable for such work. Divert drainage or dewater trenches during construction as necessary. Deflection of installed flexible pipe must not exceed the following limits:

TYPE OF PIPE	MAXIMUM ALLOWABLE DEFLECTION (percent)
Corrugated Steel and Aluminum	5
Ductile Iron Culvert	3
Plastic (PVC, HDPE, SRPE, and PP)	5

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3.3.1 Concrete, Clay, PVC, Ribbed PVC, Ductile Iron Pipe

Lay pipe proceeding upgrade with spigot ends of bell-and-spigot pipe and tongue ends of tongue-and-groove pipe pointing in the direction of the flow.

3.3.2 Structural-Plate Steel

Install structural plate in accordance with ASTM A807/A807M. Assemble structural plate in accordance with instructions furnished by the manufacturer. Instructions must show the position of each plate and the order of assembly. Tighten bolts progressively and uniformly, starting at one end of the structure after all plates are in place. Repeat the operation to ensure that all bolts are tightened to meet the torque requirements of 200 foot-pounds plus or minus 50 foot-pounds. Check power wrenches used by the use of hand torque wrenches or long-handled socket or structural wrenches for amount of torque produced. Check and adjust power wrenches frequently as needed, according to type or condition, to ensure proper adjustment to supply the required torque.

3.3.3 Structural-Plate Aluminum

Assemble structural plate in accordance with instructions furnished by the manufacturer. Instructions must show the position of each plate and the order of assembly. Tighten bolts progressively and uniformly, starting at one end of the structure after all plates are in place. Repeat the operation to ensure that all bolts are torqued to a minimum of 100 foot-pounds on aluminum alloy bolts and a minimum of 150 foot-pounds on galvanized steel bolts. Check power wrenches used by the use of hand torque wrenches or long-handled socket or structural wrenches for the amount of torque produced. Check and adjust power wrenches as frequently as needed, according to type or condition, to ensure that they are in proper adjustment to supply the required torque.

3.3.4 Multiple Culverts

Where multiple lines of pipe are installed, adjacent sides of pipe must be at least half the nominal pipe diameter or 3 feet apart, whichever is less.

3.4 JOINTING

3.4.1 Concrete and Clay Pipe

3.4.1.1 Plastic Sealing Compound Joints for Tongue-and-Grooved Pipe and Box Culverts

Follow the recommendation of the particular manufacturer in regard to sealing compound special installation requirements. When lubricants, primers, or adhesives are used, only apply on surfaces that are dry and clean. Affix sealing compounds to the pipe or box culvert not more than 3 hours prior to installation of the pipe or box culvert. Protect sealing compounds from the sun, blowing dust, and other deleterious agents at all times. Inspect sealing compounds before installation of the pipe or box culvert, and remove and replace any loose or improperly affixed sealing compound. Align the pipe or box culvert with the previously installed pipe or box culvert, and pull the joint together.

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3.4.1.2 Flexible Watertight Joints

Use lubricants, cements, adhesives, and other special installation requirements for gaskets and jointing materials as recommended by the manufacturer. When lubricants, cements, or adhesives are used, only apply on surfaces that are clean and dry. Affix gaskets and jointing materials to the pipe not more than 24 hours prior to the installation of the pipe, and protect from the sun, blowing dust, and other deleterious agents at all times. Inspect gaskets and jointing materials before installing the pipe; remove and replace any loose or improperly affixed gaskets and jointing materials. Align the pipe with the previously installed pipe, and push the joint home. If the gasket becomes visibly dislocated when joining sections of pipe, remove the pipe and remake the joint.

3.5 DRAINAGE STRUCTURES

3.5.1 Manholes and Inlets

Construct manholes of precast reinforced concrete. Construct inlets of precast reinforced concrete. Provide manholes and inlets complete with frames and covers or gratings; and with fixed galvanized steel ladders as indicated.

3.5.2 Walls and Headwalls

Construct walls and headwalls as indicated.

3.6 INSTALLATION OF TRACER WIRE AND WARNING TAPE

Install a continuous length of tracer wire for the full length of each run of nonmetallic pipe in accordance with Section 31 00 00 EARTHWORK. Attach wire to top of pipe in such a manner that it will not be displaced during construction operations.

3.7 FINAL BACKFILL

Backfill trenches with satisfactory material deposited in layers of a maximum of 8 inches loose thickness and compacted to 90 percent of maximum density for cohesive soils and 95 percent of maximum density for cohesionless soils in accordance with Section 31 00 00 EARTHWORK. Testing is the responsibility of the Contractor and will be performed at no additional cost to the Government. Unless otherwise specified, determine field in-place density of final backfill at a frequency of one test per 50 linear feet, or fraction thereof, of each lift of backfill. Submit test results in accordance with Section 31 00 00 EARTHWORK. Do not displace or damage pipe or box when compacting final backfill by rolling or operating heavy equipment parallel with the pipe or box. Movement of construction machinery over a culvert or storm drain at any stage of construction will be at the Contractor's risk. Repair or replace any damaged pipe. Protect concrete pipes with a minimum of 3 feet of cover prior to permitting heavy construction equipment to pass over them during construction. Provide the minimum cover for construction loads over corrugated steel pipes as specified in Section 26, Division II of AASHTO HB-17. Provide minimum cover for construction loads over plastic pipes as specified in ASTM D2321.

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3.8 FIELD QUALITY CONTROL

3.8.1 Tests

Testing is the responsibility of the Contractor. Perform all testing and retesting at no additional cost to the Government.

3.8.1.1 Leakage Test

Test pipe lines for leakage prior to completing backfill by performing either an exfiltration test, low pressure air pipeline test or by individual pipe joint testing. Submit leakage test results to the Contracting Officer.

3.8.1.1.1 Exfiltration Test

Prior to exfiltration tests, backfill the trench up to at least the lower half of the pipe. If required, place sufficient additional backfill to prevent pipe movement during testing, leaving the joints uncovered to permit inspection. When the water table is 2 feet or more above the top of the pipe at the upper end of the pipeline section to be tested, measure infiltration using a suitable weir or other device acceptable to the Contracting Officer. Perform exfiltration test by filling the line to be tested with water so that a head of at least 2 feet is provided above both the water table and the top of the pipe at the upper end of the pipeline to be tested. Allow the filled line to stand until the pipe has reached its maximum absorption, but not less than 4 hours. After absorption, reestablish the head. Measure the amount of water required to maintain this water level during a 2-hour test period. Leakage as measured by the exfiltration test must not exceed 0.2 gallons per inch in diameter per 100 feet of pipeline per hour. Correct visible leaks encountered regardless of leakage test results.

3.8.1.1.2 Low Pressure Air Pipeline Tests

Perform low pressure air testing for plastic pipe in accordance with ASTM F1417. Perform low pressure air testing procedures for other pipe materials using the pressures and testing times prescribed in ASTM C828, after consultation with the pipe manufacturer.

3.8.1.1.3 Individual Pipe Joint Testing

Testing of individual joints for leakage by low pressure air or water must conform to ASTM C1103M ASTM C1103.

3.8.1.2 Deflection Testing

Conduct deflection test no sooner than 30 days after completion of final backfill and compaction testing. Clean or flush all lines prior to testing. Perform a deflection test on entire length of installed flexible pipeline upon completion of work adjacent to and over the pipeline, including backfilling, placement of fill, grading, paving, placement of concrete, and any other superimposed loads. Deflection of pipe in the installed pipeline under external loads must not exceed the limits in paragraph PLACING PIPE AND BOX CULVERT above as percent of the average inside diameter of pipe. Use a mandrel to determine if allowable deflection has been exceeded.

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3.8.1.2.1 Mandrel

Pass the mandrel through each run of pipe by pulling it by hand. If deflection readings in excess of the allowable deflection of average inside diameter of pipe are obtained, stop and begin test from the opposite direction. The mandrel must meet the pipe manufacturer's recommendations and the following requirements. Provide a mandrel that is rigid, nonadjustable, has a minimum of 9 fins, pulling rings at each end, and is engraved with the nominal pipe size and mandrel outside diameter. The mandrel must be 5 percent less than the certified-actual pipe diameter for plastic pipe, 5 percent less than the certified-actual pipe diameter for corrugated steel and aluminum, 3 percent less than the certified-actual pipe diameter for ductile iron culvert pipe. The Government will verify the outside diameter (OD) of the Contractor provided mandrel through the use of Contractor provided proving rings.

3.8.1.3 Tracer Wire Continuity

Test tracer wire for continuity after initial and final backfilling of pipes. Verify that tracer wire is locatable with electronic utility location equipment. Repair breaks or separations and re-test for continuity.

3.8.2 Repair of Defects

3.8.2.1 Leakage Test

When leakage exceeds the maximum amount specified, correct source of excess leakage by replacing damaged pipe and gaskets and retest.

3.8.2.2 Deflection Testing

When deflection readings are in excess of the allowable deflection of average inside diameter of pipe are obtained, remove pipe which has excessive deflection and replace with new pipe. Retest 30 days after completing backfill, leakage testing and compaction testing.

3.8.2.3 Inspection

Replace pipe or repair defects indicated in the Post-Installation Inspection Report.

3.8.2.3.1 Concrete Pipe

Replace pipes having cracks with a width greater than 0.1 inches.

3.8.2.3.2 Flexible Pipe

Replace pipes having cracks or splits.

3.9 PROTECTION

Protect storm drainage piping and adjacent areas from superimposed and external loads during construction.

3.10 WARRANTY PERIOD

Pipe segments found to have defects during the warranty period must be replaced with new pipe and retested.

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PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2014; ERTA 2015) Building Code Requirements for Structural Concrete & Commentary

ACI SP-66 (2004) ACI Detailing Manual

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B3 (2013) Standard Specification for Soft or Annealed Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM C309 (2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete

ASTM C478 (2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections

ASTM C857 (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures

ASTM C990 (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants

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ASTM F2160 (2022a) Standard Specification for Solid Wall High Density Polyethylene (HDPE) Conduit Based on Controlled Outside Diameter (OD)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 48 (2020) Test Procedures and Requirements for Alternating-Current Cable Terminations Used on Shielded Cables Having Laminated Insulation Rated 2.5 kV through 765 kV or Extruded Insulation Rated 2.5 kV through 500 kV

IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System

IEEE 400.2 (2013) Guide for Field Testing of Shielded Power Cable Systems Using Very Low Frequency (VLF)

IEEE C2 (2023) National Electrical Safety Code

IEEE Stds Dictionary (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C119.1 (2016) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts

NEMA C119.4 (2011) Electric Connectors - Connectors for Use Between Aluminum-to-Aluminum or Aluminum-to-Copper Conductors Designed for Normal Operation at or Below 93 Degrees C and Copper-to-Copper Conductors Designed for Normal Operation at or Below 100 Degrees C

NEMA RN 1 (2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit

NEMA TC 2 (2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit

NEMA TC 7 (2021) Smooth-Wall Coilable and Straight Electrical Polyethylene Conduit

NEMA TC 9 (2020) Standard for Fittings for Polyvinyl

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Chloride (PVC) Plastic Utilities Duct for Underground Installation

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2023) National Electrical Code

SOCIETY OF CABLE TELECOMMUNICATIONS ENGINEERS (SCTE)

ANSI/SCTE 77 (2013) Specification for Underground
Enclosure Integrity

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-758 (2012b) Customer-Owned Outside Plant
Telecommunications Infrastructure Standard

U.S. DEPARTMENT OF AGRICULTURE (USDA)

RUS Bull 1751F-644 (2002) Underground Plant Construction

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 1-200-01 (2022) DoD Building Code

UFC 3-301-01 (2023) Structural Engineering

U.S. GENERAL SERVICES ADMINISTRATION (GSA)

CID A-A-60005 (Basic; Notice 2) Frames, Covers,
Gratings, Steps, Sump And Catch Basin,
Manhole

UNDERWRITERS LABORATORIES (UL)

UL 6 (2022) UL Standard for Safety Electrical
Rigid Metal Conduit-Steel

UL 44 (2018; Reprint May 2021) UL Standard for
Safety Thermoset-Insulated Wires and Cables

UL 83 (2017; Reprint Mar 2020) UL Standard for
Safety Thermoplastic-Insulated Wires and
Cables

UL 94 (2023) UL Standard for Safety Tests for
Flammability of Plastic Materials for
Parts in Devices and Appliances

UL 467 (2022) UL Standard for Safety Grounding
and Bonding Equipment

UL 486A-486B (2018; Reprint May 2021) UL Standard for
Safety Wire Connectors

UL 510 (2020; Dec 2022) UL Standard for Safety
Polyvinyl Chloride, Polyethylene and
Rubber Insulating Tape

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UL 514A	(2013; Reprint Jun 2022) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 854	(2020; Reprint Jan 2022) Standard for Service-Entrance Cables
UL 1242	(2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel

1.2 SYSTEM DESCRIPTION

Items provided under this section must be specifically suitable for the following service conditions. Structural details must conform to UFC 3-301-01.

- a. Structural Design Risk Category: II.
- b. Seismic site class: D.
- c. Seismic Design Category: B.
- d. Seismic Design Spectral Response Acceleration Parameter at Short Periods (SDS): 0.071 g.
- e. Seismic Design Spectral Response Acceleration Parameter at 1-sec Periods (SDS): 0.042 g.
- f. Corrosive Areas: Environmental Severity Classification C3, per UFC 1-200-01.

1.3 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.4 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Stds Dictionary.
- b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.
- c. In the text of this section, "medium voltage cable splices," and "medium voltage cable joints" are used interchangeably and have the same meaning.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S"

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classification. Submittals not having a "G" or "S" classification are for information only. When used, a code following the "G" classification identifies the office that will review the submittal for the Government. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast Underground Structures; G, D0

SD-03 Product Data

Live End Caps; G, D0

Precast Concrete Structures; G, D0

Sealing Material

Pulling-In Irons

Manhole Frames and Covers; G, D0

Handhole Frames and Covers; G, D0

Composite/Fiberglass Handholes; G, D0

Cable Supports (racks, arms and insulators); G, D0

SD-06 Test Reports

Field Acceptance Checks and Tests; G, D0

Arc-proofing Test for cable fireproofing tape; G, D0

Cable Installation Plan and Procedure; G, D0

Six copies of the information described below in 8-1/2 by 11 inch binders having a minimum of three rings from which material may readily be removed and replaced, including a separate section for each cable pull. Separate sections by heavy plastic dividers with tabs, with all data sheets signed and dated by the person supervising the pull.

- a. Site layout drawing with cable pulls numerically identified.
- b. A list of equipment used, with calibration certifications. The manufacturer and quantity of lubricant used on pull.
- c. The cable manufacturer and type of cable.
- d. The dates of cable pulls, time of day, and ambient temperature.
- e. The length of cable pull and calculated cable pulling tensions.
- f. The actual cable pulling tensions encountered during pull.

SD-07 Certificates

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Cable Installer Qualifications; G, D0

Directional Boring Certificate of Conformance; G, D0

1.6 QUALITY ASSURANCE

1.6.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

- a. Material description (i.e., f'_c and F_y)
- b. Manufacturer's printed assembly and installation instructions
- c. Design calculations and design loads.
- d. Reinforcing shop drawings in accordance with ACI SP-66
- e. Plans and elevations showing opening and pulling-in iron locations and details

1.6.2 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

1.6.3 Directional Boring Certificate of Conformance

Provide certification of compliance with the registered Professional Engineer's design requirements for each directional bore, including: HDPE conduit size and type, bend radius, elevation changes, vertical and horizontal path deviations, conductor size and type and any conductor derating due to depth of conduit. Record location and depth of all directional-bore installed HDPE conduits using Global Positioning System (GPS) recording means with "resource grade" accuracy.

1.6.4 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and NFPA 70 unless more stringent requirements are specified or indicated.

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1.6.5 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.5.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.6.5.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable, unless specified otherwise.

PART 2 PRODUCTS

2.1 CONDUIT, DUCTS, AND FITTINGS

2.1.1 Rigid Metal Conduit

UL 6.

2.1.1.1 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz, and tensile strength must be minimum 3500 psi.

2.1.2 Intermediate Metal Conduit

UL 1242.

2.1.2.1 Intermediate Metal Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz, and tensile strength must be minimum 3500 psi.

2.1.3 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40 or EPC-80 as indicated.

2.1.4 Plastic Duct for Concrete Encasement

Provide Type EPC-40 per UL 651 and NEMA TC 2.

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2.1.5 High Density Polyethylene (HDPE) Electrical Conduit for Directional Boring

Smoothwall, approved/listed for directional boring, minimum Schedule 80, ASTM F2160, NEMA TC 7.

2.1.6 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 35 degrees F to 95 degrees F. Cured sealant must withstand temperature ranges of -20 degrees F to 200 degrees F without loss of function.

2.1.7 Fittings

2.1.7.1 Metal Fittings

UL 514B.

2.1.7.2 PVC Conduit Fittings

UL 514B, UL 651.

2.1.7.3 PVC Duct Fittings

NEMA TC 9.

2.1.7.4 Outlet Boxes for Steel Conduit

Outlet boxes for use with rigid or flexible steel conduit must be cast-metal cadmium or zinc-coated if of ferrous metal with gasketed closures and must conform to UL 514A.

2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of NFPA 70, including listing requirements. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable. Service entrance conductors must conform to UL 854, type USE.

2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and THHN/THWN unless otherwise noted. Conductors No. 10 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded. All conductors must be copper.

2.2.2 Conductor Material

Unless specified or indicated otherwise or required by NFPA 70, wires in conduit, other than service entrance, must be 600-volt, Type THWN/THHN conforming to UL 83 or Type XHHW or RHW conforming to UL 44. Copper conductors must be annealed copper complying with ASTM B3 and ASTM B8.

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2.2.3 Jackets

Provide multiconductor cables with an overall PVC outer jacket.

2.2.4 Direct Buried

Provide single-conductor cables identified for direct burial.

2.2.5 In Duct

Cables must be single-conductor cable. Cables in factory-installed, coilaible-plastic-duct assemblies must conform to NEMA TC 7.

2.2.6 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Color code conductors. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves, or colored electrical tape. Properly identify control circuit terminations. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals may be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems are as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black
 - (2) Phase B - red
 - (3) Phase C - blue
- b. 480/277 volt, three-phase
 - (1) Phase A - brown
 - (2) Phase B - orange
 - (3) Phase C - yellow
- c. 120/240 volt, single phase: Black and red

2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

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- a. For use with copper conductors: UL 486A-486B.

2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with ANSI C119.1.

2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor, outdoor, direct-burial or submerged locations.

2.5 LIVE END CAPS

Provide live end caps using a "kit" including a heat-shrinkable tube and a high dielectric strength, polymeric plug overlapping the conductor. Conform to applicable portions of IEEE 48.

2.6 TAPE

2.6.1 Insulating Tape

UL 510, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

2.6.2 Buried Warning and Identification Tape

Provide detectable tape in accordance with Section 31 00 00 EARTHWORK.

2.6.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than .030 inch thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

2.7 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of 200 pounds.

2.8 GROUNDING AND BONDING

2.8.1 Driven Ground Rods

Provide solid copper ground rods conforming to UL 467 not less than 3/4

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inch in diameter by 10 feet in length. Sectional type rods may be used for rods 20 feet or longer.

2.8.2 Grounding Conductors

Stranded-bare copper conductors must conform to ASTM B8, Class B, soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to ASTM B1 for sizes No. 8 and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

2.9 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with 3000 psi minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be 4000 psi minimum 28-day compressive strength unless specified otherwise.

2.10 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to ASTM C857 and ASTM C478. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

2.10.1 Cast-In-Place Concrete Structures

Concrete must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE. Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

2.10.2 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures may be provided in lieu of cast-in-place subject to the requirements specified below. Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

2.10.2.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day

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compressive strength of not less than 4000 psi. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

2.10.2.2 Design for Precast Structures

ACI 318M. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

- a. Angle of Internal Friction (ϕ) = 30 degrees
- b. Unit Weight of Soil (Dry) = 110 pcf, (Saturated) = 130 pcf
- c. Coefficient of Lateral Earth Pressure (K_a) = 0.50.
- d. Ground Water Level = 3 feet below ground elevation
- e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact. Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.
- f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 6000 pounds.
- g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.
- h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

2.10.2.3 Construction

Provide a uniform thickness for structure top, bottom, and wall not less than 6 inches. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 12 inches vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 6 inches from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings

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for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 12 inches in diameter and 4 inches deep for precast structures.

2.10.2.4 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

2.10.3 Manhole Frames and Covers

Provide cast iron frames and covers for manholes conforming to CID A-A-60005. Cast the words "ELECTRIC" or "TELECOMMUNICATIONS" in the top face of power and telecommunications manhole covers, respectively.

2.10.4 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Provide rolled-steel floor plate covers having an approved antislip surface. Hinges must be of wrought steel, 5 by 5 inches by approximately 3/16 inch thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have nonremovable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised antislip surface, by grinding or other approved method.

2.10.5 Composite/Fiberglass Handholes and Covers

ANSI/SCTE 77. Provide handholes and covers of polymer concrete, reinforced with heavy weave fiberglass with a design load (Tier rating) appropriate for or greater than the intended use. All covers are required to have the Tier level rating embossed on the surface which must not exceed the design load of the box.

2.11 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

Zinc coat the metal portion of racks and arms after fabrication.

2.11.1 Cable Rack Stanchions

The wall bracket or stanchion must be 4 inches by approximately 1-1/2 inch by 3/16 inch channel steel, or 4 inches by approximately 1 inch glass-reinforced nylon with recessed bolt mounting holes, 48 inches long (minimum) in manholes. Space slots for mounting cable rack arms at 8 inch intervals.

2.11.2 Rack Arms

Cable rack arms must be steel or malleable iron or glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 8 inches and a maximum of 12 inches.

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2.11.3 Insulators

Insulators for metal rack arms must be dry-process glazed porcelain. Insulators are not required for nylon arms.

2.12 CABLE TAGS IN MANHOLES

Provide polyethylene tags for each power cable located in manholes. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identifies the circuit. The next to last position denotes the phase of the circuit and include the Greek "phi" symbol. The last position denotes the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

2.12.1 Polyethylene Cable Tags

Provide tags of polyethylene having an average tensile strength of 3250 pounds per square inch; and that are 0.08 inch thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 170 degrees F. Provide 0.05 inch (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag, having a minimum loop tensile strength of 175 pounds and black block letters, numbers, and symbols one inch high on a yellow background. Letters, numbers, and symbols must not fall off or change positions regardless of the cable tags' orientation.

2.13 LOW VOLTAGE ABOVE GROUND TERMINATION PEDESTAL

Provide copolymer polypropylene, low voltage above ground termination pedestal manufactured through an injection molding process. Pedestals must resist fertilizers, salt air environments and ultra-violet radiation. Pedestal top must be imprinted with a "WARNING" and "ELECTRIC" identification. Pedestal must contain four lay-in six port connectors, NEMA C119.4, Class "A", dual rated for aluminum or copper, and capable of terminating conductors ranging from 10 AWG to 500 kcmil. Protect each connector with a clear, hard lexan (plastic) cover. Provide pedestal with rust-free material and stainless steel hardware that is lockable.

2.14 SOURCE QUALITY CONTROL

2.14.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 12 inches long with a 2 1/2 inch outside diameter, and a 1/8 inch thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. Direct the arc toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Test each sample assembly at three

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unrelated points. Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of NFPA 70 and IEEE C2 as applicable. In addition to these requirements, install telecommunications in accordance with TIA-758 and RUS Bull 1751F-644.

3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

3.3 CABLE INSTALLATION PLAN AND PROCEDURE

Obtain from the manufacturer an installation manual or set of instructions which addresses such aspects as cable construction, insulation type, cable diameter, bending radius, cable temperature limits for installation, lubricants, coefficient of friction, conduit cleaning, storage procedures, moisture seals, testing for and purging moisture, maximum allowable pulling tension, and maximum allowable sidewall bearing pressure. Perform pulling calculations and prepare a pulling plan and submit along with the manufacturer's instructions in accordance with SUBMITTALS. Install cable strictly in accordance with the cable manufacturer's recommendations and the approved installation plan.

Calculations and pulling plan must include:

- a. Site layout drawing with cable pulls identified in numeric order of expected pulling sequence and direction of cable pull.
- b. List of cable installation equipment.
- c. Lubricant manufacturer's application instructions.
- d. Procedure for resealing cable ends to prevent moisture from entering cable.
- e. Cable pulling tension calculations of all cable pulls.
- f. Cable percentage conduit fill.
- g. Cable sidewall bearing pressure.
- h. Cable minimum bend radius and minimum diameter of pulling wheels used.
- i. Cable jam ratio.

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- j. Maximum allowable pulling tension on each different type and size of conductor.
- k. Maximum allowable pulling tension on pulling device.

3.4 UNDERGROUND FEEDERS SUPPLYING BUILDINGS

Terminate underground feeders supplying building at a point 5 feet outside the building and projections thereof, except that conductors must be continuous to the terminating point indicated. Coordinate connections of the feeders to the service entrance equipment with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide PVC, Type EPC-40 conduit from the supply equipment to a point 5 feet outside the building and projections thereof. Protect ends of underground conduit with plastic plugs until connections are made.

Encase the underground portion of the conduit in a concrete envelope and bury as specified for underground duct with concrete encasement.

3.5 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to ASTM C309. Locate duct entrances and windows in the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

3.5.1 Cast-In-Place Concrete Structures

Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

3.5.2 Precast Concrete Construction

Set commercial precast structures on 6 inches of level, 90 percent compacted granular fill, 3/4 inch to 1 inch size, extending 12 inches beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally conform to the manufacturer's instructions.

3.5.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of

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pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor, locate the irons a minimum of 6 inches from the edge of the sump, and in the walls, locate the irons within 6 inches of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 6 inches of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 6 inch clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 3 foot length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 3 inches from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 4 inches and must be designed to withstand a minimum pulling-in load of 6000 pounds. Hot-dip galvanize irons after fabrication.

3.5.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 3 feet apart, and provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 16 1/2 inches apart with the end rack being no further than 12 inches from the adjacent wall. Methods of anchoring cable racks are as follows:

- a. Provide a 5/8 inch diameter by 5 inch long anchor bolt with 3 inch foot cast in structure wall with 2 inch protrusion of threaded portion of bolt into structure. Provide 5/8 inch steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.
- b. Provide concrete channel insert with a minimum load rating of 800 pounds per foot. Insert channel must be steel of the same length as "vertical rack channel;" and cast flush in structure wall. Provide 5/8 inch steel nuts in channel insert to receive 5/8 inch diameter by 3 inch long steel, square head anchor bolts.
- c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 800 pound load rating. Provide 5/8 inch diameter by 3 inch long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

3.5.5 Field Painting

Clean cast-iron frames and covers not buried in concrete or masonry of mortar, rust, grease, dirt and other deleterious materials, and coat with bituminous paint.

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3.6 UNDERGROUND CONDUIT AND DUCT SYSTEMS

3.6.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 4/0 AWG bare copper grounding conductor above medium-voltage distribution duct banks. Bond bare copper grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of 3 inches per 100 feet. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Terminate all PVC conduit end points in utility holes, switching cabinets, transform handholes and buildings with end bells. The bell end of the conduits that enter manholes and handholes must be flush with the wall.

Perform changes in ductbank direction as follows:

- a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.
- b. The minimum manufactured bend radius must be 18 inches for ducts of less than 3 inch diameter, and 36 inches for ducts 3 inches or greater in diameter.
- c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of 25 feet for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections allowed for use in field manufactured longsweep bend: 30 degrees.

3.6.2 Treatment

Keep ducts clean of concrete, dirt, or foreign substances during construction. Make field cuts requiring tapers with proper tools and match factory tapers. Use a coupling recommended by the duct manufacturer whenever an existing duct is connected to a duct of different material or shape. Store ducts to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Thoroughly clean ducts before being laid. Store plastic ducts on a flat surface and protected from the direct rays of the sun.

3.6.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 3 inches and larger, draw a flexible testing mandrel approximately 12 inches long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 3 inches, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

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3.6.4 Jacking and Drilling Under Roads and Structures

Conduits to be installed under existing paved areas which are not to be disturbed, and under roads and railroad tracks, must be zinc-coated, rigid steel, jacked into place. Where ducts are jacked under existing pavement, install rigid steel conduit because of its strength. To protect the corrosion-resistant conduit coating, predrilling or installing conduit inside a larger iron pipe sleeve (jack-and-sleeve) is required. For crossings of existing railroads and airfield pavements greater than 50 feet in length, the predrilling method or the jack-and-sleeve method will be used. Separators or spacing blocks must be made of steel, concrete, plastic, or a combination of these materials placed not farther apart than 4 feet on centers. Hydraulic jet method must not be used.

3.6.5 Galvanized Conduit Concrete Penetrations

Galvanized conduits which penetrate concrete (slabs, pavement, and walls) in wet locations must be PVC coated and extend from at least 2 inches within the concrete to the first coupling or fitting outside the concrete (minimum of 6 inches from penetration).

3.6.6 Multiple Conduits

Separate multiple conduits by a minimum distance of 3 inches, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 12 inches. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 10 feet of conduit assembly.

3.6.7 Conduit Plugs and Pull Rope

Provide new conduit indicated as being unused or empty with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 3 feet of slack at each end of unused or empty conduits.

3.6.8 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 24 inches below finished grade. Provide not less than 3 inches clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 3 inches, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 1/4 inch sieve. The first 6 inch layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 3 to 6 inch layers. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

3.6.8.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 3 inch concrete cover around ducts. Extend concrete encasement at least 5

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feet beyond the edges of paved areas and roads, and 12 feet beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 24 inches below finished grade.

3.6.9 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 18 inches below finished grade, except under roads and pavement, concrete envelope must be not less than 24 inches below finished grade. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and provide at least 3 inches of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 3 inches. Before pouring concrete, anchor duct bank assemblies, prevent floating during concrete pouring by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly. Provide steel reinforcing in the concrete envelope as indicated. Provide color, type and depth of warning tape as specified in Section 31 00 00 EARTHWORK.

3.6.9.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 12 inches in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

3.6.9.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and extend into the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

3.6.9.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and extend into the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

3.6.9.4 Connections to Existing Ducts

Where connections to existing duct banks are indicated, excavate the banks to the maximum depth necessary. Cut off the banks and remove loose concrete from the conduits before new concrete-encased ducts are installed. Provide a reinforced concrete collar, poured monolithically with the new duct bank, to take the shear at the joint of the duct banks. Abandon in place those no longer used ducts and cables which do not interfere with the work.

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3.6.9.5 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs. Fit concrete envelope of a partially completed duct bank with reinforcing steel extending a minimum of 2 feet back into the envelope and a minimum of 2 feet beyond the end of the envelope. Provide one No. 4 bar in each corner, 3 inches from the edge of the envelope. Secure corner bars with two No. 3 ties, spaced approximately one foot apart. Restrain reinforcing assembly from moving during concrete pouring.

3.6.10 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

3.7 CABLE PULLING

Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches, transformers, switchgear, switchboards, and other enclosures. Cable with tape or wire shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

3.7.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

3.8 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations. Provide one spare three-insulator rack arm for each cable rack in each underground structure.

3.8.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

3.9 CONDUCTORS INSTALLED IN PARALLEL

Group conductors such that each conduit of a parallel run contains one

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Phase A conductor, one Phase B conductor, one Phase C conductor, and one neutral conductor.

3.10 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set. Make splices in underground distribution systems only in accessible locations such as manholes, handholes, or aboveground termination pedestals.

3.11 MEDIUM VOLTAGE CABLE TERMINATIONS

Make terminations in accordance with the written instruction of the termination kit manufacturer.

3.12 MEDIUM VOLTAGE CABLE JOINTS

Provide power cable joints (splices) suitable for continuous immersion in water. Make joints only in accessible locations in manholes or handholes by using materials and methods in accordance with the written instructions of the joint kit manufacturer.

3.12.1 Joints in Shielded Cables

Cover the joined area with metallic tape, or material like the original cable shield and connect it to the cable shield on each side of the splice. Provide a bare copper ground connection brought out in a watertight manner and grounded to the manhole grounding loop as part of the splice installation. Ground conductors, connections, and rods must be as specified elsewhere in this section. Wire must be trained to the sides of the enclosure to prevent interference with the working area.

3.13 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

3.14 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

3.14.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

3.15 GROUNDING SYSTEMS

NFPA 70 and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 25 ohms.

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3.15.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus 12 inches, installed to provide an earth ground of the appropriate value for the particular equipment being grounded.

If the specified ground resistance is not met, provide an additional ground rod in accordance with the requirements of NFPA 70 (placed not less than 6 feet from the first rod). Should the resultant (combined) resistance exceed the specified resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

3.15.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has been adequately compressed on the ground wire.

3.15.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with No. 6 AWG. Ground other noncurrent-carrying metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of No. 6 AWG. Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.

3.15.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

3.15.5 Manhole Grounding

Loop a 4/0 AWG grounding conductor around the interior perimeter, approximately 12 inches above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 36 inches. Connect the conductor to the manhole grounding electrode with 4/0 AWG conductor. Connect all incoming 4/0 grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 6 AWG conductor.

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3.15.6 Fence Grounding

Provide grounding for fences as indicated. Drive ground rods until the top is 12 inches below grade. Attach a No. 4 AWG copper conductor, by exothermic weld to the ground rods and extend underground to the immediate vicinity of fence post. Lace the conductor vertically into 12 inches of fence mesh and fasten by two approved bronze compression fittings, one to bond wire to post and the other to bond wire to fence. Bond each gate section to its gatepost by a 1/8 by one inch flexible braided copper strap and ground post clamps. Clamps must be of the anti-electrolysis type.

3.16 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with NFPA 70 and Section 31 00 00 EARTHWORK.

3.16.1 Reconditioning of Surfaces

3.16.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct or direct burial cable. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching.

3.16.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists, restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified, and to match and tie into the adjacent and surrounding existing surfaces.

3.17 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.17.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 8 inches thick, reinforced with a 6 by 6 - W2.9 by W2.9 mesh, placed uniformly 4 inches from the top of the slab. Place slab on a 6 inch thick, well-compacted gravel base. Top of concrete slab must be approximately 4 inches above finished grade with gradual slope for drainage. Edges above grade must have 1/2 inch chamfer. Slab must be of adequate size to project at least 8 inches beyond the equipment.

Stub up conduits, with bushings, 2 inches into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

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3.18 FIELD QUALITY CONTROL

3.18.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.18.1.1 Medium Voltage Cables

Perform tests after installation of cable, splices, and terminators and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Inspect for proper shield grounding, cable support, and cable termination.
- (4) Verify that cable bends are not less than ICEA or manufacturer's minimum allowable bending radius.
- (5) Inspect for proper fireproofing.
- (6) Visually inspect jacket and insulation condition.
- (7) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform a shield continuity test on each power cable by ohmmeter method. Record ohmic value, resistance values in excess of 10 ohms per 1000 feet of cable must be investigated and justified.
- (2) Perform acceptance test on new cables before the new cables are connected to existing cables and placed into service, including terminations and joints. Perform maintenance test on complete cable system after the new cables are connected to existing cables and placed into service, including existing cable, terminations, and joints. Tests must be very low frequency (VLF) alternating voltage withstand tests in accordance with IEEE 400.2. VLF test frequency must be 0.05 Hz minimum for a duration of 60 minutes using a sinusoidal waveform. Test voltages must be as follows:

CABLE RATING AC TEST VOLTAGE for ACCEPTANCE TESTING	
5 kV	10kV rms(peak)
8 kV	13kV rms(peak)

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CABLE RATING AC TEST VOLTAGE for ACCEPTANCE TESTING	
15 kV	20kV rms(peak)
25 kV	31kV rms(peak)
35 kV	44kV rms(peak)

CABLE RATING AC TEST VOLTAGE for MAINTENANCE TESTING	
5 kV	7kV rms(peak)
8 kV	10kV rms(peak)
15 kV	16kV rms(peak)
25 kV	23kV rms(peak)
35 kV	33kV rms(peak)

3.18.1.2 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Verify tightness of accessible bolted electrical connections.
- (4) Inspect compression-applied connectors for correct cable match and indentation.
- (5) Visually inspect jacket and insulation condition.
- (6) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.
- (2) Perform continuity tests to insure correct cable connection.

3.18.1.3 Grounding System

a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and

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specifications.

b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.18.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

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1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2014; ERTA 2015) Building Code Requirements for Structural Concrete & Commentary

ARMY RESERVE NETWORK ENTERPRISE CENTER (ARNEC)

ARNEC Army Reserve Network Enterprise Center Infrastructure Technical Criteria

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM C478 (2018) Standard Specification for Circular Precast Reinforced Concrete Manhole Sections

ASTM C857 (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures

ASTM C990 (2009; R 2014) Standard Specification for Joints for Concrete Pipe, Manholes and Precast Box Sections Using Preformed Flexible Joint Sealants

ASTM D709 (2017) Standard Specification for Laminated Thermosetting Materials

ASTM D1557 (2012; E 2015) Standard Test Methods for

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Laboratory Compaction Characteristics of
Soil Using Modified Effort (56,000
ft-lbf/ft³) (2700 kN-m/m³)

ASTM F512

(2019) Standard Specification for
Smooth-Wall Poly (Vinyl Chloride) (PVC)
Conduit and Fittings for Underground
Installation

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 100

(2000; Archived) The Authoritative
Dictionary of IEEE Standards Terms

IEEE C2

(2017; Errata 1-2 2017; INT 1 2017)
National Electrical Safety Code

INSULATED CABLE ENGINEERS ASSOCIATION (ICEA)

ICEA S-87-640

(2016) Optical Fiber Outside Plant
Communications Cable; 4th Edition

ICEA S-98-688

(2012) Broadband Twisted Pair
Telecommunication Cable, Aircore,
Polyolefin Insulated, Copper Conductors
Technical Requirements

ICEA S-99-689

(2012) Broadband Twisted Pair
Telecommunication Cable Filled, Polyolefin
Insulated, Copper Conductors Technical
Requirements

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA RN 1

(2005; R 2013) Polyvinyl-Chloride (PVC)
Externally Coated Galvanized Rigid Steel
Conduit and Intermediate Metal Conduit

NEMA TC 2

(2013) Standard for Electrical Polyvinyl
Chloride (PVC) Conduit

NEMA TC 6 & 8

(2013) Standard for Polyvinyl Chloride
(PVC) Plastic Utilities Duct for
Underground Installations

NEMA TC 9

(2004) Standard for Fittings for Polyvinyl
Chloride (PVC) Plastic Utilities Duct for
Underground Installation

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70

(ERTA 1-2 2017; TIA 17-1; TIA 17-2; TIA
17-3; TIA 17-4; TIA 17-5; TIA 17-6; TIA
17-7; TIA 17-8; TIA 17-9; TIA 17-10; TIA
17-11; TIA 17-12; TIA 17-13; TIA 17-14;
TIA 17-15; TIA 17-16; TIA 17-17) National
Electrical Code

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TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-455-46A	(1990) FOTP-46 Spectral Attenuation Measurement for Long-Length, Graded-Index Optical Fibers
TIA-455-78-B	(2002) FOTP-78 Optical Fibres - Part 1-40: Measurement Methods and Test Procedures - Attenuation
TIA-455-107	(1999a) FOTP-107 Determination of Component Reflectance or Link/System Return Loss using a Loss Test Set
TIA-492CAAA	(1998; R 2002) Detail Specification for Class IVa Dispersion-Unshifted Single-Mode Optical Fibers
TIA-526-7	(2015a) OFSTP-7 Measurement of Optical Power Loss of Installed Single-Mode Fiber Cable Plant
TIA-526-14	(2015c) OFSTP-14A Optical Power Loss Measurements of Installed Multimode Fiber Cable Plant
TIA-568.0	Generic Telecommunications Cabling for Customer Premises
TIA-568.1	Commercial Building Telecommunications Cabling Standard
TIA-568.2	Balanced Twisted-Pair Telecommunications Cabling and Components Standards
TIA-568.3	Optical Fiber Cabling Components Standard
TIA-568.4	Broadband Coaxial Cabling and Components Standard
TIA-569	Commercial Building Standard for Telecommunications Pathways and Spaces
TIA-590	(1997a) Standard for Physical Location and Protection of Below Ground Fiber Optic Cable Plant
TIA-598	Optical Fiber Cable Color Coding
TIA-606	Administration Standard for the Telecommunications Infrastructure
TIA-607	(2015c; Addendum 1 2017) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
TIA-758	Customer-Owned Outside Plant Telecommunications Infrastructure Standard

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TIA/EIA-455 (1998b) Standard Test Procedure for Fiber Optic Fibers, Cables, Transducers, Sensors, Connecting and Terminating Devices, and Other Fiber Optic Components

U.S. ARMY

I3A Technical Criteria for the Installation Information Infrastructure Architecture

U.S. DEPARTMENT OF AGRICULTURE (USDA)

RUS 1755 Telecommunications Standards and Specifications for Materials, Equipment and Construction

RUS Bull 345-50 (1979) Trunk Carrier Systems (PE-60)

RUS Bull 345-65 (1985) Shield Bonding Connectors (PE-65)

RUS Bull 1751F-630 (1996) Design of Aerial Plant

RUS Bull 1751F-640 (1995) Design of Buried Plant, Physical Considerations

RUS Bull 1751F-643 (2002) Underground Plant Design

RUS Bull 1751F-644 (2002) Underground Plant Construction

RUS Bull 1751F-815 (1979) Electrical Protection of Outside Plant

RUS Bull 1753F-201 (1997) Acceptance Tests of Telecommunications Plant (PC-4)

RUS Bull 1753F-401 (1995) Splicing Copper and Fiber Optic Cables (PC-2)

UNDERWRITERS LABORATORIES (UL)

UL 6 (2007; Reprint Sep 2019) UL Standard for Safety Electrical Rigid Metal Conduit-Steel

UL 83 (2017) UL Standard for Safety Thermoplastic-Insulated Wires and Cables

UL 94 (2013; Reprint Sep 2017) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances

UL 351 UL Standard for Safety Rosettes

UL 467 (2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment

UL 497 (2001; Reprint Jul 2013) Protectors for Paired Conductor Communication Circuits

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UL 510	(2017) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514B	(2012; Reprint Nov 2014) Conduit, Tubing and Cable Fittings
UL 651	(2011; Reprint Nov 2018) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 1242	(2006; Reprint Mar 2014) Standard for Electrical Intermediate Metal Conduit -- Steel

1.2 GENERAL DESIGN REQUIREMENTS

Telecommunication products, product quality, and product execution must be in accordance with ARNEC and I3A. If a difference arises between these specifications or the drawings, or both, then adherence to these criteria documents will take precedence for such products.

1.3 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in this specification are as defined in TIA-568.0, TIA-568.1, TIA-568.2, TIA-568.3, TIA-569, TIA-606, and IEEE 100 and as specified.

1.3.1 Entrance Facility (EF)

An environmentally controlled telecommunications space that serves as an entrance to the building for both private and public network service cables (including wireless) including the entrance point at the building wall and continuing to the equipment room.

1.3.2 Telecommunications Equipment Room (TER) (Telecommunications)

An environmentally controlled centralized space for telecommunications equipment that serves the occupants of a building. The space will also house all backbone cable terminations that serve all other telecommunications spaces on an Army Reserve site. Equipment housed therein is considered distinct from a telecommunications room because of the nature of its complexity. The room is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.3.3 Telecommunications Room (TR)

An environmentally controlled space for housing telecommunications equipment, cable, terminations, and cross-connects. The room is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.3.4 Main Telecommunications Room (MTR)

An environmentally controlled space for housing telecommunications equipment, cable, terminations, and cross-connects. This room will be utilized when a new building or buildings with more than one TR within the new building(s) is being constructed on an existing site where the site already has an existing TER in an existing building. This room will serve

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as space where all backbone cable from other TRs within the new building will terminate. This will also be the room where new outside plant conduit will terminate. The room is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.3.5 Telecommunications Enclosure(TE)

An environmentally controlled space for housing telecommunications equipment, cable, terminations, and cross-connects. This is a non-dedicated telecommunications space. This enclosure will only be utilized when there is no opportunity to build out a dedicated telecommunications room. A lockable floor or wall mounted telecommunications cabinet will be required in the room and will house all of the telecommunications terminations and network equipment. The enclosure is the recognized cross-connect between the backbone cable and the horizontal cabling for the area served.

1.3.6 Pathway

A physical infrastructure used for the placement and routing of telecommunications cable.

1.4 SYSTEM DESCRIPTION

The telecommunications outside plant to consist of cable, conduit, manholes, poles required to provide signal paths from the closest point of presence to the facility, including free standing frames or backboards, interconnecting hardware, terminating cables, lightning, and surge protection modules at the entrance facility. The work to consist of providing, testing and making operational cabling, interconnecting hardware and lightning, and surge protection necessary to form a complete outside plant telecommunications system for continuous use.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for information only. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Telecommunications Shop Drawings; G-D0

SD-03 Product Data

Telecommunications Entrance Facility; G-D0

Closures; G-D0

Cable Splices and Connectors; G-D0

Wire and Cable; G-D0

Inner-duct; G-D0

Maintenance Holes; G-D0

Hand Holes; G-D0

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Building Protector Assemblies; G-D0

Protector Modules; G-D0

SD-06 Test Reports

Outside Plant Test Plan; G-D0

Pre-Installation Tests; G-D0

Acceptance Tests; G-D0

SD-07 Certificates

Telecommunications Contractor Qualifications; G-D0

SD-09 Manufacturer's Field Reports

Factory Reel Test Data; G-D0

SD-11 Closeout Submittals

Record Documentation; G-D0

1.6 QUALITY ASSURANCE

1.6.1 Telecommunications Shop Drawings

Include wiring diagrams and installation details of equipment indicating proposed location, layout and arrangement, control panels, accessories, piping, ductwork, and other items that are to be shown to ensure a coordinated installation. Wiring diagrams to identify circuit terminals and indicate the internal wiring for each item of equipment and the interconnection between each item of equipment. Drawings to indicate adequate clearance for operation, maintenance, and replacement of operating equipment devices. Submittals to include the nameplate data, size, capacity, applicable federal, military, industry, and technical society publication references.

1.6.1.1 Telecommunications Outside Plant

Submit Outside Plant Design in accordance with ARNEC, I3A, TIA-758, and RUS Bull 1751F-630 for aerial system design, and RUS Bull 1751F-643 for underground system design. Provide T0 shop drawings that show the physical and logical connections from the perspective of an entire campus, such as actual building locations, exterior pathways and campus backbone cabling on plan view drawings, major system nodes, and related connections on the logical system drawings in accordance with TIA-606. Drawings to include wiring and schematic diagrams for fiber optic and copper cabling and splices, copper conductor gauge and pair count, fiber pair count and type, pathway duct and inner-duct arrangement, associated construction materials, and any details required to demonstrate that cable system has been coordinated and will properly support the switching and transmission system identified as specified and indicated. Provide Registered Communications Distribution Designer (RCDD) approved drawings of the telecommunications outside plant. Update existing telecommunication Outside Plant T0 drawings to include information modified, deleted or added as a result of this installation in accordance with TIA-606. The telecommunications outside plant shop drawings to be included in the

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as-built drawings.

1.6.1.2 Telecommunications Entrance Facility

Submit T3 drawings for EF Telecommunications as specified in Section 27 10 00.00 48 BUILDING TELECOMMUNICATIONS CABLING SYSTEMS, paragraph "Telecommunications Space." The telecommunications entrance facility shop drawings to be included in the as-built drawings.

1.6.2 Telecommunications Contractor Qualifications

Work to be performed by and the equipment provided by the approved telecommunications contractor. A minimum of 30 days prior to installation submit qualifications for: the telecommunications contractor, the telecommunications installer, the supervisor (if different from the installer), and the cable splicing and terminating personnel. Submit documentation of the experience of the telecommunications contractor and the key personnel.

1.6.2.1 Telecommunications Contractor

The telecommunications contractor to be from a firm which is regularly and professionally engaged in the business of the applications, installation, and testing of the specified telecommunications systems and equipment. The telecommunications contractor to demonstrate experience in providing successful telecommunications systems that include outside plant and broadband cabling within the past 3 years. Submit documentation for a minimum of 3 and a maximum of 5 successful telecommunication system installations for the telecommunications contractor. Each of the key personnel to demonstrate experience in providing successful telecommunications systems in accordance with TIA-758 within the past 3 years.

1.6.2.2 Key Personnel Qualifications

All key personnel must be selected in accordance with the ARNEC requirements. Provide key personnel who are regularly and professionally engaged in the business of the application, installation, and testing of the specified telecommunications systems and equipment. There may be one key person or more key persons proposed depending upon how many of the key roles each has successfully provided. Each of the key personnel to demonstrate experience in providing successful telecommunications systems within the past 3 years.

Cable splicing and terminating personnel assigned to the installation of this system or any of its components to have training in the proper techniques and have a minimum of 3 years experience in splicing and terminating the specified cables. Perform modular splices by factory certified personnel or under direct supervision of factory trained personnel for products used.

Supervisors and installers assigned to the installation of this system or any of its components to have factory or factory approved certification from each equipment manufacturer indicating that they are qualified to install and test the provided products.

Submit documentation for a minimum of 3 and a maximum of 5 successful telecommunication system installations for each of the key personnel. Documentation for each key person to include at least two successful

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system installations provided that are equivalent in system size and in construction complexity to the telecommunications system proposed. Include specific experience in installing and testing telecommunications outside plant systems, including broadband cabling, and provide the names and locations of at least 2 project installations successfully completed using optical fiber and copper telecommunications cabling systems. All of the existing telecommunications system installations offered by the key persons as successful experience to have been in successful full-time service for at least 18 months prior to the issuance date for this solicitation.

Provide the name and role of the key person, the title, location, and completed installation date of the referenced project, the referenced project owner point of contact information including name, organization, title, and telephone number, and generally, the referenced project description including system size and construction complexity.

Indicate that all key persons are currently employed by the telecommunications contractor, or have a commitment to the telecommunications contractor to work. All key persons to be employed by the telecommunications contractor at the date of issuance of this solicitation, or if not, have a commitment to the telecommunications contractor to work by the date of Bid Opening.

Note that only the key personnel approved by the Contracting Officer in the successful proposal will do the work for the telecommunications system. Key personnel to function in the same roles, as they functioned in the offered successful experience. Any substitutions for the telecommunications contractor's key personnel requires approval from the Contracting Officer.

1.6.2.3 Minimum Manufacturer's Qualifications

Cabling, equipment, and hardware manufacturers to have a minimum of 3 years of experience in the manufacturing, assembly, and factory testing of components in accordance with TIA-568.0, TIA-568.1, TIA-568.2, and TIA-568.3. In addition, cabling manufacturers to have a minimum of 3 years of experience in the manufacturing and factory testing of cabling in accordance with ICEA S-87-640, ICEA S-98-688, and ICEA S-99-689.

1.6.3 Outside Plant Test Plan

Submit a complete and detailed test plan for field tests of the outside plant including a complete list of test equipment for the copper conductor and optical fiber cables, components, and accessories at least 30 days prior to tests. Provide outside plant testing and performance measurement criteria in accordance with the ARNEC, I3A, TIA-568.1 and RUS Bull 1753F-201. Include procedures for certification, validation, and testing that includes fiber optic link performance criteria, and equipment calibration data and type.

1.6.4 Standard Products

Provide materials and equipment that are standard products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and the manufacturer's latest standard design that has been in satisfactory commercial or industrial use for at least 1 year prior to bid opening. The 1-year period to include applications of equipment and materials under similar circumstances and of similar size. The product to have been on sale on the

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commercial market through advertisements, manufacturers' catalogs, or brochures during the 1-year period. Products supplied to be specifically designed and manufactured for use with outside plant telecommunications systems. Where 2 or more items of the same class of equipment are required, these items will be products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless otherwise stated.

1.6.4.1 Alternative Qualifications

Products having less than a 1-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 3000 hours, exclusive of the manufacturers' factory or laboratory tests, is provided.

1.6.4.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site will not be used, unless specified otherwise.

1.6.5 Regulatory Requirements

Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated.

1.6.5.1 Independent Testing Organization Certificate

In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. The certificate to state that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.7 DELIVERY, STORAGE, AND HANDLING

Ship cable on reels in 500 or 1000 feet length with a minimum overage of 10 percent. Radius of the reel drum not to be smaller than the minimum bend radius of the cable. Wind cable on the reel so that unwinding can be completed without kinking the cable. Two meters of cable at both ends of the cable to be accessible for testing. Attach permanent label on each reel showing length, cable identification number, cable size, cable type, and date of manufacture. Provide water resistant label and the indelible writing on the labels. Apply end seals to each end of the cables to prevent moisture from entering the cable. Reels with cable to be suitable for outside storage conditions when temperature ranges from minus 40 degrees C to plus 65 degrees C, with relative humidity from 0 to 100 percent. Equipment, other than cable, delivered and placed in storage to be stored with protection from weather, humidity, and temperature variation, dirt and dust, or other contaminants in accordance with manufacturer's requirements.

1.8 RECORD DOCUMENTATION

Provide the activity responsible for telecommunications system maintenance and administration a single complete and accurate set of record

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documentation for the entire telecommunications system.

Provide T5 drawings including documentation on cables and termination hardware in accordance with TIA-606. T5 drawings to include schedules to show information for cut-overs and cable plant management, patch panel layouts, cross-connect information and connecting terminal layout as a minimum. Provide T5 drawings in electric format in .pdf and CAD. Provide the following T5 drawing documentation as a minimum:

- a. Cables - Provide a record of installed cable in accordance with TIA-606. The cable records to include the required data fields for each cable and complete end-to-end circuit report for each complete circuit from the assigned outlet to the entry facility in accordance with ANSI/TIA-606. Include manufacture date of cable with submittal.
- b. Termination Hardware - Provide a record of installed patch panels, cross-connect points, campus distributor and terminating block arrangements and type in accordance with TIA-606. Documentation to include the required data fields as a minimum in accordance with TIA-606.

Provide record documentation as specified in Section 27 10 00.00 48
BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

1.9 WARRANTY

The equipment items to be supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the Contract.

PART 2 PRODUCTS

2.1 TELECOMMUNICATIONS OSP TERMINATIONS

2.1.1 Building Protector Assemblies

Submit self-contained 5 pin unit supplied with a field cable stub factory connected to protector socket blocks to terminate and accept protector modules for all pairs of outside cable. Building protector assembly shall have interconnecting hardware for connection to interior cabling at full capacity. Submit manufacturer's instructions for building protector assembly installation. Provide copper cable interconnecting hardware as specified in Section 27 10 00.00 48 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

2.1.2 Protector Modules

Submit in accordance with UL 497 solid state type 5 pin rated for the application. Submit the number of surge protection modules equal to the number of pairs of exterior cable of the building protector assembly.

2.1.3 Fiber Optic Terminations

Provide fiber optic cable terminations as specified in section
27 10 00.00 48 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

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2.2 CLOSURES

2.2.1 Copper Conductor Closures

2.2.1.1 Aerial Cable Closures

Provide cable closure assembly consisting of a frame with clamps, a lift-off polyethylene cover, cable nozzles, and drop wire rings. Closure to be suitable for use on Figure 8 cables. Closures to be free breathing and suitable for housing of the type indicated splices of non-pressurized communications cables and sized as indicated. Construct the closure with ultraviolet resistant PVC.

2.2.2 Fiber Optic Closures

2.2.2.1 Aerial

Provide an aerial closure that is free breathing and suitable for housing splice organizer of non -pressurized cables. Construct the closure with ultraviolet resistant PVC.

2.2.2.2 In Vault or Manhole

Provide underground closure suitable to house splice organizer in a protective housing into which can be poured an encapsulating compound. Closure to be of thermoplastic, thermoset, or stainless steel material supplying structural strength necessary to pass the mechanical and electrical requirements in a vault or manhole environment. Encapsulating compound to be re-enterable and not alter the chemical stability of the closure.

2.3 CABLE SPLICES AND CONNECTORS

2.3.1 Fiber Optic Cable Splices

In addition to the requirements below, all fiber optic cable splices shall be in accordance with I3A, and TIA-758. Provide fiber optic cable splices and splicing materials for fusion methods at locations shown. The splice insertion loss to be 0.3 dB maximum when measured in accordance with TIA-455-78-B using an Optical Time Domain Reflectometer (OTDR). Design splices for a return loss of 40.0 db max for single mode fiber when tested in accordance with TIA-455-107. Physically protect each fiber optic splice by a splice kit specially designed for the splice.

2.3.2 Fiber Optic Splice Organizer

In addition to the requirements below, all fiber optic cable splices shall be in accordance with I3A, and TIA-758. Provide splice organizer suitable for housing fiber optic splices in a neat and orderly fashion. Splice organizer to allow for a minimum of 3 feet of fiber for each fiber within the cable to be neatly stored without kinks or twists. Splice organizer to accommodate individual strain relief for each splice and allow for future maintenance or modification, without damage to the cable or splices. Provide splice organizer hardware, such as splice trays, protective glass shelves, and shield bond connectors in a splice organizer kit.

2.3.3 Shield Connectors

Provide connectors with a stable, low-impedance electrical connection

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between the cable shield and the bonding conductor in accordance with
RUS Bull 345-65.

2.4 CONDUIT, DUCTS, AND FITTINGS

2.4.1 Rigid Metal Conduit

UL 6

2.4.2 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A
durometer, dielectric strength must be minimum 400 volts per mil at 60 Hz,
and tensile strength must be minimum 3500 psi.

2.4.3 Intermediate Metal Conduit, PVC Coated

UL 1242

2.4.4 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40 or EPC-80 as indicated.

2.4.5 Plastic Conduit for Concrete Encasement

Provide Type EB-20 per UL 351, ASTM F512, and NEMA TC 6 & 8.

2.4.6 Inner-duct

Provide corrugated or solid wall polyethylene (PE) or PVC inner-ducts, or
fabric-mesh inner-ducts, with pull wire. Size as indicated.

2.4.7 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that
expands and hardens to form a closed, chemically and water resistant,
rigid structure. Sealant must be compatible with common cable and wire
jackets and capable of adhering to metals, plastics and concrete. Sealant
must be of curing in temperature ranges of 35 degrees F to 95 degrees F.
Cured sealant must withstand temperature ranges of -20 degrees F to
200 degrees without loss of function.

2.4.8 Metal Conduit Fittings

UL 514B

2.4.9 PVC Conduit Fittings

UL 514B, UL 651.

2.4.10 PVC Duct Fittings

NEMA TC 9.

2.5 PLASTIC INSULATING TAPE

Provide in accordance with UL 510.

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2.6 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of 200 pounds.

2.7 WIRE AND CABLE

2.7.1 Copper Conductor Cable

Provide solid copper conductors, covered with an extruded solid insulating compound. Twist insulated conductors into pairs which are then stranded or oscillated to form a cylindrical core. For special high frequency applications, separate the cable core into compartments. Complete cable by the application of a suitable core wrapping material, a corrugated copper or plastic coated aluminum shield, and an overall extruded jacket. Telecommunications contractor to verify distances between splice points prior to ordering cable in specific cut lengths. Gauge of conductor to determine the range of numbers of pairs specified; 19 gauge (6 to 400 pairs), 22 gauge (6 to 1200 pairs), 24 gauge (6 to 2100 pairs), and 26 gauge (6 to 3000 pairs).

2.7.1.1 Underground

Provide filled cable in accordance with ICEA S-99-689 and RUS 1755.

2.7.1.2 Screen

Provide screen-compartmental core filled cable in accordance with ICEA S-99-689 and RUS 1755.

2.7.1.3 Fiber Optic Cable

Provide single-mode fiber optic cable in accordance with TIA-492CAAA. Multimode fiber provided to match existing multimode fiber. Provide optical fibers as indicated. Design fiber optic cable with loose tube construction specifically for outside use. Provide fiber optic color code in accordance with TIA-598.

2.7.1.4 Strength Members

Provide central, non-metallic strength members with sufficient tensile strength for installation and residual rated loads in accordance with ICEA S-87-640. The strength member is included to serve as a cable core foundation to reduce strain on the fibers, and not to serve as a pulling strength member.

2.7.1.5 Performance Requirements

Provide fiber optic cable with optical and mechanical performance requirements in accordance with ICEA S-87-640.

2.7.2 Grounding and Bonding Conductors

Provide grounding and bonding conductors in accordance with RUS 1755, TIA-607, ARNEC, I3A, IEEE C2, and NFPA 70. Solid bare copper wire meeting the requirements of ASTM B1 for sizes No. 8 AWG and smaller and stranded bare copper wire in accordance with ASTM B8, for sizes No. 6 AWG and larger. Insulated conductors to have 600-volt, Type TW insulation in accordance with UL 83.

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2.8 MAINTENANCE HOLES and HAND HOLES

Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes. Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to ASTM C857 and ASTM C478. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Concrete for precast work must have a 28-day compressive strength of not less than 4000 psi. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable. Structures must be identified with the manufactures name embedded in or otherwise permanently attached to an interior wall face.

2.8.1 Design for Precast Structures

ACI 318M. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

- a. Angle of Internal Friction (ϕ) = 30 degrees
- b. Unit Weight of Soil (Dry) = 110 pcf, (Saturated) = 130 pcf
- c. Coefficient of Lateral Earth Pressure (K_a) = 0.33
- d. Ground Water Level = 3 feet below ground elevation
- e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact. Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.
- f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 6000 pounds.
- g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.
- h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

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2.8.2 Construction

Structure top, bottom, and wall must be of a uniform thickness of not less than 6 inches. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the pre-caster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 12 inches vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 6 inches from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 12 inches in diameter and 4 inches deep for precast structures.

2.8.3 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

2.8.4 Handhole Frames and Covers

Frames and covers of steel must be welded by qualified welders in accordance with standard commercial practice. Steel covers must be rolled-steel floor plate having an approved anti-slip surface. Hinges must be of stainless steel with bronze hinge pin, 5 by 5 inches by approximately 3/16 inch thick, without screw holes, and must be for full surface application by fillet welding. Hinges must have non-removable pins and five knuckles. The surfaces of plates under hinges must be true after the removal of raised anti-slip surface, by grinding or other approved method.

2.8.5 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

The metal portion of the racks and arms must be zinc-coated after fabrication.

2.8.5.1 Cable Rack Stanchions

The wall bracket or stanchion must be 4 inches by approximately 1-1/2 inch by 3/16 inch channel steel. Cable racks must contain a minimum of 47 hook spaces. All racks and hooks shall be of galvanized metal.

2.8.5.2 Rack Arms

Cable rack arms must be galvanized steel. Rack arm length must be a minimum of 8 inches and a maximum of 12 inches.

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2.8.6 Driven Ground Rods

Provide copper-clad steel ground rods conforming to UL 467 not less than 3/4 inch in diameter by 10 feet in length. Sectional type rods may be used for rods 20 feet or longer.

2.8.7 Cable Tags In Maintenance Holes, Handholes, And Vaults

Provide polyethylene tags for each telecommunications cable or wire located in maintenance holes, hand holes, and vaults and labeled in accordance with ARNEC and TIA-606. Handwritten labeling is prohibited.

2.8.7.1 POLYETHYLENE CABLE TAGS

Provide polyethylene tags that have an average tensile strength of 3250 pounds per square inch; and that are 0.08 inch thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 170 degrees F. Provide 0.05 inch (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag. Provide minimum loop tensile strength ties of 175 pounds. The cable tags to have black block letters, numbers, and symbols one inch high on a yellow background. Letters, numbers, and symbols shall not fall off or change positions regardless of the cable tags' orientation.

2.9 T-SPAN LINE TREATMENT REPEATERS

Provide pedestal mounted repeaters with non-pressurized housings, sized as indicated, and in accordance with RUS Bull 345-50.

2.10 BURIED WARNING AND IDENTIFICATION TAPE

Provide fiber optic media marking and protection in accordance with TIA-590. Provide color, type, and depth of tape in accordance with I3A.

2.11 GROUNDING BRAID

Provide grounding braid that provides low electrical impedance connections for dependable shield bonding in accordance with RUS 1755. Make braid from flat tin-plated copper.

2.12 MANUFACTURER'S NAMEPLATE

Each item of equipment to have a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent is prohibited for use.

2.13 FIELD FABRICATED NAMEPLATES

Provide laminated plastic nameplates in accordance with ASTM D709 for each patch panel, protector assembly, rack, cabinet, and other equipment. Each nameplate inscription to identify the function and, when applicable, the position. Provide melamine plastic nameplates, 0.125 inch thick, white with black center core, and matte finish surfaces with square corners. Accurately align lettering and engrave into the core. Minimum size of nameplates to be 1 x 2-1/2 inches. Minimum lettering size 1/4 inch high normal block style.

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2.14 FACTORY REEL TEST DATA

Test 100 percent OTDR test of FO media at the factory in accordance with TIA-568.1 and TIA-568.3. Use TIA-526-7 for single mode fiber and TIA-526-14, Method B, for multimode fiber measurements. Calibrate OTDR to show anomalies of 0.2 dB minimum. Enhanced performance filled OSP copper cables, referred to as Broadband Outside Plant (BBOSP), in accordance with ICEA S-99-689. Enhanced performance air core OSP copper cables in accordance with ICEA S-98-688. Submit test reports, including manufacture date for each cable reel and receive approval before delivery of cable to the project site.

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of NFPA 70 and IEEE C2 as applicable. In addition to these requirements install in accordance with the ARNEC, I3A, TIA-758 and RUS Bull 1751F-644.

3.1.1 Damage

Promptly repair indicated utility lines or systems damaged during site preparation and construction. Damages to lines or systems not indicated, which are caused by Contractor operations, will be treated as "Changes" under the terms of the Contract Clauses. When Contractor is advised in writing of the location of a non-indicated line or system, such notice to provide that portion of the line or system with "indicated" status in determining liability for damages. In every event, immediately notify the Contracting Officer of any damage.

3.1.2 Cable Inspection and Repair

Handle cable and wire provided in the construction with care. Inspect cable reels for cuts, nicks or other damage. Replace or repair damaged cable to the satisfaction of the Contracting Officer. Reel wraps to remain intact on the reel until the cable is placed.

3.1.3 Direct Burial System

Installation in accordance with I3A, and RUS Bull 1751F-640. Under paved areas and roadways install cable in conduit encased in concrete. Slope ducts to drain. Excavate trenches by hand or mechanical trenching equipment. Provide a minimum cable cover of 24 inches below finished grade. Trenches not to be less than 6 inches wide and in straight lines between cable markers. Do not use cable plows. Bends in trenches to have a radius of not less than 36 inches. Where two or more cables are laid parallel in the same trench, space laterally at least 3 inches apart. When rock is encountered, remove rock to a depth of at least 3 inches below the cable and fill the space with sand or clean earth free from particles larger than 1/4 inch. Do not unreel and pull cables into the trench from one end. Cable may be unreeled on grade and lifted into position. Provide color, type, and depth of warning tape as specified in paragraph "BURIED WARNING AND IDENTIFICATION TAPE."

3.1.3.1 Cable Placement

- a. Separate cables crossing other cables or metal piping from the other

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cables or pipe in accordance with I3A. Do not install circuits for communications under or above traffic signal loops.

- b. Cables to be in one piece without splices between connections except where the distance exceeds the lengths in which the cable is furnished.
- c. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.
- d. Leave a horizontal slack of approximately 3 feet in the ground on each end of cable runs, on each side of connection boxes, and at points where connections are brought aboveground. Where cable is brought aboveground, leave additional slack to make necessary connections.

3.1.3.2 Backfill for Rocky Soil

When placing cable in a rocky soil trench, fill the trench with sand or selected soil at least 2 inches thick on the floor of the trench before placement. The backfill for at least 4 inches above the wire or cable to be free from stones, rocks, or other hard or sharp materials which might damage the cable or wire. If the buried cable is placed less than 24 inches in depth, use a protective cover of concrete.

3.1.4 Cable Protection

Provide direct burial cable protection in accordance with NFPA 70. Galvanized conduits which penetrate concrete (slabs, pavement, and walls) with PVC coated and to extend from the first coupling or fitting outside either side of the concrete minimum of 6 inches per 12 inches burial depth beyond the edge of the surface where cable protection is required; seal all conduits on each end. Where additional protection is required, cable may be placed in galvanized iron pipe (GIP) sized on a maximum fill of 40 percent of cross-sectional area, or in concrete encased 4 inches PVC pipe. Conduit may be installed by jacking or trenching. Backfill trenches with earth and mechanically tamped at 6 inches lift so that the earth is restored to the same density, grade, and vegetation as adjacent undisturbed material.

Seal cable ends at all times with coated heat shrinkable end caps (when delivered to the job site, while the cable is stored, and during installation of the cable). The caps to remain in place until the cable is spliced or terminated. Sealing compounds and tape are prohibited substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

3.1.5 UNDERGROUND STRUCTURE CONSTRUCTION

3.1.5.1 Precast Concrete Construction

Set commercial precast structures on 6 inches of level, 90 percent compacted granular fill, 3/4 inch to 1 inch size, extending 12 inches beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally conform to the manufacturer's instructions.

3.1.5.2 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors.

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Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor the irons must be a minimum of 6 inches from the edge of the sump, and in the walls the irons must be located within 6 inches of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 6 inches of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 6 inch clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 3 foot length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 3 inches from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 4 inches and must be designed to withstand a minimum pulling-in load of 6000 pounds. Irons must be hot-dipped galvanized after fabrication. In addition, pulling-in irons shall be constructed in accordance with I3A.

3.1.5.3 Cable Racks, Arms, and Insulators

All cable racks, arms, and insulators shall be installed in accordance with the requirements listed in I3A. Methods of anchoring cable racks must be as follows:

- a. Provide a 5/8 inch diameter by 5 inch long anchor bolt with 3 inch foot cast in structure wall with 2 inch protrusion of threaded portion of bolt into structure. Provide 5/8 inch steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.
- b. Provide concrete channel insert with a minimum load rating of 800 pounds per foot. Insert channel must be steel of the same length as "vertical rack channel;" channel insert must be cast flush in structure wall. Provide 5/8 inch steel nuts in channel insert to receive 5/8 inch diameter by 3 inch long steel, square head anchor bolts.
- c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 800 pound load rating. Provide 5/8 inch diameter by 3 inch long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

3.1.6 Underground Conduit And Duct Systems

3.1.6.1 Requirements

Run conduit in straight lines except where a change of direction is

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necessary. Provide numbers and sizes of ducts as indicated. Ducts must have a continuous slow downward toward underground structures and away from buildings, laid with a minimum slope of 3 inches per 100 feet. Depending on the contour of the finished grade, the high-point may be at a terminal, a maintenance hole, a hand hole, or between maintenance holes or hand holes. Provide ducts with end bells whenever duct lines terminate in structures.

Perform changes in duct bank direction as follows:

- a. The minimum manufactured bend radius must be 18 inches for ducts of less than 3 inch diameter, and 36 inches for ducts 3 inches or greater in diameter.
- b. As an exception to the bend radius required above, provide field manufactured long sweep bends having a minimum radius of 25 feet for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections: 30 degrees.

3.1.6.2 Treatment

Ducts must be kept clean of concrete, dirt, or foreign substances during construction. Field cuts requiring tapers must be made with proper tools and factory tapers. A coupling recommended by the duct manufacturer must be used whenever an existing duct is connected to a duct of different material or shape. Ducts must be stored to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Ducts must be thoroughly cleaned before being laid. Plastic ducts must be stored on a flat surface and protected from direct rays of the sun.

3.1.6.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 3 inches and larger, draw a flexible testing mandrel approximately 12 inches long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For conduit sizes less than 3 inches, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

3.1.6.4 Galvanized Conduit Concrete Penetrations

Galvanized conduits which penetrate concrete (slabs, pavement, and walls) in wet locations must be PVC coated and must extend 4 to 6 inches from the point of entry to the termination point.

3.1.6.5 Multiple Conduits

Separate multiple conduits by a minimum distance of 3 inches, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 12 inches. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install

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spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 10 feet of conduit assembly.

3.1.6.6 Conduit Plugs and Pull Rope

New conduit indicated as being unused or empty must be provided with plugs on each end. Provide a plastic pull rope having 3 feet of slack at each end of unused or empty conduits.

3.1.6.7 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 24 inches below finished grade. Provide not less than 3 inches clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 3 inches, fill and tamp level with original bottom with sand or earth free from particles that would be retained on a 1/4 inch sieve. The first 6 inch layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 3 to 6 inch layers. Provide color, type and depth of warning tape as specified in I3A and in section 31 00 00 EARTHWORK.

3.1.6.8 Ducts Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 18 inches below finished grade, except under roads and pavement, concrete envelope must be not less than 24 inches below finished grade. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and must provide at least 3 inches of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 3 inches. Before pouring concrete, anchor duct bank assemblies to prevent the assemblies from floating during concrete pouring. Anchoring must be done by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly. Provide steel reinforcing in the concrete envelope as indicated.

3.1.6.8.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 3 inch concrete cover around ducts. Concrete encasement must extend at least 6 feet beyond the edges of paved areas and roads, and 12 feet beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 24 inches below finished grade.

3.1.6.9 Connections to Maintenance Holes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 12 inches in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

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3.1.6.10 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs.

3.1.7 Excavating, Backfilling, And Compacting

Provide in accordance with I3A, NFPA 70 and Section 31 00 00 EARTHWORK.

3.1.8 Reconditioning Of Surfaces

3.1.8.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct or direct burial cable. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide top-soiling, fertilizing, liming, seeding, sodding, sprigging, or mulching.

3.1.8.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists Make repairs as specified in Sections 32 11 23 AGGREGATE BASE COURSE FOR FLEXIBLE PAVING, 32 12 13 BITUMINOUS TACK AND PRIME COATS, and 32 12 16.16 ROAD-MIX ASPHALT PAVING.

3.1.9 Building Penetrations

Caulk and seal cable access penetrations in walls, ceilings, and other parts of the building.

3.1.10 Cable Pulling

Test duct lines with a mandrel and clean out to remove foreign material before pulling cables. All mandrel tests shall be in accordance with I3A. Avoid damaging cables in setting up pulling apparatus or in placing tools or hardware. Do not step on cables when entering or leaving the manhole. Do not place cables in ducts other than those shown without prior written approval of the Contracting Officer. Roll cable reels in the direction indicated by the arrows painted on the reel flanges. Set up cable reels on the same side of the manhole as the conduit section in which the cable is to be placed. Level the reel and bring into proper alignment with the conduit section so that the cable pays off from the top of the reel in a long smooth bend into the duct without twisting. Under no circumstances will the cable be paid off from the bottom of a reel. Check the equipment set up prior to beginning the cable pulling to avoid an interruption once pulling has started. Use a cable feeder guide of suitable dimensions between cable reel and face of duct to protect cable and guide cable into the duct as it is paid off the reel. As cable is paid off the reel, lubricate and inspect cable for sheath defects. When defects are noticed, stop pulling operations and notify the Contracting Officer to determine required corrective action. Cable pulling to cease when reel binds or does not pay off freely. Rectify cause of binding before resuming pulling

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operations. Provide cable lubricants recommended by the cable manufacturer. Avoid bends in cables of small radii and twists that might cause damage. Do not bend cable and wire in a radius less than 10 times the outside diameter of the cable or wire.

3.1.10.1 Cable Tensions

Submit from the cable manufacturer and the maximum allowable pulling tension. This tension not to be exceeded.

3.1.10.2 Pulling Eyes

Equip cables, 1.25 inches in diameter and larger, with cable manufacturer's factory installed pulling-in eyes. Provide cables with diameter smaller than 1.25 inches with heat shrinkable type end caps or seals on cable ends when using cable pulling grips. Rings to prevent grip from slipping not to be beaten into the cable sheath. Use a swivel of 3/4 inch links between pulling-in eyes or grips and pulling strand.

3.1.10.3 Installation of Cables in Manholes, Hand holes, and Vaults

Do not install cables using the shortest route, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support cables on brackets, and cable insulators at a maximum of 4 feet.

In existing manholes, hand holes, and vaults where ducts are to be terminated, or where cables are to be installed, modify the existing installation of cables, cable supports, and grounding as required with cables arranged and supported as specified for cables.

3.1.10.4 Fiber Optic Cable Slack

Install 20 feet of slack in maintenance holes if the cable is not spliced in maintenance hole. If the fiber optic cable is spliced in the maintenance hole, then 50 feet of slack is required in the maintenance hole on each side of the splice case. Install 20 feet of slack within the building telecommunications spaces on each side of the cable.

3.1.11 Aerial Cable Installation

3.1.11.1 Figure 8 Distribution Wire

Perform spiraling of the wire within 24 hours of the tensioning operation. Perform spiraling operations at alternate poles with the approximate length of the spiral being 15 feet. Do not remove insulation from support members except at bonding and grounding points and at points where ends of support members are terminated in splicing and dead-end devices. Ground support wire at poles to the pole ground.

3.1.11.2 Suspension Strand

Place suspension strand as indicated. Tension in accordance with the data indicated. When tensioning strand, loosen cable suspension clamps enough to allow free movement of the strand. Place suspension strand on the road side of the pole line. In tangent construction, point the lip of the suspension strand clamp toward the pole. At angles in the line, point the suspension strand clamp lip away from the load. In level construction place the suspension strand clamp in such a manner to hold the strand

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below the through-bolt. At points where there is an up-pull on the strand, place clamp to support the strand above the through-bolt. Make the suspension strand electrically continuous throughout the entire length, bond to other bare cables suspension strands, and connect to pole ground at each pole.

3.1.11.3 Aerial Cable

Keep cable ends sealed at all times using cable end caps. Take cable from reel only as it is placed. During placing operations, do not bend cables in a radius less than 10 times the outside diameter of cable. Place temporary supports sufficiently close together and properly tension the cable where necessary to prevent excessive bending. In those instances where spiraling of cabling is involved, accomplish mounting of enclosures for purposes of loading, splicing, and distribution after the spiraling operation has been completed.

3.1.12 Cable Splicing

3.1.12.1 Copper Conductor Splices

Perform splicing in accordance with I3A, and RUS Bull 1753F-401 except that direct buried splices and twisted and soldered splices are prohibited. Exception does not apply for pairs assigned for carrier application.

3.1.12.2 Fiber Optic Splices

Install all fiber optic cables splices in accordance with I3A and TIA-758.

3.1.13 Surge Protection

All cables and conductors, except fiber optic cable, which serve as communication lines through off-premise lines, to have surge protection installed at each end in accordance with ARNEC, I3A, RUS Bull 1751F-815.

3.1.14 Grounding

Provide grounding and bonding in accordance with RUS 1755, TIA-607, I3A, IEEE C2, and NFPA 70. Ground exposed noncurrent carrying metallic parts of telephone equipment, cable sheaths, cable splices, and terminals.

3.1.14.1 Primary Bus-Bar (PBB)

The PBB is the hub of the basic telecommunications grounding system providing a common point of connection for ground from outside cable, CD, and equipment. Establish a PBB for connection point for cable stub shields to connector blocks and CD protector assemblies as specified in Section 27 10 00.00 48 BUILDING TELECOMMUNICATIONS CABLING SYSTEM.

3.1.14.2 Incoming Cable Shields

Accomplish all necessary transfers and cut-overs by the telecommunications contractor as indicated.

3.1.15 Cut-Over

Accomplish all necessary transfers and cut-overs by the telecommunications contractor as indicated.

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3.2 LABELING

3.2.1 Labels

Provide labeling for cabling and termination hardware installed in accordance with ARNEC, I3A, and TIA-606. Handwritten labeling is prohibited. Provide stenciled lettering for cable and termination hardware using thermal ink transfer process.

3.2.2 Cable Tag Installation

Install cable tags for each telecommunications cable or wire located in manholes, hand holes, and vaults including each splice. Tag wire and cable installed and existing wire and cable which are indicated to have splices and terminations provided. The labeling of telecommunications cable tag identifiers in accordance with ARNEC, I3A, and TIA-606. Tag legend, as indicated. Do not provide handwritten letters. Install cable tags so that they are clearly visible without disturbing any cabling or wiring in the manholes, hand holes, and vaults.

3.2.3 Termination Hardware

Label patch panels, distribution panels, connector blocks, and protection modules using color coded labels with identifiers in accordance with ARNEC and TIA-606.

3.3 FIELD APPLIED PAINTING

Provide ferrous metallic enclosure finishes as specified in Section 09 90 00 PAINTS AND COATINGS.

3.4 FIELD QUALITY CONTROL

Provide 10 working days notice prior to each test. Correct defective material and workmanship disclosed as the results of the tests. Furnish a signed copy of the test results to the Contracting Officer within 3 working days after the tests for each segment of construction are completed.

Perform testing as construction progresses and do not wait until all construction is complete before starting field tests.

3.4.1 Pre-Installation Tests

Perform the following tests on cable prior to being removed from the cable reel. For cables with factory installed pulling eyes, perform these tests at the factory and certified test results to accompany the cable.

3.4.1.1 Cable Capacitance

Perform capacitance tests on at least 10 percent of the pairs within a cable to determine if cable capacitance is within the limits specified.

3.4.1.2 Loop Resistance

Perform DC-loop resistance on at least 10 percent of the pairs within a cable to determine if DC-loop resistance is within the manufacturer's calculated resistance.

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3.4.1.3 Pre-Installation Tests

Provide results of pre-installation tests to the Contracting Officer at least 5 working days before installation is to start. Results to indicate reel number of the cable, manufacturer, size of cable, pairs tested, and recorded readings. When pre-installation tests indicate that cable does not meet specifications, remove cable.

3.4.2 Acceptance Tests

Perform acceptance testing of all installed cabling in accordance with ARNEC, TIA-568.0, TIA-568.2, TIA-568.3, TIA-568.4, RUS Bull 1753F-201, and as specified. Provide personnel, equipment, instrumentation, and supplies necessary to perform required testing. Notification of any planned testing to be given to the Contracting Officer at least 14 days prior to any test unless specified otherwise. Testing not to proceed until after the Contractor has received written Contracting Officer's approval of the test plans as specified. Test plans to define the tests required to ensure that the system meets technical, operational, and performance specifications. The test plans to define milestones for the tests, equipment, personnel, facilities, and supplies required. The test plans to identify the capabilities and functions to be tested. Provide test reports in booklet form showing all field tests performed, upon completion and testing of the installed system. Measurements to be tabulated on a pair by pair or strand by strand basis.

3.4.2.1 Copper Conductor Cable

Perform the following acceptance tests in accordance with TIA-568.0 and TIA-568.2:

- a. Wire map (pin to pin continuity);
- b. Continuity to remote end;
- c. Crossed pairs;
- d. Reversed pairs;
- e. Split pairs;
- f. Shorts between two or more conductors

3.4.2.2 Fiber Optic Cable

Test fiber optic cable in accordance with ARNEC, TIA-568.0, TIA-568.3, TIA-526-7, TIA/EIA-455, and as specified. Perform 2 optical tests on all optical fibers: Optical Time Domain Reflectometry (OTDR) Test and Attenuation Test. In addition, perform a width test on all multimode optical fibers. These tests to be performed on the completed end-to-end spans which include the near-end pre-connectorized single fiber cable assembly, outside plant as specified, and the far-end pre-connectorized single fiber cable.

- a. OTDR Test: Used to determine the adequacy of the cable installations by showing any irregularities, such as discontinuities, micro-bendings or improper splices for the cable span under test. A reference length of fiber, 150 feet minimum, used as the delay line to be placed before the new end connector and after the far end patch panel connectors for

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inspection of connector signature. Conduct OTDR test and provide calculation or interpretation of results in accordance with TIA-526-7 for single-mode fiber and TIA-526-14 for multimode fiber. Splice losses not to exceed 0.3 db.

- b. Attenuation Test: End-to-end attenuation measurements to be made on all fibers, in both directions, using an 850 and 1300 (multimode fiber) or 1310 and 1550 (single-mode) nanometer light source at one end and the optical power meter on the other end to verify that the cable system attenuation requirements are met in accordance with TIA-455-46A for multimode and TIA-526-7 for single-mode fiber optic cables. The measurement method in accordance with TIA-455-78-B. Attenuation losses not to exceed 0.5 db/km at 1310 nm and 1550 nm for single-mode fiber. Attenuation losses not to exceed 5.0 db/km at 850 nm and 1.5 db/km at 1300 nm for multimode fiber.

3.4.3 Soil Density Tests

Determine soil-density compaction of backfill material in accordance with ASTM D1557, Method D.

-- End of Section --

Section 25 05 11 - Control Correlation Identifiers
CCIs for AMI Moderate-Moderate-Moderate

Reference: UFC 4-010-06 Cybersecurity of
Facility-Related Control Systems, with
Change 1
<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc/ufc-4-010-06>

CCI Number	800-53/82 Control Text Indicator	CCI Definition	Designer Implementation	Classification	Responsibility
CCI-000054	AC-10	The information system limits the number of concurrent sessions for each organization-defined account and/or account type to an organization-defined number of sessions.	The organization being inspected/assessed configures the information system to limit the number of concurrent sessions for all accounts and/or account types to a number of sessions defined in AC-10, CCI 55. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 54. DoD has defined the account types and/or accounts as all account types and/or accounts.	Applicable	DoD Defined
CCI-000060	AC-11(1)	The information system conceals, via the session lock, information previously visible on the display with a publicly viewable image.	The organization being inspected/assessed configures the information system to conceal, via the session lock, information previously visible on the display with a publicly viewable image. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 60.	Impractical	Contractor
CCI-000057	AC-11(a)	The information system initiates a session lock after the organization-defined time period of inactivity.	The organization being inspected/assessed configures the information system to initiate a session lock after the organization-defined time period defined in CCI 59.	Impractical	Contractor
CCI-000058	AC-11(a)	The information system provides the capability for users to directly initiate session lock mechanisms.	The organization being inspected/assessed configures the information system to provide the capability for users to directly initiate session lock mechanisms. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 58.	Impractical	Contractor
CCI-000056	AC-11(b)	The information system retains the session lock until the user reestablishes access using established identification and authentication procedures.	The organization being inspected/assessed configures the information system to retain the session lock until the user reestablishes access using established identification and authentication procedures. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 56.	Impractical	Contractor
CCI-000067	AC-17(1)	The information system monitors remote access methods.	The organization being inspected/assessed configures the information system to monitor remote access methods. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 67.	Applicable	Contractor
CCI-002314	AC-17(1)	The information system controls remote access methods.	The organization being inspected/assessed configures the information system to control remote access methods. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2314.	Applicable	Contractor
CCI-000068	AC-17(2)	The information system implements cryptographic mechanisms to protect the confidentiality of remote access sessions.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the confidentiality of remote access sessions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 68.	Applicable	Contractor
CCI-001453	AC-17(2)	The information system implements cryptographic mechanisms to protect the integrity of remote access sessions.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the integrity of remote access sessions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1453.	Applicable	Contractor
CCI-000069	AC-17(3)	The information system routes all remote accesses through organization-defined number managed network access control points.	The organization being inspected/assessed configures the information system to route all remote accesses through the number of managed network access control points defined in AC-17 (3), CCI 2315. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 69.	Applicable	Contractor, Owner
CCI-002322	AC-17(9)	The organization provides the capability to expeditiously disconnect or disable remote access to the information system within the organization-defined time period.	The organization being inspected/assessed configures the information system to provide the capability to expeditiously disconnect or disable remote access to the information system immediately. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2322. DoD has defined the time period as immediately.	Applicable	DoD Defined
CCI-000065	AC-17(b)	The organization authorizes remote access to the information system prior to allowing such connections	The organization being inspected/assessed authorizes remote access to the information system prior to allowing such connections. The organization must maintain an audit trail of authorizations.	Applicable	Contractor, Owner
CCI-001443	AC-18(1)	The information system protects wireless access to the system using authentication of users and/or devices.	The organization being inspected/assessed configures the information system to protect wireless access to the system using authentication of users and/or devices. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1443.	Applicable	Contractor, Owner
CCI-001444	AC-18(1)	The information system protects wireless access to the system using encryption.	The organization being inspected/assessed configures the information system to protect wireless access to the system using encryption. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1444.	Applicable	Contractor
CCI-000015	AC-2(1)	The organization employs automated mechanisms to support the information system account management functions.	The organization being inspected/assessed configures the information system to employ automated mechanisms to support the information system account management functions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 000015.	Impractical	Owner
CCI-002142	AC-2(10)	The information system terminates shared/group account credentials when members leave the group.	The organization being inspected/assessed configures the information system to terminate shared/group account credentials when members leave the group. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2142.	Impractical	Contractor
CCI-000016	AC-2(2)	The information system automatically removes or disables temporary accounts after an organization-defined time period for each type of account.	The organization being inspected/assessed configures the information system to automatically remove or disable temporary accounts after 72 hours. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 000016. DoD has defined the time period as 72 hours.	Impractical	DoD Defined

CCI-001682	AC-2(2)	The information system automatically removes or disables emergency accounts after an organization-defined time period for each type of account.	The organization being inspected/assessed configures the information system to never automatically remove or disable emergency accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1682. DoD has defined the time period as never.	Impractical	DoD Defined
CCI-000018	AC-2(4)	The information system automatically audits account creation actions.	The organization being inspected/assessed configures the information system to automatically audit account creation actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 18.	Impractical	Contractor, Owner
CCI-001403	AC-2(4)	The information system automatically audits account modification actions.	The organization being inspected/assessed configures the information system to automatically audit account modification actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1403.	Impractical	Contractor, Owner
CCI-001404	AC-2(4)	The information system automatically audits account disabling actions.	The organization being inspected/assessed configures the information system to automatically audit account disabling actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1404.	Impractical	Contractor, Owner
CCI-001405	AC-2(4)	The information system automatically audits account removal actions.	The organization being inspected/assessed configures the information system to automatically audit account removal actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1405.	Impractical	Owner, Contractor
CCI-001683	AC-2(4)	The information system notifies organization-defined personnel or roles for account creation actions.	The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account creation actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1683. DoD has defined the personnel or roles as the system administrator and ISSO.	Impractical	DoD Defined
CCI-001684	AC-2(4)	The information system notifies organization-defined personnel or roles for account modification actions.	The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account modification actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1684. DoD has defined the personnel or roles as the system administrator and ISSO.	Impractical	DoD Defined
CCI-001685	AC-2(4)	The information system notifies organization-defined personnel or roles for account disabling actions.	The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account disabling actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1685. DoD has defined the personnel or roles as the system administrator and ISSO.	Impractical	DoD Defined
CCI-001686	AC-2(4)	The information system notifies organization-defined personnel or roles for account removal actions.	The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account removal actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1686. DoD has defined the personnel or roles as the system administrator and ISSO.	Impractical	DoD Defined
CCI-002130	AC-2(4)	The information system automatically audits account enabling actions.	The organization being inspected/assessed configures the information system to automatically audit account enabling actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2130.	Impractical	Contractor, Owner
CCI-002131	AC-2(4)	The organization defines the personnel or roles to be notified on account creation, modification, enabling, disabling, and removal actions.	DoD has defined the personnel or roles as the system administrator and ISSO.	Impractical	DoD Defined
CCI-002132	AC-2(4)	The information system notifies organization-defined personnel or roles for account enabling actions.	The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account enabling actions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2132. DoD has defined the personnel or roles as the system administrator and ISSO.	Impractical	DoD Defined
CCI-000213	AC-3	The information system enforces approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	The organization being inspected/assessed configures the information system to enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 213.	Applicable	Contractor, Owner
CCI-002165	AC-3(4)	The information system enforces organization-defined discretionary access control policies over defined subjects and objects.	The organization being inspected/assessed configures the information system to enforce the discretionary access control policies defined in AC-3 (4), CCI 2163 over defined subjects and objects. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2165.	Applicable	Contractor
CCI-000225	AC-6	The organization employs the concept of least privilege, allowing only authorized accesses for users (and processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.	The organization being inspected/assessed documents and implements the concept of least privilege, allowing only authorized accesses for users (and processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.	Applicable	Contractor, Owner
CCI-002235	AC-6(10)	The information system prevents non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures.	The organization being inspected/assessed configures the information system to prevent non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2235.	Impractical	Contractor, Owner

CCI-002233	AC-6(8)	The information system prevents organization-defined software from executing at higher privilege levels than users executing the software.	The organization being inspected/assessed configures the information system to any software except software explicitly documented from executing at higher privilege levels than users executing the software. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2233. DoD has defined the software as any software except software explicitly documented.	Impractical	DoD Defined
CCI-002234	AC-6(9)	The information system audits the execution of privileged functions.	The organization being inspected/assessed configures the information system to audit the execution of privileged functions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2234.	Impractical	Contractor
CCI-000044	AC-7(a)	The information system enforces the organization-defined limit of consecutive invalid login attempts by a user during the organization-defined time period.	The organization being inspected/assessed configures the information system to limit invalid login attempts by a user to three attempts during a 15 minute time period. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 44. DoD has defined the maximum number as three. DoD has defined the time period as 15 minutes.	Applicable	DoD Defined
CCI-002238	AC-7(b)	The information system automatically locks the account or node for either an organization-defined time period, until the locked account or node is released by an administrator, or delays the next login prompt according to the organization-defined delay algorithm when the maximum number of unsuccessful attempts is exceeded.	The organization being inspected/assessed configures the information system to automatically lock the account or node until the locked account is released by an administrator and delays the next login prompt for a minimum of 5 seconds when the maximum number of unsuccessful attempts is exceeded. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2238. DoD has defined the delay algorithm as a minimum of 5 seconds. DoD has defined the time period as until released by an administrator.	Applicable	DoD Defined
CCI-000048	AC-8(a)	The information system displays an organization-defined system use notification message or banner before granting access to the system that provides privacy and security notices consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.	The organization being inspected/assessed configures the information system to display the DoD Information Systems - Standard Consent Banner and User Agreement before granting access to the system that provides privacy and security notices consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 48. DoD has defined the use notification message or banner as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013.	Impractical	DoD Defined
CCI-002247	AC-8(a)	The organization defines the use notification message or banner the information system displays to users before granting access to the system.	DoD has defined the use notification message or banner as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013.	Impractical	DoD Defined
CCI-000050	AC-8(b)	The information system retains the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access.	The organization being inspected/assessed configures the information system to retain the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 50.	Impractical	Contractor
CCI-001384	AC-8(c)(1)	The information system, for publicly accessible systems, displays system use information organization-defined conditions before granting further access.	The organization being inspected/assessed configures the information system to display the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013 before granting further access for publicly accessible systems For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1384. DoD has defined the conditions as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013.	Impractical	DoD Defined
CCI-001385	AC-8(c)(2)	The information system, for publicly accessible systems, displays references, if any, to monitoring that are consistent with privacy accommodations for such systems that generally prohibit those activities.	The organization being inspected/assessed configures the information system to display references, if any, to monitoring that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1385.	Impractical	Contractor
CCI-001386	AC-8(c)(2)	The information system for publicly accessible systems displays references, if any, to recording that are consistent with privacy accommodations for such systems that generally prohibit those activities.	The organization being inspected/assessed configures the information system to display references, if any, to recording that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1386.	Impractical	Contractor
CCI-001387	AC-8(c)(2)	The information system for publicly accessible systems displays references, if any, to auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities.	The organization being inspected/assessed configures the information system to display references, if any, to auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1387.	Impractical	Contractor
CCI-001388	AC-8(c)(3)	The information system, for publicly accessible systems, includes a description of the authorized uses of the system.	The organization being inspected/assessed configures the information system to include a description of the authorized uses of the system for publicly accessible systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1388.	Impractical	Contractor
CCI-000174	AU-12(1)	The information system compiles audit records from organization-defined information system components into a system-wide (logical or physical) audit trail that is time-correlated to within organization defined level of tolerance for relationship between time stamps of individual records in the audit trail.	The organization being inspected/assessed configures the information system to compile audit records from information system components defined in AU-12 (1), CCI 1577 into a system-wide (logical or physical) audit trail that is time-correlated to within the level of tolerance defined in AU-12 (1), CCI-000173 for relationship between time stamps of individual records in the audit trail. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 174.	Impractical	Contractor, Owner

CCI-001914	AU-12(3)	The information system provides the capability for organization-defined individuals or roles to change the auditing to be performed on organization-defined information system components based on organization-defined selectable event criteria within organization-defined time thresholds.	The organization being inspected/assessed configures the information system to provide the capability for individuals or roles defined in AU-12 (3), CCI 1913 to change the auditing to be performed on information system components defined in AU-12 (3), CCI 2047 based on selectable event criteria defined in AU-12 (3), CCI 1911 within time thresholds defined in AU-12 (3), CCI 1912. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1914.	Impractical	Contractor, Owner
CCI-000169	AU-12(a)	The information system provides audit record generation capability for the auditable events defined in AU-2 a at organization defined information system components.	The organization being inspected/assessed acquires or designs all information system and network components that provide audit record generation capability for the auditable events defined in AU-2 a. DoD has defined the information system components as all information system and network components.	Impractical	DoD Defined
CCI-000171	AU-12(b)	The information system allows organization-defined personnel or roles to select which auditable events are to be audited by specific components of the information system.	The organization being inspected/assessed configures the information system to ensure that only the ISSM or individuals appointed by the ISSM select which auditable events are to be audited by specific components of the information system. DoD has defined the personnel or roles as the ISSM or individuals appointed by the ISSM.	Applicable	DoD Defined
CCI-000172	AU-12(c)	The information system generates audit records for the events defined in AU-2 d with the content defined in AU-3.	The organization being inspected/assessed configures the information system to generate audit records for the events defined in AU-2 d with the content defined in AU-3. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 172.	Applicable	Contractor, Owner
CCI-001464	AU-14(1)	The information system initiates session audits at system start-up.	The organization being inspected/assessed configures the information system to initiate session audits at system start-up. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1464.	Impractical	Contractor
CCI-001462	AU-14(2)	The information system provides the capability for authorized users to capture/record and log content related to a user session.	The organization being inspected/assessed configures the information system to provide the capability for authorized users to capture/record and log content related to a user session. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1462.	Impractical	Contractor
CCI-001920	AU-14(3)	The information system provides the capability for authorized users to remotely view/hear all content related to an established user session in real time.	The organization being inspected/assessed configures the information system to provide the capability for authorized users to remotely view/hear all content related to an established user session in real time. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1920.	Impractical	Contractor
CCI-001919	AU-14(b)	The information system provides the capability for authorized users to select a user session to capture/record or view/hear.	The organization being inspected/assessed configures the information system to provide the capability for authorized users to select a user session to capture/record or view/hear. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1919.	Impractical	Contractor
CCI-000130	AU-3	The information system generates audit records containing information that establishes what type of event occurred.	The organization being inspected/assessed configures the information system to generate audit records containing information that establishes what type of event occurred For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 130.	Applicable	Contractor, Owner
CCI-000131	AU-3	The information system generates audit records containing information that establishes when an event occurred.	The organization being inspected/assessed configures the information system to generate audit records containing information that establishes when an event occurred For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 131.	Applicable	Contractor, Owner
CCI-000132	AU-3	The information system generates audit records containing information that establishes where the event occurred.	The organization being inspected/assessed configures the information system to generate audit records containing information that establishes where the event occurred For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 132.	Applicable	Contractor, Owner
CCI-000133	AU-3	The information system generates audit records containing information that establishes the source of the event.	The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the source of the event. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 133.	Applicable	Contractor, Owner
CCI-000134	AU-3	The information system generates audit records containing information that establishes the outcome of the event.	The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the outcome of the event. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 134.	Applicable	Contractor, Owner
CCI-001487	AU-3	The information system generates audit records containing information that establishes the identity of any individuals or subjects associated with the event.	The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the identity of any individuals or subjects associated with the event. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1487.	Applicable	Contractor, Owner
CCI-000135	AU-3(1)	The information system generates audit records containing the organization-defined additional, more detailed information that is to be included in the audit records.	The organization being inspected/assessed configures the information system to generate audit records containing the organization defined additional, more detailed information as defined in AU-3 (1), CCI 1488 that is to be included in the audit records. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 135.	Impractical	Contractor, Owner
CCI-001849	AU-4	The organization allocates audit record storage capacity in accordance with organization-defined audit record storage requirements.	The organization being inspected/assessed allocates, and configures the information system to allocate audit record storage capacity as defined in AU-4, CCI 1848. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1849.	Applicable	Contractor, Owner
CCI-001851	AU-4(1)	The information system off-loads audit records per organization-defined frequency onto a different system or media than the system being audited.	The organization being inspected/assessed configures the information system to off-load audit records at a minimum, in real-time for interconnected systems and weekly for stand-alone systems onto a different system or media than the system being audited. DoD has defined the frequency as at a minimum, real-time for interconnected systems and weekly for stand-alone systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1851.	Impractical	DoD Defined

CCI-001855	AU-5(1)	The information system provides a warning to organization-defined personnel, roles, and/or locations within organization-defined time period when allocated audit record storage volume reaches organization-defined percentage of repository maximum audit record storage capacity.	The organization being inspected/assessed configures the information system to immediately provide a warning to personnel, roles, and/or locations defined in AU-5 (1), CCI 1852 when allocated audit record storage volume reaches 75 percent of repository maximum audit record storage capacity. DoD has defined the time period as immediate. DoD has defined the percentage as 75 percent. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1855.	Impractical	DoD Defined
CCI-001864	AU-6(1)	The organization employs automated mechanisms to integrate audit review and analysis to support organizational processes for investigation of response to suspicious activities.	The organization being inspected/assessed identifies and implements automated mechanisms to integrate audit review and analysis. The goal is to support organizational investigation of and response to suspicious activities.	Impractical	Contractor
CCI-001865	AU-6(1)	The organization employs automated mechanisms to integrate reporting processes to support organizational investigation of and response to suspicious activities.	The organization being inspected/assessed identifies and implements automated mechanisms to integrate reporting processes (e.g., centralized log analysis tools). The goal is to support organizational investigation of and response to suspicious activities.	Impractical	Contractor, Owner
CCI-000154	AU-6(4)	The information system provides the capability to centrally review and analyze audit records from multiple components within the system.	The organization being inspected/assessed configures the information system to provide a capability to centrally review and analyze audit records from multiple components within the system. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 154.	Impractical	Contractor
CCI-000159	AU-8(a)	The information system uses internal system clocks to generate time stamps for audit records.	The organization being inspected/assessed configures the information system to use internal system clocks to generate time stamps for audit records. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 159.	Applicable	Contractor, Owner
CCI-001889	AU-8(b)	The information system records time stamps for audit records that meets organization-defined granularity of time measurement.	The organization being inspected/assessed configures the information system to generate time in the time stamps for audit records that meets one second granularity of time measurement. DoD has defined the granularity of time measurement as one second. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1889.	Applicable	DoD Defined
CCI-001890	AU-8(b)	The information system records time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).	The organization being inspected/assessed configures the information system to generate time stamps for audit records that contain time zones or time offsets that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1890.	Applicable	Contractor, Owner
CCI-000162	AU-9	The information system protects audit information from unauthorized access.	The organization being inspected/assessed configures the information system to disallow unauthorized access to audit information. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 162.	Impractical	Contractor
CCI-000163	AU-9	The information system protects audit information from unauthorized modification.	The organization being inspected/assessed configures the information system to disallow unauthorized modification of audit information. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 163.	Impractical	Contractor
CCI-000164	AU-9	The information system protects audit information from unauthorized deletion.	The organization being inspected/assessed configures the information system to disallow unauthorized deletion of audit information. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 164.	Impractical	Contractor
CCI-001493	AU-9	The information system protects audit tools from unauthorized access.	The organization being inspected/assessed configures the information system to disallow unauthorized access to audit tools. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1493.	Impractical	Contractor
CCI-001494	AU-9	The information system protects audit tools from unauthorized modification.	The organization being inspected/assessed configures the information system to disallow unauthorized modification of audit tools. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1494.	Impractical	Contractor
CCI-001495	AU-9	The information system protects audit tools from unauthorized deletion.	The organization being inspected/assessed configures the information system to disallow unauthorized deletion of audit tools. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1495.	Impractical	Contractor
CCI-002080	CA-3(5)	The organization employs either an allow-all, deny-by exception or deny-all, permit by exception policy for allowing organization-defined information systems to connect to external information systems.	The organization being inspected/assessed configures the information system to employ a deny-all, permit by exception policy for allowing any systems requiring external connectivity to connect to external information systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2080. DoD has defined the information systems as any systems requiring external connectivity.	Applicable	DoD Defined
CCI-001726	CM-10(a)	The organization uses software in accordance with contract agreements.	The organization being inspected/assessed uses software in accordance with contract agreements.	Applicable	Contractor, Owner
CCI-001728	CM-10(a)	The organization uses software in accordance with copyright laws.	The organization being inspected/assessed uses software in accordance with copyright laws.	Applicable	Contractor, Owner
CCI-001812	CM-11(2)	The information system prohibits user installation of software without explicit privileged status.	The organization being inspected/assessed must configure the information system to prevent the installation of software by non-privileged users. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1812.	Applicable	Contractor
CCI-001807	CM-11(b)	The organization enforces software installation policies through organization-defined methods.	The organization being inspected/assessed must enforce software installation policies as defined in CM-11, CCI 1804 through methods defined in CM-11, CCI 1806.	Applicable	Owner, Contractor
CCI-000327	CM-3(2)	The organization tests changes to the information system before implementing the changes on the operational system.	The organization being inspected/assessed documents and implements a process to test changes to the information system before implementing the changes on the operational system. The organization must maintain an audit trail of testing activity.	Impractical	DoD Defined

CCI-000328	CM-3(2)	The organization validates changes to the information system before implementing the changes on the operational system.	The organization being inspected/assessed documents and implements a process to validate changes to the information system before implementing the changes on the operational system. The organization must maintain an audit trail of validation activity.	Impractical	Contractor, Owner
CCI-001746	CM-3(6)	The organization ensures that cryptographic mechanisms used to provide organization-defined security safeguards are under configuration management.	The organization being inspected/assessed ensures that cryptographic mechanisms used to provide all security safeguards are under configuration management. DoD has defined the security safeguards as all security safeguards.	Impractical	DoD Defined
CCI-000340	CM-5	The organization approves physical access restrictions associated with changes to the information system.	The organization being inspected/assessed documents and implements a process to approve physical access restrictions associated with changes to the information system. The organization must maintain an audit trail of approvals.	Applicable	Owner
CCI-000341	CM-5	The organization enforces physical access restrictions associated with changes to the information system.	The organization being inspected/assessed documents and implements a process to enforce physical access restrictions associated with changes to the information system.	Applicable	Owner
CCI-001762	CM-7(1)(b)	The organization disables organization-defined functions, ports, protocols, and services within the information system deemed to be unnecessary and/or nonsecure.	The organization being inspected/assessed must disable functions, ports, protocols, and services within the information system deemed to be unnecessary and/or nonsecure as defined in CM-7 (1), CCI 1761.	Applicable	Contractor
CCI-001764	CM-7(2)	The information system prevents program execution in accordance with organization-defined policies regarding software program usage and restrictions, and/or rules authorizing the terms and conditions of software program usage.	The organization being inspected/assessed configures the information system to prevent the execution of programs not authorized in accordance with CM-7 (2) CCI 1592 and 1763.	Impractical	Contractor
CCI-001774	CM-7(5)(b)	The organization employs an deny-all, permit-by-exception policy to allow the execution of authorized software programs on the information system.	The organization being inspected/assessed configures the information system to deny-all and only permit by exception the execution of authorized software programs on the information system.	Impractical	Contractor, Owner
CCI-000382	CM-7(b)	The organization configures the information system to prohibit or restrict the use of organization-defined functions, ports, protocols, and/or services.	The organization being inspected/assessed configures the information system to prohibit or restrict the use of functions, ports, protocols, and/or services IAW DoDI 8551.01. DoD has defined the information system prohibited or restricted functions, ports, protocols, and/or services as IAW DoDI 8551.01.	Applicable	DoD Defined
CCI-000411	CM-8(2)	The organization employs automated mechanisms to help maintain an up-to-date inventory of information system components.	The organization being inspected/assessed documents and implements automated mechanisms to help maintain an up-to-date inventory of information system components. An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains an up-to-date inventory.	Impractical	Owner
CCI-000412	CM-8(2)	The organization employs automated mechanisms to help maintain a complete inventory of information system components.	The organization being inspected/assessed documents and implements automated mechanisms to help maintain a complete inventory of information system components. An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains a complete inventory.	Impractical	Owner
CCI-000413	CM-8(2)	The organization employs automated mechanisms to help maintain an accurate inventory of information system components.	The organization being inspected/assessed documents and implements automated mechanisms to help maintain an accurate inventory of information system components. An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains an accurate inventory.	Impractical	Owner
CCI-000414	CM-8(2)	The organization employs automated mechanisms to help maintain a readily available inventory of information system components.	The organization being inspected/assessed documents and implements automated mechanisms to help maintain a readily available inventory of information system components. An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains a readily available inventory.	Impractical	Owner
CCI-000416	CM-8(3)(a)	The organization employs automated mechanisms, per organization defined frequency, to detect the presence of unauthorized hardware, software, and firmware components within the information system.	The organization being inspected/assessed documents and implements automated mechanisms to detect the presence of unauthorized hardware, software, and firmware components within the information system continuously. DoD has defined the frequency as continuously.	Impractical	DoD Defined
CCI-001784	CM-8(3)(b)	When unauthorized hardware, software, and firmware components are detected within the information system, the organization takes action to disable network access by such components, isolates the components, and/or notifies organization-defined personnel or roles.	The organization being inspected/assessed documents and implements a process to take action to disable network access by unauthorized software, hardware, and firmware components, isolate the components, and/or notify the ISSO and ISSM and others as the local organization deems appropriate. The organization must maintain an audit trail of actions taken upon detection of unauthorized software, hardware, and firmware components. DoD has defined the personnel or roles as the ISSO and ISSM and others as the local organization deems appropriate.	Applicable	DoD Defined
CCI-002855	CP-12	The information system, when organization-defined conditions are detected, enters a safe mode of operation with organization-defined restrictions of safe mode of operation.	The organization being inspected/assessed configures the information system to enter a safe mode of operation with restrictions of safe mode of operation defined in CP-12, CCI 2857 when conditions defined in CP-12, CCI 2856 are detected. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2855.	Applicable	Contractor
CCI-000505	CP-6(a)	The organization establishes an alternate storage site including necessary agreements to permit the storage and retrieval of information system backup information.	The organization being inspected/assessed establishes an alternate storage site and documents relevant information within the contingency plan.	Impractical	Owner, Contractor
CCI-000542	CP-9(1)	The organization tests backup information per organization-defined frequency to verify media reliability and information integrity.	The organization being inspected/assessed tests and logs backup information at least monthly in accordance with contingency plan to verify media reliability and information integrity.	Impractical	Owner
CCI-000540	CP-9(d)	The organization protects the confidentiality, integrity, and availability of backup information at storage locations.	The organization being inspected/assessed will protect the confidentiality, integrity, and availability of backup information at the storage location IAW the system security plan.	Impractical	Owner
CCI-000764	IA-2	The information system uniquely identifies and authenticates organizational users (or processes acting on behalf of organizational users).	The organization being inspected/assessed configures the information system to uniquely identify and authenticate organizational users (or processes acting on behalf of organizational users). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 764.	Applicable	Contractor, Owner
CCI-000765	IA-2(1)	The information system implements multifactor authentication for network access to privileged accounts.	The organization being inspected/assessed configures the information system to implement multifactor authentication for network access to privileged accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 765.	Applicable	Contractor, Owner
CCI-001948	IA-2(11)	The information system implements multifactor authentication for remote access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access.	The organization being inspected/assessed configures the information system to implement multifactor authentication for remote access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1948.	Applicable	Contractor

CCI-001949	IA-2(11)	The device used in the information system implementation of multifactor authentication for remote access to privileged accounts meets organization-defined strength of mechanism requirements.	The organization being inspected/assessed will use DoD PKI or a technology approved by their Authorizing Official that meet Federal standards for authentication such as FIPS 140-2, NIAP Certification, or NSA approval. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1949.	Applicable	Contractor, Owner
CCI-001951	IA-2(11)	The information system implements multifactor authentication for remote access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access.	The organization being inspected/assessed configures the information system to implement multifactor authentication for remote access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1951.	Applicable	Contractor, Owner
CCI-001952	IA-2(11)	The device used in the information system implementation of multifactor authentication for remote access to non-privileged accounts meets organization-defined strength of mechanism requirements.	The organization being inspected/assessed will use DoD PKI or a technology approved by their Authorizing Official that meet Federal standards for authentication such as FIPS 140-2, NIAP Certification, or NSA approval. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1952. The organization being inspected/assessed configures the information system to accept PIV/CAC authentication.	Applicable	Contractor, Owner
CCI-001953	IA-2(12)	The information system accepts Personal Identity Verification (PIV) credentials.	This control enhancement applies to organizations implementing logical access control systems (LACS) and physical access control systems (PACS). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1953 The organization being inspected/assessed configures the information system to verify PIV/CAC authentication.	Impractical	Contractor, Owner
CCI-001954	IA-2(12)	The information system electronically verifies Personal Identity Verification (PIV) credentials.	This control enhancement applies to organizations implementing logical access control systems (LACS) and physical access control systems (PACS). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1954.	Impractical	Contractor, Owner
CCI-000766	IA-2(2)	The information system implements multifactor authentication for network access to non-privileged accounts.	The organization being inspected/assessed configures the information system to implement multifactor authentication for network access to non-privileged accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 766.	Applicable	Contractor, Owner
CCI-000767	IA-2(3)	The information system implements multifactor authentication for local access to privileged accounts.	The organization being inspected/assessed configures the information system to implement multifactor authentication for local access to privileged accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 767.	Applicable	Contractor, Owner
CCI-000768	IA-2(4)	The information system implements multifactor authentication for local access to non-privileged accounts.	The organization being inspected/assessed configures the information system to implement multifactor authentication for local access to non-privileged accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 768.	Applicable	Contractor, Owner
CCI-000770	IA-2(5)	The organization requires individuals to be authenticated with an individual authenticator when a group authenticator is employed.	The organization being inspected/assessed requires individuals or configures the information system to require individuals to be authenticated with an individual authenticator when a group authenticator is employed. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 770.	Applicable	Contractor, Owner
CCI-001941	IA-2(8)	The information system implements replay-resistant authentication mechanisms for network access to privileged accounts.	The organization being inspected/assessed configures the information system to implement replay-resistant authentication mechanisms for network access to privileged accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1941.	Applicable	Contractor, Owner
CCI-001942	IA-2(9)	The information system implements replay-resistant authentication mechanisms for network access to non-privileged accounts.	The organization being inspected/assessed configures the information system to implement replay-resistant authentication mechanisms for network access to non-privileged accounts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1942.	Applicable	Contractor, Owner
CCI-000778	IA-3	The information system uniquely identifies an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.	The organization being inspected/assessed configures the network infrastructure to identify all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 778. DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).	Applicable	DoD Defined
CCI-001958	IA-3	The information system authenticates an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.	The organization being inspected/assessed configures the network infrastructure to authenticate all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1958. DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).	Applicable	DoD Defined
CCI-001967	IA-3(1)	The information system authenticates organization-defined devices and/or types of devices before establishing a local, remote and/or network connection using bidirectional authentication that is cryptographically based.	The organization being inspected/assessed configures the information system to use cryptographically based bidirectional authentication. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1967.	Impractical	Contractor, Owner
CCI-000192	IA-5(1)(a)	The information system enforces password complexity by the minimum number of upper case characters used.	The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of upper case characters used. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 192.	Impractical	Owner

CCI-000193	IA-5(1)(a)	The information system enforces password complexity by the minimum number of lower case characters used.	The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of lower case characters used. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 193.	Impractical	Owner
CCI-000194	IA-5(1)(a)	The information system enforces password complexity by the minimum number of numeric characters used.	The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of numeric characters used. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 194.	Impractical	Owner
CCI-000205	IA-5(1)(a)	The information system enforces minimum password length.	The organization being inspected/assessed configures the information system to enforce minimum password length. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 205.	Impractical	Owner
CCI-001619	IA-5(1)(a)	The information system enforces password complexity by the minimum number of special characters used.	The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of special characters used. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1619.	Impractical	Owner
CCI-000195	IA-5(1)(b)	The information system, for password-based authentication, when new passwords are created, enforces that at least an organization-defined number of characters are changed.	The organization being inspected/assessed configures the information system to enforce that at least 50% of the minimum password length is changed. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 195. DoD has defined the minimum number of characters as 50% of the minimum password length.	Impractical	DoD Defined
CCI-000196	IA-5(1)(c)	The information system, for password-based authentication, stores only cryptographically-protected passwords.	The organization being inspected/assessed configures the information system to store only encrypted representations of passwords. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 196.	Applicable	Contractor, Owner
CCI-000197	IA-5(1)(c)	The information system, for password-based authentication, transmits only cryptographically-protected passwords.	The organization being inspected/assessed configures the information system to transmit only encrypted representations of passwords. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 197.	Applicable	Contractor, Owner
CCI-000198	IA-5(1)(d)	The information system enforces minimum password lifetime restrictions.	The organization being inspected/assessed configures the information system to enforce minimum password lifetime restrictions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 198.	Impractical	Contractor, Owner
CCI-000199	IA-5(1)(d)	The information system enforces maximum password lifetime restrictions.	The organization being inspected/assessed configures the information system to enforce maximum password lifetime restrictions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 199.	Impractical	Contractor, Owner
CCI-000200	IA-5(1)(e)	The information system prohibits password reuse for the organization defined number of generations.	The organization being inspected/assessed configures the information system to prohibit reuse for a minimum of 5 generations. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 200. DoD has defined the number of generations as a minimum of 5.	Impractical	DoD Defined
CCI-002041	IA-5(1)(f)	The information system allows the use of a temporary password for system logons with an immediate change to a permanent password.	The organization being inspected/assessed configures the information system to allow the use of a temporary password for system logons with an immediate change to a permanent password. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2041.	Impractical	Owner, Contractor
CCI-002003	IA-5(11)	The information system, for token-based authentication, employs mechanisms that satisfy organization-defined token quality requirements.	The information system performing hardware token-based authentication must be configured to accept only DoD-approved PKI credentials in accordance with DoDI 8520.02 and DoDI 8520.03. For unclassified systems, DoD-approved PKI credentials include DoD PKI credentials, External Certification Authority (ECA) PKI credentials, and DoD-approved external PKI credentials. For SIPRNet, DoD-approved PKI credentials include DoD PKI credentials and NSS PKI credentials. If the information system accepts DoD-approved external PKI credentials, the information system must be configured to accept only certificates at approved assurance levels, as represented by the Certificate Policy Object Identifiers (OIDs) asserted in the certificate. The current list of DoD-approved external PKIs and acceptable Object Identifiers (OIDs) for each approved external PKI is available at http://isse.disa.mil/pki-pke/interoperability.	Applicable	DoD Defined
CCI-002007	IA-5(13)	The information system prohibits the use of cached authenticators after an organization defined time period.	The organization being inspected/assessed configures the information system to prohibit the use of cached authenticators after an organization defined time period. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2007.	Impractical	Contractor, Owner
CCI-002008	IA-5(14)	The organization, for PKI-based authentication, employs a deliberate organization-wide methodology for managing the content of PKI trust stores installed across all platforms including networks, operating systems, browsers, and applications.	DoD trust store management requirements are defined in information system components' applicable STIGs and SRGs. All information systems are required to undergo a STIG compliance review as part of their certification and accreditation process prior to being granted an authority to operate. DoD Components are automatically compliant with this CCI because they are covered by the DoD-level STIGs and SRGs.	Impractical	DoD Defined
CCI-000185	IA-5(2)(a)	The information system, for PKI-based authentication validates certifications by constructing and verifying a certification path to an accepted trust anchor including checking certificate status information.	The information system performing hardware token-based authentication must be configured to validate DoD-approved PKI credentials in accordance with RFC 5280. The information system must be configured to perform a revocation check as part of the certificate validation process. Revocation checking may be performed using certificate revocation lists (CRLs) published by the issuing PKI or Online Certificate Status Protocol (OCSP) services. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 185.	Applicable	Contractor

CCI-000186	IA-5(2)(b)	The information system, for PKI-based authentication enforces authorized access to the corresponding private key.	Information systems must not have access to users' private keys. The cryptographic container in which the private keys are stored (e.g. smart card or software module) implements access controls and protections to ensure that only the authorized user can activate the private key. DoD users agree to protect their PKI credentials in accordance with the DD-2842 agreement that is executed for each credential. They are reminded of these responsibilities in annual IA training. The private key identifying the information system must be stored in a cryptographic container that is FIPS 140-2 validated. Only authorized information system operators should have access to activation data (e.g. password or PIN) for the private key.	Applicable	Contractor
CCI-000187	IA-5(2)(c)	The information system, for PKI-based authentication, maps the authenticated identity to the account of the individual or group.	The information system performing PKI-based authentication must be configured to map the authenticated PKI credential to a corresponding network or information system account or role in accordance with DoDI 8520.03. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 187.	Applicable	Contractor
CCI-001997	IA-5(4)	The organization employs automated tools to determine if password authenticators are sufficiently strong to satisfy organization-defined requirements.	The organization being inspected/assessed implements automated tools to check passwords strength per the complexity requirements defined in IA-5 (1) Part A.	Impractical	Contractor
CCI-000803	IA-7	The information system implements mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication.	The organization being inspected/assessed configures the information system to implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 803.	Applicable	Contractor, Owner
CCI-000804	IA-8	The information system uniquely identifies and authenticates non-organizational users (or processes acting on behalf of non-organizational users).	The organization being inspected/assessed configures the information system to uniquely identify and authenticate non-organizational users (or processes acting on behalf of non-organizational users). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 804.	Impractical	Contractor, Owner
CCI-002009	IA-8(1)	The information system accepts Personal Identity Verification (PIV) credentials from other federal agencies.	The information system performing hardware token-based authentication must be configured to accept DoD-approved external PKI PIV credentials to authenticate federal agency users in accordance with DoDI 8520.02 and DoDI 8520.03. The information system must be configured to accept only certificates at approved assurance levels, as represented by the Certificate Policy Object Identifiers (OIDs) asserted in the certificate. The current list of DoD-approved external PKIs and acceptable Object Identifiers (OIDs) for each approved external PKI is available at http://ase.dsa.mil/pki-pke/interoperability . For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2009.	Impractical	Contractor, Owner
CCI-002010	IA-8(1)	The information system electronically verifies Personal Identity Verification (PIV) credentials from other federal agencies.	The information system performing hardware token-based authentication must be configured to validate DoD-approved external PKI PIV credentials to authenticate federal agency users in accordance with RFC 5280. The information system must be configured to perform a revocation check as part of the certificate validation process. Revocation checking may be performed using certificate revocation lists (CRLs) published by the issuing PKI or Online Certificate Status Protocol (OCSP) services. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2010.	Impractical	Contractor, Owner
CCI-002011	IA-8(2)	The information system accepts FICAM-approved third-party credentials.	The organization being inspected/assessed configures the information system to accept Federal Identity, Credential, and Access Management (FICAM)-approved third-party credentials. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2011. FICAM Guidance is available at http://www.idmanagement.gov .	Impractical	Contractor, Owner
CCI-002013	IA-8(3)	The organization employs only FICAM-approved information system components in organization-defined information systems to accept third-party credentials.	The organization being inspected/assessed employs only Federal Identity, Credential, and Access Management (FICAM)-approved information system components to accept third-party credentials in information systems defined in IA-8 (3), CCI 2012. FICAM Guidance is available at http://www.idmanagement.gov .	Impractical	Owner, Contractor
CCI-002014	IA-8(4)	The information system conforms to FICAM-issued profiles.	The organization being inspected/assessed configures the information system to conform to Federal Identity, Credential, and Access Management (FICAM)-issued profiles. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2014. FICAM Guidance is available at http://www.idmanagement.gov .	Impractical	Contractor, Owner
CCI-002818	IR-9(3)	The organization implements organization-defined procedures to ensure that organizational personnel impacted by information spills can continue to carry out assigned tasks while contaminated systems are undergoing corrective actions.	The organization being inspected/assessed implements procedures defined in IR-9 (3), CCI 2819 to ensure that organizational personnel impacted by information spills can continue to carry out assigned tasks while contaminated systems are undergoing corrective actions.	Impractical	Owner
CCI-002866	MA-2(a)	The organization schedules maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	The organization being inspected/assessed schedules maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	Impractical	Contractor, Owner
CCI-002867	MA-2(a)	The organization performs maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	The organization must maintain a record of maintenance. The organization being inspected/assessed implements a process to perform maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	Impractical	Contractor, Owner
CCI-002871	MA-2(a)	The organization performs repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	The organization must maintain a record of maintenance procedures followed. The organization being inspected/assessed implements a process to perform repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	Impractical	Contractor, Owner
CCI-002872	MA-2(a)	The organization documents repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	The organization being inspected/assessed documents repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	Impractical	Contractor, Owner
CCI-002873	MA-2(a)	The organization reviews records of repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.	The organization being inspected/assessed documents and implements a process to review records of repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements. The organization must maintain a record of reviews.	Applicable	Owner

CCI-000859	MA-2(b)	The organization approves and monitors all maintenance activities, whether performed on site or remotely and whether the equipment is serviced on site or removed to another location.	The organization being inspected/assessed approves and monitors all maintenance activities, whether performed on site or remotely and whether the equipment is serviced on site or removed to another location.	Applicable	Owner, Contractor
CCI-000861	MA-2(d)	The organization sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs.	The organization being inspected/assessed sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs IAW DoDM 5200.01-V3 for classified media and DoDM 5200.01-V4 for unclassified media.	Impractical	Owner
CCI-002884	MA-4(1)(a)	The organization audits nonlocal maintenance and diagnostic sessions' organization-defined audit events.	The organization being inspected/assessed configures the information system to audit nonlocal maintenance and diagnostic sessions' organization-defined audit events.	Impractical	Contractor
CCI-002890	MA-4(6)	The information system implements cryptographic mechanisms to protect the integrity of nonlocal maintenance and diagnostic communications.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the integrity of nonlocal maintenance and diagnostic communications.	Impractical	Contractor
CCI-003123	MA-4(6)	The information system implements cryptographic mechanisms to protect the confidentiality of nonlocal maintenance and diagnostic communications.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the confidentiality of nonlocal maintenance and diagnostic communications.	Impractical	Contractor
CCI-002891	MA-4(7)	The information system implements remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.	The organization being inspected/assessed configures the information system to implement remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.	Impractical	Contractor
CCI-000877	MA-4(c)	The organization employs strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions.	The organization being inspected/assessed configures the information system to employ strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions.	Impractical	Contractor
CCI-000879	MA-4(c)	The organization terminates sessions and network connections when nonlocal maintenance is completed.	The organization being inspected/assessed terminates session and network connections when nonlocal maintenance is completed.	Impractical	Contractor
CCI-001018	MP-4(b)	The organization protects information system media until the media are destroyed or sanitized using approved equipment, techniques, and procedures.	The organization being inspected/assessed protects information system media IAW DoDM 5200.01 M Vol. 1-4.	Impractical	Owner
CCI-001020	MP-5(a)	The organization protects and controls organization-defined types of information system media during transport outside of controlled areas using organization-defined security safeguards.	The organization being inspected/assessed protects and controls information system media during transport outside of controlled areas using security measures defined in DoDM 5200.01 M Vol. 1-4 and DoDD 5015.2.	Applicable	Contractor
CCI-002585	MP-7(1)	The organization prohibits the use of portable storage devices in organization information systems when such devices have no identifiable owner.	The organization being inspected/assessed does not use portable storage devices in organization information systems when such devices have no identifiable owner.	Applicable	Owner, Contractor
CCI-000959	PE-10(c)	The organization protects emergency power shutoff capability from unauthorized activation.	The organization being inspected/assessed will protect emergency power shutoff capability.	Applicable	DoD Defined
CCI-002955	PE-11	The organization provides a short-term uninterruptible power supply to facilitate an orderly shutdown of the information system and/or transition of the information system to long-term alternate power in the event of a primary power source loss.	The organization being inspected/assessed implements an uninterruptible power supply with sufficient capacity to support orderly shutdown of the system or transition the system to long-term alternate power in the event of a primary power source loss.	Applicable	Contractor, Owner
CCI-000963	PE-12	The organization employs and maintains automatic emergency lighting for the information system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility.	The organization being inspected/assessed must install and maintain automatic emergency lighting for the information system that activates in the event of a power outage or disruption and covers emergency exits and evacuation routes within the facility in compliance with established OSHA requirements.	Applicable	Contractor
CCI-000965	PE-13	The organization employs and maintains fire suppression and detection devices/systems for the information system that are supported by an independent energy source.	The organization being inspected/assessed must implement and maintain fire suppression and detection devices/systems for the information system that are supported by an independent energy source.	Applicable	Contractor
CCI-000971	PE-14(a)	The organization maintains temperature and humidity levels within the facility where the information system resides at organization-defined acceptable levels.	The organization being inspected/assessed must maintain temperature and where applicable humidity levels of for commercial grade information systems: 64.4 – 80.6 degrees F; 45% – 60% Relative Humidity; Dew Point 41.9 ° – 59°F; measured at the air intake inlet of the IT equipment casing; for other systems, levels within manufacturer specifications.	Applicable	DoD Defined
			Humidity controls are not required for general office areas where information system components may be in use and are only required where there are concentrations of information systems such as server farms, mainframes, etc.		
			DoD has defined the acceptable levels as for commercial grade information systems: 64.4 – 80.6 degrees F; 45% – 60% Relative Humidity; Dew Point 41.9 ° – 59°F; measured at the air intake inlet of the IT equipment casing; for other systems, levels within manufacturer specifications.		

CCI-000973	PE-14(b)	The organization monitors temperature and humidity levels in accordance with organization-defined frequency.	The organization being inspected/assessed will maintain an independent monitor device for temperature and humidity levels not located in the immediate vicinity of the controller continuously unless manufacturer specifications allow for a wide enough tolerance that control is not required. Records of monitoring must be maintained as an audit trail within the authorization lifecycle. DoD has defined the frequency as continuously unless manufacturer specifications allow for a wide enough tolerance that control is not required.	Applicable	DoD Defined
CCI-000977	PE-15	The organization protects the information system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible.	The organization being inspected/assessed must provide master shutoff valves that are accessible to protect the information system from damage resulting from water leakage.	Applicable	Contractor
CCI-000978	PE-15	The organization protects the information system from damage resulting from water leakage by providing master shutoff or isolation valves that are working properly.	The organization being inspected/assessed will ensure that master shutoff valves are working properly and have been inspected by the appropriate organization (e.g., fire marshal, department of public works).	Applicable	Contractor
CCI-000919	PE-3(a)	The organization enforces physical access authorizations at organization-defined entry/exit points to the facility where the information system resides.	The organization being inspected/assessed will implement physical access authorizations at entry/exit points defined in PE-3, CCI 2915 and secure those physical access points (i.e. doors and/or windows) that are not intended for normal access.	Applicable	Owner
CCI-000921	PE-3(a)(2)	The organization controls ingress/egress to the facility using one or more organization-defined physical access control systems/devices or guards.	The organization being inspected/assessed will control ingress/egress to the facility using the physical access control devices and/or guards defined in PE-3, CCI 2916.	Impractical	Owner
CCI-002917	PE-3(b)	The organization maintains physical access audit logs for organization-defined entry/exit points.	The organization being inspected/assessed maintains physical access audit logs for entry/exit points defined in PE-3, CCI 2918.	Applicable	Owner
CCI-002919	PE-3(c)	The organization provides organization-defined security safeguards to control access to areas within the facility officially designated as publicly accessible.	The organization being inspected/assessed provides security safeguards defined in PE-3, CCI 2920 to control access to areas within the facility officially designated as publicly accessible. The organization must document which areas are officially designated as publicly accessible.	Applicable	Owner
CCI-002921	PE-3(d)	The organization escorts visitors during organization-defined circumstances requiring visitor escorts.	The organization being inspected/assessed documents and implements a process to escort visitors during circumstances defined in PE-3, CCI 2922 requiring visitor escorts.	Applicable	Owner
CCI-002923	PE-3(d)	The organization monitors visitor activity during organization-defined circumstances requiring visitor monitoring.	The organization being inspected/assessed documents and implements a process to monitor visitor activity during circumstances defined in PE-3, CCI 2924 requiring visitor monitoring.	Impractical	Owner
CCI-000923	PE-3(e)	The organization secures keys, combinations, and other physical access devices.	The organization being inspected/assessed will secure as appropriate (in safes or secure containers) items used for physical access control such as keys, combinations, portable locks, etc. Fixed access control devices such as card readers, installed locks, key pads, etc. should be protected from tampering.	Applicable	Owner, Contractor
CCI-002950	PE-6(4)	The organization monitors physical access to the information system in addition to the physical access monitoring of the facility as organization-defined physical spaces containing one or more components of the information system.	The organization being inspected/assessed documents and implements a process to monitor physical access to the information system in addition to the physical access monitoring of the facility as physical spaces containing one or more components of the information system defined in PE-6 (4), CCI 2951. The organization must maintain records of monitoring.	Applicable	Owner, Contractor
CCI-003031	PS-5(c)	The organization modifies access authorization as needed to correspond with any changes in operational need due to reassignment or transfer.	The organization being inspected/assessed documents and implements a process to modify access authorization as needed to correspond with any changes in operational need due to reassignment or transfer.	Applicable	Owner
CCI-001541	PS-7(e)	The organization monitors third-party provider compliance with personnel security requirements.	The organization being inspected/assessed monitors third-party provider compliance with personnel security requirements. The organization must maintain an audit trail of monitoring activity.	Applicable	Owner
CCI-003360	SA-19(a)	The organization implements anti-counterfeit policy that include the means to detect counterfeit components from entering the information system.	The organization being inspected/assessed implements the policy defined in SA-19, CCI 3356 that include the means to detect counterfeit components from entering the information system.	Impractical	Owner
CCI-003361	SA-19(a)	The organization implements anti-counterfeit policy that include the means to prevent counterfeit components from entering the information system.	The organization being inspected/assessed implements the policy defined in SA-19, CCI 3357 that include the means to prevent counterfeit components from entering the information system.	Impractical	Owner
CCI-003362	SA-19(a)	The organization implements anti-counterfeit procedures that include the means to detect counterfeit components from entering the information system.	The organization being inspected/assessed implements the procedures defined in SA-19, CCI 3358 that include the means to detect counterfeit components from entering the information system.	Impractical	Owner
CCI-003363	SA-19(a)	The organization implements anti-counterfeit procedures that include the means to prevent counterfeit components from entering the information system.	The organization being inspected/assessed implements the procedures defined in SA-19, CCI 3359 that include the means to prevent counterfeit components from entering the information system.	Impractical	Owner
CCI-003093	SA-3(d)	The organization integrates the organizational information security risk management process into system development life cycle activities.	The organization being inspected/assessed documents and implements a process to integrate the organizational information security risk management process into system development life cycle activities.	Applicable	Owner, Contractor
CCI-003140	SA-9(1)(a)	The organization conducts an organizational assessment of risk prior to the acquisition or outsourcing of dedicated information security services.	The organization being inspected/assessed documents and implements a process to conduct an organizational assessment of risk prior to the acquisition or outsourcing of dedicated information security services. The organization must maintain a record of risk assessment.	Applicable	Owner, Contractor
CCI-003141	SA-9(1)(b)	The organization ensures that the acquisition or outsourcing of dedicated information security services is approved by organization-defined personnel or roles.	The organization being inspected/assessed ensures that the acquisition or outsourcing of dedicated information security services is approved by the DoD Component CIO or their delegate(s). The organization must maintain a record of approvals. DoD has defined the personnel or roles the DoD Component CIO or their delegate(s).	Applicable	DoD Defined
CCI-001133	SC-10	The information system terminates the network connection associated with a communications session at the end of the session or after 10 minutes in band management and 15 minutes for user sessions.	The organization being inspected/assessed configures the information system to terminate the network connection associated with a communications session at the end of the session or after 10 minutes in band management and 15 minutes for user sessions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1133. DoD has defined the time period as 10 minutes in band management and 15 minutes for user sessions.	Applicable	DoD Defined

CCI-001152	SC-15(b)	The information system provides an explicit indication of use to users physically present at collaborative computing devices.	The organization being inspected/assessed configures the information system to provide an explicit indication of use to users physically present at collaborative computing devices. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1152.	Impractical	Contractor
CCI-002465	SC-21	The information system requests data origin authentication verification on the name/address resolution responses the system receives from authoritative sources.	The organization being inspected/assessed configures the: 1. recursive/caching name server software to enable DNSSEC; 2. software to enable DNSSEC validation; and 3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2465.	Applicable	Contractor
CCI-002466	SC-21	The information system requests data integrity verification on the name/address resolution responses the system receives from authoritative sources.	The organization being inspected/assessed configures the: 1. recursive/caching name server software to enable DNSSEC; 2. software to enable DNSSEC validation; and 3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2466.	Applicable	Contractor
CCI-002467	SC-21	The information system performs data integrity verification on the name/address resolution responses the system receives from authoritative sources.	The organization being inspected/assessed configures the: 1. recursive/caching name server software to enable DNSSEC; 2. software to enable DNSSEC validation; and 3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2467.	Applicable	Contractor
CCI-002468	SC-21	The information system performs data origin verification authentication on the name/address resolution responses the system receives from authoritative sources.	The organization being inspected/assessed configures the: 1. recursive/caching name server software to enable DNSSEC; 2. software to enable DNSSEC validation; and 3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2468.	Applicable	Contractor
CCI-001182	SC-22	The information systems that collectively provide name/address resolution service for an organization are fault-tolerant.	The organization being inspected/assessed implements a name service resolution architecture consisting of primary and secondary servers. The organization must document the architecture in the site security plan. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1182.	Impractical	Contractor, Owner
CCI-001183	SC-22	The information systems that collectively provide name/address resolution service for an organization implement internal/external role separation.	The organization being inspected/assessed implements a name service resolution architecture where recursive and authoritative server software is not installed on the same information system. The organization must document the architecture in the site security plan. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1183.	Applicable	Contractor
CCI-001184	SC-23	The information system protects the authenticity of communications sessions.	The organization being inspected/assessed configures the information system to protect the authenticity of communications sessions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1184.	Applicable	Contractor
CCI-001185	SC-23(1)	The information system invalidates session identifiers upon user logout or other session termination.	The organization being inspected/assessed configures the information system to invalidate session identifiers upon user logout or other session termination. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1185.	Applicable	Contractor
CCI-001188	SC-23(3)	The information system generates unique session identifiers for each session with organization-defined randomness requirements.	The organization being inspected/assessed configures the information system to generate unique session identifiers for each session with randomness requirements defined in SC-23 (3), CCI 1189. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1188.	Impractical	Contractor
CCI-001664	SC-23(3)	The information system recognizes only session identifiers that are system-generated.	The organization being inspected/assessed configures the information system to recognize only session identifiers that are system-generated. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1664.	Applicable	Contractor
CCI-002470	SC-23(5)	The information system only allows the use of organization-defined certificate authorities for verification of the establishment of protected sessions.	The organization being inspected/assessed configures the information system to allow the use of DoD PKI established certificate authorities for verification of the establishment of protected sessions. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2470. DoD has defined the certificate authorities as DoD PKI established certificate authorities.	Impractical	DoD Defined
CCI-001190	SC-24	The information system fails to an organization-defined known-state for organization-defined types of failures.	The organization being inspected/assessed configures the information system to fail to a secure state for failures during system initialization, shutdown, and aborts. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1190. DoD has defined the known state as secure state. DoD has defined the types of failures as failures during system initialization, shutdown, and aborts.	Applicable	DoD Defined
CCI-001665	SC-24	The information system preserves organization-defined system state information in the event of a system failure.	The organization being inspected/assessed configures the information system to preserve information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes in the event of a system failure. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1665. DoD has defined system state information as information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes.	Applicable	DoD Defined

CCI-001199	SC-28	The information system protects the confidentiality and/or integrity of organization-defined information at rest.	The organization being inspected/assessed configures the information system to protect the confidentiality and/or integrity of organization-defined information at rest. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1199.	Applicable	Contractor
CCI-002475	SC-28(1)	The information system implements cryptographic mechanisms to prevent unauthorized modification of organization-defined information at rest on organization-defined information system components.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized modification of information at rest defined in SC-28 (1), CCI 2473 on any information system components storing data defined in SC-28 (1), 2473. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2475. DoD has defined the information system components as any information system components storing data defined in SC-28 (1), 2473.	Applicable	DoD Defined
CCI-002476	SC-28(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of organization-defined information at rest on organization-defined information system components.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information at rest defined in SC-28 (1), CCI 2473 on any information system components storing data defined in SC-28 (1), 2473. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2476. DoD has defined the information system components as any information system components storing data defined in SC-28 (1), 2473.	Applicable	DoD Defined
CCI-002529	SC-38	The organization employs organization-defined operations security safeguards to protect key organizational information throughout the system development life cycle.	The organization being inspected/assessed implements operations security safeguards defined in SC-38, CCI 2528 to protect key organizational information throughout the system development life cycle. The organization must maintain an audit trail of security safeguard implementation.	Applicable	Owner, Contractor
CCI-002530	SC-39	The information system maintains a separate execution domain for each executing process.	The organization being inspected/assessed configures the information system to maintain a separate execution domain for each executing process. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2530.	Applicable	Contractor
CCI-002546	SC-41	The organization physically disables or removes organization-defined connection ports or input/output devices on organization-defined information systems or information system components.	The organization being inspected/assessed physically disables or removes connection ports or input/output devices defined in SC-41, CCI 2545 on information systems or information system components defined in SC-41, CCI 2544.	Applicable	Contractor
CCI-002385	SC-5	The information system protects against or limits the effects of organization-defined types of denial of service attacks by employing organization-defined security safeguards.	The organization being inspected/assessed configures the information system to protect against or limits the effects of types of denial of service attacks defined in SC-5, CCI 1093 by employing security safeguards defined in SC-5, CCI 2386. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2385.	Applicable	DoD Defined
CCI-001094	SC-5(1)	The information system restricts the ability of individuals to launch organization-defined denial of service attacks against other information systems.	The organization being inspected/assessed configures the information system to restrict the ability of individuals to launch denial of service attacks defined in SC-5 (1), CCI 2387 against other information systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1094.	Applicable	DoD Defined
CCI-001095	SC-5(2)	The information system manages excess capacity, bandwidth, or other redundancy to limit the effects of information flooding types of denial of service attacks.	The organization being inspected/assessed configures the information system to manage excess capacity, bandwidth, or other redundancy to limit the effects of information flooding types of denial of service attacks. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1095.	Applicable	Contractor
CCI-002389	SC-5(3)(a)	The organization employs organization-defined list of monitoring tools to detect indicators of denial of service attacks against the information system.	The organization being inspected/assessed implements the monitoring tools defined in SC-5 (3), CCI 2388 to detect indicators of denial of service attacks against the information system.	Impractical	DoD Defined
CCI-001116	SC-7(10)	The organization prevents the unauthorized exfiltration of information across managed interfaces.	The organization being inspected/assessed documents and implements mechanisms to prevent the unauthorized exfiltration of information across managed interfaces.	Applicable	Contractor, Owner
CCI-002403	SC-7(11)	The information system only allows incoming communications from organization-defined authorized sources routed to organization-defined authorized destinations.	The organization being inspected/assessed configures the information system to allow incoming communications from authorized sources defined in SC-7 (11), CCI 2401 routed to authorized destinations defined in SC-7 (11), CCI 2402. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2403.	Applicable	Contractor
CCI-002406	SC-7(12)	The organization implements organization-defined host-based boundary protection mechanisms at organization-defined information system components.	The organization being inspected/assessed implements McAfee Host Intrusion Prevention (HIPS) on all information system components. DoD has defined the host-based boundary protection mechanisms as McAfee Host Intrusion Prevention (HIPS). DoD has defined the information system components as all information system components.	Impractical	DoD Defined
CCI-001119	SC-7(13)	The organization isolates organization-defined information security tools, mechanisms, and support components from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.	The organization being inspected/assessed designs the information system to isolate key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc. from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system. DoD has defined the key information security tools, mechanisms, and support components as key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc.	Impractical	DoD Defined
CCI-001121	SC-7(14)	The organization protects against unauthorized physical connections at organization-defined managed interfaces.	The organization being inspected/assessed documents and implements mechanisms to protect against unauthorized physical connections at internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways. DoD has defined the managed interfaces as internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.	Applicable	DoD Defined
CCI-001101	SC-7(3)	The organization limits the number of external network connections to the information system.	The organization being inspected/assessed documents and implements information system access control mechanisms to limit the number of external connections to the information system.	Applicable	Contractor, Owner

CCI-001102	SC-7(4)(a)	The organization implements a managed interface for each external telecommunication service.	The organization being inspected/assessed designs the information system to have a managed interface for each telecommunication service.	Impractical	Contractor, Owner
CCI-002396	SC-7(4)(c)	The organization protects the confidentiality and integrity of the information being transmitted across each interface for each external telecommunication service.	The organization being inspected/assessed documents and implements mechanisms to protect the confidentiality and integrity of the information being transmitted across each interface for each external telecommunication service.	Applicable	Contractor, Owner
CCI-001109	SC-7(5)	The information system at managed interfaces denies network communications traffic by default and allows network communications traffic by exception (i.e., deny all, permit by exception).	The organization being inspected/assessed configures the information system to deny network communications traffic at managed interfaces by default and allows network communications traffic by exception (i.e., deny all, permit by exception). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1109.	Impractical	Contractor
CCI-002397	SC-7(7)	The information system, in conjunction with a remote device, prevents the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks.	The organization being inspected/assessed configures the information system to prevent the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2397.	Applicable	Contractor
CCI-002398	SC-7(9)(a)	The information system detects outgoing communications traffic posing a threat to external information systems.	The organization being inspected/assessed configures the information system to detect outgoing communications traffic posing a threat to external information systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2398.	Impractical	Contractor
CCI-002399	SC-7(9)(a)	The information system denies outgoing communications traffic posing a threat to external information systems.	The organization being inspected/assessed configures the information system to deny outgoing communications traffic posing a threat to external information systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2399.	Impractical	Contractor
CCI-002400	SC-7(9)(b)	The information system audits the identity of internal users associated with denied outgoing communications traffic posing a threat to external information systems.	The organization being inspected/assessed configures the information system to audit the identity of internal users associated with denied outgoing communications traffic posing a threat to external information systems. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2400.	Impractical	Contractor
CCI-001097	SC-7(a)	The information system monitors and controls communications at the external boundary of the system and at key internal boundaries within the system.	The organization being inspected/assessed documents and implements processes to monitor and control communications at the external boundary of the system and at key internal boundaries within the system. The organization must maintain an audit trail of monitoring activities.	Applicable	Contractor, Owner
CCI-002395	SC-7(b)	The information system implements subnetworks for publicly accessible system components that are physically and/or logically separated from internal organizational networks.	The organization being inspected/assessed designs the information system to leverage subnetworks so that publicly accessible system components are physically and/or logically separated from internal organizational networks.	Applicable	Designer, Owner
CCI-001098	SC-7(c)	The information system connects to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.	The organization being inspected/assessed designs the information system to enforce requirements that components connect to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.	Applicable	Designer, Owner
CCI-002418	SC-8	The information system protects the confidentiality and/or integrity of transmitted information.	The organization being inspected/assessed configures the information system to protect the confidentiality and/or integrity of transmitted information. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2418.	Applicable	Contractor
CCI-002421	SC-8(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of information and/or detect changes to information during transmission unless otherwise protected by organization-defined alternative physical safeguards.	The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information and detect changes to information during transmission unless otherwise protected by Protected Distribution System (PDS). For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2421. DoD has defined the selection as both prevention of unauthorized disclosure and detection of changes to information. DoD has defined the alternative physical safeguards as Protected Distribution System (PDS).	Impractical	DoD Defined
CCI-002420	SC-8(2)	The information system maintains the confidentiality and/or integrity of information during preparation for transmission.	The organization being inspected/assessed configures the information system to maintain the confidentiality and integrity of information during preparation for transmission. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2420. DoD has defined the selection as both confidentiality and integrity.	Applicable	DoD Defined
CCI-002422	SC-8(2)	The information system maintains the confidentiality and/or integrity of information during reception.	The organization being inspected/assessed configures the information system to maintain the confidentiality and integrity of information during reception. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2422. DoD has defined the selection as both confidentiality and integrity.	Applicable	DoD Defined
CCI-001310	SI-10	The information system checks the validity of organization-defined inputs.	The organization being inspected/assessed configures the information system to check the validity of all inputs except those identified specifically by the organization. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1310. DoD has defined the information inputs as all inputs except those identified specifically by the organization.	Impractical	DoD Defined
CCI-001312	SI-11(a)	The information system generates error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.	The organization being inspected/assessed configures the information system to generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1312.	Applicable	Contractor

CCI-001314	SI-11(b)	The information system reveals error messages only to organization-defined personnel or roles.	The organization being inspected/assessed configures the information system to reveal error messages only to the ISSO, ISSM, and SCA. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1314. DoD has defined the personnel or roles as the ISSO, ISSM, and SCA.	Impractical	DoD Defined
CCI-002824	SI-16	The information system implements organization-defined security safeguards to protect its memory from unauthorized code execution.	The organization being inspected/assessed configures the information system to implement security safeguards defined in SI-16, CCI 2823 to protect its memory from unauthorized code execution. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2824.	Impractical	Contractor
CCI-002775	SI-17	The information system implements organization-defined fail-safe procedures when organization-defined failure conditions occur.	The organization being inspected/assessed configures the information system to implement fail-safe procedures defined in SI-17, CCI 2773 when failure conditions defined in SI-17, CCI 2774 occur. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2775.	Applicable	Contractor
CCI-002617	SI-2(6)	The organization removes organization-defined software components (e.g., previous versions) after updated versions have been installed.	The organization being inspected/assessed configures the information system to remove all upgraded/replaced software components that are no longer required for operation (e.g., previous versions) after updated versions have been installed. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2617. DoD has defined the software components as all upgraded/replaced software components that are no longer required for operation.	Impractical	DoD Defined
CCI-002618	SI-2(6)	The organization removes organization-defined firmware components (e.g., previous versions) after updated versions have been installed.	The organization being inspected/assessed configures the information system to remove all upgraded/replaced firmware components that are no longer required for operation e.g., previous versions) after updated versions have been installed. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2618. DoD has defined the firmware components as all upgraded/replaced firmware components that are no longer required for operation.	Impractical	DoD Defined
CCI-001225	SI-2(a)	The organization identifies information system flaws.	The organization being inspected/assessed documents and implements a process to identify information system flaws. The process shall include review of the system through automated scans and manual checks to determine the existence of flaws such as IAVM, CVE, or other resources.	Applicable	Contractor, Owner
CCI-001226	SI-2(a)	The organization reports information system flaws.	The organization being inspected/assessed reports information system flaws according to DoD Cybersecurity policy and organizational roles and responsibilities. The organization must report information system flaws in their POA&M.	Applicable	DoD Defined
CCI-002605	SI-2(c)	The organization installs security-relevant software updates within organization-defined time period of the release of the updates	The organization being inspected/assessed configures the information system to install security-relevant software updates within 30 days of the release of the updates For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2605. DoD has defined the time period as 30 days.	Impractical	DoD Defined
CCI-002607	SI-2(c)	The organization installs security-relevant firmware updates within organization-defined time period of the release of the updates	The organization being inspected/assessed configures the information system to install security-relevant firmware updates within 30 days of the release of the updates. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2607. DoD has defined the time period as 30 days.	Impractical	DoD Defined
CCI-002636	SI-3(10)(a)	The organization employs organization-defined tools to analyze the characteristics and behavior of malicious code.	The organization being inspected/assessed documents and implements tools defined in SI-3 (10), CCI 2634 to analyze the characteristics and behavior of malicious code.	Impractical	Contractor
CCI-002638	SI-3(10)(a)	The organization employs organization-defined techniques to analyze the characteristics and behavior of malicious code.	The organization being inspected/assessed documents and implements techniques defined in SI-3 (10), CCI 2635 to analyze the characteristics and behavior of malicious code.	Impractical	Contractor
CCI-001247	SI-3(2)	The information system automatically updates malicious code protection mechanisms.	The organization being inspected/assessed configures the information system to automatically update malicious code protection mechanisms. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1247.	Impractical	Contractor
CCI-002621	SI-3(a)	The organization employs malicious code protection mechanisms at information system entry points to eradicate malicious code.	The organization being inspected/assessed configures the malicious code protection mechanisms identified in SI-3, CCI 2619 to eradicate malicious code.	Impractical	Contractor, Owner
CCI-002622	SI-3(a)	The organization employs malicious code protection mechanisms at information system exit points to eradicate malicious code.	The organization being inspected/assessed configures the malicious code protection mechanisms identified in SI-3, CCI 2620 to eradicate malicious code.	Impractical	Contractor, Owner
CCI-001240	SI-3(b)	The organization updates malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures.	The organization being inspected/assessed configures the information system to update malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1240.	Impractical	Contractor, Owner
CCI-001241	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform periodic scans of the information system on an organization-defined frequency.	The organization being inspected/assessed configures malicious code protection mechanisms to perform periodic scans of the information system on every 7 days. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1241. DoD has defined the frequency as every 7 days.	Applicable	DoD Defined
CCI-001242	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform real-time scans of files from external sources at endpoints as the files are downloaded, opened, or executed in accordance with organizational security policy.	The organization being inspected/assessed configures malicious code protection mechanisms to perform real-time scans of files from external sources at endpoints as the files are downloaded, opened, or executed in accordance with organizational security policy. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1242.	Impractical	Contractor

CCI-002624	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform real-time scans of files from external sources at network entry/exit points as the files are downloaded, opened, or executed in accordance with organizational security policy.	The organization being inspected/assessed configures the malicious code protection mechanisms to perform real-time scans of files from external sources at network entry/exit points as the files are downloaded, opened, or executed in accordance with organizational security policy. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2624.	Impractical	Contractor
CCI-001243	SI-3(c)(2)	The organization configures malicious code protection mechanisms to perform organization-defined action(s) in response to malicious code detection.	The organization being inspected/assessed configures malicious code protection mechanisms to perform block and quarantine malicious code and then send an alert to the administrator immediately in near real-time in response to malicious code detection. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1243. DoD has defined the actions as block and quarantine malicious code and then send an alert to the administrator immediately in near real-time.	Applicable	DoD Defined
CCI-001245	SI-3(d)	The organization addresses the receipt of false positives during malicious code detection and eradication, and the resulting potential impact on the availability of the information system.	The organization being inspected/assessed configures the information system to address the receipt of false positives during malicious code detection and eradication, and the resulting potential impact on the availability of the information system. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1245.	Applicable	Contractor, Owner
CCI-001273	SI-4(11)	The organization analyzes outbound communications traffic at the external boundary of the information system to discover anomalies.	The organization being inspected/assessed documents and implements a process to analyze outbound communications traffic at the external boundary of the information system to discover anomalies. The organization must maintain a record of any discovered anomalies.	Impractical	Contractor
CCI-001671	SI-4(11)	The organization analyzes outbound communications traffic at selected organization-defined interior points within the system (e.g., subnetworks, subsystems) to discover anomalies.	The organization being inspected/assessed documents and implements a process to analyze outbound communications traffic at selected interior points defined in SI-4 (11), CCI 2668 within the system (e.g., subnetworks, subsystems) to discover anomalies. The organization must maintain a record of any discovered anomalies.	Impractical	Contractor
CCI-001274	SI-4(12)	The organization employs automated mechanisms to alert security personnel of an organization-defined inappropriate or unusual activities with security implications.	The organization being inspected/assessed documents and implements automated mechanisms to alert security personnel when there are threats identified by authoritative sources (e.g. CTOs) and IAW with CJCSM 6510.01B. For automated alert mechanisms that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1274. DoD has defined the activities that trigger alerts as when there are threats identified by authoritative sources (e.g. CTOs) and IAW with CJCSM 6510.01B.	Impractical	DoD Defined
CCI-001282	SI-4(15)	The organization employs an intrusion detection system to monitor wireless communications traffic as the traffic passes from wireless to wireline networks.	The organization being inspected/assessed documents and implements an intrusion detection system to monitor wireless communications traffic as the traffic passes from wireless to wireline networks.	Applicable	Contractor, Owner
CCI-002675	SI-4(19)	The organization implements organization-defined additional monitoring of individuals who have been identified by organization-defined sources as posing an increased level of risk.	The organization being inspected/assessed implements additional monitoring defined in SI-4 (19), CCI 2673 of individuals who have been identified by sources defined in SI-4 (19), CCI 2674 as posing an increased level of risk. The organization must maintain an audit trail of additional monitoring.	Impractical	Contractor
CCI-001260	SI-4(2)	The organization employs automated tools to support near real-time analysis of events.	The organization being inspected/assessed documents and implements automated tools to support near real-time analysis of events.	Impractical	Contractor
CCI-002677	SI-4(20)	The organization implements organization-defined additional monitoring of privileged users.	The organization being inspected/assessed implements additional monitoring defined in SI-4 (20), CCI 2676 of privileged users. The organization must maintain an audit trail of additional monitoring.	Impractical	Contractor
CCI-002683	SI-4(22)	The information system detects network services that have not been authorized or approved by the organization-defined authorization or approval processes.	The organization being inspected/assessed documents and implements a process to detect network services that have not been authorized or approved by at a minimum, the ISSO and ISSM. For network service detection mechanisms that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2683. DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.	Impractical	DoD Defined
CCI-002684	SI-4(22)	The information system audits and/or alerts organization-defined personnel when unauthorized network services are detected.	The organization being inspected/assessed configures the information system to audit and/or alert at a minimum, the ISSO and ISSM when unauthorized network services are detected. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2684. DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.	Impractical	DoD Defined
CCI-002687	SI-4(23)	The organization implements organization-defined host-based monitoring mechanisms at organization-defined information system components.	The organization being inspected/assessed documents and implements HBSS at all components. DoD has defined the host-based monitoring mechanisms as HBSS. DoD has defined the information system components as all components.	Impractical	DoD Defined
CCI-002661	SI-4(4)	The information system monitors inbound communications traffic per organization-defined frequency for unusual or unauthorized activities or conditions.	The organization being inspected/assessed configures the information system to monitor inbound communications traffic continuously for unusual or unauthorized activities or conditions. DoD has defined the frequency as continuously.	Impractical	DoD Defined
CCI-002662	SI-4(4)	The information system monitors outbound communications traffic per organization-defined frequency for unusual or unauthorized activities or conditions.	The organization being inspected/assessed configures the information system to monitor outbound communications traffic continuously for unusual or unauthorized activities or conditions. DoD has defined the frequency as continuously.	Impractical	DoD Defined
CCI-002664	SI-4(5)	The information system alerts organization-defined personnel or roles when organization-defined compromise indicators reflect the occurrence of a compromise or a potential compromise.	The organization being inspected/assessed configures the information system to alert at a minimum, the ISSM and ISSO when real time intrusion detection and when there are threats identified by authoritative sources (e.g. CTOs) and IAW incident categories I, II, IV, & VII within CJCSM 6510.01B reflect the occurrence of a compromise or a potential compromise. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2664. DoD has defined the personnel or roles as at a minimum, the ISSM and ISSO. DoD has defined the compromise indicators as real time intrusion detection and when there are threats identified by authoritative sources (e.g. CTOs) and IAW incident categories I, II, IV, & VII within CJCSM 6510.01B.	Impractical	DoD Defined

CCI-002641	SI-4(a)(1)	The organization monitors the information system to detect attacks and indicators of potential attacks in accordance with organization-defined monitoring objectives.	The organization being inspected/assessed documents and implements a process to monitor the information system to detect attacks and indicators of potential attacks in accordance with sensor placement and monitoring requirements within CJCSI 6510.01F. The organization must maintain an audit trail of monitoring. DoD has defined the monitoring objectives as sensor placement and monitoring requirements within CJCSI 6510.01F.	Impractical	DoD Defined
CCI-002642	SI-4(a)(2)	The organization monitors the information system to detect unauthorized local connections.	The organization being inspected/assessed documents and implements a process to monitor the information system to detect unauthorized local connections. The organization must maintain an audit trail of monitoring.	Applicable	Contractor
CCI-002643	SI-4(a)(2)	The organization monitors the information system to detect unauthorized network connections.	The organization being inspected/assessed documents and implements a process to monitor the information system to detect unauthorized network connections. The organization must maintain an audit trail of monitoring.	Applicable	Contractor
CCI-002644	SI-4(a)(2)	The organization monitors the information system to detect unauthorized remote connections.	The organization being inspected/assessed documents and implements a process to monitor information system to detect unauthorized remote connections. The organization must maintain an audit trail of monitoring.	Applicable	Contractor
CCI-001255	SI-4(c)	The organization deploys monitoring devices strategically within the information system to collect organization determined essential information.	The organization being inspected/assessed documents and implements a process to deploy monitoring devices strategically within the information system to collect organization determined essential information.	Impractical	Contractor, Owner
CCI-001256	SI-4(c)	The organization deploys monitoring devices at ad hoc locations within the system to track specific types of transactions of interest to the organization.	The organization being inspected/assessed documents and implements a process to deploy monitoring devices at ad hoc locations within the system to track specific types of transactions of interest to the organization.	Impractical	Contractor, Owner
CCI-002647	SI-4(d)	The organization protects information obtained from intrusion-monitoring tools from unauthorized access.	The organization being inspected/assessed documents and implements a process to protect information obtained from intrusion-monitoring tools from unauthorized access.	Impractical	Contractor, Owner
CCI-002648	SI-4(d)	The organization protects information obtained from intrusion-monitoring tools from unauthorized modification.	The organization being inspected/assessed documents and implements a process to protect information obtained from intrusion-monitoring tools from unauthorized modification.	Impractical	Contractor, Owner
CCI-002649	SI-4(d)	The organization protects information obtained from intrusion-monitoring tools from unauthorized deletion.	The organization being inspected/assessed documents and implements a process to protect information obtained from intrusion-monitoring tools from unauthorized deletion.	Impractical	Contractor, Owner
CCI-001285	SI-5(a)	The organization receives information system security alerts, advisories, and directives from organization-defined external organizations on an ongoing basis.	The organization being inspected/assessed receives information system security alerts, advisories, and directives from at a minimum, USCYBERCOM on an ongoing basis. DoD has defined the external organizations as at a minimum, USCYBERCOM.	Impractical	DoD Defined
CCI-001289	SI-5(d)	The organization implements security directives in accordance with established time frames, or notifies the issuing organization of the degree of noncompliance.	The organization being inspected/assessed implements security directives in accordance with established time frames, or notifies the issuing organization of the degree of noncompliance.	Applicable	Owner
CCI-002737	SI-7(14)(a)	The organization prohibits the use of binary or machine-executable code from sources with limited or no warranty and without the provision of source code.	The organization being inspected/assessed prohibits the use of binary or machine-executable code obtained from sources without vendor support or with no warranty and without the provision of source code.	Applicable	Contractor, Owner
CCI-002723	SI-7(8)	The information system, upon detection of a potential integrity violation, provides the capability to audit the event.	The organization being inspected/assessed configures the information system to provide the capability to audit the event upon detection of a potential integrity violation. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2723.	Applicable	Contractor
CCI-002724	SI-7(8)	The information system, upon detection of a potential integrity violation, initiates one or more of following actions: generates an audit record; alerts current user; alerts organization-defined personnel or roles; and/or organization-defined other actions.	The organization being inspected/assessed configures the information system to initiate one or more of following actions: generates an audit record; alerts current user; alerts at a minimum, the ISSO and ISSM; and/or other actions defined in SI-7 (8), CCI 2722 upon detection of a potential integrity violation. For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2724. DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.	Applicable	DoD Defined

Section 2

CCIs for Elec

CCI Number	800-53/82 Control Text Indicator	CCI Definition
CCI-000054	AC-10	The information system limits the number of concurrent sessions for each organization-defined account and/or account type to an organization-defined number of sessions.
CCI-000060	AC-11(1)	The information system conceals, via the session lock, information previously visible on the display with a publicly viewable image.
CCI-000057	AC-11(a)	The information system initiates a session lock after the organization-defined time period of inactivity.
CCI-000058	AC-11(a)	The information system provides the capability for users to directly initiate session lock mechanisms.
CCI-000056	AC-11(b)	The information system retains the session lock until the user reestablishes access using established identification and authentication procedures.
CCI-000067	AC-17(1)	The information system monitors remote access methods.

CCI-002314	AC-17(1)	The information system controls remote access methods.
CCI-000068	AC-17(2)	The information system implements cryptographic mechanisms to protect the confidentiality of remote access sessions.
CCI-001453	AC-17(2)	The information system implements cryptographic mechanisms to protect the integrity of remote access sessions.
CCI-000069	AC-17(3)	The information system routes all remote accesses through organization-defined number managed network access control points.
CCI-002322	AC-17(9)	The organization provides the capability to expeditiously disconnect or disable remote access to the information system within the organization-defined time period.
CCI-000065	AC-17(b)	The organization authorizes remote access to the information system prior to allowing such connections
CCI-001443	AC-18(1)	The information system protects wireless access to the system using authentication of users and/or devices.

CCI-001444	AC-18(1)	The information system protects wireless access to the system using encryption.
CCI-000015	AC-2(1)	The organization employs automated mechanisms to support the information system account management functions.
CCI-002142	AC-2(10)	The information system terminates shared/group account credentials when members leave the group.
CCI-000016	AC-2(2)	The information system automatically removes or disables temporary accounts after an organization-defined time period for each type of account.
CCI-001682	AC-2(2)	The information system automatically removes or disables emergency accounts after an organization-defined time period for each type of account.
CCI-000018	AC-2(4)	The information system automatically audits account creation actions.

CCI-001403	AC-2(4)	The information system automatically audits account modification actions.
CCI-001404	AC-2(4)	The information system automatically audits account disabling actions.
CCI-001405	AC-2(4)	The information system automatically audits account removal actions.
CCI-001683	AC-2(4)	The information system notifies organization-defined personnel or roles for account creation actions.
CCI-001684	AC-2(4)	The information system notifies organization-defined personnel or roles for account modification actions.
CCI-001685	AC-2(4)	The information system notifies organization-defined personnel or roles for account disabling actions.

CCI-001686	AC-2(4)	The information system notifies organization-defined personnel or roles for account removal actions.
CCI-002130	AC-2(4)	The information system automatically audits account enabling actions.
CCI-002131	AC-2(4)	The organization defines the personnel or roles to be notified on account creation, modification, enabling, disabling, and removal actions.
CCI-002132	AC-2(4)	The information system notifies organization-defined personnel or roles for account enabling actions.
CCI-000213	AC-3	The information system enforces approved authorizations for logical access to information and system resources in accordance with applicable access control policies.
CCI-002165	AC-3(4)	The information system enforces organization-defined discretionary access control policies over defined subjects and objects.

CCI-000225	AC-6	The organization employs the concept of least privilege, allowing only authorized accesses for users (and processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.
CCI-002223	AC-6(1)	The organization explicitly authorizes access to organization-defined security-relevant information.
CCI-002235	AC-6(10)	The information system prevents non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures.
CCI-002233	AC-6(8)	The information system prevents organization-defined software from executing at higher privilege levels than users executing the software.
CCI-002234	AC-6(9)	The information system audits the execution of privileged functions.

CCI-000044	AC-7(a)	The information system enforces the organization-defined limit of consecutive invalid logon attempts by a user during the organization-defined time period.
CCI-002238	AC-7(b)	The information system automatically locks the account or node for either an organization-defined time period, until the locked account or node is released by an administrator, or delays the next login prompt according to the organization-defined delay algorithm when the maximum number of unsuccessful attempts is exceeded.
CCI-000048	AC-8(a)	The information system displays an organization-defined system use notification message or banner before granting access to the system that provides privacy and security notices consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.
CCI-002247	AC-8(a)	The organization defines the use notification message or banner the information system displays to users before granting access to the system.
CCI-000050	AC-8(b)	The information system retains the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access.

CCI-001384	AC-8(c)(1)	The information system, for publicly accessible systems, displays system use information organization-defined conditions before granting further access.
CCI-001385	AC-8(c)(2)	The information system, for publicly accessible systems, displays references, if any, to monitoring that are consistent with privacy accommodations for such systems that generally prohibit those activities.
CCI-001386	AC-8(c)(2)	The information system for publicly accessible systems displays references, if any, to recording that are consistent with privacy accommodations for such systems that generally prohibit those activities.
CCI-001387	AC-8(c)(2)	The information system for publicly accessible systems displays references, if any, to auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities.
CCI-001388	AC-8(c)(3)	The information system, for publicly accessible systems, includes a description of the authorized uses of the system.

CCI-000174	AU-12(1)	The information system compiles audit records from organization-defined information system components into a system-wide (logical or physical) audit trail that is time-correlated to within organization defined level of tolerance for relationship between time stamps of individual records in the audit trail.
CCI-001914	AU-12(3)	The information system provides the capability for organization-defined individuals or roles to change the auditing to be performed on organization-defined information system components based on organization-defined selectable event criteria within organization-defined time thresholds.
CCI-000169	AU-12(a)	The information system provides audit record generation capability for the auditable events defined in AU-2 a at organization defined information system components.
CCI-000171	AU-12(b)	The information system allows organization-defined personnel or roles to select which auditable events are to be audited by specific components of the information system.
CCI-000172	AU-12(c)	The information system generates audit records for the events defined in AU-2 d with the content defined in AU-3.
CCI-001464	AU-14(1)	The information system initiates session audits at system start-up.

CCI-001462	AU-14(2)	The information system provides the capability for authorized users to capture/record and log content related to a user session.
CCI-001920	AU-14(3)	The information system provides the capability for authorized users to remotely view/hear all content related to an established user session in real time.
CCI-001919	AU-14(b)	The information system provides the capability for authorized users to select a user session to capture/record or view/hear.
CCI-000130	AU-3	The information system generates audit records containing information that establishes what type of event occurred.
CCI-000131	AU-3	The information system generates audit records containing information that establishes when an event occurred.
CCI-000132	AU-3	The information system generates audit records containing information that establishes where the event occurred.
CCI-000133	AU-3	The information system generates audit records containing information that establishes the source of the event.

CCI-000134	AU-3	The information system generates audit records containing information that establishes the outcome of the event.
CCI-001487	AU-3	The information system generates audit records containing information that establishes the identity of any individuals or subjects associated with the event.
CCI-000135	AU-3(1)	The information system generates audit records containing the organization-defined additional, more detailed information that is to be included in the audit records.
CCI-001849	AU-4	The organization allocates audit record storage capacity in accordance with organization-defined audit record storage requirements.
CCI-001851	AU-4(1)	The information system off-loads audit records per organization-defined frequency onto a different system or media than the system being audited.

CCI-001855	AU-5(1)	The information system provides a warning to organization-defined personnel, roles, and/or locations within organization-defined time period when allocated audit record storage volume reaches organization-defined percentage of repository maximum audit record storage capacity.
CCI-001864	AU-6(1)	The organization employs automated mechanisms to integrate audit review and analysis to support organizational processes for investigation of response to suspicious activities.
CCI-001865	AU-6(1)	The organization employs automated mechanisms to integrate reporting processes to support organizational investigation of and response to suspicious activities.
CCI-000154	AU-6(4)	The information system provides the capability to centrally review and analyze audit records from multiple components within the system.
CCI-000148	AU-6(a)	The organization reviews and analyzes information system audit records on an organization defined frequency for indications of organization-defined inappropriate or unusual activity.
CCI-000159	AU-8(a)	The information system uses internal system clocks to generate time stamps for audit records.

CCI-001889	AU-8(b)	The information system records time stamps for audit records that meets organization-defined granularity of time measurement.
CCI-001890	AU-8(b)	The information system records time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).
CCI-000162	AU-9	The information system protects audit information from unauthorized access.
CCI-000163	AU-9	The information system protects audit information from unauthorized modification.
CCI-000164	AU-9	The information system protects audit information from unauthorized deletion.
CCI-001493	AU-9	The information system protects audit tools from unauthorized access.
CCI-001494	AU-9	The information system protects audit tools from unauthorized modification.

CCI-001495	AU-9	The information system protects audit tools from unauthorized deletion.
CCI-002080	CA-3(5)	The organization employs either an allow-all, deny-by exception or deny-all, permit by exception policy for allowing organization-defined information systems to connect to external information systems.
CCI-001726	CM-10(a)	The organization uses software in accordance with contract agreements.
CCI-001728	CM-10(a)	The organization uses software in accordance with copyright laws.
CCI-001812	CM-11(2)	The information system prohibits user installation of software without explicit privileged status.
CCI-001807	CM-11(b)	The organization enforces software installation policies through organization-defined methods.
CCI-000327	CM-3(2)	The organization tests changes to the information system before implementing the changes on the operational system.
CCI-000328	CM-3(2)	The organization validates changes to the information system before implementing the changes on the operational system.
CCI-001746	CM-3(6)	The organization ensures that cryptographic mechanisms used to provide organization-defined security safeguards are under configuration management.
CCI-000340	CM-5	The organization approves physical access restrictions associated with changes to the information system.

CCI-000341	CM-5	The organization enforces physical access restrictions associated with changes to the information system.
CCI-001762	CM-7(1)(b)	The organization disables organization-defined functions, ports, protocols, and services within the information system deemed to be unnecessary and/or nonsecure.
CCI-001764	CM-7(2)	The information system prevents program execution in accordance with organization-defined policies regarding software program usage and restrictions, and/or rules authorizing the terms and conditions of software program usage.
CCI-001774	CM-7(5)b	The organization employs a deny-all, permit-by-exception policy to allow the execution of authorized software programs on the information system.
CCI-000381	CM-7(a)	The organization configures the information system to provide only essential capabilities.
CCI-000382	CM-7(b)	The organization configures the information system to prohibit or restrict the use of organization-defined functions, ports, protocols, and/or services.
CCI-000411	CM-8(2)	The organization employs automated mechanisms to help maintain an up-to-date inventory of information system components.
CCI-000412	CM-8(2)	The organization employs automated mechanisms to help maintain a complete inventory of information system components.
CCI-000413	CM-8(2)	The organization employs automated mechanisms to help maintain an accurate inventory of information system components.
CCI-000414	CM-8(2)	The organization employs automated mechanisms to help maintain a readily available inventory of information system components.

CCI-000416	CM-8(3)(a)	The organization employs automated mechanisms, per organization defined frequency, to detect the presence of unauthorized hardware, software, and firmware components within the information system.
CCI-001784	CM-8(3)(b)	When unauthorized hardware, software, and firmware components are detected within the information system, the organization takes action to disable network access by such components, isolates the components, and/or notifies organization-defined personnel or roles.
CCI-001792	CM-9(b)	The organization implements a configuration management plan for the information system that establishes a process for identifying configuration items throughout the system development life cycle.
CCI-001795	CM-9(b)	The organization implements a configuration management plan for the information system that establishes a process for managing the configuration of the configuration items.
CCI-000426	CM-9(c)	The organization implements a configuration management plan for the information system that defines the configuration items for the information system.
CCI-001801	CM-9(d)	The organization implements a configuration management plan for the information system that protects the configuration management plan from unauthorized disclosure and modification.
CCI-002855	CP-12	The information system, when organization-defined conditions are detected, enters a safe mode of operation with organization-defined restrictions of safe mode of operation.

CCI-000505	CP-6(a)	The organization establishes an alternate storage site including necessary agreements to permit the storage and retrieval of information system backup information.
CCI-000542	CP-9(1)	The organization tests backup information per organization-defined frequency to verify media reliability and information integrity.
CCI-000540	CP-9(d)	The organization protects the confidentiality, integrity, and availability of backup information at storage locations.
CCI-000764	IA-2	The information system uniquely identifies and authenticates organizational users (or processes acting on behalf of organizational users).
CCI-000765	IA-2(1)	The information system implements multifactor authentication for network access to privileged accounts.
CCI-001948	IA-2(11)	The information system implements multifactor authentication for remote access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access.
CCI-001949	IA-2(11)	The device used in the information system implementation of multifactor authentication for remote access to privileged accounts meets organization-defined strength of mechanism requirements.
CCI-001951	IA-2(11)	The information system implements multifactor authentication for remote access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access.

CCI-001952	IA-2(11)	The device used in the information system implementation of multifactor authentication for remote access to non-privileged accounts meets organization-defined strength of mechanism requirements.
CCI-001953	IA-2(12)	The information system accepts Personal Identity Verification (PIV) credentials.
CCI-001954	IA-2(12)	The information system electronically verifies Personal Identity Verification (PIV) credentials.
CCI-000766	IA-2(2)	The information system implements multifactor authentication for network access to non-privileged accounts.
CCI-000767	IA-2(3)	The information system implements multifactor authentication for local access to privileged accounts.
CCI-000768	IA-2(4)	The information system implements multifactor authentication for local access to non-privileged accounts.

CCI-000770	IA-2(5)	The organization requires individuals to be authenticated with an individual authenticator when a group authenticator is employed.
CCI-001941	IA-2(8)	The information system implements replay-resistant authentication mechanisms for network access to privileged accounts.
CCI-001942	IA-2(9)	The information system implements replay-resistant authentication mechanisms for network access to non-privileged accounts.
CCI-000778	IA-3	The information system uniquely identifies an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.
CCI-001958	IA-3	The information system authenticates an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.

CCI-001967	IA-3(1)	The information system authenticates organization-defined devices and/or types of devices before establishing a local, remote and/or network connection using bidirectional authentication that is cryptographically based.
CCI-000192	IA-5(1)(a)	The information system enforces password complexity by the minimum number of upper case characters used.
CCI-000193	IA-5(1)(a)	The information system enforces password complexity by the minimum number of lower case characters used.
CCI-000194	IA-5(1)(a)	The information system enforces password complexity by the minimum number of numeric characters used.
CCI-000205	IA-5(1)(a)	The information system enforces minimum password length.
CCI-001619	IA-5(1)(a)	The information system enforces password complexity by the minimum number of special characters used.
CCI-000195	IA-5(1)(b)	The information system, for password-based authentication, when new passwords are created, enforces that at least an organization-defined number of characters are changed.

CCI-000196	IA-5(1)(c)	The information system, for password-based authentication, stores only cryptographically-protected passwords.
CCI-000197	IA-5(1)(c)	The information system, for password-based authentication, transmits only cryptographically-protected passwords.
CCI-000198	IA-5(1)(d)	The information system enforces minimum password lifetime restrictions.
CCI-000199	IA-5(1)(d)	The information system enforces maximum password lifetime restrictions.
CCI-000200	IA-5(1)(e)	The information system prohibits password reuse for the organization defined number of generations.
CCI-002041	IA-5(1)(f)	The information system allows the use of a temporary password for system logons with an immediate change to a permanent password.

CCI-002003	IA-5(11)	The information system, for token-based authentication, employs mechanisms that satisfy organization-defined token quality requirements.
CCI-002007	IA-5(13)	The information system prohibits the use of cached authenticators after an organization defined time period.
CCI-002008	IA-5(14)	The organization, for PKI-based authentication, employs a deliberate organization-wide methodology for managing the content of PKI trust stores installed across all platforms including networks, operating systems, browsers, and applications.
CCI-000185	IA-5(2)(a)	The information system, for PKI-based authentication validates certifications by constructing and verifying a certification path to an accepted trust anchor including checking certificate status information.
CCI-000186	IA-5(2)(b)	The information system, for PKI-based authentication enforces authorized access to the corresponding private key.

CCI-000187	IA-5(2)(c)	The information system, for PKI-based authentication, maps the authenticated identity to the account of the individual or group.
CCI-001997	IA-5(4)	The organization employs automated tools to determine if password authenticators are sufficiently strong to satisfy organization-defined requirements.
CCI-000803	IA-7	The information system implements mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication.
CCI-000804	IA-8	The information system uniquely identifies and authenticates non-organizational users (or processes acting on behalf of non-organizational users).
CCI-002009	IA-8(1)	The information system accepts Personal Identity Verification (PIV) credentials from other federal agencies.

CCI-002010	IA-8(1)	The information system electronically verifies Personal Identity Verification (PIV) credentials from other federal agencies.
CCI-002011	IA-8(2)	The information system accepts FICAM-approved third-party credentials.
CCI-002013	IA-8(3)	The organization employs only FICAM-approved information system components in organization-defined information systems to accept third-party credentials.
CCI-002014	IA-8(4)	The information system conforms to FICAM-issued profiles.
CCI-000840	IR-7(1)	The organization employs automated mechanisms to increase the availability of incident response-related information and support.

CCI-000841	IR-7(2)(a)	The organization establishes a direct, cooperative relationship between its incident response capability and external providers of information system protection capability.
CCI-000849	IR-8(d)	The organization updates the incident response plan to address system/organizational changes or problems encountered during plan implementation, execution, or testing.
CCI-002804	IR-8(f)	The organization protects the incident response plan from unauthorized disclosure and modification.
CCI-002813	IR-9(1)	The organization assigns organization-defined personnel or roles with responsibility for responding to information spills.
CCI-002818	IR-9(3)	The organization implements organization-defined procedures to ensure that organizational personnel impacted by information spills can continue to carry out assigned tasks while contaminated systems are undergoing corrective actions.
CCI-002866	MA-2(a)	The organization schedules maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.
CCI-002867	MA-2(a)	The organization performs maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.
CCI-002868	MA-2(a)	The organization documents maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.
CCI-002869	MA-2(a)	The organization reviews records of maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

CCI-002870	MA-2(a)	The organization schedules repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements
CCI-002871	MA-2(a)	The organization performs repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.
CCI-002872	MA-2(a)	The organization documents repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.
CCI-002873	MA-2(a)	The organization reviews records of repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.
CCI-000859	MA-2(b)	The organization approves and monitors all maintenance activities, whether performed on site or remotely and whether the equipment is serviced on site or removed to another location.
CCI-000860	MA-2(c)	The organization requires that organization-defined personnel or roles explicitly approve the removal of the information system or system components from organizational facilities for off-site maintenance or repairs.
CCI-000861	MA-2(d)	The organization sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs.
CCI-002884	MA-4(1)(a)	The organization audits nonlocal maintenance and diagnostic sessions' organization-defined audit events.

CCI-002886	MA-4(1)(b)	The organization reviews the records of the nonlocal maintenance and diagnostic sessions.
CCI-002890	MA-4(6)	The information system implements cryptographic mechanisms to protect the integrity of nonlocal maintenance and diagnostic communications.
CCI-003123	MA-4(6)	The information system implements cryptographic mechanisms to protect the confidentiality of nonlocal maintenance and diagnostic communications.
CCI-002891	MA-4(7)	The information system implements remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.
CCI-000877	MA-4(c)	The organization employs strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions.
CCI-000879	MA-4(e)	The organization terminates sessions and network connections when nonlocal maintenance is completed.
CCI-001018	MP-4(b)	The organization protects information system media until the media are destroyed or sanitized using approved equipment, techniques, and procedures.

CCI-001027	MP-5(4)	The information system implements cryptographic mechanisms to protect the confidentiality and integrity of information stored on digital media during transport outside of controlled areas.
CCI-001020	MP-5(a)	The organization protects and controls organization-defined types of information system media during transport outside of controlled areas using organization-defined security safeguards.
CCI-001025	MP-5(c)	The organization documents activities associated with the transport of information system media.
CCI-002585	MP-7(1)	The organization prohibits the use of portable storage devices in organization information systems when such devices have no identifiable owner.
CCI-000959	PE-10(c)	The organization protects emergency power shutoff capability from unauthorized activation.
CCI-002955	PE-11	The organization provides a short-term uninterruptible power supply to facilitate an orderly shutdown of the information system and/or transition of the information system to long-term alternate power in the event of a primary power source loss.
CCI-000963	PE-12	The organization employs and maintains automatic emergency lighting for the information system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility.

CCI-000965	PE-13	The organization employs and maintains fire suppression and detection devices/systems for the information system that are supported by an independent energy source.
CCI-000971	PE-14(a)	The organization maintains temperature and humidity levels within the facility where the information system resides at organization-defined acceptable levels.
CCI-000973	PE-14(b)	The organization monitors temperature and humidity levels in accordance with organization-defined frequency.
CCI-000977	PE-15	The organization protects the information system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible.
CCI-000978	PE-15	The organization protects the information system from damage resulting from water leakage by providing master shutoff or isolation valves that are working properly.
CCI-000979	PE-15	Key personnel have knowledge of the master water shutoff or isolation valves.

CCI-000981	PE-16	The organization authorizes organization-defined types of information system components entering and exiting the facility.
CCI-000912	PE-2(a)	The organization develops a list of individuals with authorized access to the facility where the information system resides.
CCI-002910	PE-2(a)	The organization approves a list of individuals with authorized access to the facility where the information system resides.
CCI-002911	PE-2(a)	The organization maintains a list of individuals with authorized access to the facility where the information system resides.
CCI-000913	PE-2(b)	The organization issues authorization credentials for facility access.
CCI-001635	PE-2(c)	The organization removes individuals from the facility access list when access is no longer required.
CCI-000919	PE-3(a)	The organization enforces physical access authorizations at organization-defined entry/exit points to the facility where the information system resides.
CCI-000921	PE-3(a)(2)	The organization controls ingress/egress to the facility using one or more organization-defined physical access control systems/devices or guards.
CCI-002917	PE-3(b)	The organization maintains physical access audit logs for organization-defined entry/exit points.
CCI-002919	PE-3(c)	The organization provides organization-defined security safeguards to control access to areas within the facility officially designated as publicly accessible.

CCI-002921	PE-3(d)	The organization escorts visitors during organization-defined circumstances requiring visitor escorts.
CCI-002923	PE-3(d)	The organization monitors visitor activity during organization-defined circumstances requiring visitor monitoring.
CCI-000923	PE-3(e)	The organization secures keys, combinations, and other physical access devices.
CCI-002950	PE-6(4)	The organization monitors physical access to the information system in addition to the physical access monitoring of the facility as organization-defined physical spaces containing one or more components of the information system.
CCI-000939	PE-6(b)	The organization reviews physical access logs in accordance with organization-defined frequency.
CCI-002940	PE-6(b)	The organization reviews physical access logs upon occurrence of organization-defined events or potential indications of events
CCI-000941	PE-6(c)	The organization coordinates results of reviews and investigations with the organizations incident response capability.
CCI-000947	PE-8(a)	The organization maintains visitor access records to the facility where the information system resides for organization-defined time period.
CCI-002996	PM-12	The organization implements an insider threat program that includes a cross-discipline insider threat incident handling team.

CCI-002999	PM-14(a)(1)	The organization implements a process for ensuring that organizational plans for conducting security testing activities associated with organizational information systems are maintained.
CCI-003001	PM-14(a)(1)	The organization implements a process for ensuring that organizational plans for conducting security training activities associated with organizational information systems are maintained.
CCI-003003	PM-14(a)(1)	The organization implements a process for ensuring that organizational plans for conducting security monitoring activities associated with organizational information systems are maintained.
CCI-003004	PM-14(a)(2)	The organization implements a process for ensuring that organizational plans for conducting security testing associated with organizational information systems continue to be executed in a timely manner.
CCI-003005	PM-14(a)(2)	The organization implements a process for ensuring that organizational plans for conducting security training associated with organizational information systems continue to be executed in a timely manner.
CCI-003006	PM-14(a)(2)	The organization implements a process for ensuring that organizational plans for conducting security monitoring activities associated with organizational information systems continue to be executed in a timely manner.
CCI-003007	PM-14(b)	The organization reviews testing plans for consistency with the organizational risk management strategy and organization-wide priorities for risk response actions.

CCI-003009	PM-14(b)	The organization reviews monitoring plans for consistency with the organizational risk management strategy and organization-wide priorities for risk response actions.
CCI-003013	PM-16	The organization implements a threat awareness program that includes a cross-organization information-sharing capability.
CCI-001514	PS-2(c)	The organization reviews and updates position risk designations in accordance with organization-defined frequency.
CCI-001516	PS-3(a)	The organization screens individuals prior to authorizing access to the information system.
CCI-001517	PS-3(b)	The organization rescreens individuals with authorized access to the information system according to organization-defined conditions requiring rescreening, and where rescreening is so indicated, the organization-defined frequency of such rescreening.
CCI-003027	PS-4(1)(a)	The organization notifies terminated individuals of applicable, legally binding post-employment requirements for the protection of organizational information.
CCI-003028	PS-4(1)(b)	The organization requires terminated individuals to sign an acknowledgment of post-employment requirements as part of the organizational termination process.
CCI-001523	PS-4(c)	The organization, upon termination of individual employment, conducts exit interviews that include a discussion of organization-defined information security topics.
CCI-001524	PS-4(d)	The organization, upon termination of individual employment, retrieves all security-related organizational information systems-related property.

CCI-001525	PS-4(e)	The organization, upon termination of individual employment, retains access to organizational information formerly controlled by terminated individual.
CCI-001526	PS-4(e)	The organization, upon termination of individual employment, retains access to organizational information systems formerly controlled by terminated individual.
CCI-003016	PS-4(f)	The organization, upon termination of individual employment, notifies organization-defined personnel or roles within an organization-defined time period.
CCI-001528	PS-5(b)	The organization initiates organization-defined transfer or reassignment actions within an organization-defined time period following the formal personnel transfer action.
CCI-003031	PS-5(c)	The organization modifies access authorization as needed to correspond with any changes in operational need due to reassignment or transfer.
CCI-003032	PS-5(d)	The organization notifies organization-defined personnel or roles within an organization-defined time period when individuals are transferred or reassigned to other positions within the organization.

CCI-003039	PS-6(3)(b)	The organization requires individuals to sign an acknowledgement of legally binding post-employment requirements for protection of organizational information, if applicable, as part of granting initial access to covered information.
CCI-001531	PS-6(c)(1)	The organization ensures that individuals requiring access to organizational information and information systems sign appropriate access agreements prior to being granted access.
CCI-003036	PS-6(c)(2)	The organization ensures that individuals requiring access to organizational information and information systems re-sign access agreements to maintain access to organizational information systems when access agreements have been updated or in accordance with organization-defined frequency.
CCI-003040	PS-7(b)	The organization requires third-party providers to comply with personnel security policies and procedures established by the organization.
CCI-003041	PS-7(d)	The organization requires third-party providers to notify organization-defined personnel or roles of any personnel transfers or terminations of third-party personnel who possess organizational credentials and/or badges, or who have information system privileges within an organization-defined time period.
CCI-001541	PS-7(e)	The organization monitors third-party provider compliance with personnel security requirements.

CCI-003044	PS-8(b)	The organization notifies organization-defined personnel or roles within an organization-defined time period when a formal employee sanctions process is initiated, identifying the individual sanctioned and the reason for the sanction.
CCI-001045	RA-2(a)	The organization categorizes information and the information system in accordance with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.
CCI-001047	RA-2(c)	The organization ensures the security categorization decision is reviewed and approved by the authorizing official or authorizing official designated representative.
CCI-001067	RA-5(5)	The information system implements privileged access authorization to organization-identified information system components for selected organization-defined vulnerability scanning activities.
CCI-000723	SA-12	The organization protects against supply chain threats to the information system, system component, or information system service by employing organization-defined security safeguards as part of a comprehensive, defense-in-breadth information security strategy.

CCI-003233	SA-15	The organization requires the developer of the information system, system component, or information system service to follow a documented development process.
CCI-003234	SA-15(a)(1)	The documented information system, system component, or information system service development process explicitly addresses security requirements.
CCI-003235	SA-15(a)(2)	The documented information system, system component, or information system service development process identifies the standards used in the development process.
CCI-003236	SA-15(a)(2)	The documented information system, system component, or information system service development process identifies the tools used in the development process.
CCI-003237	SA-15(a)(3)	The documented information system, system component, or information system service development process documents the specific tool options and tool configurations used in the development process.
CCI-003238	SA-15(a)(4)	The documented information system, system component, or information system service development process documents changes to the process and/or tools used in development.
CCI-003239	SA-15(a)(4)	The documented information system, system component, or information system service development process manages changes to the process and/or tools used in development.
CCI-003240	SA-15(a)(4)	The documented information system, system component, or information system service development process ensures the integrity of changes to the process and/or tools used in development.

CCI-003241	SA-15(b)	The organization reviews the development process in accordance with organization-defined frequency to determine if the development process selected and employed can satisfy organization-defined security requirements.
CCI-003242	SA-15(b)	The organization reviews the development standards in accordance with organization-defined frequency to determine if the development standards selected and employed can satisfy organization-defined security requirements.
CCI-003243	SA-15(b)	The organization reviews the development tools in accordance with organization-defined frequency to determine if the development tools selected and employed can satisfy organization-defined security requirements.
CCI-003244	SA-15(b)	The organization reviews the development tool options/configurations in accordance with organization-defined frequency to determine if the development tool options/configurations selected and employed can satisfy organization-defined security requirements.
CCI-003356	SA-19(a)	The organization develops anti-counterfeit policy that include the means to detect counterfeit components from entering the information system.
CCI-003357	SA-19(a)	The organization develops anti-counterfeit policy that include the means to prevent counterfeit components from entering the information system.

CCI-003358	SA-19(a)	The organization develops anti-counterfeit procedures that include the means to detect counterfeit components from entering the information system.
CCI-003359	SA-19(a)	The organization develops anti-counterfeit procedures that include the means to prevent counterfeit components from entering the information system.
CCI-003360	SA-19(a)	The organization implements anti-counterfeit policy that include the means to detect counterfeit components from entering the information system.
CCI-003361	SA-19(a)	The organization implements anti-counterfeit policy that include the means to prevent counterfeit components from entering the information system.
CCI-003362	SA-19(a)	The organization implements anti-counterfeit procedures that include the means to detect counterfeit components from entering the information system.
CCI-003363	SA-19(a)	The organization implements anti-counterfeit procedures that include the means to prevent counterfeit components from entering the information system.
CCI-003364	SA-19(b)	The organization reports counterfeit information system components to source of counterfeit component, organization-defined external reporting organizations and/or organization-defined personnel or roles.
CCI-000610	SA-2(b)	The organization determines the resources required to protect the information system or information system service as part of its capital planning and investment control process.

CCI-000611	SA-2(b)	The organization documents the resources required to protect the information system or information system service as part of its capital planning and investment control process.
CCI-000612	SA-2(b)	The organization allocates the resources required to protect the information system or information system service as part of its capital planning and investment control process.
CCI-000613	SA-2(c)	The organization establishes a discrete line item for information security in organizational programming documentation.
CCI-000614	SA-2(c)	The organization establishes a discrete line item for information security in organizational budgeting documentation.
CCI-000615	SA-3(a)	The organization manages the information system using organization-defined system development life cycle that incorporates information security considerations.
CCI-003092	SA-3(a)	The organization defines a system development life cycle that is used to manage the information system.
CCI-000616	SA-3(b)	The organization defines and documents information system security roles and responsibilities throughout the system development life cycle.
CCI-000618	SA-3(c)	The organization identifies individuals having information system security roles and responsibilities.
CCI-003093	SA-3(d)	The organization integrates the organizational information security risk management process into system development life cycle activities.
CCI-003140	SA-9(1)(a)	The organization conducts an organizational assessment of risk prior to the acquisition or outsourcing of dedicated information security services.

CCI-003141 SA-9(1)(b) The organization ensures that the acquisition or outsourcing of dedicated information security services is approved by organization-defined personnel or roles.

CCI-001133 SC-10 The information system terminates the network connection associated with a communications session at the end of the session or after an organization-defined time period of inactivity.

CCI-001152 SC-15(b) The information system provides an explicit indication of use to users physically present at collaborative computing devices.

CCI-002465 SC-21 The information system requests data origin authentication verification on the name/address resolution responses the system receives from authoritative sources.

CCI-002466 SC-21 The information system requests data integrity verification on the name/address resolution responses the system receives from authoritative sources.

CCI-002467 SC-21 The information system performs data integrity verification on the name/address resolution responses the system receives from authoritative sources.

CCI-002468 SC-21 The information system performs data origin verification authentication on the name/address resolution responses the system receives from authoritative sources.

CCI-001182 SC-22 The information systems that collectively provide name/address resolution service for an organization are fault-tolerant.

CCI-001183 SC-22 The information systems that collectively provide name/address resolution service for an organization implement internal/external role separation.

CCI-001184 SC-23 The information system protects the authenticity of communications sessions.

CCI-001185	SC-23(1)	The information system invalidates session identifiers upon user logout or other session termination.
CCI-001188	SC-23(3)	The information system generates unique session identifiers for each session with organization-defined randomness requirements.
CCI-001664	SC-23(3)	The information system recognizes only session identifiers that are system-generated.
CCI-002470	SC-23(5)	The information system only allows the use of organization-defined certificate authorities for verification of the establishment of protected sessions.
CCI-001190	SC-24	The information system fails to an organization-defined known-state for organization-defined types of failures.

CCI-001665	SC-24	The information system preserves organization-defined system state information in the event of a system failure.
CCI-001199	SC-28	The information system protects the confidentiality and/or integrity of organization-defined information at rest.
CCI-002475	SC-28(1)	The information system implements cryptographic mechanisms to prevent unauthorized modification of organization-defined information at rest on organization-defined information system components.
CCI-002476	SC-28(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of organization-defined information at rest on organization-defined information system components.
CCI-002529	SC-38	The organization employs organization-defined operations security safeguards to protect key organizational information throughout the system development life cycle.

CCI-002530	SC-39	The information system maintains a separate execution domain for each executing process.
CCI-002546	SC-41	The organization physically disables or removes organization-defined connection ports or input/output devices on organization-defined information systems or information system components.
CCI-002385	SC-5	The information system protects against or limits the effects of organization-defined types of denial of service attacks by employing organization-defined security safeguards.
CCI-001094	SC-5(1)	The information system restricts the ability of individuals to launch organization-defined denial of service attacks against other information systems.
CCI-001095	SC-5(2)	The information system manages excess capacity, bandwidth, or other redundancy to limit the effects of information flooding types of denial of service attacks.
CCI-002389	SC-5(3)(a)	The organization employs organization-defined list of monitoring tools to detect indicators of denial of service attacks against the information system.
CCI-001116	SC-7(10)	The organization prevents the unauthorized exfiltration of information across managed interfaces.

CCI-002403	SC-7(11)	The information system only allows incoming communications from organization-defined authorized sources routed to organization-defined authorized destinations.
CCI-002406	SC-7(12)	The organization implements organization-defined host-based boundary protection mechanisms at organization-defined information system components.
CCI-001119	SC-7(13)	The organization isolates organization-defined information security tools, mechanisms, and support components from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.
CCI-001121	SC-7(14)	The organization protects against unauthorized physical connections at organization-defined managed interfaces.
CCI-001101	SC-7(3)	The organization limits the number of external network connections to the information system.
CCI-001102	SC-7(4)(a)	The organization implements a managed interface for each external telecommunication service.
CCI-001103	SC-7(4)(b)	The organization establishes a traffic flow policy for each managed interface for each external telecommunication service.

CCI-002396	SC-7(4)(c)	The organization protects the confidentiality and integrity of the information being transmitted across each interface for each external telecommunication service.
CCI-001109	SC-7(5)	The information system at managed interfaces denies network communications traffic by default and allows network communications traffic by exception (i.e., deny all, permit by exception).
CCI-002397	SC-7(7)	The information system, in conjunction with a remote device, prevents the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks.
CCI-001112	SC-7(8)	The information system routes organization-defined internal communications traffic to organization-defined external networks through authenticated proxy servers at managed interfaces.
CCI-002398	SC-7(9)(a)	The information system detects outgoing communications traffic posing a threat to external information systems.
CCI-002399	SC-7(9)(a)	The information system denies outgoing communications traffic posing a threat to external information systems.

CCI-002400	SC-7(9)(b)	The information system audits the identity of internal users associated with denied outgoing communications traffic posing a threat to external information systems.
CCI-001097	SC-7(a)	The information system monitors and controls communications at the external boundary of the system and at key internal boundaries within the system.
CCI-002395	SC-7(b)	The information system implements subnetworks for publicly accessible system components that are physically and/or logically separated from internal organizational networks.
CCI-001098	SC-7(c)	The information system connects to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.
CCI-002418	SC-8	The information system protects the confidentiality and/or integrity of transmitted information.
CCI-002421	SC-8(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of information and/or detect changes to information during transmission unless otherwise protected by organization-defined alternative physical safeguards.

CCI-002420	SC-8(2)	The information system maintains the confidentiality and/or integrity of information during preparation for transmission.
CCI-002422	SC-8(2)	The information system maintains the confidentiality and/or integrity of information during reception.
CCI-001310	SI-10	The information system checks the validity of organization-defined inputs.
CCI-001312	SI-11(a)	The information system generates error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.
CCI-001314	SI-11(b)	The information system reveals error messages only to organization-defined personnel or roles.
CCI-002824	SI-16	The information system implements organization-defined security safeguards to protect its memory from unauthorized code execution.

CCI-002775	SI-17	The information system implements organization-defined fail-safe procedures when organization-defined failure conditions occur.
CCI-001231	SI-2(1)	The organization centrally manages the flaw remediation process.
CCI-002617	SI-2(6)	The organization removes organization-defined software components (e.g., previous versions) after updated versions have been installed.
CCI-002618	SI-2(6)	The organization removes organization-defined firmware components (e.g., previous versions) after updated versions have been installed.
CCI-001225	SI-2(a)	The organization identifies information system flaws.
CCI-001226	SI-2(a)	The organization reports information system flaws.
CCI-001228	SI-2(b)	The organization tests software updates related to flaw remediation for effectiveness before installation.

CCI-001229	SI-2(b)	The organization tests software updates related to flaw remediation for potential side effects before installation.
CCI-002602	SI-2(b)	The organization tests firmware updates related to flaw remediation for effectiveness before installation.
CCI-002603	SI-2(b)	The organization tests firmware updates related to flaw remediation for potential side effects before installation.
CCI-002605	SI-2(c)	The organization installs security-relevant software updates within organization-defined time period of the release of the updates
CCI-002607	SI-2(c)	The organization installs security-relevant firmware updates within organization-defined time period of the release of the updates
CCI-002636	SI-3(10)(a)	The organization employs organization-defined tools to analyze the characteristics and behavior of malicious code.
CCI-002638	SI-3(10)(a)	The organization employs organization-defined techniques to analyze the characteristics and behavior of malicious code.
CCI-001247	SI-3(2)	The information system automatically updates malicious code protection mechanisms.

CCI-002621	SI-3(a)	The organization employs malicious code protection mechanisms at information system entry points to eradicate malicious code.
CCI-002622	SI-3(a)	The organization employs malicious code protection mechanisms at information system exit points to eradicate malicious code.
CCI-001240	SI-3(b)	The organization updates malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures.
CCI-001241	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform periodic scans of the information system on an organization-defined frequency.
CCI-001242	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform real-time scans of files from external sources at endpoints as the files are downloaded, opened, or executed in accordance with organizational security policy.
CCI-002624	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform real-time scans of files from external sources at network entry/exit points as the files are downloaded, opened, or executed in accordance with organizational security policy.

CCI-001243	SI-3(c)(2)	The organization configures malicious code protection mechanisms to perform organization-defined action(s) in response to malicious code detection.
CCI-001245	SI-3(d)	The organization addresses the receipt of false positives during malicious code detection and eradication, and the resulting potential impact on the availability of the information system.
CCI-001273	SI-4(11)	The organization analyzes outbound communications traffic at the external boundary of the information system to discover anomalies.
CCI-001671	SI-4(11)	The organization analyzes outbound communications traffic at selected organization-defined interior points within the system (e.g., subnetworks, subsystems) to discover anomalies.
CCI-001274	SI-4(12)	The organization employs automated mechanisms to alert security personnel of an organization-defined inappropriate or unusual activities with security implications.
CCI-001282	SI-4(15)	The organization employs an intrusion detection system to monitor wireless communications traffic as the traffic passes from wireless to wireline networks.
CCI-001283	SI-4(16)	The organization correlates information from monitoring tools employed throughout the information system.

CCI-002675	SI-4(19)	The organization implements organization-defined additional monitoring of individuals who have been identified by organization-defined sources as posing an increased level of risk.
CCI-001260	SI-4(2)	The organization employs automated tools to support near real-time analysis of events.
CCI-002677	SI-4(20)	The organization implements organization-defined additional monitoring of privileged users.
CCI-002683	SI-4(22)	The information system detects network services that have not been authorized or approved by the organization-defined authorization or approval processes.
CCI-002684	SI-4(22)	The information system audits and/or alerts organization-defined personnel when unauthorized network services are detected.
CCI-002687	SI-4(23)	The organization implements organization-defined host-based monitoring mechanisms at organization-defined information system components.
CCI-002661	SI-4(4)	The information system monitors inbound communications traffic per organization-defined frequency for unusual or unauthorized activities or conditions.
CCI-002662	SI-4(4)	The information system monitors outbound communications traffic per organization-defined frequency for unusual or unauthorized activities or conditions.

CCI-002664	SI-4(5)	The information system alerts organization-defined personnel or roles when organization-defined compromise indicators reflect the occurrence of a compromise or a potential compromise.
CCI-002641	SI-4(a)(1)	The organization monitors the information system to detect attacks and indicators of potential attacks in accordance with organization-defined monitoring objectives.
CCI-002642	SI-4(a)(2)	The organization monitors the information system to detect unauthorized local connections.
CCI-002643	SI-4(a)(2)	The organization monitors the information system to detect unauthorized network connections.
CCI-002644	SI-4(a)(2)	The organization monitors the information system to detect unauthorized remote connections.
CCI-001255	SI-4(c)	The organization deploys monitoring devices strategically within the information system to collect organization determined essential information.
CCI-001256	SI-4(c)	The organization deploys monitoring devices at ad hoc locations within the system to track specific types of transactions of interest to the organization.

CCI-002647	SI-4(d)	The organization protects information obtained from intrusion-monitoring tools from unauthorized access.
CCI-002648	SI-4(d)	The organization protects information obtained from intrusion-monitoring tools from unauthorized modification.
CCI-002649	SI-4(d)	The organization protects information obtained from intrusion-monitoring tools from unauthorized deletion.
CCI-001285	SI-5(a)	The organization receives information system security alerts, advisories, and directives from organization-defined external organizations on an ongoing basis.
CCI-001289	SI-5(d)	The organization implements security directives in accordance with established time frames, or notifies the issuing organization of the degree of noncompliance.
CCI-002737	SI-7(14)(a)	The organization prohibits the use of binary or machine-executable code from sources with limited or no warranty and without the provision of source code.
CCI-002723	SI-7(8)	The information system, upon detection of a potential integrity violation, provides the capability to audit the event.
CCI-002724	SI-7(8)	The information system, upon detection of a potential integrity violation, initiates one or more of following actions: generates an audit record; alerts current user; alerts organization-defined personnel or roles; and/or organization-defined other actions.

5 05 11 - Control Correlation Identifiers

ritical Systems Moderate-Moderate-Moderate

Designer Implementation

The organization being inspected/assessed configures the information system to limit the number of concurrent sessions for all accounts and/or account types to a number of sessions defined in AC-10, CCI 55.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 54.

DoD has defined the account types and/or accounts as all account types and/or accounts.

The organization being inspected/assessed configures the information system to conceal, via the session lock, information previously visible on the display with a publicly viewable image.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 60.

The organization being inspected/assessed configures the information system to initiate a session lock after the organization-defined time period defined in CCI 59.

The organization being inspected/assessed configures the information system to provide the capability for users to directly initiate session lock mechanisms.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 58.

The organization being inspected/assessed configures the information system to retain the session lock until the user reestablishes access using established identification and authentication procedures.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 56.

The organization being inspected/assessed configures the information system to monitor remote access methods.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 67.

The organization being inspected/assessed configures the information system to control remote access methods.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2314.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the confidentiality of remote access sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 68.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the integrity of remote access sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1453.

The organization being inspected/assessed configures the information system to route all remote accesses through the number of managed network access control points defined in AC-17 (3), CCI 2315.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 69.

The organization being inspected/assessed configures the information system to provide the capability to expeditiously disconnect or disable remote access to the information system immediately.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2322.

DoD has defined the time period as immediately.

The organization being inspected/assessed authorizes remote access to the information system prior to allowing such connections.

The organization must maintain an audit trail of authorizations.

The organization being inspected/assessed configures the information system to protect wireless access to the system using authentication of users and/or devices.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1443.

The organization being inspected/assessed configures the information system to protect wireless access to the system using encryption.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1444.

The organization being inspected/assessed configures the information system to employ automated mechanisms to support the information system account management functions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 000015.

The organization being inspected/assessed configures the information system to terminate shared/group account credentials when members leave the group.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2142.

The organization being inspected/assessed configures the information system to automatically remove or disable temporary accounts after 72 hours.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 000016.

DoD has defined the time period as 72 hours.

The organization being inspected/assessed configures the information system to never automatically remove or disable emergency accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1682.

DoD has defined the time period as never.

The organization being inspected/assessed configures the information system to automatically audit account creation actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 18.

The organization being inspected/assessed configures the information system to automatically audit account modification actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1403.

The organization being inspected/assessed configures the information system to automatically audit account disabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1404.

The organization being inspected/assessed configures the information system to automatically audit account removal actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1405.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account creation actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1683.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account modification actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1684.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account disabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1685.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account removal actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1686.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to automatically audit account enabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2130.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account enabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2132.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 213.

The organization being inspected/assessed configures the information system to enforce the discretionary access control policies defined in AC-3 (4), CCI 2163 over defined subjects and objects.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2165.

The organization being inspected/assessed documents and implements the concept of least privilege, allowing only authorized accesses for users (and processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.

The organization being inspected/assessed documents and implements a process to explicitly authorize access to all security-relevant information not publicly available.

Explicit authorization can be in the form of an acceptable use policy signed by the user at the time of access being granted.

DoD has defined the security-relevant information as all security-relevant information not publicly available.

The organization being inspected/assessed configures the information system to prevent non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2235.

The organization being inspected/assessed configures the information system to any software except software explicitly documented from executing at higher privilege levels than users executing the software.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2233.

DoD has defined the software as any software except software explicitly documented.

The organization being inspected/assessed configures the information system to audit the execution of privileged functions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2234.

The organization being inspected/assessed configures the information system to limit invalid logon attempts by a user to three attempts during a 15 minute time period.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 44.

DoD has defined the maximum number as three.

DoD has defined the time period as 15 minutes.

The organization being inspected/assessed configures the information system to automatically lock the account or node until the locked account is released by an administrator and delays the next login prompt for a minimum of 5 seconds when the maximum number of unsuccessful attempts is exceeded.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2238.

DoD has defined the delay algorithm as a minimum of 5 seconds.

DoD has defined the time period as until released by an administrator.

The organization being inspected/assessed configures the information system to display the DoD Information Systems – Standard Consent Banner and User Agreement before granting access to the system that provides privacy and security notices consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 48.

DoD has defined the use notification message or banner as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems – Standard Consent Banner and User Agreement," March 2013.

DoD has defined the use notification message or banner as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems – Standard Consent Banner and User Agreement," March 2013.

The organization being inspected/assessed configures the information system to retain the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 50.

The organization being inspected/assessed configures the information system to display the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013 before granting further access for publicly accessible systems

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1384.

DoD has defined the conditions as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013.

The organization being inspected/assessed configures the information system to display references, if any, to monitoring that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1385.

The organization being inspected/assessed configures the information system to display references, if any, to recording that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1386.

The organization being inspected/assessed configures the information system to display references, if any, to auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1387.

The organization being inspected/assessed configures the information system to include a description of the authorized uses of the system for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1388.

The organization being inspected/assessed configures the information system to compile audit records from information system components defined in AU-12 (1), CCI 1577 into a system-wide (logical or physical) audit trail that is time-correlated to within the level of tolerance defined in AU-12 (1), CCI-000173 for relationship between time stamps of individual records in the audit trail.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 174.

The organization being inspected/assessed configures the information system to provide the capability for individuals or roles defined in AU-12 (3), CCI 1913 to change the auditing to be performed on information system components defined in AU-12 (3), CCI 2047 based on selectable event criteria defined in AU-12 (3), CCI 1911 within time thresholds defined in AU-12 (3), CCI 1912.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1914.

The organization being inspected/assessed acquires or designs all information system and network components that provide audit record generation capability for the auditable events defined in AU-2 a.

DoD has defined the information system components as all information system and network components.

The organization being inspected/assessed configures the information system to ensure that only the ISSM or individuals appointed by the ISSM select which auditable events are to be audited by specific components of the information system.

DoD has defined the personnel or roles as the ISSM or individuals appointed by the ISSM.

The organization being inspected/assessed configures the information system to generate audit records for the events defined in AU-2 d with the content defined in AU-3.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 172.

The organization being inspected/assessed configures the information system to initiate session audits at system start-up.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1464.

The organization being inspected/assessed configures the information system to provide the capability for authorized users to capture/record and log content related to a user session.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1462.

The organization being inspected/assessed configures the information system to provide the capability for authorized users to remotely view/hear all content related to an established user session in real time.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1920.

The organization being inspected/assessed configures the information system to provide the capability for authorized users to select a user session to capture/record or view/hear.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1919.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes what type of event occurred

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 130.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes when an event occurred

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 131.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes where the event occurred

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 132.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the source of the event.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 133.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the outcome of the event.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 134.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the identity of any individuals or subjects associated with the event.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1487.

The organization being inspected/assessed configures the information system to generate audit records containing the organization defined additional, more detailed information as defined in AU-3 (1), CCI 1488 that is to be included in the audit records.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 135.

The organization being inspected/assessed allocates, and configures the information system to allocate audit record storage capacity as defined in AU-4, CCI 1848.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1849.

The organization being inspected/assessed configures the information system to off-load audit records at a minimum, in real-time for interconnected systems and weekly for stand-alone systems onto a different system or media than the system being audited.

DoD has defined the frequency as at a minimum, real-time for interconnected systems and weekly for stand-alone systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1851.

The organization being inspected/assessed configures the information system to immediately provide a warning to personnel, roles, and/or locations defined in AU-5 (1), CCI 1852 when allocated audit record storage volume reaches 75 percent of repository maximum audit record storage capacity.

DoD has defined the time period as immediate.

DoD has defined the percentage as 75 percent.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1855.

The organization being inspected/assessed identifies and implements automated mechanisms to integrate audit review and analysis.

The goal is to support organizational investigation of and response to suspicious activities.

The organization being inspected/assessed identifies and implements automated mechanisms to integrate reporting processes (e.g., centralized log analysis tools).

The goal is to support organizational investigation of and response to suspicious activities.

The organization being inspected/assessed configures the information system to provide a capability to centrally review and analyze audit records from multiple components within the system.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 154.

The organization being inspected/assessed documents and implements a process to review and analyze information system audit records every seven days or more frequently if required by an alarm event or anomaly for indications of activity defined in AU-6, CCI 1862.

The organization must maintain an audit trail of the reviews.

DoD has defined the frequency as every seven days or more frequently if required by an alarm event or anomaly.

The organization being inspected/assessed configures the information system to use internal system clocks to generate time stamps for audit records.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 159.

The organization being inspected/assessed configures the information system to generate time in the time stamps for audit records that meets one second granularity of time measurement.

DoD has defined the granularity of time measurement as one second.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1889.

The organization being inspected/assessed configures the information system to generate time stamps for audit records that contain time zones or time offsets that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1890.

The organization being inspected/assessed configures the information system to disallow unauthorized access to audit information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 162.

The organization being inspected/assessed configures the information system to disallow unauthorized modification of audit information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 163.

The organization being inspected/assessed configures the information system to disallow unauthorized deletion of audit information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 164.

The organization being inspected/assessed configures the information system to disallow unauthorized access to audit tools.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1493.

The organization being inspected/assessed configures the information system to disallow unauthorized modification of audit tools.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1494.

The organization being inspected/assessed configures the information system to disallow unauthorized deletion of audit tools.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1495.

The organization being inspected/assessed configures the information system to employ a deny-all, permit by exception policy for allowing any systems requiring external connectivity to connect to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2080.

DoD has defined the information systems as any systems requiring external connectivity.

The organization being inspected/assessed uses software in accordance with contract agreements.

The organization being inspected/assessed uses software in accordance with copyright laws.

The organization being inspected/assessed must configure the information system to prevent the installation of software by non-privileged users.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1812.

The organization being inspected/assessed must enforce software installation policies as defined in CM-11, CCI 1804 through methods defined in CM-11, CCI 1806.

The organization being inspected/assessed documents and implements a process to test changes to the information system before implementing the changes on the operational system.

The organization must maintain an audit trail of testing activity.

The organization being inspected/assessed documents and implements a process to validate changes to the information system before implementing the changes on the operational system.

The organization must maintain an audit trail of validation activity.

The organization being inspected/assessed ensures that cryptographic mechanisms used to provide all security safeguards are under configuration management.

DoD has defined the security safeguards as all security safeguards.

The organization being inspected/assessed documents and implements a process to approve physical access restrictions associated with changes to the information system.

The organization must maintain an audit trail of approvals.

The organization being inspected/assessed documents and implements a process to enforce physical access restrictions associated with changes to the information system.

The organization being inspected/assessed must disable functions, ports, protocols, and services within the information system deemed to be unnecessary and/or nonsecure as defined in CM-7 (1), CCI 1761.

The organization being inspected/assessed configures the information system to prevent the execution of programs not authorized in accordance with CM-7 (2) CCIs 1592 and 1763.

The organization being inspected/assessed configures the information system to deny-all and only permit by exception the execution of authorized software programs on the information system.

The organization being inspected/assessed documents in the security plan, essential capabilities which the information system must provide. The organization being inspected/assessed configures the information system to provide only those documented essential capabilities.

The organization being inspected/assessed configures the information system to prohibit or restrict the use of functions, ports, protocols, and/or services IAW DoDI 8551.01.

DoD has defined the information system prohibited or restricted functions, ports, protocols, and/or services as IAW DoDI 8551.01.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain an up-to-date inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains an up-to-date inventory.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain a complete inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains a complete inventory.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain an accurate inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains an accurate inventory.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain a readily available inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains a readily available inventory.

The organization being inspected/assessed documents and implements automated mechanisms to detect the presence of unauthorized hardware, software, and firmware components within the information system continuously.

DoD has defined the frequency as continuously.

The organization being inspected/assessed documents and implements a process to take action to disable network access by unauthorized software, hardware, and firmware components, isolate the components, and/or notify the ISSO and ISSM and others as the local organization deems appropriate.

The organization must maintain an audit trail of actions taken upon detection of unauthorized software, hardware, and firmware components.

DoD has defined the personnel or roles as the ISSO and ISSM and others as the local organization deems appropriate.

The organization being inspected/assessed will implement a configuration management plan for the information system that establishes a process for identifying configuration items throughout the system development life cycle.

The organization being inspected/assessed will implement a configuration management plan that has a process for controlling changes to configuration items.

The organization being inspected/assessed will implement a configuration management plan for the information system that defines the configuration items.

The organization being inspected/assessed must implement a plan to protect the configuration management plan from unauthorized disclosure and modification.

Measures must include marking, labeling, and handling to prevent improper disclosure. The organization being inspected/assessed must ensure that all changes to the CM plan are approved.

The organization being inspected/assessed configures the information system to enter a safe mode of operation with restrictions of safe mode of operation defined in CP-12, CCI 2857 when conditions defined in CP-12, CCI 2856 are detected.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2855.

The organization being inspected/assessed establishes an alternate storage site and documents relevant information within the contingency plan.

The organization being inspected/assessed tests and logs backup information at least monthly in accordance with contingency plan to verify media reliability and information integrity.

The organization being inspected/assessed will protect the confidentiality, integrity, and availability of backup information at the storage location IAW the system security plan.

The organization being inspected/assessed configures the information system to uniquely identify and authenticate organizational users (or processes acting on behalf of organizational users).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 764.

The organization being inspected/assessed configures the information system to implement multifactor authentication for network access to privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 765.

The organization being inspected/assessed configures the information system to implement multifactor authentication for remote access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1948.

The organization being inspected/assessed will use DoD PKI or a technology approved by their Authorizing Official that meet Federal standards for authentication such as FIPS 140-2, NIAP Certification, or NSA approval.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1949.

The organization being inspected/assessed configures the information system to implement multifactor authentication for remote access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1951.

The organization being inspected/assessed will use DoD PKI or a technology approved by their Authorizing Official that meet Federal standards for authentication such as FIPS 140-2, NIAP Certification, or NSA approval.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1952.

The organization being inspected/assessed configures the information system to accept PIV/CAC authentication.

This control enhancement applies to organizations implementing logical access control systems (LACS) and physical access control systems (PACS).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1953

The organization being inspected/assessed configures the information system to verify PIV/CAC authentication.

This control enhancement applies to organizations implementing logical access control systems (LACS) and physical access control systems (PACS).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1954.

The organization being inspected/assessed configures the information system to implement multifactor authentication for network access to non-privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 766.

The organization being inspected/assessed configures the information system to implement multifactor authentication for local access to privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 767.

The organization being inspected/assessed configures the information system to implement multifactor authentication for local access to non-privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 768.

The organization being inspected/assessed requires individuals or configures the information system to require individuals to be authenticated with an individual authenticator when a group authenticator is employed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 770.

The organization being inspected/assessed configures the information system to implement replay-resistant authentication mechanisms for network access to privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1941.

The organization being inspected/assessed configures the information system to implement replay-resistant authentication mechanisms for network access to non-privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1942.

The organization being inspected/assessed configures the network infrastructure to identify all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 778.

DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).

The organization being inspected/assessed configures the network infrastructure to authenticate all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1958.

DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).

The organization being inspected/assessed configures the information system to use cryptographically based bidirectional authentication.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1967.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of upper case characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 192.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of lower case characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 193.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of numeric characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 194.

The organization being inspected/assessed configures the information system to enforce minimum password length.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 205.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of special characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1619.

The organization being inspected/assessed configures the information system to enforce that at least 50% of the minimum password length is changed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 195.

DoD has defined the minimum number of characters as 50% of the minimum password length.

The organization being inspected/assessed configures the information system to store only encrypted representations of passwords.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 196.

The organization being inspected/assessed configures the information system to transmit only encrypted representations of passwords.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 197.

The organization being inspected/assessed configures the information system to enforce minimum password lifetime restrictions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 198.

The organization being inspected/assessed configures the information system to enforce maximum password lifetime restrictions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 199.

The organization being inspected/assessed configures the information system to prohibit reuse for a minimum of 5 generations.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 200.

DoD has defined the number of generations as a minimum of 5.

The organization being inspected/assessed configures the information system to allow the use of a temporary password for system logons with an immediate change to a permanent password.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2041.

The information system performing hardware token-based authentication must be configured to accept only DoD-approved PKI credentials in accordance with DoDI 8520.02 and DoDI 8520.03. For unclassified systems, DoD-approved PKI credentials include DoD PKI credentials, External Certification Authority (ECA) PKI credentials, and DoD-approved external PKI credentials. For SIPRNet, DoD-approved PKI credentials include DoD PKI credentials and NSS PKI credentials.

If the information system accepts DoD-approved external PKI credentials, the information system must be configured to accept only certificates at approved assurance levels, as represented by the Certificate Policy Object Identifiers (OIDs) asserted in the certificate. The current list of DoD-approved external PKIs and acceptable Object Identifiers (OIDs) for each approved external PKI is available at <http://iase.disa.mil/pki-pke/interoperability>.

The organization being inspected/assessed configures the information system to prohibit the use of cached authenticators after an organization defined time period.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2007.

DoD trust store management requirements are defined in information system components' applicable STIGs and SRGs. All information systems are required to undergo a STIG compliance review as part of their certification and accreditation process prior to being granted an authority to operate.

DoD Components are automatically compliant with this CCI because they are covered by the DoD-level STIGs and SRGs.

The information system performing hardware token-based authentication must be configured to validate DoD-approved PKI credentials in accordance with RFC 5280.

The information system must be configured to perform a revocation check as part of the certificate validation process. Revocation checking may be performed using certificate revocation lists (CRLs) published by the issuing PKI or Online Certificate Status Protocol (OCSP) services.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 185.

Information systems must not have access to users' private keys. The cryptographic container in which the private keys are stored (e.g. smart card or software module) implements access controls and protections to ensure that only the authorized user can activate the private key. DoD users agree to protect their PKI credentials in accordance with the DD-2842 agreement that is executed for each credential. They are reminded of these responsibilities in annual IA training.

The private key identifying the information system must be stored in a cryptographic container that is FIPS 140-2 validated. Only authorized information system operators should have access to activation data (e.g. password or PIN) for the private key.

The information system performing PKI-based authentication must be configured to map the authenticated PKI credential to a corresponding network or information system account or role in accordance with DoDI 8520.03.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 187.

The organization being inspected/assessed implements automated tools to check passwords strength per the complexity requirements defined in IA-5 (1) Part A.

The organization being inspected/assessed configures the information system to implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 803.

The organization being inspected/assessed configures the information system to uniquely identify and authenticate non-organizational users (or processes acting on behalf of non-organizational users).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 804.

The information system performing hardware token-based authentication must be configured to accept DoD-approved external PKI PIV credentials to authenticate federal agency users in accordance with DoDI 8520.02 and DoDI 8520.03. The information system must be configured to accept only certificates at approved assurance levels, as represented by the Certificate Policy Object Identifiers (OIDs) asserted in the certificate. The current list of DoD-approved external PKIs and acceptable Object Identifiers (OIDs) for each approved external PKI is available at <http://iase.disa.mil/pki-pke/interoperability>.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2009.

The information system performing hardware token-based authentication must be configured to validate DoD-approved external PKI PIV credentials to authenticate federal agency users in accordance with RFC 5280.

The information system must be configured to perform a revocation check as part of the certificate validation process. Revocation checking may be performed using certificate revocation lists (CRLs) published by the issuing PKI or Online Certificate Status Protocol (OCSP) services.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2010.

The organization being inspected/assessed configures the information system to accept Federal Identity, Credential, and Access Management (FICAM)-approved third-party credentials.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2011.

FICAM Guidance is available at <http://www.idmanagement.gov>.

The organization being inspected/assessed employs only Federal Identity, Credential, and Access Management (FICAM)-approved information system components to accept third-party credentials in information systems defined in IA-8 (3), CCI 2012.

FICAM Guidance is available at <http://www.idmanagement.gov>.

The organization being inspected/assessed configures the information system to conform to Federal Identity, Credential, and Access Management (FICAM)-issued profiles.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2014.

FICAM Guidance is available at <http://www.idmanagement.gov>.

The organization being inspected/assessed will implement an automated intra-organization incident response information sharing capability to provide the following incident related information and support, for example:

1. SOP for incident reporting
2. Incident handling FAQ
3. Current incident activity awareness information
4. Incident response contact information
5. Incident report submission

The organization being inspected/assessed must establish a formal agreement with a computer network defense service provider (CNDSP).

The organization being inspected/assessed must update the incident response plan to address system/organizational changes or problems encountered during plan implementation, execution, or testing and incorporate lessons learned from past incidents (IR-4a).

The organization must document the update actions as an audit trail.

The organization being inspected/assessed protects the incident response plan from unauthorized disclosure and modification.

The organization being inspected/assessed appoints personnel or roles defined in IR-9 (1), CCI 2815 as having the responsibility for responding to information spills.

The organization being inspected/assessed implements procedures defined in IR-9 (3), CCI 2819 to ensure that organizational personnel impacted by information spills can continue to carry out assigned tasks while contaminated systems are undergoing corrective actions.

The organization being inspected/assessed schedules maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization must maintain a record of maintenance.

The organization being inspected/assessed implements a process to perform maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization must maintain a record of maintenance procedures followed.

The organization being inspected/assessed documents maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization being inspected/assessed documents and implements a process to review records of maintenance on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization must maintain a record of reviews.

The organization being inspected/assessed schedules repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization must maintain a record of repairs.

The organization being inspected/assessed implements a process to perform repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization must maintain a record of repair procedures followed.

The organization being inspected/assessed documents repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization being inspected/assessed documents and implements a process to review records of repairs on information system components in accordance with manufacturer or vendor specifications and/or organizational requirements.

The organization must maintain a record of reviews.

The organization being inspected/assessed approves and monitors all maintenance activities, whether performed on site or remotely and whether the equipment is serviced on site or removed to another location.

The organization must maintain records of all approvals and monitoring activities.

The organization being inspected/assessed documents within their risk management strategy personnel or roles defined in MA-2, CCI 2874 who must explicitly approve the removal of the information system or system components from organizational facilities for off-site maintenance or repairs.

The organization must maintain written records of approval for the removal of the information system or system components from organizational facilities for off-site maintenance or repairs.

The organization being inspected/assessed sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs IAW DoDM 5200.01-V3 for classified media and DoDM 5200.01-V4 for unclassified media.

The organization must maintain written records of media sanitization.

The organization being inspected/assessed configures the information system to audit nonlocal maintenance and diagnostic sessions' organization-defined audit events.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2884.

The organization being inspected/assessed documents and implements a process to review the records of the nonlocal maintenance and diagnostic sessions.

The organization must maintain a record of reviews.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the integrity of nonlocal maintenance and diagnostic communications.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2890.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the confidentiality of nonlocal maintenance and diagnostic communications.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 3123.

The organization being inspected/assessed configures the information system to implement remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2891.

The organization being inspected/assessed configures the information system to employ strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 877.

The organization being inspected/assessed terminates session and network connections when non-local maintenance is completed.

The organization must retain audit logs of session and network connections termination for non-local maintenance.

The organization being inspected/assessed protects information system media IAW DoDM 5200.01 M Vol. 1-4.

The organization being inspected/assessed shall document within their Security Plan, and implement, FIPS 140-2 or other NSA approved cryptographic mechanisms to protect the confidentiality and integrity of information stored on digital media during transport outside of controlled areas.

The organization being inspected/assessed protects and controls information system media during transport outside of controlled areas using security measures defined in DoDM 5200.01 M Vol. 1-4 and DoDD 5015.2.

The organization being inspected/assessed documents activities associated with the transport of information system media.

The organization being inspected/assessed does not use portable storage devices in organization information systems when such devices have no identifiable owner.

The organization being inspected/assessed will protect emergency power shutoff capability.

DoD has defined the location as near more than one egress point of the IT area and ensures it is labeled and protected by a cover to prevent accidental shut-off.

The organization being inspected/assessed implements an uninterruptible power supply with sufficient capacity to support orderly shutdown of the system or transition the system to long-term alternate power in the event of a primary power source loss.

The organization being inspected/assessed must install and maintain automatic emergency lighting for the information system that activates in the event of a power outage or disruption and covers emergency exits and evacuation routes within the facility in compliance with established OSHA requirements.

The organization being inspected/assessed must implement and maintain fire suppression and detection devices/systems for the information system that are supported by an independent energy source.

An independent energy source is some source other than the primary energy source for that facility.

Examples include sprinkler systems, hand held fire extinguishers, fixed fire hoses, and smoke detectors.

Humidity controls are not required for general office areas where information system components may be in use and are only required where there are concentrations of information systems such as server farms, mainframes, etc.

The organization being inspected/assessed must maintain temperature and where applicable humidity levels of for commercial grade information systems: 64.4 – 80.6 degrees F; 45% – 60% Relative Humidity; Dew Point 41.9 ° – 59°F; measured at the air intake inlet of the IT equipment casing; for other systems, levels within manufacturer specifications.

DoD has defined the acceptable levels as for commercial grade information systems: 64.4 – 80.6 degrees F; 45% – 60% Relative Humidity; Dew Point 41.9 ° – 59°F; measured at the air intake inlet of the IT equipment casing; for other systems, levels within manufacturer specifications.

The organization being inspected/assessed will maintain an independent monitor device for temperature and humidity levels not located in the immediate vicinity of the controller continuously unless manufacturer specifications allow for a wide enough tolerance that control is not required.

Records of monitoring must be maintained as an audit trail within the authorization lifecycle.

DoD has defined the frequency as continuously unless manufacturer specifications allow for a wide enough tolerance that control is not required.

The organization being inspected/assessed must provide master shutoff valves that are accessible to protect the information system from damage resulting from water leakage.

The organization being inspected/assessed will ensure that master shutoff valves are working properly and have been inspected by the appropriate organization (e.g., fire marshal, department of public works).

The organization being inspected/assessed will identify and document key personnel and will provide training on the location and procedures for use of master shutoff valves.

The organization being inspected/assessed authorizes and maintains authorization records of all system components entering and exiting the facility.

DoD has defined the types of information system components as all system components.

The organization being inspected/assessed will develop and maintain a list of personnel with authorized access to the facilities where information systems reside.

The organization will also take action to identify and officially designate its publicly accessible areas where access authorization is not required.

The organization being inspected/assessed formally approves a list of individuals currently authorized to access the facility where the information system resides.

The organization being inspected/assessed maintains a list of individuals currently authorized to access the facility where the information system resides.

The organization being inspected/assessed utilizes the list of personnel with authorized access (IAW PE-2, CCI-000912) and issues credentials accordingly.

The organization must document the credential issuing activity as an audit trail.

The organization being inspected/assessed will remove personnel from the authorized access list who no longer have approved access and revoke their credentials, as identified in actions per PE-2, CCI 914.

The organization must document each removal and revocation action as an audit trail.

The organization being inspected/assessed will implement physical access authorizations at entry/exit points defined in PE-3, CCI 2915 and secure those physical access points (i.e. doors and/or windows) that are not intended for normal access.

The organization being inspected/assessed will control ingress/egress to the facility using the physical access control devices and/or guards defined in PE-3, CCI 2916.

The organization being inspected/assessed maintains physical access audit logs for entry/exit points defined in PE-3, CCI 2918.

The organization being inspected/assessed provides security safeguards defined in PE-3, CCI 2920 to control access to areas within the facility officially designated as publicly accessible.

The organization must document which areas are officially designated as publicly accessible.

The organization being inspected/assessed documents and implements a process to escort visitors during circumstances defined in PE-3, CCI 2922 requiring visitor escorts.

The organization being inspected/assessed documents and implements a process to monitor visitor activity during circumstances defined in PE-3, CCI 2924 requiring visitor monitoring.

The organization being inspected/assessed will secure as appropriate (in safes or secure containers) items used for physical access control such as keys, combinations, portable locks, etc. Fixed access control devices such as card readers, installed locks, key pads, etc. should be protected from tampering.

The organization being inspected/assessed documents and implements a process to monitor physical access to the information system in addition to the physical access monitoring of the facility as physical spaces containing one or more components of the information system defined in PE-6 (4), CCI 2951.

The organization must maintain records of monitoring.

The organization being inspected/assessed will review physical access logs every 30 days.

The organization must document each occurrence the physical access log review, with results of any necessary incident analysis and action taken, as an audit trail for future reference.

DoD has defined the frequency as every 30 days.

The organization being inspected/assessed documents and implements a process to review physical access logs upon occurrence of events or potential indications of events defined in PE-6, CCI 2941.

The organization must maintain records of reviews.

The organization being inspected/assessed will coordinate the results of reviews and investigations of physical security incidents with the organization's incident response capability (for physical security incidents).

The organization being inspected/assessed must maintain visitor access records for their facilities for at least one year.

DoD has defined the time period as at least one year.

The organization being inspected/assessed documents and implements an insider threat program that includes a cross-discipline insider threat incident handling team.

The organization being inspected/assessed documents and implements a process for ensuring that organizational plans for conducting security testing activities associated with organizational information systems are maintained.

The organization being inspected/assessed documents and implements a process for ensuring that organizational plans for conducting security training activities associated with organizational information systems are maintained.

The organization being inspected/assessed documents and implements a process for conducting security monitoring activities associated with organizational information systems are maintained.

The organization being inspected/assessed documents and implements a process for ensuring that organizational plans for conducting security testing associated with organizational information systems continue to be executed in a timely manner.

The organization must maintain records of execution.

The organization being inspected/assessed documents and implements a process for ensuring that organizational plans for conducting security training associated with organizational information systems continue to be executed in a timely manner.

The organization must maintain records of execution.

The organization being inspected/assessed documents and implements a process for ensuring that organizational plans for conducting security monitoring activities associated with organizational information systems continue to be executed in a timely manner.

The organization must maintain records of execution.

The organization being inspected/assessed documents and implements a process to review testing plans for consistency with the organizational risk management strategy and organization-wide priorities for risk response actions.

The organization must maintain a record of reviews.

The organization being inspected/assessed reviews monitoring plans for consistency with the organizational risk management strategy and organization-wide priorities for risk response actions.

The organization must maintain a record of reviews.

The organization being inspected/assessed documents and implements a threat awareness program that includes a cross-organization information-sharing capability.

The organization being inspected/assessed reviews position risk designations annually and revises designations as required based on the reviews.

Records of these reviews must be maintained as an audit trail.

DoD has defined the frequency as annually.

The organization being inspected/assessed will screen all government and contract personnel to ensure they meet the appropriate ADP/IT level designation requirements IAW DoD 5200.2-R prior to authorizing access to the information system.

The information system owner will rescreen individuals according to system owner defined list of conditions requiring rescreening (CCI-001518) individuals for access to the information system and frequency (CCI - 001519) of such rescreening.

Rescreening actions will be maintained as an audit trail (AU-2).

The organization being inspected/assessed notifies terminated individuals of applicable, legally binding post-employment requirements for the protection of organizational information.

The organization must maintain a record of notifications of post-employment requirements.

The organization being inspected/assessed documents within their personnel security procedures the requirement for terminated individuals to sign an acknowledgment of post-employment requirements as part of the organizational termination process.

The organization being inspected/assessed, conducts exit interviews that include a discussion of information security topics defined in PS-4, CCI 3024 upon termination of individual employment IAW organization security policy and procedures. The organization must retain an audit trail of conducted exit interviews (AU-2)

The organization being inspected/assessed upon termination of individual employment retrieves all security-related organizational information systems-related property IAW organization security policy and procedures. The organization must retain an audit trail of all retrieved security-related organizational information systems-related property (AU-2).

The organization being inspected/assessed upon termination of individual employment retains access to organizational information formerly controlled by terminated individual IAW organization security policy and procedures.

Organizational information formerly controlled by terminated individuals generally refers to online work-product including email files.

The organization being inspected/assessed upon termination of individual employment retains access to organizational information systems formerly controlled by terminated individual IAW organization security policy and procedures.

Organizational information systems formerly controlled by terminated individuals generally refers to issued hardware (e.g. laptops, BlackBerrys, PEDs, removable media, etc.)

The organization being inspected/assessed notifies at a minimum, the ISSO and personnel responsible for revoking credentials immediately or within 24 hours upon termination of individual employment.

The organization must maintain records of termination notification.

DoD has defined the personnel or roles as at a minimum, the ISSO and personnel responsible for revoking credentials.

DoD has defined the time period as immediately or within 24 hours.

The organization being inspected/assessed initiates transfer or reassignment actions to ensure all system accesses no longer required are removed and actions to ensure all system accesses required due to the individual's new position are granted immediately when personnel are reassigned or transferred to other positions.

DoD defines transfer or reassignment actions as actions to ensure all system accesses no longer required are removed.

DoD defines the time period as immediately.

The organization being inspected/assessed documents and implements a process to modify access authorization as needed to correspond with any changes in operational need due to reassignment or transfer.

The organization being inspected/assessed notifies at a minimum, the ISSO and personnel responsible for transferring credentials within 24 hours when individuals are transferred or reassigned to other positions within the organization.

The organization must maintain records of transfer/reassignment notifications.

DoD has defined the personnel or roles as at a minimum, the ISSO and personnel responsible for transferring credentials.

DoD has defined the time period as within 24 hours.

The organization being inspected/assessed documents and implements a process to require individuals to sign an acknowledgement of legally binding post-employment requirements for protection of organizational information, if applicable, as part of granting initial access to covered information.

The organization being inspected/assessed will ensure all individuals have appropriate access agreements in place prior to being granted access to information and information systems.

DD Form 2875 is the accepted DoD methodology of requesting and granting of access to information and information systems.

The organization being inspected/assessed requires that individuals re-sign access agreements to maintain access to organizational information systems when access agreements have been updated or when there is a change to the user's level of access.

DoD has defined the frequency as when there is a change to the user's level of access.

The organization being inspected/assessed documents and implements a process to require third-party providers to comply with personnel security policies and procedures established by the organization.

The organization being inspected/assessed documents and implements a process to require third-party providers to notify at a minimum, the ISSO and personnel responsible for transferring credentials of any personnel transfers or terminations of third-party personnel who possess organizational credentials and/or badges, or who have information system privileges immediately.

DoD has defined the personnel or roles as at a minimum, the ISSO and personnel responsible for transferring credentials.

DoD has defined the time period as immediately.

The organization being inspected/assessed monitors third-party provider compliance with personnel security requirements.

The organization must maintain an audit trail of monitoring activity.

The organization being inspected/assessed notifies at a minimum, the ISSO immediately when a formal employee sanctions process is initiated, identifying the individual sanctioned and the reason for the sanction.

The organization must maintain records of notifications of employee sanctions.

DoD has defined the personnel or roles as at a minimum, the ISSO.

DoD has defined the time period as immediately.

The organization being inspected/assessed documents and implements a process to categorize information and the information system in accordance with CNSSI 1253 and applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.

The organization being inspected/assessed documents and implements a process IAW CNSSI 1253 to ensure the security categorization decision is reviewed and approved by the authorizing official or authorizing official designated representative.

The organization being inspected/assessed configures the information system to implement privileged access authorization to all information systems and infrastructure components for selected vulnerability scanning activities defined in RA-5 (5), CCI 2906.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1067.

DoD has defined the information system components as all information systems and infrastructure components.

The organization being inspected/assessed must identify and document in the Security Plan whether the system is a “covered system” IAW DoDI 5200.44. If it is a covered system, the organization must implement the requirements below:

1. Conduct a criticality analysis to identify mission critical functions and critical components and reduce the vulnerability of such functions and components through secure system design;
2. Request threat analysis of suppliers of critical components from the TSN focal point and manage access to and control of threat analysis products containing U.S. person information;
3. Engage TSN focal points for guidance on managing identified risk using DoD Components and Enterprise risk management resources; and
4. Apply TSN best practices, processes, techniques, and procurement tools prior to the acquisition of critical components or their integration into applicable systems, at any point in the system lifecycle. Such tools and practices include contract requirements and the SCRM key practices Guide.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service to follow a documented development process.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service explicitly addresses security requirements.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service identifies the standards used in the development process, for example, programming languages and computer-aided design (CAD) systems.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service identifies the tools used in the development process, for example, programming languages and computer-aided design (CAD) systems.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service documents the specific tool options and tool configurations used in the development process.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service documents changes to the process and/or tools used in development.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service document a process to manage changes to the process and/or tools used in development.

The organization being inspected/assessed requires within contracts/agreements that the developer of the information system, system component, or information system service document the integrity of changes to the process and/or tools used in development.

The organization being inspected/assessed documents and implements a process to review the development process before first use and annually thereafter to determine if the development process selected and employed can satisfy the security requirements defined in SA-15, CCI 3246. Reviews of development processes can include, for example, the use of capability maturity model integration (CMMI) to determine the potential effectiveness of such processes.

The organization must maintain a record of reviews.

DoD has defined the frequency as before first use and annually thereafter.

The organization being inspected/assessed documents and implements a process to review the development standards before first use and annually thereafter to determine if the development standards selected and employed can satisfy the security requirements defined in SA-15, CCI 3246.

The organization must maintain a record of reviews.

DoD has defined the frequency as before first use and annually thereafter.

The organization being inspected/assessed documents and implements a process to review the development tools before first use and annually thereafter to determine if the development tools selected and employed can satisfy the security requirements defined in SA-15, CCI 3246.

The organization must maintain a record of reviews.

DoD has defined the frequency as before first use and annually thereafter.

The organization being inspected/assessed documents and implements a process to review the development tool options/configurations before first use and annually thereafter to determine if the development tool options/configurations selected and employed can satisfy the security requirements defined in SA-15, CCI 3246.

The organization must maintain a record of reviews.

DoD has defined the frequency as before first use and annually thereafter.

The organization being inspected/assessed develops and documents an anti-counterfeit policy that include the means to detect counterfeit components from entering the information system.

The organization being inspected/assessed develops and documents an anti-counterfeit policy that include the means to prevent counterfeit components from entering the information system.

The organization being inspected/assessed develops and documents anti-counterfeit procedures that include the means to detect counterfeit components from entering the information system.

The organization being inspected/assessed develops and documents anti-counterfeit procedures that include the means to prevent counterfeit components from entering the information system.

The organization being inspected/assessed implements the policy defined in SA-19, CCIs 3356 that include the means to detect counterfeit components from entering the information system.

The organization being inspected/assessed implements the policy defined in SA-19, CCIs 3357 that include the means to prevent counterfeit components from entering the information system.

The organization being inspected/assessed implements the procedures defined in SA-19, CCIs 3358 that include the means to detect counterfeit components from entering the information system.

The organization being inspected/assessed implements the procedures defined in SA-19, CCIs 3359 that include the means to prevent counterfeit components from entering the information system.

The organization being inspected/assessed documents and implements a process to report counterfeit information system components to source of counterfeit component, at a minimum, USCYBERCOM. And/or at a minimum, the ISSO, ISSM, and PM.

The organization must maintain a record of reporting.

DoD has defined the personnel or roles as at a minimum, the ISSO, ISSM, and PM.

DoD has defined the external reporting organizations as at a minimum, USCYBERCOM.

The organization being inspected/assessed determines the resources (funding, staffing, etc.) required for the cybersecurity requirements to protect the information system or information system service as part of its planning, programming, and budget process (PPBE).

The organization being inspected/assessed documents the resources (funding, staffing, etc.) required for the cybersecurity requirements to protect the information system or information system service as part of its planning, programming, and budget process (PPBE).

The organization being inspected/assessed allocates the resources (funding, staffing, etc.) required for the cybersecurity requirements to protect the information system or information system service as part of its planning, programming, and budget process (PPBE).

The organization being inspected/assessed identifies and establishes an individual line item for cybersecurity requirements to protect the information system as part of the planning, programming, and budget process (PPBE).

The organization being inspected/assessed identifies and establishes an individual line item for cybersecurity requirements to protect the information system as part of the planning, programming, and budget process (PPBE).

The organization being inspected/assessed documents and implements a process to manage the information system using the system development life cycle defined in SA-3, CCI 3092 that incorporates information security considerations IAW DoDI 8580.1.

The organization being inspected/assessed defines and documents a system development life cycle that is used to manage the information system.

DoD has determined the system development life cycle is not appropriate to define at the Enterprise level.

The organization being inspected/assessed defines and documents information system security roles and responsibilities throughout the system development life cycle IAW DoDI 8580.1 .

The organization being inspected/assessed identifies and documents individuals having information system security roles and responsibilities.

The organization being inspected/assessed documents and implements a process to integrate the organizational information security risk management process into system development life cycle activities.

The organization being inspected/assessed documents and implements a process to conduct an organizational assessment of risk prior to the acquisition or outsourcing of dedicated information security services.

The organization must maintain a record of risk assessment.

The organization being inspected/assessed ensures that the acquisition or outsourcing of dedicated information security services is approved by the DoD Component CIO or their delegate(s).

The organization must maintain a record of approvals.

DoD has defined the personnel or roles the DoD Component CIO or their delegate(s).

The organization being inspected/assessed configures the information system to terminate the network connection associated with a communications session at the end of the session or after 10 minutes in band management and 15 minutes for user sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1133.

DoD has defined the time period as 10 minutes in band management and 15 minutes for user sessions.

The organization being inspected/assessed configures the information system to provide an explicit indication of use to users physically present at collaborative computing devices.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1152.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2465.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2466.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2467.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2468.

The organization being inspected/assessed implements a name service resolution architecture consisting of primary and secondary servers.

The organization must document the architecture in the site security plan.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1182.

The organization being inspected/assessed implements a name service resolution architecture where recursive and authoritative server software is not installed on the same information system.

The organization must document the architecture in the site security plan.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1183.

The organization being inspected/assessed configures the information system to protect the authenticity of communications sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1184.

The organization being inspected/assessed configures the information system to invalidate session identifiers upon user logout or other session termination.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1185.

The organization being inspected/assessed configures the information system to generate unique session identifiers for each session with randomness requirements defined in SC-23 (3), CCI 1189.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1188.

The organization being inspected/assessed configures the information system to recognize only session identifiers that are system-generated.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1664.

The organization being inspected/assessed configures the information system to allow the use of DoD PKI established certificate authorities for verification of the establishment of protected sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2470.

DoD has defined the certificate authorities as DoD PKI established certificate authorities.

The organization being inspected/assessed configures the information system to fail to a secure state for failures during system initialization, shutdown, and aborts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1190.

DoD has defined the known state as secure state.

DoD has defined the types of failures as failures during system initialization, shutdown, and aborts.

The organization being inspected/assessed configures the information system to preserve information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes in the event of a system failure.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1665.

DoD has defined system state information as information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes.

The organization being inspected/assessed configures the information system to protect the confidentiality and/or integrity of organization-defined information at rest.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1199.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized modification of information at rest defined in SC-28 (1), CCI 2473 on any information system components storing data defined in SC-28 (1), 2473.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2475.

DoD has defined the information system components as any information system components storing data defined in SC-28 (1), 2473.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information at rest defined in SC-28 (1), CCI 2473 on any information system components storing data defined in SC-28 (1), 2473.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2476.

DoD has defined the information system components as any information system components storing data defined in SC-28 (1), 2473.

The organization being inspected/assessed implements operations security safeguards defined in SC-38, CCI 2528 to protect key organizational information throughout the system development life cycle.

The organization must maintain an audit trail of security safeguard implementation.

The organization being inspected/assessed configures the information system to maintain a separate execution domain for each executing process.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2530.

The organization being inspected/assessed physically disables or removes connection ports or input/output devices defined in SC-41, CCI 2545 on information systems or information system components defined in SC-41, CCI 2544.

The organization being inspected/assessed configures the information system to protect against or limits the effects of types of denial of service attacks defined in SC-5, CCI 1093 by employing security safeguards defined in SC-5, CCI 2386.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2385.

The organization being inspected/assessed configures the information system to restrict the ability of individuals to launch denial of service attacks defined in SC-5 (1), CCI 2387 against other information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1094.

The organization being inspected/assessed configures the information system to manage excess capacity, bandwidth, or other redundancy to limit the effects of information flooding types of denial of service attacks.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1095.

The organization being inspected/assessed implements the monitoring tools defined in SC-5 (3), CCI 2388 to detect indicators of denial of service attacks against the information system.

The organization being inspected/assessed documents and implements mechanisms to prevent the unauthorized exfiltration of information across managed interfaces.

The organization being inspected/assessed configures the information system to allow incoming communications from authorized sources defined in SC-7 (11), CCI 2401 routed to authorized destinations defined in SC-7 (11), CCI 2402.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2403.

The organization being inspected/assessed implements McAfee Host Intrusion Prevention (HIPS) on all information system components.

DoD has defined the host-based boundary protection mechanisms as McAfee Host Intrusion Prevention (HIPS).

DoD has defined the information system components as all information system components.

The organization being inspected/assessed designs the information system to isolate key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc. from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.

DoD has defined the key information security tools, mechanisms, and support components as key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc.

The organization being inspected/assessed documents and implements mechanisms to protect against unauthorized physical connections at internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.

DoD has defined the managed interfaces as internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.

The organization being inspected/assessed documents and implements information system access control mechanisms to limit the number of external connections to the information system.

The organization being inspected/assessed designs the information system to have a managed interface for each telecommunication service.

The organization being inspected/assessed defines and documents a traffic flow policy for each managed interface for each external telecommunication service.

The organization being inspected/assessed documents and implements mechanisms to protect the confidentiality and integrity of the information being transmitted across each interface for each external telecommunication service.

The organization being inspected/assessed configures the information system to deny network communications traffic at managed interfaces by default and allows network communications traffic by exception (i.e., deny all, permit by exception).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1109.

The organization being inspected/assessed configures the information system to prevent the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2397.

The organization being inspected/assessed configures the information system to route protocols as designated by PPSM guidance (e.g. HTTPS, HTTP, FTP, SNMP) to any network external to the authorization boundary through authenticated proxy servers at managed interfaces.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1112.

DoD has defined the internal communications traffic as protocols as designated by PPSM guidance (e.g. HTTPS, HTTP, FTP, SNMP).

DoD has defined the external networks as any network external to the authorization boundary.

The organization being inspected/assessed configures the information system to detect outgoing communications traffic posing a threat to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2398.

The organization being inspected/assessed configures the information system to deny outgoing communications traffic posing a threat to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2399.

The organization being inspected/assessed configures the information system to audit the identity of internal users associated with denied outgoing communications traffic posing a threat to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2400.

The organization being inspected/assessed documents and implements processes to monitor and control communications at the external boundary of the system and at key internal boundaries within the system.

The organization must maintain an audit trail of monitoring activities.

The organization being inspected/assessed designs the information system to leverage subnetworks so that publicly accessible system components are physically and/or logically separated from internal organizational networks.

The organization being inspected/assessed designs the information system to enforce requirements that components connect to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.

The organization being inspected/assessed configures the information system to protect the confidentiality and/or integrity of transmitted information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2418.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information and detect changes to information during transmission unless otherwise protected by Protected Distribution System (PDS).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2421.

DoD has defined the selection as both prevention of unauthorized disclosure and detection of changes to information.

DoD has defined the alternative physical safeguards as Protected Distribution System (PDS).

The organization being inspected/assessed configures the information system to maintain the confidentiality and integrity of information during preparation for transmission.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2420.

DoD has defined the selection as both confidentiality and integrity.

The organization being inspected/assessed configures the information system to maintain the confidentiality and integrity of information during reception.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2422.

DoD has defined the selection as both confidentiality and integrity.

The organization being inspected/assessed configures the information system to check the validity of all inputs except those identified specifically by the organization.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1310.

DoD has defined the information inputs as all inputs except those identified specifically by the organization.

The organization being inspected/assessed configures the information system to generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1312.

The organization being inspected/assessed configures the information system to reveal error messages only to the ISSO, ISSM, and SCA.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1314.

DoD has defined the personnel or roles as the ISSO, ISSM, and SCA.

The organization being inspected/assessed configures the information system to implement security safeguards defined in SI-16, CCI 2823 to protect its memory from unauthorized code execution.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2824.

The organization being inspected/assessed configures the information system to implement fail-safe procedures defined in SI-17, CCI 2773 when failure conditions defined in SI-17, CCI 2774 occur.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2775.

The organization being inspected/assessed documents and implements a process to centrally manage the flaw remediation process.

The organization being inspected/assessed configures the information system to remove all upgraded/replaced software components that are no longer required for operation (e.g., previous versions) after updated versions have been installed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2617.

DoD has defined the software components as all upgraded/replaced software components that are no longer required for operation.

The organization being inspected/assessed configures the information system to remove all upgraded/replaced firmware components that are no longer required for operation e.g., previous versions) after updated versions have been installed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2618.

DoD has defined the firmware components as all upgraded/replaced firmware components that are no longer required for operation.

The organization being inspected/assessed documents and implements a process to identify information system flaws.

The process shall include review of the system through automated scans and manual checks to determine the existence of flaws such as IAVM, CVE, or other resources.

The organization being inspected/assessed reports information system flaws according to DoD Cybersecurity policy and organizational roles and responsibilities.

The organization must report information system flaws in their POA&M.

The organization being inspected/assessed documents and implements a process to test software updates related to flaw remediation for effectiveness before installation.

If the software update is being provided by a vendor who has documented the effectiveness of the update in fixing the affected IAVM/CVE, further testing by the organization may not be required.

The organization being inspected/assessed documents and implements a process for regression testing IAW CM-4 to identify any potential side effects before installation of software updates.

The organization being inspected/assessed documents and implements a process to test firmware updates related to flaw remediation for effectiveness before installation.

If the firmware update is being provided by a vendor who has documented the effectiveness of the update in fixing the affected IAVM/CVE, further testing by the organization may not be required.

The organization being inspected/assessed documents and implements a process for regression testing IAW CM-4 to identify any potential side effects before installation of software updates.

The organization being inspected/assessed configures the information system to install security-relevant software updates within 30 days of the release of the updates

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2605.

DoD has defined the time period as 30 days.

The organization being inspected/assessed configures the information system to install security-relevant firmware updates within 30 days of the release of the updates.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2607.

DoD has defined the time period as 30 days.

The organization being inspected/assessed documents and implements tools defined in SI-3 (10), CCI 2634 to analyze the characteristics and behavior of malicious code.

The organization being inspected/assessed documents and implements techniques defined in SI-3 (10), CCI 2635 to analyze the characteristics and behavior of malicious code.

The organization being inspected/assessed configures the information system to automatically update malicious code protection mechanisms.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1247.

The organization being inspected/assessed configures the malicious code protection mechanisms identified in SI-3, CCI 2619 to eradicate malicious code.

The organization being inspected/assessed configures the malicious code protection mechanisms identified in SI-3, CCI 2620 to eradicate malicious code.

The organization being inspected/assessed configures the information system to update malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1240.

The organization being inspected/assessed configures malicious code protection mechanisms to perform periodic scans of the information system on every 7 days.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1241.

DoD has defined the frequency as every 7 days.

The organization being inspected/assessed configures malicious code protection mechanisms to perform real-time scans of files from external sources at endpoints as the files are downloaded, opened, or executed in accordance with organizational security policy.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1242.

The organization being inspected/assessed configures the malicious code protection mechanisms to perform real-time scans of files from external sources at network entry/exit points as the files are downloaded, opened, or executed in accordance with organizational security policy.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2624.

The organization being inspected/assessed configures malicious code protection mechanisms to perform block and quarantine malicious code and then send an alert to the administrator immediately in near real-time in response to malicious code detection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1243.

DoD has defined the actions as block and quarantine malicious code and then send an alert to the administrator immediately in near real-time.

The organization being inspected/assessed configures the information system to address the receipt of false positives during malicious code detection and eradication, and the resulting potential impact on the availability of the information system.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1245.

The organization being inspected/assessed documents and implements a process to analyze outbound communications traffic at the external boundary of the information system to discover anomalies.

The organization must maintain a record of any discovered anomalies.

The organization being inspected/assessed documents and implements a process to analyze outbound communications traffic at selected interior points defined in SI-4 (11), CCI 2668 within the system (e.g., subnetworks, subsystems) to discover anomalies.

The organization must maintain a record of any discovered anomalies.

The organization being inspected/assessed documents and implements automated mechanisms to alert security personnel when there are threats identified by authoritative sources (e.g. CTOs) and IAW with CJCSM 6510.01B.

For automated alert mechanisms that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1274.

DoD has defined the activities that trigger alerts as when there are threats identified by authoritative sources (e.g. CTOs) and IAW with CJCSM 6510.01B.

The organization being inspected/assessed documents and implements an intrusion detection system to monitor wireless communications traffic as the traffic passes from wireless to wireline networks.

The organization being inspected/assessed documents and implements a process to correlate information from monitoring tools employed throughout the information system.

The organization being inspected/assessed implements additional monitoring defined in SI-4 (19), CCI 2673 of individuals who have been identified by sources defined in SI-4 (19), CCI 2674 as posing an increased level of risk.

The organization must maintain an audit trail of additional monitoring.

The organization being inspected/assessed documents and implements automated tools to support near real-time analysis of events.

The organization being inspected/assessed implements additional monitoring defined in SI-4 (20), CCI 2676 of privileged users.

The organization must maintain an audit trail of additional monitoring.

The organization being inspected/assessed documents and implements a process to detect network services that have not been authorized or approved by at a minimum, the ISSO and ISSM.

For network service detection mechanisms that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2683.

DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.

The organization being inspected/assessed configures the information system to audit and/or alert at a minimum, the ISSO and ISSM when unauthorized network services are detected.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2684.

DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.

The organization being inspected/assessed documents and implements HBSS at all components.

DoD has defined the host-based monitoring mechanisms as HBSS.

DoD has defined the information system components as all components.

The organization being inspected/assessed configures the information system to monitor inbound communications traffic continuously for unusual or unauthorized activities or conditions.

DoD has defined the frequency as continuously.

The organization being inspected/assessed configures the information system to monitor outbound communications traffic continuously for unusual or unauthorized activities or conditions.

DoD has defined the frequency as continuously.

The organization being inspected/assessed configures the information system to alert at a minimum, the ISSM and ISSO when real time intrusion detection and when there are threats identified by authoritative sources (e.g. CTOs) and IAW incident categories I, II, IV, & VII within CJCSM 6510.01B reflect the occurrence of a compromise or a potential compromise.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2664.

DoD has defined the personnel or roles as at a minimum, the ISSM and ISSO.

DoD has defined the compromise indicators as real time intrusion detection and when there are threats identified by authoritative sources (e.g. CTOs) and IAW incident categories I, II, IV, & VII within CJCSM 6510.01B.

The organization being inspected/assessed documents and implements a process to monitor the information system to detect attacks and indicators of potential attacks in accordance with sensor placement and monitoring requirements within CJCSI 6510.01F.

The organization must maintain an audit trail of monitoring.

DoD has defined the monitoring objectives as sensor placement and monitoring requirements within CJCSI 6510.01F.

The organization being inspected/assessed documents and implements a process to monitor the information system to detect unauthorized local connections.

The organization must maintain an audit trail of monitoring.

The organization being inspected/assessed documents and implements a process to monitor the information system to detect unauthorized network connections.

The organization must maintain an audit trail of monitoring.

The organization being inspected/assessed documents and implements a process to monitor information system to detect unauthorized remote connections.

The organization must maintain an audit trail of monitoring.

The organization being inspected/assessed documents and implements a process to deploy monitoring devices strategically within the information system to collect organization determined essential information.

The organization being inspected/assessed documents and implements a process to deploy monitoring devices at ad hoc locations within the system to track specific types of transactions of interest to the organization.

The organization being inspected/assessed documents and implements a process to protect information obtained from intrusion-monitoring tools from unauthorized access.

The organization being inspected/assessed documents and implements a process to protect information obtained from intrusion-monitoring tools from unauthorized modification.

The organization being inspected/assessed documents and implements a process to protect information obtained from intrusion-monitoring tools from unauthorized deletion.

The organization being inspected/assessed receives information system security alerts, advisories, and directives from at a minimum, USCYBERCOM on an ongoing basis.

DoD has defined the external organizations as at a minimum, USCYBERCOM.

The organization being inspected/assessed implements security directives in accordance with established time frames, or notifies the issuing organization of the degree of noncompliance.

The organization being inspected/assessed prohibits the use of binary or machine-executable code obtained from sources without vendor support or with no warranty and without the provision of source code.

The organization being inspected/assessed configures the information system to provide the capability to audit the event upon detection of a potential integrity violation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2723.

The organization being inspected/assessed configures the information system to initiate one or more of following actions: generates an audit record; alerts current user; alerts at a minimum, the ISSO and ISSM; and/or other actions defined in SI-7 (8), CCI 2722 upon detection of a potential integrity violation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2724.

DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.

Classification	Responsibility
Impractical	DoD Defined
Impractical	Contractor
Impractical	Contractor
Impractical	Contractor
Impractical	Contractor
Applicable	Contractor

Reference: UFC 4-010-06 Cybersecurity of Facility-Related Control Systems, with Change 1
<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc/ufc-4-010-06>

Applicable	Contractor	
Impractical	Contractor	
Applicable	Contractor	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	

Applicable	Contractor	
Impractical	Owner	
Impractical	Contractor	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor, Owner	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Owner, Contractor	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	Contractor, Owner	
N/A-Policy	DoD Defined	
Impractical	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor	

Applicable	Contractor, Owner	
Impractical	DoD Defined	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Impractical	Contractor	

Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor	

Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Impractical	Contractor	

Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	

Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Impractical	Contractor, Owner	
Applicable	Contractor, Owner	
Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor, Owner	
Impractical	Contractor	
Impractical	DoD Defined	
Applicable	Contractor, Owner	

Impractical	Contractor
Applicable	DoD Defined
Applicable	Contractor, Owner
Applicable	Contractor, Owner
Applicable	Contractor
Applicable	Owner, Contractor
N/A-Policy	DoD Defined
N/A-Policy	Contractor, Owner
Impractical	DoD Defined
N/A-Policy	Owner

N/A-Policy	Owner	
Applicable	Contractor	
Impractical	Contractor	
Impractical	Contractor, Owner	
N/A-Policy	Contractor	
Applicable	DoD Defined	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	

Impractical	DoD Defined	
Applicable	DoD Defined	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
Applicable	Contractor	

Impractical	Owner, Contractor	
Impractical	Owner	
Applicable	Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	

Applicable	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	

Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	DoD Defined	

Impractical	Contractor, Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	DoD Defined	

Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Impractical	Owner, Contractor	

Impractical	DoD Defined	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Impractical	Contractor	
Applicable	Contractor	

Applicable	Contractor	
Impractical	Contractor	
Applicable	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Owner, Contractor	
Impractical	Contractor, Owner	
N/A-Policy	Owner, Contractor	

N/A-Policy	Owner, Contractor	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
Applicable	Owner	
Applicable	Contractor, Owner	
N/A-Policy	Contractor, Owner	
N/A-Policy	Contractor, Owner	
N/A-Policy	Contractor, Owner	

N/A-Policy	Contractor, Owner	
Applicable	Contractor, Owner	
N/A-Policy	Contractor, Owner	
N/A-Policy	Owner	
Applicable	Owner, Contractor	
N/A-Policy	Owner	
Impractical	Owner	
Impractical	Contractor	

N/A-Policy	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Owner	

N/A-Policy	Owner	
Applicable	Contractor	
N/A-Policy	Owner	
Applicable	Owner, Contractor	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor	

Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor	
Applicable	Contractor	
N/A-Policy	Contractor, Owner	

N/A-Policy	DoD Defined	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
Applicable	Owner	
Impractical	Owner	
N/A-Policy	Owner	
Applicable	Owner	

Applicable	Owner	
Impractical	Owner	
Applicable	Owner, Contractor	
Applicable	Owner, Contractor	
N/A-Policy	DoD Defined	
N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	DoD Defined	
N/A-Policy	Owner	

[illegible]

N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	DoD Defined	
N/A-Policy	DoD Defined	
N/A-Policy	Owner	
N/A-Policy	DoD Defined	

N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	DoD Defined	
Applicable	Owner	
N/A-Policy	DoD Defined	
Applicable	Owner	

N/A-Policy	DoD Defined	
N/A-Policy	Owner	
N/A-Policy	Owner	
Impractical	DoD Defined	
Applicable	Contractor, Owner	

N/A-Policy	DoD Defined	
N/A-Policy	DoD Defined	
N/A-Policy	DoD Defined	
N/A-Policy	DoD Defined	
N/A-Policy	Owner, Contractor	
N/A-Policy	Owner, Contractor	

N/A-Policy	Owner, Contractor	
N/A-Policy	Owner, Contractor	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
N/A-Policy	DoD Defined	
N/A-Policy	Owner	

N/A-Policy	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor	
Applicable	Contractor	
Applicable	Contractor	

Applicable	Contractor	
Applicable	Contractor	
Impractical	Contractor, Owner	
Applicable	Contractor	
Applicable	Contractor	

Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	

Applicable	DoD Defined	
Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Owner, Contractor	

Applicable	Contractor	
Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor	
Impractical	DoD Defined	
Applicable	Contractor, Owner	

Applicable	Contractor	
Impractical	DoD Defined	
Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Impractical	Contractor, Owner	
N/A-Policy	Owner	

Applicable	Contractor, Owner	
Impractical	Contractor	
Applicable	Contractor	
Applicable	DoD Defined	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	

Impractical	Contractor	
N/A-Policy	Contractor, Owner	
Applicable	Designer, Owner	
Applicable	Designer, Owner	
Applicable	Contractor	
Impractical	DoD Defined	

Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	DoD Defined	
Applicable	Contractor	
Impractical	DoD Defined	
Impractical	Contractor	

Applicable	Contractor	
N/A-Policy	Owner	
Impractical	DoD Defined	
Impractical	DoD Defined	
N/A-Policy	Contractor, Owner	
N/A-Policy	DoD Defined	
N/A-Policy	Owner	

N/A-Policy	Owner	
N/A-Policy	Owner	
N/A-Policy	Owner	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor	

Applicable	DoD Defined	
Applicable	Contractor, Owner	
N/A-Policy	Contractor	
N/A-Policy	Contractor	
Impractical	DoD Defined	
Applicable	Contractor, Owner	
Impractical	Contractor, Owner	

Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
N/A-Policy	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	DoD Defined	
Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Applicable	Owner	
Applicable	Contractor, Owner	
Applicable	Contractor	
Applicable	DoD Defined	

Section 2

CCIs 1

CCI Number	800-53/82 Control Text Indicator	CCI Definition
CCI-000069	AC-17(3)	The information system routes all remote accesses through organization-defined number managed network access control points.
CCI-002322	AC-17(9)	The organization provides the capability to expeditiously disconnect or disable remote access to the information system within the organization-defined time period.
CCI-000063	AC-17(a)	The organization defines allowed methods of remote access to the information system.
CCI-000018	AC-2(4)	The information system automatically audits account creation actions.
CCI-001403	AC-2(4)	The information system automatically audits account modification actions.

CCI-001404	AC-2(4)	The information system automatically audits account disabling actions.
CCI-001405	AC-2(4)	The information system automatically audits account removal actions.
CCI-001683	AC-2(4)	The information system notifies organization-defined personnel or roles for account creation actions.
CCI-001684	AC-2(4)	The information system notifies organization-defined personnel or roles for account modification actions.
CCI-001685	AC-2(4)	The information system notifies organization-defined personnel or roles for account disabling actions.
CCI-001686	AC-2(4)	The information system notifies organization-defined personnel or roles for account removal actions.

CCI-002130	AC-2(4)	The information system automatically audits account enabling actions.
CCI-002132	AC-2(4)	The information system notifies organization-defined personnel or roles for account enabling actions.
CCI-000019	AC-2(5)(a)	The organization requires that users log out in accordance with the organization-defined time period of inactivity and/or description of when to log out.
CCI-002136	AC-2(7)(c)	The organization defines the actions to be taken when privileged role assignments are no longer appropriate
CCI-002141	AC-2(9)	The organization only permits the use of shared/group accounts that meet organization-defined conditions for establishing shared/group accounts.
CCI-001368	AC-4	The information system enforces approved authorizations for controlling the flow of information within the system based on organization-defined information flow control policies.
CCI-001414	AC-4	The information system enforces approved authorizations for controlling the flow of information between interconnected systems based on organization-defined information flow control policies.

CCI-002233	AC-6(8)	The information system prevents organization-defined software from executing at higher privilege levels than users executing the software.
CCI-000044	AC-7(a)	The information system enforces the organization-defined limit of consecutive invalid logon attempts by a user during the organization-defined time period.
CCI-002238	AC-7(b)	The information system automatically locks the account or node for either an organization-defined time period, until the locked account or node is released by an administrator, or delays the next login prompt according to the organization-defined delay algorithm when the maximum number of unsuccessful attempts is exceeded.
CCI-000166	AU-10	The information system protects against an individual (or process acting on behalf of an individual) falsely denying having performed organization-defined actions to be covered by non-repudiation.
CCI-000169	AU-12(a)	The information system provides audit record generation capability for the auditable events defined in AU-2 a at organization defined information system components.

CCI-000123	AU-2(a)	The organization determines the information system must be capable of auditing an organization-defined list of auditable events.
CCI-000126	AU-2(d)	The organization determines that the organization-defined subset of the auditable events defined in AU-2 are to be audited within the information system.
CCI-000135	AU-3(1)	The information system generates audit records containing the organization-defined additional, more detailed information that is to be included in the audit records.
CCI-001851	AU-4(1)	The information system off-loads audit records per organization-defined frequency onto a different system or media than the system being audited.

CCI-001855	AU-5(1)	The information system provides a warning to organization-defined personnel, roles, and/or locations within organization-defined time period when allocated audit record storage volume reaches organization-defined percentage of repository maximum audit record storage capacity.
CCI-000139	AU-5(a)	The information system alerts designated organization-defined personnel or roles in the event of an audit processing failure.
CCI-000158	AU-7(1)	The information system provides the capability to process audit records for events of interest based on organization-defined audit fields within audit records.
CCI-001891	AU-8(1)	The information system compares internal information system clocks on an organization-defined frequency with an organization-defined authoritative time source.

CCI-002046	AU-8(1)	The information system synchronizes the internal system clocks to the authoritative time source when the time difference is greater than the organization-defined time period.
CCI-001889	AU-8(b)	The information system records time stamps for audit records that meets organization-defined granularity of time measurement.
CCI-001890	AU-8(b)	The information system records time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).
CCI-001726	CM-10(a)	The organization uses software in accordance with contract agreements.
CCI-001728	CM-10(a)	The organization uses software in accordance with copyright laws.
CCI-000296	CM-2(1)(a)	The organization reviews and updates the baseline configuration of the information system at an organization-defined frequency.
CCI-000297	CM-2(1)(b)	The organization reviews and updates the baseline configuration of the information system when required due to organization-defined circumstances.

CCI-001764	CM-7(2)	The information system prevents program execution in accordance with organization-defined policies regarding software program usage and restrictions, and/or rules authorizing the terms and conditions of software program usage.
CCI-000388	CM-7(3)	The organization ensures compliance with organization-defined registration requirements for functions, ports, protocols, and services.
CCI-001777	CM-7(5)c	The organization reviews and updates the list of authorized software programs per organization-defined frequency.
CCI-000382	CM-7(b)	The organization configures the information system to prohibit or restrict the use of organization-defined functions, ports, protocols, and/or services.
CCI-000416	CM-8(3)(a)	The organization employs automated mechanisms, per organization defined frequency, to detect the presence of unauthorized hardware, software, and firmware components within the information system.
CCI-000399	CM-8(a)(4)	The organization develops and documents an inventory of information system components that includes organization defined information deemed necessary to achieve effective information system component accountability.
CCI-001780	CM-8(b)	The organization reviews and updates the information system component inventory per organization-defined frequency.

CCI-002855	CP-12	The information system, when organization-defined conditions are detected, enters a safe mode of operation with organization-defined restrictions of safe mode of operation.
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CCI-000542	CP-9(1)	The organization tests backup information per organization-defined frequency to verify media reliability and information integrity.
CCI-000539	CP-9(c)	The organization conducts backups of information system documentation including security-related documentation per organization-defined frequency that is consistent with recovery time and recovery point objectives.
CCI-000778	IA-3	The information system uniquely identifies an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.
CCI-001958	IA-3	The information system authenticates an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.
CCI-001967	IA-3(1)	The information system authenticates organization-defined devices and/or types of devices before establishing a local, remote and/or network connection using bidirectional authentication that is cryptographically based.

CCI-000192	IA-5(1)(a)	The information system enforces password complexity by the minimum number of upper case characters used.
CCI-000193	IA-5(1)(a)	The information system enforces password complexity by the minimum number of lower case characters used.
CCI-000194	IA-5(1)(a)	The information system enforces password complexity by the minimum number of numeric characters used.
CCI-000205	IA-5(1)(a)	The information system enforces minimum password length.
CCI-001619	IA-5(1)(a)	The information system enforces password complexity by the minimum number of special characters used.
CCI-000195	IA-5(1)(b)	The information system, for password-based authentication, when new passwords are created, enforces that at least an organization-defined number of characters are changed.
CCI-000198	IA-5(1)(d)	The information system enforces minimum password lifetime restrictions.

CCI-000199	IA-5(1)(d)	The information system enforces maximum password lifetime restrictions.
CCI-000200	IA-5(1)(e)	The information system prohibits password reuse for the organization defined number of generations.
CCI-002007	IA-5(13)	The information system prohibits the use of cached authenticators after an organization defined time period.
CCI-001014	MP-4(a)	The organization physically controls and securely stores organization-defined types of digital and/or non-digital media within organization-defined controlled areas.
CCI-001020	MP-5(a)	The organization protects and controls organization-defined types of information system media during transport outside of controlled areas using organization-defined security safeguards.
CCI-000977	PE-15	The organization protects the information system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible.
CCI-000978	PE-15	The organization protects the information system from damage resulting from water leakage by providing master shutoff or isolation valves that are working properly.

CCI-000984	PE-16	The organization maintains records of information system components entering and exiting the facility.
CCI-000919	PE-3(a)	The organization enforces physical access authorizations at organization-defined entry/exit points to the facility where the information system resides.
CCI-000924	PE-3(f)	The organization inventories organization-defined physical access devices every organization-defined frequency.
CCI-000936	PE-4	The organization controls physical access to organization-defined information system distribution and transmission lines within organizational facilities using organization-defined security safeguards.
CCI-001054	RA-5(a)	The organization scans for vulnerabilities in the information system and hosted applications on an organization-defined frequency.
CCI-001056	RA-5(a)	The organization scans for vulnerabilities in the information system and hosted applications when new vulnerabilities potentially affecting the system/applications are identified and reported.

CCI-001643	RA-5(a)	The organization scans for vulnerabilities in the information system and hosted applications in accordance with the organization-defined process for random scans.
CCI-001059	RA-5(d)	The organization remediates legitimate vulnerabilities in organization-defined response times in accordance with an organizational assessment risk.
CCI-003364	SA-19(b)	The organization reports counterfeit information system components to source of counterfeit component, organization-defined external reporting organizations and/or organization-defined personnel or roles.
CCI-001133	SC-10	The information system terminates the network connection associated with a communications session at the end of the session or after an organization-defined time period of inactivity.

CCI-001150	SC-15(a)	The information system prohibits remote activation of collaborative computing devices excluding the organization-defined exceptions where remote activation is to be allowed.
CCI-001188	SC-23(3)	The information system generates unique session identifiers for each session with organization-defined randomness requirements.
CCI-001664	SC-23(3)	The information system recognizes only session identifiers that are system-generated.
CCI-002470	SC-23(5)	The information system only allows the use of organization-defined certificate authorities for verification of the establishment of protected sessions.
CCI-001190	SC-24	The information system fails to an organization-defined known-state for organization-defined types of failures.

CCI-001665	SC-24	The information system preserves organization-defined system state information in the event of a system failure.
CCI-001094	SC-5(1)	The information system restricts the ability of individuals to launch organization-defined denial of service attacks against other information systems.
CCI-001119	SC-7(13)	The organization isolates organization-defined information security tools, mechanisms, and support components from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.
CCI-001121	SC-7(14)	The organization protects against unauthorized physical connections at organization-defined managed interfaces.

CCI-002421	SC-8(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of information and/or detect changes to information during transmission unless otherwise protected by organization-defined alternative physical safeguards.
CCI-001310	SI-10	The information system checks the validity of organization-defined inputs.
CCI-001314	SI-11(b)	The information system reveals error messages only to organization-defined personnel or roles.
CCI-002824	SI-16	The information system implements organization-defined security safeguards to protect its memory from unauthorized code execution.

CCI-001233	SI-2(2)	The organization employs automated mechanisms on an organization-defined frequency to determine the state of information system components with regard to flaw remediation.
CCI-002617	SI-2(6)	The organization removes organization-defined software components (e.g., previous versions) after updated versions have been installed.
CCI-002618	SI-2(6)	The organization removes organization-defined firmware components (e.g., previous versions) after updated versions have been installed.
CCI-002605	SI-2(c)	The organization installs security-relevant software updates within organization-defined time period of the release of the updates

CCI-002607	SI-2(c)	The organization installs security-relevant firmware updates within organization-defined time period of the release of the updates
CCI-001241	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform periodic scans of the information system on an organization-defined frequency.
CCI-002684	SI-4(22)	The information system audits and/or alerts organization-defined personnel when unauthorized network services are detected.
CCI-002646	SI-4(b)	The organization identifies unauthorized use of the information system through organization-defined techniques and methods.
CCI-002710	SI-7(1)	The information system performs an integrity check of organization-defined software at startup, at organization-defined transitional states or security-relevant events, or on organization-defined frequency.
CCI-002711	SI-7(1)	The information system performs an integrity check of organization-defined firmware at startup, at organization-defined transitional states or security-relevant events, or on organization-defined frequency.

CCI-002712	SI-7(1)	The information system performs an integrity check of organization-defined information at startup, at organization-defined transitional states or security-relevant events, or on organization-defined frequency.
CCI-002723	SI-7(8)	The information system, upon detection of a potential integrity violation, provides the capability to audit the event.
CCI-002724	SI-7(8)	The information system, upon detection of a potential integrity violation, initiates one or more of following actions: generates an audit record; alerts current user; alerts organization-defined personnel or roles; and/or organization-defined other actions.

25 05 11 - Control Correlation Identifiers for FLS Moderate-Moderate-Moderate

Designer Implementation

The organization being inspected/assessed configures the information system to route all remote accesses through the number of managed network access control points defined in AC-17 (3), CCI 2315.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 69.

The organization being inspected/assessed configures the information system to provide the capability to expeditiously disconnect or disable remote access to the information system immediately.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2322.

DoD has defined the time period as immediately.

The organization being inspected/assessed defines and documents the allowed methods of remote access to the information system.

The methods should be defined IAW ports, protocols, and service requirements, as well as access control requirements for any STIGs applicable to the technology in use.

DoD has determined the allowed methods of remote access are not appropriate to define at the Enterprise level.

The organization being inspected/assessed configures the information system to automatically audit account creation actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 18.

The organization being inspected/assessed configures the information system to automatically audit account modification actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1403.

The organization being inspected/assessed configures the information system to automatically audit account disabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1404.

The organization being inspected/assessed configures the information system to automatically audit account removal actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1405.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account creation actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1683.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account modification actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1684.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account disabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1685.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account removal actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1686.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to automatically audit account enabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2130.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account enabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2132.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed documents in the user policies that users are required to log out at the end of the users standard work period unless otherwise defined in formal organizational policy and IAW conditions defined in AC-2 (5) CCI 2133.

DoD has defined the time period as at the end of the users standard work period unless otherwise defined in formal organizational policy.

DoD has defined the actions as disables (or revokes) privileged user account.

The organization being inspected/assessed only permits the use of shared/group accounts that meet the conditions for establishing shared/group accounts defined in AC-2 (9), CCI 2140.

The organization being inspected/assessed configures the information system to enforce approved authorizations for controlling the flow of information within the system based on information flow control policies defined in AC-4, CCI 1548.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1368.

The organization being inspected/assessed configures the information system to enforce approved authorizations for controlling the flow of information between interconnected systems based on information flow control policies defined in AC-4, CCI 1549.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1414.

The organization being inspected/assessed configures the information system to any software except software explicitly documented from executing at higher privilege levels than users executing the software.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2233.

DoD has defined the software as any software except software explicitly documented.

The organization being inspected/assessed configures the information system to limit invalid logon attempts by a user to three attempts during a 15 minute time period.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 44.

DoD has defined the maximum number as three.

DoD has defined the time period as 15 minutes.

The organization being inspected/assessed configures the information system to automatically lock the account or node until the locked account is released by an administrator and delays the next login prompt for a minimum of 5 seconds when the maximum number of unsuccessful attempts is exceeded.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2238.

DoD has defined the delay algorithm as a minimum of 5 seconds.

DoD has defined the time period as until released by an administrator.

The organization being inspected/assessed configures the information system to protect against an individual falsely denying having performed actions to be covered by non-repudiation defined in DoDI 8520.02 and DoDI 8520.03.

DoDI 8520.02 and DoDI 8520.03 meet the DoD requirement to define the actions to be covered by non-repudiation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 166.

The organization being inspected/assessed acquires or designs all information system and network components that provide audit record generation capability for the auditable events defined in AU-2 a.

DoD has defined the information system components as all information system and network components.

The organization being inspected/assessed determines whether the information system is capable of auditing:

- successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. Classification levels),
- Successful and unsuccessful logon attempts,
- Privileged activities or other system level access,
- Starting and ending time for user access to the system,
- Concurrent logons from different workstations,
- Successful and unsuccessful accesses to objects,
- All program initiations,
- All direct access to the information system,
- All account creations, modifications, disabling, and terminations,
- All kernel module load, unload, and restart. The organization must document those auditable events that are not captured.

DoD has defined the information system auditable events as successful and unsuccessful attempts to access, modify, or delete privileges, security objects, security levels, or categories of information (e.g. classification levels). Successful and unsuccessful logon attempts, Privileged activities or other system level access, Starting and ending time for user access to the system, Concurrent logons from different workstations, Successful and unsuccessful accesses to objects, All program initiations, All direct access to the information system. All account creations, modifications, disabling, and terminations. All kernel module load, unload, and restart.

The organization conducting the inspection/assessment reviews the documented audit process as well as audit logs to ensure that the organization being inspected/assessed audits all auditable events defined in AU-2 (a) per occurrence.

DoD has defined the actions as all auditable events defined in AU-2 (a) per occurrence.

The organization being inspected/assessed configures the information system to generate audit records containing the organization defined additional, more detailed information as defined in AU-3 (1), CCI 1488 that is to be included in the audit records.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 135.

The organization being inspected/assessed configures the information system to off-load audit records at a minimum, in real-time for interconnected systems and weekly for stand-alone systems onto a different system or media than the system being audited.

DoD has defined the frequency as at a minimum, real-time for interconnected systems and weekly for stand-alone systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1851.

The organization being inspected/assessed configures the information system to immediately provide a warning to personnel, roles, and/or locations defined in AU-5 (1), CCI 1852 when allocated audit record storage volume reaches 75 percent of repository maximum audit record storage capacity.

DoD has defined the time period as immediate.

DoD has defined the percentage as 75 percent.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1855.

The organization being inspected/assessed configures the information system to alert at a minimum, the SCA and ISSO in the event of an audit processing failure.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 139.

DoD has defined the personnel or roles as at a minimum, the SCA and ISSO.

The organization being inspected/assessed must employ information systems that provide the capability to process audit records for events of interest based on audit fields within audit records defined in AU-7 (1), CCI 1883.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 158.

The organization being inspected/assessed configures the information system to synchronize internal information system clocks every 24 hours for networked systems with an authoritative time server which is synchronized with redundant United States Naval Observatory (USNO) time servers as designated for the appropriate DoD network (NIPRNet / SIPRNet) and/or the Global Positioning System (GPS) when the time difference is greater than the difference defined in AU-8 (1), CCI 1892.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1891.

DoD has defined the frequency as every 24 hours for networked systems.

DoD has defined the authoritative time source as an authoritative time server which is synchronized with redundant United States Naval Observatory (USNO) time servers as designated for the appropriate DoD network (NIPRNet / SIPRNet) and/or the Global Positioning System (GPS).

The organization being inspected/assessed configures the information system to synchronize the internal system clocks to the authoritative time source when the time difference is greater than the time period defined in AU-8 (1), CCI 1892.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2046.

The organization being inspected/assessed configures the information system to generate time in the time stamps for audit records that meets one second granularity of time measurement.

DoD has defined the granularity of time measurement as one second.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1889.

The organization being inspected/assessed configures the information system to generate time stamps for audit records that contain time zones or time offsets that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1890.

The organization being inspected/assessed uses software in accordance with contract agreements.

The organization being inspected/assessed uses software in accordance with copyright laws.

The organization being inspected/assessed reviews and updates the baseline configuration of the information system annually.

The organization must document each occurrence of the reviews and update actions as an audit trail.

DoD has defined the frequency as annually.

The organization being inspected/assessed reviews and updates the baseline configuration of the information system when required due to baseline configuration changes or as events dictate such as changes due to USCYBERCOM tactical orders/ directives or cyber attacks.

The organization must document each occurrence of the reviews and update actions as an audit trail.

DoD has defined the circumstances as baseline configuration changes or as events dictate such as changes due to USCYBERCOM tactical orders/ directives or cyber attacks.

The organization being inspected/assessed configures the information system to prevent the execution of programs not authorized in accordance with CM-7 (2) CCIs 1592 and 1763.

The organization being inspected/assessed implements DoDI 8551.01.

DoD has defined the registration requirements as IAW DoDI 8551.01.

The organization being inspected/assessed documents and implements a process to review and update the list of authorized software programs monthly.

The organization must maintain an audit trail of the review and update activity.

DoD has defined the frequency as monthly.

The organization being inspected/assessed configures the information system to prohibit or restrict the use of functions, ports, protocols, and/or services IAW DoDI 8551.01.

DoD has defined the information system prohibited or restricted functions, ports, protocols, and/or services as IAW DoDI 8551.01.

The organization being inspected/assessed documents and implements automated mechanisms to detect the presence of unauthorized hardware, software, and firmware components within the information system continuously.

DoD has defined the frequency as continuously.

The organization being inspected/assessed documents inventory of information system components that includes organization defined information deemed necessary to achieve effective information system component accountability.

The organization being inspected/assessed documents and implements a process to review and update the information system component inventory at a minimum, annually.

The organization must maintain an audit trail of review and update activity.

DoD has defined the frequency as at a minimum, annually.

The organization being inspected/assessed configures the information system to enter a safe mode of operation with restrictions of safe mode of operation defined in CP-12, CCI 2857 when conditions defined in CP-12, CCI 2856 are detected.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2855.

The organization being inspected/assessed tests and logs backup information at least monthly in accordance with contingency plan to verify media reliability and information integrity.

The organization being inspected/assessed conducts backups of information system documentation including security-related documentation when created or received, when updated, and as required by system baseline configuration changes in accordance with the contingency plan.

The organization being inspected/assessed configures the network infrastructure to identify all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 778.

DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).

The organization being inspected/assessed configures the network infrastructure to authenticate all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1958.

DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).

The organization being inspected/assessed configures the information system to use cryptographically based bidirectional authentication.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1967.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of upper case characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 192.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of lower case characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 193.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of numeric characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 194.

The organization being inspected/assessed configures the information system to enforce minimum password length.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 205.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of special characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1619.

The organization being inspected/assessed configures the information system to enforce that at least 50% of the minimum password length is changed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 195.

DoD has defined the minimum number of characters as 50% of the minimum password length.

The organization being inspected/assessed configures the information system to enforce minimum password lifetime restrictions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 198.

The organization being inspected/assessed configures the information system to enforce maximum password lifetime restrictions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 199.

The organization being inspected/assessed configures the information system to prohibit reuse for a minimum of 5 generations.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 200.

DoD has defined the number of generations as a minimum of 5.

The organization being inspected/assessed configures the information system to prohibit the use of cached authenticators after an organization defined time period.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2007.

The organization being inspected/assessed physically controls and securely stores all digital and non-digital media containing sensitive, controlled, and/or classified information within areas approved for processing or storing data IAW the sensitivity and/or classification level of the information contained on/within the media.

DoD has defined the digital and non-digital media types as all digital and non-digital media containing sensitive, controlled, and/or classified information.

DoD has defined the controlled areas as areas approved for processing or storing data IAW the sensitivity and/or classification level of the information contained on/within the media.

The organization being inspected/assessed protects and controls information system media during transport outside of controlled areas using security measures defined in DoDM 5200.01 M Vol. 1-4 and DoDD 5015.2.

The organization being inspected/assessed must provide master shutoff valves that are accessible to protect the information system from damage resulting from water leakage.

The organization being inspected/assessed will ensure that master shutoff valves are working properly and have been inspected by the appropriate organization (e.g., fire marshal, department of public works).

The organization being inspected/assessed will maintain records of all information system components entering and exiting the facility.

If the organization is following General Records Schedule (GRS) 18, Section 12 they are automatically compliant.

The organization being inspected/assessed will implement physical access authorizations at entry/exit points defined in PE-3, CCI 2915 and secure those physical access points (i.e. doors and/or windows) that are not intended for normal access.

The organization being inspected/assessed conducts and documents an inventory of minimally keys or any other physical token used to gain access annually.

Inventory documents must be retained for at least one year beyond the completion of the next inventory.

DoD has defined the frequency as annually.

DoD has defined the physical access devices as minimally keys or any other physical token used to gain access.

The organization being inspected/assessed controls physical access to information system distribution and transmission lines defined in PE-4, CCI 2930 within organizational facilities using security safeguards defined in PE-4, CCI 2931.

The organization being inspected/assessed will define, document, and implement procedures for vulnerability scans of the information system and hosted applications; and scan for vulnerabilities in the information system and hosted applications every 30 days or as directed by an authoritative source (e.g. IAVM, CTOs, DTMs, STIGs).

This control is not targeted at security control compliance scanning.

DoD has defined the frequency as every 30 days or as directed by an authoritative source (e.g. IAVM, CTOs, DTMs, STIGs).

The organization being inspected/assessed will conduct vulnerability scans of the information system and hosted applications when new vulnerabilities potentially affecting the system/applications are identified and reported via authoritative sources (e.g., IAVM, CTO, DTM, STIG, product vendor).

The organization being inspected/assessed will conduct random vulnerability scans every 30 days or as directed by an authoritative source (e.g. IAVM, CTOs, DTMs, STIGs).

The organization will document the vulnerability scans as an audit trail for future reference. The audit trail must be maintained IAW DoD, CYBERCOM, or component policies.

DoD has defined the frequency as every 30 days or as directed by an authoritative source (e.g. IAVM, CTOs, DTMs, STIGs).

The organization being inspected/assessed takes corrective actions as appropriate on legitimate vulnerabilities identified in RA-5, CCI 001058 IAW an authoritative source (e.g. IAVM, CTOs, DTMs, STIGs).

Audit records of actions must be maintained IAW applicable DoD, CYBERCOM, and/or component policies.

DoD has defined the response times as IAW an authoritative source (e.g. IAVM, CTOs, DTMs, STIGs).

The organization being inspected/assessed documents and implements a process to report counterfeit information system components to source of counterfeit component, at a minimum, USCYBERCOM. And/or at a minimum, the ISSO, ISSM, and PM.

The organization must maintain a record of reporting.

DoD has defined the personnel or roles as at a minimum, the ISSO, ISSM, and PM.

DoD has defined the external reporting organizations as at a minimum, USCYBERCOM.

The organization being inspected/assessed configures the information system to terminate the network connection associated with a communications session at the end of the session or after 10 minutes in band management and 15 minutes for user sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1133.

DoD has defined the time period as 10 minutes in band management and 15 minutes for user sessions.

The organization being inspected/assessed configures the information system to prohibit remote activation of collaborative computing devices excluding dedicated VTC suites located in approved VTC locations that are centrally managed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1150.

DoD has defined the exceptions as dedicated VTC suites located in approved VTC locations that are centrally managed.

The organization being inspected/assessed configures the information system to generate unique session identifiers for each session with randomness requirements defined in SC-23 (3), CCI 1189.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1188.

The organization being inspected/assessed configures the information system to recognize only session identifiers that are system-generated.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1664.

The organization being inspected/assessed configures the information system to allow the use of DoD PKI established certificate authorities for verification of the establishment of protected sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2470.

DoD has defined the certificate authorities as DoD PKI established certificate authorities.

The organization being inspected/assessed configures the information system to fail to a secure state for failures during system initialization, shutdown, and aborts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1190.

DoD has defined the known state as secure state.

DoD has defined the types of failures as failures during system initialization, shutdown, and aborts.

The organization being inspected/assessed configures the information system to preserve information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes in the event of a system failure.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1665.

DoD has defined system state information as information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes.

The organization being inspected/assessed configures the information system to restrict the ability of individuals to launch denial of service attacks defined in SC-5 (1), CCI 2387 against other information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1094.

The organization being inspected/assessed designs the information system to isolate key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc. from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.

DoD has defined the key information security tools, mechanisms, and support components as key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc.

The organization being inspected/assessed documents and implements mechanisms to protect against unauthorized physical connections at internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.

DoD has defined the managed interfaces as internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information and detect changes to information during transmission unless otherwise protected by Protected Distribution System (PDS).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2421.

DoD has defined the selection as both prevention of unauthorized disclosure and detection of changes to information.

DoD has defined the alternative physical safeguards as Protected Distribution System (PDS).

The organization being inspected/assessed configures the information system to check the validity of all inputs except those identified specifically by the organization.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1310.

DoD has defined the information inputs as all inputs except those identified specifically by the organization.

The organization being inspected/assessed configures the information system to reveal error messages only to the ISSO, ISSM, and SCA.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1314.

DoD has defined the personnel or roles as the ISSO, ISSM, and SCA.

The organization being inspected/assessed configures the information system to implement security safeguards defined in SI-16, CCI 2823 to protect its memory from unauthorized code execution.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2824.

The organization being inspected/assessed configures the information system to employ automated mechanisms continuously with HBSS; 30 days for any additional internal network scans not covered by HBSS; annually for external scans by (Computer Network Defense Service Provider) CNDSP to determine the state of information system components with regard to flaw remediation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1233.

DoD has defined the frequency as continuously with HBSS; 30 days for any additional internal network scans not covered by HBSS; annually for external scans by (Computer Network Defense Service Provider) CNDSP.

The organization being inspected/assessed configures the information system to remove all upgraded/replaced software components that are no longer required for operation (e.g., previous versions) after updated versions have been installed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2617.

DoD has defined the software components as all upgraded/replaced software components that are no longer required for operation.

The organization being inspected/assessed configures the information system to remove all upgraded/replaced firmware components that are no longer required for operation e.g., previous versions) after updated versions have been installed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2618.

DoD has defined the firmware components as all upgraded/replaced firmware components that are no longer required for operation.

The organization being inspected/assessed configures the information system to install security-relevant software updates within 30 days of the release of the updates

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2605.

DoD has defined the time period as 30 days.

The organization being inspected/assessed configures the information system to install security-relevant firmware updates within 30 days of the release of the updates.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2607.

DoD has defined the time period as 30 days.

The organization being inspected/assessed configures malicious code protection mechanisms to perform periodic scans of the information system on every 7 days.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1241.

DoD has defined the frequency as every 7 days.

The organization being inspected/assessed configures the information system to audit and/or alert at a minimum, the ISSO and ISSM when unauthorized network services are detected.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2684.

DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.

The organization being inspected/assessed identifies unauthorized use of the information system through techniques and methods defined in SI-4, CCI 2645.

The organization must maintain an audit trail of identified instances of unauthorized use.

The organization being inspected/assessed configures the information system to perform an integrity check of software defined in SI-7 (1), CCI 2705 at startup, at transitional states or security-relevant events defined in SI-7 (1), CCI 2708, or annually.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2710.

DoD has defined the frequency as annually.

The organization being inspected/assessed configures the information system to perform an integrity check of firmware defined in SI-7 (1), CCI 2706 at startup, at transitional states or security-relevant events defined in SI-7 (1), CCI 2708, or annually.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2711.

DoD has defined the frequency as annually.

The organization being inspected/assessed configures the information system to perform an integrity check of information defined in SI-7 (1), CCI 2707 at startup, at transitional states or security-relevant events defined in SI-7 (1), CCI 2708, or annually.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2712.

DoD has defined the frequency as annually.

The organization being inspected/assessed configures the information system to provide the capability to audit the event upon detection of a potential integrity violation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2723.

The organization being inspected/assessed configures the information system to initiate one or more of following actions: generates an audit record; alerts current user; alerts at a minimum, the ISSO and ISSM; and/or other actions defined in SI-7 (8), CCI 2722 upon detection of a potential integrity violation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2724.

DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.

Classification	Responsibility
Impractical	Contractor, Owner
Impractical	DoD Defined
Applicable	DoD Defined
Impractical	Contractor, Owner
Impractical	Contractor, Owner

Reference: UFC 4-010-06 Cybersecurity of Facility-Related Control Systems, with Change 1
<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc/ufc-4-010-06>

Impractical	Contractor, Owner	
Impractical	Owner, Contractor	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	

Impractical	Contractor, Owner	
Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	Owner, Contractor	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	

Impractical	DoD Defined	
Applicable	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Applicable	DoD Defined	

Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	DoD Defined	

Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	Contractor	
Applicable	DoD Defined	

Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	DoD Defined	

Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	DoD Defined	
Applicable	Owner	
Applicable	DoD Defined	
Applicable	Contractor	

Impractical	Owner	
Applicable	Owner	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor, Owner	

Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	

Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Impractical	Contractor	
Applicable	Contractor	
Applicable	Contractor	

Applicable	Owner	
Impractical	Owner	
Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	DoD Defined	
Impractical	Owner	

Impractical	DoD Defined	
Impractical	DoD Defined	
Applicable	DoD Defined	
Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor	
Impractical	DoD Defined	
Applicable	DoD Defined	

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Impractical	DoD Defined	
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Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor	

Impractical	DoD Defined	
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Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	
Applicable	Owner	
Applicable	DoD Defined	
Applicable	DoD Defined	

Applicable	DoD Defined	
Impractical	Contractor	
Impractical	DoD Defined	

Section 2

CCIs for

CCI Number	800-53/82 Control Text Indicator	CCI Definition
CCI-000054	AC-10	The information system limits the number of concurrent sessions for each organization-defined account and/or account type to an organization-defined number of sessions.
CCI-000060	AC-11(1)	The information system conceals, via the session lock, information previously visible on the display with a publicly viewable image.
CCI-000057	AC-11(a)	The information system initiates a session lock after the organization-defined time period of inactivity.
CCI-000058	AC-11(a)	The information system provides the capability for users to directly initiate session lock mechanisms.
CCI-000056	AC-11(b)	The information system retains the session lock until the user reestablishes access using established identification and authentication procedures.
CCI-002295	AC-16(6)	The organization allows personnel to associate organization-defined security attributes with organization-defined subjects in accordance with organization-defined security policies.

CCI-002296	AC-16(6)	The organization allows personnel to associate organization-defined security attributes with organization-defined objects in accordance with organization-defined security policies.
CCI-002297	AC-16(6)	The organization allows personnel to maintain the association of organization-defined security attributes with organization-defined subjects in accordance with organization-defined security policies.
CCI-002298	AC-16(6)	The organization allows personnel to maintain the association of organization-defined security attributes with organization-defined objects in accordance with organization-defined security policies.
CCI-002262	AC-16(a)	The organization provides the means to associate organization-defined types of security attributes having organization-defined security attribute values with information in storage.
CCI-002263	AC-16(a)	The organization provides the means to associate organization-defined types of security attributes having organization-defined security attribute values with information in process.
CCI-002264	AC-16(a)	The organization provides the means to associate organization-defined types of security attributes having organization-defined security attribute values with information in transmission.
CCI-002265	AC-16(b)	The organization ensures that the security attribute associations are made with the information.
CCI-002266	AC-16(b)	The organization ensures that the security attribute associations are retained with the information.

CCI-000067	AC-17(1)	The information system monitors remote access methods.
CCI-002314	AC-17(1)	The information system controls remote access methods.
CCI-000068	AC-17(2)	The information system implements cryptographic mechanisms to protect the confidentiality of remote access sessions.
CCI-001453	AC-17(2)	The information system implements cryptographic mechanisms to protect the integrity of remote access sessions.
CCI-000069	AC-17(3)	The information system routes all remote accesses through organization-defined number managed network access control points.
CCI-002322	AC-17(9)	The organization provides the capability to expeditiously disconnect or disable remote access to the information system within the organization-defined time period.
CCI-000065	AC-17(b)	The organization authorizes remote access to the information system prior to allowing such connections

CCI-001443	AC-18(1)	The information system protects wireless access to the system using authentication of users and/or devices.
CCI-001444	AC-18(1)	The information system protects wireless access to the system using encryption.
CCI-001449	AC-18(3)	The organization disables, when not intended for use, wireless networking capabilities internally embedded within information system components prior to issuance and deployment.
CCI-000015	AC-2(1)	The organization employs automated mechanisms to support the information system account management functions.
CCI-002142	AC-2(10)	The information system terminates shared/group account credentials when members leave the group.
CCI-001682	AC-2(2)	The information system automatically removes or disables emergency accounts after an organization-defined time period for each type of account.
CCI-000018	AC-2(4)	The information system automatically audits account creation actions.

CCI-001403	AC-2(4)	The information system automatically audits account modification actions.
CCI-001404	AC-2(4)	The information system automatically audits account disabling actions.
CCI-001405	AC-2(4)	The information system automatically audits account removal actions.
CCI-001683	AC-2(4)	The information system notifies organization-defined personnel or roles for account creation actions.
CCI-001684	AC-2(4)	The information system notifies organization-defined personnel or roles for account modification actions.
CCI-001685	AC-2(4)	The information system notifies organization-defined personnel or roles for account disabling actions.

CCI-001686	AC-2(4)	The information system notifies organization-defined personnel or roles for account removal actions.
CCI-002130	AC-2(4)	The information system automatically audits account enabling actions.
CCI-002132	AC-2(4)	The information system notifies organization-defined personnel or roles for account enabling actions.
CCI-000213	AC-3	The information system enforces approved authorizations for logical access to information and system resources in accordance with applicable access control policies.
CCI-002165	AC-3(4)	The information system enforces organization-defined discretionary access control policies over defined subjects and objects.
CCI-000225	AC-6	The organization employs the concept of least privilege, allowing only authorized accesses for users (and processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.

CCI-002223	AC-6(1)	The organization explicitly authorizes access to organization-defined security-relevant information.
CCI-002235	AC-6(10)	The information system prevents non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures.
CCI-002233	AC-6(8)	The information system prevents organization-defined software from executing at higher privilege levels than users executing the software.
CCI-002234	AC-6(9)	The information system audits the execution of privileged functions.
CCI-000044	AC-7(a)	The information system enforces the organization-defined limit of consecutive invalid logon attempts by a user during the organization-defined time period.

CCI-002238	AC-7(b)	The information system automatically locks the account or node for either an organization-defined time period, until the locked account or node is released by an administrator, or delays the next login prompt according to the organization-defined delay algorithm when the maximum number of unsuccessful attempts is exceeded.
CCI-000048	AC-8(a)	The information system displays an organization-defined system use notification message or banner before granting access to the system that provides privacy and security notices consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.
CCI-000050	AC-8(b)	The information system retains the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access.
CCI-001384	AC-8(c)(1)	The information system, for publicly accessible systems, displays system use information organization-defined conditions before granting further access.

CCI-001385	AC-8(c)(2)	The information system, for publicly accessible systems, displays references, if any, to monitoring that are consistent with privacy accommodations for such systems that generally prohibit those activities.
CCI-001386	AC-8(c)(2)	The information system for publicly accessible systems displays references, if any, to recording that are consistent with privacy accommodations for such systems that generally prohibit those activities.
CCI-001387	AC-8(c)(2)	The information system for publicly accessible systems displays references, if any, to auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities.
CCI-001388	AC-8(c)(3)	The information system, for publicly accessible systems, includes a description of the authorized uses of the system.
CCI-000174	AU-12(1)	The information system compiles audit records from organization-defined information system components into a system-wide (logical or physical) audit trail that is time-correlated to within organization defined level of tolerance for relationship between time stamps of individual records in the audit trail.
CCI-001914	AU-12(3)	The information system provides the capability for organization-defined individuals or roles to change the auditing to be performed on organization-defined information system components based on organization-defined selectable event criteria within organization-defined time thresholds.

CCI-000169	AU-12(a)	The information system provides audit record generation capability for the auditable events defined in AU-2 a at organization defined information system components.
CCI-000171	AU-12(b)	The information system allows organization-defined personnel or roles to select which auditable events are to be audited by specific components of the information system.
CCI-000172	AU-12(c)	The information system generates audit records for the events defined in AU-2 d with the content defined in AU-3.
CCI-001464	AU-14(1)	The information system initiates session audits at system start-up.
CCI-001462	AU-14(2)	The information system provides the capability for authorized users to capture/record and log content related to a user session.
CCI-001920	AU-14(3)	The information system provides the capability for authorized users to remotely view/hear all content related to an established user session in real time.
CCI-001919	AU-14(b)	The information system provides the capability for authorized users to select a user session to capture/record or view/hear.

CCI-000130	AU-3	The information system generates audit records containing information that establishes what type of event occurred.
CCI-000131	AU-3	The information system generates audit records containing information that establishes when an event occurred.
CCI-000132	AU-3	The information system generates audit records containing information that establishes where the event occurred.
CCI-000133	AU-3	The information system generates audit records containing information that establishes the source of the event.
CCI-000134	AU-3	The information system generates audit records containing information that establishes the outcome of the event.
CCI-001487	AU-3	The information system generates audit records containing information that establishes the identity of any individuals or subjects associated with the event.
CCI-000135	AU-3(1)	The information system generates audit records containing the organization-defined additional, more detailed information that is to be included in the audit records.

CCI-001849	AU-4	The organization allocates audit record storage capacity in accordance with organization-defined audit record storage requirements.
CCI-001851	AU-4(1)	The information system off-loads audit records per organization-defined frequency onto a different system or media than the system being audited.
CCI-001855	AU-5(1)	The information system provides a warning to organization-defined personnel, roles, and/or locations within organization-defined time period when allocated audit record storage volume reaches organization-defined percentage of repository maximum audit record storage capacity.
CCI-001864	AU-6(1)	The organization employs automated mechanisms to integrate audit review and analysis to support organizational processes for investigation of response to suspicious activities.
CCI-001865	AU-6(1)	The organization employs automated mechanisms to integrate reporting processes to support organizational investigation of and response to suspicious activities.
CCI-000154	AU-6(4)	The information system provides the capability to centrally review and analyze audit records from multiple components within the system.

CCI-000148	AU-6(a)	The organization reviews and analyzes information system audit records on an organization defined frequency for indications of organization-defined inappropriate or unusual activity.
CCI-000159	AU-8(a)	The information system uses internal system clocks to generate time stamps for audit records.
CCI-001889	AU-8(b)	The information system records time stamps for audit records that meets organization-defined granularity of time measurement.
CCI-001890	AU-8(b)	The information system records time stamps for audit records that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).
CCI-000162	AU-9	The information system protects audit information from unauthorized access.
CCI-000163	AU-9	The information system protects audit information from unauthorized modification.

CCI-000164	AU-9	The information system protects audit information from unauthorized deletion.
CCI-001493	AU-9	The information system protects audit tools from unauthorized access.
CCI-001494	AU-9	The information system protects audit tools from unauthorized modification.
CCI-001495	AU-9	The information system protects audit tools from unauthorized deletion.
CCI-002080	CA-3(5)	The organization employs either an allow-all, deny-by exception or deny-all, permit by exception policy for allowing organization-defined information systems to connect to external information systems.
CCI-001726	CM-10(a)	The organization uses software in accordance with contract agreements.
CCI-001728	CM-10(a)	The organization uses software in accordance with copyright laws.
CCI-001812	CM-11(2)	The information system prohibits user installation of software without explicit privileged status.
CCI-001807	CM-11(b)	The organization enforces software installation policies through organization-defined methods.

CCI-000327	CM-3(2)	The organization tests changes to the information system before implementing the changes on the operational system.
CCI-000328	CM-3(2)	The organization validates changes to the information system before implementing the changes on the operational system.
CCI-001746	CM-3(6)	The organization ensures that cryptographic mechanisms used to provide organization-defined security safeguards are under configuration management.
CCI-000340	CM-5	The organization approves physical access restrictions associated with changes to the information system.
CCI-000341	CM-5	The organization enforces physical access restrictions associated with changes to the information system.
CCI-001762	CM-7(1)(b)	The organization disables organization-defined functions, ports, protocols, and services within the information system deemed to be unnecessary and/or nonsecure.
CCI-001764	CM-7(2)	The information system prevents program execution in accordance with organization-defined policies regarding software program usage and restrictions, and/or rules authorizing the terms and conditions of software program usage.
CCI-001774	CM-7(5)b	The organization employs an deny-all, permit-by-exception policy to allow the execution of authorized software programs on the information system.
CCI-000382	CM-7(b)	The organization configures the information system to prohibit or restrict the use of organization-defined functions, ports, protocols, and/or services.
CCI-000411	CM-8(2)	The organization employs automated mechanisms to help maintain an up-to-date inventory of information system components.

CCI-000412	CM-8(2)	The organization employs automated mechanisms to help maintain a complete inventory of information system components.
CCI-000413	CM-8(2)	The organization employs automated mechanisms to help maintain a accurate inventory of information system components.
CCI-000414	CM-8(2)	The organization employs automated mechanisms to help maintain a readily available inventory of information system components.
CCI-000416	CM-8(3)(a)	The organization employs automated mechanisms, per organization defined frequency, to detect the presence of unauthorized hardware, software, and firmware components within the information system.
CCI-001784	CM-8(3)(b)	When unauthorized hardware, software, and firmware components are detected within the information system, the organization takes action to disable network access by such components, isolates the components, and/or notifies organization-defined personnel or roles.
CCI-001792	CM-9(b)	The organization implements a configuration management plan for the information system that establishes a process for identifying configuration items throughout the system development life cycle.
CCI-001795	CM-9(b)	The organization implements a configuration management plan for the information system that establishes a process for managing the configuration of the configuration items.
CCI-000426	CM-9(c)	The organization implements a configuration management plan for the information system that defines the configuration items for the information system.

CCI-001801	CM-9(d)	The organization implements a configuration management plan for the information system that protects the configuration management plan from unauthorized disclosure and modification.
CCI-002855	CP-12	The information system, when organization-defined conditions are detected, enters a safe mode of operation with organization-defined restrictions of safe mode of operation.
CCI-000505	CP-6(a)	The organization establishes an alternate storage site including necessary agreements to permit the storage and retrieval of information system backup information.
CCI-000542	CP-9(1)	The organization tests backup information per organization-defined frequency to verify media reliability and information integrity.
CCI-000540	CP-9(d)	The organization protects the confidentiality, integrity, and availability of backup information at storage locations.
CCI-000764	IA-2	The information system uniquely identifies and authenticates organizational users (or processes acting on behalf of organizational users).
CCI-000765	IA-2(1)	The information system implements multifactor authentication for network access to privileged accounts.
CCI-001948	IA-2(11)	The information system implements multifactor authentication for remote access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access.

CCI-001949	IA-2(11)	The device used in the information system implementation of multifactor authentication for remote access to privileged accounts meets organization-defined strength of mechanism requirements.
CCI-001951	IA-2(11)	The information system implements multifactor authentication for remote access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access.
CCI-001952	IA-2(11)	The device used in the information system implementation of multifactor authentication for remote access to non-privileged accounts meets organization-defined strength of mechanism requirements.
CCI-000766	IA-2(2)	The information system implements multifactor authentication for network access to non-privileged accounts.
CCI-000767	IA-2(3)	The information system implements multifactor authentication for local access to privileged accounts.
CCI-000768	IA-2(4)	The information system implements multifactor authentication for local access to non-privileged accounts.
CCI-000770	IA-2(5)	The organization requires individuals to be authenticated with an individual authenticator when a group authenticator is employed.

CCI-001941 IA-2(8) The information system implements replay-resistant authentication mechanisms for network access to privileged accounts.

CCI-001942 IA-2(9) The information system implements replay-resistant authentication mechanisms for network access to non-privileged accounts.

CCI-000778 IA-3 The information system uniquely identifies an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.

CCI-001958 IA-3 The information system authenticates an organization defined list of specific and/or types of devices before establishing a local, remote, or network connection.

CCI-001967 IA-3(1) The information system authenticates organization-defined devices and/or types of devices before establishing a local, remote and/or network connection using bidirectional authentication that is cryptographically based.

CCI-000192	IA-5(1)(a)	The information system enforces password complexity by the minimum number of upper case characters used.
CCI-000193	IA-5(1)(a)	The information system enforces password complexity by the minimum number of lower case characters used.
CCI-000194	IA-5(1)(a)	The information system enforces password complexity by the minimum number of numeric characters used.
CCI-000205	IA-5(1)(a)	The information system enforces minimum password length.
CCI-001619	IA-5(1)(a)	The information system enforces password complexity by the minimum number of special characters used.
CCI-000195	IA-5(1)(b)	The information system, for password-based authentication, when new passwords are created, enforces that at least an organization-defined number of characters are changed.
CCI-000196	IA-5(1)(c)	The information system, for password-based authentication, stores only cryptographically-protected passwords.

CCI-000197	IA-5(1)(c)	The information system, for password-based authentication, transmits only cryptographically-protected passwords.
CCI-000198	IA-5(1)(d)	The information system enforces minimum password lifetime restrictions.
CCI-000199	IA-5(1)(d)	The information system enforces maximum password lifetime restrictions.
CCI-000200	IA-5(1)(e)	The information system prohibits password reuse for the organization defined number of generations.
CCI-002041	IA-5(1)(f)	The information system allows the use of a temporary password for system logons with an immediate change to a permanent password.
CCI-002003	IA-5(11)	The information system, for token-based authentication, employs mechanisms that satisfy organization-defined token quality requirements.

CCI-002007	IA-5(13)	The information system prohibits the use of cached authenticators after an organization defined time period.
CCI-000185	IA-5(2)(a)	The information system, for PKI-based authentication validates certifications by constructing and verifying a certification path to an accepted trust anchor including checking certificate status information.
CCI-000186	IA-5(2)(b)	The information system, for PKI-based authentication enforces authorized access to the corresponding private key.
CCI-000187	IA-5(2)(c)	The information system, for PKI-based authentication, maps the authenticated identity to the account of the individual or group.
CCI-001997	IA-5(4)	The organization employs automated tools to determine if password authenticators are sufficiently strong to satisfy organization-defined requirements.
CCI-000803	IA-7	The information system implements mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication.

CCI-000804	IA-8	The information system uniquely identifies and authenticates non-organizational users (or processes acting on behalf of non-organizational users).
CCI-002009	IA-8(1)	The information system accepts Personal Identity Verification (PIV) credentials from other federal agencies.
CCI-002010	IA-8(1)	The information system electronically verifies Personal Identity Verification (PIV) credentials from other federal agencies.
CCI-000840	IR-7(1)	The organization employs automated mechanisms to increase the availability of incident response-related information and support.
CCI-000841	IR-7(2)(a)	The organization establishes a direct, cooperative relationship between its incident response capability and external providers of information system protection capability.

CCI-000861	MA-2(d)	The organization sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs.
CCI-002884	MA-4(1)(a)	The organization audits nonlocal maintenance and diagnostic sessions' organization-defined audit events.
CCI-002886	MA-4(1)(b)	The organization reviews the records of the nonlocal maintenance and diagnostic sessions.
CCI-002890	MA-4(6)	The information system implements cryptographic mechanisms to protect the integrity of nonlocal maintenance and diagnostic communications.
CCI-003123	MA-4(6)	The information system implements cryptographic mechanisms to protect the confidentiality of nonlocal maintenance and diagnostic communications.
CCI-002891	MA-4(7)	The information system implements remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.
CCI-000877	MA-4(c)	The organization employs strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions.

CCI-000879	MA-4(e)	The organization terminates sessions and network connections when nonlocal maintenance is completed.
CCI-001020	MP-5(a)	The organization protects and controls organization-defined types of information system media during transport outside of controlled areas using organization-defined security safeguards.
CCI-002585	MP-7(1)	The organization prohibits the use of portable storage devices in organization information systems when such devices have no identifiable owner.
CCI-000959	PE-10(c)	The organization protects emergency power shutoff capability from unauthorized activation.
CCI-002955	PE-11	The organization provides a short-term uninterruptible power supply to facilitate an orderly shutdown of the information system and/or transition of the information system to long-term alternate power in the event of a primary power source loss.
CCI-000981	PE-16	The organization authorizes organization-defined types of information system components entering and exiting the facility.
CCI-000912	PE-2(a)	The organization develops a list of individuals with authorized access to the facility where the information system resides.
CCI-002910	PE-2(a)	The organization approves a list of individuals with authorized access to the facility where the information system resides.
CCI-002911	PE-2(a)	The organization maintains a list of individuals with authorized access to the facility where the information system resides.

CCI-000913	PE-2(b)	The organization issues authorization credentials for facility access.
CCI-001635	PE-2(c)	The organization removes individuals from the facility access list when access is no longer required.
CCI-000919	PE-3(a)	The organization enforces physical access authorizations at organization-defined entry/exit points to the facility where the information system resides.
CCI-000921	PE-3(a)(2)	The organization controls ingress/egress to the facility using one or more organization-defined physical access control systems/devices or guards.
CCI-002917	PE-3(b)	The organization maintains physical access audit logs for organization-defined entry/exit points.
CCI-002919	PE-3(c)	The organization provides organization-defined security safeguards to control access to areas within the facility officially designated as publicly accessible.
CCI-002921	PE-3(d)	The organization escorts visitors during organization-defined circumstances requiring visitor escorts.
CCI-002923	PE-3(d)	The organization monitors visitor activity during organization-defined circumstances requiring visitor monitoring.
CCI-000923	PE-3(e)	The organization secures keys, combinations, and other physical access devices.
CCI-002950	PE-6(4)	The organization monitors physical access to the information system in addition to the physical access monitoring of the facility as organization-defined physical spaces containing one or more components of the information system.

CCI-001517	PS-3(b)	The organization rescreens individuals with authorized access to the information system according to organization-defined conditions requiring rescreening, and where rescreening is so indicated, the organization-defined frequency of such rescreening.
CCI-001067	RA-5(5)	The information system implements privileged access authorization to organization-identified information system components for selected organization-defined vulnerability scanning activities.
CCI-001133	SC-10	The information system terminates the network connection associated with a communications session at the end of the session or after an organization-defined time period of inactivity.
CCI-001152	SC-15(b)	The information system provides an explicit indication of use to users physically present at collaborative computing devices.
CCI-002465	SC-21	The information system requests data origin authentication verification on the name/address resolution responses the system receives from authoritative sources.

CCI-002466 SC-21 The information system requests data integrity verification on the name/address resolution responses the system receives from authoritative sources.

CCI-002467 SC-21 The information system performs data integrity verification on the name/address resolution responses the system receives from authoritative sources.

CCI-002468 SC-21 The information system performs data origin verification authentication on the name/address resolution responses the system receives from authoritative sources.

CCI-001182 SC-22 The information systems that collectively provide name/address resolution service for an organization are fault-tolerant.

CCI-001183 SC-22 The information systems that collectively provide name/address resolution service for an organization implement internal/external role separation.

CCI-001184	SC-23	The information system protects the authenticity of communications sessions.
CCI-001185	SC-23(1)	The information system invalidates session identifiers upon user logout or other session termination.
CCI-001188	SC-23(3)	The information system generates unique session identifiers for each session with organization-defined randomness requirements.
CCI-001664	SC-23(3)	The information system recognizes only session identifiers that are system-generated.
CCI-002470	SC-23(5)	The information system only allows the use of organization-defined certificate authorities for verification of the establishment of protected sessions.
CCI-001190	SC-24	The information system fails to an organization-defined known-state for organization-defined types of failures.

CCI-001665	SC-24	The information system preserves organization-defined system state information in the event of a system failure.
CCI-001199	SC-28	The information system protects the confidentiality and/or integrity of organization-defined information at rest.
CCI-002475	SC-28(1)	The information system implements cryptographic mechanisms to prevent unauthorized modification of organization-defined information at rest on organization-defined information system components.
CCI-002476	SC-28(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of organization-defined information at rest on organization-defined information system components.
CCI-002529	SC-38	The organization employs organization-defined operations security safeguards to protect key organizational information throughout the system development life cycle.

CCI-002546	SC-41	The organization physically disables or removes organization-defined connection ports or input/output devices on organization-defined information systems or information system components.
CCI-002385	SC-5	The information system protects against or limits the effects of organization-defined types of denial of service attacks by employing organization-defined security safeguards.
CCI-001094	SC-5(1)	The information system restricts the ability of individuals to launch organization-defined denial of service attacks against other information systems.
CCI-001095	SC-5(2)	The information system manages excess capacity, bandwidth, or other redundancy to limit the effects of information flooding types of denial of service attacks.
CCI-002389	SC-5(3)(a)	The organization employs organization-defined list of monitoring tools to detect indicators of denial of service attacks against the information system.
CCI-001116	SC-7(10)	The organization prevents the unauthorized exfiltration of information across managed interfaces.
CCI-002403	SC-7(11)	The information system only allows incoming communications from organization-defined authorized sources routed to organization-defined authorized destinations.

CCI-002406	SC-7(12)	The organization implements organization-defined host-based boundary protection mechanisms at organization-defined information system components.
CCI-001119	SC-7(13)	The organization isolates organization-defined information security tools, mechanisms, and support components from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.
CCI-001121	SC-7(14)	The organization protects against unauthorized physical connections at organization-defined managed interfaces.
CCI-001101	SC-7(3)	The organization limits the number of external network connections to the information system.
CCI-002396	SC-7(4)(c)	The organization protects the confidentiality and integrity of the information being transmitted across each interface for each external telecommunication service.
CCI-001109	SC-7(5)	The information system at managed interfaces denies network communications traffic by default and allows network communications traffic by exception (i.e., deny all, permit by exception).

CCI-002397	SC-7(7)	The information system, in conjunction with a remote device, prevents the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks.
CCI-002398	SC-7(9)(a)	The information system detects outgoing communications traffic posing a threat to external information systems.
CCI-002399	SC-7(9)(a)	The information system denies outgoing communications traffic posing a threat to external information systems.
CCI-002400	SC-7(9)(b)	The information system audits the identity of internal users associated with denied outgoing communications traffic posing a threat to external information systems.
CCI-001097	SC-7(a)	The information system monitors and controls communications at the external boundary of the system and at key internal boundaries within the system.
CCI-002395	SC-7(b)	The information system implements subnetworks for publicly accessible system components that are physically and/or logically separated from internal organizational networks.
CCI-001098	SC-7(c)	The information system connects to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.

CCI-002418	SC-8	The information system protects the confidentiality and/or integrity of transmitted information.
CCI-002421	SC-8(1)	The information system implements cryptographic mechanisms to prevent unauthorized disclosure of information and/or detect changes to information during transmission unless otherwise protected by organization-defined alternative physical safeguards.
CCI-002420	SC-8(2)	The information system maintains the confidentiality and/or integrity of information during preparation for transmission.
CCI-002422	SC-8(2)	The information system maintains the confidentiality and/or integrity of information during reception.
CCI-001310	SI-10	The information system checks the validity of organization-defined inputs.

CCI-001312	SI-11(a)	The information system generates error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.
CCI-001314	SI-11(b)	The information system reveals error messages only to organization-defined personnel or roles.
CCI-002824	SI-16	The information system implements organization-defined security safeguards to protect its memory from unauthorized code execution.
CCI-002775	SI-17	The information system implements organization-defined fail-safe procedures when organization-defined failure conditions occur.
CCI-002617	SI-2(6)	The organization removes organization-defined software components (e.g., previous versions) after updated versions have been installed.

CCI-002618	SI-2(6)	The organization removes organization-defined firmware components (e.g., previous versions) after updated versions have been installed.
CCI-001225	SI-2(a)	The organization identifies information system flaws.
CCI-001226	SI-2(a)	The organization reports information system flaws.
CCI-001228	SI-2(b)	The organization tests software updates related to flaw remediation for effectiveness before installation.
CCI-001229	SI-2(b)	The organization tests software updates related to flaw remediation for potential side effects before installation.
CCI-002602	SI-2(b)	The organization tests firmware updates related to flaw remediation for effectiveness before installation.
CCI-002603	SI-2(b)	The organization tests firmware updates related to flaw remediation for potential side effects before installation.
CCI-002605	SI-2(c)	The organization installs security-relevant software updates within organization-defined time period of the release of the updates

CCI-002607	SI-2(c)	The organization installs security-relevant firmware updates within organization-defined time period of the release of the updates
CCI-002636	SI-3(10)(a)	The organization employs organization-defined tools to analyze the characteristics and behavior of malicious code.
CCI-002638	SI-3(10)(a)	The organization employs organization-defined techniques to analyze the characteristics and behavior of malicious code.
CCI-001247	SI-3(2)	The information system automatically updates malicious code protection mechanisms.
CCI-002621	SI-3(a)	The organization employs malicious code protection mechanisms at information system entry points to eradicate malicious code.
CCI-002622	SI-3(a)	The organization employs malicious code protection mechanisms at information system exit points to eradicate malicious code.
CCI-001240	SI-3(b)	The organization updates malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures.
CCI-001241	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform periodic scans of the information system on an organization-defined frequency.

CCI-001242	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform real-time scans of files from external sources at endpoints as the files are downloaded, opened, or executed in accordance with organizational security policy.
CCI-002624	SI-3(c)(1)	The organization configures malicious code protection mechanisms to perform real-time scans of files from external sources at network entry/exit points as the files are downloaded, opened, or executed in accordance with organizational security policy.
CCI-001243	SI-3(c)(2)	The organization configures malicious code protection mechanisms to perform organization-defined action(s) in response to malicious code detection.
CCI-001245	SI-3(d)	The organization addresses the receipt of false positives during malicious code detection and eradication, and the resulting potential impact on the availability of the information system.
CCI-002737	SI-7(14)(a)	The organization prohibits the use of binary or machine-executable code from sources with limited or no warranty and without the provision of source code.
CCI-002723	SI-7(8)	The information system, upon detection of a potential integrity violation, provides the capability to audit the event.

CCI-002724

SI-7(8)

The information system, upon detection of a potential integrity violation, initiates one or more of following actions:
generates an audit record; alerts current user; alerts organization-defined personnel or roles; and/or organization-defined other actions.

5 05 11 - Control Correlation Identifiers for MCS Moderate-Moderate-Moderate

Designer Implementation

The organization being inspected/assessed configures the information system to limit the number of concurrent sessions for all accounts and/or account types to a number of sessions defined in AC-10, CCI 55.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 54.

DoD has defined the account types and/or accounts as all account types and/or accounts.

The organization being inspected/assessed configures the information system to conceal, via the session lock, information previously visible on the display with a publicly viewable image.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 60.

The organization being inspected/assessed configures the information system to initiate a session lock after the organization-defined time period defined in CCI 59.

The organization being inspected/assessed configures the information system to provide the capability for users to directly initiate session lock mechanisms.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 58.

The organization being inspected/assessed configures the information system to retain the session lock until the user reestablishes access using established identification and authentication procedures.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 56.

The organization being inspected/assessed documents and implements a process requiring personnel to associate security attributes defined in AC-16 (6), CCI 2292 with subjects defined in AC-16 (6), CCI 2293 in accordance with security policies defined in AC-16 (6), CCI 2291.

The organization being inspected/assessed documents and implements a process requiring personnel to associate security attributes defined in AC-16 (6), CCI 2292 with objects defined in AC-16 (6), CCI 2294 in accordance with security policies defined in AC-16 (6), CCI 2291.

The organization being inspected/assessed documents and implements a process requiring personnel to maintain the association of security attributes defined in AC-16 (6), CCI 2292 with subjects defined in AC-16 (6), CCI 2293 in accordance with security policies defined in AC-16 (6), CCI 2291.

The organization being inspected/assessed documents and implements a process requiring personnel to maintain the association of security attributes defined in AC-16 (6), CCI 2292 with objects defined in AC-16 (6), CCI 2294 in accordance with security policies defined in AC-16 (6), CCI 2291.

The organization being inspected/assessed configures the information system to provide the means to associate types of security attributes in defined in AC-16, CCI 2256 having security attribute values defined in AC-16, CCI 2259 with information in storage.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2262.

The organization being inspected/assessed configures the information system to provide the means to associate types of security attributes in defined in AC-16, CCI 2257 having security attribute values defined in AC-16, CCI 2260 with information in process.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2263.

The organization being inspected/assessed configures the information system to provide the means to associate types of security attributes in defined in AC-16, CCI 2258 having security attribute values defined in AC-16, CCI 2261 with information in transmission.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2264.

The organization being inspected/assessed documents and implements a process to ensure that the security attribute associations are made with the information.

The organization being inspected/assessed documents and implements a process to ensure that the security attribute associations are retained with the information.

The organization being inspected/assessed configures the information system to monitor remote access methods.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 67.

The organization being inspected/assessed configures the information system to control remote access methods.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2314.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the confidentiality of remote access sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 68.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the integrity of remote access sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1453.

The organization being inspected/assessed configures the information system to route all remote accesses through the number of managed network access control points defined in AC-17 (3), CCI 2315.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 69.

The organization being inspected/assessed configures the information system to provide the capability to expeditiously disconnect or disable remote access to the information system immediately.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2322.

DoD has defined the time period as immediately.

The organization being inspected/assessed authorizes remote access to the information system prior to allowing such connections.

The organization must maintain an audit trail of authorizations.

The organization being inspected/assessed configures the information system to protect wireless access to the system using authentication of users and/or devices.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1443.

The organization being inspected/assessed configures the information system to protect wireless access to the system using encryption.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1444.

The organization being inspected/assessed documents and implements a process to disable wireless networking capabilities internally embedded within information system components prior to issuance and deployment when not intended for use.

The organization being inspected/assessed configures the information system to employ automated mechanisms to support the information system account management functions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 000015.

The organization being inspected/assessed configures the information system to terminate shared/group account credentials when members leave the group.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2142.

The organization being inspected/assessed configures the information system to never automatically remove or disable emergency accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1682.

DoD has defined the time period as never.

The organization being inspected/assessed configures the information system to automatically audit account creation actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 18.

The organization being inspected/assessed configures the information system to automatically audit account modification actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1403.

The organization being inspected/assessed configures the information system to automatically audit account disabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1404.

The organization being inspected/assessed configures the information system to automatically audit account removal actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1405.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account creation actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1683.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account modification actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1684.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account disabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1685.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account removal actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1686.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to automatically audit account enabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2130.

The organization being inspected/assessed configures the information system to notify the system administrator and ISSO for account enabling actions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2132.

DoD has defined the personnel or roles as the system administrator and ISSO.

The organization being inspected/assessed configures the information system to enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 213.

The organization being inspected/assessed configures the information system to enforce the discretionary access control policies defined in AC-3 (4), CCI 2163 over defined subjects and objects.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2165.

The organization being inspected/assessed documents and implements the concept of least privilege, allowing only authorized accesses for users (and processes acting on behalf of users) which are necessary to accomplish assigned tasks in accordance with organizational missions and business functions.

The organization being inspected/assessed documents and implements a process to explicitly authorize access to all security-relevant information not publicly available.

Explicit authorization can be in the form of an acceptable use policy signed by the user at the time of access being granted.

DoD has defined the security-relevant information as all security-relevant information not publicly available.

The organization being inspected/assessed configures the information system to prevent non-privileged users from executing privileged functions to include disabling, circumventing, or altering implemented security safeguards/countermeasures.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2235.

The organization being inspected/assessed configures the information system to any software except software explicitly documented from executing at higher privilege levels than users executing the software.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2233.

DoD has defined the software as any software except software explicitly documented.

The organization being inspected/assessed configures the information system to audit the execution of privileged functions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2234.

The organization being inspected/assessed configures the information system to limit invalid logon attempts by a user to three attempts during a 15 minute time period.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 44.

DoD has defined the maximum number as three.

DoD has defined the time period as 15 minutes.

The organization being inspected/assessed configures the information system to automatically lock the account or node until the locked account is released by an administrator and delays the next login prompt for a minimum of 5 seconds when the maximum number of unsuccessful attempts is exceeded.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2238.

DoD has defined the delay algorithm as a minimum of 5 seconds.

DoD has defined the time period as until released by an administrator.

The organization being inspected/assessed configures the information system to display the DoD Information Systems – Standard Consent Banner and User Agreement before granting access to the system that provides privacy and security notices consistent with applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 48.

DoD has defined the use notification message or banner as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems – Standard Consent Banner and User Agreement," March 2013.

The organization being inspected/assessed configures the information system to retain the notification message or banner on the screen until users acknowledge the usage conditions and take explicit actions to log on to or further access.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 50.

The organization being inspected/assessed configures the information system to display the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013 before granting further access for publicly accessible systems

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1384.

DoD has defined the conditions as the content of DTM 08-060, "Policy on Use of Department of Defense (DoD) Information Systems - Standard Consent Banner and User Agreement," March 2013.

The organization being inspected/assessed configures the information system to display references, if any, to monitoring that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1385.

The organization being inspected/assessed configures the information system to display references, if any, to recording that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1386.

The organization being inspected/assessed configures the information system to display references, if any, to auditing that are consistent with privacy accommodations for such systems that generally prohibit those activities for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1387.

The organization being inspected/assessed configures the information system to include a description of the authorized uses of the system for publicly accessible systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1388.

The organization being inspected/assessed configures the information system to compile audit records from information system components defined in AU-12 (1), CCI 1577 into a system-wide (logical or physical) audit trail that is time-correlated to within the level of tolerance defined in AU-12 (1), CCI-000173 for relationship between time stamps of individual records in the audit trail.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 174.

The organization being inspected/assessed configures the information system to provide the capability for individuals or roles defined in AU-12 (3), CCI 1913 to change the auditing to be performed on information system components defined in AU-12 (3), CCI 2047 based on selectable event criteria defined in AU-12 (3), CCI 1911 within time thresholds defined in AU-12 (3), CCI 1912.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1914.

The organization being inspected/assessed acquires or designs all information system and network components that provide audit record generation capability for the auditable events defined in AU-2 a.

DoD has defined the information system components as all information system and network components.

The organization being inspected/assessed configures the information system to ensure that only the ISSM or individuals appointed by the ISSM select which auditable events are to be audited by specific components of the information system.

DoD has defined the personnel or roles as the ISSM or individuals appointed by the ISSM.

The organization being inspected/assessed configures the information system to generate audit records for the events defined in AU-2 d with the content defined in AU-3.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 172.

The organization being inspected/assessed configures the information system to initiate session audits at system start-up.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1464.

The organization being inspected/assessed configures the information system to provide the capability for authorized users to capture/record and log content related to a user session.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1462.

The organization being inspected/assessed configures the information system to provide the capability for authorized users to remotely view/hear all content related to an established user session in real time.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1920.

The organization being inspected/assessed configures the information system to provide the capability for authorized users to select a user session to capture/record or view/hear.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1919.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes what type of event occurred

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 130.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes when an event occurred

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 131.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes where the event occurred

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 132.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the source of the event.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 133.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the outcome of the event.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 134.

The organization being inspected/assessed configures the information system to generate audit records containing information that establishes the identity of any individuals or subjects associated with the event.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1487.

The organization being inspected/assessed configures the information system to generate audit records containing the organization defined additional, more detailed information as defined in AU-3 (1), CCI 1488 that is to be included in the audit records.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 135.

The organization being inspected/assessed allocates, and configures the information system to allocate audit record storage capacity as defined in AU-4, CCI 1848.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1849.

The organization being inspected/assessed configures the information system to off-load audit records at a minimum, in real-time for interconnected systems and weekly for stand-alone systems onto a different system or media than the system being audited.

DoD has defined the frequency as at a minimum, real-time for interconnected systems and weekly for stand-alone systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1851.

The organization being inspected/assessed configures the information system to immediately provide a warning to personnel, roles, and/or locations defined in AU-5 (1), CCI 1852 when allocated audit record storage volume reaches 75 percent of repository maximum audit record storage capacity.

DoD has defined the time period as immediate.

DoD has defined the percentage as 75 percent.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1855.

The organization being inspected/assessed identifies and implements automated mechanisms to integrate audit review and analysis.

The goal is to support organizational investigation of and response to suspicious activities.

The organization being inspected/assessed identifies and implements automated mechanisms to integrate reporting processes (e.g., centralized log analysis tools).

The goal is to support organizational investigation of and response to suspicious activities.

The organization being inspected/assessed configures the information system to provide a capability to centrally review and analyze audit records from multiple components within the system.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 154.

The organization being inspected/assessed documents and implements a process to review and analyze information system audit records every seven days or more frequently if required by an alarm event or anomaly for indications of activity defined in AU-6, CCI 1862.

The organization must maintain an audit trail of the reviews.

DoD has defined the frequency as every seven days or more frequently if required by an alarm event or anomaly.

The organization being inspected/assessed configures the information system to use internal system clocks to generate time stamps for audit records.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 159.

The organization being inspected/assessed configures the information system to generate time in the time stamps for audit records that meets one second granularity of time measurement.

DoD has defined the granularity of time measurement as one second.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1889.

The organization being inspected/assessed configures the information system to generate time stamps for audit records that contain time zones or time offsets that can be mapped to Coordinated Universal Time (UTC) or Greenwich Mean Time (GMT).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1890.

The organization being inspected/assessed configures the information system to disallow unauthorized access to audit information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 162.

The organization being inspected/assessed configures the information system to disallow unauthorized modification of audit information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 163.

The organization being inspected/assessed configures the information system to disallow unauthorized deletion of audit information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 164.

The organization being inspected/assessed configures the information system to disallow unauthorized access to audit tools.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1493.

The organization being inspected/assessed configures the information system to disallow unauthorized modification of audit tools.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1494.

The organization being inspected/assessed configures the information system to disallow unauthorized deletion of audit tools.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1495.

The organization being inspected/assessed configures the information system to employ a deny-all, permit by exception policy for allowing any systems requiring external connectivity to connect to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2080.

DoD has defined the information systems as any systems requiring external connectivity.

The organization being inspected/assessed uses software in accordance with contract agreements.

The organization being inspected/assessed uses software in accordance with copyright laws.

The organization being inspected/assessed must configure the information system to prevent the installation of software by non-privileged users.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1812.

The organization being inspected/assessed must enforce software installation policies as defined in CM-11, CCI 1804 through methods defined in CM-11, CCI 1806.

The organization being inspected/assessed documents and implements a process to test changes to the information system before implementing the changes on the operational system.

The organization must maintain an audit trail of testing activity.

The organization being inspected/assessed documents and implements a process to validate changes to the information system before implementing the changes on the operational system.

The organization must maintain an audit trail of validation activity.

The organization being inspected/assessed ensures that cryptographic mechanisms used to provide all security safeguards are under configuration management.

DoD has defined the security safeguards as all security safeguards.

The organization being inspected/assessed documents and implements a process to approve physical access restrictions associated with changes to the information system.

The organization must maintain an audit trail of approvals.

The organization being inspected/assessed documents and implements a process to enforce physical access restrictions associated with changes to the information system.

The organization being inspected/assessed must disable functions, ports, protocols, and services within the information system deemed to be unnecessary and/or nonsecure as defined in CM-7 (1), CCI 1761.

The organization being inspected/assessed configures the information system to prevent the execution of programs not authorized in accordance with CM-7 (2) CCIs 1592 and 1763.

The organization being inspected/assessed configures the information system to deny-all and only permit by exception the execution of authorized software programs on the information system.

The organization being inspected/assessed configures the information system to prohibit or restrict the use of functions, ports, protocols, and/or services IAW DoDI 8551.01.

DoD has defined the information system prohibited or restricted functions, ports, protocols, and/or services as IAW DoDI 8551.01.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain an up-to-date inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains an up-to-date inventory.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain a complete inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains a complete inventory.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain an accurate inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains an accurate inventory.

The organization being inspected/assessed documents and implements automated mechanisms to help maintain a readily available inventory of information system components.

An automated mechanism implemented IAW CM-2 (2) satisfies the requirements of this CCI if the automated mechanism maintains a readily available inventory.

The organization being inspected/assessed documents and implements automated mechanisms to detect the presence of unauthorized hardware, software, and firmware components within the information system continuously.

DoD has defined the frequency as continuously.

The organization being inspected/assessed documents and implements a process to take action to disable network access by unauthorized software, hardware, and firmware components, isolate the components, and/or notify the ISSO and ISSM and others as the local organization deems appropriate.

The organization must maintain an audit trail of actions taken upon detection of unauthorized software, hardware, and firmware components.

DoD has defined the personnel or roles as the ISSO and ISSM and others as the local organization deems appropriate.

The organization being inspected/assessed will implement a configuration management plan for the information system that establishes a process for identifying configuration items throughout the system development life cycle.

The organization being inspected/assessed will implement a configuration management plan that has a process for controlling changes to configuration items.

The organization being inspected/assessed will implement a configuration management plan for the information system that defines the configuration items.

The organization being inspected/assessed must implement a plan to protect the configuration management plan from unauthorized disclosure and modification.

Measures must include marking, labeling, and handling to prevent improper disclosure. The organization being inspected/assessed must ensure that all changes to the CM plan are approved.

The organization being inspected/assessed configures the information system to enter a safe mode of operation with restrictions of safe mode of operation defined in CP-12, CCI 2857 when conditions defined in CP-12, CCI 2856 are detected.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2855.

The organization being inspected/assessed establishes an alternate storage site and documents relevant information within the contingency plan.

The organization being inspected/assessed tests and logs backup information at least monthly in accordance with contingency plan to verify media reliability and information integrity.

The organization being inspected/assessed will protect the confidentiality, integrity, and availability of backup information at the storage location IAW the system security plan.

The organization being inspected/assessed configures the information system to uniquely identify and authenticate organizational users (or processes acting on behalf of organizational users).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 764.

The organization being inspected/assessed configures the information system to implement multifactor authentication for network access to privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 765.

The organization being inspected/assessed configures the information system to implement multifactor authentication for remote access to privileged accounts such that one of the factors is provided by a device separate from the system gaining access.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1948.

The organization being inspected/assessed will use DoD PKI or a technology approved by their Authorizing Official that meet Federal standards for authentication such as FIPS 140-2, NIAP Certification, or NSA approval.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1949.

The organization being inspected/assessed configures the information system to implement multifactor authentication for remote access to non-privileged accounts such that one of the factors is provided by a device separate from the system gaining access.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1951.

The organization being inspected/assessed will use DoD PKI or a technology approved by their Authorizing Official that meet Federal standards for authentication such as FIPS 140-2, NIAP Certification, or NSA approval.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1952.

The organization being inspected/assessed configures the information system to implement multifactor authentication for network access to non-privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 766.

The organization being inspected/assessed configures the information system to implement multifactor authentication for local access to privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 767.

The organization being inspected/assessed configures the information system to implement multifactor authentication for local access to non-privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 768.

The organization being inspected/assessed requires individuals or configures the information system to require individuals to be authenticated with an individual authenticator when a group authenticator is employed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 770.

The organization being inspected/assessed configures the information system to implement replay-resistant authentication mechanisms for network access to privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1941.

The organization being inspected/assessed configures the information system to implement replay-resistant authentication mechanisms for network access to non-privileged accounts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1942.

The organization being inspected/assessed configures the network infrastructure to identify all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 778.

DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).

The organization being inspected/assessed configures the network infrastructure to authenticate all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs) before establishing a local, remote, network connection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1958.

DoD has defined the value as all mobile devices and network connected endpoint devices (including but not limited to: workstations, printers, servers (outside a datacenter), VoIP Phones, VTC CODECs).

The organization being inspected/assessed configures the information system to use cryptographically based bidirectional authentication.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1967.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of upper case characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 192.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of lower case characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 193.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of numeric characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 194.

The organization being inspected/assessed configures the information system to enforce minimum password length.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 205.

The organization being inspected/assessed configures the information system to enforce password complexity by the minimum number of special characters used.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1619.

The organization being inspected/assessed configures the information system to enforce that at least 50% of the minimum password length is changed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 195.

DoD has defined the minimum number of characters as 50% of the minimum password length.

The organization being inspected/assessed configures the information system to store only encrypted representations of passwords.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 196.

The organization being inspected/assessed configures the information system to transmit only encrypted representations of passwords.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 197.

The organization being inspected/assessed configures the information system to enforce minimum password lifetime restrictions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 198.

The organization being inspected/assessed configures the information system to enforce maximum password lifetime restrictions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 199.

The organization being inspected/assessed configures the information system to prohibit reuse for a minimum of 5 generations.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 200.

DoD has defined the number of generations as a minimum of 5.

The organization being inspected/assessed configures the information system to allow the use of a temporary password for system logons with an immediate change to a permanent password.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2041.

The information system performing hardware token-based authentication must be configured to accept only DoD-approved PKI credentials in accordance with DoDI 8520.02 and DoDI 8520.03. For unclassified systems, DoD-approved PKI credentials include DoD PKI credentials, External Certification Authority (ECA) PKI credentials, and DoD-approved external PKI credentials. For SIPRNet, DoD-approved PKI credentials include DoD PKI credentials and NSS PKI credentials.

If the information system accepts DoD-approved external PKI credentials, the information system must be configured to accept only certificates at approved assurance levels, as represented by the Certificate Policy Object Identifiers (OIDs) asserted in the certificate. The current list of DoD-approved external PKIs and acceptable Object Identifiers (OIDs) for each approved external PKI is available at <http://iase.disa.mil/pki-pke/interoperability>.

The organization being inspected/assessed configures the information system to prohibit the use of cached authenticators after an organization defined time period.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2007.

The information system performing hardware token-based authentication must be configured to validate DoD-approved PKI credentials in accordance with RFC 5280.

The information system must be configured to perform a revocation check as part of the certificate validation process. Revocation checking may be performed using certificate revocation lists (CRLs) published by the issuing PKI or Online Certificate Status Protocol (OCSP) services.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 185.

Information systems must not have access to users' private keys. The cryptographic container in which the private keys are stored (e.g. smart card or software module) implements access controls and protections to ensure that only the authorized user can activate the private key. DoD users agree to protect their PKI credentials in accordance with the DD-2842 agreement that is executed for each credential. They are reminded of these responsibilities in annual IA training.

The private key identifying the information system must be stored in a cryptographic container that is FIPS 140-2 validated. Only authorized information system operators should have access to activation data (e.g. password or PIN) for the private key.

The information system performing PKI-based authentication must be configured to map the authenticated PKI credential to a corresponding network or information system account or role in accordance with DoDI 8520.03.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 187.

The organization being inspected/assessed implements automated tools to check passwords strength per the complexity requirements defined in IA-5 (1) Part A.

The organization being inspected/assessed configures the information system to implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable federal laws, Executive Orders, directives, policies, regulations, standards, and guidance for such authentication.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 803.

The organization being inspected/assessed configures the information system to uniquely identify and authenticate non-organizational users (or processes acting on behalf of non-organizational users).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 804.

The information system performing hardware token-based authentication must be configured to accept DoD-approved external PKI PIV credentials to authenticate federal agency users in accordance with DoDI 8520.02 and DoDI 8520.03. The information system must be configured to accept only certificates at approved assurance levels, as represented by the Certificate Policy Object Identifiers (OIDs) asserted in the certificate. The current list of DoD-approved external PKIs and acceptable Object Identifiers (OIDs) for each approved external PKI is available at <http://iase.disa.mil/pki-pke/interoperability>.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2009.

The information system performing hardware token-based authentication must be configured to validate DoD-approved external PKI PIV credentials to authenticate federal agency users in accordance with RFC 5280.

The information system must be configured to perform a revocation check as part of the certificate validation process. Revocation checking may be performed using certificate revocation lists (CRLs) published by the issuing PKI or Online Certificate Status Protocol (OCSP) services.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2010.

The organization being inspected/assessed will implement an automated intra-organization incident response information sharing capability to provide the following incident related information and support, for example:

1. SOP for incident reporting
2. Incident handling FAQ
3. Current incident activity awareness information
4. Incident response contact information
5. Incident report submission

The organization being inspected/assessed must establish a formal agreement with a computer network defense service provider (CNDSP).

The organization being inspected/assessed sanitizes equipment to remove all information from associated media prior to removal from organizational facilities for off-site maintenance or repairs IAW DoDM 5200.01-V3 for classified media and DoDM 5200.01-V4 for unclassified media.

The organization must maintain written records of media sanitization.

The organization being inspected/assessed configures the information system to audit nonlocal maintenance and diagnostic sessions' organization-defined audit events.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2884.

The organization being inspected/assessed documents and implements a process to review the records of the nonlocal maintenance and diagnostic sessions.

The organization must maintain a record of reviews.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the integrity of nonlocal maintenance and diagnostic communications.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2890.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to protect the confidentiality of nonlocal maintenance and diagnostic communications.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 3123.

The organization being inspected/assessed configures the information system to implement remote disconnect verification at the termination of nonlocal maintenance and diagnostic sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2891.

The organization being inspected/assessed configures the information system to employ strong authenticators in the establishment of nonlocal maintenance and diagnostic sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 877.

The organization being inspected/assessed terminates session and network connections when non-local maintenance is completed.

The organization must retain audit logs of session and network connections termination for non-local maintenance.

The organization being inspected/assessed protects and controls information system media during transport outside of controlled areas using security measures defined in DoDM 5200.01 M Vol. 1-4 and DoDD 5015.2.

The organization being inspected/assessed does not use portable storage devices in organization information systems when such devices have no identifiable owner.

The organization being inspected/assessed will protect emergency power shutoff capability.

DoD has defined the location as near more than one egress point of the IT area and ensures it is labeled and protected by a cover to prevent accidental shut-off.

The organization being inspected/assessed implements an uninterruptible power supply with sufficient capacity to support orderly shutdown of the system or transition the system to long-term alternate power in the event of a primary power source loss.

The organization being inspected/assessed authorizes and maintains authorization records of all system components entering and exiting the facility.

DoD has defined the types of information system components as all system components.

The organization being inspected/assessed will develop and maintain a list of personnel with authorized access to the facilities where information systems reside.

The organization will also take action to identify and officially designate its publicly accessible areas where access authorization is not required.

The organization being inspected/assessed formally approves a list of individuals currently authorized to access the facility where the information system resides.

The organization being inspected/assessed maintains a list of individuals currently authorized to access the facility where the information system resides.

The organization being inspected/assessed utilizes the list of personnel with authorized access (IAW PE-2, CCI-000912) and issues credentials accordingly.

The organization must document the credential issuing activity as an audit trail.

The organization being inspected/assessed will remove personnel from the authorized access list who no longer have approved access and revoke their credentials, as identified in actions per PE-2, CCI 914.

The organization must document each removal and revocation action as an audit trail.

The organization being inspected/assessed will implement physical access authorizations at entry/exit points defined in PE-3, CCI 2915 and secure those physical access points (i.e. doors and/or windows) that are not intended for normal access.

The organization being inspected/assessed will control ingress/egress to the facility using the physical access control devices and/or guards defined in PE-3, CCI 2916.

The organization being inspected/assessed maintains physical access audit logs for entry/exit points defined in PE-3, CCI 2918.

The organization being inspected/assessed provides security safeguards defined in PE-3, CCI 2920 to control access to areas within the facility officially designated as publicly accessible.

The organization must document which areas are officially designated as publicly accessible.

The organization being inspected/assessed documents and implements a process to escort visitors during circumstances defined in PE-3, CCI 2922 requiring visitor escorts.

The organization being inspected/assessed documents and implements a process to monitor visitor activity during circumstances defined in PE-3, CCI 2924 requiring visitor monitoring.

The organization being inspected/assessed will secure as appropriate (in safes or secure containers) items used for physical access control such as keys, combinations, portable locks, etc. Fixed access control devices such as card readers, installed locks, key pads, etc. should be protected from tampering.

The organization being inspected/assessed documents and implements a process to monitor physical access to the information system in addition to the physical access monitoring of the facility as physical spaces containing one or more components of the information system defined in PE-6 (4), CCI 2951.

The organization must maintain records of monitoring.

The information system owner will rescreen individuals according to system owner defined list of conditions requiring rescreening (CCI-001518) individuals for access to the information system and frequency (CCI - 001519) of such rescreening.

Rescreening actions will be maintained as an audit trail (AU-2).

The organization being inspected/assessed configures the information system to implement privileged access authorization to all information systems and infrastructure components for selected vulnerability scanning activities defined in RA-5 (5), CCI 2906.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1067.

DoD has defined the information system components as all information systems and infrastructure components.

The organization being inspected/assessed configures the information system to terminate the network connection associated with a communications session at the end of the session or after 10 minutes in band management and 15 minutes for user sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1133.

DoD has defined the time period as 10 minutes in band management and 15 minutes for user sessions.

The organization being inspected/assessed configures the information system to provide an explicit indication of use to users physically present at collaborative computing devices.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1152.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2465.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2466.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2467.

The organization being inspected/assessed configures the:

1. recursive/caching name server software to enable DNSSEC;
2. software to enable DNSSEC validation; and
3. software to establish a secure entry point trust anchor by installing key signing keys in the software configuration of trusted keys.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that determines the name server software configuration files and pertains to CCI 2468.

The organization being inspected/assessed implements a name service resolution architecture consisting of primary and secondary servers.

The organization must document the architecture in the site security plan.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1182.

The organization being inspected/assessed implements a name service resolution architecture where recursive and authoritative server software is not installed on the same information system.

The organization must document the architecture in the site security plan.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1183.

The organization being inspected/assessed configures the information system to protect the authenticity of communications sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1184.

The organization being inspected/assessed configures the information system to invalidate session identifiers upon user logout or other session termination.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1185.

The organization being inspected/assessed configures the information system to generate unique session identifiers for each session with randomness requirements defined in SC-23 (3), CCI 1189.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1188.

The organization being inspected/assessed configures the information system to recognize only session identifiers that are system-generated.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1664.

The organization being inspected/assessed configures the information system to allow the use of DoD PKI established certificate authorities for verification of the establishment of protected sessions.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2470.

DoD has defined the certificate authorities as DoD PKI established certificate authorities.

The organization being inspected/assessed configures the information system to fail to a secure state for failures during system initialization, shutdown, and aborts.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1190.

DoD has defined the known state as secure state.

DoD has defined the types of failures as failures during system initialization, shutdown, and aborts.

The organization being inspected/assessed configures the information system to preserve information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes in the event of a system failure.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1665.

DoD has defined system state information as information necessary to determine cause of failure and to return to operations with least disruption to mission/ business processes.

The organization being inspected/assessed configures the information system to protect the confidentiality and/or integrity of organization-defined information at rest.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1199.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized modification of information at rest defined in SC-28 (1), CCI 2473 on any information system components storing data defined in SC-28 (1), 2473.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2475.

DoD has defined the information system components as any information system components storing data defined in SC-28 (1), 2473.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information at rest defined in SC-28 (1), CCI 2473 on any information system components storing data defined in SC-28 (1), 2473.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2476.

DoD has defined the information system components as any information system components storing data defined in SC-28 (1), 2473.

The organization being inspected/assessed implements operations security safeguards defined in SC-38, CCI 2528 to protect key organizational information throughout the system development life cycle.

The organization must maintain an audit trail of security safeguard implementation.

The organization being inspected/assessed physically disables or removes connection ports or input/output devices defined in SC-41, CCI 2545 on information systems or information system components defined in SC-41, CCI 2544.

The organization being inspected/assessed configures the information system to protect against or limits the effects of types of denial of service attacks defined in SC-5, CCI 1093 by employing security safeguards defined in SC-5, CCI 2386.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2385.

The organization being inspected/assessed configures the information system to restrict the ability of individuals to launch denial of service attacks defined in SC-5 (1), CCI 2387 against other information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1094.

The organization being inspected/assessed configures the information system to manage excess capacity, bandwidth, or other redundancy to limit the effects of information flooding types of denial of service attacks.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1095.

The organization being inspected/assessed implements the monitoring tools defined in SC-5 (3), CCI 2388 to detect indicators of denial of service attacks against the information system.

The organization being inspected/assessed documents and implements mechanisms to prevent the unauthorized exfiltration of information across managed interfaces.

The organization being inspected/assessed configures the information system to allow incoming communications from authorized sources defined in SC-7 (11), CCI 2401 routed to authorized destinations defined in SC-7 (11), CCI 2402.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2403.

The organization being inspected/assessed implements McAfee Host Intrusion Prevention (HIPS) on all information system components.

DoD has defined the host-based boundary protection mechanisms as McAfee Host Intrusion Prevention (HIPS).

DoD has defined the information system components as all information system components.

The organization being inspected/assessed designs the information system to isolate key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc. from other internal information system components by implementing physically separate subnetworks with managed interfaces to other components of the system.

DoD has defined the key information security tools, mechanisms, and support components as key information security tools, mechanisms, and support components such as, but not limited to PKI, Patching infrastructure, HBSS, CND Tools, Special Purpose Gateway, vulnerability tracking systems, honeypots, internet access points (IAPs); network element and data center administrative/management traffic; Demilitarized Zones (DMZs), Server farms/computing centers, centralized audit log servers etc.

The organization being inspected/assessed documents and implements mechanisms to protect against unauthorized physical connections at internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.

DoD has defined the managed interfaces as internet access points, enclave LAN to WAN, cross domain solutions, and any DoD Approved Alternate Gateways.

The organization being inspected/assessed documents and implements information system access control mechanisms to limit the number of external connections to the information system.

The organization being inspected/assessed documents and implements mechanisms to protect the confidentiality and integrity of the information being transmitted across each interface for each external telecommunication service.

The organization being inspected/assessed configures the information system to deny network communications traffic at managed interfaces by default and allows network communications traffic by exception (i.e., deny all, permit by exception).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1109.

The organization being inspected/assessed configures the information system to prevent the device from simultaneously establishing non-remote connections with the system and communicating via some other connection to resources in external networks.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2397.

The organization being inspected/assessed configures the information system to detect outgoing communications traffic posing a threat to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2398.

The organization being inspected/assessed configures the information system to deny outgoing communications traffic posing a threat to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2399.

The organization being inspected/assessed configures the information system to audit the identity of internal users associated with denied outgoing communications traffic posing a threat to external information systems.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2400.

The organization being inspected/assessed documents and implements processes to monitor and control communications at the external boundary of the system and at key internal boundaries within the system.

The organization must maintain an audit trail of monitoring activities.

The organization being inspected/assessed designs the information system to leverage subnetworks so that publicly accessible system components are physically and/or logically separated from internal organizational networks.

The organization being inspected/assessed designs the information system to enforce requirements that components connect to external networks or information systems only through managed interfaces consisting of boundary protection devices arranged in accordance with an organizational security architecture.

The organization being inspected/assessed configures the information system to protect the confidentiality and/or integrity of transmitted information.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2418.

The organization being inspected/assessed configures the information system to implement cryptographic mechanisms to prevent unauthorized disclosure of information and detect changes to information during transmission unless otherwise protected by Protected Distribution System (PDS).

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2421.

DoD has defined the selection as both prevention of unauthorized disclosure and detection of changes to information.

DoD has defined the alternative physical safeguards as Protected Distribution System (PDS).

The organization being inspected/assessed configures the information system to maintain the confidentiality and integrity of information during preparation for transmission.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2420.

DoD has defined the selection as both confidentiality and integrity.

The organization being inspected/assessed configures the information system to maintain the confidentiality and integrity of information during reception.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2422.

DoD has defined the selection as both confidentiality and integrity.

The organization being inspected/assessed configures the information system to check the validity of all inputs except those identified specifically by the organization.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1310.

DoD has defined the information inputs as all inputs except those identified specifically by the organization.

The organization being inspected/assessed configures the information system to generate error messages that provide information necessary for corrective actions without revealing information that could be exploited by adversaries.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1312.

The organization being inspected/assessed configures the information system to reveal error messages only to the ISSO, ISSM, and SCA.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1314.

DoD has defined the personnel or roles as the ISSO, ISSM, and SCA.

The organization being inspected/assessed configures the information system to implement security safeguards defined in SI-16, CCI 2823 to protect its memory from unauthorized code execution.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2824.

The organization being inspected/assessed configures the information system to implement fail-safe procedures defined in SI-17, CCI 2773 when failure conditions defined in SI-17, CCI 2774 occur.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2775.

The organization being inspected/assessed configures the information system to remove all upgraded/replaced software components that are no longer required for operation (e.g., previous versions) after updated versions have been installed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2617.

DoD has defined the software components as all upgraded/replaced software components that are no longer required for operation.

The organization being inspected/assessed configures the information system to remove all upgraded/replaced firmware components that are no longer required for operation e.g., previous versions) after updated versions have been installed.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2618.

DoD has defined the firmware components as all upgraded/replaced firmware components that are no longer required for operation.

The organization being inspected/assessed documents and implements a process to identify information system flaws.

The process shall include review of the system through automated scans and manual checks to determine the existence of flaws such as IAVM, CVE, or other resources.

The organization being inspected/assessed reports information system flaws according to DoD Cybersecurity policy and organizational roles and responsibilities.

The organization must report information system flaws in their POA&M.

The organization being inspected/assessed documents and implements a process to test software updates related to flaw remediation for effectiveness before installation.

If the software update is being provided by a vendor who has documented the effectiveness of the update in fixing the affected IAVM/CVE, further testing by the organization may not be required.

The organization being inspected/assessed documents and implements a process for regression testing IAW CM-4 to identify any potential side effects before installation of software updates.

The organization being inspected/assessed documents and implements a process to test firmware updates related to flaw remediation for effectiveness before installation.

If the firmware update is being provided by a vendor who has documented the effectiveness of the update in fixing the affected IAVM/CVE, further testing by the organization may not be required.

The organization being inspected/assessed documents and implements a process for regression testing IAW CM-4 to identify any potential side effects before installation of software updates.

The organization being inspected/assessed configures the information system to install security-relevant software updates within 30 days of the release of the updates

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2605.

DoD has defined the time period as 30 days.

The organization being inspected/assessed configures the information system to install security-relevant firmware updates within 30 days of the release of the updates.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2607.

DoD has defined the time period as 30 days.

The organization being inspected/assessed documents and implements tools defined in SI-3 (10), CCI 2634 to analyze the characteristics and behavior of malicious code.

The organization being inspected/assessed documents and implements techniques defined in SI-3 (10), CCI 2635 to analyze the characteristics and behavior of malicious code.

The organization being inspected/assessed configures the information system to automatically update malicious code protection mechanisms.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1247.

The organization being inspected/assessed configures the malicious code protection mechanisms identified in SI-3, CCI 2619 to eradicate malicious code.

The organization being inspected/assessed configures the malicious code protection mechanisms identified in SI-3, CCI 2620 to eradicate malicious code.

The organization being inspected/assessed configures the information system to update malicious code protection mechanisms whenever new releases are available in accordance with organizational configuration management policy and procedures.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1240.

The organization being inspected/assessed configures malicious code protection mechanisms to perform periodic scans of the information system on every 7 days.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1241.

DoD has defined the frequency as every 7 days.

The organization being inspected/assessed configures malicious code protection mechanisms to perform real-time scans of files from external sources at endpoints as the files are downloaded, opened, or executed in accordance with organizational security policy.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1242.

The organization being inspected/assessed configures the malicious code protection mechanisms to perform real-time scans of files from external sources at network entry/exit points as the files are downloaded, opened, or executed in accordance with organizational security policy.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2624.

The organization being inspected/assessed configures malicious code protection mechanisms to perform block and quarantine malicious code and then send an alert to the administrator immediately in near real-time in response to malicious code detection.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1243.

DoD has defined the actions as block and quarantine malicious code and then send an alert to the administrator immediately in near real-time.

The organization being inspected/assessed configures the information system to address the receipt of false positives during malicious code detection and eradication, and the resulting potential impact on the availability of the information system.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 1245.

The organization being inspected/assessed prohibits the use of binary or machine-executable code obtained from sources without vendor support or with no warranty and without the provision of source code.

The organization being inspected/assessed configures the information system to provide the capability to audit the event upon detection of a potential integrity violation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2723.

The organization being inspected/assessed configures the information system to initiate one or more of following actions: generates an audit record; alerts current user; alerts at a minimum, the ISSO and ISSM; and/or other actions defined in SI-7 (8), CCI 2722 upon detection of a potential integrity violation.

For information system components that have applicable STIGs or SRGs, the organization being inspected/assessed must comply with the STIG/SRG guidance that pertains to CCI 2724.

DoD has defined the personnel or roles as at a minimum, the ISSO and ISSM.

Classification	Responsibility
Applicable	DoD Defined
Impractical	Contractor
Impractical	Contractor
Impractical	Contractor
Impractical	Contractor
Applicable	Owner

Reference: UFC 4-010-06 Cybersecurity of Facility-Related Control Systems, with Change 1
<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc/ufc-4-010-06>

Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Applicable	Owner	
Applicable	Owner	

Impractical	Contractor	
Impractical	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	

Applicable	Contractor, Owner	
Applicable	Contractor	
Applicable	Contractor	
Impractical	Owner	
Impractical	Contractor	
Impractical	DoD Defined	
Impractical	Contractor, Owner	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Owner, Contractor	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor	
Applicable	Contractor, Owner	

Impractical	DoD Defined	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Impractical	Contractor	
Applicable	DoD Defined	

Applicable	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor	
Impractical	DoD Defined	

Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	

Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	

Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Impractical	Contractor, Owner	

Applicable	Contractor, Owner	
Impractical	DoD Defined	
Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor, Owner	
Impractical	Contractor	

Impractical	DoD Defined	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Impractical	Contractor	
Impractical	Contractor	

Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor	
Applicable	Owner, Contractor	

Impractical	DoD Defined	
Impractical	Contractor, Owner	
Impractical	DoD Defined	
Applicable	Owner	
Applicable	Owner	
Applicable	Contractor	
Impractical	Contractor	
Impractical	Contractor, Owner	
Applicable	DoD Defined	
Impractical	Owner	

Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	

Applicable	Owner	
Applicable	Contractor	
Impractical	Owner, Contractor	
Impractical	Owner	
Applicable	Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor	

Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	Contractor, Owner	

Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	DoD Defined	
Applicable	Contractor, Owner	

Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Owner, Contractor	
Applicable	DoD Defined	

Impractical	Contractor, Owner	
Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Impractical	Contractor	
Applicable	Contractor, Owner	

Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Applicable	Owner, Contractor	
Impractical	Owner, Contractor	

Impractical	Contractor	
Applicable	Contractor	
Applicable	Owner, Contractor	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	

Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Impractical	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner	
Applicable	Owner, Contractor	
Applicable	Owner, Contractor	

Applicable	Owner	
Impractical	DoD Defined	
Applicable	DoD Defined	
Impractical	Contractor	
Applicable	Contractor	

Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Impractical	Contractor, Owner	
Applicable	Contractor	

Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	

Applicable	DoD Defined	
Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Owner, Contractor	

Applicable	Contractor	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor	
Impractical	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor	

Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	DoD Defined	
Applicable	Contractor, Owner	
Applicable	Contractor, Owner	
Impractical	Contractor	

Applicable	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Applicable	Contractor, Owner	
Applicable	Designer, Owner	
Applicable	Designer, Owner	

Applicable	Contractor	
Impractical	DoD Defined	
Applicable	DoD Defined	
Applicable	DoD Defined	
Impractical	DoD Defined	

Applicable	Contractor	
Impractical	DoD Defined	
Impractical	Contractor	
Applicable	Contractor	
Impractical	DoD Defined	

Impractical	DoD Defined	
Applicable	Contractor, Owner	
Applicable	DoD Defined	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	Owner	
Impractical	DoD Defined	

Impractical	DoD Defined	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	DoD Defined	

Impractical	Contractor	
Impractical	Contractor	
Applicable	DoD Defined	
Impractical	Contractor, Owner	
Impractical	Contractor, Owner	
Impractical	Contractor	

Impractical	DoD Defined	
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